Fluidum: the derivations of the generated matrices

Each section is written by the derivation script it describes, from the matrices, sources and check results that script computed:

- `deriv/derive_2p1d_viscous.wls`
- `deriv/derive_hq_2p1d.wls`
- `deriv/derive_two_way.wls`
- `deriv/derive_1d_matrices.wls`
- `deriv/derive_upstream.wls`
- `deriv/derive_drivers.wls`

Regenerate with `deriv/render_all.sh`.

Conventions: boost-invariant Milne coordinates; every system is written source on the left, $A_t\,\partial_\tau\phi + A_x\,\partial_x\phi + A_y\,\partial_y\phi + S = 0$; only nonzero entries are listed; $u^\tau = \sqrt{1+u_\perp^2}$ is written for the root wherever it occurs.

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1 Viscous hydrodynamics: Israel–Stewart with $\delta_{\pi\pi}$

Script: `deriv/derive_2p1d_viscous.wls`

Writes: `two_d_viscous_matrix_derived` in `src/Matrix/2d_viscous.jl`; the 1+1D system is the control, compared entry by entry with `one_d_viscous_matrix` (V-G4).

1.1 Setup

Milne coordinates, boost invariant. Two charts:

$$x^\mu = (\tau, r, \phi, \eta), \qquad g_{\mu\nu} = \mathrm{diag}(-1, 1, r^2, \tau^2)$$
$$x^\mu = (\tau, x, y, \eta), \qquad g_{\mu\nu} = \mathrm{diag}(-1, 1, 1, \tau^2)$$

$$T^{\mu\nu} = (e + p + \Pi)\, u^\mu u^\nu + (p + \Pi)\, g^{\mu\nu} + \pi^{\mu\nu}, \qquad e = T\, p'(T) - p(T)$$
$$\nabla_\mu T^{\mu\nu} = 0$$
$$\tau_\pi\, \Delta^{\mu\nu}_{\alpha\beta}\, D\pi^{\alpha\beta} + \pi^{\mu\nu} + 2\eta\, \sigma^{\mu\nu} + \delta_{\pi\pi}\, \pi^{\mu\nu}\theta = 0$$
$$\tau_\Pi\, D\Pi + \Pi + \zeta\, \theta = 0$$

with

$$D = u^\mu \nabla_\mu, \qquad \theta = \nabla_\mu u^\mu, \qquad \Delta^{\mu\nu} = g^{\mu\nu} + u^\mu u^\nu$$
$$\sigma^{\mu\nu} = \Delta^{\mu\nu}_{\alpha\beta} \nabla^\alpha u^\beta, \qquad \Delta^{\mu\nu}_{\alpha\beta} = \tfrac{1}{2}\left(\Delta^\mu_\alpha \Delta^\nu_\beta + \Delta^\mu_\beta \Delta^\nu_\alpha\right) - \tfrac{1}{3}\Delta^{\mu\nu}\Delta_{\alpha\beta}$$

η, τ_π, ζ, τ_Π are not differentiated. Quasi-linear form, source on the left:

$$A_t\, \partial_\tau \phi + A_r\, \partial_r \phi + S = 0 \quad \text{(1+1D)}$$
$$A_t\, \partial_\tau \phi + A_x\, \partial_x \phi + A_y\, \partial_y \phi + S = 0 \quad \text{(2+1D)}$$

The $\delta_{\pi\pi}$ term is linear, so each matrix and the source split as

$$A = A^{\mathrm{IS}} + \delta_{\pi\pi}\, A^{(\delta)}, \qquad S = S^{\mathrm{IS}} + \delta_{\pi\pi}\, S^{(\delta)}$$

1.2 Steps

1.2.1 1+1D: Christoffels and constraints

$$\Gamma^\tau{}_{\eta\eta} = \tau, \quad \Gamma^r{}_{\phi\phi} = -r, \quad \Gamma^\phi{}_{r\phi} = \frac{1}{r}, \quad \Gamma^\eta{}_{\tau\eta} = \frac{1}{\tau}$$

$$u^\tau = \sqrt{(u^r)^2 + 1}$$

$u_\mu \pi^{\mu\nu} = 0$, $\pi^\mu{}_\mu = 0$:

$$\pi^{rr} = -\left((u^\tau)^2\, (\pi^\eta{}_\eta + \pi^\phi{}_\phi)\right)$$
$$\pi^{\tau r} = -(u^r u^\tau (\pi^\eta{}_\eta + \pi^\phi{}_\phi))$$
$$\pi^{\tau\tau} = -\left((u^r)^2\, (\pi^\eta{}_\eta + \pi^\phi{}_\phi)\right)$$
$$\pi^{\phi\phi} = \frac{\pi^\phi{}_\phi}{r^2}$$
$$\pi^{\eta\eta} = \frac{\pi^\eta{}_\eta}{\tau^2}$$

1.2.2 1+1D: velocity gradients

$$
\begin{aligned}
\theta &= \frac{(\partial_\tau u^r)\tau u^r + (u^r)^2 + 1}{\tau u^\tau} + (\partial_r u^r) + \frac{u^r}{r} \\
\nabla_\tau u^\tau &= \frac{(\partial_\tau u^r)u^r}{u^\tau} \\
\nabla_\tau u^r &= (\partial_\tau u^r) \\
\nabla_r u^\tau &= \frac{(\partial_r u^r)u^r}{u^\tau} \\
\nabla_r u^r &= (\partial_r u^r) \\
\nabla_\phi u^\phi &= \frac{u^r}{r} \\
\nabla_\eta u^\eta &= \frac{u^\tau}{\tau} \\
\sigma^{\tau\tau} &= -\frac{(u^r)^2\left(-2(\partial_\tau u^r)r\tau u^r - 2(\partial_r u^r)r\tau u^\tau + r(u^r)^2 + r + \tau u^r u^\tau\right)}{3r\tau u^\tau} \\
\sigma^{\tau r} &= -\frac{u^r\left(r\left(-2(\partial_\tau u^r)\tau u^r - 2(\partial_r u^r)\tau u^\tau + (u^r)^2 + 1\right) + \tau u^r u^\tau\right)}{3r\tau} \\
\sigma^{rr} &= \frac{2(\partial_\tau u^r)r\tau u^r u^\tau + 2(\partial_r u^r)r\tau (u^\tau)^2 - r(u^r)^2 u^\tau - r u^\tau - \tau (u^r)^3 - \tau u^r}{3r\tau} \\
\sigma^{\phi\phi} &= \frac{-(\partial_\tau u^r)r\tau u^r - (\partial_r u^r)r\tau u^\tau - r(u^r)^2 - r + 2\tau u^r u^\tau}{3r^3\tau u^\tau} \\
\sigma^{\eta\eta} &= \frac{-(\partial_\tau u^r)r\tau u^r - (\partial_r u^r)r\tau u^\tau + 2r(u^r)^2 + 2r - \tau u^r u^\tau}{3r\tau^3 u^\tau}
\end{aligned}
$$

1.2.3 1+1D: conservation

$$
\begin{aligned}
T^{\tau\tau} &= p'(T)(u^r)^2 T + p'(T)T + \Pi (u^r)^2 - \pi^\eta{}_\eta (u^r)^2 - \pi^\phi{}_\phi (u^r)^2 - p \\
T^{\tau r} &= u^r u^\tau (p'(T)T + \Pi - \pi^\eta{}_\eta - \pi^\phi{}_\phi) \\
T^{rr} &= (u^r)^2 (p'(T)T + \Pi) + \Pi - (u^\tau)^2 (\pi^\eta{}_\eta + \pi^\phi{}_\phi) + p \\
T^{\phi\phi} &= \frac{\Pi + \pi^\phi{}_\phi + p}{r^2} \\
T^{\eta\eta} &= \frac{\Pi + \pi^\eta{}_\eta + p}{\tau^2}
\end{aligned}
$$

$$
\begin{aligned}
\nabla_\mu T^{\mu\tau} &= (\partial_\tau T^{\tau\tau}) + (\partial_r T^{\tau r}) + \frac{T^{\tau r}}{r} + \frac{T^{\tau\tau}}{\tau} + \tau T^{\eta\eta} \\
\nabla_\mu T^{\mu r} &= (\partial_\tau T^{\tau r}) + (\partial_r T^{rr}) + \frac{T^{rr}}{r} - r T^{\phi\phi} + \frac{T^{\tau r}}{\tau}
\end{aligned}
$$

1.2.4 1+1D: shear and bulk

$$
\begin{aligned}
(D\pi)^{\tau\tau} &= -u^r\Big(2(\partial_r u^r)u^r(\pi^\eta{}_\eta + \pi^\phi{}_\phi) + (\partial_x \pi^\eta{}_\eta)(u^r)^2 + (\partial_r \pi^\phi{}_\phi)(u^r)^2 + 2(\partial_\tau u^r)\pi^\eta{}_\eta u^\tau \\
&\quad + 2(\partial_\tau u^r)\pi^\phi{}_\phi u^\tau + (\partial_\tau \pi^\eta{}_\eta)u^r u^\tau + (\partial_\tau \pi^\phi{}_\phi)u^r u^\tau\Big) \\
(D\pi)^{\tau r} &= -(\partial_\tau u^r)\left(2(u^r)^2 + 1\right)(\pi^\eta{}_\eta + \pi^\phi{}_\phi) \\
&\quad - \frac{1}{u^\tau}u^r\Big((\partial_r u^r)\left(2(u^r)^2 + 1\right)(\pi^\eta{}_\eta + \pi^\phi{}_\phi) \\
&\quad + (u^r)^2\left(u^\tau((\partial_\tau \pi^\eta{}_\eta) + (\partial_\tau \pi^\phi{}_\phi)) + (\partial_x \pi^\eta{}_\eta)u^r + (\partial_r \pi^\phi{}_\phi)u^r\right)\Big) \\
(D\pi)^{rr} &= u^\tau\left(-\left((u^\tau)^2((\partial_\tau \pi^\eta{}_\eta) + (\partial_\tau \pi^\phi{}_\phi))\right) - 2(\partial_\tau u^r)u^r(\pi^\eta{}_\eta + \pi^\phi{}_\phi)\right) \\
&\quad + u^r\left(-\left((u^\tau)^2((\partial_x \pi^\eta{}_\eta) + (\partial_r \pi^\phi{}_\phi))\right) - 2(\partial_r u^r)u^r(\pi^\eta{}_\eta + \pi^\phi{}_\phi)\right)
\end{aligned}
$$

$$(D\pi)^{\phi\phi} = \frac{(\partial_\tau \pi^\phi{}_\phi)u^\tau + (\partial_r \pi^\phi{}_\phi)u^r}{r^2}$$
$$(D\pi)^{\eta\eta} = \frac{(\partial_\tau \pi^\eta{}_\eta)u^\tau + (\partial_x \pi^\eta{}_\eta)u^r}{\tau^2}$$

$$\Delta^{\phi\phi}_{\alpha\beta} D\pi^{\alpha\beta} = \frac{(\partial_\tau \pi^\phi{}_\phi)u^\tau + (\partial_r \pi^\phi{}_\phi)u^r}{r^2}$$
$$\Delta^{\eta\eta}_{\alpha\beta} D\pi^{\alpha\beta} = \frac{(\partial_\tau \pi^\eta{}_\eta)u^\tau + (\partial_x \pi^\eta{}_\eta)u^r}{\tau^2}$$

$$E^{\phi\phi} = \tau_\pi \Delta^{\phi\phi}_{\alpha\beta} D\pi^{\alpha\beta} + \pi^{\phi\phi} + 2\eta\sigma^{\phi\phi} + \delta_{\pi\pi}\pi^{\phi\phi}\theta, \quad \phi \to \eta$$
$$E_\Pi = \tau_\Pi(u^\tau\partial_\tau + u^r\partial_r)\Pi + \Pi + \zeta\theta$$

1.2.5 1+1D: from the rows to the matrices

$$A^{(k)}_{ij} = \frac{\partial E_i}{\partial(\partial_k \phi_j)}, \qquad S_i = E_i\big|_{\partial\phi=0}, \qquad k = \tau, r$$

$$\begin{aligned}
E_1 = &-(\partial_\tau\Pi) - (\partial_\tau T)p'(T) + \frac{\Pi + \pi^\eta{}_\eta + p}{\tau} - 2(\partial_\tau u^r)u^r(\pi^\eta{}_\eta + \pi^\phi{}_\phi) \\
&- (u^r)^2((\partial_\tau \pi^\eta{}_\eta) + (\partial_\tau \pi^\phi{}_\phi)) + 2(\partial_\tau u^r)u^r(p'(T)T + \Pi) - \frac{(\partial_r u^r)(u^r)^2(\pi^\eta{}_\eta + \pi^\phi{}_\phi)}{u^\tau} \\
&+ \frac{(\partial_r u^r)(u^r)^2(p'(T)T + \Pi)}{u^\tau} - (\partial_r u^r)u^\tau(\pi^\eta{}_\eta + \pi^\phi{}_\phi) - u^r u^\tau((\partial_x \pi^\eta{}_\eta) + (\partial_r \pi^\phi{}_\phi)) \\
&+ (\partial_r u^r)u^\tau(p'(T)T + \Pi) + \frac{u^r u^\tau(p'(T)T + \Pi - \pi^\eta{}_\eta - \pi^\phi{}_\phi)}{r} \\
&+ u^r u^\tau((\partial_x\Pi) + (\partial_x T)(p''(T)T + p'(T))) \\
&+ (u^\tau)^2((\partial_\tau\Pi) + (\partial_\tau T)(p''(T)T + p'(T))) \\
&- \frac{-(u^\tau)^2(p'(T)T + \Pi) + \Pi + (u^r)^2(\pi^\eta{}_\eta + \pi^\phi{}_\phi) + p}{\tau}
\end{aligned}$$

$$E_5 = \tau_\Pi((\partial_\tau\Pi)u^\tau + (\partial_x\Pi)u^r) + \zeta\left(\frac{(\partial_\tau u^r)\tau u^r + (u^r)^2 + 1}{\tau u^\tau} + (\partial_r u^r) + \frac{u^r}{r}\right) + \Pi$$

1.2.6 2+1D

$$\Gamma^\tau{}_{\eta\eta} = \tau, \quad \Gamma^\eta{}_{\tau\eta} = \frac{1}{\tau}$$

$u_\mu \pi^{\mu\nu} = 0$, $\pi^\mu{}_\mu = 0$:

$$\begin{aligned}
\pi^{xx} &= -\frac{\pi^\eta{}_\eta (u^\tau)^2 - 2\pi^{xy}u^x u^y + \pi^{yy}(u^x)^2 + \pi^{yy}}{(u^y)^2 + 1} \\
\pi^{\tau x} &= \frac{-\pi^\eta{}_\eta (u^\tau)^2 u^x + 2\pi^{xy}(u^x)^2 u^y + \pi^{xy}(u^y)^3 + \pi^{xy}u^y - \pi^{yy}(u^x)^3 - \pi^{yy}u^x}{u^\tau\left((u^y)^2 + 1\right)} \\
\pi^{\tau y} &= \frac{\pi^{xy}u^x + \pi^{yy}u^y}{u^\tau} \\
\pi^{\tau\tau} &= \frac{-\pi^\eta{}_\eta (u^x)^2 + 2\pi^{xy}u^x u^y + \pi^{yy}\left((u^y)^2 - (u^x)^2\right)}{(u^y)^2 + 1}
\end{aligned}$$

$$\theta = \frac{(\partial_\tau u^x)u^x + (\partial_\tau u^y)u^y}{u^\tau} + (\partial_x u^x) + (\partial_y u^y) + \frac{u^\tau}{\tau}$$
$$\nabla_\tau u^\tau = \frac{(\partial_\tau u^x)u^x + (\partial_\tau u^y)u^y}{u^\tau}$$

$$
\begin{aligned}
\nabla_\tau u^x &= (\partial_\tau u^x) \\
\nabla_\tau u^y &= (\partial_\tau u^y) \\
\nabla_x u^\tau &= \frac{(\partial_x u^x)u^x + (\partial_x u^y)u^y}{u^\tau} \\
\nabla_x u^x &= (\partial_x u^x) \\
\nabla_x u^y &= (\partial_x u^y) \\
\nabla_y u^\tau &= \frac{(\partial_y u^x)u^x + (\partial_y u^y)u^y}{u^\tau} \\
\nabla_y u^x &= (\partial_y u^x) \\
\nabla_y u^y &= (\partial_y u^y) \\
\nabla_\eta u^\eta &= \frac{u^\tau}{\tau} \\
\sigma^{\tau\tau} &= \frac{1}{3\tau u^\tau}\Big(2(\partial_\tau u^x)\tau\left(u^x\right)^3 - \left(u^x\right)^4 + \tau u^x u^y(2(\partial_\tau u^x)u^y + 3u^\tau((\partial_x u^y) + (\partial_y u^x))) \\
&\quad + \left(u^x\right)^2\Big(2(\partial_\tau u^y)\tau u^y + 2(\partial_x u^x)\tau u^\tau - (\partial_y u^y)\tau u^\tau - 2\left(u^y\right)^2 - 1\Big) \\
&\quad - \left(u^y\right)^2\Big(-2(\partial_\tau u^y)\tau u^y + (\partial_x u^x)\tau u^\tau - 2(\partial_y u^y)\tau u^\tau + \left(u^y\right)^2 + 1\Big)\Big) \\
\sigma^{\tau x} &= \frac{1}{6\tau u^\tau}\Big(-2u^\tau u^x + 4(\partial_\tau u^x)\tau u^\tau\left(u^x\right)^2 - 2u^\tau\left(u^x\right)^3 + 3(\partial_x u^y)\tau u^y + 3(\partial_y u^x)\tau u^y \\
&\quad + (\partial_\tau u^y)\tau u^\tau u^x u^y + 3(\partial_x u^y)\tau\left(u^x\right)^2 u^y + 6(\partial_y u^x)\tau\left(u^x\right)^2 u^y + 3(\partial_\tau u^x)\tau u^\tau\left(u^y\right)^2 \\
&\quad - 2u^\tau u^x\left(u^y\right)^2 + 3(\partial_y u^x)\tau\left(u^y\right)^3 + (\partial_y u^y)\tau u^x\Big(-2\left(u^x\right)^2 + \left(u^y\right)^2 - 2\Big) \\
&\quad + (\partial_x u^x)\tau u^x\Big(4\left(u^x\right)^2 + \left(u^y\right)^2 + 4\Big)\Big) \\
\sigma^{\tau y} &= \frac{1}{6\tau u^\tau}\Big(3(\partial_\tau u^y)\tau u^\tau\left(u^x\right)^2 - 2(\partial_x u^x)\tau u^y + 4(\partial_y u^y)\tau u^y - 2u^\tau u^y + (\partial_\tau u^x)\tau u^\tau u^x u^y \\
&\quad + (\partial_x u^x)\tau\left(u^x\right)^2 u^y + (\partial_y u^y)\tau\left(u^x\right)^2 u^y - 2u^\tau\left(u^x\right)^2 u^y + 4(\partial_\tau u^y)\tau u^\tau\left(u^y\right)^2 \\
&\quad - 2(\partial_x u^x)\tau\left(u^y\right)^3 + 4(\partial_y u^y)\tau\left(u^y\right)^3 - 2u^\tau\left(u^y\right)^3 + 3(\partial_y u^x)\tau u^x\Big(\left(u^y\right)^2 + 1\Big) \\
&\quad + 3(\partial_x u^y)\tau u^x\Big(\left(u^x\right)^2 + 2\left(u^y\right)^2 + 1\Big)\Big) \\
\sigma^{xx} &= \frac{1}{3\tau u^\tau}\Big(-1 + 2(\partial_x u^x)\tau u^\tau - (\partial_y u^y)\tau u^\tau - \left(u^x\right)^4 - (\partial_\tau u^y)\tau u^y + 3(\partial_y u^x)\tau u^\tau u^x u^y \\
&\quad - \left(u^y\right)^2 - \left(u^x\right)^2\Big((\partial_\tau u^y)\tau u^y - 2(\partial_x u^x)\tau u^\tau + (\partial_y u^y)\tau u^\tau + \left(u^y\right)^2 + 2\Big) \\
&\quad + (\partial_\tau u^x)\tau u^x\Big(2\left(u^x\right)^2 + 3\left(u^y\right)^2 + 2\Big)\Big) \\
\sigma^{xy} &= \frac{1}{6\tau u^\tau}\Big(3(\partial_x u^y)\tau u^\tau + 3(\partial_y u^x)\tau u^\tau + 3(\partial_x u^y)\tau u^\tau\left(u^x\right)^2 - 2u^x u^y + (\partial_x u^x)\tau u^\tau u^x u^y \\
&\quad + (\partial_y u^y)\tau u^\tau u^x u^y - 2\left(u^x\right)^3 u^y + 3(\partial_y u^x)\tau u^\tau\left(u^y\right)^2 - 2u^x\left(u^y\right)^3 \\
&\quad + (\partial_\tau u^y)\tau u^x\Big(3\left(u^x\right)^2 + \left(u^y\right)^2 + 3\Big) + (\partial_\tau u^x)\tau u^y\Big(\left(u^x\right)^2 + 3\left(u^y\right)^2 + 3\Big)\Big) \\
\sigma^{yy} &= \frac{1}{3\tau u^\tau}\Big(3(\partial_x u^y)\tau u^\tau u^x u^y + \left(u^x\right)^2\Big(3(\partial_\tau u^y)\tau u^y - \left(u^y\right)^2 - 1\Big) \\
&\quad - (\partial_\tau u^x)\tau u^x\Big(\left(u^y\right)^2 + 1\Big) \\
&\quad - \Big(\left(u^y\right)^2 + 1\Big)\Big(-2(\partial_\tau u^y)\tau u^y + (\partial_x u^x)\tau u^\tau - 2(\partial_y u^y)\tau u^\tau + \left(u^y\right)^2 + 1\Big)\Big) \\
\sigma^{\eta\eta} &= \frac{-(\partial_\tau u^x)\tau u^x - (\partial_\tau u^y)\tau u^y - (\partial_x u^x)\tau u^\tau - (\partial_y u^y)\tau u^\tau + 2\left(u^x\right)^2 + 2\left(u^y\right)^2 + 2}{3\tau^3 u^\tau}
\end{aligned}
$$

$$
\begin{aligned}
T^{\tau\tau} &= \frac{1}{\left(u^y\right)^2 + 1}\Big(\Pi\left(u^x\right)^2 - \pi^\eta{}_\eta\left(u^x\right)^2 - \pi^{yy}\left(u^x\right)^2 + 2\pi^{xy}u^x u^y + \Pi\left(u^y\right)^2 + \pi^{yy}\left(u^y\right)^2 \\
&\quad + \Pi\left(u^x\right)^2\left(u^y\right)^2 + \Pi\left(u^y\right)^4 - p\Big(\left(u^y\right)^2 + 1\Big) + p'(T)T\left(u^\tau\right)^2\Big(\left(u^y\right)^2 + 1\Big)\Big) \\
T^{\tau x} &= \frac{1}{u^\tau\Big(\left(u^y\right)^2 + 1\Big)}\Big(-\pi^{yy}u^x + \Pi\left(u^\tau\right)^2 u^x - \pi^\eta{}_\eta\left(u^\tau\right)^2 u^x + p'(T)T\left(u^\tau\right)^2 u^x - \pi^{yy}\left(u^x\right)^3
\end{aligned}
$$

$$
\begin{aligned}
&+ \pi^{xy}u^y + 2\pi^{xy}\left(u^x\right)^2 u^y + \Pi\left(u^\tau\right)^2 u^x\left(u^y\right)^2 + p'(T)T\left(u^\tau\right)^2 u^x\left(u^y\right)^2 \\
&+ \pi^{xy}\left(u^y\right)^3\Big) \\
T^{\tau y} &= \frac{\left(u^\tau\right)^2 u^y(p'(T)T+\Pi) + \pi^{xy}u^x + \pi^{yy}u^y}{u^\tau} \\
T^{xx} &= \left(u^x\right)^2(p'(T)T+\Pi) + \Pi - \frac{\pi^\eta{}_\eta\left(u^\tau\right)^2 - 2\pi^{xy}u^xu^y + \pi^{yy}\left(u^x\right)^2 + \pi^{yy}}{\left(u^y\right)^2+1} + p \\
T^{xy} &= u^xu^y(p'(T)T+\Pi) + \pi^{xy} \\
T^{yy} &= \left(u^y\right)^2(p'(T)T+\Pi) + \Pi + p + \pi^{yy} \\
T^{\eta\eta} &= \frac{\Pi + \pi^\eta{}_\eta + p}{\tau^2}
\end{aligned}
$$

$$
\begin{aligned}
\nabla_\mu T^{\mu\tau} &= (\partial_\tau T^{\tau\tau}) + (\partial_x T^{\tau x}) + (\partial_y T^{\tau y}) + \frac{T^{\tau\tau}}{\tau} + \tau T^{\eta\eta} \\
\nabla_\mu T^{\mu x} &= (\partial_\tau T^{\tau x}) + (\partial_x T^{xx}) + (\partial_y T^{xy}) + \frac{T^{\tau x}}{\tau} \\
\nabla_\mu T^{\mu y} &= (\partial_\tau T^{\tau y}) + (\partial_x T^{xy}) + (\partial_y T^{yy}) + \frac{T^{\tau y}}{\tau}
\end{aligned}
$$

$$
\begin{aligned}
(D\pi)^{\tau\tau} = \frac{1}{\left(\left(u^y\right)^2+1\right)^2}\Big(&-2(\partial_\tau u^x)\pi^\eta{}_\eta u^\tau u^x + 2(\partial_\tau u^y)\pi^{xy}u^\tau u^x - 2(\partial_\tau u^x)\pi^{yy}u^\tau u^x \\
&- (\partial_\tau\pi^\eta{}_\eta)u^\tau\left(u^x\right)^2 - (\partial_\tau\pi^{yy})u^\tau\left(u^x\right)^2 - (\partial_x\pi^\eta{}_\eta)\left(u^x\right)^3 - (\partial_x\pi^{yy})\left(u^x\right)^3 \\
&+ 2(\partial_\tau u^x)\pi^{xy}u^\tau u^y + 2(\partial_\tau u^y)\pi^{yy}u^\tau u^y - 2(\partial_y u^x)\pi^\eta{}_\eta u^x u^y \\
&+ 2(\partial_y u^y)\pi^{xy}u^x u^y - 2(\partial_y u^x)\pi^{yy}u^x u^y + 2(\partial_\tau\pi^{xy})u^\tau u^x u^y \\
&+ 2(\partial_x\pi^{xy})\left(u^x\right)^2 u^y - (\partial_y\pi^\eta{}_\eta)\left(u^x\right)^2 u^y - (\partial_y\pi^{yy})\left(u^x\right)^2 u^y \\
&+ 2(\partial_\tau u^y)\pi^\eta{}_\eta u^\tau\left(u^x\right)^2 u^y + 2(\partial_\tau u^y)\pi^{yy}u^\tau\left(u^x\right)^2 u^y + 2(\partial_y u^x)\pi^{xy}\left(u^y\right)^2 \\
&+ 2(\partial_y u^y)\pi^{yy}\left(u^y\right)^2 + (\partial_\tau\pi^{yy})u^\tau\left(u^y\right)^2 + (\partial_x\pi^{yy})u^x\left(u^y\right)^2 \\
&+ 2(\partial_y\pi^{xy})u^x\left(u^y\right)^2 - 2(\partial_\tau u^x)\pi^\eta{}_\eta u^\tau u^x\left(u^y\right)^2 - 2(\partial_\tau u^y)\pi^{xy}u^\tau u^x\left(u^y\right)^2 \\
&- 2(\partial_\tau u^x)\pi^{yy}u^\tau u^x\left(u^y\right)^2 + 2(\partial_y u^y)\pi^\eta{}_\eta\left(u^x\right)^2\left(u^y\right)^2 \\
&+ 2(\partial_y u^y)\pi^{yy}\left(u^x\right)^2\left(u^y\right)^2 - (\partial_\tau\pi^\eta{}_\eta)u^\tau\left(u^x\right)^2\left(u^y\right)^2 \\
&- (\partial_\tau\pi^{yy})u^\tau\left(u^x\right)^2\left(u^y\right)^2 - (\partial_x\pi^\eta{}_\eta)\left(u^x\right)^3\left(u^y\right)^2 - (\partial_x\pi^{yy})\left(u^x\right)^3\left(u^y\right)^2 \\
&+ (\partial_y\pi^{yy})\left(u^y\right)^3 + 2(\partial_\tau u^x)\pi^{xy}u^\tau\left(u^y\right)^3 - 2(\partial_y u^x)\pi^\eta{}_\eta u^x\left(u^y\right)^3 \\
&- 2(\partial_y u^y)\pi^{xy}u^x\left(u^y\right)^3 - 2(\partial_y u^x)\pi^{yy}u^x\left(u^y\right)^3 + 2(\partial_\tau\pi^{xy})u^\tau u^x\left(u^y\right)^3 \\
&+ 2(\partial_x\pi^{xy})\left(u^x\right)^2\left(u^y\right)^3 - (\partial_y\pi^\eta{}_\eta)\left(u^x\right)^2\left(u^y\right)^3 - (\partial_y\pi^{yy})\left(u^x\right)^2\left(u^y\right)^3 \\
&+ 2(\partial_y u^x)\pi^{xy}\left(u^y\right)^4 + (\partial_\tau\pi^{yy})u^\tau\left(u^y\right)^4 + (\partial_x\pi^{yy})u^x\left(u^y\right)^4 \\
&+ 2(\partial_y\pi^{xy})u^x\left(u^y\right)^4 + (\partial_y\pi^{yy})\left(u^y\right)^5 \\
&- 2(\partial_x u^x)u^x\left(\left(u^y\right)^2+1\right)(\pi^\eta{}_\eta u^x - \pi^{xy}u^y + \pi^{yy}u^x) \\
&+ 2(\partial_x u^y)u^x\left(u^y\left(\pi^\eta{}_\eta\left(u^x\right)^2 + \pi^{yy}\left(u^x\right)^2 + \pi^{yy}\right) + \pi^{xy}\left(u^x - u^x\left(u^y\right)^2\right)\right)\Big)
\end{aligned}
$$

$$
\begin{aligned}
(D\pi)^{\tau x} = \frac{1}{\left(u^\tau\right)^3\left(\left(u^y\right)^2+1\right)^2}\Big(&-u^\tau\left(\left(u^y\right)^2+1\right)((\partial_\tau u^x)u^x + (\partial_\tau u^y)u^y)\Big(-\pi^\eta{}_\eta\left(u^\tau\right)^2 u^x - \pi^{yy}\left(\left(u^x\right)^3 + u^x\right) \\
&+ \pi^{xy}u^y\left(2\left(u^x\right)^2 + \left(u^y\right)^2 + 1\right)\Big) \\
&- u^x\left(\left(u^y\right)^2+1\right)((\partial_x u^x)u^x + (\partial_x u^y)u^y)\Big(-\pi^\eta{}_\eta\left(u^\tau\right)^2 u^x - \pi^{yy}\left(\left(u^x\right)^3 + u^x\right) \\
&+ \pi^{xy}u^y\left(2\left(u^x\right)^2 + \left(u^y\right)^2 + 1\right)\Big) \\
&- u^y\left(\left(u^y\right)^2+1\right)((\partial_y u^x)u^x + (\partial_y u^y)u^y)\Big(-\pi^\eta{}_\eta\left(u^\tau\right)^2 u^x - \pi^{yy}\left(\left(u^x\right)^3 + u^x\right) \\
&+ \pi^{xy}u^y\left(2\left(u^x\right)^2 + \left(u^y\right)^2 + 1\right)\Big)
\end{aligned}
$$

$$
\begin{aligned}
&+ (u^\tau)^3 \Big(- (u^y)^2 + 1 \Big((\partial_\tau \pi^\eta{}_\eta) (u^\tau)^2 u^x + (\partial_\tau \pi^{yy}) \Big((u^x)^3 + u^x \Big) \\
&- (\partial_\tau \pi^{xy}) u^y \Big(2 (u^x)^2 + (u^y)^2 + 1 \Big) \Big) \\
&- (\partial_\tau u^x) \Big((u^y)^2 + 1 \Big) \Big(\pi^{yy} + 3 \pi^{yy} (u^x)^2 - 4 \pi^{xy} u^x u^y \\
&+ \pi^\eta{}_\eta \Big(3 (u^x)^2 + (u^y)^2 + 1 \Big) \Big) \\
&+ (\partial_\tau u^y) \Big(2 u^x u^y \Big(\pi^\eta{}_\eta (u^x)^2 + \pi^{yy} (u^x)^2 + \pi^{yy} \Big) \\
&+ \pi^{xy} \Big(\Big((u^y)^2 + 1 \Big)^2 - 2 (u^x)^2 \Big((u^y)^2 - 1 \Big) \Big) \Big) \Big) \\
&+ (u^\tau)^2 u^x \Big(- (u^y)^2 + 1 \Big((\partial_x \pi^\eta{}_\eta) (u^\tau)^2 u^x + (\partial_x \pi^{yy}) \Big((u^x)^3 + u^x \Big) \\
&- (\partial_x \pi^{xy}) u^y \Big(2 (u^x)^2 + (u^y)^2 + 1 \Big) \Big) \\
&- (\partial_x u^x) \Big((u^y)^2 + 1 \Big) \Big(\pi^{yy} + 3 \pi^{yy} (u^x)^2 - 4 \pi^{xy} u^x u^y \\
&+ \pi^\eta{}_\eta \Big(3 (u^x)^2 + (u^y)^2 + 1 \Big) \Big) \\
&+ (\partial_x u^y) \Big(2 u^x u^y \Big(\pi^\eta{}_\eta (u^x)^2 + \pi^{yy} (u^x)^2 + \pi^{yy} \Big) \\
&+ \pi^{xy} \Big(\Big((u^y)^2 + 1 \Big)^2 - 2 (u^x)^2 \Big((u^y)^2 - 1 \Big) \Big) \Big) \Big) \\
&+ (u^\tau)^2 u^y \Big(- (u^y)^2 + 1 \Big((\partial_y \pi^\eta{}_\eta) (u^\tau)^2 u^x + (\partial_y \pi^{yy}) \Big((u^x)^3 + u^x \Big) \\
&- (\partial_y \pi^{xy}) u^y \Big(2 (u^x)^2 + (u^y)^2 + 1 \Big) \Big) \\
&- (\partial_y u^x) \Big((u^y)^2 + 1 \Big) \Big(\pi^{yy} + 3 \pi^{yy} (u^x)^2 - 4 \pi^{xy} u^x u^y \\
&+ \pi^\eta{}_\eta \Big(3 (u^x)^2 + (u^y)^2 + 1 \Big) \Big) \\
&+ (\partial_y u^y) \Big(2 u^x u^y \Big(\pi^\eta{}_\eta (u^x)^2 + \pi^{yy} (u^x)^2 + \pi^{yy} \Big) \\
&+ \pi^{xy} \Big(\Big((u^y)^2 + 1 \Big)^2 - 2 (u^x)^2 \Big((u^y)^2 - 1 \Big) \Big) \Big) \Big) \Big) \\
(D\pi)^{\tau y} = &\frac{1}{(u^\tau)^3} \Big((u^\tau)^3 ((\partial_\tau \pi^{xy}) u^x + (\partial_\tau \pi^{yy}) u^y + (\partial_\tau u^x) \pi^{xy} + (\partial_\tau u^y) \pi^{yy}) \\
&+ (u^\tau)^2 u^x ((\partial_x \pi^{xy}) u^x + (\partial_x \pi^{yy}) u^y + (\partial_x u^x) \pi^{xy} + (\partial_x u^y) \pi^{yy}) \\
&+ (u^\tau)^2 u^y ((\partial_y \pi^{xy}) u^x + (\partial_y \pi^{yy}) u^y + (\partial_y u^x) \pi^{xy} + (\partial_y u^y) \pi^{yy}) \\
&- u^\tau ((\partial_\tau u^x) u^x + (\partial_\tau u^y) u^y) (\pi^{xy} u^x + \pi^{yy} u^y) \\
&- u^x ((\partial_x u^x) u^x + (\partial_x u^y) u^y) (\pi^{xy} u^x + \pi^{yy} u^y) \\
&- u^y ((\partial_y u^x) u^x + (\partial_y u^y) u^y) (\pi^{xy} u^x + \pi^{yy} u^y) \Big) \\
(D\pi)^{xx} = &\frac{1}{\Big((u^y)^2 + 1 \Big)^2} \Big(u^\tau \Big(2 (\partial_\tau u^y) u^y \Big(\pi^\eta{}_\eta (u^\tau)^2 - 2 \pi^{xy} u^x u^y + \pi^{yy} (u^x)^2 + \pi^{yy} \Big) \\
&- (u^y)^2 + 1 \Big((\partial_\tau \pi^{yy}) + (\partial_\tau \pi^\eta{}_\eta) (u^\tau)^2 + 2 (\partial_\tau u^x) \pi^\eta{}_\eta u^x \\
&- 2 (\partial_\tau u^y) \pi^{xy} u^x + 2 (\partial_\tau u^x) \pi^{yy} u^x + (\partial_\tau \pi^{yy}) (u^x)^2 \\
&+ 2 (\partial_\tau u^y) \pi^\eta{}_\eta u^y - 2 (\partial_\tau u^x) \pi^{xy} u^y - 2 (\partial_\tau \pi^{xy}) u^x u^y \Big) \Big) \\
&+ u^x \Big(2 (\partial_x u^y) u^y \Big(\pi^\eta{}_\eta (u^\tau)^2 - 2 \pi^{xy} u^x u^y + \pi^{yy} (u^x)^2 + \pi^{yy} \Big) \\
&- (u^y)^2 + 1 \Big((\partial_x \pi^{yy}) + (\partial_x \pi^\eta{}_\eta) (u^\tau)^2 + 2 (\partial_x u^x) \pi^\eta{}_\eta u^x \\
&- 2 (\partial_x u^y) \pi^{xy} u^x + 2 (\partial_x u^x) \pi^{yy} u^x + (\partial_x \pi^{yy}) (u^x)^2 \\
&+ 2 (\partial_x u^y) \pi^\eta{}_\eta u^y - 2 (\partial_x u^x) \pi^{xy} u^y - 2 (\partial_x \pi^{xy}) u^x u^y \Big) \Big) \\
&+ u^y \Big(2 (\partial_y u^y) u^y \Big(\pi^\eta{}_\eta (u^\tau)^2 - 2 \pi^{xy} u^x u^y + \pi^{yy} (u^x)^2 + \pi^{yy} \Big)
\end{aligned}
$$

$$
\begin{aligned}
&- (u^y)^2 + 1\Big(\left(\partial_y \pi^{yy}\right) + \left(\partial_y \pi^{\eta}{}_{\eta}\right)(u^\tau)^2 + 2(\partial_y u^x)\pi^{\eta}{}_{\eta} u^x \\
&- 2(\partial_y u^y)\pi^{xy} u^x + 2(\partial_y u^x)\pi^{yy} u^x + \left(\partial_y \pi^{yy}\right)(u^x)^2 \\
&+ 2(\partial_y u^y)\pi^{\eta}{}_{\eta} u^y - 2(\partial_y u^x)\pi^{xy} u^y - 2(\partial_y \pi^{xy}) u^x u^y\Big)\Big)\Big) \\
(D\pi)^{xy} &= (\partial_\tau \pi^{xy})u^\tau + (\partial_x \pi^{xy})u^x + (\partial_y \pi^{xy})u^y \\
(D\pi)^{yy} &= (\partial_\tau \pi^{yy})u^\tau + (\partial_x \pi^{yy})u^x + (\partial_y \pi^{yy})u^y \\
(D\pi)^{\eta\eta} &= \frac{(\partial_\tau \pi^{\eta}{}_{\eta})u^\tau + (\partial_x \pi^{\eta}{}_{\eta})u^x + (\partial_y \pi^{\eta}{}_{\eta})u^y}{\tau^2}
\end{aligned}
$$

$$
\begin{aligned}
\Delta^{yy}_{\alpha\beta} D\pi^{\alpha\beta} &= \frac{1}{(u^\tau)^2}\Big((\partial_\tau \pi^{yy})u^\tau + \left(\partial_x \pi^{yy}\right)(u^\tau)^2 u^x + (\partial_\tau \pi^{yy})u^\tau (u^x)^2 - 2(\partial_\tau u^x)\pi^{xy} u^\tau u^y \\
&- 2(\partial_\tau u^y)\pi^{yy} u^\tau u^y + \left(\partial_y \pi^{yy}\right)(u^\tau)^2 u^y - 2(\partial_x u^x)\pi^{xy} u^x u^y - 2(\partial_x u^y)\pi^{yy} u^x u^y \\
&- 2(\partial_\tau u^y)\pi^{yy} u^\tau (u^x)^2 u^y - 2(\partial_x u^y)\pi^{yy} (u^x)^3 u^y - 2(\partial_y u^x)\pi^{xy} (u^y)^2 \\
&- 2(\partial_y u^y)\pi^{yy} (u^y)^2 + (\partial_\tau \pi^{yy})u^\tau (u^y)^2 + 2(\partial_\tau u^y)\pi^{xy} u^\tau u^x (u^y)^2 \\
&+ 2(\partial_\tau u^x)\pi^{yy} u^\tau u^x (u^y)^2 + 2(\partial_x u^y)\pi^{xy} (u^x)^2 (u^y)^2 + 2(\partial_x u^x)\pi^{yy} (u^x)^2 (u^y)^2 \\
&- 2(\partial_y u^y)\pi^{yy} (u^x)^2 (u^y)^2 - 2(\partial_\tau u^x)\pi^{xy} u^\tau (u^y)^3 - 2(\partial_x u^x)\pi^{xy} u^x (u^y)^3 \\
&+ 2(\partial_y u^y)\pi^{xy} u^x (u^y)^3 + 2(\partial_y u^x)\pi^{yy} u^x (u^y)^3 - 2(\partial_y u^x)\pi^{xy} (u^y)^4\Big) \\
\Delta^{\eta\eta}_{\alpha\beta} D\pi^{\alpha\beta} &= \frac{(\partial_\tau \pi^{\eta}{}_{\eta})u^\tau + (\partial_x \pi^{\eta}{}_{\eta})u^x + (\partial_y \pi^{\eta}{}_{\eta})u^y}{\tau^2} \\
\Delta^{xy}_{\alpha\beta} D\pi^{\alpha\beta} &= \frac{1}{(u^\tau)^2\left((u^y)^2+1\right)}\Big((\partial_\tau \pi^{xy})u^\tau - (\partial_\tau u^x)\pi^{xy} u^\tau u^x - (\partial_\tau u^y)\pi^{yy} u^\tau u^x - (\partial_x u^y)\pi^{yy} (u^x)^2 \\
&+ (\partial_\tau \pi^{xy})u^\tau (u^x)^2 - (\partial_\tau u^y)\pi^{yy} u^\tau (u^x)^3 - (\partial_x u^y)\pi^{yy} (u^x)^4 \\
&+ (\partial_y \pi^{xy})u^y + (\partial_\tau u^x)\pi^{\eta}{}_{\eta} u^\tau u^y - (\partial_\tau u^y)\pi^{xy} u^\tau u^y + (\partial_\tau u^x)\pi^{yy} u^\tau u^y \\
&- (\partial_x u^y)\pi^{xy} u^x u^y - (\partial_y u^x)\pi^{xy} u^x u^y - (\partial_y u^y)\pi^{yy} u^x u^y \\
&+ (\partial_y \pi^{xy})(u^x)^2 u^y + (\partial_\tau u^x)\pi^{\eta}{}_{\eta} u^\tau (u^x)^2 u^y + 2(\partial_\tau u^x)\pi^{yy} u^\tau (u^x)^2 u^y \\
&- (\partial_y u^y)\pi^{yy} (u^x)^3 u^y + (\partial_y u^x)\pi^{\eta}{}_{\eta} (u^y)^2 - (\partial_y u^y)\pi^{xy} (u^y)^2 \\
&+ (\partial_y u^x)\pi^{yy} (u^y)^2 + 2(\partial_\tau \pi^{xy})u^\tau (u^y)^2 - (\partial_\tau u^y)\pi^{\eta}{}_{\eta} u^\tau u^x (u^y)^2 \\
&- 3(\partial_\tau u^x)\pi^{xy} u^\tau u^x (u^y)^2 - 2(\partial_\tau u^y)\pi^{yy} u^\tau u^x (u^y)^2 \\
&- (\partial_x u^y)\pi^{\eta}{}_{\eta} (u^x)^2 (u^y)^2 + (\partial_y u^x)\pi^{\eta}{}_{\eta} (u^x)^2 (u^y)^2 \\
&- 2(\partial_x u^y)\pi^{yy} (u^x)^2 (u^y)^2 + 2(\partial_y u^x)\pi^{yy} (u^x)^2 (u^y)^2 \\
&+ (\partial_\tau \pi^{xy})u^\tau (u^x)^2 (u^y)^2 - (\partial_\tau u^y)\pi^{\eta}{}_{\eta} u^\tau (u^x)^3 (u^y)^2 \\
&- 2(\partial_\tau u^y)\pi^{yy} u^\tau (u^x)^3 (u^y)^2 - (\partial_x u^y)\pi^{\eta}{}_{\eta} (u^x)^4 (u^y)^2 \\
&- 2(\partial_x u^y)\pi^{yy} (u^x)^4 (u^y)^2 + 2(\partial_y \pi^{xy})(u^y)^3 + 2(\partial_\tau u^x)\pi^{\eta}{}_{\eta} u^\tau (u^y)^3 \\
&- (\partial_\tau u^y)\pi^{xy} u^\tau (u^y)^3 + (\partial_\tau u^x)\pi^{yy} u^\tau (u^y)^3 - (\partial_y u^y)\pi^{\eta}{}_{\eta} u^x (u^y)^3 \\
&- (\partial_x u^y)\pi^{xy} u^x (u^y)^3 - 3(\partial_y u^x)\pi^{xy} u^x (u^y)^3 - 2(\partial_y u^y)\pi^{yy} u^x (u^y)^3 \\
&+ (\partial_y \pi^{xy})(u^x)^2 (u^y)^3 + (\partial_\tau u^x)\pi^{\eta}{}_{\eta} u^\tau (u^x)^2 (u^y)^3 \\
&+ 2(\partial_\tau u^y)\pi^{xy} u^\tau (u^x)^2 (u^y)^3 + 2(\partial_\tau u^x)\pi^{yy} u^\tau (u^x)^2 (u^y)^3 \\
&- (\partial_y u^y)\pi^{\eta}{}_{\eta} (u^x)^3 (u^y)^3 + 2(\partial_x u^y)\pi^{xy} (u^x)^3 (u^y)^3 \\
&- 2(\partial_y u^y)\pi^{yy} (u^x)^3 (u^y)^3 + 2(\partial_y u^x)\pi^{\eta}{}_{\eta} (u^y)^4 - (\partial_y u^y)\pi^{xy} (u^y)^4 \\
&+ (\partial_y u^x)\pi^{yy} (u^y)^4 + (\partial_\tau \pi^{xy})u^\tau (u^y)^4 - (\partial_\tau u^y)\pi^{\eta}{}_{\eta} u^\tau u^x (u^y)^4 \\
&- 2(\partial_\tau u^x)\pi^{xy} u^\tau u^x (u^y)^4 - (\partial_x u^y)\pi^{\eta}{}_{\eta} (u^x)^2 (u^y)^4 \\
&+ (\partial_y u^x)\pi^{\eta}{}_{\eta} (u^x)^2 (u^y)^4 + 2(\partial_y u^y)\pi^{xy} (u^x)^2 (u^y)^4 \\
&+ 2(\partial_y u^x)\pi^{yy} (u^x)^2 (u^y)^4 + (\partial_y \pi^{xy})(u^y)^5 + (\partial_\tau u^x)\pi^{\eta}{}_{\eta} u^\tau (u^y)^5 \\
&- (\partial_y u^y)\pi^{\eta}{}_{\eta} u^x (u^y)^5 - 2(\partial_y u^x)\pi^{xy} u^x (u^y)^5 + (\partial_y u^x)\pi^{\eta}{}_{\eta} (u^y)^6 \\
&+ (\partial_x \pi^{xy})(u^\tau)^2 u^x \left((u^y)^2+1\right)
\end{aligned}
$$

$$+ (\partial_x u^x) u^x \left((u^y)^2 + 1 \right) \left(u^y \left(\pi^\eta{}_\eta \, (u^\tau)^2 + 2\pi^{yy} \, (u^x)^2 + \pi^{yy} \right) \right.$$
$$\left. - \pi^{xy} \left(2u^x \, (u^y)^2 + u^x \right) \right) \Big)$$

$$E^{ab} = \tau_\pi \Delta^{ab}_{\alpha\beta} D\pi^{\alpha\beta} + \pi^{ab} + 2\eta\sigma^{ab} + \delta_{\pi\pi}\pi^{ab}\theta, \qquad ab = yy,\ \eta\eta,\ xy$$
$$E_\Pi = \tau_\Pi (u^\tau \partial_\tau + u^x \partial_x + u^y \partial_y)\Pi + \Pi + \zeta\theta$$
$$A^{(k)}_{ij} = \frac{\partial E_i}{\partial(\partial_k \phi_j)}, \qquad S_i = E_i\big|_{\partial\phi=0}, \qquad k = \tau, x, y$$

$$\begin{aligned}
E_1 = & -(\partial_\tau \Pi) - (\partial_\tau T) p'(T) + \frac{\Pi + \pi^\eta{}_\eta + p}{\tau} + (\partial_x u^x) u^\tau (p'(T) T + \Pi) \\
& + (\partial_y u^y) u^\tau (p'(T) T + \Pi) + (u^\tau)^2 \left((\partial_\tau \Pi) + (\partial_\tau T)(p''(T) T + p'(T)) \right) \\
& + u^\tau u^x ((\partial_x \Pi) + (\partial_x T)(p''(T) T + p'(T))) \\
& + u^\tau u^y ((\partial_y \Pi) + (\partial_y T)(p''(T) T + p'(T))) \\
& + 2(p'(T) T + \Pi)((\partial_\tau u^x) u^x + (\partial_\tau u^y) u^y) + \frac{u^x (p'(T) T + \Pi)((\partial_x u^x) u^x + (\partial_x u^y) u^y)}{u^\tau} \\
& + \frac{(\partial_y \pi^{xy}) u^x + (\partial_y \pi^{yy}) u^y + (\partial_y u^x) \pi^{xy} + (\partial_y u^y) \pi^{yy}}{u^\tau} \\
& + \frac{u^y (p'(T) T + \Pi)((\partial_y u^x) u^x + (\partial_y u^y) u^y)}{u^\tau} - \frac{((\partial_y u^x) u^x + (\partial_y u^y) u^y)(\pi^{xy} u^x + \pi^{yy} u^y)}{(u^\tau)^3} \\
& + \frac{((\partial_\tau u^x) u^x + (\partial_\tau u^y) u^y) \left(\pi^\eta{}_\eta \, (u^x)^2 - 2\pi^{xy} u^x u^y + \pi^{yy} \left((u^x)^2 - (u^y)^2 \right) \right)}{(u^\tau)^2 \left((u^y)^2 + 1 \right)} \\
& + \frac{1}{\tau \left((u^y)^2 + 1 \right)} \Big(\Pi \, (u^x)^2 - \pi^\eta{}_\eta \, (u^x)^2 - \pi^{yy} \, (u^x)^2 + 2\pi^{xy} u^x u^y + \Pi \, (u^y)^2 + \pi^{yy} \, (u^y)^2 \\
& + \Pi \, (u^x)^2 \, (u^y)^2 + \Pi \, (u^y)^4 - p \left((u^y)^2 + 1 \right) \\
& + p'(T) T \, (u^\tau)^2 \left((u^y)^2 + 1 \right) \Big) \\
& + \frac{1}{(u^\tau)^3 \left((u^y)^2 + 1 \right)} (\partial_x u^x) u^x + (\partial_x u^y) u^y \Big(\pi^\eta{}_\eta \, (u^\tau)^2 \, u^x + \pi^{yy} \left((u^x)^3 + u^x \right) \\
& - \pi^{xy} u^y \left(2 \, (u^x)^2 + (u^y)^2 + 1 \right) \Big) \\
& + \frac{1}{u^\tau \left((u^y)^2 + 1 \right)^2} \Big(- (u^y)^2 + 1 \Big((\partial_x \pi^\eta{}_\eta) \, (u^\tau)^2 \, u^x + (\partial_x \pi^{yy}) \left((u^x)^3 + u^x \right) \\
& - (\partial_x \pi^{xy}) u^y \left(2 \, (u^x)^2 + (u^y)^2 + 1 \right) \Big) \\
& - (\partial_x u^x) \left((u^y)^2 + 1 \right) \left(\pi^{yy} + 3\pi^{yy} \, (u^x)^2 - 4\pi^{xy} u^x u^y \right. \\
& \left. + \pi^\eta{}_\eta \left(3 \, (u^x)^2 + (u^y)^2 + 1 \right) \right) \\
& + (\partial_x u^y) \Big(2u^x u^y \left(\pi^\eta{}_\eta \, (u^x)^2 + \pi^{yy} \, (u^x)^2 + \pi^{yy} \right) \\
& + \pi^{xy} \left(\left((u^y)^2 + 1 \right)^2 - 2 \, (u^x)^2 \left((u^y)^2 - 1 \right) \right) \Big) \Big) \\
& + \frac{1}{(u^\tau)^2 \left((u^y)^2 + 1 \right)^2} \Big(2(\partial_\tau u^y) \pi^{xy} u^x \left(- \left((u^x)^2 \left((u^y)^2 - 1 \right) \right) + (u^y)^2 + 1 \right) \\
& + (u^\tau)^2 \left((u^y)^2 + 1 \right) \left(- (\partial_\tau \pi^\eta{}_\eta) \, (u^x)^2 + 2(\partial_\tau \pi^{xy}) u^x u^y \right. \\
& \left. + (\partial_\tau \pi^{yy}) \left((u^y)^2 - (u^x)^2 \right) \right) \\
& - (\partial_\tau u^x) \left((u^y)^2 + 1 \right) \left(- 2\pi^{xy} u^y \left(2 \, (u^x)^2 + (u^y)^2 + 1 \right) \right. \\
& \left. + \pi^{yy} u^x \left(3 \, (u^x)^2 + (u^y)^2 + 2 \right) \right.
\end{aligned}$$

$$
\begin{aligned}
&+ \pi^{\eta}{}_{\eta} u^x \left(3\left(u^x\right)^2 + 2\left(u^y\right)^2 + 2\right)\Big) \\
&+ (\partial_\tau u^y) u^y \Big(\pi^{\eta}{}_{\eta} \left(u^x\right)^2 \left(2\left(u^x\right)^2 + \left(u^y\right)^2 + 1\right) \\
&+ \pi^{yy} \Big(2 + 2\left(u^x\right)^4 + 3\left(u^y\right)^2 + \left(u^y\right)^4 \\
&+ \left(u^x\right)^2 \left(\left(u^y\right)^2 + 3\right)\Big)\Big)\Big) \\
E_7 = \Pi &+ \tau_\Pi((\partial_\tau \Pi) u^\tau + (\partial_x \Pi) u^x + (\partial_y \Pi) u^y) \\
&+ \zeta \left(\frac{(\partial_\tau u^x) u^x + (\partial_\tau u^y) u^y}{u^\tau} + (\partial_x u^x) + (\partial_y u^y) + \frac{u^\tau}{\tau}\right)
\end{aligned}
$$

1.3 1+1D cylindrical

Fields, with π^{rr} and the τ row of $\pi^{\mu\nu}$ fixed by $u_\mu \pi^{\mu\nu} = 0$, $\pi^{\mu}{}_{\mu} = 0$:

$$
\begin{aligned}
\phi = (T, u^r, \pi^{\phi}{}_{\phi}, \pi^{\eta}{}_{\eta}, \Pi), \qquad \pi^{\phi}{}_{\phi} = r^2 \pi^{\phi\phi}, \quad \pi^{\eta}{}_{\eta} = \tau^2 \pi^{\eta\eta} \\
\pi^{rr} = -\left(\left(u^\tau\right)^2 \left(\pi^{\eta}{}_{\eta} + \pi^{\phi}{}_{\phi}\right)\right), \qquad u^\tau = \sqrt{\left(u^r\right)^2 + 1}
\end{aligned}
$$

Rows: $\nabla_\mu T^{\mu\tau}$, $\nabla_\mu T^{\mu r}$, $E^{\phi\phi}$, $E^{\eta\eta}$, E_Π, where $E^{\mu\nu}$ and E_Π are the left-hand sides of the shear and bulk equations.

1.3.1 IS part

$$
\begin{aligned}
(A_t^{\mathrm{IS}})_{11} &= \left(u^r\right)^2 \left(p''(T) T + p'(T)\right) + p''(T) T \\
(A_t^{\mathrm{IS}})_{12} &= 2u^r (p'(T) T + \Pi - \pi^{\eta}{}_{\eta} - \pi^{\phi}{}_{\phi}) \\
(A_t^{\mathrm{IS}})_{13} &= -\left(u^r\right)^2 \\
(A_t^{\mathrm{IS}})_{14} &= -\left(u^r\right)^2 \\
(A_t^{\mathrm{IS}})_{15} &= \left(u^r\right)^2 \\
(A_t^{\mathrm{IS}})_{21} &= u^r u^\tau (p''(T) T + p'(T)) \\
(A_t^{\mathrm{IS}})_{22} &= \frac{\left(2\left(u^r\right)^2 + 1\right)(p'(T) T + \Pi - \pi^{\eta}{}_{\eta} - \pi^{\phi}{}_{\phi})}{u^\tau} \\
(A_t^{\mathrm{IS}})_{23} &= -u^r u^\tau \\
(A_t^{\mathrm{IS}})_{24} &= -u^r u^\tau \\
(A_t^{\mathrm{IS}})_{25} &= u^r u^\tau \\
(A_t^{\mathrm{IS}})_{32} &= -\frac{2u^r \eta}{3r^2 u^\tau} \\
(A_t^{\mathrm{IS}})_{33} &= \frac{\tau_\pi u^\tau}{r^2} \\
(A_t^{\mathrm{IS}})_{42} &= -\frac{2u^r \eta}{3\tau^2 u^\tau} \\
(A_t^{\mathrm{IS}})_{44} &= \frac{\tau_\pi u^\tau}{\tau^2} \\
(A_t^{\mathrm{IS}})_{52} &= \frac{u^r \zeta}{u^\tau} \\
(A_t^{\mathrm{IS}})_{55} &= \tau_\Pi u^\tau
\end{aligned}
$$

$$
\begin{aligned}
(A_r^{\mathrm{IS}})_{11} &= u^r u^\tau (p''(T) T + p'(T)) \\
(A_r^{\mathrm{IS}})_{12} &= \frac{\left(2\left(u^r\right)^2 + 1\right)(p'(T) T + \Pi - \pi^{\eta}{}_{\eta} - \pi^{\phi}{}_{\phi})}{u^\tau} \\
(A_r^{\mathrm{IS}})_{13} &= -u^r u^\tau \\
(A_r^{\mathrm{IS}})_{14} &= -u^r u^\tau
\end{aligned}
$$

$$
\begin{aligned}
(A_r^{\mathrm{IS}})_{15} &= u^r u^\tau \\
(A_r^{\mathrm{IS}})_{21} &= p''(T)\,(u^r)^2\,T + p'(T)\,(u^r)^2 + p'(T) \\
(A_r^{\mathrm{IS}})_{22} &= 2u^r(p'(T)T + \Pi - \pi^\eta{}_\eta - \pi^\phi{}_\phi) \\
(A_r^{\mathrm{IS}})_{23} &= -\,(u^r)^2 - 1 \\
(A_r^{\mathrm{IS}})_{24} &= -\,(u^r)^2 - 1 \\
(A_r^{\mathrm{IS}})_{25} &= (u^\tau)^2 \\
(A_r^{\mathrm{IS}})_{32} &= -\frac{2\eta}{3r^2} \\
(A_r^{\mathrm{IS}})_{33} &= \frac{\tau_\pi u^r}{r^2} \\
(A_r^{\mathrm{IS}})_{42} &= -\frac{2\eta}{3\tau^2} \\
(A_r^{\mathrm{IS}})_{44} &= \frac{\tau_\pi u^r}{\tau^2} \\
(A_r^{\mathrm{IS}})_{52} &= \zeta \\
(A_r^{\mathrm{IS}})_{55} &= \tau_\Pi u^r
\end{aligned}
$$

$$
\begin{aligned}
S_1^{\mathrm{IS}} &= \frac{1}{r\tau u^\tau}\Big((u^\tau)^2\,(p'(T)T+\Pi)(ru^\tau+\tau u^r) - \pi^\phi{}_\phi u^r\left(ru^r u^\tau + \tau\,(u^r)^2 + \tau\right) \\
&\quad - \pi^\eta{}_\eta\left(r\left((u^r)^2-1\right)u^\tau + \tau\left((u^r)^3+u^r\right)\right)\Big) \\
S_2^{\mathrm{IS}} &= \frac{1}{r\tau u^\tau}\Big(-\pi^\eta{}_\eta\,(u^\tau)^2\,(ru^r+\tau u^\tau) + u^r(p'(T)T+\Pi)\left(r\,(u^r)^2 + r + \tau u^r u^\tau\right) \\
&\quad - \pi^\phi{}_\phi\left(r\left((u^r)^3+u^r\right) + \tau\left((u^r)^2+2\right)u^\tau\right)\Big) \\
S_3^{\mathrm{IS}} &= \frac{\pi^\phi{}_\phi}{r^2} - \frac{2\eta\left(r\,(u^r)^2 + r - 2\tau u^r u^\tau\right)}{3r^3\tau u^\tau} \\
S_4^{\mathrm{IS}} &= \frac{3\pi^\eta{}_\eta r\tau + 4ru^\tau\eta - 2\tau u^r\eta}{3r\tau^3} \\
S_5^{\mathrm{IS}} &= \Pi + \frac{u^r\zeta}{r} + \frac{u^\tau\zeta}{\tau}
\end{aligned}
$$

1.3.2 $\delta_{\pi\pi}$ part

$$
\begin{aligned}
(A_t^{(\delta)})_{32} &= \frac{\pi^\phi{}_\phi u^r}{r^2u^\tau} \\
(A_t^{(\delta)})_{42} &= \frac{\pi^\eta{}_\eta u^r}{\tau^2u^\tau}
\end{aligned}
$$

$$
\begin{aligned}
(A_r^{(\delta)})_{32} &= \frac{\pi^\phi{}_\phi}{r^2} \\
(A_r^{(\delta)})_{42} &= \frac{\pi^\eta{}_\eta}{\tau^2}
\end{aligned}
$$

$$
\begin{aligned}
S_3^{(\delta)} &= \frac{\pi^\phi{}_\phi(ru^\tau+\tau u^r)}{r^3\tau} \\
S_4^{(\delta)} &= \frac{\pi^\eta{}_\eta(ru^\tau+\tau u^r)}{r\tau^3}
\end{aligned}
$$

1.4 2+1D Cartesian

Fields, with π^{xx} and the τ row fixed by the same constraints:

$$
\phi = (T, u^x, u^y, \pi^{yy}, \pi^\eta{}_\eta, \pi^{xy}, \Pi)
$$

$$\pi^{xx} = -\frac{\pi^{\eta}{}_{\eta}\left(u^{\tau}\right)^2 - 2\pi^{xy}u^x u^y + \pi^{yy}\left(u^x\right)^2 + \pi^{yy}}{\left(u^y\right)^2 + 1}, \qquad u^{\tau} = \sqrt{\left(u^x\right)^2 + \left(u^y\right)^2 + 1}$$

Rows: $\nabla_\mu T^{\mu\tau}$, $\nabla_\mu T^{\mu x}$, $\nabla_\mu T^{\mu y}$, E^{yy}, $E^{\eta\eta}$, E^{xy}, E_Π.

1.4.1 IS part

$$\begin{aligned}
(A_t^{\mathrm{IS}})_{11} &= \left(u^{\tau}\right)^2\left(p''(T)T + p'(T)\right) - p'(T)\\
(A_t^{\mathrm{IS}})_{12} &= \frac{2\left(p'(T)Tu^x\left(u^y\right)^2 + p'(T)Tu^x + \Pi u^x\left(\left(u^y\right)^2+1\right) - \pi^{\eta}{}_{\eta}u^x + \pi^{xy}u^y - \pi^{yy}u^x\right)}{\left(u^y\right)^2+1}\\
(A_t^{\mathrm{IS}})_{13} &= \frac{2}{\left(\left(u^y\right)^2+1\right)^2}\Big(\pi^{xy}\left(u^x - u^x\left(u^y\right)^2\right)\\
&\quad + u^y\Big(\pi^{yy} + p'(T)T + \pi^{\eta}{}_{\eta}\left(u^x\right)^2 + \pi^{yy}\left(u^x\right)^2 + 2p'(T)T\left(u^y\right)^2\\
&\quad + p'(T)T\left(u^y\right)^4 + \Pi\left(\left(u^y\right)^2+1\right)^2\Big)\Big)\\
(A_t^{\mathrm{IS}})_{14} &= \frac{\left(u^y\right)^2 - \left(u^x\right)^2}{\left(u^y\right)^2+1}\\
(A_t^{\mathrm{IS}})_{15} &= -\frac{\left(u^x\right)^2}{\left(u^y\right)^2+1}\\
(A_t^{\mathrm{IS}})_{16} &= \frac{2u^xu^y}{\left(u^y\right)^2+1}\\
(A_t^{\mathrm{IS}})_{17} &= \left(u^x\right)^2 + \left(u^y\right)^2\\
(A_t^{\mathrm{IS}})_{21} &= u^{\tau}u^x(p''(T)T + p'(T))\\
(A_t^{\mathrm{IS}})_{22} &= \frac{1}{\left(u^{\tau}\right)^3\left(\left(u^y\right)^2+1\right)}\Big(-\pi^{\eta}{}_{\eta}\left(u^{\tau}\right)^2 - \pi^{yy}\left(u^{\tau}\right)^2 + \Pi\left(u^{\tau}\right)^4 + p'(T)T\left(u^{\tau}\right)^4 + \pi^{yy}\left(u^x\right)^2\\
&\quad + \Pi\left(u^{\tau}\right)^2\left(u^x\right)^2 - 2\pi^{\eta}{}_{\eta}\left(u^{\tau}\right)^2\left(u^x\right)^2 - 3\pi^{yy}\left(u^{\tau}\right)^2\left(u^x\right)^2\\
&\quad + p'(T)T\left(u^{\tau}\right)^2\left(u^x\right)^2 + \pi^{yy}\left(u^x\right)^4 - \pi^{xy}u^xu^y + 4\pi^{xy}\left(u^{\tau}\right)^2u^xu^y\\
&\quad - 2\pi^{xy}\left(u^x\right)^3u^y - \pi^{\eta}{}_{\eta}\left(u^{\tau}\right)^2\left(u^y\right)^2 + \Pi\left(u^{\tau}\right)^4\left(u^y\right)^2\\
&\quad + p'(T)T\left(u^{\tau}\right)^4\left(u^y\right)^2 + \Pi\left(u^{\tau}\right)^2\left(u^x\right)^2\left(u^y\right)^2\\
&\quad + p'(T)T\left(u^{\tau}\right)^2\left(u^x\right)^2\left(u^y\right)^2 - \pi^{xy}u^x\left(u^y\right)^3\Big)\\
(A_t^{\mathrm{IS}})_{23} &= \frac{1}{\left(u^{\tau}\right)^3\left(\left(u^y\right)^2+1\right)^2}\Big(\pi^{xy}\left(u^{\tau}\right)^2 + 2\pi^{xy}\left(u^{\tau}\right)^2\left(u^x\right)^2 + \pi^{yy}u^xu^y + \Pi\left(u^{\tau}\right)^2u^xu^y\\
&\quad + \pi^{\eta}{}_{\eta}\left(u^{\tau}\right)^2u^xu^y + 2\pi^{yy}\left(u^{\tau}\right)^2u^xu^y + p'(T)T\left(u^{\tau}\right)^2u^xu^y\\
&\quad + \pi^{yy}\left(u^x\right)^3u^y + 2\pi^{\eta}{}_{\eta}\left(u^{\tau}\right)^2\left(u^x\right)^3u^y + 2\pi^{yy}\left(u^{\tau}\right)^2\left(u^x\right)^3u^y\\
&\quad - \pi^{xy}\left(u^y\right)^2 + 2\pi^{xy}\left(u^{\tau}\right)^2\left(u^y\right)^2 - 2\pi^{xy}\left(u^x\right)^2\left(u^y\right)^2\\
&\quad - 2\pi^{xy}\left(u^{\tau}\right)^2\left(u^x\right)^2\left(u^y\right)^2 + \pi^{yy}u^x\left(u^y\right)^3 + 2\Pi\left(u^{\tau}\right)^2u^x\left(u^y\right)^3\\
&\quad + \pi^{\eta}{}_{\eta}\left(u^{\tau}\right)^2u^x\left(u^y\right)^3 + 2p'(T)T\left(u^{\tau}\right)^2u^x\left(u^y\right)^3 + \pi^{yy}\left(u^x\right)^3\left(u^y\right)^3\\
&\quad - 2\pi^{xy}\left(u^y\right)^4 + \pi^{xy}\left(u^{\tau}\right)^2\left(u^y\right)^4 - 2\pi^{xy}\left(u^x\right)^2\left(u^y\right)^4\\
&\quad + \Pi\left(u^{\tau}\right)^2u^x\left(u^y\right)^5 + p'(T)T\left(u^{\tau}\right)^2u^x\left(u^y\right)^5 - \pi^{xy}\left(u^y\right)^6\Big)\\
(A_t^{\mathrm{IS}})_{24} &= -\frac{u^x\left(\left(u^x\right)^2+1\right)}{u^{\tau}\left(\left(u^y\right)^2+1\right)}\\
(A_t^{\mathrm{IS}})_{25} &= -\frac{u^{\tau}u^x}{\left(u^y\right)^2+1}
\end{aligned}$$

$$(A_t^{\mathrm{IS}})_{26} = \frac{u^y\left(2\left(u^x\right)^2+\left(u^y\right)^2+1\right)}{u^\tau\left(\left(u^y\right)^2+1\right)}$$
$$(A_t^{\mathrm{IS}})_{27} = u^\tau u^x$$
$$(A_t^{\mathrm{IS}})_{31} = u^\tau u^y(p''(T)T+p'(T))$$
$$(A_t^{\mathrm{IS}})_{32} = \frac{u^x u^y\left(p'(T)T\left(u^\tau\right)^2+\Pi\left(u^\tau\right)^2-\pi^{yy}\right)+\pi^{xy}\left(\left(u^y\right)^2+1\right)}{\left(u^\tau\right)^3}$$
$$(A_t^{\mathrm{IS}})_{33} = \frac{\left(u^\tau\right)^4\left(p'(T)T+\Pi\right)+\left(u^\tau\right)^2\left(u^y\right)^2\left(p'(T)T+\Pi\right)-u^y(\pi^{xy}u^x+\pi^{yy}u^y)+\pi^{yy}\left(u^\tau\right)^2}{\left(u^\tau\right)^3}$$
$$(A_t^{\mathrm{IS}})_{34} = \frac{u^y}{u^\tau}$$
$$(A_t^{\mathrm{IS}})_{36} = \frac{u^x}{u^\tau}$$
$$(A_t^{\mathrm{IS}})_{37} = u^\tau u^y$$
$$(A_t^{\mathrm{IS}})_{42} = -\frac{2\left(3\pi^{xy}\tau_\pi u^y\left(\left(u^y\right)^2+1\right)+u^x\left(\left(u^y\right)^2(\eta-3\pi^{yy}\tau_\pi)+\eta\right)\right)}{3u^\tau}$$
$$(A_t^{\mathrm{IS}})_{43} = \frac{2u^y\left(3\pi^{xy}\tau_\pi u^x u^y-3\pi^{yy}\tau_\pi\left(\left(u^x\right)^2+1\right)+\eta\left(3\left(u^x\right)^2+2\left(u^y\right)^2+2\right)\right)}{3u^\tau}$$
$$(A_t^{\mathrm{IS}})_{44} = \tau_\pi u^\tau$$
$$(A_t^{\mathrm{IS}})_{52} = -\frac{2u^x\eta}{3\tau^2u^\tau}$$
$$(A_t^{\mathrm{IS}})_{53} = -\frac{2u^y\eta}{3\tau^2u^\tau}$$
$$(A_t^{\mathrm{IS}})_{55} = \frac{\tau_\pi u^\tau}{\tau^2}$$
$$\begin{aligned}(A_t^{\mathrm{IS}})_{62} = \frac{1}{3u^\tau}\Big(&-3\pi^{xy}\tau_\pi u^x\left(2\left(u^y\right)^2+1\right)\\&+u^y\left(3\pi^\eta{}_\eta\tau_\pi\left(u^\tau\right)^2+3\pi^{yy}\left(2\tau_\pi\left(u^x\right)^2+\tau_\pi\right)+\eta\left(\left(u^x\right)^2+3\left(u^y\right)^2+3\right)\right)\Big)\end{aligned}$$
$$\begin{aligned}(A_t^{\mathrm{IS}})_{63} = \frac{1}{3u^\tau\left(\left(u^y\right)^2+1\right)}\Big(&-3\pi^{yy}\tau_\pi u^x\left(\left(u^x\right)^2+1\right)\left(2\left(u^y\right)^2+1\right)\\&+3\pi^{xy}\tau_\pi u^y\left(\left(2\left(u^x\right)^2-1\right)\left(u^y\right)^2-1\right)\\&+u^x\left(\left(\left(u^y\right)^2+1\right)\eta\left(3\left(u^x\right)^2+\left(u^y\right)^2+3\right)-3\pi^\eta{}_\eta\tau_\pi\left(u^\tau\right)^2\left(u^y\right)^2\right)\Big)\end{aligned}$$
$$(A_t^{\mathrm{IS}})_{66} = \tau_\pi u^\tau$$
$$(A_t^{\mathrm{IS}})_{72} = \frac{u^x\zeta}{u^\tau}$$
$$(A_t^{\mathrm{IS}})_{73} = \frac{u^y\zeta}{u^\tau}$$
$$(A_t^{\mathrm{IS}})_{77} = \tau_\Pi u^\tau$$

$$(A_x^{\mathrm{IS}})_{11} = u^\tau u^x(p''(T)T+p'(T))$$
$$\begin{aligned}(A_x^{\mathrm{IS}})_{12} = \frac{1}{\left(u^\tau\right)^3\left(\left(u^y\right)^2+1\right)}\Big(&-\pi^\eta{}_\eta\left(u^\tau\right)^2-\pi^{yy}\left(u^\tau\right)^2+\Pi\left(u^\tau\right)^4+p'(T)T\left(u^\tau\right)^4+\pi^{yy}\left(u^x\right)^2\\&+\Pi\left(u^\tau\right)^2\left(u^x\right)^2-2\pi^\eta{}_\eta\left(u^\tau\right)^2\left(u^x\right)^2-3\pi^{yy}\left(u^\tau\right)^2\left(u^x\right)^2\\&+p'(T)T\left(u^\tau\right)^2\left(u^x\right)^2+\pi^{yy}\left(u^x\right)^4-\pi^{xy}u^xu^y+4\pi^{xy}\left(u^\tau\right)^2u^xu^y\\&-2\pi^{xy}\left(u^x\right)^3u^y-\pi^\eta{}_\eta\left(u^\tau\right)^2\left(u^y\right)^2+\Pi\left(u^\tau\right)^4\left(u^y\right)^2\\&+p'(T)T\left(u^\tau\right)^4\left(u^y\right)^2+\Pi\left(u^\tau\right)^2\left(u^x\right)^2\left(u^y\right)^2\\&+p'(T)T\left(u^\tau\right)^2\left(u^x\right)^2\left(u^y\right)^2-\pi^{xy}u^x\left(u^y\right)^3\Big)\end{aligned}$$

$$
\begin{aligned}
(A_x^{\mathrm{IS}})_{13} = & \frac{1}{(u^\tau)^3\left((u^y)^2+1\right)^2}\Big(\pi^{xy}(u^\tau)^2+2\pi^{xy}(u^\tau)^2(u^x)^2+\pi^{yy}u^xu^y+\Pi(u^\tau)^2u^xu^y \\
& +\pi^\eta{}_\eta(u^\tau)^2u^xu^y+2\pi^{yy}(u^\tau)^2u^xu^y+p'(T)T(u^\tau)^2u^xu^y \\
& +\pi^{yy}(u^x)^3u^y+2\pi^\eta{}_\eta(u^\tau)^2(u^x)^3u^y+2\pi^{yy}(u^\tau)^2(u^x)^3u^y \\
& -\pi^{xy}(u^y)^2+2\pi^{xy}(u^\tau)^2(u^y)^2-2\pi^{xy}(u^x)^2(u^y)^2 \\
& -2\pi^{xy}(u^\tau)^2(u^x)^2(u^y)^2+\pi^{yy}u^x(u^y)^3+2\Pi(u^\tau)^2u^x(u^y)^3 \\
& +\pi^\eta{}_\eta(u^\tau)^2u^x(u^y)^3+2p'(T)T(u^\tau)^2u^x(u^y)^3+\pi^{yy}(u^x)^3(u^y)^3 \\
& -2\pi^{xy}(u^y)^4+\pi^{xy}(u^\tau)^2(u^y)^4-2\pi^{xy}(u^x)^2(u^y)^4 \\
& +\Pi(u^\tau)^2u^x(u^y)^5+p'(T)T(u^\tau)^2u^x(u^y)^5-\pi^{xy}(u^y)^6\Big) \\
(A_x^{\mathrm{IS}})_{14} = & -\frac{u^x\left((u^x)^2+1\right)}{u^\tau\left((u^y)^2+1\right)} \\
(A_x^{\mathrm{IS}})_{15} = & -\frac{u^\tau u^x}{(u^y)^2+1} \\
(A_x^{\mathrm{IS}})_{16} = & \frac{u^y\left(2(u^x)^2+(u^y)^2+1\right)}{u^\tau\left((u^y)^2+1\right)} \\
(A_x^{\mathrm{IS}})_{17} = & u^\tau u^x \\
(A_x^{\mathrm{IS}})_{21} = & p''(T)T(u^x)^2+p'(T)(u^x)^2+p'(T) \\
(A_x^{\mathrm{IS}})_{22} = & 2u^x(p'(T)T+\Pi)-\frac{2(\pi^\eta{}_\eta u^x-\pi^{xy}u^y+\pi^{yy}u^x)}{(u^y)^2+1} \\
(A_x^{\mathrm{IS}})_{23} = & \frac{2\left(u^y\left(\pi^\eta{}_\eta(u^x)^2+\pi^{yy}(u^x)^2+\pi^{yy}\right)+\pi^{xy}\left(u^x-u^x(u^y)^2\right)\right)}{\left((u^y)^2+1\right)^2} \\
(A_x^{\mathrm{IS}})_{24} = & -\frac{(u^x)^2+1}{(u^y)^2+1} \\
(A_x^{\mathrm{IS}})_{25} = & -\frac{(u^\tau)^2}{(u^y)^2+1} \\
(A_x^{\mathrm{IS}})_{26} = & \frac{2u^xu^y}{(u^y)^2+1} \\
(A_x^{\mathrm{IS}})_{27} = & (u^x)^2+1 \\
(A_x^{\mathrm{IS}})_{31} = & u^xu^y(p''(T)T+p'(T)) \\
(A_x^{\mathrm{IS}})_{32} = & u^y(p'(T)T+\Pi) \\
(A_x^{\mathrm{IS}})_{33} = & u^x(p'(T)T+\Pi) \\
(A_x^{\mathrm{IS}})_{36} = & 1 \\
(A_x^{\mathrm{IS}})_{37} = & u^xu^y \\
(A_x^{\mathrm{IS}})_{42} = & -\frac{2\left(3\pi^{xy}\tau_\pi u^xu^y\left((u^y)^2+1\right)-3\pi^{yy}\tau_\pi(u^x)^2(u^y)^2+(u^\tau)^2\left((u^y)^2+1\right)\eta\right)}{3(u^\tau)^2} \\
(A_x^{\mathrm{IS}})_{43} = & \frac{2u^xu^y\left(\pi^{xy}\tau_\pi u^xu^y-\pi^{yy}\tau_\pi\left((u^x)^2+1\right)+(u^\tau)^2\eta\right)}{(u^\tau)^2} \\
(A_x^{\mathrm{IS}})_{44} = & \tau_\pi u^x \\
(A_x^{\mathrm{IS}})_{52} = & -\frac{2\eta}{3\tau^2} \\
(A_x^{\mathrm{IS}})_{55} = & \frac{\tau_\pi u^x}{\tau^2}
\end{aligned}
$$

$$(A_x^{\mathrm{IS}})_{62} = \frac{u^x\left(u^y\left(3\pi^\eta{}_\eta\tau_\pi\left(u^\tau\right)^2+3\pi^{yy}\left(2\tau_\pi\left(u^x\right)^2+\tau_\pi\right)+\left(u^\tau\right)^2\eta\right)-3\pi^{xy}\tau_\pi u^x\left(2\left(u^y\right)^2+1\right)\right)}{3\left(u^\tau\right)^2}$$

$$\begin{aligned}(A_x^{\mathrm{IS}})_{63} = {}& \frac{1}{\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right)}\Big(-\pi^{yy}\tau_\pi\left(u^x\right)^2\left(\left(u^x\right)^2+1\right)\left(2\left(u^y\right)^2+1\right)\\ &+\pi^{xy}\tau_\pi u^x u^y\left(\left(2\left(u^x\right)^2-1\right)\left(u^y\right)^2-1\right)\\ &-\left(u^\tau\right)^2\left(\pi^\eta{}_\eta\tau_\pi\left(u^x\right)^2\left(u^y\right)^2-\left(\left(u^x\right)^2+1\right)\left(\left(u^y\right)^2+1\right)\eta\right)\Big)\end{aligned}$$

$$(A_x^{\mathrm{IS}})_{66} = \tau_\pi u^x$$

$$(A_x^{\mathrm{IS}})_{72} = \zeta$$

$$(A_x^{\mathrm{IS}})_{77} = \tau_\Pi u^x$$

$$(A_y^{\mathrm{IS}})_{11} = u^\tau u^y(p''(T)T+p'(T))$$

$$(A_y^{\mathrm{IS}})_{12} = \frac{u^x u^y\left(p'(T)T\left(u^\tau\right)^2+\Pi\left(u^\tau\right)^2-\pi^{yy}\right)+\pi^{xy}\left(\left(u^y\right)^2+1\right)}{\left(u^\tau\right)^3}$$

$$(A_y^{\mathrm{IS}})_{13} = \frac{\left(u^\tau\right)^4\left(p'(T)T+\Pi\right)+\left(u^\tau\right)^2\left(u^y\right)^2\left(p'(T)T+\Pi\right)-u^y(\pi^{xy}u^x+\pi^{yy}u^y)+\pi^{yy}\left(u^\tau\right)^2}{\left(u^\tau\right)^3}$$

$$(A_y^{\mathrm{IS}})_{14} = \frac{u^y}{u^\tau}$$

$$(A_y^{\mathrm{IS}})_{16} = \frac{u^x}{u^\tau}$$

$$(A_y^{\mathrm{IS}})_{17} = u^\tau u^y$$

$$(A_y^{\mathrm{IS}})_{21} = u^x u^y(p''(T)T+p'(T))$$

$$(A_y^{\mathrm{IS}})_{22} = u^y(p'(T)T+\Pi)$$

$$(A_y^{\mathrm{IS}})_{23} = u^x(p'(T)T+\Pi)$$

$$(A_y^{\mathrm{IS}})_{26} = 1$$

$$(A_y^{\mathrm{IS}})_{27} = u^x u^y$$

$$(A_y^{\mathrm{IS}})_{31} = p''(T)T\left(u^y\right)^2+p'(T)\left(u^y\right)^2+p'(T)$$

$$(A_y^{\mathrm{IS}})_{33} = 2u^y(p'(T)T+\Pi)$$

$$(A_y^{\mathrm{IS}})_{34} = 1$$

$$(A_y^{\mathrm{IS}})_{37} = \left(u^y\right)^2+1$$

$$(A_y^{\mathrm{IS}})_{42} = -\frac{2\tau_\pi\left(u^y\right)^2\left(\pi^{xy}\left(u^y\right)^2+\pi^{xy}-\pi^{yy}u^x u^y\right)}{\left(u^\tau\right)^2}$$

$$(A_y^{\mathrm{IS}})_{43} = \frac{6\pi^{xy}\tau_\pi u^x\left(u^y\right)^3-6\pi^{yy}\tau_\pi\left(\left(u^x\right)^2+1\right)\left(u^y\right)^2+4\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right)\eta}{3\left(u^\tau\right)^2}$$

$$(A_y^{\mathrm{IS}})_{44} = \tau_\pi u^y$$

$$(A_y^{\mathrm{IS}})_{53} = -\frac{2\eta}{3\tau^2}$$

$$(A_y^{\mathrm{IS}})_{55} = \frac{\tau_\pi u^y}{\tau^2}$$

$$\begin{aligned}(A_y^{\mathrm{IS}})_{62} = {}& \frac{1}{\left(u^\tau\right)^2}\Big(-\pi^{xy}\tau_\pi u^x u^y\left(2\left(u^y\right)^2+1\right)+\eta+\left(u^y\right)^4\left(\pi^\eta{}_\eta\tau_\pi+\eta\right)\\ &+\left(u^y\right)^2\left(\pi^\eta{}_\eta\tau_\pi+\pi^{yy}\tau_\pi+2\eta\right)+\left(u^x\right)^2\left(\left(u^y\right)^2\left(\pi^\eta{}_\eta\tau_\pi+2\pi^{yy}\tau_\pi+\eta\right)+\eta\right)\Big)\end{aligned}$$

$$\begin{aligned}(A_y^{\mathrm{IS}})_{63} = {}& -\frac{1}{3\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right)}u^y\Big(3\pi^{yy}\tau_\pi u^x\left(\left(u^x\right)^2+1\right)\left(2\left(u^y\right)^2+1\right)\\ &+3\pi^{xy}\tau_\pi u^y\left(\left(1-2\left(u^x\right)^2\right)\left(u^y\right)^2+1\right)\end{aligned}$$

$$+ \left(u^{\tau}\right)^{2} u^{x} \left(3\pi^{\eta}{}_{\eta}\tau_{\pi} \left(u^{y}\right)^{2} - \left(\left(u^{y}\right)^{2}+1\right)\eta\right)\Big)$$

$$(A_y^{\mathrm{IS}})_{66} = \tau_{\pi} u^{y}$$
$$(A_y^{\mathrm{IS}})_{73} = \zeta$$
$$(A_y^{\mathrm{IS}})_{77} = \tau_{\Pi} u^{y}$$

$$\begin{aligned}
S_1^{\mathrm{IS}} = \frac{1}{\tau\left(\left(u^{y}\right)^{2}+1\right)}\Big(& p'(T)T - \pi^{yy}\left(u^{x}\right)^{2} + p'(T)T\left(u^{x}\right)^{2} + 2\pi^{xy}u^{x}u^{y} + \pi^{yy}\left(u^{y}\right)^{2} \\
& + 2p'(T)T\left(u^{y}\right)^{2} + p'(T)T\left(u^{x}\right)^{2}\left(u^{y}\right)^{2} + p'(T)T\left(u^{y}\right)^{4} \\
& + \Pi\left(u^{\tau}\right)^{2}\left(\left(u^{y}\right)^{2}+1\right) + \pi^{\eta}{}_{\eta}\left(-\left(u^{x}\right)^{2} + \left(u^{y}\right)^{2} + 1\right)\Big) \\
S_2^{\mathrm{IS}} = \frac{1}{\tau u^{\tau}\left(\left(u^{y}\right)^{2}+1\right)}\Big(& -\pi^{yy}u^{x} + p'(T)Tu^{x} - \pi^{\eta}{}_{\eta}\left(u^{\tau}\right)^{2}u^{x} - \pi^{yy}\left(u^{x}\right)^{3} + p'(T)T\left(u^{x}\right)^{3} \\
& + \pi^{xy}u^{y} + 2\pi^{xy}\left(u^{x}\right)^{2}u^{y} + 2p'(T)Tu^{x}\left(u^{y}\right)^{2} + p'(T)T\left(u^{x}\right)^{3}\left(u^{y}\right)^{2} \\
& + \pi^{xy}\left(u^{y}\right)^{3} + p'(T)Tu^{x}\left(u^{y}\right)^{4} + \Pi\left(u^{\tau}\right)^{2}u^{x}\left(\left(u^{y}\right)^{2}+1\right)\Big) \\
S_3^{\mathrm{IS}} = \frac{u^{y}\left(p'(T)T\left(u^{\tau}\right)^{2} + \Pi\left(u^{\tau}\right)^{2} + \pi^{yy}\right) + \pi^{xy}u^{x}}{\tau u^{\tau}} \\
S_4^{\mathrm{IS}} = \pi^{yy} - \frac{2u^{\tau}\left(\left(u^{y}\right)^{2}+1\right)\eta}{3\tau} \\
S_5^{\mathrm{IS}} = \frac{3\pi^{\eta}{}_{\eta}\tau + 4u^{\tau}\eta}{3\tau^{3}} \\
S_6^{\mathrm{IS}} = \pi^{xy} - \frac{2u^{\tau}u^{x}u^{y}\eta}{3\tau} \\
S_7^{\mathrm{IS}} = \Pi + \frac{u^{\tau}\zeta}{\tau}
\end{aligned}$$

1.4.2 $\delta_{\pi\pi}$ part

$$\begin{aligned}
(A_t^{(\delta)})_{42} &= \frac{\pi^{yy}u^{x}}{u^{\tau}} \\
(A_t^{(\delta)})_{43} &= \frac{\pi^{yy}u^{y}}{u^{\tau}} \\
(A_t^{(\delta)})_{52} &= \frac{\pi^{\eta}{}_{\eta}u^{x}}{\tau^{2}u^{\tau}} \\
(A_t^{(\delta)})_{53} &= \frac{\pi^{\eta}{}_{\eta}u^{y}}{\tau^{2}u^{\tau}} \\
(A_t^{(\delta)})_{62} &= \frac{\pi^{xy}u^{x}}{u^{\tau}} \\
(A_t^{(\delta)})_{63} &= \frac{\pi^{xy}u^{y}}{u^{\tau}}
\end{aligned}$$

$$\begin{aligned}
(A_x^{(\delta)})_{42} &= \pi^{yy} \\
(A_x^{(\delta)})_{52} &= \frac{\pi^{\eta}{}_{\eta}}{\tau^{2}} \\
(A_x^{(\delta)})_{62} &= \pi^{xy}
\end{aligned}$$

$$\begin{aligned}
(A_y^{(\delta)})_{43} &= \pi^{yy} \\
(A_y^{(\delta)})_{53} &= \frac{\pi^{\eta}{}_{\eta}}{\tau^{2}} \\
(A_y^{(\delta)})_{63} &= \pi^{xy}
\end{aligned}$$

$$S_4^{(\delta)} = \frac{\pi^{yy}u^{\tau}}{\tau}$$

$$S_5^{(\delta)} = \frac{\pi^\eta{}_\eta u^\tau}{\tau^3}$$
$$S_6^{(\delta)} = \frac{\pi^{xy} u^\tau}{\tau}$$

1.5 Checks

id	what it tests	
`V-G0cyl`	cylindrical Milne: $\nabla_\lambda g_{\mu\nu} = 0$	PASS
`V-G0car`	Cartesian Milne: $\nabla_\lambda g_{\mu\nu} = 0$	PASS
`V-G1cyl`	cylindrical: $\nabla_\mu V^\mu = \partial_\tau X + X/\tau$	PASS
`V-G1car`	Cartesian: $\nabla_\mu V^\mu = \partial_\tau X + X/\tau$	PASS
`V-G2cyl-uu`	$u \cdot u$ = -1	PASS
`V-G2cyl-un`	$u_\mu \pi^{\mu\nu} = 0$	PASS
`V-G2cyl-tr`	$\pi^\mu{}_\mu = 0$	PASS
`V-G3cyl`	cylindrical system is quasi-linear	PASS
`V-G6cyl`	cylindrical: (IS + $\delta_{\pi\pi}\pi\theta$ in eq. 2) = IS + $\delta_{\pi\pi}\times$ (the exported $\delta_{\pi\pi}$ part), equation by equation	PASS
`V-G4`	1-D: derived = shipped one_d_viscous_matrix, A_t, A_r, S entry by entry, read from the .jl, symbolic $\delta_{\pi\pi}$ (control)	PASS
`V-G4-0`	1-D at $\delta_{\pi\pi} = 0$ (plain IS): plain is == the shipped is part	PASS
`V-G4-1`	1-D at $\delta_{\pi\pi} = \frac{4}{3}\tau_\pi$ (default)	PASS
`V-G4m`	1-D at $\delta_{\pi\pi} = \frac{4}{3}\tau_\pi$ = the medium block of the 7-field :MIS one_d_viscous_matrix_fugacity, raw, read from the .jl	PASS
`V-G4m-ctl`	control: at $\delta_{\pi\pi} = 0$ it is not the :MIS medium block	PASS
`V-G2car-uu`	$u \cdot u$ = -1	PASS
`V-G2car-un`	$u_\mu \pi^{\mu\nu} = 0$	PASS
`V-G2car-tr`	$\pi^\mu{}_\mu = 0$	PASS
`V-G3car`	Cartesian system is quasi-linear	PASS
`V-G6car`	Cartesian: (IS + $\delta_{\pi\pi}\pi\theta$ in eq. 2) = IS + $\delta_{\pi\pi}\times$ (the exported $\delta_{\pi\pi}$ part), equation by equation	PASS
`V-G5-e`	energy row on Bjorken = $\partial_\tau e + (e + p + \Pi + \pi^\eta{}_\eta)/\tau$	PASS
`V-G5-pi`	$\pi^\eta{}_\eta$ row on Bjorken = $\tau_\pi \partial_\tau \pi + \pi + 4\eta/(3\tau) + \delta_{\pi\pi}\pi/\tau$ ($\theta = 1/\tau$)	PASS
`V-G5-Pi`	bulk row on Bjorken = $\tau_\Pi \partial_\tau \Pi + \Pi + \zeta/\tau$	PASS
`V-G5-mom`	the x, y and xy rows vanish identically on Bjorken flow	PASS

23 passed, 0 failed (23 checks evaluated).

Not checked in this script: the slab limit (2+1D at $u^y = 0$, $\partial_y = 0$ against 1+1D at $r \to \infty$); it is a numerical comparison of the two generated routines.

Re-run with `wolframscript -file deriv/derive_2p1d_viscous.wls`.

2 Charm first moment on a 2+1D background ($\nabla\alpha$ closure)

Script: `deriv/derive_hq_2p1d.wls`

Writes: `two_d_HQ_BG_matrix` and `two_d_HQ_live_rows` in `src/Matrix/2d_viscous_HQ_BG.jl`

2.1 Setup

Cartesian Milne coordinates, boost invariant, on a prescribed background T, u^x, u^y:

$$x^\mu = (\tau, x, y, \eta), \qquad g_{\mu\nu} = \mathrm{diag}(-1, 1, 1, \tau^2), \qquad u^\mu = (u^\tau, u^x, u^y, 0), \quad u^\tau = \sqrt{(u^x)^2 + (u^y)^2 + 1}$$

Charm is a passive tracer:

$$N^\mu = n\, u^\mu + \nu^\mu, \qquad \nabla_\mu N^\mu = 0$$
$$\tau_n\, \Delta^\mu{}_\beta\, D\nu^\beta + \nu^\mu + \kappa\, \Delta^{\mu\beta}\partial_\beta\alpha = 0$$
$$n = n(T, \alpha), \qquad \kappa = D_s\, \partial_\alpha n, \qquad D = u^\mu\nabla_\mu, \qquad \Delta^{\mu\nu} = g^{\mu\nu} + u^\mu u^\nu$$

Navier–Stokes limit: Fick's law $\nu^\mu = -\kappa\, \nabla_\perp^\mu \alpha$. Fields, with ν^τ from $u_\mu \nu^\mu = 0$:

$$\phi = (\alpha, \nu^x, \nu^y), \qquad \nu^\tau = \frac{\nu^x u^x + \nu^y u^y}{u^\tau}$$

Rows: $\nabla_\mu N^\mu$ and the x, y components of the current equation, in

$$A_t\, \partial_\tau\phi + A_x\, \partial_x\phi + A_y\, \partial_y\phi + S = 0$$

2.2 Steps

$$\Gamma^\tau{}_{\eta\eta} = \tau, \quad \Gamma^\eta{}_{\tau\eta} = \frac{1}{\tau}$$

$$\begin{aligned}
a^\tau &= \frac{(\partial_\tau u^x)u^\tau u^x + (\partial_\tau u^y)u^\tau u^y + (\partial_x u^x)\,(u^x)^2 + (\partial_x u^y)u^x u^y + (\partial_y u^x)u^x u^y + (\partial_y u^y)\,(u^y)^2}{u^\tau} \\
a^x &= (\partial_\tau u^x)u^\tau + (\partial_x u^x)u^x + (\partial_y u^x)u^y \\
a^y &= (\partial_\tau u^y)u^\tau + (\partial_x u^y)u^x + (\partial_y u^y)u^y \\
\nu_\lambda a^\lambda &= \nu^x((\partial_\tau u^x)u^\tau + (\partial_x u^x)u^x + (\partial_y u^x)u^y) + \nu^y((\partial_\tau u^y)u^\tau + (\partial_x u^y)u^x + (\partial_y u^y)u^y) \\
&\quad - \frac{1}{(u^\tau)^2}\nu^x u^x + \nu^y u^y\Big(u^\tau((\partial_\tau u^x)u^x + (\partial_\tau u^y)u^y) + u^x((\partial_x u^x)u^x + (\partial_x u^y)u^y) \\
&\quad + u^y((\partial_y u^x)u^x + (\partial_y u^y)u^y)\Big) \\
D\nu^\tau &= \frac{1}{(u^\tau)^3}\Big((u^\tau)^3((\partial_\tau\nu^x)u^x + (\partial_\tau\nu^y)u^y + (\partial_\tau u^x)\nu^x + (\partial_\tau u^y)\nu^y) \\
&\quad + (u^\tau)^2 u^x((\partial_x\nu^x)u^x + (\partial_x\nu^y)u^y + (\partial_x u^x)\nu^x + (\partial_x u^y)\nu^y) \\
&\quad + (u^\tau)^2 u^y((\partial_y\nu^x)u^x + (\partial_y\nu^y)u^y + (\partial_y u^x)\nu^x + (\partial_y u^y)\nu^y) \\
&\quad - u^\tau((\partial_\tau u^x)u^x + (\partial_\tau u^y)u^y)(\nu^x u^x + \nu^y u^y) \\
&\quad - u^x((\partial_x u^x)u^x + (\partial_x u^y)u^y)(\nu^x u^x + \nu^y u^y) \\
&\quad - u^y((\partial_y u^x)u^x + (\partial_y u^y)u^y)(\nu^x u^x + \nu^y u^y)\Big) \\
D\nu^x &= (\partial_\tau\nu^x)u^\tau + (\partial_x\nu^x)u^x + (\partial_y\nu^x)u^y \\
D\nu^y &= (\partial_\tau\nu^y)u^\tau + (\partial_x\nu^y)u^x + (\partial_y\nu^y)u^y
\end{aligned}$$

$\Delta^\mu{}_\beta D\nu^\beta = D\nu^\mu - u^\mu\nu_\lambda a^\lambda$, because $u_\beta\nu^\beta = 0$. Rows:

$$E_1 = \nabla_\mu N^\mu = (\partial_x\nu^x) + (\partial_y\nu^y) + (\partial_x u^x)n + (\partial_y u^y)n + u^\tau((\partial_\tau\alpha)(\partial_\alpha n) + (\partial_\tau T)(\partial_T n))$$

$$
\begin{aligned}
&+ u^x((\partial_x\alpha)(\partial_\alpha n) + (\partial_x T)(\partial_T n)) + u^y((\partial_y\alpha)(\partial_\alpha n) + (\partial_y T)(\partial_T n))\\
&+ \frac{(\partial_\tau\nu^x)u^x + (\partial_\tau\nu^y)u^y + (\partial_\tau u^x)\nu^x + (\partial_\tau u^y)\nu^y}{u^\tau} + \frac{n((\partial_\tau u^x)u^x + (\partial_\tau u^y)u^y)}{u^\tau}\\
&- \frac{((\partial_\tau u^x)u^x + (\partial_\tau u^y)u^y)(\nu^x u^x + \nu^y u^y)}{(u^\tau)^3} + \frac{n\,(u^\tau)^2 + \nu^x u^x + \nu^y u^y}{\tau u^\tau}\\
E_2 = \nu^x &+ \kappa\left((\partial_\tau\alpha)u^\tau u^x + (\partial_x\alpha)\,(u^x)^2 + (\partial_x\alpha) + (\partial_y\alpha)u^x u^y\right)\\
&+ \tau_n\Big((\partial_\tau\nu^x)u^\tau + (\partial_x\nu^x)u^x + (\partial_y\nu^x)u^y\\
&- u^x\Big(\nu^x((\partial_\tau u^x)u^\tau + (\partial_x u^x)u^x + (\partial_y u^x)u^y)\\
&+ \nu^y((\partial_\tau u^y)u^\tau + (\partial_x u^y)u^x + (\partial_y u^y)u^y)\\
&- \frac{1}{(u^\tau)^2}\nu^x u^x + \nu^y u^y\Big(u^\tau((\partial_\tau u^x)u^x + (\partial_\tau u^y)u^y)\\
&+ u^x((\partial_x u^x)u^x + (\partial_x u^y)u^y)\\
&+ u^y((\partial_y u^x)u^x + (\partial_y u^y)u^y)\Big)\Big)\Big)\\
E_3 = \nu^y &+ \kappa\left((\partial_\tau\alpha)u^\tau u^y + (\partial_x\alpha)u^x u^y + (\partial_y\alpha)\,(u^y)^2 + (\partial_y\alpha)\right)\\
&+ \tau_n\Big((\partial_\tau\nu^y)u^\tau + (\partial_x\nu^y)u^x + (\partial_y\nu^y)u^y\\
&- u^y\Big(\nu^x((\partial_\tau u^x)u^\tau + (\partial_x u^x)u^x + (\partial_y u^x)u^y)\\
&+ \nu^y((\partial_\tau u^y)u^\tau + (\partial_x u^y)u^x + (\partial_y u^y)u^y)\\
&- \frac{1}{(u^\tau)^2}\nu^x u^x + \nu^y u^y\Big(u^\tau((\partial_\tau u^x)u^x + (\partial_\tau u^y)u^y)\\
&+ u^x((\partial_x u^x)u^x + (\partial_x u^y)u^y)\\
&+ u^y((\partial_y u^x)u^x + (\partial_y u^y)u^y)\Big)\Big)\Big)
\end{aligned}
$$

$$
A_{ij}^{(k)} = \frac{\partial E_i}{\partial(\partial_k\phi_j)}, \qquad S_i = E_i\big|_{\partial\phi=0}, \qquad k = \tau, x, y
$$

2.3 Result

$$
\begin{aligned}
(A_t)_{11} &= (\partial_\alpha n)u^\tau\\
(A_t)_{12} &= \frac{u^x}{u^\tau}\\
(A_t)_{13} &= \frac{u^y}{u^\tau}\\
(A_t)_{21} &= \kappa u^\tau u^x\\
(A_t)_{22} &= \tau_n u^\tau\\
(A_t)_{31} &= \kappa u^\tau u^y\\
(A_t)_{33} &= \tau_n u^\tau
\end{aligned}
$$

$$
\begin{aligned}
(A_x)_{11} &= (\partial_\alpha n)u^x\\
(A_x)_{12} &= 1\\
(A_x)_{21} &= \kappa\left((u^x)^2 + 1\right)\\
(A_x)_{22} &= \tau_n u^x\\
(A_x)_{31} &= \kappa u^x u^y\\
(A_x)_{33} &= \tau_n u^x
\end{aligned}
$$

$$
\begin{aligned}
(A_y)_{11} &= (\partial_\alpha n)u^y\\
(A_y)_{13} &= 1
\end{aligned}
$$

$$
\begin{aligned}
(A_y)_{21} &= \kappa u^x u^y \\
(A_y)_{22} &= \tau_n u^y \\
(A_y)_{31} &= \kappa \left((u^y)^2 + 1 \right) \\
(A_y)_{33} &= \tau_n u^y
\end{aligned}
$$

$$
\begin{aligned}
S_1 = \frac{1}{\tau (u^\tau)^3} \Big(& (\partial_\tau u^x) \nu^x \tau (u^\tau)^2 + (\partial_\tau u^y) \nu^y \tau (u^\tau)^2 + (\partial_x u^x) n \tau (u^\tau)^3 + (\partial_y u^y) n \tau (u^\tau)^3 \\
& + n (u^\tau)^4 + (\partial_\tau T)(\partial_T n) \tau (u^\tau)^4 + \nu^x (u^\tau)^2 u^x + (\partial_\tau u^x) n \tau (u^\tau)^2 u^x \\
& + (\partial_x T)(\partial_T n) \tau (u^\tau)^3 u^x - (\partial_\tau u^x) \nu^x \tau (u^x)^2 + \nu^y (u^\tau)^2 u^y + (\partial_\tau u^y) n \tau (u^\tau)^2 u^y \\
& + (\partial_y T)(\partial_T n) \tau (u^\tau)^3 u^y - (\partial_\tau u^y) \nu^x \tau u^x u^y - (\partial_\tau u^x) \nu^y \tau u^x u^y \\
& - (\partial_\tau u^y) \nu^y \tau (u^y)^2 \Big) \\
S_2 = \frac{1}{(u^\tau)^2} \Big(& -\nu^y \tau_n u^x \Big((\partial_\tau u^y) u^\tau + (\partial_x u^y) (u^x)^3 + (\partial_y u^y) u^y \\
& + (u^x)^2 ((\partial_\tau u^y) u^\tau - (\partial_x u^x) u^y + (\partial_y u^y) u^y) \\
& + u^x ((\partial_x u^y) - u^y ((\partial_\tau u^x) u^\tau + (\partial_y u^x) u^y)) \Big) \\
& + \nu^x \Big(1 + (\partial_x u^y) \tau_n (u^x)^3 u^y + (u^y)^2 \\
& - \tau_n u^x \left((u^y)^2 + 1 \right) ((\partial_\tau u^x) u^\tau + (\partial_y u^x) u^y) \\
& + (u^x)^2 \Big(1 + (\partial_\tau u^y) \tau_n u^\tau u^y + (\partial_y u^y) \tau_n (u^y)^2 \\
& - (\partial_x u^x) \tau_n \left((u^y)^2 + 1 \right) \Big) \Big) \Big) \\
S_3 = \frac{1}{(u^\tau)^2} \Big(& -\nu^x \tau_n u^y \Big((\partial_\tau u^x) u^\tau + (\partial_x u^x) u^x \\
& + u^y ((\partial_y u^x) - u^x ((\partial_\tau u^y) u^\tau + (\partial_x u^y) u^x)) \\
& + (u^y)^2 ((\partial_\tau u^x) u^\tau + (\partial_x u^x) u^x - (\partial_y u^y) u^x) + (\partial_y u^x) (u^y)^3 \Big) \\
& + \nu^y \Big(1 - (\partial_\tau u^y) \tau_n u^\tau u^y - (\partial_x u^y) \tau_n (u^x)^3 u^y + (u^y)^2 (1 - (\partial_y u^y) \tau_n) \\
& + (u^x)^2 \Big(-(\partial_\tau u^y) \tau_n u^\tau u^y + \tau_n (u^y)^2 ((\partial_x u^x) - (\partial_y u^y)) + 1 \Big) \\
& + \tau_n u^x u^y (u^y ((\partial_\tau u^x) u^\tau + (\partial_y u^x) u^y) - (\partial_x u^y)) \Big) \Big)
\end{aligned}
$$

2.3.1 Live background

The same three rows with the medium promoted to fields, $\phi_L = (T, u^x, u^y, \alpha, \nu^x, \nu^y)$. Columns 4–6 are the matrices above (H-G6a); columns 1–3 and the remaining source:

$$
\begin{aligned}
(A_t^L)_{11} &= (\partial_T n) u^\tau \\
(A_t^L)_{12} &= \frac{n (u^\tau)^2 u^x + \nu^x \left((u^y)^2 + 1 \right) - \nu^y u^x u^y}{(u^\tau)^3} \\
(A_t^L)_{13} &= \frac{n (u^\tau)^2 u^y - \nu^x u^x u^y + \nu^y \left((u^x)^2 + 1 \right)}{(u^\tau)^3} \\
(A_t^L)_{22} &= \frac{\tau_n u^x \left(\nu^y u^x u^y - \nu^x \left((u^y)^2 + 1 \right) \right)}{u^\tau} \\
(A_t^L)_{23} &= -\frac{\tau_n u^x \left(-\nu^x u^x u^y + \nu^y (u^x)^2 + \nu^y \right)}{u^\tau} \\
(A_t^L)_{32} &= -\frac{\tau_n u^y \left(\nu^x (u^y)^2 + \nu^x - \nu^y u^x u^y \right)}{u^\tau}
\end{aligned}
$$

$$(A_t^L)_{33} = \frac{\tau_n u^y \left(\nu^x u^x u^y - \nu^y \left(\left(u^x\right)^2 + 1\right)\right)}{u^\tau}$$

$$(A_x^L)_{11} = (\partial_T n) u^x$$
$$(A_x^L)_{12} = n$$
$$(A_x^L)_{22} = \frac{\tau_n \left(u^x\right)^2 \left(\nu^y u^x u^y - \nu^x \left(\left(u^y\right)^2 + 1\right)\right)}{\left(u^\tau\right)^2}$$
$$(A_x^L)_{23} = -\frac{\tau_n \left(u^x\right)^2 \left(-\nu^x u^x u^y + \nu^y \left(u^x\right)^2 + \nu^y\right)}{\left(u^\tau\right)^2}$$
$$(A_x^L)_{32} = \frac{\tau_n u^x u^y \left(\nu^y u^x u^y - \nu^x \left(\left(u^y\right)^2 + 1\right)\right)}{\left(u^\tau\right)^2}$$
$$(A_x^L)_{33} = \frac{\tau_n u^x u^y \left(\nu^x u^x u^y - \nu^y \left(\left(u^x\right)^2 + 1\right)\right)}{\left(u^\tau\right)^2}$$

$$(A_y^L)_{11} = (\partial_T n) u^y$$
$$(A_y^L)_{13} = n$$
$$(A_y^L)_{22} = \frac{\tau_n u^x u^y \left(\nu^y u^x u^y - \nu^x \left(\left(u^y\right)^2 + 1\right)\right)}{\left(u^\tau\right)^2}$$
$$(A_y^L)_{23} = -\frac{\tau_n u^x u^y \left(-\nu^x u^x u^y + \nu^y \left(u^x\right)^2 + \nu^y\right)}{\left(u^\tau\right)^2}$$
$$(A_y^L)_{32} = -\frac{\tau_n \left(u^y\right)^2 \left(\nu^x \left(u^y\right)^2 + \nu^x - \nu^y u^x u^y\right)}{\left(u^\tau\right)^2}$$
$$(A_y^L)_{33} = \frac{\tau_n \left(u^y\right)^2 \left(\nu^x u^x u^y - \nu^y \left(\left(u^x\right)^2 + 1\right)\right)}{\left(u^\tau\right)^2}$$

$$S_1^L = \frac{n \left(u^\tau\right)^2 + \nu^x u^x + \nu^y u^y}{\tau u^\tau}$$
$$S_2^L = \nu^x$$
$$S_3^L = \nu^y$$

2.4 Checks

id	what it tests	
`H-G0cyl`	cylindrical Milne: $\nabla_\lambda g_{\mu\nu} = 0$	PASS
`H-G0car`	Cartesian Milne: $\nabla_\lambda g_{\mu\nu} = 0$	PASS
`H-G1cyl`	cylindrical: $\nabla_\mu V^\mu = \partial_\tau X + X/\tau$ for $V^\mu = (X(\tau), 0, 0, 0)$	PASS
`H-G1car`	Cartesian: $\nabla_\mu V^\mu = \partial_\tau X + X/\tau$	PASS
`H-G2cyl-uu`	$u \cdot u$ = -1	PASS
`H-G2cyl-un`	$u \cdot \nu = 0$	PASS
`H-G3cyl`	cylindrical system is quasi-linear	PASS
`H-G4-At`	cylindrical $A_t \times (u^\tau)^3 r\tau$ = Fluidum's shipped A_t	PASS
`H-G4-Ax`	cylindrical $A_x \times (u^\tau)^3 r\tau$ = Fluidum's shipped A_x	PASS
`H-G4-src`	cylindrical S $\times (u^\tau)^3 r\tau$ = Fluidum's shipped source	PASS
`H-G4src`	cylindrical $A_t, A_x, S \times (u^\tau)^3 r\tau$ = 1d_viscous_HQ_cilindrical_BG.jl read from the file	PASS
`H-G2car-uu`	$u \cdot u$ = -1	PASS
`H-G2car-un`	$u \cdot \nu = 0$	PASS
`H-G3car`	Cartesian system is quasi-linear	PASS

id	what it tests	
`H-G5`	slab limit ($u^y \to 0$, $\partial_y \to 0$) decouples ν^y from (α, ν^x)	PASS
`H-G6a`	the (α, ν^x, ν^y) columns of the 10-field rows = the background-mode matrices	PASS
`H-G6b`	S^L + (medium columns).(background derivatives) = S: same system, re-sorted	PASS

17 passed, 0 failed (17 checks evaluated).

Re-run with `wolframscript -file deriv/derive_hq_2p1d.wls`.

3 Medium, charm and the two-way coupled system

Script: `deriv/derive_two_way.wls`, with `derive_two_way_gates.wls` and `derive_two_way_export.wls`
Writes: the charm and two-way matrices of `src/Matrix` (one function per system, listed below)

3.1 Setup

Three charts, boost invariant: 0+1D Bjorken (fields of τ only), 1+1D cylindrical, 2+1D Cartesian Milne:

$$x^\mu = (\tau, x, y, \eta), \qquad g_{\mu\nu} = \mathrm{diag}(-1, 1, 1, \tau^2), \quad u^\mu = (1, 0, 0, 0)$$
$$x^\mu = (\tau, r, \phi, \eta), \qquad g_{\mu\nu} = \mathrm{diag}(-1, 1, r^2, \tau^2)$$
$$x^\mu = (\tau, x, y, \eta), \qquad g_{\mu\nu} = \mathrm{diag}(-1, 1, 1, \tau^2)$$

Medium, with the EOS $P(T)$ kept symbolic:

$$T_m^{\mu\nu} = (e + P + \Pi)\, u^\mu u^\nu + (P + \Pi)\, g^{\mu\nu} + \pi^{\mu\nu}, \qquad e = TP'(T) - P(T)$$
$$\tau_\pi\, \Delta^{\mu\nu}_{\alpha\beta} D\pi^{\alpha\beta} + \pi^{\mu\nu} + 2\eta\, \sigma^{\mu\nu} + \delta_{\pi\pi}\, \pi^{\mu\nu}\theta = 0, \qquad \tau_\Pi D\Pi + \Pi + \zeta\theta + \delta_{\Pi\Pi}\Pi\theta = 0$$

(D, θ, $\Delta^{\mu\nu}$, $\Delta^{\mu\nu}_{\alpha\beta}$, $\sigma^{\mu\nu}$ as in the viscous section.) Charm, with $n = n(T, \alpha)$, $h = h(T)$, $P_0 = nT$ and $\delta M^{\mu\nu} = \pi_Q^{\mu\nu} + \Pi_Q \Delta^{\mu\nu}$:

$$\nabla_\mu (n\, u^\mu + \nu^\mu) = 0, \qquad \frac{D_s}{T}\, \Delta^\mu{}_\lambda \nabla_\nu T_Q^{\nu\lambda} + \nu^\mu = 0$$
$$T_Q^{\mu\nu} = (nh - P_0)\, u^\mu u^\nu + P_0\, \Delta^{\mu\nu} + h\, (u^\mu \nu^\nu + \nu^\mu u^\nu)$$
$$S^{\lambda\alpha\beta} = I_{31}\, (u^\lambda \Delta^{\alpha\beta} + 2 \text{ perm.}) + \mathcal{A}\, (u^\lambda u^\alpha \nu^\beta + 2 \text{ perm.}) + \mathcal{B}\, (\Delta^{\lambda\alpha} \nu^\beta + 2 \text{ perm.}) + \mathcal{C}\, (u^\lambda \delta M^{\alpha\beta} + 2 \text{ perm.})$$
$$\mathcal{E}^{\mu\nu} = \Delta^\mu{}_\alpha \Delta^\nu{}_\beta \nabla_\lambda S^{\lambda\alpha\beta}: \qquad \frac{D_s}{2T}\Big(\mathcal{E}^{\mu\nu} - \tfrac{1}{3}\Delta^{\mu\nu}\mathcal{E}^\lambda{}_\lambda\Big) + \pi_Q^{\mu\nu} = 0, \qquad \frac{D_s}{2T}\, \tfrac{1}{3}\mathcal{E}^\lambda{}_\lambda + \Pi_Q = 0$$

"2 perm." are the two other placements of the lone index. $I_{31}(T, \alpha)$, $\mathcal{A}(T)$, $\mathcal{B}(T)$, $\mathcal{C}(T)$ stay symbolic (Boltzmann: $I_{31} = nTh$, $\mathcal{A} = m^2 + 5Th$, $\mathcal{B} = Th$, $\mathcal{C} = mK_4/K_3$). In S1–S5 the second moment δM is evolved but does not enter T_Q. In 2+1D the traceless rows also carry

$$-(1 - w_\omega)\, \tau_M W^{\mu\nu}, \qquad \tau_M = \frac{D_s \mathcal{C}}{2T}, \qquad W^{\mu\nu} = \pi_{Q\lambda}^\mu \omega^{\lambda\nu} + \pi_{Q\lambda}^\nu \omega^{\lambda\mu}, \qquad \omega^{\mu\nu} = \tfrac{1}{2}(\Delta^{\mu\lambda}\nabla_\lambda u^\nu - \Delta^{\nu\lambda}\nabla_\lambda u^\mu)$$

so $w_\omega = 1$ is the covariant row, which is exported; $w_\omega = 0$ removes the corotation coupling. The five systems (S1, S2 and every 0+1D system are derived for the checks; S3, S4, S5 in 1+1D and 2+1D are exported, with $\delta_{\Pi\Pi} = 0$):

$$\text{S1}: \ \nabla_\mu T_m^{\mu\nu} = 0, \ \pi = \Pi = 0 \qquad \text{S2}: \ \text{S1} + \text{the } \pi, \Pi \text{ equations}$$
$$\text{S3}: \ \text{charm number} + \text{first moment, background } T, u \qquad \text{S4}: \ \text{S3} + \text{second moment, background } T, u$$
$$\text{S5}: \ \nabla_\mu (T_m + T_Q)^{\mu\nu} = 0, \ \text{the } \pi, \Pi \text{ equations, the charm rows of S4; all fields evolved}$$
$$\text{S6}: \ \nabla_\mu (T_m + T_Q + \delta M)^{\mu\nu} = 0, \ \frac{D_s}{T}\Delta^\mu{}_\lambda \nabla_\nu (T_Q + \delta M)^{\nu\lambda} + \nu^\mu = 0, \ \text{the rest as S5}$$

Conformal: $P = a_{\mathrm{SB}} T^4$, no bulk. Source on the left, $A_t\, \partial_\tau \phi + A_r\, \partial_r \phi + S = 0$ in 1+1D and $A_t\, \partial_\tau \phi + A_x\, \partial_x \phi + A_y\, \partial_y \phi + S = 0$ in 2+1D.

3.1.1 Fields

Medium, charm first moment, charm second moment, chart by chart:

$$\text{0+1D}: \ (T, \pi^\eta{}_\eta, \Pi), \ (\alpha), \ (\pi_Q{}^\eta{}_\eta, \Pi_Q)$$

$$
\begin{aligned}
&1{+}1\text{D}: \ (T, u^r, {\pi^\phi}_\phi, {\pi^\eta}_\eta, \Pi), \ (\alpha, \nu^r), \ ({\pi_Q}^\phi{}_\phi, {\pi_Q}^\eta{}_\eta, \Pi_Q)\\
&2{+}1\text{D}: \ (T, u^x, u^y, \pi^{yy}, {\pi^\eta}_\eta, \pi^{xy}, \Pi), \ (\alpha, \nu^x, \nu^y)\\
&\qquad\qquad (\pi_Q^{yy}, {\pi_Q}^\eta{}_\eta, \pi_Q^{xy}, \Pi_Q)
\end{aligned}
$$

Rows, in the same order: the fluid rows $\nabla_\mu T^{\mu\tau}$, $\nabla_\mu T^{\mu i}$ ($T_m + T_Q$ in S5); the shear rows $E^{\eta\eta}$ (0+1D), $E^{\phi\phi}$, $E^{\eta\eta}$ (1+1D), E^{yy}, $E^{\eta\eta}$, E^{xy} (2+1D); the bulk row; $\nabla_\mu N^\mu$; the current components; the traceless second-moment rows, components $\eta\eta$ (0+1D), $\phi\phi$, $\eta\eta$ (1+1D), yy, $\eta\eta$, xy (2+1D), as the medium's shear rows; the trace row.

In 0+1D the tensors are diagonal and $\nu = 0$:

$$
\pi^{\mu\nu} = \mathrm{diag}(0, -\frac{{\pi^\eta}_\eta}{2}, -\frac{{\pi^\eta}_\eta}{2}, \frac{{\pi^\eta}_\eta}{\tau^2}), \qquad \pi_Q^{\mu\nu} = \mathrm{diag}(0, -\frac{{\pi_Q}^\eta{}_\eta}{2}, -\frac{{\pi_Q}^\eta{}_\eta}{2}, \frac{{\pi_Q}^\eta{}_\eta}{\tau^2})
$$

In 1+1D the medium $\pi^{\mu\nu}$ is as in the viscous section, and so is the charm stress: ${\pi_Q}^\phi{}_\phi$ and ${\pi_Q}^\eta{}_\eta$ are independent, the radial component follows from the trace. With the orthonormal triad transverse to u,

$$
\begin{aligned}
\pi_Q^{\mu\nu} &= -({\pi_Q}^\phi{}_\phi + {\pi_Q}^\eta{}_\eta)\, \ell^\mu \ell^\nu + {\pi_Q}^\phi{}_\phi\, \hat{\phi}^\mu \hat{\phi}^\nu + {\pi_Q}^\eta{}_\eta\, \hat{\eta}^\mu \hat{\eta}^\nu\\
\ell^\mu &= (u^r, u^\tau, 0, 0), \quad \hat{\phi}^\mu = (0, 0, \frac{1}{r}, 0), \quad \hat{\eta}^\mu = (0, 0, 0, \frac{1}{\tau})
\end{aligned}
$$

In 2+1D the medium $\pi^{\mu\nu}$ is as in the viscous section, and so is the charm stress: π_Q^{yy}, ${\pi_Q}^\eta{}_\eta$, π_Q^{xy} are independent, the τ row follows from $u_\mu \pi_Q^{\mu\nu} = 0$ and

$$
\pi_Q^{xx} = -\frac{-2\pi_Q^{xy} u^x u^y + \pi_Q^{yy} \left(u^x\right)^2 + \pi_Q^{yy} + {\pi_Q}^\eta{}_\eta \left(u^x\right)^2 + {\pi_Q}^\eta{}_\eta \left(u^y\right)^2 + {\pi_Q}^\eta{}_\eta}{\left(u^y\right)^2 + 1}
$$

3.2 Steps

3.2.1 S3: charm first moment

$$
{\Gamma^\tau}_{\eta\eta} = \tau, \quad {\Gamma^r}_{\phi\phi} = -r, \quad {\Gamma^\phi}_{r\phi} = \frac{1}{r}, \quad {\Gamma^\eta}_{\tau\eta} = \frac{1}{\tau}
$$

$$
\begin{aligned}
N^\tau &= \frac{n \left(u^r\right)^2 + n + \nu^r u^r}{u^\tau}\\
N^r &= n u^r + \nu^r
\end{aligned}
$$

$$
\begin{aligned}
E_\alpha = \nabla_\mu N^\mu = {} & (\partial_r \nu^r) + (\partial_r u^r) n + u^r((\partial_\alpha n)(\partial_r \alpha) + (\partial_T n)(\partial_r T)) + \frac{n u^r + \nu^r}{r} - \frac{(\partial_\tau u^r)\nu^r \left(u^r\right)^2}{\left(u^\tau\right)^3}\\
& + \frac{(\partial_\tau u^r)\nu^r}{u^\tau} + \frac{(\partial_\tau \nu^r) u^r}{u^\tau} + \frac{(\partial_\tau u^r) n u^r}{u^\tau} + \frac{n \left(u^r\right)^2 + n + \nu^r u^r}{\tau u^\tau}\\
& + u^\tau((\partial_\alpha n)(\partial_\tau \alpha) + (\partial_T n)(\partial_\tau T))
\end{aligned}
$$

$$
\begin{aligned}
T_Q^{\tau\tau} &= h\left(n \left(u^r\right)^2 + n + 2\nu^r u^r\right) - nT\\
T_Q^{\tau r} &= \frac{h\left(n u^r \left(u^\tau\right)^2 + 2\nu^r \left(u^r\right)^2 + \nu^r\right)}{u^\tau}\\
T_Q^{rr} &= n\left(h \left(u^r\right)^2 + T\right) + 2h\nu^r u^r\\
T_Q^{\phi\phi} &= \frac{nT}{r^2}\\
T_Q^{\eta\eta} &= \frac{nT}{\tau^2}
\end{aligned}
$$

$$
\begin{aligned}
\nabla_\mu T_Q^{\mu\tau} &= (\partial_\tau T_Q^{\tau\tau}) + (\partial_r T_Q^{\tau r}) + \frac{T_Q^{\tau r}}{r} + \frac{T_Q^{\tau\tau}}{\tau} + \tau T_Q^{\eta\eta}\\
\nabla_\mu T_Q^{\mu r} &= (\partial_\tau T_Q^{\tau r}) + (\partial_r T_Q^{rr}) + \frac{T_Q^{rr}}{r} - r T_Q^{\phi\phi} + \frac{T_Q^{\tau r}}{\tau}
\end{aligned}
$$

$$
\begin{aligned}
\nabla_\mu T_Q^{\mu\tau} = \frac{1}{r\tau\left(u^\tau\right)^3}\Big(&r\tau\left(u^\tau\right)^2\Big(-u^\tau((\partial_\alpha n)(\partial_\tau\alpha)T+(\partial_\tau T)((\partial_T n)T+n))\\
&+h'(T)\Big((\partial_r T)\left(n\left(u^r\right)^3+nu^r+2\nu^r\left(u^r\right)^2+\nu^r\right)\\
&+(\partial_\tau T)u^\tau\left(n\left(u^r\right)^2+n+2\nu^r u^r\right)\Big)\Big)\\
&+h\Big(\left(u^\tau\right)^2\Big(n\tau u^r+(\partial_\alpha n)(\partial_r\alpha)r\tau u^r+(\partial_T n)(\partial_r T)r\tau u^r\\
&+n\tau\left(u^r\right)^3+(\partial_\alpha n)(\partial_r\alpha)r\tau\left(u^r\right)^3+(\partial_T n)(\partial_r T)r\tau\left(u^r\right)^3\\
&+(\partial_r\nu^r)r\tau\left(2\left(u^r\right)^2+1\right)+(\partial_r u^r)nr\tau\left(2\left(u^r\right)^2+1\right)\\
&+nru^\tau+(\partial_\alpha n)(\partial_\tau\alpha)r\tau u^\tau+(\partial_T n)(\partial_\tau T)r\tau u^\tau\\
&+2(\partial_\tau\nu^r)r\tau u^r u^\tau+2(\partial_\tau u^r)nr\tau u^r u^\tau+nr\left(u^r\right)^2u^\tau\\
&+(\partial_\alpha n)(\partial_\tau\alpha)r\tau\left(u^r\right)^2u^\tau+(\partial_T n)(\partial_\tau T)r\tau\left(u^r\right)^2u^\tau\Big)\\
&+\nu^r\Big((\partial_r u^r)r\tau u^r\left(2\left(u^r\right)^2+3\right)+2ru^r\left(u^\tau\right)^3\\
&+\tau\left(u^\tau\right)^2\left(2(\partial_\tau u^r)ru^\tau+2\left(u^r\right)^2+1\right)\Big)\Big)\Big)\\
\nabla_\mu T_Q^{\mu r} = \frac{1}{r\tau\left(u^\tau\right)^3}\Big(&r\tau\left(u^\tau\right)^2\Big(u^\tau((\partial_\alpha n)(\partial_r\alpha)T+(\partial_r T)((\partial_T n)T+n))\\
&+h'(T)\Big((\partial_\tau T)\left(n\left(u^r\right)^3+nu^r+2\nu^r\left(u^r\right)^2+\nu^r\right)\\
&+(\partial_r T)u^r u^\tau(nu^r+2\nu^r)\Big)\Big)\\
&+h\Big(\nu^r\Big((\partial_\tau u^r)r\tau u^r\left(2\left(u^r\right)^2+3\right)+2\tau u^r\left(u^\tau\right)^3\\
&+r\left(u^\tau\right)^2\left(2(\partial_r u^r)\tau u^\tau+2\left(u^r\right)^2+1\right)\Big)\\
&+\left(u^\tau\right)^2\Big((\partial_\tau\nu^r)r\tau\left(2\left(u^r\right)^2+1\right)+(\partial_\tau u^r)nr\tau\left(2\left(u^r\right)^2+1\right)\\
&+nu^r\left(r\left(2(\partial_r u^r)\tau u^\tau+\left(u^r\right)^2+1\right)+\tau u^r u^\tau\right)\\
&+r\tau u^r\Big(2(\partial_r\nu^r)u^\tau\\
&+(\partial_\alpha n)\left((\partial_r\alpha)u^r u^\tau+(\partial_\tau\alpha)\left(u^r\right)^2+(\partial_\tau\alpha)\right)\\
&+(\partial_T n)\left((\partial_r T)u^r u^\tau+(\partial_\tau T)\left(u^r\right)^2+(\partial_\tau T)\right)\Big)\Big)\Big)\Big)\\
\Delta^r{}_\tau &= -u^r u^\tau\\
\Delta^r{}_r &= \left(u^\tau\right)^2
\end{aligned}
$$

$$
E_\nu = \frac{D_s}{T}\left(\Delta^r{}_\tau\nabla_\mu T_Q^{\mu\tau}+\Delta^r{}_r\nabla_\mu T_Q^{\mu r}\right)+\nu^r
$$

$$
\begin{aligned}
E_\nu = \nu^r&\\
&+\frac{1}{Tu^\tau}D_s\Big(h'(T)\nu^r\left((\partial_r T)u^r u^\tau+(\partial_\tau T)\left(u^r\right)^2+(\partial_\tau T)\right)\\
&+\left(u^\tau\right)^2\Big((\partial_\tau T)u^r((\partial_T n)T+n)+(\partial_r T)u^\tau((\partial_T n)T+n)\\
&+(\partial_\alpha n)T((\partial_r\alpha)u^\tau+(\partial_\tau\alpha)u^r)\Big)\Big)\\
&+\frac{1}{rT\tau\left(u^\tau\right)^3}D_s h\Big(\nu^r\tau u^r\left(u^\tau\right)^3\\
&+r\tau\left(u^\tau\right)^2\Big(u^r u^\tau((\partial_r\nu^r)+(\partial_r u^r)n)+(\partial_\tau\nu^r)\left(u^\tau\right)^2\\
&+(\partial_\tau u^r)n\left(u^\tau\right)^2\Big)\\
&+\nu^r r\Big(1+\left(u^r\right)^4+2(\partial_r u^r)\tau u^\tau+(\partial_\tau u^r)\tau u^r\left(u^\tau\right)^2\\
&+\left(u^r\right)^2\left((\partial_r u^r)\tau u^\tau+2\right)\Big)\Big)
\end{aligned}
$$

3.2.2 S4: charm second moment

$\mathcal{A}$ enters $S^{\lambda\alpha\beta}$ only through $u^\lambda u^\alpha \nu^\beta$: after the projection its derivative has a vanishing coefficient (G-aQ).

$$\nabla_\lambda S^{\lambda\alpha\beta} = \partial_\lambda S^{\lambda\alpha\beta} + \Gamma^\lambda{}_{\lambda\sigma} S^{\sigma\alpha\beta} + \Gamma^\alpha{}_{\lambda\sigma} S^{\lambda\sigma\beta} + \Gamma^\beta{}_{\lambda\sigma} S^{\lambda\alpha\sigma}$$

$$\begin{aligned}
\mathcal{E}^{\phi\phi} = \frac{1}{r^3\tau\left(u^\tau\right)^3}\Big(&-\mathcal{B}(\partial_\tau u^r)\nu^r r\tau\left(u^r\right)^2 + \mathcal{B}(\partial_\tau u^r)\nu^r r\tau\left(u^\tau\right)^2 + \mathcal{B}\nu^r r u^r\left(u^\tau\right)^2 \\
&+ \mathcal{B}(\partial_\tau \nu^r) r\tau u^r\left(u^\tau\right)^2 + (\partial_\tau u^r) I_{31} r\tau u^r\left(u^\tau\right)^2 + \mathcal{B}'(T)(\partial_\tau T)\nu^r r\tau u^r\left(u^\tau\right)^2 \\
&+ \mathcal{C}(\partial_\tau u^r)\Pi_Q r\tau u^r\left(u^\tau\right)^2 + \mathcal{C}(\partial_\tau u^r)\pi_Q{}^\phi{}_\phi r\tau u^r\left(u^\tau\right)^2 + 3\mathcal{B}\nu^r\tau\left(u^\tau\right)^3 \\
&+ \mathcal{B}(\partial_r \nu^r) r\tau\left(u^\tau\right)^3 + (\partial_r u^r) I_{31} r\tau\left(u^\tau\right)^3 + \mathcal{B}'(T)(\partial_r T)\nu^r r\tau\left(u^\tau\right)^3 \\
&+ \mathcal{C}(\partial_r u^r)\Pi_Q r\tau\left(u^\tau\right)^3 + \mathcal{C}(\partial_r u^r)\pi_Q{}^\phi{}_\phi r\tau\left(u^\tau\right)^3 + 3I_{31}\tau u^r\left(u^\tau\right)^3 \\
&+ 3\mathcal{C}\Pi_Q\tau u^r\left(u^\tau\right)^3 + 3\mathcal{C}\pi_Q{}^\phi{}_\phi\tau u^r\left(u^\tau\right)^3 + (\partial_\alpha I_{31})(\partial_r\alpha) r\tau u^r\left(u^\tau\right)^3 \\
&+ \mathcal{C}(\partial_r\Pi_Q) r\tau u^r\left(u^\tau\right)^3 + \mathcal{C}(\partial_r\pi_Q{}^\phi{}_\phi) r\tau u^r\left(u^\tau\right)^3 + (\partial_T I_{31})(\partial_r T) r\tau u^r\left(u^\tau\right)^3 \\
&+ \mathcal{C}'(T)(\partial_r T)\Pi_Q r\tau u^r\left(u^\tau\right)^3 + \mathcal{C}'(T)(\partial_r T)\pi_Q{}^\phi{}_\phi r\tau u^r\left(u^\tau\right)^3 + I_{31} r\left(u^\tau\right)^4 \\
&+ \mathcal{C}\Pi_Q r\left(u^\tau\right)^4 + \mathcal{C}\pi_Q{}^\phi{}_\phi r\left(u^\tau\right)^4 + (\partial_\alpha I_{31})(\partial_\tau\alpha) r\tau\left(u^\tau\right)^4 + \mathcal{C}(\partial_\tau\Pi_Q) r\tau\left(u^\tau\right)^4 \\
&+ \mathcal{C}(\partial_\tau\pi_Q{}^\phi{}_\phi) r\tau\left(u^\tau\right)^4 + (\partial_T I_{31})(\partial_\tau T) r\tau\left(u^\tau\right)^4 + \mathcal{C}'(T)(\partial_\tau T)\Pi_Q r\tau\left(u^\tau\right)^4 \\
&+ \mathcal{C}'(T)(\partial_\tau T)\pi_Q{}^\phi{}_\phi r\tau\left(u^\tau\right)^4\Big)
\end{aligned}$$

$$\begin{aligned}
\mathcal{E}^{\eta\eta} = \frac{1}{r\tau^3\left(u^\tau\right)^3}\Big(&-\mathcal{B}(\partial_\tau u^r)\nu^r r\tau\left(u^r\right)^2 + \mathcal{B}(\partial_\tau u^r)\nu^r r\tau\left(u^\tau\right)^2 - 2\mathcal{B}\nu^r r u^r\left(u^\tau\right)^2 \\
&+ \mathcal{B}(\partial_\tau \nu^r) r\tau u^r\left(u^\tau\right)^2 + (\partial_\tau u^r) I_{31} r\tau u^r\left(u^\tau\right)^2 + \mathcal{B}'(T)(\partial_\tau T)\nu^r r\tau u^r\left(u^\tau\right)^2 \\
&+ \mathcal{C}(\partial_\tau u^r) r\tau u^r\left(u^\tau\right)^2\left(\Pi_Q + \pi_Q{}^\eta{}_\eta\right) + \mathcal{B}(\partial_r\nu^r) r\tau\left(u^\tau\right)^3 + (\partial_r u^r) I_{31} r\tau\left(u^\tau\right)^3 \\
&+ \mathcal{B}'(T)(\partial_r T)\nu^r r\tau\left(u^\tau\right)^3 + \mathcal{C}(\partial_r u^r) r\tau\left(u^\tau\right)^3\left(\Pi_Q + \pi_Q{}^\eta{}_\eta\right) \\
&+ \mathcal{C} r\tau u^r\left(u^\tau\right)^3\left((\partial_r\Pi_Q) + (\partial_r\pi_Q{}^\eta{}_\eta)\right) \\
&+ r\tau u^r\left(u^\tau\right)^3\left((\partial_\alpha I_{31})(\partial_r\alpha) + (\partial_T I_{31})(\partial_r T)\right) \\
&+ \mathcal{C}'(T)(\partial_r T) r\tau u^r\left(u^\tau\right)^3\left(\Pi_Q + \pi_Q{}^\eta{}_\eta\right) \\
&+ \tau\left(u^\tau\right)^3\left(\mathcal{B}\nu^r + u^r(\mathcal{C}(\Pi_Q + \pi_Q{}^\eta{}_\eta) + I_{31})\right) - 2I_{31} r\left(u^\tau\right)^4 \\
&+ r\tau\left(u^\tau\right)^4\left((\partial_\alpha I_{31})(\partial_\tau\alpha) + (\partial_T I_{31})(\partial_\tau T)\right) \\
&+ \mathcal{C}'(T)(\partial_\tau T) r\tau\left(u^\tau\right)^4\left(\Pi_Q + \pi_Q{}^\eta{}_\eta\right) \\
&- \mathcal{C} r\left(u^\tau\right)^4\left(-\tau((\partial_\tau\Pi_Q) + (\partial_\tau\pi_Q{}^\eta{}_\eta)) + 2\Pi_Q + 2\pi_Q{}^\eta{}_\eta\right) \\
&+ 5r\left(u^\tau\right)^2\left(\mathcal{B}\nu^r u^r + \mathcal{C}\left(u^\tau\right)^2\left(\Pi_Q + \pi_Q{}^\eta{}_\eta\right) + I_{31}\left(u^\tau\right)^2\right)\Big)
\end{aligned}$$

$$\begin{aligned}
\mathcal{E}^\lambda{}_\lambda = \frac{1}{r\tau\left(u^\tau\right)^3}\Big(&3(\partial_\alpha I_{31})(\partial_\tau\alpha) r\tau + 3(\partial_T I_{31})(\partial_\tau T) r\tau + 2\mathcal{A}(\partial_\tau u^r)\nu^r r\tau + 5\mathcal{B}(\partial_\tau u^r)\nu^r r\tau \\
&+ 3\mathcal{C}'(T)(\partial_\tau T)\Pi_Q r\tau + 5\mathcal{B}\nu^r r u^r + 5\mathcal{B}(\partial_\tau\nu^r) r\tau u^r + 5\mathcal{B}'(T)(\partial_\tau T)\nu^r r\tau u^r \\
&+ 6(\partial_\alpha I_{31})(\partial_\tau\alpha) r\tau\left(u^r\right)^2 + 6(\partial_T I_{31})(\partial_\tau T) r\tau\left(u^r\right)^2 + 2\mathcal{A}(\partial_\tau u^r)\nu^r r\tau\left(u^r\right)^2 \\
&+ 6\mathcal{C}'(T)(\partial_\tau T)\Pi_Q r\tau\left(u^r\right)^2 + 5\mathcal{B}\nu^r r\left(u^r\right)^3 + 5\mathcal{B}(\partial_\tau\nu^r) r\tau\left(u^r\right)^3 \\
&+ 5\mathcal{B}'(T)(\partial_\tau T)\nu^r r\tau\left(u^r\right)^3 + 3(\partial_\alpha I_{31})(\partial_\tau\alpha) r\tau\left(u^r\right)^4 \\
&+ 3(\partial_T I_{31})(\partial_\tau T) r\tau\left(u^r\right)^4 + 3\mathcal{C}'(T)(\partial_\tau T)\Pi_Q r\tau\left(u^r\right)^4 + 5\mathcal{B}\nu^r\tau u^\tau \\
&+ 5\mathcal{B}(\partial_r\nu^r) r\tau u^\tau + 5\mathcal{B}'(T)(\partial_r T)\nu^r r\tau u^\tau + 3(\partial_\alpha I_{31})(\partial_r\alpha) r\tau u^r u^\tau \\
&+ 3(\partial_T I_{31})(\partial_r T) r\tau u^r u^\tau + 2\mathcal{A}(\partial_r u^r)\nu^r r\tau u^r u^\tau + 3\mathcal{C}'(T)(\partial_r T)\Pi_Q r\tau u^r u^\tau \\
&+ 5\mathcal{B}\nu^r\tau\left(u^r\right)^2 u^\tau + 5\mathcal{B}(\partial_r\nu^r) r\tau\left(u^r\right)^2 u^\tau + 5\mathcal{B}'(T)(\partial_r T)\nu^r r\tau\left(u^r\right)^2 u^\tau \\
&+ 3(\partial_\alpha I_{31})(\partial_r\alpha) r\tau\left(u^r\right)^3 u^\tau + 3(\partial_T I_{31})(\partial_r T) r\tau\left(u^r\right)^3 u^\tau \\
&+ 3\mathcal{C}'(T)(\partial_r T)\Pi_Q r\tau\left(u^r\right)^3 u^\tau \\
&+ 5I_{31}\left(u^\tau\right)^2\left(r\left((\partial_r u^r)\tau u^\tau + (\partial_\tau u^r)\tau u^r + \left(u^r\right)^2 + 1\right) + \tau u^r u^\tau\right) \\
&+ \mathcal{C}\left(u^\tau\right)^2\left(-2\pi_Q{}^\eta{}_\eta r\left((\partial_r u^r)\tau u^\tau + (\partial_\tau u^r)\tau u^r - \left(u^r\right)^2 - 1\right)\right.
\end{aligned}$$

$$
\begin{aligned}
&+ \tau\Big(-2(\partial_\tau u^r)\pi_Q{}^\phi{}_\phi r u^r \\
&+ u^\tau(3(\partial_r \Pi_Q) r u^r - 2(\partial_r u^r)\pi_Q{}^\phi{}_\phi r + 2\pi_Q{}^\phi{}_\phi u^r) \\
&+ 3(\partial_\tau \Pi_Q) r\left(u^\tau\right)^2\Big) \\
&+ 5\Pi_Q\left(r\left((\partial_r u^r)\tau u^\tau + (\partial_\tau u^r)\tau u^r + \left(u^r\right)^2 + 1\right) + \tau u^r u^\tau\right)\Big)
\end{aligned}
$$

$$
E_\phi = \frac{D_s}{2T}\Big(r^2\mathcal{E}^{\phi\phi} - \tfrac{1}{3}\mathcal{E}^\lambda{}_\lambda\Big) + \pi_Q{}^\phi{}_\phi, \quad E_\eta = \frac{D_s}{2T}\Big(\tau^2\mathcal{E}^{\eta\eta} - \tfrac{1}{3}\mathcal{E}^\lambda{}_\lambda\Big) + \pi_Q{}^\eta{}_\eta, \quad E_{\Pi_Q} = \frac{D_s}{6T}\mathcal{E}^\lambda{}_\lambda + \Pi_Q
$$

3.2.3 S5: two-way

$$
\begin{aligned}
T_m^{\tau\tau} &= P'(T)T\left(u^\tau\right)^2 + \left(u^r\right)^2(\Pi - \pi^\eta{}_\eta - \pi^\phi{}_\phi) - P \\
T_m^{\tau r} &= u^r u^\tau(P'(T)T + \Pi - \pi^\eta{}_\eta - \pi^\phi{}_\phi) \\
T_m^{rr} &= \left(u^r\right)^2(P'(T)T + \Pi) + \Pi - \left(u^\tau\right)^2(\pi^\eta{}_\eta + \pi^\phi{}_\phi) + P \\
T_m^{\phi\phi} &= \frac{\Pi + \pi^\phi{}_\phi + P}{r^2} \\
T_m^{\eta\eta} &= \frac{\Pi + \pi^\eta{}_\eta + P}{\tau^2}
\end{aligned}
$$

$$
E_\tau = \nabla_\mu T_m^{\mu\tau} + \nabla_\mu T_Q^{\mu\tau}, \qquad E_r = \nabla_\mu T_m^{\mu r} + \nabla_\mu T_Q^{\mu r}
$$

The shear and bulk rows as in the viscous section, the charm rows as S4.

$$
\begin{aligned}
E_\tau = \frac{1}{r\tau\left(u^\tau\right)^3}\Big(&\left(u^\tau\right)^2\Big(\Pi\tau u^r - \pi^\eta{}_\eta\tau u^r - \pi^\phi{}_\phi\tau u^r + (\partial_r\Pi)r\tau u^r - (\partial_r\pi^\eta{}_\eta)r\tau u^r \\
&- (\partial_r\pi^\phi{}_\phi)r\tau u^r + (\partial_r T)P'(T)r\tau u^r + P'(T)T\tau u^r \\
&+ (\partial_r T)P''(T)rT\tau u^r + \Pi\tau\left(u^r\right)^3 - \pi^\eta{}_\eta\tau\left(u^r\right)^3 - \pi^\phi{}_\phi\tau\left(u^r\right)^3 \\
&+ (\partial_r\Pi)r\tau\left(u^r\right)^3 - (\partial_r\pi^\eta{}_\eta)r\tau\left(u^r\right)^3 - (\partial_r\pi^\phi{}_\phi)r\tau\left(u^r\right)^3 \\
&+ (\partial_r T)P'(T)r\tau\left(u^r\right)^3 + P'(T)T\tau\left(u^r\right)^3 \\
&+ (\partial_r T)P''(T)rT\tau\left(u^r\right)^3 \\
&+ (\partial_r u^r)r\tau\left(2\left(u^r\right)^2 + 1\right)(P'(T)T + \Pi - \pi^\eta{}_\eta - \pi^\phi{}_\phi) + \Pi r u^\tau \\
&+ \pi^\eta{}_\eta r u^\tau + P'(T)rTu^\tau - (\partial_\tau T)nr\tau u^\tau - (\partial_\alpha n)(\partial_\tau\alpha)rT\tau u^\tau \\
&- (\partial_T n)(\partial_\tau T)rT\tau u^\tau + P''(T)(\partial_\tau T)rT\tau u^\tau + 2(\partial_\tau u^r)\Pi r\tau u^r u^\tau \\
&- 2(\partial_\tau u^r)\pi^\eta{}_\eta r\tau u^r u^\tau - 2(\partial_\tau u^r)\pi^\phi{}_\phi r\tau u^r u^\tau \\
&+ 2P'(T)(\partial_\tau u^r)rT\tau u^r u^\tau + \Pi r\left(u^r\right)^2 u^\tau - \pi^\eta{}_\eta r\left(u^r\right)^2 u^\tau \\
&- \pi^\phi{}_\phi r\left(u^r\right)^2 u^\tau + P'(T)rT\left(u^r\right)^2 u^\tau + (\partial_\tau\Pi)r\tau\left(u^r\right)^2 u^\tau \\
&- (\partial_\tau\pi^\eta{}_\eta)r\tau\left(u^r\right)^2 u^\tau - (\partial_\tau\pi^\phi{}_\phi)r\tau\left(u^r\right)^2 u^\tau \\
&+ P'(T)(\partial_\tau T)r\tau\left(u^r\right)^2 u^\tau + P''(T)(\partial_\tau T)rT\tau\left(u^r\right)^2 u^\tau \\
&+ h'(T)r\tau\Big((\partial_r T)\left(n\left(u^r\right)^3 + nu^r + 2\nu^r\left(u^r\right)^2 + \nu^r\right) \\
&+ (\partial_\tau T)u^\tau\left(n\left(u^r\right)^2 + n + 2\nu^r u^r\right)\Big)\Big) \\
&+ h\Big(\left(u^\tau\right)^2\Big(n\tau u^r + (\partial_\alpha n)(\partial_r\alpha)r\tau u^r + (\partial_T n)(\partial_r T)r\tau u^r \\
&+ n\tau\left(u^r\right)^3 + (\partial_\alpha n)(\partial_r\alpha)r\tau\left(u^r\right)^3 + (\partial_T n)(\partial_r T)r\tau\left(u^r\right)^3 \\
&+ (\partial_r\nu^r)r\tau\left(2\left(u^r\right)^2 + 1\right) + (\partial_r u^r)nr\tau\left(2\left(u^r\right)^2 + 1\right) \\
&+ nru^\tau + (\partial_\alpha n)(\partial_\tau\alpha)r\tau u^\tau + (\partial_T n)(\partial_\tau T)r\tau u^\tau \\
&+ 2(\partial_\tau\nu^r)r\tau u^r u^\tau + 2(\partial_\tau u^r)nr\tau u^r u^\tau + nr\left(u^r\right)^2 u^\tau \\
&+ (\partial_\alpha n)(\partial_\tau\alpha)r\tau\left(u^r\right)^2 u^\tau + (\partial_T n)(\partial_\tau T)r\tau\left(u^r\right)^2 u^\tau\Big)
\end{aligned}
$$

$$+\nu^r\Big((\partial_r u^r)r\tau u^r\left(2\left(u^r\right)^2+3\right)+2ru^r\left(u^\tau\right)^3$$
$$+\tau\left(u^\tau\right)^2\left(2(\partial_\tau u^r)ru^\tau+2\left(u^r\right)^2+1\right)\Big)\Big)$$

3.2.4 S6: fully coupled

$$\begin{aligned}
\delta M^{\tau\tau} &= \left(u^r\right)^2\left(\Pi_Q-{\pi_Q}^\eta{}_\eta-{\pi_Q}^\phi{}_\phi\right)\\
\delta M^{\tau r} &= u^r u^\tau(\Pi_Q-{\pi_Q}^\eta{}_\eta-{\pi_Q}^\phi{}_\phi)\\
\delta M^{rr} &= \left(u^\tau\right)^2\left(\Pi_Q-{\pi_Q}^\eta{}_\eta-{\pi_Q}^\phi{}_\phi\right)\\
\delta M^{\phi\phi} &= \frac{\Pi_Q+{\pi_Q}^\phi{}_\phi}{r^2}\\
\delta M^{\eta\eta} &= \frac{\Pi_Q+{\pi_Q}^\eta{}_\eta}{\tau^2}
\end{aligned}$$

$$\begin{aligned}
\nabla_\mu\delta M^{\mu\tau} &= (\partial_\tau\delta M^{\tau\tau})+(\partial_r\delta M^{\tau r})+\frac{\delta M^{\tau r}}{r}+\frac{\delta M^{\tau\tau}}{\tau}+\tau\delta M^{\eta\eta}\\
\nabla_\mu\delta M^{\mu r} &= (\partial_\tau\delta M^{\tau r})+(\partial_r\delta M^{rr})+\frac{\delta M^{rr}}{r}-r\delta M^{\phi\phi}+\frac{\delta M^{\tau r}}{\tau}
\end{aligned}$$

$$\begin{aligned}
\nabla_\mu\delta M^{\mu\tau} = {}& \frac{\Pi_Q+{\pi_Q}^\eta{}_\eta}{\tau}+2(\partial_\tau u^r)\Pi_Q u^r-2(\partial_\tau u^r)u^r({\pi_Q}^\eta{}_\eta+{\pi_Q}^\phi{}_\phi)+(\partial_\tau\Pi_Q)\left(u^r\right)^2\\
&-\left(u^r\right)^2\left((\partial_\tau{\pi_Q}^\eta{}_\eta)+(\partial_\tau{\pi_Q}^\phi{}_\phi)\right)+\frac{\left(u^r\right)^2\left(\Pi_Q-{\pi_Q}^\eta{}_\eta-{\pi_Q}^\phi{}_\phi\right)}{\tau}+\frac{(\partial_r u^r)\Pi_Q\left(u^r\right)^2}{u^\tau}\\
&-\frac{(\partial_r u^r)\left(u^r\right)^2\left({\pi_Q}^\eta{}_\eta+{\pi_Q}^\phi{}_\phi\right)}{u^\tau}+(\partial_r u^r)\Pi_Q u^\tau-(\partial_r u^r)u^\tau({\pi_Q}^\eta{}_\eta+{\pi_Q}^\phi{}_\phi)+(\partial_r\Pi_Q)u^r u^\tau\\
&-u^r u^\tau((\partial_r{\pi_Q}^\eta{}_\eta)+(\partial_r{\pi_Q}^\phi{}_\phi))+\frac{u^r u^\tau(\Pi_Q-{\pi_Q}^\eta{}_\eta-{\pi_Q}^\phi{}_\phi)}{r}\\
\nabla_\mu\delta M^{\mu r} = {}& -\frac{\Pi_Q+{\pi_Q}^\phi{}_\phi}{r}+2(\partial_r u^r)\Pi_Q u^r-2(\partial_r u^r)u^r({\pi_Q}^\eta{}_\eta+{\pi_Q}^\phi{}_\phi)+\frac{(\partial_\tau u^r)\Pi_Q\left(u^r\right)^2}{u^\tau}\\
&-\frac{(\partial_\tau u^r)\left(u^r\right)^2\left({\pi_Q}^\eta{}_\eta+{\pi_Q}^\phi{}_\phi\right)}{u^\tau}+(\partial_\tau u^r)\Pi_Q u^\tau-(\partial_\tau u^r)u^\tau({\pi_Q}^\eta{}_\eta+{\pi_Q}^\phi{}_\phi)+(\partial_\tau\Pi_Q)u^r u^\tau\\
&-u^r u^\tau((\partial_\tau{\pi_Q}^\eta{}_\eta)+(\partial_\tau{\pi_Q}^\phi{}_\phi))+\frac{u^r u^\tau(\Pi_Q-{\pi_Q}^\eta{}_\eta-{\pi_Q}^\phi{}_\phi)}{\tau}+(\partial_r\Pi_Q)\left(u^\tau\right)^2\\
&-\left(u^\tau\right)^2\left((\partial_r{\pi_Q}^\eta{}_\eta)+(\partial_r{\pi_Q}^\phi{}_\phi)\right)+\frac{\left(u^\tau\right)^2\left(\Pi_Q-{\pi_Q}^\eta{}_\eta-{\pi_Q}^\phi{}_\phi\right)}{r}
\end{aligned}$$

$E_\tau^{\mathrm{S6}}=E_\tau^{\mathrm{S5}}+\nabla_\mu\delta M^{\mu\tau}$, $E_r^{\mathrm{S6}}=E_r^{\mathrm{S5}}+\nabla_\mu\delta M^{\mu r}$, $E_\nu^{\mathrm{S6}}=E_\nu+c_M\big(\Delta^r{}_\tau\nabla_\mu\delta M^{\mu\tau}+\Delta^r{}_r\nabla_\mu\delta M^{\mu r}\big)$, $c_M=\dfrac{D_s}{T}$

Every other row as S5.

3.2.5 From the rows to the matrices

$$A_{ij}^{(k)}=\frac{\partial E_i}{\partial(\partial_k\phi_j)},\qquad S_i=E_i\big|_{\partial\phi=0},\qquad k=\tau,r$$

3.2.6 2+1D, S3

$$\Gamma^\tau{}_{\eta\eta}=\tau,\quad \Gamma^\eta{}_{\tau\eta}=\frac{1}{\tau}$$

$$\begin{aligned}
N^\tau &= \frac{n\left(u^\tau\right)^2+\nu^x u^x+\nu^y u^y}{u^\tau}\\
N^x &= nu^x+\nu^x\\
N^y &= nu^y+\nu^y\\
E_\alpha=\nabla_\mu N^\mu &= (\partial_x\nu^x)+(\partial_y\nu^y)+(\partial_x u^x)n+(\partial_y u^y)n+u^\tau((\partial_\alpha n)(\partial_\tau\alpha)+(\partial_T n)(\partial_\tau T))
\end{aligned}$$

$$
\begin{aligned}
&+ u^x((\partial_\alpha n)(\partial_x \alpha) + (\partial_T n)(\partial_x T)) + u^y((\partial_\alpha n)(\partial_y \alpha) + (\partial_T n)(\partial_y T)) \\
&+ \frac{(\partial_\tau \nu^x)u^x + (\partial_\tau \nu^y)u^y + (\partial_\tau u^x)\nu^x + (\partial_\tau u^y)\nu^y}{u^\tau} + \frac{n((\partial_\tau u^x)u^x + (\partial_\tau u^y)u^y)}{u^\tau} \\
&- \frac{((\partial_\tau u^x)u^x + (\partial_\tau u^y)u^y)(\nu^x u^x + \nu^y u^y)}{(u^\tau)^3} + \frac{n\left(u^\tau\right)^2 + \nu^x u^x + \nu^y u^y}{\tau u^\tau}
\end{aligned}
$$

$$
\begin{aligned}
T_Q^{\tau\tau} &= hn\left(u^\tau\right)^2 + 2h(\nu^x u^x + \nu^y u^y) - nT \\
T_Q^{\tau x} &= \frac{h\left(n\left(u^\tau\right)^2 u^x + \nu^x\left(2\left(u^x\right)^2 + \left(u^y\right)^2 + 1\right) + \nu^y u^x u^y\right)}{u^\tau} \\
T_Q^{\tau y} &= \frac{h\left(n\left(u^\tau\right)^2 u^y + \nu^x u^x u^y + \nu^y\left(\left(u^x\right)^2 + 2\left(u^y\right)^2 + 1\right)\right)}{u^\tau} \\
T_Q^{xx} &= n\left(h\left(u^x\right)^2 + T\right) + 2h\nu^x u^x \\
T_Q^{xy} &= h(u^y(nu^x + \nu^x) + \nu^y u^x) \\
T_Q^{yy} &= n\left(h\left(u^y\right)^2 + T\right) + 2h\nu^y u^y \\
T_Q^{\eta\eta} &= \frac{nT}{\tau^2}
\end{aligned}
$$

$$
\begin{aligned}
\nabla_\mu T_Q^{\mu\tau} &= (\partial_\tau T_Q^{\tau\tau}) + (\partial_x T_Q^{\tau x}) + (\partial_y T_Q^{\tau y}) + \frac{T_Q^{\tau\tau}}{\tau} + \tau T_Q^{\eta\eta} \\
\nabla_\mu T_Q^{\mu x} &= (\partial_\tau T_Q^{\tau x}) + (\partial_x T_Q^{xx}) + (\partial_y T_Q^{xy}) + \frac{T_Q^{\tau x}}{\tau} \\
\nabla_\mu T_Q^{\mu y} &= (\partial_\tau T_Q^{\tau y}) + (\partial_x T_Q^{xy}) + (\partial_y T_Q^{yy}) + \frac{T_Q^{\tau y}}{\tau}
\end{aligned}
$$

$$
\begin{aligned}
\nabla_\mu T_Q^{\mu\tau} &= h((\partial_\alpha n)(\partial_\tau \alpha) + (\partial_T n)(\partial_\tau T)) - (\partial_\tau T)n + h'(T)(\partial_\tau T)n \\
&- T((\partial_\alpha n)(\partial_\tau \alpha) + (\partial_T n)(\partial_\tau T)) + \frac{nT}{\tau} + (\partial_x \nu^x)hu^\tau + (\partial_y \nu^y)hu^\tau + h'(T)(\partial_x T)\nu^x u^\tau \\
&+ h'(T)(\partial_y T)\nu^y u^\tau + (\partial_x u^x)nu^\tau(h - T) + (\partial_y u^y)nu^\tau(h - T) + (\partial_x u^x)nTu^\tau \\
&+ (\partial_y u^y)nTu^\tau + \frac{n\left(u^\tau\right)^2(h - T)}{\tau} + hu^\tau u^x((\partial_\alpha n)(\partial_x \alpha) + (\partial_T n)(\partial_x T)) \\
&+ h'(T)(\partial_x T)nu^\tau u^x + h\left(u^x\right)^2((\partial_\alpha n)(\partial_\tau \alpha) + (\partial_T n)(\partial_\tau T)) - (\partial_\tau T)n\left(u^x\right)^2 \\
&+ h'(T)(\partial_\tau T)n\left(u^x\right)^2 - T\left(u^x\right)^2((\partial_\alpha n)(\partial_\tau \alpha) + (\partial_T n)(\partial_\tau T)) \\
&+ hu^\tau u^y((\partial_\alpha n)(\partial_y \alpha) + (\partial_T n)(\partial_y T)) + h'(T)(\partial_y T)nu^\tau u^y \\
&+ h\left(u^y\right)^2((\partial_\alpha n)(\partial_\tau \alpha) + (\partial_T n)(\partial_\tau T)) - (\partial_\tau T)n\left(u^y\right)^2 + h'(T)(\partial_\tau T)n\left(u^y\right)^2 \\
&- T\left(u^y\right)^2((\partial_\alpha n)(\partial_\tau \alpha) + (\partial_T n)(\partial_\tau T)) \\
&+ 2h((\partial_\tau \nu^x)u^x + (\partial_\tau \nu^y)u^y + (\partial_\tau u^x)\nu^x + (\partial_\tau u^y)\nu^y) \\
&+ 2n(h - T)((\partial_\tau u^x)u^x + (\partial_\tau u^y)u^y) + 2nT((\partial_\tau u^x)u^x + (\partial_\tau u^y)u^y) \\
&+ \frac{hu^x((\partial_x \nu^x)u^x + (\partial_x \nu^y)u^y + (\partial_x u^x)\nu^x + (\partial_x u^y)\nu^y)}{u^\tau} + \frac{h\nu^x((\partial_x u^x)u^x + (\partial_x u^y)u^y)}{u^\tau} \\
&+ \frac{nu^x(h - T)((\partial_x u^x)u^x + (\partial_x u^y)u^y)}{u^\tau} + \frac{nTu^x((\partial_x u^x)u^x + (\partial_x u^y)u^y)}{u^\tau} \\
&+ \frac{hu^y((\partial_y \nu^x)u^x + (\partial_y \nu^y)u^y + (\partial_y u^x)\nu^x + (\partial_y u^y)\nu^y)}{u^\tau} + \frac{h\nu^y((\partial_y u^x)u^x + (\partial_y u^y)u^y)}{u^\tau} \\
&+ \frac{nu^y(h - T)((\partial_y u^x)u^x + (\partial_y u^y)u^y)}{u^\tau} + \frac{nTu^y((\partial_y u^x)u^x + (\partial_y u^y)u^y)}{u^\tau} \\
&+ 2h'(T)(\partial_\tau T)(\nu^x u^x + \nu^y u^y) + \frac{2h(\nu^x u^x + \nu^y u^y)}{\tau} + \frac{(\partial_x u^x)h(\nu^x u^x + \nu^y u^y)}{u^\tau} \\
&+ \frac{(\partial_y u^y)h(\nu^x u^x + \nu^y u^y)}{u^\tau} + \frac{h'(T)(\partial_x T)u^x(\nu^x u^x + \nu^y u^y)}{u^\tau} \\
&+ \frac{h'(T)(\partial_y T)u^y(\nu^x u^x + \nu^y u^y)}{u^\tau} - \frac{hu^x((\partial_x u^x)u^x + (\partial_x u^y)u^y)(\nu^x u^x + \nu^y u^y)}{(u^\tau)^3}
\end{aligned}
$$

$$
\begin{aligned}
&- \frac{hu^y((\partial_y u^x)u^x + (\partial_y u^y)u^y)(\nu^x u^x + \nu^y u^y)}{(u^\tau)^3} + (\partial_\tau T)n\left((u^x)^2 + (u^y)^2\right) \\
&+ T\left((u^x)^2 + (u^y)^2\right)((\partial_\alpha n)(\partial_\tau \alpha) + (\partial_T n)(\partial_\tau T)) + \frac{nT\left((u^x)^2 + (u^y)^2\right)}{\tau} \\
\nabla_\mu T_Q^{\mu x} &= (\partial_y u^y)h\nu^x + (\partial_y u^x)h\nu^y + (\partial_\alpha n)(\partial_x \alpha)T + (\partial_\tau \nu^x)hu^\tau + (\partial_\tau u^x)hnu^\tau \\
&+ h'(T)(\partial_\tau T)\nu^x u^\tau + \frac{h\nu^x u^\tau}{\tau} + 2(\partial_x \nu^x)hu^x + (\partial_y \nu^y)hu^x + (\partial_y u^y)hnu^x \\
&+ h'(T)(\partial_y T)\nu^y u^x + \frac{3(\partial_\tau u^x)h\nu^x u^x}{u^\tau} + \frac{(\partial_\tau u^y)h\nu^y u^x}{u^\tau} + (\partial_\alpha n)(\partial_\tau \alpha)hu^\tau u^x \\
&+ (\partial_T n)(\partial_\tau T)hu^\tau u^x + h'(T)(\partial_\tau T)nu^\tau u^x + \frac{hnu^\tau u^x}{\tau} + (\partial_\alpha n)(\partial_x \alpha)h\left(u^x\right)^2 \\
&+ \frac{(\partial_\tau \nu^x)h\left(u^x\right)^2}{u^\tau} + \frac{(\partial_\tau u^x)hn\left(u^x\right)^2}{u^\tau} + \frac{h'(T)(\partial_\tau T)\nu^x\left(u^x\right)^2}{u^\tau} + \frac{h\nu^x\left(u^x\right)^2}{\tau u^\tau} \\
&- \frac{(\partial_\tau u^x)h\nu^x\left(u^x\right)^3}{\left(u^\tau\right)^3} + 2(\partial_x u^x)h(nu^x + \nu^x) \\
&+ (\partial_x T)\left(h'(T)n\left(u^x\right)^2 + 2h'(T)\nu^x u^x + (\partial_T n)h\left(u^x\right)^2 + (\partial_T n)T + n\right) + (\partial_y \nu^x)hu^y \\
&+ (\partial_y u^x)hnu^y + h'(T)(\partial_y T)\nu^x u^y + \frac{(\partial_\tau u^y)h\nu^x u^y}{u^\tau} + \frac{(\partial_\tau u^x)h\nu^y u^y}{u^\tau} + (\partial_\alpha n)(\partial_y \alpha)hu^x u^y \\
&+ (\partial_T n)(\partial_y T)hu^x u^y + h'(T)(\partial_y T)nu^x u^y + \frac{(\partial_\tau \nu^y)hu^x u^y}{u^\tau} + \frac{(\partial_\tau u^y)hnu^x u^y}{u^\tau} \\
&+ \frac{h'(T)(\partial_\tau T)\nu^y u^x u^y}{u^\tau} + \frac{h\nu^y u^x u^y}{\tau u^\tau} - \frac{(\partial_\tau u^y)h\nu^x\left(u^x\right)^2 u^y}{\left(u^\tau\right)^3} - \frac{(\partial_\tau u^x)h\nu^y\left(u^x\right)^2 u^y}{\left(u^\tau\right)^3} \\
&- \frac{(\partial_\tau u^y)h\nu^y u^x\left(u^y\right)^2}{\left(u^\tau\right)^3} \\
\nabla_\mu T_Q^{\mu y} &= (\partial_x u^x)h\nu^y + 2(\partial_y u^y)h\nu^y + (\partial_\alpha n)(\partial_y \alpha)T + (\partial_\tau \nu^y)hu^\tau + (\partial_\tau u^y)hnu^\tau \\
&+ h'(T)(\partial_\tau T)\nu^y u^\tau + \frac{h\nu^y u^\tau}{\tau} + (\partial_x \nu^y)hu^x + h'(T)(\partial_x T)\nu^y u^x + \frac{(\partial_\tau u^y)h\nu^x u^x}{u^\tau} \\
&+ \frac{(\partial_\tau u^x)h\nu^y u^x}{u^\tau} + (\partial_x u^y)h(nu^x + \nu^x) + (\partial_x \nu^x)hu^y + 2(\partial_y \nu^y)hu^y + (\partial_x u^x)hnu^y \\
&+ 2(\partial_y u^y)hnu^y + h'(T)(\partial_x T)\nu^x u^y + \frac{(\partial_\tau u^x)h\nu^x u^y}{u^\tau} + \frac{3(\partial_\tau u^y)h\nu^y u^y}{u^\tau} \\
&+ (\partial_\alpha n)(\partial_\tau \alpha)hu^\tau u^y + (\partial_T n)(\partial_\tau T)hu^\tau u^y + h'(T)(\partial_\tau T)nu^\tau u^y + \frac{hnu^\tau u^y}{\tau} \\
&+ (\partial_\alpha n)(\partial_x \alpha)hu^x u^y + (\partial_T n)(\partial_x T)hu^x u^y + h'(T)(\partial_x T)nu^x u^y + \frac{(\partial_\tau \nu^x)hu^x u^y}{u^\tau} \\
&+ \frac{(\partial_\tau u^x)hnu^x u^y}{u^\tau} + \frac{h'(T)(\partial_\tau T)\nu^x u^x u^y}{u^\tau} + \frac{h\nu^x u^x u^y}{\tau u^\tau} - \frac{(\partial_\tau u^x)h\nu^x\left(u^x\right)^2 u^y}{\left(u^\tau\right)^3} \\
&+ (\partial_\alpha n)(\partial_y \alpha)h\left(u^y\right)^2 + \frac{(\partial_\tau \nu^y)h\left(u^y\right)^2}{u^\tau} + \frac{(\partial_\tau u^y)hn\left(u^y\right)^2}{u^\tau} + \frac{h'(T)(\partial_\tau T)\nu^y\left(u^y\right)^2}{u^\tau} \\
&+ \frac{h\nu^y\left(u^y\right)^2}{\tau u^\tau} - \frac{(\partial_\tau u^y)h\nu^x u^x\left(u^y\right)^2}{\left(u^\tau\right)^3} - \frac{(\partial_\tau u^x)h\nu^y u^x\left(u^y\right)^2}{\left(u^\tau\right)^3} - \frac{(\partial_\tau u^y)h\nu^y\left(u^y\right)^3}{\left(u^\tau\right)^3} \\
&+ (\partial_y T)\left(h'(T)n\left(u^y\right)^2 + 2h'(T)\nu^y u^y + (\partial_T n)h\left(u^y\right)^2 + (\partial_T n)T + n\right)
\end{aligned}
$$

$$
\begin{aligned}
\Delta^x{}_\tau &= -u^\tau u^x \\
\Delta^x{}_x &= (u^x)^2 + 1 \\
\Delta^x{}_y &= u^x u^y \\
\Delta^y{}_\tau &= -u^\tau u^y \\
\Delta^y{}_x &= u^x u^y \\
\Delta^y{}_y &= (u^y)^2 + 1
\end{aligned}
$$

$$
E_{\nu^i} = \frac{D_s}{T}\Delta^i{}_\lambda \nabla_\mu T_Q^{\mu\lambda} + \nu^i, \qquad i = x, y
$$

$$
\begin{aligned}
E_{\nu^x} = {} & (\partial_\alpha n) D_s(\partial_x \alpha) + (\partial_T n) D_s(\partial_x T) + \nu^x + \frac{D_s(\partial_x T) n}{T} + \frac{(\partial_\alpha n) D_s(\partial_\tau \alpha) u^x}{u^\tau} \\
& + \frac{D_s(\partial_\tau T) u^\tau u^x ((\partial_T n) T + n)}{T} + (\partial_\alpha n) D_s(\partial_x \alpha) \left(u^x\right)^2 + (\partial_T n) D_s(\partial_x T) \left(u^x\right)^2 \\
& + \frac{D_s(\partial_x T) n \left(u^x\right)^2}{T} + \frac{(\partial_\alpha n) D_s(\partial_\tau \alpha) \left(u^x\right)^3}{u^\tau} + (\partial_\alpha n) D_s(\partial_y \alpha) u^x u^y + (\partial_T n) D_s(\partial_y T) u^x u^y \\
& + \frac{D_s(\partial_y T) n u^x u^y}{T} + \frac{(\partial_\alpha n) D_s(\partial_\tau \alpha) u^x \left(u^y\right)^2}{u^\tau} \\
& + \frac{h'(T) D_s \nu^x ((\partial_\tau T) u^\tau + (\partial_x T) u^x + (\partial_y T) u^y)}{T} \\
& + \frac{1}{T \left(u^\tau\right)^3} D_s h \Big((\partial_y u^x) \nu^y u^\tau + (\partial_\tau \nu^x) \left(u^\tau\right)^4 - (\partial_\tau u^y) \nu^y u^x + (\partial_x \nu^x) u^\tau u^x \\
& + (\partial_x u^x) n u^\tau u^x - (\partial_x u^y) \nu^y u^\tau \left(u^x\right)^2 + (\partial_y u^x) \nu^y u^\tau \left(u^x\right)^2 \\
& - 2 (\partial_\tau u^y) \nu^y \left(u^x\right)^3 + (\partial_x \nu^x) u^\tau \left(u^x\right)^3 + (\partial_x u^x) n u^\tau \left(u^x\right)^3 \\
& - (\partial_x u^y) \nu^y u^\tau \left(u^x\right)^4 - (\partial_\tau u^y) \nu^y \left(u^x\right)^5 + (\partial_y \nu^x) u^\tau u^y + (\partial_y u^x) n u^\tau u^y \\
& - (\partial_y u^y) \nu^y u^\tau u^x u^y + (\partial_y \nu^x) u^\tau \left(u^x\right)^2 u^y + (\partial_y u^x) n u^\tau \left(u^x\right)^2 u^y \\
& + (\partial_x u^x) \nu^y u^\tau \left(u^x\right)^3 u^y - (\partial_y u^y) \nu^y u^\tau \left(u^x\right)^3 u^y + (\partial_y u^x) \nu^y u^\tau \left(u^y\right)^2 \\
& - (\partial_\tau u^y) \nu^y u^x \left(u^y\right)^2 + (\partial_x \nu^x) u^\tau u^x \left(u^y\right)^2 + (\partial_x u^x) n u^\tau u^x \left(u^y\right)^2 \\
& + (\partial_y u^x) \nu^y u^\tau \left(u^x\right)^2 \left(u^y\right)^2 - (\partial_\tau u^y) \nu^y \left(u^x\right)^3 \left(u^y\right)^2 + (\partial_y \nu^x) u^\tau \left(u^y\right)^3 \\
& + (\partial_y u^x) n u^\tau \left(u^y\right)^3 + (\partial_\tau u^x) \left(u^\tau\right)^2 \left(n \left(u^\tau\right)^2 + \nu^y \left(\left(u^x\right)^2 + 1 \right) u^y \right) \Big) \\
& + \frac{1}{T \tau \left(u^\tau\right)^3} D_s h \nu^x \Big((\partial_x u^y) \tau u^\tau \left(u^x\right)^3 u^y + \left(u^x\right)^4 ((\partial_\tau u^y) \tau u^y + 1) \\
& - (\partial_\tau u^x) \tau \left(u^\tau\right)^2 u^x \left(\left(u^y\right)^2 - 1 \right) - (\partial_y u^x) \tau u^\tau u^x u^y \left(\left(u^y\right)^2 + 1 \right) \\
& + \left(u^y\right)^2 + 1 \Big(1 + 2 (\partial_x u^x) \tau u^\tau + (\partial_y u^y) \tau u^\tau + (\partial_\tau u^y) \tau u^y \\
& + \left(u^y\right)^2 \Big) \\
& + \left(u^x\right)^2 \Big(2 + (\partial_x u^x) \tau u^\tau + (\partial_y u^y) \tau u^\tau \\
& + \left(u^y\right)^2 \left(- (\partial_x u^x) \tau u^\tau + (\partial_y u^y) \tau u^\tau + 2 \right) \\
& + (\partial_\tau u^y) \tau u^y \left(\left(u^y\right)^2 + 2 \right) \Big) \Big)
\end{aligned}
$$

$$
\begin{aligned}
E_{\nu^y} = {} & (\partial_\alpha n) D_s(\partial_y \alpha) + (\partial_T n) D_s(\partial_y T) + \nu^y + \frac{D_s(\partial_y T) n}{T} + \frac{(\partial_\alpha n) D_s(\partial_\tau \alpha) u^y}{u^\tau} \\
& + \frac{D_s(\partial_\tau T) u^\tau u^y ((\partial_T n) T + n)}{T} + (\partial_\alpha n) D_s(\partial_x \alpha) u^x u^y + (\partial_T n) D_s(\partial_x T) u^x u^y \\
& + \frac{D_s(\partial_x T) n u^x u^y}{T} + \frac{(\partial_\alpha n) D_s(\partial_\tau \alpha) \left(u^x\right)^2 u^y}{u^\tau} + (\partial_\alpha n) D_s(\partial_y \alpha) \left(u^y\right)^2 \\
& + (\partial_T n) D_s(\partial_y T) \left(u^y\right)^2 + \frac{D_s(\partial_y T) n \left(u^y\right)^2}{T} + \frac{(\partial_\alpha n) D_s(\partial_\tau \alpha) \left(u^y\right)^3}{u^\tau} \\
& + \frac{h'(T) D_s \nu^y ((\partial_\tau T) u^\tau + (\partial_x T) u^x + (\partial_y T) u^y)}{T} \\
& + \frac{1}{T \tau \left(u^\tau\right)^3} D_s h \nu^y \Big(1 + (\partial_x u^x) \tau u^\tau + 2 (\partial_y u^y) \tau u^\tau + (\partial_\tau u^y) \tau u^y - (\partial_x u^y) \tau u^\tau \left(u^x\right)^3 u^y \\
& + 2 \left(u^y\right)^2 + (\partial_x u^x) \tau u^\tau \left(u^y\right)^2 + (\partial_y u^y) \tau u^\tau \left(u^y\right)^2 + (\partial_\tau u^y) \tau \left(u^y\right)^3 \\
& + \left(u^y\right)^4 + \left(u^x\right)^4 (1 - (\partial_\tau u^y) \tau u^y) + (\partial_\tau u^x) \tau \left(u^\tau\right)^2 u^x \left(\left(u^y\right)^2 + 1 \right) \\
& + \tau u^\tau u^x u^y \left((\partial_y u^x) \left(u^y\right)^2 - (\partial_x u^y) \right) \\
& + \left(u^x\right)^2 \Big(2 + (\partial_x u^x) \tau u^\tau + 2 (\partial_y u^y) \tau u^\tau \\
& + \left(u^y\right)^2 ((\partial_x u^x) \tau u^\tau - (\partial_y u^y) \tau u^\tau + 2) \\
& - (\partial_\tau u^y) \tau \left(u^y\right)^3 \Big) \Big)
\end{aligned}
$$

$$
\begin{aligned}
&+\frac{1}{T\left(u^{\tau}\right)^{3}}D_{s}h\Big(\left(\partial_{x}u^{y}\right)\nu^{x}u^{\tau}+\left(\partial_{\tau}\nu^{y}\right)\left(u^{\tau}\right)^{4}+\left(\partial_{x}\nu^{y}\right)u^{\tau}u^{x}+\left(\partial_{x}u^{y}\right)nu^{\tau}u^{x}\\
&+\left(\partial_{x}u^{y}\right)\nu^{x}u^{\tau}\left(u^{x}\right)^{2}+\left(\partial_{x}\nu^{y}\right)u^{\tau}\left(u^{x}\right)^{3}+\left(\partial_{x}u^{y}\right)nu^{\tau}\left(u^{x}\right)^{3}-\left(\partial_{\tau}u^{x}\right)\nu^{x}u^{y}\\
&+\left(\partial_{y}\nu^{y}\right)u^{\tau}u^{y}+\left(\partial_{y}u^{y}\right)nu^{\tau}u^{y}-\left(\partial_{x}u^{x}\right)\nu^{x}u^{\tau}u^{x}u^{y}-\left(\partial_{\tau}u^{x}\right)\nu^{x}\left(u^{x}\right)^{2}u^{y}\\
&+\left(\partial_{y}\nu^{y}\right)u^{\tau}\left(u^{x}\right)^{2}u^{y}+\left(\partial_{y}u^{y}\right)nu^{\tau}\left(u^{x}\right)^{2}u^{y}+\left(\partial_{x}u^{y}\right)\nu^{x}u^{\tau}\left(u^{y}\right)^{2}\\
&-\left(\partial_{y}u^{x}\right)\nu^{x}u^{\tau}\left(u^{y}\right)^{2}+\left(\partial_{x}\nu^{y}\right)u^{\tau}u^{x}\left(u^{y}\right)^{2}+\left(\partial_{x}u^{y}\right)nu^{\tau}u^{x}\left(u^{y}\right)^{2}\\
&+\left(\partial_{x}u^{y}\right)\nu^{x}u^{\tau}\left(u^{x}\right)^{2}\left(u^{y}\right)^{2}-2(\partial_{\tau}u^{x})\nu^{x}\left(u^{y}\right)^{3}+\left(\partial_{y}\nu^{y}\right)u^{\tau}\left(u^{y}\right)^{3}\\
&+\left(\partial_{y}u^{y}\right)nu^{\tau}\left(u^{y}\right)^{3}-\left(\partial_{x}u^{x}\right)\nu^{x}u^{\tau}u^{x}\left(u^{y}\right)^{3}+\left(\partial_{y}u^{y}\right)\nu^{x}u^{\tau}u^{x}\left(u^{y}\right)^{3}\\
&-\left(\partial_{\tau}u^{x}\right)\nu^{x}\left(u^{x}\right)^{2}\left(u^{y}\right)^{3}-\left(\partial_{y}u^{x}\right)\nu^{x}u^{\tau}\left(u^{y}\right)^{4}-\left(\partial_{\tau}u^{x}\right)\nu^{x}\left(u^{y}\right)^{5}\\
&+\left(\partial_{\tau}u^{y}\right)\left(u^{\tau}\right)^{2}\left(n\left(u^{\tau}\right)^{2}+\nu^{x}u^{x}\left(\left(u^{y}\right)^{2}+1\right)\right)\Big)
\end{aligned}
$$

3.2.7 2+1D, S4

$$
\begin{aligned}
S^{\tau\tau\tau}=&\frac{3}{u^{\tau}\left(\left(u^{y}\right)^{2}+1\right)}\Big(\mathcal{A}\nu^{x}\left(u^{\tau}\right)^{2}u^{x}-\mathcal{C}\pi_{Q}^{yy}\left(u^{x}\right)^{2}+I_{31}\left(u^{\tau}\right)^{2}\left(u^{x}\right)^{2}+\mathcal{C}\Pi_{Q}\left(u^{\tau}\right)^{2}\left(u^{x}\right)^{2}\\
&-\mathcal{C}{\pi_{Q}}^{\eta}{}_{\eta}\left(u^{\tau}\right)^{2}\left(u^{x}\right)^{2}+\mathcal{B}\nu^{x}\left(u^{x}\right)^{3}-\mathcal{C}\pi_{Q}^{yy}\left(u^{x}\right)^{4}+\mathcal{A}\nu^{y}\left(u^{\tau}\right)^{2}u^{y}\\
&+2\mathcal{C}\pi_{Q}^{xy}u^{x}u^{y}+\mathcal{B}\nu^{y}\left(u^{x}\right)^{2}u^{y}+2\mathcal{C}\pi_{Q}^{xy}\left(u^{x}\right)^{3}u^{y}+\mathcal{C}\pi_{Q}^{yy}\left(u^{y}\right)^{2}\\
&+I_{31}\left(u^{\tau}\right)^{2}\left(u^{y}\right)^{2}+\mathcal{C}\Pi_{Q}\left(u^{\tau}\right)^{2}\left(u^{y}\right)^{2}+\mathcal{B}\nu^{x}u^{x}\left(u^{y}\right)^{2}+\mathcal{A}\nu^{x}\left(u^{\tau}\right)^{2}u^{x}\left(u^{y}\right)^{2}\\
&+I_{31}\left(u^{\tau}\right)^{2}\left(u^{x}\right)^{2}\left(u^{y}\right)^{2}+\mathcal{C}\Pi_{Q}\left(u^{\tau}\right)^{2}\left(u^{x}\right)^{2}\left(u^{y}\right)^{2}+\mathcal{B}\nu^{x}\left(u^{x}\right)^{3}\left(u^{y}\right)^{2}\\
&+\mathcal{B}\nu^{y}\left(u^{y}\right)^{3}+\mathcal{A}\nu^{y}\left(u^{\tau}\right)^{2}\left(u^{y}\right)^{3}+2\mathcal{C}\pi_{Q}^{xy}u^{x}\left(u^{y}\right)^{3}+\mathcal{B}\nu^{y}\left(u^{x}\right)^{2}\left(u^{y}\right)^{3}\\
&+\mathcal{C}\pi_{Q}^{yy}\left(u^{y}\right)^{4}+I_{31}\left(u^{\tau}\right)^{2}\left(u^{y}\right)^{4}+\mathcal{C}\Pi_{Q}\left(u^{\tau}\right)^{2}\left(u^{y}\right)^{4}+\mathcal{B}\nu^{x}u^{x}\left(u^{y}\right)^{4}\\
&+\mathcal{B}\nu^{y}\left(u^{y}\right)^{5}\Big)\\
S^{\tau\tau x}=&\frac{1}{\left(u^{y}\right)^{2}+1}\Big(2\mathcal{C}\Pi_{Q}u^{x}-2\mathcal{C}\pi_{Q}^{yy}u^{x}-2\mathcal{C}{\pi_{Q}}^{\eta}{}_{\eta}u^{x}+3\mathcal{B}\nu^{x}\left(u^{x}\right)^{2}+3\mathcal{C}\Pi_{Q}\left(u^{x}\right)^{3}-3\mathcal{C}\pi_{Q}^{yy}\left(u^{x}\right)^{3}\\
&-3\mathcal{C}{\pi_{Q}}^{\eta}{}_{\eta}\left(u^{x}\right)^{3}+2\mathcal{C}\pi_{Q}^{xy}u^{y}+2\mathcal{B}\nu^{y}u^{x}u^{y}+6\mathcal{C}\pi_{Q}^{xy}\left(u^{x}\right)^{2}u^{y}+\mathcal{B}\nu^{x}\left(u^{y}\right)^{2}\\
&+5\mathcal{C}\Pi_{Q}u^{x}\left(u^{y}\right)^{2}+\mathcal{C}\pi_{Q}^{yy}u^{x}\left(u^{y}\right)^{2}-2\mathcal{C}{\pi_{Q}}^{\eta}{}_{\eta}u^{x}\left(u^{y}\right)^{2}+3\mathcal{B}\nu^{x}\left(u^{x}\right)^{2}\left(u^{y}\right)^{2}\\
&+3\mathcal{C}\Pi_{Q}\left(u^{x}\right)^{3}\left(u^{y}\right)^{2}+2\mathcal{C}\pi_{Q}^{xy}\left(u^{y}\right)^{3}+2\mathcal{B}\nu^{y}u^{x}\left(u^{y}\right)^{3}+\mathcal{B}\nu^{x}\left(u^{y}\right)^{4}\\
&+3\mathcal{C}\Pi_{Q}u^{x}\left(u^{y}\right)^{4}+I_{31}u^{x}\left(\left(u^{y}\right)^{2}+1\right)\left(3\left(u^{x}\right)^{2}+3\left(u^{y}\right)^{2}+2\right)\\
&+\mathcal{A}\left(\left(u^{y}\right)^{2}+1\right)\left(\nu^{x}\left(3\left(u^{x}\right)^{2}+\left(u^{y}\right)^{2}+1\right)+2\nu^{y}u^{x}u^{y}\right)\Big)\\
S^{\tau\tau y}=&2\mathcal{B}\nu^{x}u^{x}u^{y}+\mathcal{B}\nu^{y}\left(\left(u^{x}\right)^{2}+3\left(u^{y}\right)^{2}\right)+I_{31}u^{y}\left(3\left(u^{x}\right)^{2}+3\left(u^{y}\right)^{2}+2\right)\\
&+\mathcal{A}\left(2\nu^{x}u^{x}u^{y}+\nu^{y}\left(\left(u^{x}\right)^{2}+3\left(u^{y}\right)^{2}+1\right)\right)\\
&+\frac{1}{\left(u^{y}\right)^{2}+1}\mathcal{C}\Big(2\pi_{Q}^{xy}\left(2u^{x}\left(u^{y}\right)^{2}+u^{x}\right)\\
&+u^{y}\Big(-{\pi_{Q}}^{\eta}{}_{\eta}\left(u^{x}\right)^{2}+\pi_{Q}^{yy}\left(-\left(u^{x}\right)^{2}+3\left(u^{y}\right)^{2}+2\right)\\
&+\Pi_{Q}\left(\left(u^{y}\right)^{2}+1\right)\left(3\left(u^{x}\right)^{2}+3\left(u^{y}\right)^{2}+2\right)\Big)\Big)\\
S^{\tau xx}=&\frac{1}{u^{\tau}\left(\left(u^{y}\right)^{2}+1\right)}\Big(I_{31}\left(u^{\tau}\right)^{2}\left(3\left(u^{x}\right)^{2}+1\right)\left(\left(u^{y}\right)^{2}+1\right)\\
&+\mathcal{C}\Big(-{\pi_{Q}}^{\eta}{}_{\eta}-4{\pi_{Q}}^{\eta}{}_{\eta}\left(u^{x}\right)^{2}-3{\pi_{Q}}^{\eta}{}_{\eta}\left(u^{x}\right)^{4}+4\pi_{Q}^{xy}u^{x}u^{y}+6\pi_{Q}^{xy}\left(u^{x}\right)^{3}u^{y}\\
&-2{\pi_{Q}}^{\eta}{}_{\eta}\left(u^{y}\right)^{2}-4{\pi_{Q}}^{\eta}{}_{\eta}\left(u^{x}\right)^{2}\left(u^{y}\right)^{2}+4\pi_{Q}^{xy}u^{x}\left(u^{y}\right)^{3}-{\pi_{Q}}^{\eta}{}_{\eta}\left(u^{y}\right)^{4}\\
&+\Pi_{Q}\left(u^{\tau}\right)^{2}\left(3\left(u^{x}\right)^{2}+1\right)\left(\left(u^{y}\right)^{2}+1\right)\\
&-\pi_{Q}^{yy}\left(\left(u^{x}\right)^{2}+1\right)\left(3\left(u^{x}\right)^{2}+\left(u^{y}\right)^{2}+1\right)\Big)
\end{aligned}
$$

$$
\begin{aligned}
&+ (u^y)^2 + 1\Big(\mathcal{A}u^x\left(\nu^x\left(3\,(u^x)^2 + 2\,(u^y)^2 + 2\right) + \nu^y u^x u^y\right)\\
&+ \mathcal{B}\Big(\nu^x u^x\left(3\,(u^x)^2 + 2\,(u^y)^2 + 3\right) + \nu^y\left((u^x)^2 + 1\right)u^y\Big)\Big)\Big)\\
S^{\tau xy} = &\frac{1}{u^\tau\left((u^y)^2 + 1\right)}\Big((u^y)^2 + 1\Big(3I_{31}\,(u^\tau)^2\,u^x u^y + \mathcal{A}\nu^x u^y\left(2\,(u^x)^2 + (u^y)^2 + 1\right)\\
&+ \mathcal{B}\nu^x u^y\left(2\,(u^x)^2 + (u^y)^2 + 1\right)\\
&+ \mathcal{A}\nu^y\left((u^x)^3 + 2u^x\,(u^y)^2 + u^x\right)\\
&+ \mathcal{B}\nu^y\left((u^x)^3 + 2u^x\,(u^y)^2 + u^x\right)\Big)\\
&+ \mathcal{C}\Big(\pi_Q^{xy}\left(2\,(u^y)^2 + 1\right)\left(2\,(u^x)^2 + (u^y)^2 + 1\right)\\
&+ u^x u^y\Big(-{\pi_Q}^\eta{}_\eta\,(u^\tau)^2 + 3\Pi_Q\,(u^\tau)^2\left((u^y)^2 + 1\right)\\
&+ \pi_Q^{yy}\left((u^y)^2 - (u^x)^2\right)\Big)\Big)\Big)\\
S^{\tau yy} = &\frac{1}{u^\tau}\Big(\mathcal{B}\nu^x u^x + 2\mathcal{A}\nu^y u^y + 3\mathcal{B}\nu^y u^y + 2\mathcal{A}\nu^y\,(u^x)^2\,u^y + 2\mathcal{B}\nu^y\,(u^x)^2\,u^y + \mathcal{A}\nu^x u^x\,(u^y)^2\\
&+ \mathcal{B}\nu^x u^x\,(u^y)^2 + 3\mathcal{A}\nu^y\,(u^y)^3 + 3\mathcal{B}\nu^y\,(u^y)^3 + I_{31}\,(u^\tau)^2\left(3\,(u^y)^2 + 1\right)\\
&+ \mathcal{C}\left(\Pi_Q\,(u^\tau)^2\left(3\,(u^y)^2 + 1\right) + 2\pi_Q^{xy}u^x u^y + \pi_Q^{yy}\,(u^x)^2 + 3\pi_Q^{yy}\,(u^y)^2 + \pi_Q^{yy}\right)\Big)\\
S^{\tau\eta\eta} = &\frac{\mathcal{B}(\nu^x u^x + \nu^y u^y) + \mathcal{C}\,(u^\tau)^2\,(\Pi_Q + {\pi_Q}^\eta{}_\eta) + I_{31}\,(u^\tau)^2}{\tau^2 u^\tau}\\
S^{xxx} = &3\Big(\mathcal{A}\nu^x\,(u^x)^2 + \mathcal{B}\nu^x\left((u^x)^2 + 1\right) + I_{31}u^x\left((u^x)^2 + 1\right)\\
&+ \mathcal{C}u^x\Big(\Pi_Q + \Pi_Q\,(u^x)^2\\
&- \frac{-2\pi_Q^{xy}u^x u^y + \pi_Q^{yy}\,(u^x)^2 + \pi_Q^{yy} + {\pi_Q}^\eta{}_\eta\,(u^x)^2 + {\pi_Q}^\eta{}_\eta\,(u^y)^2 + {\pi_Q}^\eta{}_\eta}{(u^y)^2 + 1}\Big)\Big)\\
S^{xxy} = &\mathcal{B}\nu^y\left((u^x)^2 + 1\right) + 2\mathcal{B}\nu^x u^x u^y + 2I_{31}\,(u^x)^2\,u^y + I_{31}\left((u^x)^2 + 1\right)u^y\\
&+ \mathcal{A}u^x(2\nu^x u^y + \nu^y u^x) + 2\mathcal{C}u^x(\Pi_Q u^x u^y + \pi_Q^{xy})\\
&+ \mathcal{C}u^y\Big(\Pi_Q + \Pi_Q\,(u^x)^2\\
&- \frac{-2\pi_Q^{xy}u^x u^y + \pi_Q^{yy}\,(u^x)^2 + \pi_Q^{yy} + {\pi_Q}^\eta{}_\eta\,(u^x)^2 + {\pi_Q}^\eta{}_\eta\,(u^y)^2 + {\pi_Q}^\eta{}_\eta}{(u^y)^2 + 1}\Big)\\
S^{xyy} = &\mathcal{C}\Pi_Q u^x + \mathcal{C}\pi_Q^{yy}u^x + 2\mathcal{C}\pi_Q^{xy}u^y + 2\mathcal{A}\nu^y u^x u^y + \mathcal{A}\nu^x\,(u^y)^2 + 3\mathcal{C}\Pi_Q u^x\,(u^y)^2\\
&+ \mathcal{B}\left(\nu^x\,(u^y)^2 + \nu^x + 2\nu^y u^x u^y\right) + I_{31}\left(3u^x\,(u^y)^2 + u^x\right)\\
S^{x\eta\eta} = &\frac{\mathcal{B}\nu^x + u^x(\mathcal{C}(\Pi_Q + {\pi_Q}^\eta{}_\eta) + I_{31})}{\tau^2}\\
S^{yyy} = &\,3\left(u^y\left(\mathcal{A}\nu^y u^y + \mathcal{C}\left(\Pi_Q\,(u^y)^2 + \Pi_Q + \pi_Q^{yy}\right) + I_{31}\left((u^y)^2 + 1\right)\right) + \mathcal{B}\nu^y\left((u^y)^2 + 1\right)\right)\\
S^{y\eta\eta} = &\frac{\mathcal{B}\nu^y + u^y(\mathcal{C}(\Pi_Q + {\pi_Q}^\eta{}_\eta) + I_{31})}{\tau^2}
\end{aligned}
$$

$$
\begin{aligned}
\nabla_\lambda S^{\lambda xx} &= (\partial_\tau S^{\tau xx}) + (\partial_x S^{xxx}) + (\partial_y S^{xxy}) + \frac{S^{\tau xx}}{\tau}\\
\nabla_\lambda S^{\lambda xy} &= (\partial_\tau S^{\tau xy}) + (\partial_x S^{xxy}) + (\partial_y S^{xyy}) + \frac{S^{\tau xy}}{\tau}\\
\nabla_\lambda S^{\lambda yy} &= (\partial_\tau S^{\tau yy}) + (\partial_x S^{xyy}) + (\partial_y S^{yyy}) + \frac{S^{\tau yy}}{\tau}\\
\nabla_\lambda S^{\lambda\eta\eta} &= (\partial_\tau S^{\tau\eta\eta}) + (\partial_x S^{x\eta\eta}) + (\partial_y S^{y\eta\eta}) + \frac{5S^{\tau\eta\eta}}{\tau}\\
\mathcal{E}^{ab} &= \Delta^a{}_\alpha\Delta^b{}_\beta\nabla_\lambda S^{\lambda\alpha\beta}, \qquad \mathcal{E}^\lambda{}_\lambda = \Delta_{ab}\mathcal{E}^{ab}
\end{aligned}
$$

$$
E_{ab} = \frac{D_s}{2T}\Big(\mathcal{E}^{ab} - \tfrac{1}{3}\Delta^{ab}\mathcal{E}^\lambda{}_\lambda\Big) + \pi_Q^{ab}, \quad ab = yy,\ xy, \qquad E_\eta = \tau^2\frac{D_s}{2T}\Big(\mathcal{E}^{\eta\eta} - \tfrac{1}{3}\Delta^{\eta\eta}\mathcal{E}^\lambda{}_\lambda\Big) + {\pi_Q}^\eta{}_\eta
$$

$$E_{\Pi_Q} = \frac{D_s}{6T}\mathcal{E}^{\lambda}{}_{\lambda} + \Pi_Q$$

3.2.8 2+1D, S5

$$\begin{aligned}
T_m^{\tau\tau} &= \frac{1}{(u^y)^2+1}\Big(\Pi\,(u^x)^2 - \pi^{\eta}{}_{\eta}\,(u^x)^2 - \pi^{yy}\,(u^x)^2 + 2\pi^{xy}u^x u^y + \Pi\,(u^y)^2 + \pi^{yy}\,(u^y)^2 \\
&\quad + \Pi\,(u^x)^2\,(u^y)^2 + \Pi\,(u^y)^4 - P\left((u^y)^2+1\right) + P'(T)T\,(u^\tau)^2\left((u^y)^2+1\right)\Big) \\
T_m^{\tau x} &= \frac{1}{u^\tau\left((u^y)^2+1\right)}\Big(-\pi^{yy}u^x + \Pi\,(u^\tau)^2\,u^x - \pi^{\eta}{}_{\eta}\,(u^\tau)^2\,u^x + P'(T)T\,(u^\tau)^2\,u^x - \pi^{yy}\,(u^x)^3 \\
&\quad + \pi^{xy}u^y + 2\pi^{xy}\,(u^x)^2\,u^y + \Pi\,(u^\tau)^2\,u^x\,(u^y)^2 + P'(T)T\,(u^\tau)^2\,u^x\,(u^y)^2 \\
&\quad + \pi^{xy}\,(u^y)^3\Big) \\
T_m^{\tau y} &= \frac{(u^\tau)^2\,u^y(P'(T)T+\Pi) + \pi^{xy}u^x + \pi^{yy}u^y}{u^\tau} \\
T_m^{xx} &= (u^x)^2\,(P'(T)T+\Pi) + \Pi - \frac{\pi^{\eta}{}_{\eta}\,(u^\tau)^2 - 2\pi^{xy}u^x u^y + \pi^{yy}\,(u^x)^2 + \pi^{yy}}{(u^y)^2+1} + P \\
T_m^{xy} &= u^x u^y(P'(T)T+\Pi) + \pi^{xy} \\
T_m^{yy} &= (u^y)^2\,(P'(T)T+\Pi) + \Pi + \pi^{yy} + P \\
T_m^{\eta\eta} &= \frac{\Pi + \pi^{\eta}{}_{\eta} + P}{\tau^2}
\end{aligned}$$

$$E_\tau = \nabla_\mu(T_m+T_Q)^{\mu\tau}, \quad E_x = \nabla_\mu(T_m+T_Q)^{\mu x}, \quad E_y = \nabla_\mu(T_m+T_Q)^{\mu y}$$
$$A_{ij}^{(k)} = \frac{\partial E_i}{\partial(\partial_k\phi_j)}, \qquad S_i = E_i\big|_{\partial\phi=0}, \qquad k = \tau, x, y$$

$$\begin{aligned}
E_\tau &= -(\partial_\tau\Pi) - P'(T)(\partial_\tau T) + \frac{\Pi + \pi^{\eta}{}_{\eta} + P}{\tau} + \frac{nT}{\tau} + (\partial_x u^x)nu^\tau(h-T) \\
&\quad + (\partial_y u^y)nu^\tau(h-T) + (\partial_x u^x)nTu^\tau + (\partial_y u^y)nTu^\tau + (\partial_x u^x)u^\tau(P'(T)T+\Pi) \\
&\quad + (\partial_y u^y)u^\tau(P'(T)T+\Pi) \\
&\quad + (u^\tau)^2\left((h'(T)-1)(\partial_\tau T)n + (\partial_\alpha n)(\partial_\tau\alpha)(h-T) + (\partial_T n)(\partial_\tau T)(h-T)\right) \\
&\quad + (u^\tau)^2\left((\partial_\tau T)(P''(T)T+P'(T)) + (\partial_\tau\Pi)\right) + (\partial_x T)nu^\tau u^x \\
&\quad + u^\tau u^x\left((h'(T)-1)(\partial_x T)n + (\partial_\alpha n)(\partial_x\alpha)(h-T) + (\partial_T n)(\partial_x T)(h-T)\right) \\
&\quad + Tu^\tau u^x\left((\partial_\alpha n)(\partial_x\alpha) + (\partial_T n)(\partial_x T)\right) + u^\tau u^x\left((\partial_x T)(P''(T)T+P'(T)) + (\partial_x\Pi)\right) \\
&\quad + (\partial_y T)nu^\tau u^y \\
&\quad + u^\tau u^y\left((h'(T)-1)(\partial_y T)n + (\partial_\alpha n)(\partial_y\alpha)(h-T) + (\partial_T n)(\partial_y T)(h-T)\right) \\
&\quad + Tu^\tau u^y\left((\partial_\alpha n)(\partial_y\alpha) + (\partial_T n)(\partial_y T)\right) + u^\tau u^y\left((\partial_y T)(P''(T)T+P'(T)) + (\partial_y\Pi)\right) \\
&\quad + 2h\left((\partial_\tau\nu^x)u^x + (\partial_\tau\nu^y)u^y + (\partial_\tau u^x)\nu^x + (\partial_\tau u^y)\nu^y\right) \\
&\quad + 2n(h-T)\left((\partial_\tau u^x)u^x + (\partial_\tau u^y)u^y\right) + 2nT\left((\partial_\tau u^x)u^x + (\partial_\tau u^y)u^y\right) \\
&\quad + 2(P'(T)T+\Pi)\left((\partial_\tau u^x)u^x + (\partial_\tau u^y)u^y\right) + \frac{nu^x(h-T)\left((\partial_x u^x)u^x + (\partial_x u^y)u^y\right)}{u^\tau} \\
&\quad + \frac{nTu^x\left((\partial_x u^x)u^x + (\partial_x u^y)u^y\right)}{u^\tau} + \frac{u^x(P'(T)T+\Pi)\left((\partial_x u^x)u^x + (\partial_x u^y)u^y\right)}{u^\tau} \\
&\quad + \frac{(\partial_y\pi^{xy})u^x + (\partial_y\pi^{yy})u^y + (\partial_y u^x)\pi^{xy} + (\partial_y u^y)\pi^{yy}}{u^\tau} \\
&\quad + \frac{nu^y(h-T)\left((\partial_y u^x)u^x + (\partial_y u^y)u^y\right)}{u^\tau} + \frac{nTu^y\left((\partial_y u^x)u^x + (\partial_y u^y)u^y\right)}{u^\tau} \\
&\quad + \frac{u^y(P'(T)T+\Pi)\left((\partial_y u^x)u^x + (\partial_y u^y)u^y\right)}{u^\tau} + 2h'(T)(\partial_\tau T)(\nu^x u^x + \nu^y u^y) \\
&\quad - \frac{\left((\partial_y u^x)u^x + (\partial_y u^y)u^y\right)(\pi^{xy}u^x + \pi^{yy}u^y)}{(u^\tau)^3} + (\partial_\tau T)n\left((u^x)^2 + (u^y)^2\right)
\end{aligned}$$

$$
\begin{aligned}
&+ T\left(\left(u^x\right)^2+\left(u^y\right)^2\right)\left((\partial_\alpha n)(\partial_\tau \alpha)+(\partial_T n)(\partial_\tau T)\right)\\
&+\frac{hn\left(u^\tau\right)^2+2h(\nu^x u^x+\nu^y u^y)-nT}{\tau}\\
&+\frac{\left((\partial_\tau u^x)u^x+(\partial_\tau u^y)u^y\right)\left(\pi^\eta{}_\eta\left(u^x\right)^2-2\pi^{xy}u^x u^y+\pi^{yy}\left(\left(u^x\right)^2-\left(u^y\right)^2\right)\right)}{\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right)}\\
&+\frac{1}{\tau\left(\left(u^y\right)^2+1\right)}\Big(\Pi\left(u^x\right)^2-\pi^\eta{}_\eta\left(u^x\right)^2-\pi^{yy}\left(u^x\right)^2+2\pi^{xy}u^x u^y+\Pi\left(u^y\right)^2+\pi^{yy}\left(u^y\right)^2\\
&+\Pi\left(u^x\right)^2\left(u^y\right)^2+\Pi\left(u^y\right)^4-P\left(\left(u^y\right)^2+1\right)\\
&+P'(T)T\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right)\Big)+\frac{h'(T)(\partial_x T)\left(\nu^x\left(2\left(u^x\right)^2+\left(u^y\right)^2+1\right)+\nu^y u^x u^y\right)}{u^\tau}\\
&+\frac{1}{\left(u^\tau\right)^3\left(\left(u^y\right)^2+1\right)}(\partial_x u^x)u^x+(\partial_x u^y)u^y\Big(\pi^\eta{}_\eta\left(u^\tau\right)^2u^x+\pi^{yy}\left(\left(u^x\right)^3+u^x\right)\\
&-\pi^{xy}u^y\left(2\left(u^x\right)^2+\left(u^y\right)^2+1\right)\Big)+\frac{h'(T)(\partial_y T)\left(\nu^x u^x u^y+\nu^y\left(\left(u^x\right)^2+2\left(u^y\right)^2+1\right)\right)}{u^\tau}\\
&+\frac{1}{\left(u^\tau\right)^3}h\Big((\partial_x u^y)\nu^y u^x+(\partial_x u^y)\nu^y\left(u^x\right)^3+(\partial_x u^y)\nu^x u^y+(\partial_x\nu^y)u^x u^y\\
&+(\partial_x\nu^y)\left(u^x\right)^3u^y+(\partial_x u^y)\nu^x\left(u^y\right)^3+(\partial_x\nu^y)u^x\left(u^y\right)^3\\
&+(\partial_x u^x)\nu^y u^y\left(\left(u^y\right)^2+1\right)+(\partial_x u^x)\nu^x u^x\left(2\left(u^x\right)^2+3\left(u^y\right)^2+3\right)\\
&+(\partial_x\nu^x)\left(2\left(u^x\right)^4+3\left(u^x\right)^2\left(\left(u^y\right)^2+1\right)+\left(\left(u^y\right)^2+1\right)^2\right)\Big)\\
&+\frac{1}{\left(u^\tau\right)^3}h\Big((\partial_y u^x)\nu^y u^x+(\partial_y u^x)\nu^y\left(u^x\right)^3+(\partial_y u^y)\nu^x\left(\left(u^x\right)^3+u^x\right)+(\partial_y u^x)\nu^x u^y\\
&+(\partial_y\nu^x)u^x u^y+(\partial_y\nu^x)\left(u^x\right)^3u^y+(\partial_y u^x)\nu^x\left(u^y\right)^3+(\partial_y\nu^x)u^x\left(u^y\right)^3\\
&+(\partial_y u^y)\nu^y u^y\left(3\left(u^x\right)^2+2\left(u^y\right)^2+3\right)\\
&+(\partial_y\nu^y)\left(\left(u^x\right)^4+\left(u^x\right)^2\left(3\left(u^y\right)^2+2\right)+2\left(u^y\right)^4+3\left(u^y\right)^2+1\right)\Big)\\
&+\frac{1}{u^\tau\left(\left(u^y\right)^2+1\right)^2}\Big(-\left(u^y\right)^2+1\Big(\left(\partial_x\pi^\eta{}_\eta\right)\left(u^\tau\right)^2u^x+\left(\partial_x\pi^{yy}\right)\left(\left(u^x\right)^3+u^x\right)\\
&-(\partial_x\pi^{xy})u^y\left(2\left(u^x\right)^2+\left(u^y\right)^2+1\right)\Big)\\
&-(\partial_x u^x)\left(\left(u^y\right)^2+1\right)\left(\pi^{yy}+3\pi^{yy}\left(u^x\right)^2-4\pi^{xy}u^x u^y\right.\\
&\left.+\pi^\eta{}_\eta\left(3\left(u^x\right)^2+\left(u^y\right)^2+1\right)\right)\\
&+(\partial_x u^y)\Big(2u^x u^y\left(\pi^\eta{}_\eta\left(u^x\right)^2+\pi^{yy}\left(u^x\right)^2+\pi^{yy}\right)\\
&+\pi^{xy}\left(\left(\left(u^y\right)^2+1\right)^2-2\left(u^x\right)^2\left(\left(u^y\right)^2-1\right)\right)\Big)\Big)\\
&+\frac{1}{\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right)^2}\Big(2(\partial_\tau u^y)\pi^{xy}u^x\left(-\left(\left(u^x\right)^2\left(\left(u^y\right)^2-1\right)\right)+\left(u^y\right)^2+1\right)\\
&+\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right)\Big(-(\partial_\tau\pi^\eta{}_\eta)\left(u^x\right)^2+2(\partial_\tau\pi^{xy})u^x u^y\\
&+(\partial_\tau\pi^{yy})\left(\left(u^y\right)^2-\left(u^x\right)^2\right)\Big)\\
&-(\partial_\tau u^x)\left(\left(u^y\right)^2+1\right)\Big(-2\pi^{xy}u^y\left(2\left(u^x\right)^2+\left(u^y\right)^2+1\right)\\
&+\pi^{yy}u^x\left(3\left(u^x\right)^2+\left(u^y\right)^2+2\right)\\
&+\pi^\eta{}_\eta u^x\left(3\left(u^x\right)^2+2\left(u^y\right)^2+2\right)\Big)
\end{aligned}
$$

$$
\begin{aligned}
&+ (\partial_\tau u^y) u^y \Big({\pi^\eta}_\eta \left(u^x\right)^2 \left(2\left(u^x\right)^2 + \left(u^y\right)^2 + 1\right) \\
&+ \pi^{yy} \Big(2 + 2\left(u^x\right)^4 + 3\left(u^y\right)^2 + \left(u^y\right)^4 \\
&+ \left(u^x\right)^2 \left(\left(u^y\right)^2 + 3\right)\Big)\Big)\Big)
\end{aligned}
$$

3.2.9 2+1D, S6

$$
\begin{aligned}
\delta M^{\tau\tau} &= \frac{\Pi_Q \left(\left(u^y\right)^2 + 1\right)\left(\left(u^x\right)^2 + \left(u^y\right)^2\right) + u^x (2\pi_Q^{xy} u^y - {\pi_Q}^\eta{}_\eta u^x) + \pi_Q^{yy}\left(\left(u^y\right)^2 - \left(u^x\right)^2\right)}{\left(u^y\right)^2 + 1} \\
\delta M^{\tau x} &= \frac{1}{u^\tau \left(\left(u^y\right)^2 + 1\right)} \Big(-\pi_Q^{yy} u^x + \Pi_Q \left(u^\tau\right)^2 u^x - {\pi_Q}^\eta{}_\eta \left(u^\tau\right)^2 u^x - \pi_Q^{yy} \left(u^x\right)^3 + \pi_Q^{xy} u^y \\
&\quad + 2\pi_Q^{xy} \left(u^x\right)^2 u^y + \Pi_Q \left(u^\tau\right)^2 u^x \left(u^y\right)^2 + \pi_Q^{xy} \left(u^y\right)^3 \Big) \\
\delta M^{\tau y} &= \frac{u^y \left(\Pi_Q \left(u^\tau\right)^2 + \pi_Q^{yy}\right) + \pi_Q^{xy} u^x}{u^\tau} \\
\delta M^{xx} &= \Pi_Q \left(u^x\right)^2 + \Pi_Q - \frac{-2\pi_Q^{xy} u^x u^y + \pi_Q^{yy} \left(u^x\right)^2 + \pi_Q^{yy} + {\pi_Q}^\eta{}_\eta \left(u^x\right)^2 + {\pi_Q}^\eta{}_\eta \left(u^y\right)^2 + {\pi_Q}^\eta{}_\eta}{\left(u^y\right)^2 + 1} \\
\delta M^{xy} &= \Pi_Q u^x u^y + \pi_Q^{xy} \\
\delta M^{yy} &= \Pi_Q \left(u^y\right)^2 + \Pi_Q + \pi_Q^{yy} \\
\delta M^{\eta\eta} &= \frac{\Pi_Q + {\pi_Q}^\eta{}_\eta}{\tau^2}
\end{aligned}
$$

$$
\begin{aligned}
\nabla_\mu \delta M^{\mu\tau} &= (\partial_\tau \delta M^{\tau\tau}) + (\partial_x \delta M^{\tau x}) + (\partial_y \delta M^{\tau y}) + \frac{\delta M^{\tau\tau}}{\tau} + \tau \delta M^{\eta\eta} \\
\nabla_\mu \delta M^{\mu x} &= (\partial_\tau \delta M^{\tau x}) + (\partial_x \delta M^{x x}) + (\partial_y \delta M^{x y}) + \frac{\delta M^{\tau x}}{\tau} \\
\nabla_\mu \delta M^{\mu y} &= (\partial_\tau \delta M^{\tau y}) + (\partial_x \delta M^{x y}) + (\partial_y \delta M^{y y}) + \frac{\delta M^{\tau y}}{\tau}
\end{aligned}
$$

$$
\begin{aligned}
\nabla_\mu \delta M^{\mu\tau} = &\frac{\Pi_Q + {\pi_Q}^\eta{}_\eta}{\tau} + (\partial_x u^x)\Pi_Q u^\tau + (\partial_y u^y)\Pi_Q u^\tau + (\partial_x \Pi_Q) u^\tau u^x + (\partial_y \Pi_Q) u^\tau u^y \\
&+ 2\Pi_Q((\partial_\tau u^x)u^x + (\partial_\tau u^y)u^y) + \frac{\Pi_Q u^x((\partial_x u^x)u^x + (\partial_x u^y)u^y)}{u^\tau} \\
&+ \frac{(\partial_y \pi_Q^{xy})u^x + (\partial_y \pi_Q^{yy})u^y + (\partial_y u^x)\pi_Q^{xy} + (\partial_y u^y)\pi_Q^{yy}}{u^\tau} + \frac{\Pi_Q u^y((\partial_y u^x)u^x + (\partial_y u^y)u^y)}{u^\tau} \\
&- \frac{((\partial_y u^x)u^x + (\partial_y u^y)u^y)(\pi_Q^{xy} u^x + \pi_Q^{yy} u^y)}{\left(u^\tau\right)^3} + (\partial_\tau \Pi_Q)\left(\left(u^x\right)^2 + \left(u^y\right)^2\right) \\
&+ \frac{((\partial_\tau u^x)u^x + (\partial_\tau u^y)u^y)\left(u^x({\pi_Q}^\eta{}_\eta u^x - 2\pi_Q^{xy} u^y) + \pi_Q^{yy}\left(\left(u^x\right)^2 - \left(u^y\right)^2\right)\right)}{\left(u^\tau\right)^2 \left(\left(u^y\right)^2 + 1\right)} \\
&+ \frac{\Pi_Q \left(\left(u^y\right)^2 + 1\right)\left(\left(u^x\right)^2 + \left(u^y\right)^2\right) + u^x (2\pi_Q^{xy} u^y - {\pi_Q}^\eta{}_\eta u^x) + \pi_Q^{yy}\left(\left(u^y\right)^2 - \left(u^x\right)^2\right)}{\tau \left(\left(u^y\right)^2 + 1\right)} \\
&+ \frac{1}{\left(u^\tau\right)^3 \left(\left(u^y\right)^2 + 1\right)} (\partial_x u^x) u^x + (\partial_x u^y) u^y \Big({\pi_Q}^\eta{}_\eta \left(u^\tau\right)^2 u^x + \pi_Q^{yy}\left(\left(u^x\right)^3 + u^x\right) \\
&- \pi_Q^{xy} u^y \left(2\left(u^x\right)^2 + \left(u^y\right)^2 + 1\right)\Big) \\
&+ \frac{1}{u^\tau \left(\left(u^y\right)^2 + 1\right)^2} \Big(-\left(u^y\right)^2 + 1 \Big(\left(\partial_x {\pi_Q}^\eta{}_\eta\right) u^x + \left(\partial_x {\pi_Q}^\eta{}_\eta\right) \left(u^x\right)^3 \\
&+ \left(\partial_x \pi_Q^{yy}\right)\left(\left(u^x\right)^3 + u^x\right) - \left(\partial_x \pi_Q^{xy}\right) u^y
\end{aligned}
$$

$$
\begin{aligned}
&- 2(\partial_x \pi_Q^{xy}) (u^x)^2 u^y + (\partial_x \pi_Q{}^\eta{}_\eta) u^x (u^y)^2 \\
&- (\partial_x \pi_Q^{xy}) (u^y)^3 \\
&+ (\partial_x u^x) \Big(\pi_Q^{yy} + \pi_Q{}^\eta{}_\eta + 3\pi_Q^{yy} (u^x)^2 + 3\pi_Q{}^\eta{}_\eta (u^x)^2 \\
&- 4\pi_Q^{xy} u^x u^y + \pi_Q{}^\eta{}_\eta (u^y)^2 \Big) \Big) \\
&+ (\partial_x u^y) \Big(2u^x u^y \left(\pi_Q^{yy} (u^x)^2 + \pi_Q^{yy} + \pi_Q{}^\eta{}_\eta (u^x)^2 \right) \\
&+ \pi_Q^{xy} \left(\left((u^y)^2 + 1 \right)^2 - 2 (u^x)^2 \left((u^y)^2 - 1 \right) \right) \Big) \Big) \\
&+ \frac{1}{(u^\tau)^2 \left((u^y)^2 + 1 \right)^2} \Big(- (u^y)^2 + 1 \Big((u^\tau)^2 \left(u^x ((\partial_\tau \pi_Q{}^\eta{}_\eta) u^x - 2(\partial_\tau \pi_Q^{xy}) u^y) \right. \right. \\
&+ (\partial_\tau \pi_Q^{yy}) \left((u^x)^2 - (u^y)^2 \right) \Big) \\
&+ (\partial_\tau u^x) \Big(-2\pi_Q^{xy} u^y \left(2 (u^x)^2 + (u^y)^2 + 1 \right) \\
&+ \pi_Q^{yy} u^x \left(3 (u^x)^2 + (u^y)^2 + 2 \right) \\
&+ \pi_Q{}^\eta{}_\eta u^x \left(3 (u^x)^2 + 2 (u^y)^2 + 2 \right) \Big) \Big) \\
&+ (\partial_\tau u^y) \Big(2\pi_Q^{xy} u^x \left(- \left((u^x)^2 \left((u^y)^2 - 1 \right) \right) + (u^y)^2 + 1 \right) \\
&+ u^y \Big(\pi_Q{}^\eta{}_\eta (u^x)^2 \left(2 (u^x)^2 + (u^y)^2 + 1 \right) \\
&+ \pi_Q^{yy} \Big(2 + 2 (u^x)^4 + 3 (u^y)^2 + (u^y)^4 \\
&+ (u^x)^2 \left((u^y)^2 + 3 \right) \Big) \Big) \Big) \Big)
\end{aligned}
$$

$$
\begin{aligned}
\nabla_\mu \delta M^{\mu x} = \frac{1}{\tau (u^\tau)^3 \left((u^y)^2 + 1 \right)^2} \Big(& (\partial_\tau u^y) \pi_Q^{xy} \tau (u^\tau)^2 - (\partial_\tau u^x) \pi_Q^{yy} \tau (u^\tau)^2 - (\partial_\tau u^x) \pi_Q{}^\eta{}_\eta \tau (u^\tau)^2 \\
&+ (\partial_x \Pi_Q) \tau (u^\tau)^3 - (\partial_x \pi_Q^{yy}) \tau (u^\tau)^3 - (\partial_x \pi_Q{}^\eta{}_\eta) \tau (u^\tau)^3 \\
&+ (\partial_y \pi_Q^{xy}) \tau (u^\tau)^3 + (\partial_\tau u^x) \Pi_Q \tau (u^\tau)^4 - \pi_Q^{yy} (u^\tau)^2 u^x \\
&- (\partial_\tau \pi_Q^{yy}) \tau (u^\tau)^2 u^x - (\partial_\tau \pi_Q{}^\eta{}_\eta) \tau (u^\tau)^2 u^x + 2(\partial_x u^x) \Pi_Q \tau (u^\tau)^3 u^x \\
&+ (\partial_y u^y) \Pi_Q \tau (u^\tau)^3 u^x + 2(\partial_x u^y) \pi_Q^{xy} \tau (u^\tau)^3 u^x \\
&- 2(\partial_x u^x) \pi_Q^{yy} \tau (u^\tau)^3 u^x - 2(\partial_x u^x) \pi_Q{}^\eta{}_\eta \tau (u^\tau)^3 u^x + \Pi_Q (u^\tau)^4 u^x \\
&- \pi_Q{}^\eta{}_\eta (u^\tau)^4 u^x + (\partial_\tau \Pi_Q) \tau (u^\tau)^4 u^x + (\partial_\tau u^x) \pi_Q^{yy} \tau (u^x)^2 \\
&+ (\partial_\tau u^x) \Pi_Q \tau (u^\tau)^2 (u^x)^2 + 2(\partial_\tau u^y) \pi_Q^{xy} \tau (u^\tau)^2 (u^x)^2 \\
&- 3(\partial_\tau u^x) \pi_Q^{yy} \tau (u^\tau)^2 (u^x)^2 - 2(\partial_\tau u^x) \pi_Q{}^\eta{}_\eta \tau (u^\tau)^2 (u^x)^2 \\
&+ (\partial_x \Pi_Q) \tau (u^\tau)^3 (u^x)^2 - (\partial_x \pi_Q^{yy}) \tau (u^\tau)^3 (u^x)^2 \\
&- (\partial_x \pi_Q{}^\eta{}_\eta) \tau (u^\tau)^3 (u^x)^2 - \pi_Q^{yy} (u^\tau)^2 (u^x)^3 - (\partial_\tau \pi_Q^{yy}) \tau (u^\tau)^2 (u^x)^3 \\
&- (\partial_\tau \pi_Q{}^\eta{}_\eta) \tau (u^\tau)^2 (u^x)^3 + (\partial_\tau u^x) \pi_Q^{yy} \tau (u^x)^4 + \pi_Q^{xy} (u^\tau)^2 u^y \\
&+ (\partial_\tau \pi_Q^{xy}) \tau (u^\tau)^2 u^y + (\partial_y u^x) \Pi_Q \tau (u^\tau)^3 u^y + 2(\partial_x u^x) \pi_Q^{xy} \tau (u^\tau)^3 u^y \\
&+ 2(\partial_x u^y) \pi_Q^{yy} \tau (u^\tau)^3 u^y - 2(\partial_x u^y) \pi_Q{}^\eta{}_\eta \tau (u^\tau)^3 u^y \\
&+ 2(\partial_x u^y) \pi_Q{}^\eta{}_\eta \tau (u^\tau)^5 u^y - (\partial_\tau u^x) \pi_Q^{xy} \tau u^x u^y + (\partial_\tau u^y) \pi_Q^{yy} \tau u^x u^y \\
&+ (\partial_\tau u^y) \Pi_Q \tau (u^\tau)^2 u^x u^y + 4(\partial_\tau u^x) \pi_Q^{xy} \tau (u^\tau)^2 u^x u^y \\
&+ 2(\partial_\tau u^y) \pi_Q^{yy} \tau (u^\tau)^2 u^x u^y + (\partial_\tau u^y) \pi_Q{}^\eta{}_\eta \tau (u^\tau)^2 u^x u^y \\
&+ 2(\partial_x \pi_Q^{xy}) \tau (u^\tau)^3 u^x u^y + (\partial_y \Pi_Q) \tau (u^\tau)^3 u^x u^y \\
&+ 2\pi_Q^{xy} (u^\tau)^2 (u^x)^2 u^y + 2(\partial_\tau \pi_Q^{xy}) \tau (u^\tau)^2 (u^x)^2 u^y \\
&+ 2(\partial_x u^y) \pi_Q^{yy} \tau (u^\tau)^3 (u^x)^2 u^y - 2(\partial_\tau u^x) \pi_Q^{xy} \tau (u^x)^3 u^y \\
&+ (\partial_\tau u^y) \pi_Q^{yy} \tau (u^x)^3 u^y + 2(\partial_\tau u^y) \pi_Q^{yy} \tau (u^\tau)^2 (u^x)^3 u^y
\end{aligned}
$$

$$
\begin{aligned}
&+ 2(\partial_\tau u^y)\pi_Q{}^\eta{}_\eta \tau \left(u^\tau\right)^2 \left(u^x\right)^3 u^y - (\partial_\tau u^y)\pi_Q^{xy}\tau \left(u^y\right)^2 \\
&+ 2(\partial_\tau u^y)\pi_Q^{xy}\tau \left(u^\tau\right)^2 \left(u^y\right)^2 - (\partial_\tau u^x)\pi_Q^{yy}\tau \left(u^\tau\right)^2 \left(u^y\right)^2 \\
&- 2(\partial_\tau u^x)\pi_Q{}^\eta{}_\eta \tau \left(u^\tau\right)^2 \left(u^y\right)^2 + 2(\partial_x \Pi_Q)\tau \left(u^\tau\right)^3 \left(u^y\right)^2 \\
&- (\partial_x \pi_Q^{yy})\tau \left(u^\tau\right)^3 \left(u^y\right)^2 - 2(\partial_x \pi_Q{}^\eta{}_\eta)\tau \left(u^\tau\right)^3 \left(u^y\right)^2 \\
&+ 2(\partial_y \pi_Q^{xy})\tau \left(u^\tau\right)^3 \left(u^y\right)^2 + 2(\partial_\tau u^x)\Pi_Q \tau \left(u^\tau\right)^4 \left(u^y\right)^2 \\
&- \pi_Q^{yy} \left(u^\tau\right)^2 u^x \left(u^y\right)^2 - (\partial_\tau \pi_Q^{yy})\tau \left(u^\tau\right)^2 u^x \left(u^y\right)^2 \\
&- 2(\partial_\tau \pi_Q{}^\eta{}_\eta)\tau \left(u^\tau\right)^2 u^x \left(u^y\right)^2 + 4(\partial_x u^x)\Pi_Q \tau \left(u^\tau\right)^3 u^x \left(u^y\right)^2 \\
&+ 2(\partial_y u^y)\Pi_Q \tau \left(u^\tau\right)^3 u^x \left(u^y\right)^2 - 2(\partial_x u^y)\pi_Q^{xy}\tau \left(u^\tau\right)^3 u^x \left(u^y\right)^2 \\
&- 2(\partial_x u^x)\pi_Q^{yy}\tau \left(u^\tau\right)^3 u^x \left(u^y\right)^2 - 2(\partial_x u^x)\pi_Q{}^\eta{}_\eta \tau \left(u^\tau\right)^3 u^x \left(u^y\right)^2 \\
&+ 2\Pi_Q \left(u^\tau\right)^4 u^x \left(u^y\right)^2 - \pi_Q{}^\eta{}_\eta \left(u^\tau\right)^4 u^x \left(u^y\right)^2 \\
&+ 2(\partial_\tau \Pi_Q)\tau \left(u^\tau\right)^4 u^x \left(u^y\right)^2 - 2(\partial_\tau u^y)\pi_Q^{xy}\tau \left(u^x\right)^2 \left(u^y\right)^2 \\
&+ (\partial_\tau u^x)\pi_Q^{yy}\tau \left(u^x\right)^2 \left(u^y\right)^2 + 2(\partial_\tau u^x)\Pi_Q \tau \left(u^\tau\right)^2 \left(u^x\right)^2 \left(u^y\right)^2 \\
&- 2(\partial_\tau u^y)\pi_Q^{xy}\tau \left(u^\tau\right)^2 \left(u^x\right)^2 \left(u^y\right)^2 - 3(\partial_\tau u^x)\pi_Q^{yy}\tau \left(u^\tau\right)^2 \left(u^x\right)^2 \left(u^y\right)^2 \\
&- 2(\partial_\tau u^x)\pi_Q{}^\eta{}_\eta \tau \left(u^\tau\right)^2 \left(u^x\right)^2 \left(u^y\right)^2 + 2(\partial_x \Pi_Q)\tau \left(u^\tau\right)^3 \left(u^x\right)^2 \left(u^y\right)^2 \\
&- (\partial_x \pi_Q^{yy})\tau \left(u^\tau\right)^3 \left(u^x\right)^2 \left(u^y\right)^2 - (\partial_x \pi_Q{}^\eta{}_\eta)\tau \left(u^\tau\right)^3 \left(u^x\right)^2 \left(u^y\right)^2 \\
&- \pi_Q^{yy} \left(u^\tau\right)^2 \left(u^x\right)^3 \left(u^y\right)^2 - (\partial_\tau \pi_Q^{yy})\tau \left(u^\tau\right)^2 \left(u^x\right)^3 \left(u^y\right)^2 \\
&- (\partial_\tau \pi_Q{}^\eta{}_\eta)\tau \left(u^\tau\right)^2 \left(u^x\right)^3 \left(u^y\right)^2 + (\partial_\tau u^x)\pi_Q^{yy}\tau \left(u^x\right)^4 \left(u^y\right)^2 \\
&+ 2\pi_Q^{xy} \left(u^\tau\right)^2 \left(u^y\right)^3 + 2(\partial_\tau \pi_Q^{xy})\tau \left(u^\tau\right)^2 \left(u^y\right)^3 \\
&+ 2(\partial_y u^x)\Pi_Q \tau \left(u^\tau\right)^3 \left(u^y\right)^3 + 2(\partial_x u^x)\pi_Q^{xy}\tau \left(u^\tau\right)^3 \left(u^y\right)^3 \\
&- 2(\partial_x u^y)\pi_Q{}^\eta{}_\eta \tau \left(u^\tau\right)^3 \left(u^y\right)^3 - 2(\partial_\tau u^x)\pi_Q^{xy}\tau u^x \left(u^y\right)^3 \\
&+ (\partial_\tau u^y)\pi_Q^{yy}\tau u^x \left(u^y\right)^3 + 2(\partial_\tau u^y)\Pi_Q \tau \left(u^\tau\right)^2 u^x \left(u^y\right)^3 \\
&+ 4(\partial_\tau u^x)\pi_Q^{xy}\tau \left(u^\tau\right)^2 u^x \left(u^y\right)^3 + (\partial_\tau u^y)\pi_Q{}^\eta{}_\eta \tau \left(u^\tau\right)^2 u^x \left(u^y\right)^3 \\
&+ 2(\partial_x \pi_Q^{xy})\tau \left(u^\tau\right)^3 u^x \left(u^y\right)^3 + 2(\partial_y \Pi_Q)\tau \left(u^\tau\right)^3 u^x \left(u^y\right)^3 \\
&+ 2\pi_Q^{xy} \left(u^\tau\right)^2 \left(u^x\right)^2 \left(u^y\right)^3 + 2(\partial_\tau \pi_Q^{xy})\tau \left(u^\tau\right)^2 \left(u^x\right)^2 \left(u^y\right)^3 \\
&- 2(\partial_\tau u^x)\pi_Q^{xy}\tau \left(u^x\right)^3 \left(u^y\right)^3 + (\partial_\tau u^y)\pi_Q^{yy}\tau \left(u^x\right)^3 \left(u^y\right)^3 \\
&- 2(\partial_\tau u^y)\pi_Q^{xy}\tau \left(u^y\right)^4 + (\partial_\tau u^y)\pi_Q^{xy}\tau \left(u^\tau\right)^2 \left(u^y\right)^4 \\
&- (\partial_\tau u^x)\pi_Q{}^\eta{}_\eta \tau \left(u^\tau\right)^2 \left(u^y\right)^4 + (\partial_x \Pi_Q)\tau \left(u^\tau\right)^3 \left(u^y\right)^4 \\
&- (\partial_x \pi_Q{}^\eta{}_\eta)\tau \left(u^\tau\right)^3 \left(u^y\right)^4 + (\partial_y \pi_Q^{xy})\tau \left(u^\tau\right)^3 \left(u^y\right)^4 \\
&+ (\partial_\tau u^x)\Pi_Q \tau \left(u^\tau\right)^4 \left(u^y\right)^4 - (\partial_\tau \pi_Q{}^\eta{}_\eta)\tau \left(u^\tau\right)^2 u^x \left(u^y\right)^4 \\
&+ 2(\partial_x u^x)\Pi_Q \tau \left(u^\tau\right)^3 u^x \left(u^y\right)^4 + (\partial_y u^y)\Pi_Q \tau \left(u^\tau\right)^3 u^x \left(u^y\right)^4 \\
&+ \Pi_Q \left(u^\tau\right)^4 u^x \left(u^y\right)^4 + (\partial_\tau \Pi_Q)\tau \left(u^\tau\right)^4 u^x \left(u^y\right)^4 \\
&- 2(\partial_\tau u^y)\pi_Q^{xy}\tau \left(u^x\right)^2 \left(u^y\right)^4 + (\partial_\tau u^x)\Pi_Q \tau \left(u^\tau\right)^2 \left(u^x\right)^2 \left(u^y\right)^4 \\
&+ (\partial_x \Pi_Q)\tau \left(u^\tau\right)^3 \left(u^x\right)^2 \left(u^y\right)^4 + \pi_Q^{xy} \left(u^\tau\right)^2 \left(u^y\right)^5 \\
&+ (\partial_\tau \pi_Q^{xy})\tau \left(u^\tau\right)^2 \left(u^y\right)^5 + (\partial_y u^x)\Pi_Q \tau \left(u^\tau\right)^3 \left(u^y\right)^5 \\
&- (\partial_\tau u^x)\pi_Q^{xy}\tau u^x \left(u^y\right)^5 + (\partial_\tau u^y)\Pi_Q \tau \left(u^\tau\right)^2 u^x \left(u^y\right)^5 \\
&+ (\partial_y \Pi_Q)\tau \left(u^\tau\right)^3 u^x \left(u^y\right)^5 - (\partial_\tau u^y)\pi_Q^{xy}\tau \left(u^y\right)^6 \Big)
\end{aligned}
$$

$$
\begin{aligned}
\nabla_\mu \delta M^{\mu y} = {} & (\partial_x \pi_Q^{xy}) + (\partial_y \pi_Q^{yy}) + (\partial_\tau u^y)\Pi_Q u^\tau + (\partial_x u^y)\Pi_Q u^x + (\partial_x u^x)\Pi_Q u^y + 2(\partial_y u^y)\Pi_Q u^y \\
& + (\partial_\tau \Pi_Q)u^\tau u^y + (\partial_x \Pi_Q)u^x u^y + \frac{(\partial_\tau \pi_Q^{xy})u^x + (\partial_\tau \pi_Q^{yy})u^y + (\partial_\tau u^x)\pi_Q^{xy} + (\partial_\tau u^y)\pi_Q^{yy}}{u^\tau} \\
& + \frac{\Pi_Q u^y((\partial_\tau u^x)u^x + (\partial_\tau u^y)u^y)}{u^\tau} - \frac{((\partial_\tau u^x)u^x + (\partial_\tau u^y)u^y)(\pi_Q^{xy}u^x + \pi_Q^{yy}u^y)}{\left(u^\tau\right)^3}
\end{aligned}
$$

$$+\frac{u^y\left(\Pi_Q\left(u^\tau\right)^2+\pi_Q^{yy}\right)+\pi_Q^{xy}u^x}{\tau u^\tau}+\left(\partial_y\Pi_Q\right)\left(\left(u^y\right)^2+1\right)$$

$$E_a^{\mathrm{S6}}=E_a^{\mathrm{S5}}+\nabla_\mu\delta M^{\mu a},\quad a=\tau,x,y,\qquad E_{\nu^i}^{\mathrm{S6}}=E_{\nu^i}+c_M\Delta^i{}_\lambda\nabla_\mu\delta M^{\mu\lambda}$$

Conformal:

$$P=a_{\mathrm{SB}}T^4,\quad P'=4a_{\mathrm{SB}}T^3,\quad P''=12a_{\mathrm{SB}}T^2,\quad \Pi=\zeta=0,\quad \text{drop the }\Pi\text{ row and column}$$

3.3 Result

The 1+1D charm systems and the 1+1D conformal two-way system in full; the generic 1+1D two-way system and every 2+1D system as the pattern of nonzero entries (•), rows and columns in the field order given, with their entries in the appendix.

3.4 1+1D cylindrical

3.4.1 S3, charm first moment on a background

`one_d_HQ_BG_consistent_matrix` in `src/Matrix/1d_viscous_HQ_cilindrical_BG.jl`, fields:

$$\phi=(\alpha,\nu^r)$$

$$\begin{aligned}
(A_t)_{11}&=(\partial_\alpha n)u^\tau\\
(A_t)_{12}&=\frac{u^r}{u^\tau}\\
(A_t)_{21}&=(\partial_\alpha n)D_s u^r u^\tau\\
(A_t)_{22}&=\frac{D_s h u^\tau}{T}\\
(A_r)_{11}&=(\partial_\alpha n)u^r\\
(A_r)_{12}&=1\\
(A_r)_{21}&=(\partial_\alpha n)D_s\left(u^\tau\right)^2\\
(A_r)_{22}&=\frac{D_s h u^r}{T}\\
S_1&=(\partial_r u^r)n+(\partial_T n)(\partial_r T)u^r+\frac{nu^r+\nu^r}{r}-\frac{(\partial_\tau u^r)\nu^r\left(u^r\right)^2}{\left(u^\tau\right)^3}+\frac{(\partial_\tau u^r)\nu^r}{u^\tau}+\frac{(\partial_\tau u^r)nu^r}{u^\tau}\\
&\quad+\frac{n\left(u^r\right)^2+n+\nu^r u^r}{\tau u^\tau}+(\partial_T n)(\partial_\tau T)u^\tau\\
S_2&=\nu^r\\
&\quad+\frac{1}{Tu^\tau}D_s\Big(\left(u^\tau\right)^2\left((\partial_T n)T+n\right)\left((\partial_r T)u^\tau+(\partial_\tau T)u^r\right)\\
&\quad+h'(T)\nu^r\left((\partial_r T)u^r u^\tau+(\partial_\tau T)\left(u^r\right)^2+(\partial_\tau T)\right)\Big)\\
&\quad+\frac{1}{rT\tau\left(u^\tau\right)^3}D_s h\Big(\nu^r\tau u^r\left(u^\tau\right)^3+nr\tau\left(u^\tau\right)^2\left((\partial_r u^r)u^r u^\tau+(\partial_\tau u^r)\left(u^r\right)^2+(\partial_\tau u^r)\right)\\
&\quad+\nu^r r\Big(1+\left(u^r\right)^4+2(\partial_r u^r)\tau u^\tau+(\partial_\tau u^r)\tau u^r\left(u^\tau\right)^2\\
&\quad+\left(u^r\right)^2\left((\partial_r u^r)\tau u^\tau+2\right)\Big)\Big)
\end{aligned}$$

3.4.2 S4, charm first and second moment on a background

`one_d_HQ_BG_m2_matrix` in `src/Matrix/1d_viscous_HQ_cilindrical_BG_m2.jl`, fields:

$$\phi=(\alpha,\nu^r,{\pi_Q}^\phi{}_\phi,{\pi_Q}^\eta{}_\eta,\Pi_Q)$$

$$
\begin{aligned}
(A_t)_{11} &= (\partial_\alpha n)u^\tau \\
(A_t)_{12} &= \frac{u^r}{u^\tau} \\
(A_t)_{21} &= (\partial_\alpha n)D_s u^r u^\tau \\
(A_t)_{22} &= \frac{D_s h u^\tau}{T} \\
(A_t)_{32} &= -\frac{\mathcal{B}D_s u^r}{3Tu^\tau} \\
(A_t)_{33} &= \frac{\mathcal{C}D_s u^\tau}{2T} \\
(A_t)_{42} &= -\frac{\mathcal{B}D_s u^r}{3Tu^\tau} \\
(A_t)_{44} &= \frac{\mathcal{C}D_s u^\tau}{2T} \\
(A_t)_{51} &= \frac{(\partial_\alpha I_{31})D_s u^\tau}{2T} \\
(A_t)_{52} &= \frac{5\mathcal{B}D_s u^r}{6Tu^\tau} \\
(A_t)_{55} &= \frac{\mathcal{C}D_s u^\tau}{2T} \\
(A_r)_{11} &= (\partial_\alpha n)u^r \\
(A_r)_{12} &= 1 \\
(A_r)_{21} &= (\partial_\alpha n)D_s \left(u^\tau\right)^2 \\
(A_r)_{22} &= \frac{D_s h u^r}{T} \\
(A_r)_{32} &= -\frac{\mathcal{B}D_s}{3T} \\
(A_r)_{33} &= \frac{\mathcal{C}D_s u^r}{2T} \\
(A_r)_{42} &= -\frac{\mathcal{B}D_s}{3T} \\
(A_r)_{44} &= \frac{\mathcal{C}D_s u^r}{2T} \\
(A_r)_{51} &= \frac{(\partial_\alpha I_{31})D_s u^r}{2T} \\
(A_r)_{52} &= \frac{5\mathcal{B}D_s}{6T} \\
(A_r)_{55} &= \frac{\mathcal{C}D_s u^r}{2T}
\end{aligned}
$$

$$
\begin{aligned}
S_1 = {} & (\partial_r u^r)n + (\partial_T n)(\partial_r T)u^r + \frac{nu^r + \nu^r}{r} - \frac{(\partial_\tau u^r)\nu^r \left(u^r\right)^2}{\left(u^\tau\right)^3} + \frac{(\partial_\tau u^r)\nu^r}{u^\tau} + \frac{(\partial_\tau u^r)nu^r}{u^\tau} \\
& + \frac{n\left(u^r\right)^2 + n + \nu^r u^r}{\tau u^\tau} + (\partial_T n)(\partial_\tau T)u^\tau \\
S_2 = {} & \nu^r \\
& + \frac{1}{Tu^\tau} D_s \Big(\left(u^\tau\right)^2 ((\partial_T n)T + n)((\partial_r T)u^\tau + (\partial_\tau T)u^r) \\
& + h'(T)\nu^r \left((\partial_r T)u^r u^\tau + (\partial_\tau T)\left(u^r\right)^2 + (\partial_\tau T) \right) \Big) \\
& + \frac{1}{rT\tau\left(u^\tau\right)^3} D_s h \Big(\nu^r \tau u^r \left(u^\tau\right)^3 + nr\tau\left(u^\tau\right)^2 \left((\partial_r u^r)u^r u^\tau + (\partial_\tau u^r)\left(u^r\right)^2 + (\partial_\tau u^r) \right) \Big) \\
& + \nu^r r \Big(1 + \left(u^r\right)^4 + 2(\partial_r u^r)\tau u^\tau + (\partial_\tau u^r)\tau u^r \left(u^\tau\right)^2 \\
& + \left(u^r\right)^2 ((\partial_r u^r)\tau u^\tau + 2) \Big) \Big) \\
S_3 = {} & \pi_Q{}^\phi{}_\phi \\
& - \frac{1}{6rT\tau\left(u^\tau\right)^3} D_s \Big(2\mathcal{A}(\partial_\tau u^r)\nu^r r\tau + 2\mathcal{B}(\partial_\tau u^r)\nu^r r\tau - 3\mathcal{C}'(T)(\partial_\tau T)\pi_Q{}^\phi{}_\phi r\tau
\end{aligned}
$$

$$
\begin{aligned}
&+ 2\mathcal{B}\nu^r r u^r + 2\mathcal{B}'(T)(\partial_\tau T)\nu^r r\tau u^r + 2\mathcal{A}(\partial_\tau u^r)\nu^r r\tau\left(u^r\right)^2 \\
&- 6\mathcal{C}'(T)(\partial_\tau T)\pi_Q{}^\phi{}_\phi r\tau\left(u^r\right)^2 + 2\mathcal{B}\nu^r r\left(u^r\right)^3 \\
&+ 2\mathcal{B}'(T)(\partial_\tau T)\nu^r r\tau\left(u^r\right)^3 - 3\mathcal{C}'(T)(\partial_\tau T)\pi_Q{}^\phi{}_\phi r\tau\left(u^r\right)^4 - 4\mathcal{B}\nu^r\tau u^\tau \\
&+ 2\mathcal{B}'(T)(\partial_\tau T)\nu^r r\tau u^\tau + 2\mathcal{A}(\partial_r u^r)\nu^r r\tau u^r u^\tau \\
&- 3\mathcal{C}'(T)(\partial_r T)\pi_Q{}^\phi{}_\phi r\tau u^r u^\tau - 4\mathcal{B}\nu^r\tau\left(u^r\right)^2 u^\tau \\
&+ 2\mathcal{B}'(T)(\partial_r T)\nu^r r\tau\left(u^r\right)^2 u^\tau - 3\mathcal{C}'(T)(\partial_r T)\pi_Q{}^\phi{}_\phi r\tau\left(u^r\right)^3 u^\tau \\
&+ 2I_{31}\left(u^\tau\right)^2\Big(r\Big((\partial_r u^r)\tau u^\tau + (\partial_\tau u^r)\tau u^r + \left(u^r\right)^2 + 1\Big) - 2\tau u^r u^\tau\Big) \\
&+ \mathcal{C}\left(u^\tau\right)^2\Big(-4\Pi_Q\tau u^r u^\tau - 7\pi_Q{}^\phi{}_\phi\tau u^r u^\tau \\
&- 2\pi_Q{}^\eta{}_\eta r\Big((\partial_r u^r)\tau u^\tau + (\partial_\tau u^r)\tau u^r - \left(u^r\right)^2 - 1\Big) \\
&+ 2\Pi_Q r\Big((\partial_r u^r)\tau u^\tau + (\partial_\tau u^r)\tau u^r + \left(u^r\right)^2 + 1\Big) \\
&- \pi_Q{}^\phi{}_\phi r\Big(5(\partial_r u^r)\tau u^\tau + 5(\partial_\tau u^r)\tau u^r + 3\left(u^r\right)^2 + 3\Big)\Big)\Big) \\
S_4 = \pi_Q{}^\eta{}_\eta& \\
&+ \frac{1}{6rT\tau\left(u^\tau\right)^3} D_s\Big(-2\mathcal{A}(\partial_\tau u^r)\nu^r r\tau - 2\mathcal{B}(\partial_\tau u^r)\nu^r r\tau + 3\mathcal{C}'(T)(\partial_\tau T)\pi_Q{}^\eta{}_\eta r\tau \\
&+ 4\mathcal{B}\nu^r r u^r - 2\mathcal{B}'(T)(\partial_\tau T)\nu^r r\tau u^r - 2\mathcal{A}(\partial_\tau u^r)\nu^r r\tau\left(u^r\right)^2 \\
&+ 6\mathcal{C}'(T)(\partial_\tau T)\pi_Q{}^\eta{}_\eta r\tau\left(u^r\right)^2 + 4\mathcal{B}\nu^r r\left(u^r\right)^3 \\
&- 2\mathcal{B}'(T)(\partial_\tau T)\nu^r r\tau\left(u^r\right)^3 + 3\mathcal{C}'(T)(\partial_\tau T)\pi_Q{}^\eta{}_\eta r\tau\left(u^r\right)^4 - 2\mathcal{B}\nu^r\tau u^\tau \\
&- 2\mathcal{B}'(T)(\partial_r T)\nu^r r\tau u^\tau - 2\mathcal{A}(\partial_r u^r)\nu^r r\tau u^r u^\tau \\
&+ 3\mathcal{C}'(T)(\partial_r T)\pi_Q{}^\eta{}_\eta r\tau u^r u^\tau - 2\mathcal{B}\nu^r\tau\left(u^r\right)^2 u^\tau \\
&- 2\mathcal{B}'(T)(\partial_r T)\nu^r r\tau\left(u^r\right)^2 u^\tau + 3\mathcal{C}'(T)(\partial_r T)\pi_Q{}^\eta{}_\eta r\tau\left(u^r\right)^3 u^\tau \\
&- 2I_{31}\left(u^\tau\right)^2\Big(r\Big((\partial_r u^r)\tau u^\tau + (\partial_\tau u^r)\tau u^r - 2\left(u^r\right)^2 - 2\Big) + \tau u^r u^\tau\Big) \\
&+ \mathcal{C}\left(u^\tau\right)^2\Big(2\pi_Q{}^\phi{}_\phi\tau(u^\tau((\partial_r u^r)r - u^r) + (\partial_\tau u^r)ru^r) \\
&- 2\Pi_Q\Big(\tau u^r u^\tau \\
&+ r\Big((\partial_r u^r)\tau u^\tau + (\partial_\tau u^r)\tau u^r - 2\left(u^r\right)^2 - 2\Big)\Big) \\
&+ \pi_Q{}^\eta{}_\eta\Big(3\tau u^r u^\tau \\
&+ r\Big(5(\partial_r u^r)\tau u^\tau + 5(\partial_\tau u^r)\tau u^r + 7\left(u^r\right)^2 + 7\Big)\Big)\Big)\Big) \\
S_5 = \Pi_Q& \\
&+ \frac{1}{6rT\tau\left(u^\tau\right)^3} D_s\Big(3(\partial_T I_{31})(\partial_\tau T)r\tau + 2\mathcal{A}(\partial_\tau u^r)\nu^r r\tau + 5\mathcal{B}(\partial_\tau u^r)\nu^r r\tau \\
&+ 3\mathcal{C}'(T)(\partial_\tau T)\Pi_Q r\tau + 5\mathcal{B}\nu^r r u^r + 5\mathcal{B}'(T)(\partial_\tau T)\nu^r r\tau u^r \\
&+ 6(\partial_T I_{31})(\partial_\tau T)r\tau\left(u^r\right)^2 + 2\mathcal{A}(\partial_\tau u^r)\nu^r r\tau\left(u^r\right)^2 \\
&+ 6\mathcal{C}'(T)(\partial_\tau T)\Pi_Q r\tau\left(u^r\right)^2 + 5\mathcal{B}\nu^r r\left(u^r\right)^3 + 5\mathcal{B}'(T)(\partial_\tau T)\nu^r r\tau\left(u^r\right)^3 \\
&+ 3(\partial_T I_{31})(\partial_\tau T)r\tau\left(u^r\right)^4 + 3\mathcal{C}'(T)(\partial_\tau T)\Pi_Q r\tau\left(u^r\right)^4 + 5\mathcal{B}\nu^r\tau u^\tau \\
&+ 5\mathcal{B}'(T)(\partial_r T)\nu^r r\tau u^\tau + 3(\partial_T I_{31})(\partial_r T)r\tau u^r u^\tau \\
&+ 2\mathcal{A}(\partial_r u^r)\nu^r r\tau u^r u^\tau + 3\mathcal{C}'(T)(\partial_r T)\Pi_Q r\tau u^r u^\tau + 5\mathcal{B}\nu^r\tau\left(u^r\right)^2 u^\tau \\
&+ 5\mathcal{B}'(T)(\partial_r T)\nu^r r\tau\left(u^r\right)^2 u^\tau + 3(\partial_T I_{31})(\partial_r T)r\tau\left(u^r\right)^3 u^\tau \\
&+ 3\mathcal{C}'(T)(\partial_r T)\Pi_Q r\tau\left(u^r\right)^3 u^\tau \\
&+ 5I_{31}\left(u^\tau\right)^2\Big(r\Big((\partial_r u^r)\tau u^\tau + (\partial_\tau u^r)\tau u^r + \left(u^r\right)^2 + 1\Big) + \tau u^r u^\tau\Big) \\
&+ \mathcal{C}\left(u^\tau\right)^2\Big(5\Pi_Q\Big(r\Big((\partial_r u^r)\tau u^\tau + (\partial_\tau u^r)\tau u^r + \left(u^r\right)^2 + 1\Big) + \tau u^r u^\tau\Big) \\
&- 2\Big(\pi_Q{}^\eta{}_\eta r\Big((\partial_r u^r)\tau u^\tau + (\partial_\tau u^r)\tau u^r - \left(u^r\right)^2 - 1\Big)
\end{aligned}
$$

$$+ \pi_Q{}^{\phi}{}_{\phi}\tau(u^\tau((\partial_r u^r)r - u^r) + (\partial_\tau u^r)ru^r)\Big)\Big)\Big)$$

3.4.3 S5, two-way

`one_d_two_way_matrix` in `src/Matrix/1d_two_way_cilindrical.jl`, fields:

$$\phi = (T, u^r, \pi^{\phi}{}_{\phi}, \pi^{\eta}{}_{\eta}, \Pi, \alpha, \nu^r, \pi_Q{}^{\phi}{}_{\phi}, \pi_Q{}^{\eta}{}_{\eta}, \Pi_Q)$$

$$A_t = \begin{pmatrix}
\bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot \\
\bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot \\
\cdot & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \bullet & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \bullet & \cdot & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot \\
\bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot & \cdot \\
\bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot & \cdot \\
\bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot \\
\bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \cdot & \bullet & \cdot \\
\bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot & \bullet
\end{pmatrix}, \qquad A_r = \begin{pmatrix}
\bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot \\
\bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot \\
\cdot & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \bullet & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \bullet & \cdot & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot \\
\bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot & \cdot \\
\bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot & \cdot \\
\bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot \\
\bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \cdot & \bullet & \cdot \\
\bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot & \bullet
\end{pmatrix}$$

$$S = \begin{pmatrix}
\bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet
\end{pmatrix}$$

3.4.4 S5, two-way, conformal

`one_d_two_way_matrix_conformal` in `src/Matrix/1d_two_way_cilindrical.jl`, fields:

$$\phi = (T, u^r, \pi^{\phi}{}_{\phi}, \pi^{\eta}{}_{\eta}, \alpha, \nu^r, \pi_Q{}^{\phi}{}_{\phi}, \pi_Q{}^{\eta}{}_{\eta}, \Pi_Q)$$

$$\begin{aligned}
(A_t)_{11} &= 2h'(T)\nu^r u^r + 4a_{\mathrm{SB}}T^3\left(4\left(u^r\right)^2 + 3\right) + n\left(h'(T)\left(u^r\right)^2 + h'(T) - 1\right) \\
&\quad + (\partial_T n)\left(h\left(u^r\right)^2 + h - T\right) \\
(A_t)_{12} &= 2h(nu^r + \nu^r) - 2u^r\left(-4a_{\mathrm{SB}}T^4 + \pi^{\eta}{}_{\eta} + \pi^{\phi}{}_{\phi}\right) \\
(A_t)_{13} &= -\left(u^r\right)^2 \\
(A_t)_{14} &= -\left(u^r\right)^2 \\
(A_t)_{15} &= (\partial_\alpha n)\left(h\left(u^r\right)^2 + h - T\right) \\
(A_t)_{16} &= 2hu^r \\
(A_t)_{21} &= \frac{u^r\left(u^\tau\right)^2\left(16a_{\mathrm{SB}}T^3 + (\partial_T n)h\right) + h'(T)\left(nu^r\left(u^\tau\right)^2 + 2\nu^r\left(u^r\right)^2 + \nu^r\right)}{u^\tau} \\
(A_t)_{22} &= \frac{1}{\left(u^\tau\right)^3}\Big(-\left(2\left(u^r\right)^4 + 3\left(u^r\right)^2 + 1\right)\left(-4a_{\mathrm{SB}}T^4 + \pi^{\eta}{}_{\eta} + \pi^{\phi}{}_{\phi}\right) \\
&\quad + h\left(2n\left(u^r\right)^4 + 3n\left(u^r\right)^2 + n + 2\nu^r\left(u^r\right)^3 + 3\nu^r u^r\right)\Big) \\
(A_t)_{23} &= -u^r u^\tau \\
(A_t)_{24} &= -u^r u^\tau \\
(A_t)_{25} &= (\partial_\alpha n)hu^r u^\tau \\
(A_t)_{26} &= \frac{h\left(2\left(u^r\right)^2 + 1\right)}{u^\tau}
\end{aligned}$$

$$(A_t)_{32} = \frac{u^r(3\delta_{\pi\pi}\pi^\phi{}_\phi - 2\eta)}{3r^2u^\tau}$$
$$(A_t)_{33} = \frac{\tau_\pi u^\tau}{r^2}$$
$$(A_t)_{42} = \frac{u^r(3\delta_{\pi\pi}\pi^\eta{}_\eta - 2\eta)}{3\tau^2u^\tau}$$
$$(A_t)_{44} = \frac{\tau_\pi u^\tau}{\tau^2}$$
$$(A_t)_{51} = (\partial_T n)u^\tau$$
$$(A_t)_{52} = \frac{nu^r\left(u^\tau\right)^2 + \nu^r}{\left(u^\tau\right)^3}$$
$$(A_t)_{55} = (\partial_\alpha n)u^\tau$$
$$(A_t)_{56} = \frac{u^r}{u^\tau}$$
$$(A_t)_{61} = \frac{D_s u^\tau(h'(T)\nu^r + u^r((\partial_T n)T + n))}{T}$$
$$(A_t)_{62} = \frac{D_s h\left(n\left(u^r\right)^2 + n + \nu^r u^r\right)}{Tu^\tau}$$
$$(A_t)_{65} = (\partial_\alpha n)D_s u^r u^\tau$$
$$(A_t)_{66} = \frac{D_s h u^\tau}{T}$$
$$(A_t)_{71} = \frac{D_s\left(3\mathcal{C}'(T)\pi_Q{}^\phi{}_\phi\left(u^\tau\right)^2 - 2\mathcal{B}'(T)\nu^r u^r\right)}{6Tu^\tau}$$
$$(A_t)_{72} = -\frac{D_s\left(2\mathcal{A}\nu^r\left(u^\tau\right)^2 + 2\mathcal{B}\nu^r + u^r\left(u^\tau\right)^2(2\mathcal{C}\Pi_Q - 2\mathcal{C}\pi_Q{}^\eta{}_\eta - 5\mathcal{C}\pi_Q{}^\phi{}_\phi + 2I_{31})\right)}{6T\left(u^\tau\right)^3}$$
$$(A_t)_{76} = -\frac{\mathcal{B}D_s u^r}{3Tu^\tau}$$
$$(A_t)_{77} = \frac{\mathcal{C}D_s u^\tau}{2T}$$
$$(A_t)_{81} = \frac{D_s\left(3\mathcal{C}'(T)\pi_Q{}^\eta{}_\eta\left(u^\tau\right)^2 - 2\mathcal{B}'(T)\nu^r u^r\right)}{6Tu^\tau}$$
$$(A_t)_{82} = -\frac{D_s\left(2\mathcal{A}\nu^r\left(u^\tau\right)^2 + 2\mathcal{B}\nu^r + u^r\left(u^\tau\right)^2(2\mathcal{C}\Pi_Q - 5\mathcal{C}\pi_Q{}^\eta{}_\eta - 2\mathcal{C}\pi_Q{}^\phi{}_\phi + 2I_{31})\right)}{6T\left(u^\tau\right)^3}$$
$$(A_t)_{86} = -\frac{\mathcal{B}D_s u^r}{3Tu^\tau}$$
$$(A_t)_{88} = \frac{\mathcal{C}D_s u^\tau}{2T}$$
$$(A_t)_{91} = \frac{D_s\left(5\mathcal{B}'(T)\nu^r u^r + 3\mathcal{C}'(T)\Pi_Q\left(u^\tau\right)^2 + 3(\partial_T I_{31})\left(u^\tau\right)^2\right)}{6Tu^\tau}$$
$$(A_t)_{92} = \frac{D_s\left(2\mathcal{A}\nu^r\left(u^\tau\right)^2 + 5\mathcal{B}\nu^r + u^r\left(u^\tau\right)^2(5\mathcal{C}\Pi_Q - 2\mathcal{C}(\pi_Q{}^\eta{}_\eta + \pi_Q{}^\phi{}_\phi) + 5I_{31})\right)}{6T\left(u^\tau\right)^3}$$
$$(A_t)_{95} = \frac{(\partial_\alpha I_{31})D_s u^\tau}{2T}$$
$$(A_t)_{96} = \frac{5\mathcal{B}D_s u^r}{6Tu^\tau}$$
$$(A_t)_{99} = \frac{\mathcal{C}D_s u^\tau}{2T}$$
$$(A_r)_{11} = \frac{u^r\left(u^\tau\right)^2\left(16a_{\mathrm{SB}}T^3 + (\partial_T n)h\right) + h'(T)\left(nu^r\left(u^\tau\right)^2 + 2\nu^r\left(u^\tau\right)^2 + \nu^r\right)}{u^\tau}$$
$$(A_r)_{12} = \frac{1}{\left(u^\tau\right)^3}\Big(-\left(2\left(u^r\right)^4 + 3\left(u^r\right)^2 + 1\right)\left(-4a_{\mathrm{SB}}T^4 + \pi^\eta{}_\eta + \pi^\phi{}_\phi\right)$$

$$+ h\left(2n\left(u^r\right)^4 + 3n\left(u^r\right)^2 + n + 2\nu^r\left(u^r\right)^3 + 3\nu^r u^r\right)\Big)$$

$$(A_r)_{13} = -u^r u^\tau$$

$$(A_r)_{14} = -u^r u^\tau$$

$$(A_r)_{15} = (\partial_\alpha n) h u^r u^\tau$$

$$(A_r)_{16} = \frac{h\left(2\left(u^r\right)^2 + 1\right)}{u^\tau}$$

$$(A_r)_{21} = 16 a_{\mathrm{SB}} T^3 \left(u^r\right)^2 + 4 a_{\mathrm{SB}} T^3 + h'(T) n \left(u^r\right)^2 + 2h'(T)\nu^r u^r + (\partial_T n) h \left(u^r\right)^2 + (\partial_T n) T + n$$

$$(A_r)_{22} = 2h(n u^r + \nu^r) - 2u^r\left(-4 a_{\mathrm{SB}} T^4 + \pi^\eta{}_\eta + \pi^\phi{}_\phi\right)$$

$$(A_r)_{23} = -\left(u^r\right)^2 - 1$$

$$(A_r)_{24} = -\left(u^r\right)^2 - 1$$

$$(A_r)_{25} = (\partial_\alpha n)\left(h\left(u^r\right)^2 + T\right)$$

$$(A_r)_{26} = 2 h u^r$$

$$(A_r)_{32} = \frac{3\delta_{\pi\pi}\pi^\phi{}_\phi - 2\eta}{3r^2}$$

$$(A_r)_{33} = \frac{\tau_\pi u^r}{r^2}$$

$$(A_r)_{42} = \frac{3\delta_{\pi\pi}\pi^\eta{}_\eta - 2\eta}{3\tau^2}$$

$$(A_r)_{44} = \frac{\tau_\pi u^r}{\tau^2}$$

$$(A_r)_{51} = (\partial_T n) u^r$$

$$(A_r)_{52} = n$$

$$(A_r)_{55} = (\partial_\alpha n) u^r$$

$$(A_r)_{56} = 1$$

$$(A_r)_{61} = \frac{D_s\left(h'(T)\nu^r u^r + (\partial_T n) T \left(u^\tau\right)^2 + n\left(u^\tau\right)^2\right)}{T}$$

$$(A_r)_{62} = \frac{D_s h\left(n u^r \left(u^\tau\right)^2 + \nu^r\left(\left(u^r\right)^2 + 2\right)\right)}{T\left(u^\tau\right)^2}$$

$$(A_r)_{65} = (\partial_\alpha n) D_s \left(u^\tau\right)^2$$

$$(A_r)_{66} = \frac{D_s h u^r}{T}$$

$$(A_r)_{71} = -\frac{D_s(2\mathcal{B}'(T)\nu^r - 3\mathcal{C}'(T)\pi_Q{}^\phi{}_\phi u^r)}{6T}$$

$$(A_r)_{72} = -\frac{D_s\left(2\mathcal{A}\nu^r u^r + \mathcal{C}\left(u^\tau\right)^2\left(2\Pi_Q - 2\pi_Q{}^\eta{}_\eta - 5\pi_Q{}^\phi{}_\phi\right) + 2I_{31}\left(u^\tau\right)^2\right)}{6T\left(u^\tau\right)^2}$$

$$(A_r)_{76} = -\frac{\mathcal{B} D_s}{3T}$$

$$(A_r)_{77} = \frac{\mathcal{C} D_s u^r}{2T}$$

$$(A_r)_{81} = -\frac{D_s(2\mathcal{B}'(T)\nu^r - 3\mathcal{C}'(T)\pi_Q{}^\eta{}_\eta u^r)}{6T}$$

$$(A_r)_{82} = -\frac{D_s\left(2\mathcal{A}\nu^r u^r + \mathcal{C}\left(u^\tau\right)^2\left(2\Pi_Q - 5\pi_Q{}^\eta{}_\eta - 2\pi_Q{}^\phi{}_\phi\right) + 2I_{31}\left(u^\tau\right)^2\right)}{6T\left(u^\tau\right)^2}$$

$$(A_r)_{86} = -\frac{\mathcal{B} D_s}{3T}$$

$$(A_r)_{88} = \frac{\mathcal{C} D_s u^r}{2T}$$

$$(A_r)_{91} = \frac{D_s(5\mathcal{B}'(T)\nu^r + 3u^r(\mathcal{C}'(T)\Pi_Q + (\partial_T I_{31})))}{6T}$$

$$
\begin{aligned}
(A_r)_{92} &= \frac{D_s\left(2\mathcal{A}\nu^r u^r + \mathcal{C}\left(u^\tau\right)^2\left(5\Pi_Q - 2(\pi_Q{}^\eta{}_\eta + \pi_Q{}^\phi{}_\phi)\right) + 5I_{31}\left(u^\tau\right)^2\right)}{6T\left(u^\tau\right)^2}\\
(A_r)_{95} &= \frac{(\partial_\alpha I_{31})D_s u^r}{2T}\\
(A_r)_{96} &= \frac{5\mathcal{B}D_s}{6T}\\
(A_r)_{99} &= \frac{\mathcal{C}D_s u^r}{2T}\\
S_1 &= \frac{1}{r\tau u^\tau}\Big(-\pi^\phi{}_\phi \tau u^r + 4a_{\mathrm{SB}}T^4\tau u^r - \pi^\phi{}_\phi \tau\left(u^r\right)^3 + 4a_{\mathrm{SB}}T^4\tau\left(u^r\right)^3 + 4a_{\mathrm{SB}}rT^4u^\tau\\
&\quad - \pi^\phi{}_\phi r\left(u^r\right)^2 u^\tau + 4a_{\mathrm{SB}}rT^4\left(u^r\right)^2 u^\tau - \pi^\eta{}_\eta\left(r\left(\left(u^r\right)^2 - 1\right)u^\tau + \tau\left(\left(u^r\right)^3 + u^r\right)\right)\\
&\quad + h\left(n\left(u^\tau\right)^2\left(ru^\tau + \tau u^r\right) + \nu^r\left(2ru^r u^\tau + 2\tau\left(u^r\right)^2 + \tau\right)\right)\Big)\\
S_2 &= \frac{1}{r\tau u^\tau}\Big(-\pi^\phi{}_\phi r u^r + 4a_{\mathrm{SB}}rT^4u^r - \pi^\phi{}_\phi r\left(u^r\right)^3 + 4a_{\mathrm{SB}}rT^4\left(u^r\right)^3 - 2\pi^\phi{}_\phi\tau u^\tau - \pi^\phi{}_\phi\tau\left(u^r\right)^2 u^\tau\\
&\quad + 4a_{\mathrm{SB}}T^4\tau\left(u^r\right)^2 u^\tau - \pi^\eta{}_\eta\left(u^\tau\right)^2\left(ru^r + \tau u^\tau\right) + hnu^r\left(r\left(u^r\right)^2 + r + \tau u^r u^\tau\right)\\
&\quad + h\nu^r\left(2r\left(u^r\right)^2 + r + 2\tau u^r u^\tau\right)\Big)\\
S_3 &= \frac{3\pi^\phi{}_\phi(r(\delta_{\pi\pi}u^\tau + \tau) + \delta_{\pi\pi}\tau u^r) - 2ru^\tau\eta + 4\tau u^r\eta}{3r^3\tau}\\
S_4 &= \frac{3\pi^\eta{}_\eta(r(\delta_{\pi\pi}u^\tau + \tau) + \delta_{\pi\pi}\tau u^r) + 4ru^\tau\eta - 2\tau u^r\eta}{3r\tau^3}\\
S_5 &= \frac{nu^r + \nu^r}{r} + \frac{n\left(u^r\right)^2 + n + \nu^r u^r}{\tau u^\tau}\\
S_6 &= \frac{D_s h\nu^r\left(r\left(u^r\right)^2 + r + \tau u^r u^\tau\right)}{rT\tau u^\tau} + \nu^r\\
S_7 &= \pi_Q{}^\phi{}_\phi\\
&\quad + \frac{1}{6rT\tau u^\tau}D_s\Big(-2\mathcal{B}\nu^r(ru^r - 2\tau u^\tau) - 2I_{31}\left(r\left(u^r\right)^2 + r - 2\tau u^r u^\tau\right)\\
&\quad + \mathcal{C}\Big(7\pi_Q{}^\phi{}_\phi\tau u^r u^\tau - 2\pi_Q{}^\eta{}_\eta r\left(u^\tau\right)^2 + 3\pi_Q{}^\phi{}_\phi r\left(u^\tau\right)^2\\
&\quad - 2\Pi_Q\left(r\left(u^r\right)^2 + r - 2\tau u^r u^\tau\right)\Big)\Big)\\
S_8 &= \pi_Q{}^\eta{}_\eta\\
&\quad + \frac{1}{6rT\tau u^\tau}D_s\Big(-2I_{31}\tau u^r u^\tau + 4I_{31}r\left(u^\tau\right)^2 + 2\mathcal{B}\nu^r(2ru^r - \tau u^\tau)\\
&\quad + \mathcal{C}\Big(-2\Pi_Q\tau u^r u^\tau + 3\pi_Q{}^\eta{}_\eta\tau u^r u^\tau - 2\pi_Q{}^\phi{}_\phi\tau u^r u^\tau + 4\Pi_Q r\left(u^\tau\right)^2\\
&\quad + 7\pi_Q{}^\eta{}_\eta r\left(u^\tau\right)^2\Big)\Big)\\
S_9 &= \Pi_Q\\
&\quad + \frac{1}{6rT\tau u^\tau}D_s\Big(5\mathcal{B}\nu^r(ru^r + \tau u^\tau) + 5I_{31}\left(r\left(u^r\right)^2 + r + \tau u^r u^\tau\right)\\
&\quad + \mathcal{C}\left(5\Pi_Q\left(r\left(u^r\right)^2 + r + \tau u^r u^\tau\right) + 2\pi_Q{}^\eta{}_\eta r\left(u^\tau\right)^2 + 2\pi_Q{}^\phi{}_\phi\tau u^r u^\tau\right)\Big)
\end{aligned}
$$

3.4.5 S6, fully coupled

`one_d_two_way_cM_matrix` in `src/Matrix/1d_two_way_cM_cilindrical.jl`, fields:

$$\phi = (T, u^r, \pi^\phi{}_\phi, \pi^\eta{}_\eta, \Pi, \alpha, \nu^r, \pi_Q{}^\phi{}_\phi, \pi_Q{}^\eta{}_\eta, \Pi_Q)$$

$$A_t = \begin{pmatrix} \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet \\ \cdot & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \bullet & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \bullet & \cdot & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot \\ \bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot \\ \bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot \\ \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \cdot & \bullet \\ \bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot \end{pmatrix}, \qquad A_r = \begin{pmatrix} \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet \\ \cdot & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \bullet & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \bullet & \cdot & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot \\ \bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot \\ \bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot \\ \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \cdot & \bullet \\ \bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot \end{pmatrix}$$

$$S = \begin{pmatrix} \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \end{pmatrix}$$

3.4.6 S6, fully coupled, conformal

`one_d_two_way_cM_matrix_conformal` in `src/Matrix/1d_two_way_cM_cilindrical.jl`, fields:

$$\phi = (T, u^r, \pi^\phi{}_\phi, \pi^\eta{}_\eta, \alpha, \nu^r, \pi_Q{}^\phi{}_\phi, \pi_Q{}^\eta{}_\eta, \Pi_Q)$$

$$\begin{aligned}
(A_t)_{11} &= 2h'(T)\nu^r u^r + 4a_{\mathrm{SB}}T^3\left(4\left(u^r\right)^2+3\right) + n\left(h'(T)\left(u^r\right)^2 + h'(T) - 1\right) \\
&\quad + (\partial_T n)\left(h\left(u^r\right)^2 + h - T\right) \\
(A_t)_{12} &= 2\left(u^r\left(4a_{\mathrm{SB}}T^4 + \Pi_Q - \pi^\eta{}_\eta - \pi_Q{}^\eta{}_\eta - \pi^\phi{}_\phi - \pi_Q{}^\phi{}_\phi\right) + h(nu^r + \nu^r)\right) \\
(A_t)_{13} &= -\left(u^r\right)^2 \\
(A_t)_{14} &= -\left(u^r\right)^2 \\
(A_t)_{15} &= (\partial_\alpha n)\left(h\left(u^r\right)^2 + h - T\right) \\
(A_t)_{16} &= 2hu^r \\
(A_t)_{17} &= -\left(u^r\right)^2 \\
(A_t)_{18} &= -\left(u^r\right)^2 \\
(A_t)_{19} &= \left(u^r\right)^2 \\
(A_t)_{21} &= \frac{u^r\left(u^\tau\right)^2\left(16a_{\mathrm{SB}}T^3 + (\partial_T n)h\right) + h'(T)\left(nu^r\left(u^\tau\right)^2 + 2\nu^r\left(u^r\right)^2 + \nu^r\right)}{u^\tau} \\
(A_t)_{22} &= \frac{1}{\left(u^\tau\right)^3}\Big(\left(2\left(u^r\right)^4 + 3\left(u^r\right)^2 + 1\right)\left(4a_{\mathrm{SB}}T^4 + \Pi_Q - \pi^\eta{}_\eta - \pi_Q{}^\eta{}_\eta - \pi^\phi{}_\phi - \pi_Q{}^\phi{}_\phi\right) \\
&\quad + h\left(2n\left(u^r\right)^4 + 3n\left(u^r\right)^2 + n + 2\nu^r\left(u^r\right)^3 + 3\nu^r u^r\right)\Big) \\
(A_t)_{23} &= -u^r u^\tau \\
(A_t)_{24} &= -u^r u^\tau \\
(A_t)_{25} &= (\partial_\alpha n)hu^r u^\tau \\
(A_t)_{26} &= \frac{h\left(2\left(u^r\right)^2 + 1\right)}{u^\tau} \\
(A_t)_{27} &= -u^r u^\tau \\
(A_t)_{28} &= -u^r u^\tau \\
(A_t)_{29} &= u^r u^\tau
\end{aligned}$$

$$(A_t)_{32} = \frac{u^r(3\delta_{\pi\pi}\pi^{\phi}{}_{\phi} - 2\eta)}{3r^2u^\tau}$$
$$(A_t)_{33} = \frac{\tau_\pi u^\tau}{r^2}$$
$$(A_t)_{42} = \frac{u^r(3\delta_{\pi\pi}\pi^{\eta}{}_{\eta} - 2\eta)}{3\tau^2u^\tau}$$
$$(A_t)_{44} = \frac{\tau_\pi u^\tau}{\tau^2}$$
$$(A_t)_{51} = (\partial_T n)u^\tau$$
$$(A_t)_{52} = \frac{nu^r\left(u^\tau\right)^2 + \nu^r}{\left(u^\tau\right)^3}$$
$$(A_t)_{55} = (\partial_\alpha n)u^\tau$$
$$(A_t)_{56} = \frac{u^r}{u^\tau}$$
$$(A_t)_{61} = \frac{D_s u^\tau(h'(T)\nu^r + u^r((\partial_T n)T + n))}{T}$$
$$(A_t)_{62} = \frac{D_s\left(h\left(n\left(u^r\right)^2 + n + \nu^r u^r\right) + \left(u^\tau\right)^2\left(\Pi_Q - \pi_Q{}^{\eta}{}_{\eta} - \pi_Q{}^{\phi}{}_{\phi}\right)\right)}{Tu^\tau}$$
$$(A_t)_{65} = (\partial_\alpha n)D_s u^r u^\tau$$
$$(A_t)_{66} = \frac{D_s h u^\tau}{T}$$
$$(A_t)_{67} = -\frac{D_s u^r u^\tau}{T}$$
$$(A_t)_{68} = -\frac{D_s u^r u^\tau}{T}$$
$$(A_t)_{69} = \frac{D_s u^r u^\tau}{T}$$
$$(A_t)_{71} = \frac{D_s\left(3\mathcal{C}'(T)\pi_Q{}^{\phi}{}_{\phi}\left(u^\tau\right)^2 - 2\mathcal{B}'(T)\nu^r u^r\right)}{6Tu^\tau}$$
$$(A_t)_{72} = -\frac{D_s\left(2\mathcal{A}\nu^r\left(u^\tau\right)^2 + 2\mathcal{B}\nu^r + u^r\left(u^\tau\right)^2\left(2\mathcal{C}\Pi_Q - 2\mathcal{C}\pi_Q{}^{\eta}{}_{\eta} - 5\mathcal{C}\pi_Q{}^{\phi}{}_{\phi} + 2I_{31}\right)\right)}{6T\left(u^\tau\right)^3}$$
$$(A_t)_{76} = -\frac{\mathcal{B}D_s u^r}{3Tu^\tau}$$
$$(A_t)_{77} = \frac{\mathcal{C}D_s u^\tau}{2T}$$
$$(A_t)_{81} = \frac{D_s\left(3\mathcal{C}'(T)\pi_Q{}^{\eta}{}_{\eta}\left(u^\tau\right)^2 - 2\mathcal{B}'(T)\nu^r u^r\right)}{6Tu^\tau}$$
$$(A_t)_{82} = -\frac{D_s\left(2\mathcal{A}\nu^r\left(u^\tau\right)^2 + 2\mathcal{B}\nu^r + u^r\left(u^\tau\right)^2\left(2\mathcal{C}\Pi_Q - 5\mathcal{C}\pi_Q{}^{\eta}{}_{\eta} - 2\mathcal{C}\pi_Q{}^{\phi}{}_{\phi} + 2I_{31}\right)\right)}{6T\left(u^\tau\right)^3}$$
$$(A_t)_{86} = -\frac{\mathcal{B}D_s u^r}{3Tu^\tau}$$
$$(A_t)_{88} = \frac{\mathcal{C}D_s u^\tau}{2T}$$
$$(A_t)_{91} = \frac{D_s\left(5\mathcal{B}'(T)\nu^r u^r + 3\mathcal{C}'(T)\Pi_Q\left(u^\tau\right)^2 + 3(\partial_T I_{31})\left(u^\tau\right)^2\right)}{6Tu^\tau}$$
$$(A_t)_{92} = \frac{D_s\left(2\mathcal{A}\nu^r\left(u^\tau\right)^2 + 5\mathcal{B}\nu^r + u^r\left(u^\tau\right)^2\left(5\mathcal{C}\Pi_Q - 2\mathcal{C}(\pi_Q{}^{\eta}{}_{\eta} + \pi_Q{}^{\phi}{}_{\phi}) + 5I_{31}\right)\right)}{6T\left(u^\tau\right)^3}$$
$$(A_t)_{95} = \frac{(\partial_\alpha I_{31})D_s u^\tau}{2T}$$
$$(A_t)_{96} = \frac{5\mathcal{B}D_s u^r}{6Tu^\tau}$$
$$(A_t)_{99} = \frac{\mathcal{C}D_s u^\tau}{2T}$$

$$(A_r)_{11} = \frac{u^r \left(u^\tau\right)^2 \left(16 a_{\mathrm{SB}} T^3 + (\partial_T n) h\right) + h'(T) \left(n u^r \left(u^\tau\right)^2 + 2\nu^r \left(u^r\right)^2 + \nu^r\right)}{u^\tau}$$

$$(A_r)_{12} = \frac{1}{\left(u^\tau\right)^3} \Big(\left(2\left(u^r\right)^4 + 3\left(u^r\right)^2 + 1\right) \left(4 a_{\mathrm{SB}} T^4 + \Pi_Q - \pi^\eta{}_\eta - \pi_Q{}^\eta{}_\eta - \pi^\phi{}_\phi - \pi_Q{}^\phi{}_\phi\right) + h\left(2n\left(u^r\right)^4 + 3n\left(u^r\right)^2 + n + 2\nu^r\left(u^r\right)^3 + 3\nu^r u^r\right)\Big)$$

$$(A_r)_{13} = -u^r u^\tau$$

$$(A_r)_{14} = -u^r u^\tau$$

$$(A_r)_{15} = (\partial_\alpha n) h u^r u^\tau$$

$$(A_r)_{16} = \frac{h\left(2\left(u^r\right)^2 + 1\right)}{u^\tau}$$

$$(A_r)_{17} = -u^r u^\tau$$

$$(A_r)_{18} = -u^r u^\tau$$

$$(A_r)_{19} = u^r u^\tau$$

$$(A_r)_{21} = 16 a_{\mathrm{SB}} T^3 \left(u^r\right)^2 + 4 a_{\mathrm{SB}} T^3 + h'(T) n \left(u^r\right)^2 + 2h'(T)\nu^r u^r + (\partial_T n) h \left(u^r\right)^2 + (\partial_T n) T + n$$

$$(A_r)_{22} = 2\left(u^r\left(4 a_{\mathrm{SB}} T^4 + \Pi_Q - \pi^\eta{}_\eta - \pi_Q{}^\eta{}_\eta - \pi^\phi{}_\phi - \pi_Q{}^\phi{}_\phi\right) + h(n u^r + \nu^r)\right)$$

$$(A_r)_{23} = -\left(u^r\right)^2 - 1$$

$$(A_r)_{24} = -\left(u^r\right)^2 - 1$$

$$(A_r)_{25} = (\partial_\alpha n)\left(h\left(u^r\right)^2 + T\right)$$

$$(A_r)_{26} = 2 h u^r$$

$$(A_r)_{27} = -\left(u^r\right)^2 - 1$$

$$(A_r)_{28} = -\left(u^r\right)^2 - 1$$

$$(A_r)_{29} = \left(u^\tau\right)^2$$

$$(A_r)_{32} = \frac{3\delta_{\pi\pi}\pi^\phi{}_\phi - 2\eta}{3r^2}$$

$$(A_r)_{33} = \frac{\tau_\pi u^r}{r^2}$$

$$(A_r)_{42} = \frac{3\delta_{\pi\pi}\pi^\eta{}_\eta - 2\eta}{3\tau^2}$$

$$(A_r)_{44} = \frac{\tau_\pi u^r}{\tau^2}$$

$$(A_r)_{51} = (\partial_T n) u^r$$

$$(A_r)_{52} = n$$

$$(A_r)_{55} = (\partial_\alpha n) u^r$$

$$(A_r)_{56} = 1$$

$$(A_r)_{61} = \frac{D_s\left(h'(T)\nu^r u^r + (\partial_T n) T \left(u^\tau\right)^2 + n\left(u^\tau\right)^2\right)}{T}$$

$$(A_r)_{62} = \frac{D_s\left(h n u^r \left(u^\tau\right)^2 + h\nu^r \left(u^r\right)^2 + 2h\nu^r + \Pi_Q u^r \left(u^\tau\right)^2 - \pi_Q{}^\eta{}_\eta u^r \left(u^\tau\right)^2 - \pi_Q{}^\phi{}_\phi u^r \left(u^\tau\right)^2\right)}{T\left(u^\tau\right)^2}$$

$$(A_r)_{65} = (\partial_\alpha n) D_s \left(u^\tau\right)^2$$

$$(A_r)_{66} = \frac{D_s h u^r}{T}$$

$$(A_r)_{67} = -\frac{D_s\left(u^\tau\right)^2}{T}$$

$$(A_r)_{68} = -\frac{D_s\left(u^\tau\right)^2}{T}$$

$$(A_r)_{69} = \frac{D_s\left(u^\tau\right)^2}{T}$$

$$(A_r)_{71} = -\frac{D_s(2\mathcal{B}'(T)\nu^r - 3\mathcal{C}'(T)\pi_Q{}^\phi{}_\phi u^r)}{6T}$$
$$(A_r)_{72} = -\frac{D_s\left(2\mathcal{A}\nu^r u^r + \mathcal{C}\left(u^\tau\right)^2(2\Pi_Q - 2\pi_Q{}^\eta{}_\eta - 5\pi_Q{}^\phi{}_\phi) + 2I_{31}\left(u^\tau\right)^2\right)}{6T\left(u^\tau\right)^2}$$
$$(A_r)_{76} = -\frac{\mathcal{B}D_s}{3T}$$
$$(A_r)_{77} = \frac{\mathcal{C}D_s u^r}{2T}$$
$$(A_r)_{81} = -\frac{D_s(2\mathcal{B}'(T)\nu^r - 3\mathcal{C}'(T)\pi_Q{}^\eta{}_\eta u^r)}{6T}$$
$$(A_r)_{82} = -\frac{D_s\left(2\mathcal{A}\nu^r u^r + \mathcal{C}\left(u^\tau\right)^2(2\Pi_Q - 5\pi_Q{}^\eta{}_\eta - 2\pi_Q{}^\phi{}_\phi) + 2I_{31}\left(u^\tau\right)^2\right)}{6T\left(u^\tau\right)^2}$$
$$(A_r)_{86} = -\frac{\mathcal{B}D_s}{3T}$$
$$(A_r)_{88} = \frac{\mathcal{C}D_s u^r}{2T}$$
$$(A_r)_{91} = \frac{D_s(5\mathcal{B}'(T)\nu^r + 3u^r(\mathcal{C}'(T)\Pi_Q + (\partial_T I_{31})))}{6T}$$
$$(A_r)_{92} = \frac{D_s\left(2\mathcal{A}\nu^r u^r + \mathcal{C}\left(u^\tau\right)^2(5\Pi_Q - 2(\pi_Q{}^\eta{}_\eta + \pi_Q{}^\phi{}_\phi)) + 5I_{31}\left(u^\tau\right)^2\right)}{6T\left(u^\tau\right)^2}$$
$$(A_r)_{95} = \frac{(\partial_\alpha I_{31})D_s u^r}{2T}$$
$$(A_r)_{96} = \frac{5\mathcal{B}D_s}{6T}$$
$$(A_r)_{99} = \frac{\mathcal{C}D_s u^r}{2T}$$
$$\begin{aligned}S_1 = \frac{1}{r\tau u^\tau}\Big(&-\pi^\eta{}_\eta\tau u^r - \pi_Q{}^\eta{}_\eta\tau u^r - \pi^\phi{}_\phi\tau u^r - \pi_Q{}^\phi{}_\phi\tau u^r + 4a_{\rm SB}T^4\tau u^r - \pi^\eta{}_\eta\tau\left(u^r\right)^3\\ &- \pi_Q{}^\eta{}_\eta\tau\left(u^r\right)^3 - \pi^\phi{}_\phi\tau\left(u^r\right)^3 - \pi_Q{}^\phi{}_\phi\tau\left(u^r\right)^3 + 4a_{\rm SB}T^4\tau\left(u^r\right)^3 + \pi^\eta{}_\eta ru^\tau + \pi_Q{}^\eta{}_\eta ru^\tau\\ &+ 4a_{\rm SB}rT^4u^\tau - \pi^\eta{}_\eta r\left(u^r\right)^2u^\tau - \pi_Q{}^\eta{}_\eta r\left(u^r\right)^2u^\tau - \pi^\phi{}_\phi r\left(u^r\right)^2u^\tau - \pi_Q{}^\phi{}_\phi r\left(u^r\right)^2u^\tau\\ &+ 4a_{\rm SB}rT^4\left(u^r\right)^2u^\tau + hn\left(u^\tau\right)^2(ru^\tau + \tau u^r) + \Pi_Q\left(u^\tau\right)^2(ru^\tau + \tau u^r)\\ &+ h\nu^r\left(2ru^ru^\tau + 2\tau\left(u^r\right)^2 + \tau\right)\Big)\end{aligned}$$
$$\begin{aligned}S_2 = &-\frac{\Pi_Q + \pi_Q{}^\phi{}_\phi}{r} - \frac{nT}{r} - \frac{a_{\rm SB}T^4 + \pi^\phi{}_\phi}{r} + \frac{hn\left(u^r\right)^2 + 2h\nu^ru^r + nT}{r}\\ &+ \frac{u^ru^\tau(\Pi_Q - \pi_Q{}^\eta{}_\eta - \pi_Q{}^\phi{}_\phi)}{\tau} - \frac{u^ru^\tau\left(-4a_{\rm SB}T^4 + \pi^\eta{}_\eta + \pi^\phi{}_\phi\right)}{\tau}\\ &+ \frac{\left(u^\tau\right)^2(\Pi_Q - \pi_Q{}^\eta{}_\eta - \pi_Q{}^\phi{}_\phi)}{r} + \frac{4a_{\rm SB}T^4\left(u^r\right)^2 + a_{\rm SB}T^4 - \left(u^\tau\right)^2(\pi^\eta{}_\eta + \pi^\phi{}_\phi)}{r}\\ &+ \frac{h\left(nu^r\left(u^\tau\right)^2 + 2\nu^r\left(u^r\right)^2 + \nu^r\right)}{\tau u^\tau}\end{aligned}$$
$$S_3 = \frac{3\pi^\phi{}_\phi(r(\delta_{\pi\pi}u^\tau + \tau) + \delta_{\pi\pi}\tau u^r) - 2ru^\tau\eta + 4\tau u^r\eta}{3r^3\tau}$$
$$S_4 = \frac{3\pi^\eta{}_\eta(r(\delta_{\pi\pi}u^\tau + \tau) + \delta_{\pi\pi}\tau u^r) + 4ru^\tau\eta - 2\tau u^r\eta}{3r\tau^3}$$
$$S_5 = \frac{nu^r + \nu^r}{r} + \frac{n\left(u^r\right)^2 + n + \nu^ru^r}{\tau u^\tau}$$
$$\begin{aligned}S_6 = \frac{1}{rT\tau u^\tau}\Big(&\nu^rrT\tau u^\tau - D_s\left(u^\tau\right)^2(2\pi_Q{}^\eta{}_\eta ru^r + \pi_Q{}^\eta{}_\eta\tau u^\tau + \pi_Q{}^\phi{}_\phi ru^r + 2\pi_Q{}^\phi{}_\phi\tau u^\tau)\\ &+ D_sh\nu^r\left(r\left(u^r\right)^2 + r + \tau u^ru^\tau\right)\Big)\end{aligned}$$
$$\begin{aligned}S_7 = &\pi_Q{}^\phi{}_\phi\\ &+ \frac{1}{6rT\tau u^\tau}D_s\Big(-2\mathcal{B}\nu^r(ru^r - 2\tau u^\tau) - 2I_{31}\left(r\left(u^r\right)^2 + r - 2\tau u^ru^\tau\right)\end{aligned}$$

$$
\begin{aligned}
&+ \mathcal{C}\Big(7{\pi_Q}^{\phi}{}_{\phi}\tau u^r u^\tau - 2{\pi_Q}^{\eta}{}_{\eta} r\left(u^\tau\right)^2 + 3{\pi_Q}^{\phi}{}_{\phi} r\left(u^\tau\right)^2 \\
&- 2\Pi_Q\left(r\left(u^r\right)^2 + r - 2\tau u^r u^\tau\right)\Big)\Big) \\
S_8 = {}& {\pi_Q}^{\eta}{}_{\eta} \\
&+ \frac{1}{6rT\tau u^\tau} D_s\Big(-2I_{31}\tau u^r u^\tau + 4I_{31} r\left(u^\tau\right)^2 + 2\mathcal{B}\nu^r(2ru^r - \tau u^\tau) \\
&+ \mathcal{C}\Big(-2\Pi_Q\tau u^r u^\tau + 3{\pi_Q}^{\eta}{}_{\eta}\tau u^r u^\tau - 2{\pi_Q}^{\phi}{}_{\phi}\tau u^r u^\tau + 4\Pi_Q r\left(u^\tau\right)^2 \\
&+ 7{\pi_Q}^{\eta}{}_{\eta} r\left(u^\tau\right)^2\Big)\Big) \\
S_9 = {}& \Pi_Q \\
&+ \frac{1}{6rT\tau u^\tau} D_s\Big(5\mathcal{B}\nu^r(ru^r + \tau u^\tau) + 5I_{31}\left(r\left(u^r\right)^2 + r + \tau u^r u^\tau\right) \\
&+ \mathcal{C}\left(5\Pi_Q\left(r\left(u^r\right)^2 + r + \tau u^r u^\tau\right) + 2{\pi_Q}^{\eta}{}_{\eta} r\left(u^\tau\right)^2 + 2{\pi_Q}^{\phi}{}_{\phi}\tau u^r u^\tau\right)\Big)
\end{aligned}
$$

3.5 2+1D Cartesian

3.5.1 S3, charm first moment on a background

`two_d_HQ_BG_consistent_matrix` in `src/Matrix/2d_viscous_HQ_BG.jl`, fields:

$$
\phi = (\alpha, \nu^x, \nu^y)
$$

$$
A_t = \begin{pmatrix} \bullet & \bullet & \bullet \\ \bullet & \bullet & \cdot \\ \bullet & \cdot & \bullet \end{pmatrix}, \qquad A_x = \begin{pmatrix} \bullet & \bullet & \cdot \\ \bullet & \bullet & \cdot \\ \bullet & \cdot & \bullet \end{pmatrix}, \qquad A_y = \begin{pmatrix} \bullet & \cdot & \bullet \\ \bullet & \bullet & \cdot \\ \bullet & \cdot & \bullet \end{pmatrix}, \qquad S = \begin{pmatrix} \bullet \\ \bullet \\ \bullet \end{pmatrix}
$$

3.5.2 S4, charm first and second moment on a background

`two_d_HQ_BG_m2_matrix` in `src/Matrix/2d_viscous_HQ_BG_m2.jl`, fields:

$$
\phi = (\alpha, \nu^x, \nu^y, \pi_Q^{yy}, {\pi_Q}^{\eta}{}_{\eta}, \pi_Q^{xy}, \Pi_Q)
$$

$$
A_t = \begin{pmatrix} \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot \\ \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot \\ \bullet & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot \\ \cdot & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot \\ \cdot & \bullet & \bullet & \cdot & \bullet & \cdot & \cdot \\ \cdot & \bullet & \bullet & \cdot & \cdot & \bullet & \cdot \\ \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \bullet \end{pmatrix}, \qquad A_x = \begin{pmatrix} \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot \\ \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot \\ \bullet & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot \\ \cdot & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot \\ \cdot & \bullet & \cdot & \cdot & \bullet & \cdot & \cdot \\ \cdot & \bullet & \bullet & \cdot & \cdot & \bullet & \cdot \\ \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet \end{pmatrix}, \qquad A_y = \begin{pmatrix} \bullet & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot \\ \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot \\ \bullet & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \bullet & \bullet & \cdot & \cdot & \cdot \\ \cdot & \cdot & \bullet & \cdot & \bullet & \cdot & \cdot \\ \cdot & \bullet & \bullet & \cdot & \cdot & \bullet & \cdot \\ \bullet & \cdot & \bullet & \cdot & \cdot & \cdot & \bullet \end{pmatrix}
$$

$$
S = \begin{pmatrix} \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \end{pmatrix}
$$

3.5.3 S5, two-way

`two_d_two_way_matrix` in `src/Matrix/2d_two_way.jl`, fields:

$$
\phi = (T, u^x, u^y, \pi^{yy}, \pi^{\eta}{}_{\eta}, \pi^{xy}, \Pi, \alpha, \nu^x, \nu^y, \pi_Q^{yy}, {\pi_Q}^{\eta}{}_{\eta}, \pi_Q^{xy}, \Pi_Q)
$$

$$A_t = \left(\begin{array}{cccccccccccccc}
\bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot \\
\bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot \\
\bullet & \bullet & \bullet & \bullet & \cdot & \bullet & \bullet & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot \\
\cdot & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \bullet & \bullet & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \bullet & \bullet & \cdot & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot \\
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\end{array}\right), \qquad A_x = \left(\begin{array}{cccccccccccccc}
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$$A_y = \left(\begin{array}{cccccccccccccc}
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\end{array}\right), \qquad S = \left(\begin{array}{c}
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\end{array}\right)$$

3.5.4 S5, two-way, conformal

`two_d_two_way_matrix_conformal` in `src/Matrix/2d_two_way.jl`, fields:

$$\phi = (T, u^x, u^y, \pi^{yy}, \pi^{\eta}{}_{\eta}, \pi^{xy}, \alpha, \nu^x, \nu^y, \pi_Q^{yy}, \pi_Q{}^{\eta}{}_{\eta}, \pi_Q^{xy}, \Pi_Q)$$

$$A_t = \left(\begin{array}{cccccccccccccc}
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\end{array}\right), \qquad A_x = \left(\begin{array}{cccccccccccccc}
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$$A_y = \left(\begin{array}{cccccccccccc}
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\end{array}\right), \qquad S = \left(\begin{array}{c}
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\end{array}\right)$$

3.5.5 S6, fully coupled

`two_d_two_way_cM_matrix` in `src/Matrix/2d_two_way_cM.jl`, fields:

$$\phi = (T, u^x, u^y, \pi^{yy}, \pi^{\eta}{}_{\eta}, \pi^{xy}, \Pi, \alpha, \nu^x, \nu^y, \pi_Q^{yy}, \pi_Q{}^{\eta}{}_{\eta}, \pi_Q^{xy}, \Pi_Q)$$

$$A_t = \left(\begin{array}{cccccccccccccc}
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\end{array}\right)$$

$$A_y = \left(\begin{array}{cccccccccccccc}
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\end{array}\right)$$

3.5.6 S6, fully coupled, conformal

`two_d_two_way_cM_matrix_conformal` in `src/Matrix/2d_two_way_cM.jl`, fields:

$$\phi = (T, u^x, u^y, \pi^{yy}, \pi^{\eta}{}_{\eta}, \pi^{xy}, \alpha, \nu^x, \nu^y, \pi_Q^{yy}, \pi_Q{}^{\eta}{}_{\eta}, \pi_Q^{xy}, \Pi_Q)$$

$$A_t = \left(\begin{array}{cccccccccccc}
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\end{array}\right), \qquad A_x = \left(\begin{array}{cccccccccccc}
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$$A_y = \left(\begin{array}{cccccccccccc}
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\end{array}\right), \qquad S = \left(\begin{array}{c}
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\end{array}\right)$$

3.6 Checks

id	what it tests	
`G0-0p1d`	$\nabla_\lambda g_{\mu\nu} = 0$	PASS
`G1-0p1d`	$\nabla_\mu (X,0,0,0)^\mu = \partial_\tau X + X/\tau$ (Christoffel and $\sqrt{-g}$ routes)	PASS
`G2-0p1d`	$u \cdot u = -1$; $u_\mu \pi^{\mu\nu} = 0$, $\pi^\mu{}_\mu = 0$ (medium); $u \cdot \nu = 0$; $u_\mu \pi_Q^{\mu\nu} = 0$, ${\pi_Q}^\mu{}_\mu = 0$; both symmetric	PASS
`G-aQ-0p1d`	$\mathcal{A}'$ has identically vanishing coefficient in the second-moment rows (symbolic); dropped	PASS
`G3-0p1d`	every row (medium, charm-in-T, first moment, second moment, δM in T_Q) is quasi-linear	PASS
`S3-free-0p1d`	S3 rows carry no second-moment field	PASS
`S34-nomedpi-0p1d`	charm rows never see the medium π or Π (so S3/S4 need only T, u as background)	PASS
`Gc-extract-0p1d`	$A_t \partial_\tau \phi + A_x \partial_x \phi + A_y \partial_y \phi + S =$ the raw rows (symbolic)	PASS
`Gc-ctrl-0p1d`	control: the exact random-point zero test rejects a 1e-9-perturbed identity and a nonzero row	PASS
`Gc-total-0p1d`	S5 fluid rows $= \nabla_\mu (T_m + T_Q)^{\mu\nu}$, $\sqrt{-g}$ route (symbolic)	PASS
`Gc-number-0p1d`	charm row $= \nabla_\mu N^\mu$, $\sqrt{-g}$ route (symbolic)	PASS
`Gc-dM-0p1d`	the second moment is passive: δM columns of the fluid + first-moment rows vanish identically, and their sources carry no δM	PASS
`Gc-off-0p1d`	charm off ($n, \nu, \delta M \to 0$): S5's medium block and medium sources = S2 exactly; the charm rows' medium columns and sources vanish, so charm = 0 is an invariant subspace on which S5 is S2	PASS
`G6-extract-0p1d`	S6: $A_t \partial_\tau \phi + A_x \partial_x \phi + A_y \partial_y \phi + S =$ the raw rows (symbolic)	PASS
`G6-total-0p1d`	S6 fluid rows $= \nabla_\mu (T_m + T_Q + \delta M)^{\mu\nu}$, $\sqrt{-g}$ route (symbolic)	PASS
`G6-off-0p1d`	S6 at $\delta M = 0$ = S5, row by row (symbolic)	PASS
`G6-m2-0p1d`	the second-moment rows of S6 are those of S5 (symbolic)	PASS
`G6-live-0p1d`	the δM columns of the fluid and current rows are live (in 0+1D only through the source)	PASS
`G0-1p1d`	$\nabla_\lambda g_{\mu\nu} = 0$	PASS
`G1-1p1d`	$\nabla_\mu (X,0,0,0)^\mu = \partial_\tau X + X/\tau$ (Christoffel and $\sqrt{-g}$ routes)	PASS
`G2-1p1d`	$u \cdot u = -1$; $u_\mu \pi^{\mu\nu} = 0$, $\pi^\mu{}_\mu = 0$ (medium); $u \cdot \nu = 0$; $u_\mu \pi_Q^{\mu\nu} = 0$, ${\pi_Q}^\mu{}_\mu = 0$; both symmetric	PASS

id	what it tests	
G2-triad	the $(\ell, \hat{\phi}, \hat{\eta})$ triad is orthonormal and transverse	PASS
G-aQ-1p1d	$\mathcal{A}'$ has identically vanishing coefficient in the second-moment rows (symbolic); dropped	PASS
G3-1p1d	every row (medium, charm-in-T, first moment, second moment, δM in T_Q) is quasi-linear	PASS
S3-free-1p1d	S3 rows carry no second-moment field	PASS
S34-nomedpi-1p1d	charm rows never see the medium π or Π (so S3/S4 need only T, u as background)	PASS
Gc-extract-1p1d	$A_t\partial_\tau\phi + A_x\partial_x\phi + A_y\partial_y\phi + S =$ the raw rows (symbolic)	PASS
Gc-ctrl-1p1d	control: the exact random-point zero test rejects a 1e-9-perturbed identity and a nonzero row	PASS
Gc-total-1p1d	S5 fluid rows $= \nabla_\mu(T_m + T_Q)^{\mu\nu}$, $\sqrt{-g}$ route (symbolic)	PASS
Gc-number-1p1d	charm row $= \nabla_\mu N^\mu$, $\sqrt{-g}$ route (symbolic)	PASS
Gc-dM-1p1d	the second moment is passive: δM columns of the fluid + first-moment rows vanish identically, and their sources carry no δM	PASS
Gc-off-1p1d	charm off $(n, \nu, \delta M \to 0)$: S5's medium block and medium sources = S2 exactly; the charm rows' medium columns and sources vanish, so charm = 0 is an invariant subspace on which S5 is S2	PASS
G6-extract-1p1d	S6: $A_t\partial_\tau\phi + A_x\partial_x\phi + A_y\partial_y\phi + S =$ the raw rows (symbolic)	PASS
G6-total-1p1d	S6 fluid rows $= \nabla_\mu(T_m + T_Q + \delta M)^{\mu\nu}$, $\sqrt{-g}$ route (symbolic)	PASS
G6-off-1p1d	S6 at $\delta M = 0 =$ S5, row by row (symbolic)	PASS
G6-m2-1p1d	the second-moment rows of S6 are those of S5 (symbolic)	PASS
G6-live-1p1d	the δM columns of the fluid and current rows are live (in 0+1D only through the source)	PASS
G0-2p1d	$\nabla_\lambda g_{\mu\nu} = 0$	PASS
G1-2p1d	$\nabla_\mu(X, 0, 0, 0)^\mu = \partial_\tau X + X/\tau$ (Christoffel and $\sqrt{-g}$ routes)	PASS
G2-2p1d	$u \cdot u = -1$; $u_\mu\pi^{\mu\nu} = 0$, $\pi^\mu{}_\mu = 0$ (medium); $u \cdot \nu = 0$; $u_\mu\pi_Q^{\mu\nu} = 0$, $\pi_Q{}^\mu{}_\mu = 0$; both symmetric	PASS
G-aQ-2p1d	$\mathcal{A}'$ has identically vanishing coefficient in the second-moment rows (symbolic); dropped	PASS
G3-2p1d	every row (medium, charm-in-T, first moment, second moment, δM in T_Q) is quasi-linear	PASS
S3-free-2p1d	S3 rows carry no second-moment field	PASS
S34-nomedpi-2p1d	charm rows never see the medium π or Π (so S3/S4 need only T, u as background)	PASS
Gc-extract-2p1d	$A_t\partial_\tau\phi + A_x\partial_x\phi + A_y\partial_y\phi + S =$ the raw rows (exact at 4 random rational points)	PASS
Gc-ctrl-2p1d	control: the exact random-point zero test rejects a 1e-9-perturbed identity and a nonzero row	PASS
Gc-total-2p1d	S5 fluid rows $= \nabla_\mu(T_m + T_Q)^{\mu\nu}$, $\sqrt{-g}$ route (symbolic)	PASS
Gc-number-2p1d	charm row $= \nabla_\mu N^\mu$, $\sqrt{-g}$ route (symbolic)	PASS
Gc-dM-2p1d	the second moment is passive: δM columns of the fluid + first-moment rows vanish identically, and their sources carry no δM	PASS
Gc-off-2p1d	charm off $(n, \nu, \delta M \to 0)$: S5's medium block and medium sources = S2 exactly; the charm rows' medium columns and sources vanish, so charm = 0 is an invariant subspace on which S5 is S2	PASS
G6-extract-2p1d	S6: $A_t\partial_\tau\phi + A_x\partial_x\phi + A_y\partial_y\phi + S =$ the raw rows (exact at 4 random rational points)	PASS
G6-total-2p1d	S6 fluid rows $= \nabla_\mu(T_m + T_Q + \delta M)^{\mu\nu}$, $\sqrt{-g}$ route (symbolic)	PASS
G6-off-2p1d	S6 at $\delta M = 0 =$ S5, row by row (symbolic)	PASS
G6-m2-2p1d	the second-moment rows of S6 are those of S5 (symbolic)	PASS
G6-live-2p1d	the δM columns of the fluid and current rows are live (in 0+1D only through the source)	PASS
R0-jl2wl	Julia -> WL conversion of the shipped matrices (0)	PASS
R-S2-1p1d-shipped	S2 1+1D generic EOS, $\delta_{\pi\pi} = \delta_{\Pi\Pi} = 0 =$ one_d_viscous_matrix (1d_viscous_cilindrical.jl), entry by entry (55 entries, symbolic)	PASS
R-S2-2p1d-generator	S2 2+1D, $\delta_{\pi\pi} = \delta_{\Pi\Pi} = 0 =$ two_d_viscous_matrix_derived (2d_viscous.jl, from derive_2p1d_viscous.wls), entry by entry (154 entries, symbolic)	PASS
R-S2-1p1d-MIS	S2 1+1D with $\delta_{\pi\pi} = \frac{4}{3}\tau_\pi$, $\delta_{\Pi\Pi} = 0 =$ medium block of one_d_viscous_matrix_fugacity (:MIS, read from the .jl), entry by entry (55 entries, symbolic)	PASS
R-S2-1p1d-MIS-ctl	control: $\delta_{\pi\pi} = 0$ does not reproduce the :MIS medium block (the two 1-D media differ by $\delta_{\pi\pi}$)	PASS

id	what it tests	
R-S4S3-1p1d	S4 rows 1-2 = S3 (so the first-moment block of S4/S5 is the consistent closure)	PASS
X6-2to1-1	S6 2+1D axisymmetric on the x axis: tau-momentum (symbolic)	PASS
X6-2to1-2	S6 2+1D axisymmetric on the x axis: x-momentum = r-momentum (symbolic)	PASS
X6-2to1-3	S6 2+1D axisymmetric on the x axis: y-momentum = 0 (symbolic)	PASS
X6-2to1-4	S6 2+1D axisymmetric on the x axis: E^{yy} = r^2 $E^{\phi\phi}$ (symbolic)	PASS
X6-2to1-5	S6 2+1D axisymmetric on the x axis: $E^{\eta\eta}$ (symbolic)	PASS
X6-2to1-6	S6 2+1D axisymmetric on the x axis: $E^{xy} = 0$ (symbolic)	PASS
X6-2to1-7	S6 2+1D axisymmetric on the x axis: bulk (symbolic)	PASS
X6-2to1-8	S6 2+1D axisymmetric on the x axis: charm number (symbolic)	PASS
X6-2to1-9	S6 2+1D axisymmetric on the x axis: ν^x row = ν^r row (symbolic)	PASS
X6-2to1-10	S6 2+1D axisymmetric on the x axis: ν^y row = 0 (symbolic)	PASS
X6-2to1-11	S6 2+1D axisymmetric on the x axis: π_Q^{yy} row = ($\phi\phi$) row (symbolic)	PASS
X6-2to1-12	S6 2+1D axisymmetric on the x axis: ${\pi_Q}^{\eta}{}_{\eta}$ row = ($\eta\eta$) row (symbolic)	PASS
X6-2to1-13	S6 2+1D axisymmetric on the x axis: π_Q^{xy} row = 0 (symbolic)	PASS
X6-2to1-14	S6 2+1D axisymmetric on the x axis: trace row (symbolic)	PASS
X-2to1-1	S5 2+1D axisymmetric on the x axis: tau-momentum (symbolic)	PASS
X-2to1-2	S5 2+1D axisymmetric on the x axis: x-momentum = r-momentum (symbolic)	PASS
X-2to1-3	S5 2+1D axisymmetric on the x axis: y-momentum = 0 (symbolic)	PASS
X-2to1-4	S5 2+1D axisymmetric on the x axis: E^{yy} = r^2 $E^{\phi\phi}$ (symbolic)	PASS
X-2to1-5	S5 2+1D axisymmetric on the x axis: $E^{\eta\eta}$ (symbolic)	PASS
X-2to1-6	S5 2+1D axisymmetric on the x axis: $E^{xy} = 0$ (symbolic)	PASS
X-2to1-7	S5 2+1D axisymmetric on the x axis: bulk (symbolic)	PASS
X-2to1-8	S5 2+1D axisymmetric on the x axis: charm number (symbolic)	PASS
X-2to1-9	S5 2+1D axisymmetric on the x axis: ν^x row = ν^r row (symbolic)	PASS
X-2to1-10	S5 2+1D axisymmetric on the x axis: ν^y row = 0 (symbolic)	PASS
X-2to1-11	S5 2+1D axisymmetric on the x axis: π_Q^{yy} row = ($\phi\phi$) row (symbolic)	PASS
X-2to1-12	S5 2+1D axisymmetric on the x axis: ${\pi_Q}^{\eta}{}_{\eta}$ row = ($\eta\eta$) row (symbolic)	PASS
X-2to1-13	S5 2+1D axisymmetric on the x axis: π_Q^{xy} row = 0 (symbolic)	PASS
X-2to1-14	S5 2+1D axisymmetric on the x axis: trace row (symbolic)	PASS
X-ctrl	control: the same test rejects a wrong pairing (2+1D tau row vs 1+1D r row)	PASS
X-vort-axis	the corotation coupling W vanishes identically on an axisymmetric state (so G-chart is blind to wv, as documented)	PASS
X-1to0-1	S5 1+1D at $u^r = 0$, no r-dependence: energy (symbolic)	PASS
X-1to0-2	S5 1+1D at $u^r = 0$, no r-dependence: r-momentum = 0 (symbolic)	PASS
X-1to0-3	S5 1+1D at $u^r = 0$, no r-dependence: $E^{\phi\phi}$ = -(tau^2/2r^2) $E^{\eta\eta}$ (symbolic)	PASS
X-1to0-4	S5 1+1D at $u^r = 0$, no r-dependence: $E^{\eta\eta}$ (symbolic)	PASS
X-1to0-5	S5 1+1D at $u^r = 0$, no r-dependence: bulk (symbolic)	PASS
X-1to0-6	S5 1+1D at $u^r = 0$, no r-dependence: charm number (symbolic)	PASS
X-1to0-7	S5 1+1D at $u^r = 0$, no r-dependence: ν^r row = 0 (symbolic)	PASS
X-1to0-8	S5 1+1D at $u^r = 0$, no r-dependence: ($\phi\phi$) row = -1/2 ($\eta\eta$) row (symbolic)	PASS
X-1to0-9	S5 1+1D at $u^r = 0$, no r-dependence: ($\eta\eta$) row (symbolic)	PASS
X-1to0-10	S5 1+1D at $u^r = 0$, no r-dependence: trace row (symbolic)	PASS
X6-1to0-1	S6 1+1D at $u^r = 0$, no r-dependence: energy (symbolic)	PASS
X6-1to0-2	S6 1+1D at $u^r = 0$, no r-dependence: r-momentum = 0 (symbolic)	PASS
X6-1to0-3	S6 1+1D at $u^r = 0$, no r-dependence: $E^{\phi\phi}$ = -(tau^2/2r^2) $E^{\eta\eta}$ (symbolic)	PASS
X6-1to0-4	S6 1+1D at $u^r = 0$, no r-dependence: $E^{\eta\eta}$ (symbolic)	PASS
X6-1to0-5	S6 1+1D at $u^r = 0$, no r-dependence: bulk (symbolic)	PASS
X6-1to0-6	S6 1+1D at $u^r = 0$, no r-dependence: charm number (symbolic)	PASS
X6-1to0-7	S6 1+1D at $u^r = 0$, no r-dependence: ν^r row = 0 (symbolic)	PASS
X6-1to0-8	S6 1+1D at $u^r = 0$, no r-dependence: ($\phi\phi$) row = -1/2 ($\eta\eta$) row (symbolic)	PASS
X6-1to0-9	S6 1+1D at $u^r = 0$, no r-dependence: ($\eta\eta$) row (symbolic)	PASS
X6-1to0-10	S6 1+1D at $u^r = 0$, no r-dependence: trace row (symbolic)	PASS
B-ideal-conf	conformal ideal Bjorken: $\partial_\tau T = -T/(3\tau)$	PASS
B-ideal-gen	generic-EOS ideal Bjorken: $\partial_\tau T = -P'/(P''\tau) = -c_s^2 T/\tau,\ c_s^2 = P'/(TP'')$	PASS

id	what it tests	
B-IS-gen	generic is Bjorken: $\partial_\tau T = -(TP' + \Pi + \pi)/(\tau TP'')$, $\tau_\pi \partial_\tau \pi = -\pi - 4\eta/(3\tau) - \delta_{\pi\pi}\pi/\tau$, $\tau_\Pi \partial_\tau \Pi = -\Pi - \zeta/\tau - \delta_{\Pi\Pi}\Pi/\tau$	PASS
B-IS-conf	conformal is Bjorken: $\partial_\tau T = -T/(3\tau) - \pi/(12aT^3\tau)$, $\tau_\pi \partial_\tau \pi = -\pi - 4\eta/(3\tau) - \delta_{\pi\pi}\pi/\tau$ [$\delta_{\pi\pi} = \frac{4}{3}\tau_\pi$: the DNMR $\delta_{\pi\pi}$ form; no $\tau_{\pi\pi}, \lambda_1$]	PASS
B-NS-conf	first-order (NS) limit $\pi = -4\eta/(3\tau)$: $\partial_\tau T = -T/(3\tau) + (4/9)(\eta/s)/\tau^2$, $s = 4aT^3$	PASS
B-twoway-energy	S5 Bjorken energy row: $\partial_\tau(e + e_Q) + (e + P + \Pi + \pi + nh)/\tau$	PASS
B-S6-energy	S6 Bjorken energy row: $\partial_\tau(e + e_Q) + (e + P + \Pi + \pi + nh + \pi_Q{}^\eta{}_\eta + \Pi_Q)/\tau$	PASS
B-charm-number	Bjorken charm number: $\tau^{-1}\partial_\tau(n\tau) = 0$	PASS
T-rest-1p1d	$\nu = 0, \delta M = 0, \nabla\alpha = 0$, medium at rest: transverse momentum exchange $\Delta^\mu{}_\nu \nabla_\lambda T_Q^{\lambda\nu} = 0$ and the nu row is satisfied	PASS
T-drag-1p1d	drag: on shell, $\nabla_\mu T_Q^{\mu i}$ = -$(T/D_s)\nu^i$ (drag on charm) and the fluid row then gives $\nabla_\mu T_m^{\mu i}$ = +$(T/D_s)\nu^i$ (Newton III), exact	PASS
T-split-1p1d	fluid momentum rows = $\nabla_\mu T_m^{\mu\nu} + \nabla_\mu T_Q^{\mu\nu}$ identically	PASS
T6-drag-1p1d	S6 drag: on shell, $\nabla_\mu(T_Q + \delta M)^{\mu i} = -(T/D_s)\nu^i$, so the medium gains $(T/D_s)\nu^i$ (Newton III), exact	PASS
T6-static	static δM at rest: $\nabla_\mu \delta M^{\mu r} = \partial_r(\pi_{rr} + \Pi_Q) + (\pi_{rr} - \pi^\phi{}_\phi)/r$	PASS
T-rest-2p1d	$\nu = 0, \delta M = 0, \nabla\alpha = 0$, medium at rest: transverse momentum exchange $\Delta^\mu{}_\nu \nabla_\lambda T_Q^{\lambda\nu} = 0$ and the nu row is satisfied	PASS
T-drag-2p1d	drag: on shell, $\nabla_\mu T_Q^{\mu i}$ = -$(T/D_s)\nu^i$ (drag on charm) and the fluid row then gives $\nabla_\mu T_m^{\mu i}$ = +$(T/D_s)\nu^i$ (Newton III), exact	PASS
T-split-2p1d	fluid momentum rows = $\nabla_\mu T_m^{\mu\nu} + \nabla_\mu T_Q^{\mu\nu}$ identically	PASS
T6-drag-2p1d	S6 drag: on shell, $\nabla_\mu(T_Q + \delta M)^{\mu i} = -(T/D_s)\nu^i$, so the medium gains $(T/D_s)\nu^i$ (Newton III), exact	PASS
S6-speeds-1p1d	S6 is hyperbolic and subluminal at all 72 (state, direction) pairs, exactly, with max speed 0.9901	PASS
S6-speeds-2p1d	S6 is hyperbolic and subluminal at all 144 (state, direction) pairs, exactly, with max speed 0.9903	PASS
E-same-one_d_HQ_BG_consistent_matrix	simplified exported entries = derived entries (symbolic)	PASS
E-free-one_d_HQ_BG_consistent_matrix	every free symbol is an argument or a field	PASS
E-same-one_d_HQ_BG_m2_matrix	simplified exported entries = derived entries (symbolic)	PASS
E-free-one_d_HQ_BG_m2_matrix	every free symbol is an argument or a field	PASS
E-same-one_d_two_way_matrix	simplified exported entries = derived entries (symbolic)	PASS
E-free-one_d_two_way_matrix	every free symbol is an argument or a field	PASS
E-same-one_d_two_way_matrix_conformal	simplified exported entries = derived entries (symbolic)	PASS
E-free-one_d_two_way_matrix_conformal	every free symbol is an argument or a field	PASS
E-same-two_d_HQ_BG_consistent_matrix	simplified exported entries = derived entries (symbolic)	PASS
E-free-two_d_HQ_BG_consistent_matrix	every free symbol is an argument or a field	PASS
E-same-two_d_HQ_BG_m2_matrix	simplified exported entries = derived entries (exact at 3 random rational points)	PASS
E-free-two_d_HQ_BG_m2_matrix	every free symbol is an argument or a field	PASS
E-same-two_d_two_way_matrix	simplified exported entries = derived entries (exact at 3 random rational points)	PASS
E-free-two_d_two_way_matrix	every free symbol is an argument or a field	PASS
E-same-two_d_two_way_matrix_conformal	simplified exported entries = derived entries (exact at 3 random rational points)	PASS
E-free-two_d_two_way_matrix_conformal	every free symbol is an argument or a field	PASS
E-same-one_d_two_way_cM_matrix	simplified exported entries = derived entries (symbolic)	PASS
E-free-one_d_two_way_cM_matrix	every free symbol is an argument or a field	PASS

id	what it tests	
E-same-one_d_two_way_cM_matrix_conformal	simplified exported entries = derived entries (symbolic)	PASS
E-free-one_d_two_way_cM_matrix_conformal	every free symbol is an argument or a field	PASS
E-same-two_d_two_way_cM_matrix	simplified exported entries = derived entries (exact at 3 random rational points)	PASS
E-free-two_d_two_way_cM_matrix	every free symbol is an argument or a field	PASS
E-same-two_d_two_way_cM_matrix_conformal	simplified exported entries = derived entries (exact at 3 random rational points)	PASS
E-free-two_d_two_way_cM_matrix_conformal	every free symbol is an argument or a field	PASS
E-jl	the generated Julia (12 functions) parses, runs, and reproduces its 24-state CSVs: max rel err 1.4298582449629782e-15	PASS

155 passed, 0 failed (155 checks evaluated).

Not a check: the characteristic speeds of S5 are computed and reported (`two_way_char_speeds.tsv` in the scratch directory).

Re-run with `wolframscript -file deriv/derive_two_way.wls`.

4 The hand-written 1+1D matrices

Script: `deriv/derive_1d_matrices.wls`
Writes nothing into `src/Matrix`: it compares the derivation with the hand-written `.jl` files, read from the file.

4.1 Setup

$$x^\mu = (\tau, r, \phi, \eta), \qquad g_{\mu\nu} = \mathrm{diag}(-1, 1, r^2, \tau^2), \qquad u^\mu = (u^\tau, u^r, 0, 0)$$

Medium as in the viscous section. Charm:

$$\nabla_\mu(n\,u^\mu + \nu^\mu) = 0, \qquad \tau_n\,\Delta^\mu{}_\beta D\nu^\beta + \nu^\mu + \kappa\,\Delta^{\mu\beta}\partial_\beta\alpha = 0, \qquad \nu^\tau = \frac{u^r\nu^r}{u^\tau}$$

Rows $\nabla_\mu N^\mu$ and the r component of the current equation. Variants:

$$\begin{gathered}\text{MIS}: \;\; n = n(T,\alpha),\;\; \delta_{\pi\pi} = \tfrac{4}{3}\tau_\pi, \quad \phi = (T, u^r, \pi^\phi{}_\phi, \pi^\eta{}_\eta, \Pi, \alpha, \nu^r)\\ \text{gamma}: \;\; n = \gamma\,n(T),\;\; \kappa\partial\alpha \to \kappa\partial\gamma,\;\; \delta_{\pi\pi} = \tfrac{4}{3}\tau_\pi, \quad \phi = (T, u^r, \pi^\phi{}_\phi, \pi^\eta{}_\eta, \Pi, \gamma, \nu^r)\\ \text{ideal}: \;\; \pi = \Pi = 0,\;\; \nu = 0, \quad \phi = (T, u^r, n) \text{ or } (T, u^r, \alpha)\end{gathered}$$

4.2 Steps

$$\Gamma^\tau{}_{\eta\eta} = \tau, \quad \Gamma^r{}_{\phi\phi} = -r, \quad \Gamma^\phi{}_{r\phi} = \frac{1}{r}, \quad \Gamma^\eta{}_{\tau\eta} = \frac{1}{\tau}$$

$$\begin{aligned}
a^\tau &= u^r\left((\partial_\tau u^r) + \frac{(\partial_r u^r)u^r}{u^\tau}\right)\\
a^r &= (\partial_\tau u^r)u^\tau + (\partial_r u^r)u^r\\
\nu_\lambda a^\lambda &= \frac{\nu^r((\partial_\tau u^r)u^\tau + (\partial_r u^r)u^r)}{(u^\tau)^2}\\
D\nu^\tau &= \frac{u^\tau\left((\partial_\tau\nu^r)(u^r)^3 + (\partial_\tau\nu^r)u^r + (\partial_\tau u^r)\nu^r\right) + (\partial_r\nu^r)\left((u^r)^4 + (u^r)^2\right) + (\partial_r u^r)\nu^r u^r}{(u^\tau)^3}\\
D\nu^r &= (\partial_\tau\nu^r)u^\tau + (\partial_r\nu^r)u^r\\
E_\nu &= \tau_n\big(D\nu^r - u^r\nu_\lambda a^\lambda\big) + \nu^r + \kappa\big(\Delta^{r\tau}\partial_\tau\alpha + \Delta^{rr}\partial_r\alpha\big), \qquad \Delta^{r\tau} = u^r u^\tau,\;\; \Delta^{rr} = 1 + (u^r)^2
\end{aligned}$$

$$\begin{aligned}
E_\alpha &= (\partial_r\nu^r) + (\partial_r u^r)n + u^r((\partial_r\alpha)(\partial_\alpha n) + (\partial_r T)(\partial_T n)) + \frac{nu^r + \nu^r}{r} - \frac{(\partial_\tau u^r)\nu^r\,(u^r)^2}{(u^\tau)^3}\\
&\quad + \frac{(\partial_\tau u^r)\nu^r}{u^\tau} + \frac{(\partial_\tau\nu^r)u^r}{u^\tau} + \frac{(\partial_\tau u^r)nu^r}{u^\tau} + \frac{n\,(u^r)^2 + n + \nu^r u^r}{\tau u^\tau}\\
&\quad + u^\tau((\partial_\tau\alpha)(\partial_\alpha n) + (\partial_\tau T)(\partial_T n))\\
E_\nu &= (\partial_r\nu^r)\tau_n u^r + u^\tau((\partial_\tau\alpha)\kappa u^r + (\partial_\tau\nu^r)\tau_n) + (\partial_r\alpha)\kappa\,(u^\tau)^2\\
&\quad + \frac{\nu^r\left(-(\partial_\tau u^r)\tau_n u^r u^\tau + (u^r)^2\,(1 - (\partial_r u^r)\tau_n) + 1\right)}{(u^\tau)^2}
\end{aligned}$$

$$A^{(k)}_{ij} = \frac{\partial E_i}{\partial(\partial_k\phi_j)}, \qquad S_i = E_i\big|_{\partial\phi=0}, \qquad k = \tau, r$$

4.3 Result

4.3.1 `one_d_viscous_matrix_fugacity_BG_only` in `src/Matrix/1d_viscous_HQ_cilindrical_BG.jl`

Background T, u^r; $\phi = (\alpha, \nu^r)$; the file multiplies every row by $(u^\tau)^3 r\tau$ (M-BG). As derived:

$$
\begin{aligned}
(A_t)_{11} &= (\partial_\alpha n)u^\tau \\
(A_t)_{12} &= \frac{u^r}{u^\tau} \\
(A_t)_{21} &= \kappa u^r u^\tau \\
(A_t)_{22} &= \tau_n u^\tau \\
(A_r)_{11} &= (\partial_\alpha n)u^r \\
(A_r)_{12} &= 1 \\
(A_r)_{21} &= \kappa \left(u^\tau\right)^2 \\
(A_r)_{22} &= \tau_n u^r \\
S_1 &= \frac{1}{r\tau \left(u^\tau\right)^3}\Big((\partial_\tau u^r)\nu^r r\tau + \nu^r \left(u^\tau\right)^2 (ru^r + \tau u^\tau) \\
&\quad + (\partial_T n) r\tau \left(u^\tau\right)^2 \left((\partial_r T)u^r u^\tau + (\partial_\tau T)\left(u^r\right)^2 + (\partial_\tau T)\right) \\
&\quad + n\left(u^\tau\right)^2 \left(r\left((\partial_r u^r)\tau u^\tau + (\partial_\tau u^r)\tau u^r + \left(u^r\right)^2 + 1\right) + \tau u^r u^\tau\right)\Big) \\
S_2 &= \frac{\nu^r\left(\left(u^r\right)^2\left(1 - (\partial_r u^r)\tau_n\right) - (\partial_\tau u^r)\tau_n u^r u^\tau + 1\right)}{\left(u^\tau\right)^2}
\end{aligned}
$$

4.3.2 `one_d_viscous_matrix_fugacity` in `src/Matrix/1d_viscous_HQ_cilindrical_fugacity.jl`

Derived, charm rows:

$$
\begin{aligned}
(A_t)_{61} &= (\partial_T n)u^\tau \\
(A_t)_{62} &= \frac{nu^r\left(u^\tau\right)^2 + \nu^r}{\left(u^\tau\right)^3} \\
(A_t)_{66} &= (\partial_\alpha n)u^\tau \\
(A_t)_{67} &= \frac{u^r}{u^\tau} \\
(A_t)_{72} &= -\frac{\nu^r \tau_n u^r}{u^\tau} \\
(A_t)_{76} &= \kappa u^r u^\tau \\
(A_t)_{77} &= \tau_n u^\tau \\
(A_r)_{61} &= (\partial_T n)u^r \\
(A_r)_{62} &= n \\
(A_r)_{66} &= (\partial_\alpha n)u^r \\
(A_r)_{67} &= 1 \\
(A_r)_{72} &= -\frac{\nu^r \tau_n \left(u^r\right)^2}{\left(u^\tau\right)^2} \\
(A_r)_{76} &= \kappa\left(u^\tau\right)^2 \\
(A_r)_{77} &= \tau_n u^r \\
S_6 &= \frac{n\left(r\left(u^r\right)^2 + r + \tau u^r u^\tau\right) + \nu^r(ru^r + \tau u^\tau)}{r\tau u^\tau} \\
S_7 &= \nu^r
\end{aligned}
$$

With c the charm rows and m the medium rows, $\mathrm{row}_c \to \mathrm{row}_c - L\,\mathrm{row}_m$, $L = A_t^{cm}(A_t^{mm})^{-1}$, removes ∂_τ of the medium from the charm rows:

$$
\begin{aligned}
\mathcal{D}_1 &= 3\left(p'(T)\right)^2 \tau_\Pi \tau_\pi \left(u^r\right)^2 T \\
&\quad + 3p'(T)\tau_\Pi\tau_\pi\left(-p''(T)\left(u^r\right)^2 T^2 - p''(T)T^2 + \Pi\left(u^r\right)^2 - \pi^\eta{}_\eta\left(u^r\right)^2 - \pi^\phi{}_\phi\left(u^r\right)^2\right)
\end{aligned}
$$

$$
\begin{aligned}
&- p''(T)T\Big(-3\pi^{\phi}{}_{\phi}\tau_{\Pi}\tau_{\pi} + \pi^{\phi}{}_{\phi}\tau_{\Pi}\tau_{\pi}\left(u^{r}\right)^{2} + \pi^{\eta}{}_{\eta}\tau_{\Pi}\tau_{\pi}\left(\left(u^{r}\right)^{2} - 3\right) + 3\Pi\tau_{\Pi}\tau_{\pi}\left(u^{\tau}\right)^{2} \\
&- 4\tau_{\Pi}\left(u^{r}\right)^{2}\eta - 3\tau_{\pi}\left(u^{r}\right)^{2}\zeta\Big) \\
\mathcal{D}_{2} = &-3\left(p'(T)\right)^{2}\tau_{\Pi}\tau_{\pi}\left(u^{r}\right)^{2}T \\
&+ 3p'(T)\tau_{\Pi}\tau_{\pi}\left(p''(T)\left(u^{r}\right)^{2}T^{2} + p''(T)T^{2} - \Pi\left(u^{r}\right)^{2} + \pi^{\eta}{}_{\eta}\left(u^{r}\right)^{2} + \pi^{\phi}{}_{\phi}\left(u^{r}\right)^{2}\right) \\
&+ p''(T)T\Big(-3\pi^{\phi}{}_{\phi}\tau_{\Pi}\tau_{\pi} + \pi^{\phi}{}_{\phi}\tau_{\Pi}\tau_{\pi}\left(u^{r}\right)^{2} + \pi^{\eta}{}_{\eta}\tau_{\Pi}\tau_{\pi}\left(\left(u^{r}\right)^{2} - 3\right) + 3\Pi\tau_{\Pi}\tau_{\pi}\left(u^{\tau}\right)^{2} \\
&- 4\tau_{\Pi}\left(u^{r}\right)^{2}\eta - 3\tau_{\pi}\left(u^{r}\right)^{2}\zeta\Big) \\
\mathcal{D}_{3} = &\,3p'(T)\tau_{\Pi}\tau_{\pi}\left(u^{r}\right)^{2}\left(-p'(T)T - \Pi + \pi^{\eta}{}_{\eta} + \pi^{\phi}{}_{\phi}\right) \\
&+ p''(T)T\Big(3p'(T)\tau_{\Pi}\tau_{\pi}T\left(u^{\tau}\right)^{2} \\
&+ \tau_{\Pi}\left(\left(u^{r}\right)^{2}\left(\tau_{\pi}(3\Pi + \pi^{\eta}{}_{\eta} + \pi^{\phi}{}_{\phi}) - 4\eta\right) + 3\Pi\tau_{\pi} - 3\tau_{\pi}(\pi^{\eta}{}_{\eta} + \pi^{\phi}{}_{\phi})\right) \\
&- 3\tau_{\pi}\left(u^{r}\right)^{2}\zeta\Big) \\
\mathcal{D}_{4} = &\,3p'(T)\tau_{\Pi}\tau_{\pi}\left(u^{r}\right)^{2}\left(p'(T)T + \Pi - \pi^{\eta}{}_{\eta} - \pi^{\phi}{}_{\phi}\right) \\
&- p''(T)T\Big(3p'(T)\tau_{\Pi}\tau_{\pi}T\left(u^{\tau}\right)^{2} \\
&+ \tau_{\Pi}\left(\left(u^{r}\right)^{2}\left(\tau_{\pi}(3\Pi + \pi^{\eta}{}_{\eta} + \pi^{\phi}{}_{\phi}) - 4\eta\right) + 3\Pi\tau_{\pi} - 3\tau_{\pi}(\pi^{\eta}{}_{\eta} + \pi^{\phi}{}_{\phi})\right) \\
&- 3\tau_{\pi}\left(u^{r}\right)^{2}\zeta\Big)
\end{aligned}
$$

$$
\begin{aligned}
L_{61} = &\frac{1}{u^{\tau}\mathcal{D}_{1}}\Big(3\tau_{\Pi}\tau_{\pi}u^{r}(p''(T)T + p'(T))\left(nu^{r}\left(u^{\tau}\right)^{2} + \nu^{r}\right) \\
&+ \left(\partial_{T}n\right)\left(u^{\tau}\right)^{2}\Big(3\pi^{\phi}{}_{\phi}\tau_{\Pi}\tau_{\pi} + 2\pi^{\phi}{}_{\phi}\tau_{\Pi}\tau_{\pi}\left(u^{r}\right)^{2} - 3\Pi\tau_{\Pi}\tau_{\pi}\left(2\left(u^{r}\right)^{2} + 1\right) \\
&+ \pi^{\eta}{}_{\eta}\tau_{\Pi}\tau_{\pi}\left(2\left(u^{r}\right)^{2} + 3\right) - 3p'(T)\tau_{\Pi}\tau_{\pi}T - 6p'(T)\tau_{\Pi}\tau_{\pi}\left(u^{r}\right)^{2}T \\
&+ 4\tau_{\Pi}\left(u^{r}\right)^{2}\eta + 3\tau_{\pi}\left(u^{r}\right)^{2}\zeta\Big)\Big) \\
L_{62} = &\frac{1}{\left(u^{\tau}\right)^{2}\mathcal{D}_{1}}\Big(-3\tau_{\Pi}\tau_{\pi}\left(nu^{r}\left(u^{\tau}\right)^{2} + \nu^{r}\right)\left(p''(T)T\left(u^{\tau}\right)^{2} + p'(T)\left(u^{r}\right)^{2}\right) \\
&+ \left(\partial_{T}n\right)u^{r}\left(u^{\tau}\right)^{2}\Big(-6\pi^{\phi}{}_{\phi}\tau_{\Pi}\tau_{\pi} - 2\pi^{\phi}{}_{\phi}\tau_{\Pi}\tau_{\pi}\left(u^{r}\right)^{2} - 2\pi^{\eta}{}_{\eta}\tau_{\Pi}\tau_{\pi}\left(\left(u^{r}\right)^{2} + 3\right) \\
&+ 6p'(T)\tau_{\Pi}\tau_{\pi}T + 6p'(T)\tau_{\Pi}\tau_{\pi}\left(u^{r}\right)^{2}T + 6\Pi\tau_{\Pi}\tau_{\pi}\left(u^{\tau}\right)^{2} \\
&- 4\tau_{\Pi}\left(u^{r}\right)^{2}\eta - 3\tau_{\pi}\left(u^{r}\right)^{2}\zeta\Big)\Big) \\
L_{63} = &\frac{3r^{2}\tau_{\Pi}u^{r}\left(\left(\partial_{T}n\right)u^{r}\left(u^{\tau}\right)^{2}\left(p'(T)T + \Pi - \pi^{\eta}{}_{\eta} - \pi^{\phi}{}_{\phi}\right) - p''(T)T\left(nu^{r}\left(u^{\tau}\right)^{2} + \nu^{r}\right)\right)}{\left(u^{\tau}\right)^{2}\mathcal{D}_{1}} \\
L_{64} = &\frac{3\tau^{2}\tau_{\Pi}u^{r}\left(\left(\partial_{T}n\right)u^{r}\left(u^{\tau}\right)^{2}\left(p'(T)T + \Pi - \pi^{\eta}{}_{\eta} - \pi^{\phi}{}_{\phi}\right) - p''(T)T\left(nu^{r}\left(u^{\tau}\right)^{2} + \nu^{r}\right)\right)}{\left(u^{\tau}\right)^{2}\mathcal{D}_{1}} \\
L_{65} = &\frac{3\tau_{\pi}u^{r}\left(p''(T)T\left(nu^{r}\left(u^{\tau}\right)^{2} + \nu^{r}\right) - \left(\partial_{T}n\right)u^{r}\left(u^{\tau}\right)^{2}\left(p'(T)T + \Pi - \pi^{\eta}{}_{\eta} - \pi^{\phi}{}_{\phi}\right)\right)}{\left(u^{\tau}\right)^{2}\mathcal{D}_{1}} \\
L_{71} = &-\frac{3\nu^{r}\tau_{\Pi}\tau_{n}\tau_{\pi}\left(u^{r}\right)^{2}u^{\tau}(p''(T)T + p'(T))}{\mathcal{D}_{1}} \\
L_{72} = &\frac{3\nu^{r}\tau_{\Pi}\tau_{n}\tau_{\pi}u^{r}\left(p''(T)T\left(u^{\tau}\right)^{2} + p'(T)\left(u^{r}\right)^{2}\right)}{\mathcal{D}_{1}} \\
L_{73} = &-\frac{3p''(T)\nu^{r}r^{2}\tau_{\Pi}\tau_{n}\left(u^{r}\right)^{2}T}{\mathcal{D}_{2}} \\
L_{74} = &-\frac{3p''(T)\nu^{r}\tau^{2}\tau_{\Pi}\tau_{n}\left(u^{r}\right)^{2}T}{\mathcal{D}_{2}}
\end{aligned}
$$

$$L_{75} = \frac{3p''(T)\nu^r\tau_n\tau_\pi \left(u^r\right)^2 T}{\mathcal{D}_2}$$

The file, charm rows (M-V7e, M-G7e, M-I3ne, M-I3ae):

$$\begin{aligned}
(A_t)_{66} &= (\partial_\alpha n)u^\tau \\
(A_t)_{67} &= \frac{u^r}{u^\tau} \\
(A_t)_{76} &= \kappa u^r u^\tau \\
(A_t)_{77} &= \tau_n u^\tau \\
(A_r)_{61} &= -\frac{3}{\left(u^\tau\right)^2 \mathcal{D}_3} p'(T)\tau_\Pi\tau_\pi\Big((\partial_T n)u^r \left(u^\tau\right)^2 (-p'(T)T - \Pi + \pi^\eta{}_\eta + \pi^\phi{}_\phi) \\
&\quad + p''(T)T\left(nu^r \left(u^\tau\right)^2 + \nu^r\right)\Big) \\
(A_r)_{62} &= \frac{1}{\left(u^\tau\right)^2 \mathcal{D}_3}\Big(3p'(T)\nu^r\tau_\Pi\tau_\pi u^r(p'(T)T + \Pi - \pi^\eta{}_\eta - \pi^\phi{}_\phi) \\
&\quad - 3(\partial_T n)\tau_\Pi\tau_\pi \left(u^\tau\right)^2 (-p'(T)T - \Pi + \pi^\eta{}_\eta + \pi^\phi{}_\phi)^2 \\
&\quad + p''(T)T\Big(3n\tau_\Pi\tau_\pi \left(u^\tau\right)^2 (p'(T)T + \Pi - \pi^\eta{}_\eta - \pi^\phi{}_\phi) \\
&\quad - \nu^r u^r\Big(3\Pi\tau_\Pi\tau_\pi + \pi^\eta{}_\eta\tau_\Pi\tau_\pi + \pi^\phi{}_\phi\tau_\Pi\tau_\pi + 3p'(T)\tau_\Pi\tau_\pi T \\
&\quad - 4\tau_\Pi\eta - 3\tau_\pi\zeta\Big)\Big)\Big) \\
(A_r)_{63} &= \frac{3\tau_\Pi\tau_\pi\left(p''(T)T\left(nu^r \left(u^\tau\right)^2 + \nu^r\right) + (\partial_T n)u^r \left(u^\tau\right)^2 (-p'(T)T - \Pi + \pi^\eta{}_\eta + \pi^\phi{}_\phi)\right)}{\left(u^\tau\right)^2 \mathcal{D}_3} \\
(A_r)_{64} &= \frac{3\tau_\Pi\tau_\pi\left(p''(T)T\left(nu^r \left(u^\tau\right)^2 + \nu^r\right) + (\partial_T n)u^r \left(u^\tau\right)^2 (-p'(T)T - \Pi + \pi^\eta{}_\eta + \pi^\phi{}_\phi)\right)}{\left(u^\tau\right)^2 \mathcal{D}_3} \\
(A_r)_{65} &= -\frac{3\tau_\Pi\tau_\pi\left(p''(T)T\left(nu^r \left(u^\tau\right)^2 + \nu^r\right) + (\partial_T n)u^r \left(u^\tau\right)^2 (-p'(T)T - \Pi + \pi^\eta{}_\eta + \pi^\phi{}_\phi)\right)}{\left(u^\tau\right)^2 \mathcal{D}_3} \\
(A_r)_{66} &= (\partial_\alpha n)u^r \\
(A_r)_{67} &= 1 \\
(A_r)_{71} &= \frac{3p''(T)p'(T)\nu^r\tau_\Pi\tau_n\tau_\pi u^r T}{\mathcal{D}_3} \\
(A_r)_{72} &= \frac{1}{\left(u^\tau\right)^2 \mathcal{D}_3}\nu^r\tau_n \left(u^r\right)^2 \Big(3p'(T)\tau_\Pi\tau_\pi(-\Pi + \pi^\eta{}_\eta + \pi^\phi{}_\phi) - 3\left(p'(T)\right)^2\tau_\Pi\tau_\pi T \\
&\quad + p''(T)T(4\tau_\Pi\tau_\pi(\pi^\eta{}_\eta + \pi^\phi{}_\phi) - 4\tau_\Pi\eta - 3\tau_\pi\zeta)\Big) \\
(A_r)_{73} &= -\frac{3p''(T)\nu^r\tau_\Pi\tau_n\tau_\pi u^r T}{\mathcal{D}_3} \\
(A_r)_{74} &= -\frac{3p''(T)\nu^r\tau_\Pi\tau_n\tau_\pi u^r T}{\mathcal{D}_3} \\
(A_r)_{75} &= \frac{3p''(T)\nu^r\tau_\Pi\tau_n\tau_\pi u^r T}{\mathcal{D}_3} \\
(A_r)_{76} &= \kappa \left(u^\tau\right)^2 \\
(A_r)_{77} &= \tau_n u^r \\
S_6 &= \frac{1}{r\tau \left(u^\tau\right)^3 \mathcal{D}_4}\Big(3p'(T)\tau_\Pi\tau_\pi u^r\Big(p'(T)\nu^r T\left(r\left(\left(u^r\right)^4 - 1\right) + \tau\left(u^r\right)^3 u^\tau\right) \\
&\quad - nu^r \left(u^\tau\right)^2\left(r\left(u^\tau\right)^2(2\pi^\eta{}_\eta + \pi^\phi{}_\phi) + \tau u^r u^\tau(\pi^\eta{}_\eta + 2\pi^\phi{}_\phi)\right) \\
&\quad + \nu^r\Big(\tau u^r u^\tau\left(\left(u^r\right)^2(\Pi - \pi^\eta{}_\eta - \pi^\phi{}_\phi) - \pi^\eta{}_\eta - 2\pi^\phi{}_\phi\right) \\
&\quad - r\left(u^\tau\right)^2\left(-\Pi\left(u^r\right)^2 + \Pi + \pi^\eta{}_\eta\left(u^r\right)^2 + \pi^\eta{}_\eta + \pi^\phi{}_\phi\left(u^r\right)^2\right)\Big)\Big)
\end{aligned}$$

$$
\begin{aligned}
&+ (\partial_T n)\left(u^\tau\right)^2 \Big(3\left(p'(T)\right)^2 \tau_\Pi \tau_\pi T^2 \left(u^\tau\right)^2 \Big(r\left(u^r\right)^2 + r + \tau u^r u^\tau\Big) \\
&- 3p'(T)T\Big(2\tau\tau_\Pi u^r u^\tau \Big(\left(u^r\right)^2 (\eta - \tau_\pi(\Pi + {\pi^\phi}_\phi)) - \tau_\pi(\Pi + {\pi^\phi}_\phi)\Big) \\
&+ r\Big(-\Pi\tau\tau_\pi \left(u^r\right)^2 u^\tau \\
&+ \tau_\Pi\Big(-2\Pi\tau_\pi + {\pi^\phi}_\phi\tau_\pi + 2\left(u^r\right)^4 (\eta - \tau_\pi(\Pi + {\pi^\eta}_\eta)) \\
&+ \left(u^r\right)^2 \Big(\tau_\pi(-4\Pi - 2{\pi^\eta}_\eta + {\pi^\phi}_\phi) + \tau u^\tau({\pi^\eta}_\eta + {\pi^\phi}_\phi) + 2\eta\Big)\Big)\Big)\Big) \\
&- \tau u^r u^\tau\Big(-3\tau_\Pi\tau_\pi(\Pi - {\pi^\eta}_\eta - {\pi^\phi}_\phi)(\Pi + {\pi^\eta}_\eta + 3{\pi^\phi}_\phi) \\
&+ \tau_\Pi \left(u^r\right)^2 \Big(-\tau_\pi \Big(3\Pi^2 + 6\Pi{\pi^\phi}_\phi + \left({\pi^\eta}_\eta\right)^2 - \left({\pi^\phi}_\phi\right)^2\Big) \\
&+ 2\eta(3\Pi - {\pi^\eta}_\eta + {\pi^\phi}_\phi)\Big) \\
&+ 3\tau_\pi \left(u^r\right)^2 \zeta({\pi^\eta}_\eta + 2{\pi^\phi}_\phi)\Big) \\
&- r\Big(\tau_\Pi\Big(-3\tau_\pi(\Pi + {\pi^\eta}_\eta)(\Pi - {\pi^\eta}_\eta - {\pi^\phi}_\phi) \\
&+ \left(u^r\right)^2 \Big(-\tau_\pi\Big(6\Pi^2 - 3\Pi({\pi^\phi}_\phi - 2{\pi^\eta}_\eta) \\
&+ ({\pi^\phi}_\phi - 4{\pi^\eta}_\eta)({\pi^\eta}_\eta + {\pi^\phi}_\phi)\Big) \\
&+ 3\tau u^\tau({\pi^\eta}_\eta + {\pi^\phi}_\phi)(\Pi - {\pi^\eta}_\eta - {\pi^\phi}_\phi) \\
&+ 2\eta(3\Pi + {\pi^\eta}_\eta - {\pi^\phi}_\phi)\Big) \\
&+ \left(u^r\right)^4 \Big(-3\Pi^2\tau_\pi + ({\pi^\eta}_\eta - {\pi^\phi}_\phi)({\pi^\eta}_\eta\tau_\pi + {\pi^\phi}_\phi\tau_\pi + 2\eta) \\
&+ \Pi(6\eta - 6{\pi^\eta}_\eta\tau_\pi)\Big)\Big) \\
&+ 3\tau_\pi \left(u^r\right)^2 \Big(\Pi\tau u^\tau(-\Pi + {\pi^\eta}_\eta + {\pi^\phi}_\phi) + \left(u^\tau\right)^2 \zeta(2{\pi^\eta}_\eta + {\pi^\phi}_\phi)\Big)\Big)\Big) \\
&- p''(T)T\Big(3p'(T)\tau_\Pi\tau_\pi T \left(u^\tau\right)^4 \Big(nr\left(u^\tau\right)^2 + n\tau u^r u^\tau + \nu^r r u^r + \nu^r\tau u^\tau\Big) \\
&- 3n\left(u^\tau\right)^2 \Big(\tau\tau_\Pi u^r u^\tau\Big(\left(u^r\right)^2 (2\eta - \tau_\pi(\Pi + {\pi^\phi}_\phi)) - \tau_\pi(\Pi + {\pi^\phi}_\phi)\Big) \\
&+ r\Big(-\Pi\tau\tau_\pi \left(u^r\right)^2 u^\tau \\
&+ \tau_\Pi\Big(\tau_\pi(-\Pi + {\pi^\eta}_\eta + {\pi^\phi}_\phi) \\
&+ \left(u^r\right)^4 (2\eta - \tau_\pi(\Pi + {\pi^\eta}_\eta)) \\
&+ \left(u^r\right)^2 (\tau_\pi({\pi^\phi}_\phi - 2\Pi) + \tau u^\tau({\pi^\eta}_\eta + {\pi^\phi}_\phi) + 2\eta)\Big)\Big)\Big) \\
&+ \nu^r\Big(\tau u^\tau\Big(\tau_\Pi\Big(3\tau_\pi(\Pi + {\pi^\phi}_\phi) \\
&+ \left(u^r\right)^4 (\tau_\pi(3\Pi + {\pi^\eta}_\eta + {\pi^\phi}_\phi) - 4\eta) \\
&- 3\left(u^r\right)^2 (-2\Pi\tau_\pi + {\pi^\eta}_\eta\tau_\pi + 2\eta)\Big) - 3\tau_\pi \left(u^r\right)^4 \zeta\Big) \\
&+ r u^r\Big(-3\tau\tau_\Pi u^\tau({\pi^\eta}_\eta + {\pi^\phi}_\phi) \\
&+ \tau_\Pi\tau_\pi \left(u^\tau\right)^2 \Big(3\Pi\left(u^\tau\right)^2 + \left(u^r\right)^2({\pi^\eta}_\eta + {\pi^\phi}_\phi) - {\pi^\eta}_\eta - 4{\pi^\phi}_\phi\Big) \\
&- 2\tau_\Pi\left(2\left(u^r\right)^4 + 3\left(u^r\right)^2 + 1\right)\eta \\
&+ 3\tau_\pi\left(\Pi\tau u^\tau + \left(u^r\right)^4(-\zeta) + \zeta\right)\Big)\Big)\Big)\Big) \\
S_7 = \frac{1}{r\tau\mathcal{D}_3}&\nu^r\Big(-3p'(T)\tau_\Pi\tau_\pi \left(u^r\right)^2 \Big(r\tau(\Pi - {\pi^\eta}_\eta - {\pi^\phi}_\phi) + \tau\tau_n u^r(\Pi + {\pi^\phi}_\phi) \\
&+ r\tau_n u^\tau(\Pi + {\pi^\eta}_\eta) + p'(T)T(r(\tau + \tau_n u^\tau) + \tau\tau_n u^r)\Big) \\
&+ p''(T)T\Big(3p'(T)r\tau\tau_\Pi\tau_\pi T \left(u^\tau\right)^2 \\
&+ \tau\tau_n u^r\Big({\pi^\eta}_\eta\tau_\Pi\tau_\pi \Big(\left(u^r\right)^2 - 3\Big) - 2{\pi^\phi}_\phi\tau_\Pi\tau_\pi \Big(\left(u^r\right)^2 + 3\Big)
\end{aligned}
$$

$$
\begin{aligned}
&+ (u^r)^2 (2\tau_\Pi \eta - 3\tau_\pi \zeta)\Big) \\
&+ r\Big(\tau\tau_\Pi\Big(3\Pi\tau_\pi - 3\tau_\pi(\pi^\eta{}_\eta + \pi^\phi{}_\phi) \\
&+ (u^r)^2 \Big(\tau_\pi(3\Pi + \pi^\eta{}_\eta + \pi^\phi{}_\phi) + 3\tau_n(\pi^\eta{}_\eta + \pi^\phi{}_\phi) - 4\eta\Big)\Big) \\
&- 3\tau\tau_\pi (u^r)^2 (\Pi\tau_n + \zeta) \\
&+ \tau_n (u^r)^2 u^\tau(\tau_\Pi\tau_\pi(\pi^\phi{}_\phi - 2\pi^\eta{}_\eta) + 2\tau_\Pi\eta - 3\tau_\pi\zeta)\Big)\Big)\Big)
\end{aligned}
$$

Rows 1–5 are the viscous section's system at $\delta_{\pi\pi} = \frac{4}{3}\tau_\pi$ (M-V7m).

4.3.3 `one_d_viscous_matrix_gamma` in `src/Matrix/1d_viscous_HQ_cilindrical_gamma.jl`

Derived, charm rows:

$$
\begin{aligned}
(A_t)_{61} &= (\partial_T n)\gamma u^\tau \\
(A_t)_{62} &= \frac{\gamma n u^r (u^\tau)^2 + \nu^r}{(u^\tau)^3} \\
(A_t)_{66} &= n u^\tau \\
(A_t)_{67} &= \frac{u^r}{u^\tau} \\
(A_t)_{72} &= -\frac{\nu^r \tau_n u^r}{u^\tau} \\
(A_t)_{76} &= \kappa u^r u^\tau \\
(A_t)_{77} &= \tau_n u^\tau \\
(A_r)_{61} &= (\partial_T n)\gamma u^r \\
(A_r)_{62} &= \gamma n \\
(A_r)_{66} &= n u^r \\
(A_r)_{67} &= 1 \\
(A_r)_{72} &= -\frac{\nu^r \tau_n (u^r)^2}{(u^\tau)^2} \\
(A_r)_{76} &= \kappa (u^\tau)^2 \\
(A_r)_{77} &= \tau_n u^r \\
S_6 &= \frac{\gamma n \left(r (u^r)^2 + r + \tau u^r u^\tau\right) + \nu^r (r u^r + \tau u^\tau)}{r\tau u^\tau} \\
S_7 &= \nu^r
\end{aligned}
$$

With c the charm rows and m the medium rows, $\mathrm{row}_c \to \mathrm{row}_c - L\,\mathrm{row}_m$, $L = A_t^{cm}(A_t^{mm})^{-1}$, removes ∂_τ of the medium from the charm rows:

$$
\begin{aligned}
\mathcal{D}_1 = {} & 3 (p'(T))^2 \tau_\Pi \tau_\pi (u^r)^2 T \\
& + 3p'(T)\tau_\Pi\tau_\pi \left(-p''(T) (u^r)^2 T^2 - p''(T)T^2 + \Pi (u^r)^2 - \pi^\eta{}_\eta (u^r)^2 - \pi^\phi{}_\phi (u^r)^2\right) \\
& - p''(T)T\Big(-3\pi^\phi{}_\phi\tau_\Pi\tau_\pi + \pi^\phi{}_\phi\tau_\Pi\tau_\pi (u^r)^2 + \pi^\eta{}_\eta\tau_\Pi\tau_\pi \left((u^r)^2 - 3\right) + 3\Pi\tau_\Pi\tau_\pi (u^\tau)^2 \\
& - 4\tau_\Pi (u^r)^2 \eta - 3\tau_\pi (u^r)^2 \zeta\Big) \\
\mathcal{D}_2 = {} & -3 (p'(T))^2 \tau_\Pi \tau_\pi (u^r)^2 T \\
& + 3p'(T)\tau_\Pi\tau_\pi \left(p''(T) (u^r)^2 T^2 + p''(T)T^2 - \Pi (u^r)^2 + \pi^\eta{}_\eta (u^r)^2 + \pi^\phi{}_\phi (u^r)^2\right) \\
& + p''(T)T\Big(-3\pi^\phi{}_\phi\tau_\Pi\tau_\pi + \pi^\phi{}_\phi\tau_\Pi\tau_\pi (u^r)^2 + \pi^\eta{}_\eta\tau_\Pi\tau_\pi \left((u^r)^2 - 3\right) + 3\Pi\tau_\Pi\tau_\pi (u^\tau)^2 \\
& - 4\tau_\Pi (u^r)^2 \eta - 3\tau_\pi (u^r)^2 \zeta\Big)
\end{aligned}
$$

$$
\begin{aligned}
\mathcal{D}_3 &= 3p'(T)\tau_\Pi\tau_\pi \left(u^r\right)^2 \left(-p'(T)T - \Pi + \pi^\eta{}_\eta + \pi^\phi{}_\phi\right) \\
&\quad + p''(T)T\Big(3p'(T)\tau_\Pi\tau_\pi T \left(u^\tau\right)^2 \\
&\quad + \tau_\Pi \left(\left(u^r\right)^2 \left(\tau_\pi(3\Pi + \pi^\eta{}_\eta + \pi^\phi{}_\phi) - 4\eta\right) + 3\Pi\tau_\pi - 3\tau_\pi(\pi^\eta{}_\eta + \pi^\phi{}_\phi)\right) \\
&\quad - 3\tau_\pi \left(u^r\right)^2 \zeta\Big) \\
\mathcal{D}_4 &= 3p'(T)\tau_\Pi\tau_\pi \left(u^r\right)^2 \left(p'(T)T + \Pi - \pi^\eta{}_\eta - \pi^\phi{}_\phi\right) \\
&\quad - p''(T)T\Big(3p'(T)\tau_\Pi\tau_\pi T \left(u^\tau\right)^2 \\
&\quad + \tau_\Pi \left(\left(u^r\right)^2 \left(\tau_\pi(3\Pi + \pi^\eta{}_\eta + \pi^\phi{}_\phi) - 4\eta\right) + 3\Pi\tau_\pi - 3\tau_\pi(\pi^\eta{}_\eta + \pi^\phi{}_\phi)\right) \\
&\quad - 3\tau_\pi \left(u^r\right)^2 \zeta\Big)
\end{aligned}
$$

$$
\begin{aligned}
L_{61} &= \frac{1}{u^\tau \mathcal{D}_1}\Big(3\tau_\Pi\tau_\pi u^r (p''(T)T + p'(T)) \left(\gamma n u^r \left(u^\tau\right)^2 + \nu^r\right) \\
&\quad + (\partial_T n)\gamma \left(u^\tau\right)^2 \Big(3\pi^\phi{}_\phi\tau_\Pi\tau_\pi + 2\pi^\phi{}_\phi\tau_\Pi\tau_\pi \left(u^r\right)^2 - 3\Pi\tau_\Pi\tau_\pi \left(2\left(u^r\right)^2 + 1\right) \\
&\quad + \pi^\eta{}_\eta\tau_\Pi\tau_\pi \left(2\left(u^r\right)^2 + 3\right) - 3p'(T)\tau_\Pi\tau_\pi T \\
&\quad - 6p'(T)\tau_\Pi\tau_\pi \left(u^r\right)^2 T + 4\tau_\Pi \left(u^r\right)^2 \eta + 3\tau_\pi \left(u^r\right)^2 \zeta\Big)\Big) \\
L_{62} &= \frac{1}{\left(u^\tau\right)^2 \mathcal{D}_1}\Big(-3\tau_\Pi\tau_\pi \left(p''(T)T \left(u^\tau\right)^2 + p'(T) \left(u^r\right)^2\right) \left(\gamma n u^r \left(u^\tau\right)^2 + \nu^r\right) \\
&\quad + (\partial_T n)\gamma u^r \left(u^\tau\right)^2 \Big(-6\pi^\phi{}_\phi\tau_\Pi\tau_\pi - 2\pi^\phi{}_\phi\tau_\Pi\tau_\pi \left(u^r\right)^2 - 2\pi^\eta{}_\eta\tau_\Pi\tau_\pi \left(\left(u^r\right)^2 + 3\right) \\
&\quad + 6p'(T)\tau_\Pi\tau_\pi T + 6p'(T)\tau_\Pi\tau_\pi \left(u^r\right)^2 T + 6\Pi\tau_\Pi\tau_\pi \left(u^\tau\right)^2 \\
&\quad - 4\tau_\Pi \left(u^r\right)^2 \eta - 3\tau_\pi \left(u^r\right)^2 \zeta\Big)\Big) \\
L_{63} &= \frac{3r^2\tau_\Pi u^r \left((\partial_T n)\gamma u^r \left(u^\tau\right)^2 \left(p'(T)T + \Pi - \pi^\eta{}_\eta - \pi^\phi{}_\phi\right) - p''(T)T \left(\gamma n u^r \left(u^\tau\right)^2 + \nu^r\right)\right)}{\left(u^\tau\right)^2 \mathcal{D}_1} \\
L_{64} &= \frac{3\tau^2\tau_\Pi u^r \left((\partial_T n)\gamma u^r \left(u^\tau\right)^2 \left(p'(T)T + \Pi - \pi^\eta{}_\eta - \pi^\phi{}_\phi\right) - p''(T)T \left(\gamma n u^r \left(u^\tau\right)^2 + \nu^r\right)\right)}{\left(u^\tau\right)^2 \mathcal{D}_1} \\
L_{65} &= \frac{3\tau_\pi u^r \left(p''(T)T \left(\gamma n u^r \left(u^\tau\right)^2 + \nu^r\right) - (\partial_T n)\gamma u^r \left(u^\tau\right)^2 \left(p'(T)T + \Pi - \pi^\eta{}_\eta - \pi^\phi{}_\phi\right)\right)}{\left(u^\tau\right)^2 \mathcal{D}_1} \\
L_{71} &= -\frac{3\nu^r\tau_\Pi\tau_n\tau_\pi \left(u^r\right)^2 u^\tau (p''(T)T + p'(T))}{\mathcal{D}_1} \\
L_{72} &= \frac{3\nu^r\tau_\Pi\tau_n\tau_\pi u^r \left(p''(T)T \left(u^\tau\right)^2 + p'(T) \left(u^r\right)^2\right)}{\mathcal{D}_1} \\
L_{73} &= -\frac{3p''(T)\nu^r r^2\tau_\Pi\tau_n \left(u^r\right)^2 T}{\mathcal{D}_2} \\
L_{74} &= -\frac{3p''(T)\nu^r\tau^2\tau_\Pi\tau_n \left(u^r\right)^2 T}{\mathcal{D}_2} \\
L_{75} &= \frac{3p''(T)\nu^r\tau_n\tau_\pi \left(u^r\right)^2 T}{\mathcal{D}_2}
\end{aligned}
$$

The file, charm rows (M-V7e, M-G7e, M-I3ne, M-I3ae):

$$
\begin{aligned}
(A_t)_{66} &= nu^\tau \\
(A_t)_{67} &= \frac{u^r}{u^\tau} \\
(A_t)_{76} &= \kappa u^r u^\tau \\
(A_t)_{77} &= \tau_n u^\tau
\end{aligned}
$$

$$
\begin{aligned}
(A_r)_{61} &= -\frac{3}{(u^\tau)^2 \mathcal{D}_3} p'(T)\tau_\Pi\tau_\pi\Big((\partial_T n)\gamma u^r (u^\tau)^2 (-p'(T)T - \Pi + \pi^\eta{}_\eta + \pi^\phi{}_\phi) \\
&\quad + p''(T)T\left(\gamma n u^r (u^\tau)^2 + \nu^r\right)\Big) \\
(A_r)_{62} &= \frac{1}{(u^\tau)^2 \mathcal{D}_3}\Big(3p'(T)\nu^r\tau_\Pi\tau_\pi u^r(p'(T)T + \Pi - \pi^\eta{}_\eta - \pi^\phi{}_\phi) \\
&\quad - 3(\partial_T n)\gamma\tau_\Pi\tau_\pi (u^\tau)^2 (-p'(T)T - \Pi + \pi^\eta{}_\eta + \pi^\phi{}_\phi)^2 \\
&\quad + p''(T)T\Big(3\gamma n\tau_\Pi\tau_\pi (u^\tau)^2 (p'(T)T + \Pi - \pi^\eta{}_\eta - \pi^\phi{}_\phi) \\
&\quad - \nu^r u^r\Big(3\Pi\tau_\Pi\tau_\pi + \pi^\eta{}_\eta\tau_\Pi\tau_\pi + \pi^\phi{}_\phi\tau_\Pi\tau_\pi + 3p'(T)\tau_\Pi\tau_\pi T \\
&\quad - 4\tau_\Pi\eta - 3\tau_\pi\zeta\Big)\Big)\Big) \\
(A_r)_{63} &= \frac{3\tau_\Pi\tau_\pi\left(p''(T)T\left(\gamma n u^r (u^\tau)^2 + \nu^r\right) + (\partial_T n)\gamma u^r (u^\tau)^2 (-p'(T)T - \Pi + \pi^\eta{}_\eta + \pi^\phi{}_\phi)\right)}{(u^\tau)^2 \mathcal{D}_3} \\
(A_r)_{64} &= \frac{3\tau_\Pi\tau_\pi\left(p''(T)T\left(\gamma n u^r (u^\tau)^2 + \nu^r\right) + (\partial_T n)\gamma u^r (u^\tau)^2 (-p'(T)T - \Pi + \pi^\eta{}_\eta + \pi^\phi{}_\phi)\right)}{(u^\tau)^2 \mathcal{D}_3} \\
(A_r)_{65} &= -\frac{3\tau_\Pi\tau_\pi\left(p''(T)T\left(\gamma n u^r (u^\tau)^2 + \nu^r\right) + (\partial_T n)\gamma u^r (u^\tau)^2 (-p'(T)T - \Pi + \pi^\eta{}_\eta + \pi^\phi{}_\phi)\right)}{(u^\tau)^2 \mathcal{D}_3} \\
(A_r)_{66} &= n u^r \\
(A_r)_{67} &= 1 \\
(A_r)_{71} &= \frac{3p''(T)p'(T)\nu^r\tau_\Pi\tau_n\tau_\pi u^r T}{\mathcal{D}_3} \\
(A_r)_{72} &= \frac{1}{(u^\tau)^2 \mathcal{D}_3}\nu^r\tau_n (u^r)^2\Big(3p'(T)\tau_\Pi\tau_\pi(-\Pi + \pi^\eta{}_\eta + \pi^\phi{}_\phi) - 3(p'(T))^2\tau_\Pi\tau_\pi T \\
&\quad + p''(T)T(4\tau_\Pi\tau_\pi(\pi^\eta{}_\eta + \pi^\phi{}_\phi) - 4\tau_\Pi\eta - 3\tau_\pi\zeta)\Big) \\
(A_r)_{73} &= -\frac{3p''(T)\nu^r\tau_\Pi\tau_n\tau_\pi u^r T}{\mathcal{D}_3} \\
(A_r)_{74} &= -\frac{3p''(T)\nu^r\tau_\Pi\tau_n\tau_\pi u^r T}{\mathcal{D}_3} \\
(A_r)_{75} &= \frac{3p''(T)\nu^r\tau_\Pi\tau_n\tau_\pi u^r T}{\mathcal{D}_3} \\
(A_r)_{76} &= \kappa (u^\tau)^2 \\
(A_r)_{77} &= \tau_n u^r \\
S_6 &= \frac{1}{r\tau (u^\tau)^3 \mathcal{D}_4}\Big(3p'(T)\tau_\Pi\tau_\pi u^r\Big(p'(T)\nu^r T\left(r\left((u^r)^4 - 1\right) + \tau (u^r)^3 u^\tau\right) \\
&\quad - \gamma n u^r (u^\tau)^2\left(r (u^\tau)^2 (2\pi^\eta{}_\eta + \pi^\phi{}_\phi) + \tau u^r u^\tau(\pi^\eta{}_\eta + 2\pi^\phi{}_\phi)\right) \\
&\quad + \nu^r\Big(\tau u^r u^\tau\left((u^r)^2(\Pi - \pi^\eta{}_\eta - \pi^\phi{}_\phi) - \pi^\eta{}_\eta - 2\pi^\phi{}_\phi\right) \\
&\quad - r (u^\tau)^2\left(-\Pi (u^r)^2 + \Pi + \pi^\eta{}_\eta (u^r)^2 + \pi^\eta{}_\eta + \pi^\phi{}_\phi (u^r)^2\right)\Big)\Big) \\
&\quad + (\partial_T n)\gamma (u^\tau)^2\Big(3(p'(T))^2\tau_\Pi\tau_\pi T^2 (u^\tau)^2\left(r (u^r)^2 + r + \tau u^r u^\tau\right) \\
&\quad - 3p'(T)T\Big(2\tau\tau_\Pi u^r u^\tau\left((u^r)^2(\eta - \tau_\pi(\Pi + \pi^\phi{}_\phi)) - \tau_\pi(\Pi + \pi^\phi{}_\phi)\right) \\
&\quad + r\Big(-\Pi\tau\tau_\pi (u^r)^2 u^\tau \\
&\quad + \tau_\Pi\Big(-2\Pi\tau_\pi + \pi^\phi{}_\phi\tau_\pi + 2(u^r)^4(\eta - \tau_\pi(\Pi + \pi^\eta{}_\eta)) \\
&\quad + (u^r)^2\left(\tau_\pi(-4\Pi - 2\pi^\eta{}_\eta + \pi^\phi{}_\phi) + \tau u^\tau(\pi^\eta{}_\eta + \pi^\phi{}_\phi) + 2\eta\right)\Big)\Big)\Big) \\
&\quad - \tau u^r u^\tau\Big(-3\tau_\Pi\tau_\pi(\Pi - \pi^\eta{}_\eta - \pi^\phi{}_\phi)(\Pi + \pi^\eta{}_\eta + 3\pi^\phi{}_\phi) \\
&\quad + \tau_\Pi (u^r)^2\left(-\tau_\pi\left(3\Pi^2 + 6\Pi\pi^\phi{}_\phi + (\pi^\eta{}_\eta)^2 - \left(\pi^\phi{}_\phi\right)^2\right)\right.
\end{aligned}
$$

$$
\begin{aligned}
&+ 2\eta(3\Pi - \pi^{\eta}{}_{\eta} + \pi^{\phi}{}_{\phi})\Big) \\
&+ 3\tau_\pi \left(u^r\right)^2 \zeta(\pi^{\eta}{}_{\eta} + 2\pi^{\phi}{}_{\phi})\Big) \\
&- r\Big(\tau_\Pi\Big(-3\tau_\pi(\Pi + \pi^{\eta}{}_{\eta})(\Pi - \pi^{\eta}{}_{\eta} - \pi^{\phi}{}_{\phi}) \\
&+ \left(u^r\right)^2 \Big(-\tau_\pi\Big(6\Pi^2 - 3\Pi(\pi^{\phi}{}_{\phi} - 2\pi^{\eta}{}_{\eta}) \\
&+ (\pi^{\phi}{}_{\phi} - 4\pi^{\eta}{}_{\eta})(\pi^{\eta}{}_{\eta} + \pi^{\phi}{}_{\phi})\Big) \\
&+ 3\tau u^\tau(\pi^{\eta}{}_{\eta} + \pi^{\phi}{}_{\phi})(\Pi - \pi^{\eta}{}_{\eta} - \pi^{\phi}{}_{\phi}) \\
&+ 2\eta(3\Pi + \pi^{\eta}{}_{\eta} - \pi^{\phi}{}_{\phi})\Big) \\
&+ \left(u^r\right)^4 \Big(-3\Pi^2\tau_\pi + (\pi^{\eta}{}_{\eta} - \pi^{\phi}{}_{\phi})(\pi^{\eta}{}_{\eta}\tau_\pi + \pi^{\phi}{}_{\phi}\tau_\pi + 2\eta) \\
&+ \Pi(6\eta - 6\pi^{\eta}{}_{\eta}\tau_\pi)\Big)\Big) \\
&+ 3\tau_\pi \left(u^r\right)^2 \Big(\Pi\tau u^\tau(-\Pi + \pi^{\eta}{}_{\eta} + \pi^{\phi}{}_{\phi}) + \left(u^\tau\right)^2 \zeta(2\pi^{\eta}{}_{\eta} + \pi^{\phi}{}_{\phi})\Big)\Big)\Big) \\
&- p''(T)T\Big(3p'(T)\tau_\Pi\tau_\pi T \left(u^\tau\right)^4 \Big(\gamma n \Big(r \left(u^r\right)^2 + r + \tau u^r u^\tau\Big) + \nu^r(ru^r + \tau u^\tau)\Big) \\
&- 3\gamma n \left(u^\tau\right)^2 \Big(\tau\tau_\Pi u^r u^\tau\Big(\left(u^r\right)^2 (2\eta - \tau_\pi(\Pi + \pi^{\phi}{}_{\phi})) - \tau_\pi(\Pi + \pi^{\phi}{}_{\phi})\Big) \\
&+ r\Big(-\Pi\tau\tau_\pi \left(u^r\right)^2 u^\tau \\
&+ \tau_\Pi\Big(\tau_\pi(-\Pi + \pi^{\eta}{}_{\eta} + \pi^{\phi}{}_{\phi}) \\
&+ \left(u^r\right)^4 (2\eta - \tau_\pi(\Pi + \pi^{\eta}{}_{\eta})) \\
&+ \left(u^r\right)^2 (\tau_\pi(\pi^{\phi}{}_{\phi} - 2\Pi) + \tau u^\tau(\pi^{\eta}{}_{\eta} + \pi^{\phi}{}_{\phi}) + 2\eta)\Big)\Big)\Big) \\
&+ \nu^r\Big(\tau u^\tau\Big(\tau_\Pi\Big(3\tau_\pi(\Pi + \pi^{\phi}{}_{\phi}) \\
&+ \left(u^r\right)^4 (\tau_\pi(3\Pi + \pi^{\eta}{}_{\eta} + \pi^{\phi}{}_{\phi}) - 4\eta) \\
&- 3\left(u^r\right)^2 (-2\Pi\tau_\pi + \pi^{\eta}{}_{\eta}\tau_\pi + 2\eta)\Big) - 3\tau_\pi \left(u^r\right)^4 \zeta\Big) \\
&+ ru^r\Big(-3\tau\tau_\Pi u^\tau(\pi^{\eta}{}_{\eta} + \pi^{\phi}{}_{\phi}) \\
&+ \tau_\Pi\tau_\pi \left(u^\tau\right)^2 \Big(3\Pi \left(u^\tau\right)^2 + \left(u^r\right)^2 (\pi^{\eta}{}_{\eta} + \pi^{\phi}{}_{\phi}) - \pi^{\eta}{}_{\eta} - 4\pi^{\phi}{}_{\phi}\Big) \\
&- 2\tau_\Pi \Big(2\left(u^r\right)^4 + 3\left(u^r\right)^2 + 1\Big)\eta \\
&+ 3\tau_\pi \Big(\Pi\tau u^\tau + \left(u^r\right)^4 (-\zeta) + \zeta\Big)\Big)\Big)\Big)\Big) \\
S_7 = \frac{1}{r\tau\mathcal{D}_3}&\nu^r\Big(-3p'(T)\tau_\Pi\tau_\pi \left(u^r\right)^2 \Big(r\tau(\Pi - \pi^{\eta}{}_{\eta} - \pi^{\phi}{}_{\phi}) + \tau\tau_n u^r(\Pi + \pi^{\phi}{}_{\phi}) \\
&+ r\tau_n u^\tau(\Pi + \pi^{\eta}{}_{\eta}) + p'(T)T(r(\tau + \tau_n u^\tau) + \tau\tau_n u^r)\Big) \\
&+ p''(T)T\Big(3p'(T)r\tau\tau_\Pi\tau_\pi T \left(u^\tau\right)^2 \\
&+ \tau\tau_n u^r\Big(\pi^{\eta}{}_{\eta}\tau_\Pi\tau_\pi \Big(\left(u^r\right)^2 - 3\Big) - 2\pi^{\phi}{}_{\phi}\tau_\Pi\tau_\pi \Big(\left(u^r\right)^2 + 3\Big) \\
&+ \left(u^r\right)^2 (2\tau_\Pi\eta - 3\tau_\pi\zeta)\Big) \\
&+ r\Big(\tau\tau_\Pi\Big(3\Pi\tau_\pi - 3\tau_\pi(\pi^{\eta}{}_{\eta} + \pi^{\phi}{}_{\phi}) \\
&+ \left(u^r\right)^2 \Big(\tau_\pi(3\Pi + \pi^{\eta}{}_{\eta} + \pi^{\phi}{}_{\phi}) + 3\tau_n(\pi^{\eta}{}_{\eta} + \pi^{\phi}{}_{\phi}) - 4\eta\Big)\Big) \\
&- 3\tau\tau_\pi \left(u^r\right)^2 (\Pi\tau_n + \zeta) \\
&+ \tau_n \left(u^r\right)^2 u^\tau(\tau_\Pi\tau_\pi(\pi^{\phi}{}_{\phi} - 2\pi^{\eta}{}_{\eta}) + 2\tau_\Pi\eta - 3\tau_\pi\zeta)\Big)\Big)\Big)
\end{aligned}
$$

$\partial_T n$ above is γ-free: n and $\partial_T n$ are those of $n(T)$ at $\gamma = 1$.

4.3.4 Ideal medium, rows 1–2

$$
\begin{aligned}
(A_t)_{11} &= (u^r)^2 (p''(T)T + p'(T)) + p''(T)T \\
(A_t)_{12} &= 2p'(T)u^r T \\
(A_t)_{21} &= u^r u^\tau (p''(T)T + p'(T)) \\
(A_t)_{22} &= \frac{p'(T)\left(2(u^r)^2 + 1\right)T}{u^\tau} \\
(A_r)_{11} &= u^r u^\tau (p''(T)T + p'(T)) \\
(A_r)_{12} &= \frac{p'(T)\left(2(u^r)^2 + 1\right)T}{u^\tau} \\
(A_r)_{21} &= p''(T)(u^r)^2 T + p'(T)(u^r)^2 + p'(T) \\
(A_r)_{22} &= 2p'(T)u^r T \\
S_1 &= \frac{p'(T)T\left(r(u^r)^2 + r + \tau u^r u^\tau\right)}{r\tau} \\
S_2 &= \frac{p'(T)u^r T(ru^\tau + \tau u^r)}{r\tau}
\end{aligned}
$$

4.3.5 `one_d_ideal_matrix_ruwen4` in `src/Matrix/1d_ideal_HQ_cilindrical_density.jl`

Derived, charm rows:

$$
\begin{aligned}
(A_t)_{32} &= \frac{nu^r}{u^\tau} \\
(A_t)_{33} &= u^\tau \\
(A_r)_{32} &= n \\
(A_r)_{33} &= u^r \\
S_3 &= n\left(\frac{u^r}{r} + \frac{u^\tau}{\tau}\right)
\end{aligned}
$$

With c the charm rows and m the medium rows, $\mathrm{row}_c \to \mathrm{row}_c - L\,\mathrm{row}_m$, $L = A_t^{cm}(A_t^{mm})^{-1}$, removes ∂_τ of the medium from the charm rows:

$$
\begin{aligned}
L_{31} &= \frac{n(u^r)^2 u^\tau (p''(T)T + p'(T))}{p'(T)T\left(p'(T)(u^r)^2 - p''(T)T(u^\tau)^2\right)} \\
L_{32} &= \frac{nu^r\left(p''(T)T(u^\tau)^2 + p'(T)(u^r)^2\right)}{p'(T)T\left(p''(T)T(u^\tau)^2 - p'(T)(u^r)^2\right)}
\end{aligned}
$$

The file, charm rows (M-V7e, M-G7e, M-I3ne, M-I3ae):

$$
\begin{aligned}
(A_t)_{33} &= u^\tau \\
(A_r)_{31} &= -\frac{p''(T)nu^r}{p''(T)T(u^\tau)^2 - p'(T)(u^r)^2} \\
(A_r)_{32} &= \frac{p''(T)nT}{p''(T)T(u^\tau)^2 - p'(T)(u^r)^2} \\
(A_r)_{33} &= u^r \\
S_3 &= \frac{p''(T)nT(u^\tau)^2(ru^\tau + \tau u^r)}{r\tau\left(p''(T)T(u^\tau)^2 - p'(T)(u^r)^2\right)}
\end{aligned}
$$

4.3.6 `one_d_ideal_matrix_ruwen5` in `src/Matrix/1d_ideal_HQ_cilindrical_fugacity.jl`

Derived, charm rows:

$$
\begin{aligned}
(A_t)_{31} &= (\partial_T n) u^\tau \\
(A_t)_{32} &= \frac{n u^r}{u^\tau} \\
(A_t)_{33} &= (\partial_\alpha n) u^\tau \\
(A_r)_{31} &= (\partial_T n) u^r \\
(A_r)_{32} &= n \\
(A_r)_{33} &= (\partial_\alpha n) u^r \\
S_3 &= n \left(\frac{u^r}{r} + \frac{u^\tau}{\tau} \right)
\end{aligned}
$$

With c the charm rows and m the medium rows, $\mathrm{row}_c \to \mathrm{row}_c - L\,\mathrm{row}_m$, $L = A_t^{cm}(A_t^{mm})^{-1}$, removes ∂_τ of the medium from the charm rows:

$$
\begin{aligned}
L_{31} &= \frac{u^\tau \left(p''(T) n \left(u^r\right)^2 T - (\partial_T n) p'(T) \left(2\left(u^r\right)^2 + 1 \right) T + p'(T) n \left(u^r\right)^2 \right)}{p'(T) T \left(p'(T) \left(u^r\right)^2 - p''(T) T \left(u^\tau\right)^2 \right)} \\
L_{32} &= -\frac{u^r \left(p''(T) n T \left(u^\tau\right)^2 - 2(\partial_T n) p'(T) T \left(u^\tau\right)^2 + p'(T) n \left(u^r\right)^2 \right)}{p'(T) T \left(p'(T) \left(u^r\right)^2 - p''(T) T \left(u^\tau\right)^2 \right)}
\end{aligned}
$$

The file, charm rows (M-V7e, M-G7e, M-I3ne, M-I3ae):

$$
\begin{aligned}
(A_t)_{33} &= (\partial_\alpha n) u^\tau \\
(A_r)_{31} &= \frac{u^r (p''(T) n - (\partial_T n) p'(T))}{p'(T) \left(u^r\right)^2 - p''(T) T \left(u^\tau\right)^2} \\
(A_r)_{32} &= \frac{T((\partial_T n) p'(T) - p''(T) n)}{p'(T) \left(u^r\right)^2 - p''(T) T \left(u^\tau\right)^2} \\
(A_r)_{33} &= (\partial_\alpha n) u^r \\
S_3 &= \frac{T \left(u^\tau\right)^2 ((\partial_T n) p'(T) - p''(T) n)(r u^\tau + \tau u^r)}{r\tau \left(p'(T) \left(u^r\right)^2 - p''(T) T \left(u^\tau\right)^2 \right)}
\end{aligned}
$$

4.4 Checks

id	what it tests	
`M-G0`	cylindrical Milne: $\nabla_\lambda g_{\mu\nu} = 0$	PASS
`M-G1`	$\nabla_\mu V^\mu = \partial_\tau X + X/\tau$ for $V^\mu = (X(\tau), 0, 0, 0)$: the Christoffels are in	PASS
`M-G2`	$u \cdot u = -1$, pi traceless and transverse	PASS
`M-V5q`	quasi-linear	PASS
`M-V5`	derived is + $\delta_{\pi\pi}\pi\theta$ = the shipped 5-field A_t, A_r, S, raw entry by entry, read from the .jl, symbolic $\delta_{\pi\pi}$	PASS
`M-V5-0`	FLUIDUM_DELTA_PIPI=0: $\delta_{\pi\pi} = 0$, plain is == the shipped is part	PASS
`M-V5-1`	FLUIDUM_DELTA_PIPI=1 (default): $\delta_{\pi\pi} = \frac{4}{3}\tau_\pi$	PASS
`M-BGq`	quasi-linear	PASS
`M-BG`	derived first moment$\times(u^\tau)^3 r\tau$ = 1d_viscous_HQ_cilindrical_BG.jl, raw, read from the .jl	PASS
`M-V7m-dpp`	the medium carries $\delta_{\pi\pi} = \frac{4}{3}\tau_\pi$ (DNMR $\delta_{\pi\pi}$)	PASS
`M-V7m`	medium rows/cols 1-5 = IS + $\frac{4}{3}\tau_\pi\pi\theta$, raw entry by entry	PASS
`M-V7m-ctl`	control: plain is ($\delta_{\pi\pi} = 0$) does not reproduce the 7-field medium block	PASS

id	what it tests	
`M-V5=V7m`	at the default $\delta_{\pi\pi} = \frac{4}{3}\tau_\pi$ the 5-field kernel is the 7-field :MIS medium block, raw entry by entry (both read from the .jl)	PASS
`M-V7q`	quasi-linear	PASS
`M-V7c`	$A_t^{-1}A_r$ and $A_t^{-1}S$ = derived, exactly (rational arithmetic), 8 random states	PASS
`M-V7e`	charm rows 6-7 of the .jl = derived rows with $\partial_\tau T, \partial_\tau u^r$ eliminated with rows 1-5, raw, exact, 8 random states	PASS
`M-V7e-ctl`	control: without the elimination the charm rows of A_r differ	PASS
`M-G7`	$n = \gamma n_0(T)$, $\kappa 0$ d gamma: solved form = derived exactly, 8 random states	PASS
`M-G7e`	charm rows 6-7 of the .jl = derived rows with $\partial_\tau T, \partial_\tau u^r$ eliminated, raw, exact, 8 random states	PASS
`M-I3n`	ruwen4 (T, u^r, n): solved form = derived exactly	PASS
`M-I3ne`	charm row 3 of the .jl = derived row with $\partial_\tau T, \partial_\tau u^r$ eliminated, raw, exact, 8 random states	PASS
`M-I3a`	ruwen5 (T, u^r, α): solved form = derived exactly	PASS
`M-I3ae`	charm row 3 of the .jl = derived row with $\partial_\tau T, \partial_\tau u^r$ eliminated, raw, exact, 8 random states	PASS

23 passed, 0 failed (23 checks evaluated).

Re-run with `wolframscript -file deriv/derive_1d_matrices.wls`.

5 The matrices no production path relies on

Script: `deriv/derive_upstream.wls`

Writes nothing into `src/Matrix`.

5.1 `two_d_viscous_matrix (legacy = true)`

Against the first section's matrix at $\delta_{\pi\pi} = 0$, $\phi = (T, u^x, u^y, \pi^{yy}, \pi^{\eta}{}_{\eta}, \pi^{xy}, \Pi)$. The source and 137 entries agree. The ten that differ, with $\delta = \text{legacy} - \text{derived}$:

$$
\begin{aligned}
(A_t)_{43}^{\text{legacy}} &= \frac{3\pi^{xy}\left(2\tau_\pi u^x \left(u^y\right)^2 + u^x\right) + \pi^{yy}u^y\left(3 - 6\tau_\pi\left(\left(u^x\right)^2 + 1\right)\right) + 2u^y\eta\left(3\left(u^x\right)^2 + 2\left(u^y\right)^2 + 2\right)}{3u^\tau} \\
(A_t)_{43}^{\text{derived}} &= \frac{2u^y\left(3\pi^{xy}\tau_\pi u^x u^y - 3\pi^{yy}\tau_\pi\left(\left(u^x\right)^2 + 1\right) + \eta\left(3\left(u^x\right)^2 + 2\left(u^y\right)^2 + 2\right)\right)}{3u^\tau} \\
\delta(A_t)_{43} &= \frac{\pi^{xy}u^x + \pi^{yy}u^y}{u^\tau} \\
(A_t)_{62}^{\text{legacy}} &= \frac{1}{6u^\tau}\Big(-3\pi^{xy}u^x\left(\tau_\pi\left(4\left(u^y\right)^2 + 2\right) - 1\right) \\
&\quad + u^y\left(6\pi^{\eta}{}_{\eta}\tau_\pi\left(u^\tau\right)^2 + 3\pi^{yy}\left(\tau_\pi\left(4\left(u^x\right)^2 + 2\right) + 1\right) - 4\left(u^x\right)^2\eta\right)\Big) \\
(A_t)_{62}^{\text{derived}} &= \frac{1}{3u^\tau}\Big(-3\pi^{xy}\tau_\pi u^x\left(2\left(u^y\right)^2 + 1\right) \\
&\quad + u^y\left(3\pi^{\eta}{}_{\eta}\tau_\pi\left(u^\tau\right)^2 + 3\pi^{yy}\left(2\tau_\pi\left(u^x\right)^2 + \tau_\pi\right) + \eta\left(\left(u^x\right)^2 + 3\left(u^y\right)^2 + 3\right)\right)\Big) \\
\delta(A_t)_{62} &= \frac{\pi^{xy}u^x + u^y\left(\pi^{yy} - 2\left(u^\tau\right)^2\eta\right)}{2u^\tau} \\
(A_t)_{63}^{\text{legacy}} &= \frac{1}{6u^\tau\left(\left(u^y\right)^2 + 1\right)}\Big(-3\pi^{xy}(2\tau_\pi - 1)u^y\left(\left(u^y\right)^2 + 1\right) + 6\pi^{xy}\left(u^x\right)^2 u^y\left(2\tau_\pi\left(u^y\right)^2 + 1\right) \\
&\quad - 3\left(u^x\right)^3\left(\pi^{\eta}{}_{\eta} + \pi^{yy} + 2\pi^{yy}\tau_\pi + 2\tau_\pi\left(u^y\right)^2\left(\pi^{\eta}{}_{\eta} + 2\pi^{yy}\right)\right. \\
&\quad \left. - 4\left(\left(u^y\right)^2 + 1\right)\eta\right) \\
&\quad + u^x\Big(-3(\pi^{\eta}{}_{\eta} + 2\pi^{yy}\tau_\pi + \pi^{yy}) \\
&\quad - 3\left(u^y\right)^2\left(2\pi^{\eta}{}_{\eta}\tau_\pi\left(\left(u^y\right)^2 + 1\right) + \pi^{\eta}{}_{\eta} + 4\pi^{yy}\tau_\pi\right) \\
&\quad + 4\left(2\left(u^y\right)^4 + 5\left(u^y\right)^2 + 3\right)\eta\Big)\Big) \\
(A_t)_{63}^{\text{derived}} &= \frac{1}{3u^\tau\left(\left(u^y\right)^2 + 1\right)}\Big(-3\pi^{yy}\tau_\pi u^x\left(\left(u^x\right)^2 + 1\right)\left(2\left(u^y\right)^2 + 1\right) \\
&\quad + 3\pi^{xy}\tau_\pi u^y\left(\left(2\left(u^x\right)^2 - 1\right)\left(u^y\right)^2 - 1\right) \\
&\quad + u^x\left(\left(\left(u^y\right)^2 + 1\right)\eta\left(3\left(u^x\right)^2 + \left(u^y\right)^2 + 3\right) - 3\pi^{\eta}{}_{\eta}\tau_\pi\left(u^\tau\right)^2\left(u^y\right)^2\right)\Big) \\
\delta(A_t)_{63} &= \frac{1}{2u^\tau\left(\left(u^y\right)^2 + 1\right)}\Big(-\pi^{\eta}{}_{\eta}\left(u^\tau\right)^2 u^x - \pi^{yy}\left(\left(u^x\right)^3 + u^x\right) + \pi^{xy}u^y + 2\pi^{xy}\left(u^x\right)^2 u^y \\
&\quad + \pi^{xy}\left(u^y\right)^3 + 2u^x\eta + 2\left(u^x\right)^3\eta + 4u^x\left(u^y\right)^2\eta + 2\left(u^x\right)^3\left(u^y\right)^2\eta \\
&\quad + 2u^x\left(u^y\right)^4\eta\Big) \\
(A_x)_{43}^{\text{legacy}} &= \frac{2u^x u^y\left(\pi^{xy}\tau_\pi u^x u^y - \pi^{yy}\tau_\pi\left(\left(u^x\right)^2 + 1\right) + \left(u^\tau\right)^2\eta\right)}{\left(u^\tau\right)^2} + \pi^{xy} \\
(A_x)_{43}^{\text{derived}} &= \frac{2u^x u^y\left(\pi^{xy}\tau_\pi u^x u^y - \pi^{yy}\tau_\pi\left(\left(u^x\right)^2 + 1\right) + \left(u^\tau\right)^2\eta\right)}{\left(u^\tau\right)^2} \\
\delta(A_x)_{43} &= \pi^{xy}
\end{aligned}
$$

$$
\begin{aligned}
(A_x)_{62}^{\text{legacy}} &= \frac{1}{6\left(u^\tau\right)^2} u^x \Big(3\pi^{xy} u^x + 3\pi^{yy} u^y \left(\tau_\pi \left(4\left(u^x\right)^2 + 2\right) + 1\right) - 6\pi^{xy}\tau_\pi u^x \left(2\left(u^y\right)^2 + 1\right) \\
&\quad + 2\left(u^\tau\right)^2 u^y (3\pi^\eta{}_\eta \tau_\pi - 2\eta)\Big) \\
(A_x)_{62}^{\text{derived}} &= \frac{u^x \left(u^y \left(3\pi^\eta{}_\eta \tau_\pi \left(u^\tau\right)^2 + 3\pi^{yy}\left(2\tau_\pi \left(u^x\right)^2 + \tau_\pi\right) + \left(u^\tau\right)^2 \eta\right) - 3\pi^{xy}\tau_\pi u^x \left(2\left(u^y\right)^2 + 1\right)\right)}{3\left(u^\tau\right)^2} \\
\delta(A_x)_{62} &= \frac{u^x \left(\pi^{xy} u^x + u^y \left(\pi^{yy} - 2\left(u^\tau\right)^2 \eta\right)\right)}{2\left(u^\tau\right)^2} \\
(A_x)_{63}^{\text{legacy}} &= \frac{1}{2\left(u^\tau\right)^2 \left(\left(u^y\right)^2 + 1\right)} \Big(-\pi^{xy}(2\tau_\pi - 3) u^x u^y \left(\left(u^y\right)^2 + 1\right) \\
&\quad - \left(\left(u^y\right)^2 + 1\right)\left(\pi^\eta{}_\eta \left(u^y\right)^2 + \pi^\eta{}_\eta + 2\pi^{yy}\right) \\
&\quad + 2\pi^{xy}\left(u^x\right)^3 u^y \left(2\tau_\pi \left(u^y\right)^2 + 1\right) \\
&\quad - \left(u^x\right)^4 \left(2\pi^\eta{}_\eta \tau_\pi \left(u^y\right)^2 + \pi^\eta{}_\eta + \pi^{yy}\tau_\pi \left(4\left(u^y\right)^2 + 2\right) + \pi^{yy}\right) \\
&\quad - \left(u^x\right)^2 \left(2\pi^\eta{}_\eta \left(\left(u^y\right)^2 + 1\right)\left(\tau_\pi \left(u^y\right)^2 + 1\right)\right. \\
&\quad \left. + \pi^{yy}\left(2(2\tau_\pi + 1)\left(u^y\right)^2 + 2\tau_\pi + 3\right)\right) \\
&\quad + 4\left(u^\tau\right)^2 \left(\left(u^x\right)^2 + 1\right)\left(\left(u^y\right)^2 + 1\right)\eta\Big) \\
(A_x)_{63}^{\text{derived}} &= \frac{1}{\left(u^\tau\right)^2 \left(\left(u^y\right)^2 + 1\right)} \Big(-\pi^{yy}\tau_\pi \left(u^x\right)^2 \left(\left(u^x\right)^2 + 1\right)\left(2\left(u^y\right)^2 + 1\right) \\
&\quad + \pi^{xy}\tau_\pi u^x u^y \left(\left(2\left(u^x\right)^2 - 1\right)\left(u^y\right)^2 - 1\right) \\
&\quad - \left(u^\tau\right)^2 \left(\pi^\eta{}_\eta \tau_\pi \left(u^x\right)^2 \left(u^y\right)^2 - \left(\left(u^x\right)^2 + 1\right)\left(\left(u^y\right)^2 + 1\right)\eta\right)\Big) \\
\delta(A_x)_{63} &= \frac{1}{2\left(u^\tau\right)^2 \left(\left(u^y\right)^2 + 1\right)} \Big(-\pi^\eta{}_\eta \left(u^\tau\right)^4 + 3\pi^{xy} u^x u^y + 2\pi^{xy}\left(u^x\right)^3 u^y + 3\pi^{xy} u^x \left(u^y\right)^3 \\
&\quad - \pi^{yy}\left(\left(u^x\right)^2 + 1\right)\left(\left(u^x\right)^2 + 2\left(u^y\right)^2 + 2\right) + 2\eta + 4\left(u^x\right)^2 \eta \\
&\quad + 2\left(u^x\right)^4 \eta + 4\left(u^y\right)^2 \eta + 6\left(u^x\right)^2 \left(u^y\right)^2 \eta + 2\left(u^x\right)^4 \left(u^y\right)^2 \eta \\
&\quad + 2\left(u^y\right)^4 \eta + 2\left(u^x\right)^2 \left(u^y\right)^4 \eta\Big) \\
(A_y)_{42}^{\text{legacy}} &= -\frac{\left(2\tau_\pi \left(u^y\right)^2 + 1\right)\left(\pi^{xy}\left(u^y\right)^2 + \pi^{xy} - \pi^{yy} u^x u^y\right)}{\left(u^\tau\right)^2} \\
(A_y)_{42}^{\text{derived}} &= -\frac{2\tau_\pi \left(u^y\right)^2 \left(\pi^{xy}\left(u^y\right)^2 + \pi^{xy} - \pi^{yy} u^x u^y\right)}{\left(u^\tau\right)^2} \\
\delta(A_y)_{42} &= \frac{\pi^{yy} u^x u^y - \pi^{xy}\left(\left(u^y\right)^2 + 1\right)}{\left(u^\tau\right)^2} \\
(A_y)_{43}^{\text{legacy}} &= \frac{u^y \left(\pi^{xy}\left(2\tau_\pi u^x \left(u^y\right)^2 + u^x\right) + \pi^{yy} u^y \left(1 - 2\tau_\pi \left(\left(u^x\right)^2 + 1\right)\right)\right)}{\left(u^\tau\right)^2} + \frac{4}{3}\left(\left(u^y\right)^2 + 1\right)\eta \\
(A_y)_{43}^{\text{derived}} &= \frac{6\pi^{xy}\tau_\pi u^x \left(u^y\right)^3 - 6\pi^{yy}\tau_\pi \left(\left(u^x\right)^2 + 1\right)\left(u^y\right)^2 + 4\left(u^\tau\right)^2 \left(\left(u^y\right)^2 + 1\right)\eta}{3\left(u^\tau\right)^2} \\
\delta(A_y)_{43} &= \frac{u^y(\pi^{xy} u^x + \pi^{yy} u^y)}{\left(u^\tau\right)^2} \\
(A_y)_{62}^{\text{legacy}} &= \frac{1}{2\left(u^\tau\right)^2} \Big(\pi^\eta{}_\eta + 2\pi^{yy} + \left(u^x\right)^2 \left(\pi^\eta{}_\eta + 2\pi^{yy}\right)\left(2\tau_\pi \left(u^y\right)^2 + 1\right) \\
&\quad + \left(u^y\right)^2 \left(2\pi^\eta{}_\eta \tau_\pi \left(\left(u^y\right)^2 + 1\right) + \pi^\eta{}_\eta + 2\pi^{yy}\tau_\pi + \pi^{yy}\right)
\end{aligned}
$$

$$
\begin{aligned}
&- \pi^{xy} u^x u^y \left(\tau_\pi \left(4 \left(u^y\right)^2 + 2\right) + 1\right)\Big) \\
(A_y)_{62}^{\text{derived}} &= \frac{1}{\left(u^\tau\right)^2}\Big(-\pi^{xy}\tau_\pi u^x u^y \left(2 \left(u^y\right)^2 + 1\right) + \eta + \left(u^y\right)^4 \left(\pi^\eta{}_\eta \tau_\pi + \eta\right) \\
&+ \left(u^y\right)^2 \left(\pi^\eta{}_\eta \tau_\pi + \pi^{yy}\tau_\pi + 2\eta\right) + \left(u^x\right)^2 \left(\left(u^y\right)^2 \left(\pi^\eta{}_\eta \tau_\pi + 2\pi^{yy}\tau_\pi + \eta\right) + \eta\right)\Big) \\
\delta(A_y)_{62} &= \frac{1}{2\left(u^\tau\right)^2}\Big(\pi^\eta{}_\eta \left(u^\tau\right)^2 - \pi^{xy} u^x u^y + \pi^{yy}\left(2 \left(u^x\right)^2 + \left(u^y\right)^2 + 2\right) - 2\eta - 2\left(u^x\right)^2 \eta \\
&- 4\left(u^y\right)^2 \eta - 2\left(u^x\right)^2 \left(u^y\right)^2 \eta - 2\left(u^y\right)^4 \eta\Big) \\
(A_y)_{63}^{\text{legacy}} &= \frac{1}{6\left(u^\tau\right)^2 \left(\left(u^y\right)^2 + 1\right)} u^y \Big(-3\pi^{xy}(2\tau_\pi - 1) u^y \left(\left(u^y\right)^2 + 1\right) \\
&+ 6\pi^{xy} \left(u^x\right)^2 u^y \left(2\tau_\pi \left(u^y\right)^2 + 1\right) \\
&+ \left(u^x\right)^3 \Big(-3(\pi^\eta{}_\eta + 2\pi^{yy}\tau_\pi + \pi^{yy}) - 6\tau_\pi \left(u^y\right)^2 (\pi^\eta{}_\eta + 2\pi^{yy}) \\
&+ 8\left(\left(u^y\right)^2 + 1\right)\eta\Big) \\
&+ u^x \Big(-3\Big(\pi^{yy} + \pi^{yy}\tau_\pi \left(4\left(u^y\right)^2 + 2\right) \\
&+ \pi^\eta{}_\eta \left(\left(u^y\right)^2 + 1\right)\left(2\tau_\pi \left(u^y\right)^2 + 1\right)\Big) \\
&+ 8\left(\left(u^y\right)^2 + 1\right)^2 \eta\Big)\Big) \\
(A_y)_{63}^{\text{derived}} &= -\frac{1}{3\left(u^\tau\right)^2 \left(\left(u^y\right)^2 + 1\right)} u^y \Big(3\pi^{yy}\tau_\pi u^x \left(\left(u^x\right)^2 + 1\right)\left(2\left(u^y\right)^2 + 1\right) \\
&+ 3\pi^{xy}\tau_\pi u^y \left(\left(1 - 2\left(u^x\right)^2\right)\left(u^y\right)^2 + 1\right) \\
&+ \left(u^\tau\right)^2 u^x \left(3\pi^\eta{}_\eta \tau_\pi \left(u^y\right)^2 - \left(\left(u^y\right)^2 + 1\right)\eta\right)\Big) \\
\delta(A_y)_{63} &= \frac{1}{2\left(u^\tau\right)^2 \left(\left(u^y\right)^2 + 1\right)} u^y \Big(-\pi^\eta{}_\eta \left(u^\tau\right)^2 u^x - \pi^{yy}\left(\left(u^x\right)^3 + u^x\right) + \pi^{xy} u^y + 2\pi^{xy}\left(u^x\right)^2 u^y \\
&+ \pi^{xy}\left(u^y\right)^3 + 2u^x\eta + 2\left(u^x\right)^3 \eta + 4u^x \left(u^y\right)^2 \eta + 2\left(u^x\right)^3 \left(u^y\right)^2 \eta \\
&+ 2u^x \left(u^y\right)^4 \eta\Big)
\end{aligned}
$$

5.2 `two_d_viscous_HQ_matrix`

$\phi = (T, u^x, u^y, \pi^{yy}, \pi^\eta{}_\eta, \pi^{xy}, \Pi, \alpha, \nu^x, \nu^y)$. Rows 1–7: the derived matrix at $\delta_{\pi\pi} = 0$. Rows 8–10:

$$
\nabla_\mu (n u^\mu + \nu^\mu) = 0, \qquad \tau_n \Delta^\mu{}_\beta D\nu^\beta + \nu^\mu + \kappa \Delta^{\mu\beta}\partial_\beta \alpha = 0, \qquad \kappa = D_s n
$$

The steps are those of the second section. With the medium as fields, in columns 1, 2, 3, 8, 9, 10:

$$
\begin{aligned}
(A_t)_{8,1} &= (\partial_T n) u^\tau \\
(A_t)_{8,2} &= \frac{n\left(u^\tau\right)^2 u^x + \nu^x \left(\left(u^y\right)^2 + 1\right) - \nu^y u^x u^y}{\left(u^\tau\right)^3} \\
(A_t)_{8,3} &= \frac{n\left(u^\tau\right)^2 u^y - \nu^x u^x u^y + \nu^y \left(\left(u^x\right)^2 + 1\right)}{\left(u^\tau\right)^3} \\
(A_t)_{8,8} &= (\partial_\alpha n) u^\tau \\
(A_t)_{8,9} &= \frac{u^x}{u^\tau} \\
(A_t)_{8,10} &= \frac{u^y}{u^\tau} \\
(A_t)_{9,2} &= \frac{\tau_n u^x \left(\nu^y u^x u^y - \nu^x \left(\left(u^y\right)^2 + 1\right)\right)}{u^\tau}
\end{aligned}
$$

$$
\begin{aligned}
(A_t)_{9,3} &= -\frac{\tau_n u^x \left(-\nu^x u^x u^y + \nu^y \left(u^x\right)^2 + \nu^y\right)}{u^\tau} \\
(A_t)_{9,8} &= D_s n u^\tau u^x \\
(A_t)_{9,9} &= \tau_n u^\tau \\
(A_t)_{10,2} &= -\frac{\tau_n u^y \left(\nu^x \left(u^y\right)^2 + \nu^x - \nu^y u^x u^y\right)}{u^\tau} \\
(A_t)_{10,3} &= \frac{\tau_n u^y \left(\nu^x u^x u^y - \nu^y \left(\left(u^x\right)^2 + 1\right)\right)}{u^\tau} \\
(A_t)_{10,8} &= D_s n u^\tau u^y \\
(A_t)_{10,10} &= \tau_n u^\tau \\
(A_x)_{8,1} &= (\partial_T n) u^x \\
(A_x)_{8,2} &= n \\
(A_x)_{8,8} &= (\partial_\alpha n) u^x \\
(A_x)_{8,9} &= 1 \\
(A_x)_{9,2} &= \frac{\tau_n \left(u^x\right)^2 \left(\nu^y u^x u^y - \nu^x \left(\left(u^y\right)^2 + 1\right)\right)}{\left(u^\tau\right)^2} \\
(A_x)_{9,3} &= -\frac{\tau_n \left(u^x\right)^2 \left(-\nu^x u^x u^y + \nu^y \left(u^x\right)^2 + \nu^y\right)}{\left(u^\tau\right)^2} \\
(A_x)_{9,8} &= D_s n \left(\left(u^x\right)^2 + 1\right) \\
(A_x)_{9,9} &= \tau_n u^x \\
(A_x)_{10,2} &= \frac{\tau_n u^x u^y \left(\nu^y u^x u^y - \nu^x \left(\left(u^y\right)^2 + 1\right)\right)}{\left(u^\tau\right)^2} \\
(A_x)_{10,3} &= \frac{\tau_n u^x u^y \left(\nu^x u^x u^y - \nu^y \left(\left(u^x\right)^2 + 1\right)\right)}{\left(u^\tau\right)^2} \\
(A_x)_{10,8} &= D_s n u^x u^y \\
(A_x)_{10,10} &= \tau_n u^x \\
(A_y)_{8,1} &= (\partial_T n) u^y \\
(A_y)_{8,3} &= n \\
(A_y)_{8,8} &= (\partial_\alpha n) u^y \\
(A_y)_{8,10} &= 1 \\
(A_y)_{9,2} &= \frac{\tau_n u^x u^y \left(\nu^y u^x u^y - \nu^x \left(\left(u^y\right)^2 + 1\right)\right)}{\left(u^\tau\right)^2} \\
(A_y)_{9,3} &= -\frac{\tau_n u^x u^y \left(-\nu^x u^x u^y + \nu^y \left(u^x\right)^2 + \nu^y\right)}{\left(u^\tau\right)^2} \\
(A_y)_{9,8} &= D_s n u^x u^y \\
(A_y)_{9,9} &= \tau_n u^y \\
(A_y)_{10,2} &= -\frac{\tau_n \left(u^y\right)^2 \left(\nu^x \left(u^y\right)^2 + \nu^x - \nu^y u^x u^y\right)}{\left(u^\tau\right)^2} \\
(A_y)_{10,3} &= \frac{\tau_n \left(u^y\right)^2 \left(\nu^x u^x u^y - \nu^y \left(\left(u^x\right)^2 + 1\right)\right)}{\left(u^\tau\right)^2} \\
(A_y)_{10,8} &= D_s n \left(\left(u^y\right)^2 + 1\right) \\
(A_y)_{10,10} &= \tau_n u^y \\
S_8 &= \frac{n \left(u^\tau\right)^2 + \nu^x u^x + \nu^y u^y}{\tau u^\tau}
\end{aligned}
$$

$$S_9 = \nu^x$$
$$S_{10} = \nu^y$$

Its wrapper is `matrix2d_visc_HQ!` (D-HQup).

5.3 one_d_ideal_matrix_ruwen7

$$x^\mu = (\tau, r, \phi, \eta), \qquad g_{\mu\nu} = \mathrm{diag}(-1, 1, r^2, \tau^2), \qquad u^\mu = (u^\tau, u^r, 0, 0)$$
$$P_{\rm tot} = P(T) + P_h(T, \alpha), \qquad T^{\mu\nu} = (e + P_{\rm tot})u^\mu u^\nu + P_{\rm tot} g^{\mu\nu}, \qquad \nabla_\mu T^{\mu\nu} = 0, \qquad \nabla_\mu (n u^\mu) = 0$$
$$e + P_{\rm tot} = Ts + \mu n = T\partial_T P_{\rm tot}\big|_\mu + \mu \partial_\mu P_{\rm tot}\big|_T = T\partial_T P_{\rm tot}\big|_\alpha, \qquad \mu = \alpha T$$

$\phi = (T, u^r, \alpha)$, the $T^{\mu\nu}$ as written, the same extraction. The file, raw (U-P7):

$$\begin{aligned}
(A_t)_{11} &= (u^r)^2 \left(p'(T) + T(p''(T) + (\partial_T^2 P_h)) + (\partial_T P_h)\right) + T(p''(T) + (\partial_T^2 P_h)) \\
(A_t)_{12} &= 2Tu^r(p'(T) + (\partial_T P_h)) \\
(A_t)_{13} &= (\partial_T \partial_\alpha P_h) T (u^\tau)^2 - (\partial_\alpha P_h) \\
(A_t)_{21} &= u^r u^\tau (p'(T) + T(p''(T) + (\partial_T^2 P_h)) + (\partial_T P_h)) \\
(A_t)_{22} &= \frac{T\left(2(u^r)^2 + 1\right)(p'(T) + (\partial_T P_h))}{u^\tau} \\
(A_t)_{23} &= (\partial_T \partial_\alpha P_h) T u^r u^\tau \\
(A_t)_{31} &= (\partial_T n) u^\tau \\
(A_t)_{32} &= \frac{n u^r}{u^\tau} \\
(A_t)_{33} &= (\partial_\alpha n) u^\tau \\
(A_r)_{11} &= u^r u^\tau (p'(T) + T(p''(T) + (\partial_T^2 P_h)) + (\partial_T P_h)) \\
(A_r)_{12} &= \frac{T\left(2(u^r)^2 + 1\right)(p'(T) + (\partial_T P_h))}{u^\tau} \\
(A_r)_{13} &= (\partial_T \partial_\alpha P_h) T u^r u^\tau \\
(A_r)_{21} &= p'(T)(u^\tau)^2 + T(u^r)^2 (p''(T) + (\partial_T^2 P_h)) + (\partial_T P_h)(u^\tau)^2 \\
(A_r)_{22} &= 2Tu^r(p'(T) + (\partial_T P_h)) \\
(A_r)_{23} &= (\partial_\alpha P_h) + (\partial_T \partial_\alpha P_h) T (u^r)^2 \\
(A_r)_{31} &= (\partial_T n) u^r \\
(A_r)_{32} &= n \\
(A_r)_{33} &= (\partial_\alpha n) u^r \\
S_1 &= \frac{T u^\tau (p'(T) + (\partial_T P_h))(r u^\tau + \tau u^r)}{r\tau} \\
S_2 &= \frac{T u^r (p'(T) + (\partial_T P_h))\left(r(u^r)^2 + r + \tau u^r u^\tau\right)}{r\tau u^\tau} \\
S_3 &= n\left(\frac{u^r}{r} + \frac{u^\tau}{\tau}\right)
\end{aligned}$$

Its wrapper is `matrix1d_ideal_HQ_br_p_fugacity!` (D-r7). For a Boltzmann gas of unit-charge species $P_h = nT$:

$$\partial_T P_h = n + T\partial_T n, \quad \partial_\alpha P_h = T\partial_\alpha n, \quad \partial_T^2 P_h = 2\partial_T n + T\partial_T^2 n, \quad \partial_T \partial_\alpha P_h = \partial_\alpha n + T\partial_T \partial_\alpha n, \quad \partial_\alpha^2 P_h = T\partial_\alpha^2 n$$

5.4 one_d_ideal_matrix_ruwen6

The same system in $\phi = (T, u^r, n)$, with $n = \partial_\alpha P_h / T$ and $\alpha = \alpha(T, n)$:

$$\partial_n\alpha\big|_T = \frac{T}{\partial_\alpha^2 P_h}, \qquad \partial_T\alpha\big|_n = \frac{\partial_\alpha P_h/T - \partial_T\partial_\alpha P_h}{\partial_\alpha^2 P_h}, \qquad \partial_k\alpha = \partial_T\alpha\,\partial_k T + \partial_n\alpha\,\partial_k n$$

Derived:

$$\begin{aligned}
(A_t)_{11} &= -p'(T) - (\partial_T P_h) - \frac{(\partial_\alpha P_h)((\partial_\alpha P_h) - (\partial_T\partial_\alpha P_h)T)}{(\partial_\alpha^2 P_h)T} + (u^\tau)^2\,(p'(T) + (\partial_T P_h)) \\
&\quad + T\,(u^\tau)^2\left(p''(T) + \frac{(\partial_T\partial_\alpha P_h)((\partial_\alpha P_h) - (\partial_T\partial_\alpha P_h)T)}{(\partial_\alpha^2 P_h)T} + (\partial_T^2 P_h)\right) \\
(A_t)_{12} &= 2Tu^r(p'(T) + (\partial_T P_h)) \\
(A_t)_{13} &= \frac{T\left((\partial_T\partial_\alpha P_h)T\,(u^\tau)^2 - (\partial_\alpha P_h)\right)}{(\partial_\alpha^2 P_h)} \\
(A_t)_{21} &= u^r u^\tau\left(p'(T) + T(p''(T) + (\partial_T^2 P_h)) + \frac{(\partial_T\partial_\alpha P_h)((\partial_\alpha P_h) - (\partial_T\partial_\alpha P_h)T)}{(\partial_\alpha^2 P_h)} + (\partial_T P_h)\right) \\
(A_t)_{22} &= \frac{T\left(2\,(u^r)^2 + 1\right)(p'(T) + (\partial_T P_h))}{u^\tau} \\
(A_t)_{23} &= \frac{(\partial_T\partial_\alpha P_h)T^2 u^r u^\tau}{(\partial_\alpha^2 P_h)} \\
(A_t)_{32} &= \frac{u^r n}{u^\tau} \\
(A_t)_{33} &= u^\tau \\
(A_r)_{11} &= u^r u^\tau\left(p'(T) + T(p''(T) + (\partial_T^2 P_h)) + \frac{(\partial_T\partial_\alpha P_h)((\partial_\alpha P_h) - (\partial_T\partial_\alpha P_h)T)}{(\partial_\alpha^2 P_h)} + (\partial_T P_h)\right) \\
(A_r)_{12} &= \frac{T\left(2\,(u^r)^2 + 1\right)(p'(T) + (\partial_T P_h))}{u^\tau} \\
(A_r)_{13} &= \frac{(\partial_T\partial_\alpha P_h)T^2 u^r u^\tau}{(\partial_\alpha^2 P_h)} \\
(A_r)_{21} &= p'(T) + (\partial_T P_h) + \frac{(\partial_\alpha P_h)((\partial_\alpha P_h) - (\partial_T\partial_\alpha P_h)T)}{(\partial_\alpha^2 P_h)T} + (u^r)^2\,(p'(T) + (\partial_T P_h)) \\
&\quad + T\,(u^r)^2\left(p''(T) + \frac{(\partial_T\partial_\alpha P_h)((\partial_\alpha P_h) - (\partial_T\partial_\alpha P_h)T)}{(\partial_\alpha^2 P_h)T} + (\partial_T^2 P_h)\right) \\
(A_r)_{22} &= 2Tu^r(p'(T) + (\partial_T P_h)) \\
(A_r)_{23} &= \frac{T\left((\partial_\alpha P_h) + (\partial_T\partial_\alpha P_h)T\,(u^r)^2\right)}{(\partial_\alpha^2 P_h)} \\
(A_r)_{32} &= n \\
(A_r)_{33} &= u^r \\
S_1 &= \frac{Tu^\tau(p'(T) + (\partial_T P_h))(ru^\tau + \tau u^r)}{r\tau} \\
S_2 &= \frac{Tu^r(p'(T) + (\partial_T P_h))\left(r\,(u^r)^2 + r + \tau u^r u^\tau\right)}{r\tau u^\tau} \\
S_3 &= n\left(\frac{u^r}{r} + \frac{u^\tau}{\tau}\right)
\end{aligned}$$

The file agrees, raw (U-P6-cols, U-P6-row3, U-P6-c3). Its wrapper takes $P_h = nT$ from `charm_pressure` and α from `charm_alpha` (D-r6).

5.5 Checks

id	what it tests	
`U-L-count`	exactly ten of the 147 entries of A_t, A_x, A_y differ from the derived matrix ($\delta_{\pi\pi} = 0$)	PASS
`U-L-place`	they are in the velocity columns of the π^{yy} and π^{xy} rows	PASS
`U-L-src`	the source is right	PASS
`U-H-med`	its medium block = the derived matrix at $\delta_{\pi\pi} = 0$, raw (fixed 2026-10-30: it carried the ten legacy entries)	PASS
`U-H-pass`	the medium does not see charm	PASS
`U-H-charm`	rows 8-10 = N and Fick, $\tau_n \Delta D\nu + \nu + D_s n\, \Delta\nabla\alpha = 0$, raw	PASS
`U-H-charmsrc`	and their source, raw	PASS
`U-H-kappa`	the argument D_s enters as $\kappa = D_s n$, which is $D_s\ \partial_\alpha n$ for one unit-charge Boltzmann species	PASS
`U-P7`	ruwen7 = the derivation with $e + P_{\mathrm{tot}} = T\partial_T P_{\mathrm{tot}}$, raw, entry by entry	PASS
`U-P6-chain`	implicit function: $\partial_T \alpha$ at fixed n and $\partial_n \alpha$ at fixed T from $n = \partial_\alpha P_h / T$	PASS
`U-P6-cols`	ruwen6, columns 1-2: = the derivation, raw	PASS
`U-P6-row3`	ruwen6, row 3 and the source: = the derivation, raw	PASS
`U-P6-c3`	ruwen6, column 3 of rows 1-2: = the derivation, raw (fixed 2026-10-30: A_t and A_r were exchanged there)	PASS
`U-P6-ctl`	control: A_t and A_r differ in that column, so the exchange would be seen	PASS

14 passed, 0 failed (14 checks evaluated).

Re-run with `wolframscript -file deriv/derive_upstream.wls`.

6 From the matrices to the solver

Script: `deriv/derive_drivers.wls`, which runs `deriv/audit_matrices.jl`

6.1 The solved form

$$\partial_\tau \phi = -A_t^{-1} A_i \, \partial_i \phi - A_t^{-1} S$$

Each driver evaluates the EOS and the transport at the state, calls a kernel, and returns $A_t^{-1}A_i$, $A_t^{-1}S$; the solver upwinds $A_t^{-1}A_i\partial_i\phi$. η, τ_π, ζ, τ_Π are inputs, not derived. $\delta_{\pi\pi} = \frac{4}{3}\tau_\pi$ unless switched off.

6.2 κ and τ_n

From the consistent first moment of the section on the two-way system, at rest ($u^r = 0$, $\nu^r = 0$ in the coefficients):

$$\frac{D_s}{T}\Delta^r{}_\lambda \nabla_\mu T_Q^{\mu\lambda} + \nu^r \;\to\; \frac{D_s h}{T}\partial_\tau \nu^r + \nu^r + D_s \partial_\alpha n \, \partial_r \alpha + \frac{D_s}{T}(n + T\partial_T n)\partial_r T$$
$$\tau_n = \frac{D_s h}{T}, \qquad \kappa = D_s \partial_\alpha n$$

The Fick closure keeps these two coefficients and drops the $\partial_r T$ term. In the code τ_n is written with a Bessel combination:

$$K_5 = K_3 + \frac{8}{z}K_4, \quad K_4 = K_2 + \frac{6}{z}K_3, \quad K_1 = K_3 - \frac{4}{z}K_2 \;\Rightarrow\; 2K_1 - 3K_3 + K_5 = \frac{48}{z^2}K_3$$
$$\tau_n = \frac{\sum_i g_i q_i^2 \, 2\pi D_s T \, m_i^5 \, e^{q_i\alpha}(2K_1 - 3K_3 + K_5)/(192\pi^3 T^3)}{\sum_i q_i^2 n_i} = \frac{D_s}{T}\frac{\sum_i q_i^2 n_i h_i}{\sum_i q_i^2 n_i} = \frac{D_s h}{T}, \qquad h_i = m_i \frac{K_3(z_i)}{K_2(z_i)}$$

6.3 The charm closures

$$f = g\, e^{q\alpha - E/T}, \qquad dP = \frac{d^3p}{(2\pi)^3 E}, \qquad I_{nq} = \frac{1}{(2q+1)!!}\int dP \, E^{n-2q} p^{2q} f$$

Angular averages, then the matching in the rest frame:

$$p_x^2 \to \frac{p^2}{3}, \qquad p_x^4 \to \frac{p^4}{5}, \qquad p_x^2 p_y^2 \to \frac{p^4}{15}$$
$$\delta f_\nu = f\,\frac{\nu_a p^a}{P_0}: \quad \int dP\, p_x \delta f_\nu = \frac{I_{21}}{P_0}\nu^x = \nu^x, \quad S^{\tau\tau x} = \frac{I_{41}}{P_0}\nu^x = \mathcal{A}\nu^x, \quad S^{xxx} = 3\frac{I_{42}}{P_0}\nu^x = 3\mathcal{B}\nu^x$$
$$\delta f_\pi = f\,\frac{\pi_{ab}p^a p^b}{2I_{42}}: \quad \int dP\, p_x p_y \delta f_\pi = \pi^{xy}, \quad S^{\tau xy} = \frac{I_{52}}{I_{42}}\pi^{xy} = \mathcal{C}\pi^{xy}$$
$$S^{\tau xx}\big|_f = I_{31}, \qquad \delta M^{ab} = \pi_Q^{ab} + \Pi_Q \Delta^{ab} \text{ with the same } \mathcal{C} \text{ (closure)}$$

One Boltzmann species:

$$n = \frac{g}{2\pi^2} m^2 T\, K_2(z) e^{q\alpha}, \quad I_{21} = P_0 = nT, \quad I_{31} = P_0 h, \quad I_{41} = P_0(m^2 + 5Th), \quad I_{42} = P_0 Th, \quad I_{52} = I_{42}\, m\frac{K_4}{K_3}$$

A list of species at one (T, α), with $w_i = q_i^2 n_i$, as in `charm_eos`:

$$h = \frac{\sum w_i h_i}{\sum w_i}, \quad \mathcal{A} = \frac{\sum w_i m_i^2}{\sum w_i} + 5Th, \quad \mathcal{B} = Th, \quad \mathcal{C} = \frac{\sum w_i h_i \mathcal{C}_i}{\sum w_i h_i}, \quad I_{31} = nTh$$
$$\partial_T h_i = -\frac{m^2}{T^2}\frac{-\frac{1}{2}(K_2+K_4)K_2 + \frac{1}{2}K_3(K_1+K_3)}{K_2^2}, \quad \partial_T \mathcal{C}_i = -\frac{m^2}{T^2}\frac{-\frac{1}{2}(K_3+K_5)K_3 + \frac{1}{2}K_4(K_2+K_4)}{K_3^2}$$
$$\partial_T n_i = \frac{g}{2\pi^2} e^{q\alpha}\frac{m^2(mK_1 + 2TK_2 + mK_3)}{2T}, \quad \partial_\alpha I_{31} = \partial_\alpha n\, Th, \quad \partial_T I_{31} = \partial_T n\, Th + nh + nT\partial_T h$$

A state of charge $|q| = 2$ enters with these weights; a charge-correct second moment would also carry $\partial_\alpha \mathcal{C}$, $\partial_\alpha h$.

6.4 The vacuum regulators

Two forms. Two-way drivers, $\epsilon =$ `dmn_eps`:

$$E_\alpha \to \nabla_\mu N^\mu + \epsilon\, u^\mu \partial_\mu \alpha : \qquad (A_t)_{\alpha\alpha} \to (A_t)_{\alpha\alpha} + \epsilon u^\tau, \quad (A_i)_{\alpha\alpha} \to (A_i)_{\alpha\alpha} + \epsilon u^i$$

Background and 7-field drivers, $\partial_\alpha n \to \partial_\alpha n + \epsilon$: the same term in the number row. In the consistent rows it also reaches κ, since κ is formed inside the kernel:

$$\kappa \to D_s(\partial_\alpha n + \epsilon) \quad \text{(consistent, second moment)}, \qquad \kappa \text{ unchanged} \quad \text{(Fick)}$$

It is not conservative:

$$\partial_\tau(\tau r N^\tau) + \partial_r(\tau r N^r) = -\tau r\, \epsilon\, u^\mu \partial_\mu \alpha$$

`matrix1d_visc_gamma_HQ!` accepts `dmn_eps`, but its kernel has no $\partial_\alpha n$: the number row carries n, so the regulator does nothing there. α is clamped to $|\alpha| \le \alpha_{\max}$ where the EOS is read.

6.5 The cold edge

$$x = \frac{T - (T_{\text{edge}} - w_T)}{w_T}, \qquad s(x) = x^3(10 - 15x + 6x^2)$$

$$s(0) = s'(0) = s''(0) = s'(1) = s''(1) = 0, \quad s(1) = 1$$

$$D_s \to f(T) D_s, \qquad f = f_0 + (1 - f_0)s \text{ in the window}, \quad f = 1 \ (T \ge T_{\text{edge}}), \quad f = f_0 \ (T \le T_{\text{edge}} - w_T)$$

$$w = 1 - s = \frac{1 - f}{1 - f_0}$$

Only D_s changes, and it is not differentiated, as in the matrices. Its gradient is dropped; this is a decoupling prescription, not a derived term.

6.6 The corona update

Below T_{edge} the quasi-linear $\dot\alpha$ is blended with one that gives a flux-form number update. $V = r$ in 1+1D, $V = 1$ in 2+1D; rates of T, u, ν are those the matrices gave:

$$N^\tau = n u^\tau + \nu^\tau, \qquad N^i = n u^i + \nu^i, \qquad \nu^\tau = \frac{u_i \nu^i}{u^\tau}$$

$$\frac{d}{d\tau}(\tau V N^\tau) = V N^\tau + \tau V\big(\partial_T n\, \dot T\, u^\tau + \partial_\alpha n\, \dot\alpha\, u^\tau + n \dot u^\tau + \dot\nu^\tau\big) \equiv \mathcal{T}$$

$$\dot\alpha_{\text{flux}} = \frac{(\mathcal{T} - V N^\tau)/(\tau V) - \partial_T n\, \dot T\, u^\tau - n \dot u^\tau - \dot\nu^\tau}{u^\tau(\partial_\alpha n + \epsilon)}$$

$$\dot u^\tau = \frac{u_i \dot u^i}{u^\tau}, \qquad \dot\nu^\tau = \frac{\dot u_i \nu^i + u_i \dot\nu^i}{u^\tau} - \frac{\nu^\tau \dot u^\tau}{u^\tau}$$

$$\mathcal{T} = -\frac{G_{i+1/2} - G_{i-1/2}}{\Delta r}, \qquad G_{i+1/2} = \tau r_{i+1/2} F_{i+1/2}, \qquad G_{1/2} = 0 \text{ on the axis} \quad \text{(1+1D)}$$

$$\mathcal{T} = -\tau\Big(\frac{F^x_{i+1/2,j} - F^x_{i-1/2,j}}{\Delta x} + \frac{F^y_{i,j+1/2} - F^y_{i,j-1/2}}{\Delta y}\Big) \quad \text{(2+1D)}$$

$$F_{i+1/2} = \frac{N^i_L + N^i_R}{2} - a\frac{N^\tau_R - N^\tau_L}{2}, \qquad a = \min\Big(1,\ \max\Big(\frac{|N^i_L|}{N^\tau_L},\ \frac{|N^i_R|}{N^\tau_R}\Big)\Big)$$

$$\dot\alpha = (1 - w)\dot\alpha_{\text{ql}} + w\, \dot\alpha_{\text{flux}}$$

For $w = 1$ and $\epsilon = 0$ the sum over cells telescopes to the boundary fluxes:

$$\sum_i \Delta r\, \frac{d}{d\tau}(\tau r_i N^\tau_i) = -(G_{\text{end}} - G_{1/2})$$

The upwind cap and the blend are numerical choices, not limits of the equations. The update is a discretisation of E_α, so the matrices are unchanged.

6.7 Checks

R-: symbolic, in the script. A-: `audit_matrices.jl` rebuilds each system numerically with ForwardDiff and compares the solved form at random states, with a control that multiplies one coefficient by 1.2. D-: it calls the shipped drivers at physical states and evaluates the corona update on a grid.

id	what it tests	
`R-fick`	at rest the row is $\tau_n \partial_\tau \nu^r + \kappa \partial_r \alpha + (D_s/T)(n + T\partial_T n)\partial_r T$	PASS
`R-taun`	the coefficient of $D\nu^r$ is $\tau_n = D_s h/T$	PASS
`R-kappa`	the coefficient of $\partial_r \alpha$ at rest is $\kappa = D_s \partial_\alpha n$	PASS
`R-bess`	$2K_1 - 3K_3 + K_5 = 48K_3/z^2$, so the code's τ_n is $D_s h/T$	PASS
`R-tausp`	tau_diffusion_hadron, one species: the prefactor of the code gives $(D_s/T)mK_3/K_2$	PASS
`R-clos`	momentum integrals at $z = 3,\ 15/2,\ 19/2$: $n,\ I_{21} = P_0,\ I_{31} = P_0 h,\ I_{41} = P_0 \mathcal{A},\ I_{42} = P_0 \mathcal{B},\ I_{52} = I_{42}\mathcal{C}$	PASS
`R-ang`	angular averages: $p_x^2 \to p^2/3,\ p_x^4 \to p^4/5,\ p_x^2 p_y^2 \to p^4/15$	PASS
`R-match`	$\int dP\, p_x \delta f_\nu = \nu^x,\ S^{\tau\tau x} = (I_{41}/P_0)\nu^x,\ S^{xxx} = 3(I_{42}/P_0)\nu^x,\ \int dP\, p_x p_y \delta f_\pi = \pi^{xy},\ S^{\tau xy} = (I_{52}/I_{42})\pi^{xy}$	PASS
`R-dh`	$\partial_T h$ of charm_eos	PASS
`R-dC`	$\partial_T \mathcal{C}$ of charm_eos	PASS
`R-dn`	$\partial_T n$ of charm_eos, per species	PASS
`R-I31`	$\partial_T I_{31}$ and $\partial_\alpha I_{31}$ of $I_{31} = nTh$	PASS
`R-reg`	two-way: $\epsilon\, u \cdot \partial\alpha$ adds $\epsilon u^\tau, \epsilon u^r$ to the α column of the number row	PASS
`R-regbg`	background drivers: $\partial_\alpha n \to \partial_\alpha n + \epsilon$ in the number row is the same term	PASS
`R-regk`	and through $\kappa = D_s \partial_\alpha n$ it adds $D_s \epsilon\, \partial\alpha$ to the consistent current row; the two-way drivers avoid this	PASS
`R-leak`	$\tau r N$ is conserved by the number row: the regulated row gives $\partial_\tau(\tau r N^\tau) + \partial_r(\tau r N^r) = -\tau r \epsilon\, u \cdot \partial\alpha$	PASS
`R-step`	$s(x) = x^3(10 - 15x + 6x^2)$ is C2: s, s', s" fixed at both ends	PASS
`R-stepw`	$f = f_0 + (1 - f_0)s,\ w = 1 - s = (1 - f)/(1 - f_0)$	PASS
`R-cor`	corona_α_rate = the solution of $\partial_\tau(\tau V N^\tau)$ = target, with its u^τ and nu^τ rates, symbolically	PASS
`R-tele`	the flux differences telescope	PASS
`R-llf`	the LLF flux is consistent, and antisymmetric under exchanging the two sides with a	PASS
`R-pos`	with a >= |N^r|/N^tau the flux is upwind where the drift equals a: the outflow of a vacuum cell vanishes	PASS
`R-audit`	audit_matrices.jl ran and every check passed: 71 checks	PASS
`A-visc2d`	two_d_viscous_matrix_derived	PASS
`A-BGf2d`	two_d_HQ_BG_matrix	PASS
`A-live`	two_d_HQ_live_rows	PASS
`A-BGc1d`	one_d_HQ_BG_consistent_matrix	PASS
`A-BGm1d`	one_d_HQ_BG_m2_matrix	PASS
`A-BGc2d`	two_d_HQ_BG_consistent_matrix	PASS
`A-BGm2d`	two_d_HQ_BG_m2_matrix	PASS
`A-tw1d`	one_d_two_way_matrix	PASS
`A-tw1dc`	one_d_two_way_matrix_conformal	PASS
`A-tw2d`	two_d_two_way_matrix	PASS
`A-tw2dc`	two_d_two_way_matrix_conformal	PASS
`A-cm1d`	one_d_two_way_cM_matrix	PASS
`A-cm1dc`	one_d_two_way_cM_matrix_conformal	PASS
`A-cm2d`	two_d_two_way_cM_matrix	PASS
`A-cm2dc`	two_d_two_way_cM_matrix_conformal	PASS
`A-cm0`	control: the S5 kernel differs from the system with δM in T_Q	PASS
`A-visc1d`	one_d_viscous_matrix	PASS
`A-MIS`	one_d_viscous_matrix_fugacity	PASS
`A-gamma`	one_d_viscous_matrix_gamma	PASS
`A-BGf1d`	one_d_viscous_matrix_fugacity_BG_only	PASS
`A-r4`	one_d_ideal_matrix_ruwen4	PASS
`A-r5`	one_d_ideal_matrix_ruwen5	PASS
`A-legacy`	two_d_viscous_matrix differs from the equations (its ten wrong entries)	PASS
`A-HQ`	two_d_viscous_HQ_matrix: derived medium at $\delta_{\pi\pi} = 0$, Fick charm rows with $\kappa = D_s n$	PASS
`A-r7`	one_d_ideal_matrix_ruwen7, $e + P = T\partial_T P_{\rm tot}$ at fixed α	PASS

id	what it tests	
`A-r7mu`	ruwen7 differs from $e + P = T\partial_T P_{\rm tot} + \alpha\partial_\alpha P_{\rm tot}$ (the wrong enthalpy)	PASS
`A-r6`	one_d_ideal_matrix_ruwen6 $= n = \partial_\alpha P_h/T$	PASS
`A-r6s`	control: ruwen6 with column 3, rows 1-2 of A_t and A_x exchanged differs	PASS
`D-visc1`	matrix1d_visc!: EOS, $\eta, \tau_\pi, \zeta, \tau_\Pi$, $\delta_{\pi\pi} = 4/3$ tau_pi into the kernel	PASS
`D-visc2`	matrix2d_visc!: the same, derived matrix	PASS
`D-MIS`	matrix1d_visc_HQ!, $\epsilon = 0$: κ, τ_n from the list; the p slot holds the entropy and is unused	PASS
`D-MISe`	matrix1d_visc_HQ!, $\epsilon = 10^{-3}$: only $\partial_\alpha n$ of the number row, not κ	PASS
`D-gamma`	matrix1d_visc_gamma_HQ!: n, κ, τ_n at $\alpha = 0$	PASS
`D-BGf1`	matrix1d_visc_HQ_BG!, $\epsilon = 10^{-3}$: $\partial_\alpha n$ only, not κ	PASS
`D-BGc1`	matrix1d_visc_HQ_BG_consistent!, $\epsilon = 10^{-3}$: enters $\partial_\alpha n$ in both rows	PASS
`D-BGm1`	matrix1d_visc_HQ_BG_consistent_m2!, $\epsilon = 10^{-3}$	PASS
`D-BGf2`	matrix2d_HQ_BG!, $\epsilon = 10^{-3}$	PASS
`D-BGc2`	matrix2d_HQ_BG_consistent!, $\epsilon = 10^{-3}$	PASS
`D-BGm2`	matrix2d_HQ_BG_m2!, $\epsilon = 10^{-3}$	PASS
`D-live`	matrix2d_visc_HQ_BG!, $\epsilon = 0$: derived medium with $\delta_{\pi\pi}$, passive medium, live charm rows	PASS
`D-tw1`	matrix1d_two_way!, $\epsilon = 0$, no edge	PASS
`D-tw1c`	matrix1d_two_way!, conformal	PASS
`D-tw2`	matrix2d_two_way!, $\epsilon = 0$, no edge	PASS
`D-tw2c`	matrix2d_two_way!, conformal	PASS
`D-r4`	matrix1d_ideal_HQ_density!: the medium EOS into ruwen4	PASS
`D-r5`	matrix1d_ideal_HQ_fugacity!: the medium EOS and n from the list into ruwen5	PASS
`D-r7`	matrix1d_ideal_HQ_br_p_fugacity!: P_h = n T, n from the list	PASS
`D-r6`	matrix1d_ideal_HQ_br_p_density!: α from n, P_h = n T	PASS
`D-HQup`	matrix2d_visc_HQ!: phi_8 is the fugacity, P in the pressure slot, D_s n $= \kappa$	PASS
`D-hess`	the Hessian of a charm list = differences of its gradient	PASS
`D-Ph`	d_α P_h / T = n, and charm_α inverts n	PASS
`D-hqp`	hq_pressure runs: the density of one charm quark	PASS
`D-mv`	matrix1d_visc runs	PASS
`D-cm1`	matrix1d_two_way_cM!, $\epsilon = 0$, no edge	PASS
`D-cm1c`	matrix1d_two_way_cM!, conformal	PASS
`D-cm2`	matrix2d_two_way_cM!, $\epsilon = 0$, no edge	PASS
`D-cm2c`	matrix2d_two_way_cM!, conformal	PASS
`D-edge`	matrix1d_two_way! at T below $T_{\rm edge}$: $D_s \to f(T)D_s$ and nothing else	PASS
`D-edge2`	matrix2d_two_way! at T below $T_{\rm edge}$: $D_s \to f(T)D_s$ and nothing else	PASS
`D-edgew`	$w = (1-f)/(1-f_0)$ inside the window	PASS
`D-edgeb`	f = 1 above $T_{\rm edge}$, $f = f_0$ below $T_{\rm edge} - w_T$; the weight the other way round	PASS
`D-twe`	matrix1d_two_way!, $\epsilon = 10^{-3}$: $+ \epsilon\, u \cdot \partial\alpha$ on the number row's α column, nothing else	PASS
`D-kappa`	κ of the code $= D_s\partial_\alpha n$, one species and a list	PASS
`D-taun`	τ_n of the code $= D_s h/T$, one species and a list	PASS
`D-I31`	$I_{31} = nTh$, $\mathcal{B} = Th$	PASS
`D-mom`	$I_{31}/n,\ I_{41}/P_0,\ I_{42}/P_0,\ I_{52}/I_{42}$ of a Boltzmann species, momentum quadrature $= Th,\ \mathcal{A},\ \mathcal{B},\ \mathcal{C}$	PASS
`D-cor`	corona_update_1d!, T below the window: $\partial_\tau(\tau r N^\tau) = -(G_{i+1/2} - G_{i-1/2})/\Delta r$, LLF, with the updated α rate	PASS
`D-tele`	the flux form telescopes: the sum of r dr dN/dtau over the cells is the boundary flux difference	PASS
`D-cor0`	above $T_{\rm edge}$ the update changes nothing	PASS
`D-corb`	inside the window: $\dot\alpha = (1-w)\dot\alpha_{\rm ql} + w\dot\alpha_{\rm flux}$	PASS
`D-cor2`	corona_update_2d!, T below the window: $\partial_\tau(\tau N^\tau)$ = the LLF flux differences	PASS

94 passed, 0 failed (94 checks evaluated).

Re-run with `wolframscript -file deriv/derive_drivers.wls`.

A The entries of the remaining two-way and charm matrices

As computed by `deriv/derive_two_way.wls`: the entries of its generated blocks, each compared with the derived expression (E-same). Field names and u^τ as in the section on the two-way system.

A.1 1+1D cylindrical

A.1.1 S5, two-way

`one_d_two_way_matrix` in `src/Matrix/1d_two_way_cilindrical.jl`, fields:

$$\phi = (T, u^r, {\pi^\phi}_\phi, {\pi^\eta}_\eta, \Pi, \alpha, \nu^r, {\pi_Q}^\phi{}_\phi, {\pi_Q}^\eta{}_\eta, \Pi_Q)$$

$$\begin{aligned}
(A_t)_{1,1} &= P''(T)T + 2h'(T)\nu^r u^r + P'(T)\left(u^r\right)^2 + P''(T)T\left(u^r\right)^2 \\
&\quad + n\left(h'(T)\left(u^r\right)^2 + h'(T) - 1\right) + (\partial_T n)\left(h\left(u^r\right)^2 + h - T\right) \\
(A_t)_{1,2} &= 2(u^r(P'(T)T + \Pi - {\pi^\eta}_\eta - {\pi^\phi}_\phi) + h(nu^r + \nu^r)) \\
(A_t)_{1,3} &= -\left(u^r\right)^2 \\
(A_t)_{1,4} &= -\left(u^r\right)^2 \\
(A_t)_{1,5} &= \left(u^r\right)^2 \\
(A_t)_{1,6} &= (\partial_\alpha n)\left(h\left(u^r\right)^2 + h - T\right) \\
(A_t)_{1,7} &= 2hu^r \\
(A_t)_{2,1} &= \frac{h'(T)\left(nu^r\left(u^\tau\right)^2 + 2\nu^r\left(u^r\right)^2 + \nu^r\right) + u^r\left(u^\tau\right)^2((\partial_T n)h + P''(T)T + P'(T))}{u^\tau} \\
(A_t)_{2,2} &= \frac{1}{\left(u^\tau\right)^3}\Big(\left(2\left(u^r\right)^4 + 3\left(u^r\right)^2 + 1\right)(P'(T)T + \Pi - {\pi^\eta}_\eta - {\pi^\phi}_\phi) \\
&\quad + h\left(2n\left(u^r\right)^4 + 3n\left(u^r\right)^2 + n + 2\nu^r\left(u^r\right)^3 + 3\nu^r u^r\right)\Big) \\
(A_t)_{2,3} &= -u^r u^\tau \\
(A_t)_{2,4} &= -u^r u^\tau \\
(A_t)_{2,5} &= u^r u^\tau \\
(A_t)_{2,6} &= (\partial_\alpha n)hu^r u^\tau \\
(A_t)_{2,7} &= \frac{h\left(2\left(u^r\right)^2 + 1\right)}{u^\tau} \\
(A_t)_{3,2} &= \frac{u^r(3\delta_{\pi\pi}{\pi^\phi}_\phi - 2\eta)}{3r^2u^\tau} \\
(A_t)_{3,3} &= \frac{\tau_\pi u^\tau}{r^2} \\
(A_t)_{4,2} &= \frac{u^r(3\delta_{\pi\pi}{\pi^\eta}_\eta - 2\eta)}{3\tau^2u^\tau} \\
(A_t)_{4,4} &= \frac{\tau_\pi u^\tau}{\tau^2} \\
(A_t)_{5,2} &= \frac{u^r\zeta}{u^\tau} \\
(A_t)_{5,5} &= \tau_\Pi u^\tau \\
(A_t)_{6,1} &= (\partial_T n)u^\tau \\
(A_t)_{6,2} &= \frac{nu^r\left(u^\tau\right)^2 + \nu^r}{\left(u^\tau\right)^3} \\
(A_t)_{6,6} &= (\partial_\alpha n)u^\tau \\
(A_t)_{6,7} &= \frac{u^r}{u^\tau} \\
(A_t)_{7,1} &= \frac{D_s u^\tau(h'(T)\nu^r + u^r((\partial_T n)T + n))}{T}
\end{aligned}$$

$$(A_t)_{7,2} = \frac{D_s h \left(n \left(u^r\right)^2 + n + \nu^r u^r\right)}{T u^\tau}$$
$$(A_t)_{7,6} = (\partial_\alpha n) D_s u^r u^\tau$$
$$(A_t)_{7,7} = \frac{D_s h u^\tau}{T}$$
$$(A_t)_{8,1} = \frac{D_s \left(3\mathcal{C}'(T)\pi_Q{}^\phi{}_\phi \left(u^\tau\right)^2 - 2\mathcal{B}'(T)\nu^r u^r\right)}{6T u^\tau}$$
$$(A_t)_{8,2} = -\frac{D_s \left(2\mathcal{A}\nu^r \left(u^\tau\right)^2 + 2\mathcal{B}\nu^r + u^r \left(u^\tau\right)^2 \left(2\mathcal{C}\Pi_Q - 2\mathcal{C}\pi_Q{}^\eta{}_\eta - 5\mathcal{C}\pi_Q{}^\phi{}_\phi + 2I_{31}\right)\right)}{6T \left(u^\tau\right)^3}$$
$$(A_t)_{8,7} = -\frac{\mathcal{B} D_s u^r}{3T u^\tau}$$
$$(A_t)_{8,8} = \frac{\mathcal{C} D_s u^\tau}{2T}$$
$$(A_t)_{9,1} = \frac{D_s \left(3\mathcal{C}'(T)\pi_Q{}^\eta{}_\eta \left(u^\tau\right)^2 - 2\mathcal{B}'(T)\nu^r u^r\right)}{6T u^\tau}$$
$$(A_t)_{9,2} = -\frac{D_s \left(2\mathcal{A}\nu^r \left(u^\tau\right)^2 + 2\mathcal{B}\nu^r + u^r \left(u^\tau\right)^2 \left(2\mathcal{C}\Pi_Q - 5\mathcal{C}\pi_Q{}^\eta{}_\eta - 2\mathcal{C}\pi_Q{}^\phi{}_\phi + 2I_{31}\right)\right)}{6T \left(u^\tau\right)^3}$$
$$(A_t)_{9,7} = -\frac{\mathcal{B} D_s u^r}{3T u^\tau}$$
$$(A_t)_{9,9} = \frac{\mathcal{C} D_s u^\tau}{2T}$$
$$(A_t)_{10,1} = \frac{D_s \left(5\mathcal{B}'(T)\nu^r u^r + 3\mathcal{C}'(T)\Pi_Q \left(u^\tau\right)^2 + 3(\partial_T I_{31}) \left(u^\tau\right)^2\right)}{6T u^\tau}$$
$$(A_t)_{10,2} = \frac{D_s \left(2\mathcal{A}\nu^r \left(u^\tau\right)^2 + 5\mathcal{B}\nu^r + u^r \left(u^\tau\right)^2 \left(5\mathcal{C}\Pi_Q - 2\mathcal{C}(\pi_Q{}^\eta{}_\eta + \pi_Q{}^\phi{}_\phi) + 5I_{31}\right)\right)}{6T \left(u^\tau\right)^3}$$
$$(A_t)_{10,6} = \frac{(\partial_\alpha I_{31}) D_s u^\tau}{2T}$$
$$(A_t)_{10,7} = \frac{5\mathcal{B} D_s u^r}{6T u^\tau}$$
$$(A_t)_{10,10} = \frac{\mathcal{C} D_s u^\tau}{2T}$$
$$(A_r)_{1,1} = \frac{h'(T) \left(n u^r \left(u^\tau\right)^2 + 2\nu^r \left(u^r\right)^2 + \nu^r\right) + u^r \left(u^\tau\right)^2 \left((\partial_T n) h + P''(T) T + P'(T)\right)}{u^\tau}$$
$$(A_r)_{1,2} = \frac{1}{\left(u^\tau\right)^3} \Big(\left(2 \left(u^r\right)^4 + 3 \left(u^r\right)^2 + 1\right) \left(P'(T) T + \Pi - \pi^\eta{}_\eta - \pi^\phi{}_\phi\right) + h \left(2n \left(u^r\right)^4 + 3n \left(u^r\right)^2 + n + 2\nu^r \left(u^r\right)^3 + 3\nu^r u^r\right) \Big)$$
$$(A_r)_{1,3} = -u^r u^\tau$$
$$(A_r)_{1,4} = -u^r u^\tau$$
$$(A_r)_{1,5} = u^r u^\tau$$
$$(A_r)_{1,6} = (\partial_\alpha n) h u^r u^\tau$$
$$(A_r)_{1,7} = \frac{h \left(2 \left(u^r\right)^2 + 1\right)}{u^\tau}$$
$$(A_r)_{2,1} = P'(T) + n + (\partial_T n) T + 2h'(T)\nu^r u^r + P'(T) \left(u^r\right)^2 + (\partial_T n) h \left(u^r\right)^2 + h'(T) n \left(u^r\right)^2 + P''(T) T \left(u^r\right)^2$$
$$(A_r)_{2,2} = 2(u^r (P'(T) T + \Pi - \pi^\eta{}_\eta - \pi^\phi{}_\phi) + h(n u^r + \nu^r))$$
$$(A_r)_{2,3} = -\left(u^r\right)^2 - 1$$
$$(A_r)_{2,4} = -\left(u^r\right)^2 - 1$$
$$(A_r)_{2,5} = \left(u^\tau\right)^2$$

$$
\begin{aligned}
(A_r)_{2,6} &= (\partial_\alpha n)\left(h\left(u^r\right)^2 + T\right)\\
(A_r)_{2,7} &= 2hu^r\\
(A_r)_{3,2} &= \frac{3\delta_{\pi\pi}\pi^\phi{}_\phi - 2\eta}{3r^2}\\
(A_r)_{3,3} &= \frac{\tau_\pi u^r}{r^2}\\
(A_r)_{4,2} &= \frac{3\delta_{\pi\pi}\pi^\eta{}_\eta - 2\eta}{3\tau^2}\\
(A_r)_{4,4} &= \frac{\tau_\pi u^r}{\tau^2}\\
(A_r)_{5,2} &= \zeta\\
(A_r)_{5,5} &= \tau_\Pi u^r\\
(A_r)_{6,1} &= (\partial_T n)u^r\\
(A_r)_{6,2} &= n\\
(A_r)_{6,6} &= (\partial_\alpha n)u^r\\
(A_r)_{6,7} &= 1\\
(A_r)_{7,1} &= \frac{D_s\left(h'(T)\nu^r u^r + (\partial_T n)T\left(u^\tau\right)^2 + n\left(u^\tau\right)^2\right)}{T}\\
(A_r)_{7,2} &= \frac{D_s h\left(nu^r\left(u^\tau\right)^2 + \nu^r\left(\left(u^r\right)^2 + 2\right)\right)}{T\left(u^\tau\right)^2}\\
(A_r)_{7,6} &= (\partial_\alpha n)D_s\left(u^\tau\right)^2\\
(A_r)_{7,7} &= \frac{D_s h u^r}{T}\\
(A_r)_{8,1} &= -\frac{D_s(2\mathcal{B}'(T)\nu^r - 3\mathcal{C}'(T)\pi_Q{}^\phi{}_\phi u^r)}{6T}\\
(A_r)_{8,2} &= -\frac{D_s\left(2\mathcal{A}\nu^r u^r + \mathcal{C}\left(u^\tau\right)^2(2\Pi_Q - 2\pi_Q{}^\eta{}_\eta - 5\pi_Q{}^\phi{}_\phi) + 2I_{31}\left(u^\tau\right)^2\right)}{6T\left(u^\tau\right)^2}\\
(A_r)_{8,7} &= -\frac{\mathcal{B}D_s}{3T}\\
(A_r)_{8,8} &= \frac{\mathcal{C}D_s u^r}{2T}\\
(A_r)_{9,1} &= -\frac{D_s(2\mathcal{B}'(T)\nu^r - 3\mathcal{C}'(T)\pi_Q{}^\eta{}_\eta u^r)}{6T}\\
(A_r)_{9,2} &= -\frac{D_s\left(2\mathcal{A}\nu^r u^r + \mathcal{C}\left(u^\tau\right)^2(2\Pi_Q - 5\pi_Q{}^\eta{}_\eta - 2\pi_Q{}^\phi{}_\phi) + 2I_{31}\left(u^\tau\right)^2\right)}{6T\left(u^\tau\right)^2}\\
(A_r)_{9,7} &= -\frac{\mathcal{B}D_s}{3T}\\
(A_r)_{9,9} &= \frac{\mathcal{C}D_s u^r}{2T}\\
(A_r)_{10,1} &= \frac{D_s(5\mathcal{B}'(T)\nu^r + 3u^r(\mathcal{C}'(T)\Pi_Q + (\partial_T I_{31})))}{6T}\\
(A_r)_{10,2} &= \frac{D_s\left(2\mathcal{A}\nu^r u^r + \mathcal{C}\left(u^\tau\right)^2(5\Pi_Q - 2(\pi_Q{}^\eta{}_\eta + \pi_Q{}^\phi{}_\phi)) + 5I_{31}\left(u^\tau\right)^2\right)}{6T\left(u^\tau\right)^2}\\
(A_r)_{10,6} &= \frac{(\partial_\alpha I_{31})D_s u^r}{2T}\\
(A_r)_{10,7} &= \frac{5\mathcal{B}D_s}{6T}\\
(A_r)_{10,10} &= \frac{\mathcal{C}D_s u^r}{2T}\\
S_1 &= \frac{1}{r\tau u^\tau}\Big(-\pi^\eta{}_\eta\tau u^r - \pi^\phi{}_\phi\tau u^r + P'(T)T\tau u^r - \pi^\eta{}_\eta\tau\left(u^r\right)^3 - \pi^\phi{}_\phi\tau\left(u^r\right)^3 + P'(T)T\tau\left(u^r\right)^3
\end{aligned}
$$

$$
\begin{aligned}
&+ \pi^{\eta}{}_{\eta} r u^{\tau} + P'(T) r T u^{\tau} - \pi^{\eta}{}_{\eta} r \left(u^{r}\right)^{2} u^{\tau} - \pi^{\phi}{}_{\phi} r \left(u^{r}\right)^{2} u^{\tau} + P'(T) r T \left(u^{r}\right)^{2} u^{\tau} \\
&+ h n \left(u^{\tau}\right)^{2} \left(r u^{\tau} + \tau u^{r}\right) + \Pi \left(u^{\tau}\right)^{2} \left(r u^{\tau} + \tau u^{r}\right) + h \nu^{r} \left(2 r u^{r} u^{\tau} + 2 \tau \left(u^{r}\right)^{2} + \tau\right)\Big) \\
S_2 = &- \frac{\Pi + \pi^{\phi}{}_{\phi} + P}{r} - \frac{nT}{r} + \frac{h n \left(u^{r}\right)^{2} + 2 h \nu^{r} u^{r} + nT}{r} + \frac{u^{r} u^{\tau} (P'(T) T + \Pi - \pi^{\eta}{}_{\eta} - \pi^{\phi}{}_{\phi})}{\tau} \\
&+ \frac{\left(u^{r}\right)^{2} \left(P'(T) T + \Pi\right) + \Pi - \left(u^{\tau}\right)^{2} \left(\pi^{\eta}{}_{\eta} + \pi^{\phi}{}_{\phi}\right) + P}{r} + \frac{h \left(n u^{r} \left(u^{\tau}\right)^{2} + 2 \nu^{r} \left(u^{r}\right)^{2} + \nu^{r}\right)}{\tau u^{\tau}} \\
S_3 = &\frac{3 \pi^{\phi}{}_{\phi} (r(\delta_{\pi\pi} u^{\tau} + \tau) + \delta_{\pi\pi} \tau u^{r}) - 2 r u^{\tau} \eta + 4 \tau u^{r} \eta}{3 r^{3} \tau} \\
S_4 = &\frac{3 \pi^{\eta}{}_{\eta} (r(\delta_{\pi\pi} u^{\tau} + \tau) + \delta_{\pi\pi} \tau u^{r}) + 4 r u^{\tau} \eta - 2 \tau u^{r} \eta}{3 r \tau^{3}} \\
S_5 = &\Pi + \frac{u^{r} \zeta}{r} + \frac{u^{\tau} \zeta}{\tau} \\
S_6 = &\frac{n u^{r} + \nu^{r}}{r} + \frac{n \left(u^{r}\right)^{2} + n + \nu^{r} u^{r}}{\tau u^{\tau}} \\
S_7 = &\frac{D_s h \nu^{r} \left(r \left(u^{r}\right)^{2} + r + \tau u^{r} u^{\tau}\right)}{r T \tau u^{\tau}} + \nu^{r} \\
S_8 = &\pi_Q{}^{\phi}{}_{\phi} \\
&+ \frac{1}{6 r T \tau u^{\tau}} D_s \Big(-2 \mathcal{B} \nu^{r} (r u^{r} - 2 \tau u^{\tau}) - 2 I_{31} \left(r \left(u^{r}\right)^{2} + r - 2 \tau u^{r} u^{\tau}\right) \\
&+ \mathcal{C} \Big(7 \pi_Q{}^{\phi}{}_{\phi} \tau u^{r} u^{\tau} - 2 \pi_Q{}^{\eta}{}_{\eta} r \left(u^{\tau}\right)^{2} + 3 \pi_Q{}^{\phi}{}_{\phi} r \left(u^{\tau}\right)^{2} \\
&- 2 \Pi_Q \left(r \left(u^{r}\right)^{2} + r - 2 \tau u^{r} u^{\tau}\right)\Big)\Big) \\
S_9 = &\pi_Q{}^{\eta}{}_{\eta} \\
&+ \frac{1}{6 r T \tau u^{\tau}} D_s \Big(-2 I_{31} \tau u^{r} u^{\tau} + 4 I_{31} r \left(u^{\tau}\right)^{2} + 2 \mathcal{B} \nu^{r} (2 r u^{r} - \tau u^{\tau}) \\
&+ \mathcal{C} \Big(-2 \Pi_Q \tau u^{r} u^{\tau} + 3 \pi_Q{}^{\eta}{}_{\eta} \tau u^{r} u^{\tau} - 2 \pi_Q{}^{\phi}{}_{\phi} \tau u^{r} u^{\tau} + 4 \Pi_Q r \left(u^{\tau}\right)^{2} \\
&+ 7 \pi_Q{}^{\eta}{}_{\eta} r \left(u^{\tau}\right)^{2}\Big)\Big) \\
S_{10} = &\Pi_Q \\
&+ \frac{1}{6 r T \tau u^{\tau}} D_s \Big(5 \mathcal{B} \nu^{r} (r u^{r} + \tau u^{\tau}) + 5 I_{31} \left(r \left(u^{r}\right)^{2} + r + \tau u^{r} u^{\tau}\right) \\
&+ \mathcal{C} \left(5 \Pi_Q \left(r \left(u^{r}\right)^{2} + r + \tau u^{r} u^{\tau}\right) + 2 \pi_Q{}^{\eta}{}_{\eta} r \left(u^{\tau}\right)^{2} + 2 \pi_Q{}^{\phi}{}_{\phi} \tau u^{r} u^{\tau}\right)\Big)
\end{aligned}
$$

A.1.2 S6, fully coupled

`one_d_two_way_cM_matrix` in `src/Matrix/1d_two_way_cM_cilindrical.jl`, fields:

$$
\phi = (T, u^{r}, \pi^{\phi}{}_{\phi}, \pi^{\eta}{}_{\eta}, \Pi, \alpha, \nu^{r}, \pi_Q{}^{\phi}{}_{\phi}, \pi_Q{}^{\eta}{}_{\eta}, \Pi_Q)
$$

$$
\begin{aligned}
(A_t)_{1,1} = &P''(T) T + 2 h'(T) \nu^{r} u^{r} + P'(T) \left(u^{r}\right)^{2} + P''(T) T \left(u^{r}\right)^{2} \\
&+ n \left(h'(T) \left(u^{r}\right)^{2} + h'(T) - 1\right) + (\partial_T n) \left(h \left(u^{r}\right)^{2} + h - T\right) \\
(A_t)_{1,2} = &2(u^{r}(P'(T) T + \Pi + \Pi_Q - \pi^{\eta}{}_{\eta} - \pi_Q{}^{\eta}{}_{\eta} - \pi^{\phi}{}_{\phi} - \pi_Q{}^{\phi}{}_{\phi}) + h(n u^{r} + \nu^{r})) \\
(A_t)_{1,3} = &- \left(u^{r}\right)^{2} \\
(A_t)_{1,4} = &- \left(u^{r}\right)^{2} \\
(A_t)_{1,5} = &\left(u^{r}\right)^{2} \\
(A_t)_{1,6} = &(\partial_\alpha n) \left(h \left(u^{r}\right)^{2} + h - T\right) \\
(A_t)_{1,7} = &2 h u^{r} \\
(A_t)_{1,8} = &- \left(u^{r}\right)^{2} \\
(A_t)_{1,9} = &- \left(u^{r}\right)^{2}
\end{aligned}
$$

$$
\begin{aligned}
(A_t)_{1,10} &= (u^r)^2 \\
(A_t)_{2,1} &= \frac{h'(T)\left(nu^r\left(u^\tau\right)^2+2\nu^r\left(u^r\right)^2+\nu^r\right)+u^r\left(u^\tau\right)^2\left((\partial_T n)h+P''(T)T+P'(T)\right)}{u^\tau} \\
(A_t)_{2,2} &= \frac{1}{\left(u^\tau\right)^3}\Big(\left(2\left(u^r\right)^4+3\left(u^r\right)^2+1\right)\left(P'(T)T+\Pi+\Pi_Q-\pi^\eta{}_\eta-\pi_Q{}^\eta{}_\eta-\pi^\phi{}_\phi-\pi_Q{}^\phi{}_\phi\right) \\
&\quad +h\left(2n\left(u^r\right)^4+3n\left(u^r\right)^2+n+2\nu^r\left(u^r\right)^3+3\nu^r u^r\right)\Big) \\
(A_t)_{2,3} &= -u^r u^\tau \\
(A_t)_{2,4} &= -u^r u^\tau \\
(A_t)_{2,5} &= u^r u^\tau \\
(A_t)_{2,6} &= (\partial_\alpha n)hu^r u^\tau \\
(A_t)_{2,7} &= \frac{h\left(2\left(u^r\right)^2+1\right)}{u^\tau} \\
(A_t)_{2,8} &= -u^r u^\tau \\
(A_t)_{2,9} &= -u^r u^\tau \\
(A_t)_{2,10} &= u^r u^\tau \\
(A_t)_{3,2} &= \frac{u^r(3\delta_{\pi\pi}\pi^\phi{}_\phi-2\eta)}{3r^2u^\tau} \\
(A_t)_{3,3} &= \frac{\tau_\pi u^\tau}{r^2} \\
(A_t)_{4,2} &= \frac{u^r(3\delta_{\pi\pi}\pi^\eta{}_\eta-2\eta)}{3\tau^2u^\tau} \\
(A_t)_{4,4} &= \frac{\tau_\pi u^\tau}{\tau^2} \\
(A_t)_{5,2} &= \frac{u^r\zeta}{u^\tau} \\
(A_t)_{5,5} &= \tau_\Pi u^\tau \\
(A_t)_{6,1} &= (\partial_T n)u^\tau \\
(A_t)_{6,2} &= \frac{nu^r\left(u^\tau\right)^2+\nu^r}{\left(u^\tau\right)^3} \\
(A_t)_{6,6} &= (\partial_\alpha n)u^\tau \\
(A_t)_{6,7} &= \frac{u^r}{u^\tau} \\
(A_t)_{7,1} &= \frac{D_s u^\tau(h'(T)\nu^r+u^r((\partial_T n)T+n))}{T} \\
(A_t)_{7,2} &= \frac{D_s\left(h\left(n\left(u^r\right)^2+n+\nu^r u^r\right)+\left(u^\tau\right)^2\left(\Pi_Q-\pi_Q{}^\eta{}_\eta-\pi_Q{}^\phi{}_\phi\right)\right)}{Tu^\tau} \\
(A_t)_{7,6} &= (\partial_\alpha n)D_s u^r u^\tau \\
(A_t)_{7,7} &= \frac{D_s h u^\tau}{T} \\
(A_t)_{7,8} &= -\frac{D_s u^r u^\tau}{T} \\
(A_t)_{7,9} &= -\frac{D_s u^r u^\tau}{T} \\
(A_t)_{7,10} &= \frac{D_s u^r u^\tau}{T} \\
(A_t)_{8,1} &= \frac{D_s\left(3\mathcal{C}'(T)\pi_Q{}^\phi{}_\phi\left(u^\tau\right)^2-2\mathcal{B}'(T)\nu^r u^r\right)}{6Tu^\tau} \\
(A_t)_{8,2} &= -\frac{D_s\left(2\mathcal{A}\nu^r\left(u^\tau\right)^2+2\mathcal{B}\nu^r+u^r\left(u^\tau\right)^2\left(2\mathcal{C}\Pi_Q-2\mathcal{C}\pi_Q{}^\eta{}_\eta-5\mathcal{C}\pi_Q{}^\phi{}_\phi+2I_{31}\right)\right)}{6T\left(u^\tau\right)^3} \\
(A_t)_{8,7} &= -\frac{\mathcal{B}D_s u^r}{3Tu^\tau}
\end{aligned}
$$

$$(A_t)_{8,8} = \frac{\mathcal{C}D_s u^\tau}{2T}$$
$$(A_t)_{9,1} = \frac{D_s\left(3\mathcal{C}'(T)\pi_Q{}^\eta{}_\eta\left(u^\tau\right)^2 - 2\mathcal{B}'(T)\nu^r u^r\right)}{6Tu^\tau}$$
$$(A_t)_{9,2} = -\frac{D_s\left(2\mathcal{A}\nu^r\left(u^\tau\right)^2 + 2\mathcal{B}\nu^r + u^r\left(u^\tau\right)^2\left(2\mathcal{C}\Pi_Q - 5\mathcal{C}\pi_Q{}^\eta{}_\eta - 2\mathcal{C}\pi_Q{}^\phi{}_\phi + 2I_{31}\right)\right)}{6T\left(u^\tau\right)^3}$$
$$(A_t)_{9,7} = -\frac{\mathcal{B}D_s u^r}{3Tu^\tau}$$
$$(A_t)_{9,9} = \frac{\mathcal{C}D_s u^\tau}{2T}$$
$$(A_t)_{10,1} = \frac{D_s\left(5\mathcal{B}'(T)\nu^r u^r + 3\mathcal{C}'(T)\Pi_Q\left(u^\tau\right)^2 + 3(\partial_T I_{31})\left(u^\tau\right)^2\right)}{6Tu^\tau}$$
$$(A_t)_{10,2} = \frac{D_s\left(2\mathcal{A}\nu^r\left(u^\tau\right)^2 + 5\mathcal{B}\nu^r + u^r\left(u^\tau\right)^2\left(5\mathcal{C}\Pi_Q - 2\mathcal{C}(\pi_Q{}^\eta{}_\eta + \pi_Q{}^\phi{}_\phi) + 5I_{31}\right)\right)}{6T\left(u^\tau\right)^3}$$
$$(A_t)_{10,6} = \frac{(\partial_\alpha I_{31})D_s u^\tau}{2T}$$
$$(A_t)_{10,7} = \frac{5\mathcal{B}D_s u^r}{6Tu^\tau}$$
$$(A_t)_{10,10} = \frac{\mathcal{C}D_s u^\tau}{2T}$$
$$(A_r)_{1,1} = \frac{h'(T)\left(nu^r\left(u^\tau\right)^2 + 2\nu^r\left(u^r\right)^2 + \nu^r\right) + u^r\left(u^\tau\right)^2\left((\partial_T n)h + P''(T)T + P'(T)\right)}{u^\tau}$$
$$(A_r)_{1,2} = \frac{1}{\left(u^\tau\right)^3}\Big(\left(2\left(u^r\right)^4 + 3\left(u^r\right)^2 + 1\right)\left(P'(T)T + \Pi + \Pi_Q - \pi^\eta{}_\eta - \pi_Q{}^\eta{}_\eta - \pi^\phi{}_\phi - \pi_Q{}^\phi{}_\phi\right) + h\left(2n\left(u^r\right)^4 + 3n\left(u^r\right)^2 + n + 2\nu^r\left(u^r\right)^3 + 3\nu^r u^r\right)\Big)$$
$$(A_r)_{1,3} = -u^r u^\tau$$
$$(A_r)_{1,4} = -u^r u^\tau$$
$$(A_r)_{1,5} = u^r u^\tau$$
$$(A_r)_{1,6} = (\partial_\alpha n)hu^r u^\tau$$
$$(A_r)_{1,7} = \frac{h\left(2\left(u^r\right)^2 + 1\right)}{u^\tau}$$
$$(A_r)_{1,8} = -u^r u^\tau$$
$$(A_r)_{1,9} = -u^r u^\tau$$
$$(A_r)_{1,10} = u^r u^\tau$$
$$(A_r)_{2,1} = P'(T) + n + (\partial_T n)T + 2h'(T)\nu^r u^r + P'(T)\left(u^r\right)^2 + (\partial_T n)h\left(u^r\right)^2 + h'(T)n\left(u^r\right)^2 + P''(T)T\left(u^r\right)^2$$
$$(A_r)_{2,2} = 2(u^r(P'(T)T + \Pi + \Pi_Q - \pi^\eta{}_\eta - \pi_Q{}^\eta{}_\eta - \pi^\phi{}_\phi - \pi_Q{}^\phi{}_\phi) + h(nu^r + \nu^r))$$
$$(A_r)_{2,3} = -\left(u^r\right)^2 - 1$$
$$(A_r)_{2,4} = -\left(u^r\right)^2 - 1$$
$$(A_r)_{2,5} = \left(u^\tau\right)^2$$
$$(A_r)_{2,6} = (\partial_\alpha n)\left(h\left(u^r\right)^2 + T\right)$$
$$(A_r)_{2,7} = 2hu^r$$
$$(A_r)_{2,8} = -\left(u^r\right)^2 - 1$$
$$(A_r)_{2,9} = -\left(u^r\right)^2 - 1$$
$$(A_r)_{2,10} = \left(u^\tau\right)^2$$
$$(A_r)_{3,2} = \frac{3\delta_{\pi\pi}\pi^\phi{}_\phi - 2\eta}{3r^2}$$

$$(A_r)_{3,3} = \frac{\tau_\pi u^r}{r^2}$$
$$(A_r)_{4,2} = \frac{3\delta_{\pi\pi}\pi^\eta{}_\eta - 2\eta}{3\tau^2}$$
$$(A_r)_{4,4} = \frac{\tau_\pi u^r}{\tau^2}$$
$$(A_r)_{5,2} = \zeta$$
$$(A_r)_{5,5} = \tau_\Pi u^r$$
$$(A_r)_{6,1} = (\partial_T n)u^r$$
$$(A_r)_{6,2} = n$$
$$(A_r)_{6,6} = (\partial_\alpha n)u^r$$
$$(A_r)_{6,7} = 1$$
$$(A_r)_{7,1} = \frac{D_s\left(h'(T)\nu^r u^r + (\partial_T n)T\left(u^\tau\right)^2 + n\left(u^\tau\right)^2\right)}{T}$$
$$(A_r)_{7,2} = \frac{D_s\left(hnu^r\left(u^\tau\right)^2 + h\nu^r\left(u^r\right)^2 + 2h\nu^r + \Pi_Q u^r\left(u^\tau\right)^2 - \pi_Q{}^\eta{}_\eta u^r\left(u^\tau\right)^2 - \pi_Q{}^\phi{}_\phi u^r\left(u^\tau\right)^2\right)}{T\left(u^\tau\right)^2}$$
$$(A_r)_{7,6} = (\partial_\alpha n)D_s\left(u^\tau\right)^2$$
$$(A_r)_{7,7} = \frac{D_s h u^r}{T}$$
$$(A_r)_{7,8} = -\frac{D_s\left(u^\tau\right)^2}{T}$$
$$(A_r)_{7,9} = -\frac{D_s\left(u^\tau\right)^2}{T}$$
$$(A_r)_{7,10} = \frac{D_s\left(u^\tau\right)^2}{T}$$
$$(A_r)_{8,1} = -\frac{D_s(2\mathcal{B}'(T)\nu^r - 3\mathcal{C}'(T)\pi_Q{}^\phi{}_\phi u^r)}{6T}$$
$$(A_r)_{8,2} = -\frac{D_s\left(2\mathcal{A}\nu^r u^r + \mathcal{C}\left(u^\tau\right)^2(2\Pi_Q - 2\pi_Q{}^\eta{}_\eta - 5\pi_Q{}^\phi{}_\phi) + 2I_{31}\left(u^\tau\right)^2\right)}{6T\left(u^\tau\right)^2}$$
$$(A_r)_{8,7} = -\frac{\mathcal{B}D_s}{3T}$$
$$(A_r)_{8,8} = \frac{\mathcal{C}D_s u^r}{2T}$$
$$(A_r)_{9,1} = -\frac{D_s(2\mathcal{B}'(T)\nu^r - 3\mathcal{C}'(T)\pi_Q{}^\eta{}_\eta u^r)}{6T}$$
$$(A_r)_{9,2} = -\frac{D_s\left(2\mathcal{A}\nu^r u^r + \mathcal{C}\left(u^\tau\right)^2(2\Pi_Q - 5\pi_Q{}^\eta{}_\eta - 2\pi_Q{}^\phi{}_\phi) + 2I_{31}\left(u^\tau\right)^2\right)}{6T\left(u^\tau\right)^2}$$
$$(A_r)_{9,7} = -\frac{\mathcal{B}D_s}{3T}$$
$$(A_r)_{9,9} = \frac{\mathcal{C}D_s u^r}{2T}$$
$$(A_r)_{10,1} = \frac{D_s(5\mathcal{B}'(T)\nu^r + 3u^r(\mathcal{C}'(T)\Pi_Q + (\partial_T I_{31})))}{6T}$$
$$(A_r)_{10,2} = \frac{D_s\left(2\mathcal{A}\nu^r u^r + \mathcal{C}\left(u^\tau\right)^2(5\Pi_Q - 2(\pi_Q{}^\eta{}_\eta + \pi_Q{}^\phi{}_\phi)) + 5I_{31}\left(u^\tau\right)^2\right)}{6T\left(u^\tau\right)^2}$$
$$(A_r)_{10,6} = \frac{(\partial_\alpha I_{31})D_s u^r}{2T}$$
$$(A_r)_{10,7} = \frac{5\mathcal{B}D_s}{6T}$$
$$(A_r)_{10,10} = \frac{\mathcal{C}D_s u^r}{2T}$$

$$
\begin{aligned}
S_1 &= \frac{\Pi_Q + {\pi_Q}^{\eta}{}_{\eta}}{\tau} + \frac{\Pi + {\pi^{\eta}}_{\eta} + P}{\tau} + \frac{nT}{\tau} + \frac{(u^r)^2\left(\Pi_Q - {\pi_Q}^{\eta}{}_{\eta} - {\pi_Q}^{\phi}{}_{\phi}\right)}{\tau} \\
&+ \frac{h\left(n\,(u^r)^2 + n + 2\nu^r u^r\right) - nT}{\tau} + \frac{u^r u^\tau(\Pi_Q - {\pi_Q}^{\eta}{}_{\eta} - {\pi_Q}^{\phi}{}_{\phi})}{r} \\
&+ \frac{u^r u^\tau(P'(T)T + \Pi - {\pi^{\eta}}_{\eta} - {\pi^{\phi}}_{\phi})}{r} + \frac{P'(T)T\,(u^\tau)^2 + (u^r)^2\left(\Pi - {\pi^{\eta}}_{\eta} - {\pi^{\phi}}_{\phi}\right) - P}{\tau} \\
&+ \frac{h\left(nu^r\,(u^\tau)^2 + 2\nu^r\,(u^r)^2 + \nu^r\right)}{ru^\tau} \\
S_2 &= -\frac{\Pi_Q + {\pi_Q}^{\phi}{}_{\phi}}{r} - \frac{\Pi + {\pi^{\phi}}_{\phi} + P}{r} - \frac{nT}{r} + \frac{hn\,(u^r)^2 + 2h\nu^r u^r + nT}{r} \\
&+ \frac{u^r u^\tau(\Pi_Q - {\pi_Q}^{\eta}{}_{\eta} - {\pi_Q}^{\phi}{}_{\phi})}{\tau} + \frac{u^r u^\tau(P'(T)T + \Pi - {\pi^{\eta}}_{\eta} - {\pi^{\phi}}_{\phi})}{\tau} \\
&+ \frac{(u^\tau)^2\left(\Pi_Q - {\pi_Q}^{\eta}{}_{\eta} - {\pi_Q}^{\phi}{}_{\phi}\right)}{r} + \frac{(u^r)^2\left(P'(T)T + \Pi\right) + \Pi - (u^\tau)^2\left({\pi^{\eta}}_{\eta} + {\pi^{\phi}}_{\phi}\right) + P}{r} \\
&+ \frac{h\left(nu^r\,(u^\tau)^2 + 2\nu^r\,(u^r)^2 + \nu^r\right)}{\tau u^\tau} \\
S_3 &= \frac{3{\pi^{\phi}}_{\phi}(r(\delta_{\pi\pi}u^\tau + \tau) + \delta_{\pi\pi}\tau u^r) - 2ru^\tau\eta + 4\tau u^r\eta}{3r^3\tau} \\
S_4 &= \frac{3{\pi^{\eta}}_{\eta}(r(\delta_{\pi\pi}u^\tau + \tau) + \delta_{\pi\pi}\tau u^r) + 4ru^\tau\eta - 2\tau u^r\eta}{3r\tau^3} \\
S_5 &= \Pi + \frac{u^r\zeta}{r} + \frac{u^\tau\zeta}{\tau} \\
S_6 &= \frac{nu^r + \nu^r}{r} + \frac{n\,(u^r)^2 + n + \nu^r u^r}{\tau u^\tau} \\
S_7 &= \frac{1}{rT\tau u^\tau}\Big(\nu^r rT\tau u^\tau - D_s\,(u^\tau)^2\left(2{\pi_Q}^{\eta}{}_{\eta}ru^r + {\pi_Q}^{\eta}{}_{\eta}\tau u^\tau + {\pi_Q}^{\phi}{}_{\phi}ru^r + 2{\pi_Q}^{\phi}{}_{\phi}\tau u^\tau\right) \\
&+ D_s h\nu^r\left(r\,(u^r)^2 + r + \tau u^r u^\tau\right)\Big) \\
S_8 &= {\pi_Q}^{\phi}{}_{\phi} \\
&+ \frac{1}{6rT\tau u^\tau}D_s\Big(-2\mathcal{B}\nu^r(ru^r - 2\tau u^\tau) - 2I_{31}\left(r\,(u^r)^2 + r - 2\tau u^r u^\tau\right) \\
&+ \mathcal{C}\Big(7{\pi_Q}^{\phi}{}_{\phi}\tau u^r u^\tau - 2{\pi_Q}^{\eta}{}_{\eta}r\,(u^\tau)^2 + 3{\pi_Q}^{\phi}{}_{\phi}r\,(u^\tau)^2 \\
&- 2\Pi_Q\left(r\,(u^r)^2 + r - 2\tau u^r u^\tau\right)\Big)\Big) \\
S_9 &= {\pi_Q}^{\eta}{}_{\eta} \\
&+ \frac{1}{6rT\tau u^\tau}D_s\Big(-2I_{31}\tau u^r u^\tau + 4I_{31}r\,(u^\tau)^2 + 2\mathcal{B}\nu^r(2ru^r - \tau u^\tau) \\
&+ \mathcal{C}\Big(-2\Pi_Q\tau u^r u^\tau + 3{\pi_Q}^{\eta}{}_{\eta}\tau u^r u^\tau - 2{\pi_Q}^{\phi}{}_{\phi}\tau u^r u^\tau + 4\Pi_Q r\,(u^\tau)^2 \\
&+ 7{\pi_Q}^{\eta}{}_{\eta}r\,(u^\tau)^2\Big)\Big) \\
S_{10} &= \Pi_Q \\
&+ \frac{1}{6rT\tau u^\tau}D_s\Big(5\mathcal{B}\nu^r(ru^r + \tau u^\tau) + 5I_{31}\left(r\,(u^r)^2 + r + \tau u^r u^\tau\right) \\
&+ \mathcal{C}\left(5\Pi_Q\left(r\,(u^r)^2 + r + \tau u^r u^\tau\right) + 2{\pi_Q}^{\eta}{}_{\eta}r\,(u^\tau)^2 + 2{\pi_Q}^{\phi}{}_{\phi}\tau u^r u^\tau\right)\Big)
\end{aligned}
$$

A.2 2+1D Cartesian

A.2.1 S3, charm first moment on a background

`two_d_HQ_BG_consistent_matrix` in `src/Matrix/2d_viscous_HQ_BG.jl`, fields:

$$\phi = (\alpha, \nu^x, \nu^y)$$

$$(A_t)_{11} = (\partial_\alpha n)u^\tau$$

$$
\begin{aligned}
(A_t)_{12} &= \frac{u^x}{u^\tau} \\
(A_t)_{13} &= \frac{u^y}{u^\tau} \\
(A_t)_{21} &= (\partial_\alpha n) D_s u^\tau u^x \\
(A_t)_{22} &= \frac{D_s h u^\tau}{T} \\
(A_t)_{31} &= (\partial_\alpha n) D_s u^\tau u^y \\
(A_t)_{33} &= \frac{D_s h u^\tau}{T} \\
(A_x)_{11} &= (\partial_\alpha n) u^x \\
(A_x)_{12} &= 1 \\
(A_x)_{21} &= (\partial_\alpha n) D_s \left(\left(u^x\right)^2 + 1 \right) \\
(A_x)_{22} &= \frac{D_s h u^x}{T} \\
(A_x)_{31} &= (\partial_\alpha n) D_s u^x u^y \\
(A_x)_{33} &= \frac{D_s h u^x}{T} \\
(A_y)_{11} &= (\partial_\alpha n) u^y \\
(A_y)_{13} &= 1 \\
(A_y)_{21} &= (\partial_\alpha n) D_s u^x u^y \\
(A_y)_{22} &= \frac{D_s h u^y}{T} \\
(A_y)_{31} &= (\partial_\alpha n) D_s \left(\left(u^y\right)^2 + 1 \right) \\
(A_y)_{33} &= \frac{D_s h u^y}{T}
\end{aligned}
$$

$$
\begin{aligned}
S_1 = {} & (\partial_x u^x) n + (\partial_y u^y) n + \frac{(\partial_\tau u^x)\nu^x + (\partial_\tau u^y)\nu^y}{u^\tau} + (\partial_T n)(\partial_\tau T) u^\tau + (\partial_T n)(\partial_x T) u^x \\
& + (\partial_T n)(\partial_y T) u^y + \frac{n((\partial_\tau u^x) u^x + (\partial_\tau u^y) u^y)}{u^\tau} - \frac{((\partial_\tau u^x) u^x + (\partial_\tau u^y) u^y)(\nu^x u^x + \nu^y u^y)}{\left(u^\tau\right)^3} \\
& + \frac{n \left(u^\tau\right)^2 + \nu^x u^x + \nu^y u^y}{\tau u^\tau} \\
S_2 = {} & \nu^x \\
& + \frac{1}{T u^\tau} D_s \Big(h'(T) \nu^x \left((\partial_\tau T) \left(u^\tau\right)^2 + u^\tau ((\partial_x T) u^x + (\partial_y T) u^y) \right) \\
& + (\partial_T n) T + n \Big((\partial_\tau T) \left(u^\tau\right)^2 u^x + u^\tau \left((\partial_x T) \left(u^x\right)^2 + (\partial_x T) + (\partial_y T) u^x u^y \right) \Big) \Big) \\
& + \frac{1}{T \tau \left(u^\tau\right)^3} D_s h \Big(\tau \Big(-(\partial_\tau u^y) \nu^y \left(u^\tau\right)^2 u^x \left(\left(u^x\right)^2 + 1 \right) \\
& + (\partial_\tau u^x) \left(u^\tau\right)^2 \left(n \left(u^\tau\right)^2 + \nu^y \left(\left(u^x\right)^2 + 1 \right) u^y \right) \\
& + u^\tau \Big((\partial_x u^x) n \left(u^\tau\right)^2 u^x + (\partial_y u^x) n \left(u^\tau\right)^2 u^y \\
& + (\partial_x u^x) \nu^y \left(u^x\right)^3 u^y \\
& - \nu^y u^x \left(\left(u^x\right)^2 + 1 \right) ((\partial_x u^y) u^x + (\partial_y u^y) u^y) \\
& + (\partial_y u^x) \nu^y \left(\left(u^x\right)^2 + 1 \right) \left(\left(u^y\right)^2 + 1 \right) \Big) \Big) \\
& + \nu^x \Big((\partial_x u^y) \tau u^\tau \left(u^x\right)^3 u^y + \left(u^x\right)^4 ((\partial_\tau u^y) \tau u^y + 1) \\
& - (\partial_\tau u^x) \tau \left(u^\tau\right)^2 u^x \left(\left(u^y\right)^2 - 1 \right) \\
& - (\partial_y u^x) \tau u^\tau u^x u^y \left(\left(u^y\right)^2 + 1 \right) \\
& + \left(u^y\right)^2 + 1 \Big(1 + 2 (\partial_x u^x) \tau u^\tau + (\partial_y u^y) \tau u^\tau + (\partial_\tau u^y) \tau u^y
\end{aligned}
$$

$$
\begin{aligned}
&+ (u^y)^2\Big) \\
&+ (u^x)^2\Big(2 + (\partial_x u^x)\tau u^\tau + (\partial_y u^y)\tau u^\tau \\
&+ (u^y)^2\left(-(\partial_x u^x)\tau u^\tau + (\partial_y u^y)\tau u^\tau + 2\right) \\
&+ (\partial_\tau u^y)\tau u^y\Big((u^y)^2 + 2\Big)\Big)\Big)\Big) \\
S_3 = \nu^y & \\
&+ \frac{1}{Tu^\tau} D_s\Big(h'(T)\nu^y\Big((\partial_\tau T)(u^\tau)^2 + u^\tau((\partial_x T)u^x + (\partial_y T)u^y)\Big) \\
&+ (\partial_T n)T + n\Big((\partial_\tau T)(u^\tau)^2 u^y + u^\tau\Big((\partial_x T)u^x u^y + (\partial_y T)(u^y)^2 + (\partial_y T)\Big)\Big)\Big) \\
&+ \frac{1}{T\tau(u^\tau)^3} D_s h\Big(\nu^y\Big(1 + (\partial_x u^x)\tau u^\tau + 2(\partial_y u^y)\tau u^\tau + (\partial_\tau u^y)\tau u^y \\
&- (\partial_x u^y)\tau u^\tau (u^x)^3 u^y + 2(u^y)^2 + (\partial_x u^x)\tau u^\tau (u^y)^2 \\
&+ (\partial_y u^y)\tau u^\tau (u^y)^2 + (\partial_\tau u^y)\tau (u^y)^3 + (u^y)^4 \\
&+ (u^x)^4(1 - (\partial_\tau u^y)\tau u^y) + (\partial_\tau u^x)\tau (u^\tau)^2 u^x\Big((u^y)^2 + 1\Big) \\
&+ \tau u^\tau u^x u^y\Big((\partial_y u^x)(u^y)^2 - (\partial_x u^y)\Big) \\
&+ (u^x)^2\Big(2 + (\partial_x u^x)\tau u^\tau + 2(\partial_y u^y)\tau u^\tau \\
&+ (u^y)^2((\partial_x u^x)\tau u^\tau - (\partial_y u^y)\tau u^\tau + 2) \\
&- (\partial_\tau u^y)\tau (u^y)^3\Big)\Big) \\
&+ \tau\Big(-(\partial_\tau u^x)\nu^x (u^\tau)^2 u^y\Big((u^y)^2 + 1\Big) \\
&+ (\partial_\tau u^y)(u^\tau)^2\Big(n(u^\tau)^2 + \nu^x u^x\Big((u^y)^2 + 1\Big)\Big) \\
&+ u^\tau\Big((\partial_x u^y)n(u^\tau)^2 u^x + (\partial_y u^y)n(u^\tau)^2 u^y \\
&+ (\partial_y u^y)\nu^x u^x (u^y)^3 \\
&+ (\partial_x u^y)\nu^x\Big((u^x)^2 + 1\Big)\Big((u^y)^2 + 1\Big) \\
&- \nu^x u^y\Big((u^y)^2 + 1\Big)((\partial_x u^x)u^x + (\partial_y u^x)u^y)\Big)\Big)\Big)
\end{aligned}
$$

A.2.2 S4, charm first and second moment on a background

`two_d_HQ_BG_m2_matrix` in `src/Matrix/2d_viscous_HQ_BG_m2.jl`, fields:

$$\phi = (\alpha, \nu^x, \nu^y, \pi_Q^{yy}, {\pi_Q}^\eta{}_\eta, \pi_Q^{xy}, \Pi_Q)$$

$$
\begin{aligned}
(A_t)_{11} &= (\partial_\alpha n)u^\tau \\
(A_t)_{12} &= \frac{u^x}{u^\tau} \\
(A_t)_{13} &= \frac{u^y}{u^\tau} \\
(A_t)_{21} &= (\partial_\alpha n)D_s u^\tau u^x \\
(A_t)_{22} &= \frac{D_s h u^\tau}{T} \\
(A_t)_{31} &= (\partial_\alpha n)D_s u^\tau u^y \\
(A_t)_{33} &= \frac{D_s h u^\tau}{T} \\
(A_t)_{42} &= -\frac{\mathcal{B}D_s u^x\left((u^y)^2 + 1\right)}{3Tu^\tau} \\
(A_t)_{43} &= \frac{\mathcal{B}D_s u^y\left(3(u^x)^2 + 2(u^y)^2 + 2\right)}{3Tu^\tau}
\end{aligned}
$$

$$(A_t)_{44} = \frac{\mathcal{C}D_s u^\tau}{2T}$$
$$(A_t)_{52} = -\frac{\mathcal{B}D_s u^x}{3Tu^\tau}$$
$$(A_t)_{53} = -\frac{\mathcal{B}D_s u^y}{3Tu^\tau}$$
$$(A_t)_{55} = \frac{\mathcal{C}D_s u^\tau}{2T}$$
$$(A_t)_{62} = \frac{\mathcal{B}D_s u^y \left(\left(u^x\right)^2 + 3\left(u^y\right)^2 + 3\right)}{6Tu^\tau}$$
$$(A_t)_{63} = \frac{\mathcal{B}D_s u^x \left(3\left(u^x\right)^2 + \left(u^y\right)^2 + 3\right)}{6Tu^\tau}$$
$$(A_t)_{66} = \frac{\mathcal{C}D_s u^\tau}{2T}$$
$$(A_t)_{71} = \frac{(\partial_\alpha I_{31})D_s u^\tau}{2T}$$
$$(A_t)_{72} = \frac{5\mathcal{B}D_s u^x}{6Tu^\tau}$$
$$(A_t)_{73} = \frac{5\mathcal{B}D_s u^y}{6Tu^\tau}$$
$$(A_t)_{77} = \frac{\mathcal{C}D_s u^\tau}{2T}$$
$$(A_x)_{11} = (\partial_\alpha n)u^x$$
$$(A_x)_{12} = 1$$
$$(A_x)_{21} = (\partial_\alpha n)D_s \left(\left(u^x\right)^2 + 1\right)$$
$$(A_x)_{22} = \frac{D_s h u^x}{T}$$
$$(A_x)_{31} = (\partial_\alpha n)D_s u^x u^y$$
$$(A_x)_{33} = \frac{D_s h u^x}{T}$$
$$(A_x)_{42} = -\frac{\mathcal{B}D_s \left(\left(u^y\right)^2 + 1\right)}{3T}$$
$$(A_x)_{43} = \frac{\mathcal{B}D_s u^x u^y}{T}$$
$$(A_x)_{44} = \frac{\mathcal{C}D_s u^x}{2T}$$
$$(A_x)_{52} = -\frac{\mathcal{B}D_s}{3T}$$
$$(A_x)_{55} = \frac{\mathcal{C}D_s u^x}{2T}$$
$$(A_x)_{62} = \frac{\mathcal{B}D_s u^x u^y}{6T}$$
$$(A_x)_{63} = \frac{\mathcal{B}D_s \left(\left(u^x\right)^2 + 1\right)}{2T}$$
$$(A_x)_{66} = \frac{\mathcal{C}D_s u^x}{2T}$$
$$(A_x)_{71} = \frac{(\partial_\alpha I_{31})D_s u^x}{2T}$$
$$(A_x)_{72} = \frac{5\mathcal{B}D_s}{6T}$$
$$(A_x)_{77} = \frac{\mathcal{C}D_s u^x}{2T}$$
$$(A_y)_{11} = (\partial_\alpha n)u^y$$
$$(A_y)_{13} = 1$$
$$(A_y)_{21} = (\partial_\alpha n)D_s u^x u^y$$

$$
\begin{aligned}
(A_y)_{22} &= \frac{D_s h u^y}{T} \\
(A_y)_{31} &= (\partial_\alpha n) D_s \left((u^y)^2 + 1 \right) \\
(A_y)_{33} &= \frac{D_s h u^y}{T} \\
(A_y)_{43} &= \frac{2\mathcal{B} D_s \left((u^y)^2 + 1 \right)}{3T} \\
(A_y)_{44} &= \frac{\mathcal{C} D_s u^y}{2T} \\
(A_y)_{53} &= -\frac{\mathcal{B} D_s}{3T} \\
(A_y)_{55} &= \frac{\mathcal{C} D_s u^y}{2T} \\
(A_y)_{62} &= \frac{\mathcal{B} D_s \left((u^y)^2 + 1 \right)}{2T} \\
(A_y)_{63} &= \frac{\mathcal{B} D_s u^x u^y}{6T} \\
(A_y)_{66} &= \frac{\mathcal{C} D_s u^y}{2T} \\
(A_y)_{71} &= \frac{(\partial_\alpha I_{31}) D_s u^y}{2T} \\
(A_y)_{73} &= \frac{5\mathcal{B} D_s}{6T} \\
(A_y)_{77} &= \frac{\mathcal{C} D_s u^y}{2T}
\end{aligned}
$$

$$
\begin{aligned}
S_1 = {} & (\partial_x u^x) n + (\partial_y u^y) n + \frac{(\partial_\tau u^x)\nu^x + (\partial_\tau u^y)\nu^y}{u^\tau} + (\partial_T n)(\partial_\tau T) u^\tau + (\partial_T n)(\partial_x T) u^x \\
& + (\partial_T n)(\partial_y T) u^y + \frac{n((\partial_\tau u^x) u^x + (\partial_\tau u^y) u^y)}{u^\tau} - \frac{((\partial_\tau u^x) u^x + (\partial_\tau u^y) u^y)(\nu^x u^x + \nu^y u^y)}{(u^\tau)^3} \\
& + \frac{n (u^\tau)^2 + \nu^x u^x + \nu^y u^y}{\tau u^\tau} \\
S_2 = {} & \nu^x \\
& + \frac{1}{T u^\tau} D_s \Big(h'(T) \nu^x \left((\partial_\tau T) (u^\tau)^2 + u^\tau ((\partial_x T) u^x + (\partial_y T) u^y) \right) \\
& + (\partial_T n) T + n \Big((\partial_\tau T) (u^\tau)^2 u^x + u^\tau \left((\partial_x T) (u^x)^2 + (\partial_x T) + (\partial_y T) u^x u^y \right) \Big) \Big) \\
& + \frac{1}{T\tau (u^\tau)^3} D_s h \Big(\tau \Big(-(\partial_\tau u^y) \nu^y (u^\tau)^2 u^x \left((u^x)^2 + 1 \right) \\
& + (\partial_\tau u^x) (u^\tau)^2 \left(n (u^\tau)^2 + \nu^y \left((u^x)^2 + 1 \right) u^y \right) \\
& + u^\tau \Big((\partial_x u^x) n (u^\tau)^2 u^x + (\partial_y u^x) n (u^\tau)^2 u^y \\
& + (\partial_x u^x) \nu^y (u^x)^3 u^y \\
& - \nu^y u^x \left((u^x)^2 + 1 \right) ((\partial_x u^y) u^x + (\partial_y u^y) u^y) \\
& + (\partial_y u^x) \nu^y \left((u^x)^2 + 1 \right) \left((u^y)^2 + 1 \right) \Big) \Big) \\
& + \nu^x \Big((\partial_x u^y) \tau u^\tau (u^x)^3 u^y + (u^x)^4 ((\partial_\tau u^y) \tau u^y + 1) \\
& - (\partial_\tau u^x) \tau (u^\tau)^2 u^x \left((u^y)^2 - 1 \right) \\
& - (\partial_y u^x) \tau u^\tau u^x u^y \left((u^y)^2 + 1 \right) \\
& + (u^y)^2 + 1 \Big(1 + 2(\partial_x u^x) \tau u^\tau + (\partial_y u^y) \tau u^\tau + (\partial_\tau u^y) \tau u^y \\
& + (u^y)^2 \Big)
\end{aligned}
$$

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\begin{aligned}
&+ (u^x)^2 \Big(2 + (\partial_x u^x)\tau u^\tau + (\partial_y u^y)\tau u^\tau \\
&+ (u^y)^2 \left(-(\partial_x u^x)\tau u^\tau + (\partial_y u^y)\tau u^\tau + 2\right) \\
&+ (\partial_\tau u^y)\tau u^y \Big((u^y)^2 + 2\Big)\Big)\Big)\Big) \\
S_3 = \nu^y & \\
&+ \frac{1}{Tu^\tau} D_s\Big(h'(T)\nu^y \Big((\partial_\tau T)(u^\tau)^2 + u^\tau((\partial_x T)u^x + (\partial_y T)u^y)\Big) \\
&+ (\partial_T n)T + n\Big((\partial_\tau T)(u^\tau)^2 u^y + u^\tau \Big((\partial_x T)u^x u^y + (\partial_y T)(u^y)^2 + (\partial_y T)\Big)\Big)\Big) \\
&+ \frac{1}{T\tau (u^\tau)^3} D_s h\Big(\nu^y\Big(1 + (\partial_x u^x)\tau u^\tau + 2(\partial_y u^y)\tau u^\tau + (\partial_\tau u^y)\tau u^y \\
&- (\partial_x u^y)\tau u^\tau (u^x)^3 u^y + 2(u^y)^2 + (\partial_x u^x)\tau u^\tau (u^y)^2 \\
&+ (\partial_y u^y)\tau u^\tau (u^y)^2 + (\partial_\tau u^y)\tau (u^y)^3 + (u^y)^4 \\
&+ (u^x)^4 (1 - (\partial_\tau u^y)\tau u^y) + (\partial_\tau u^x)\tau (u^\tau)^2 u^x \Big((u^y)^2 + 1\Big) \\
&+ \tau u^\tau u^x u^y \Big((\partial_y u^x)(u^y)^2 - (\partial_x u^y)\Big) \\
&+ (u^x)^2 \Big(2 + (\partial_x u^x)\tau u^\tau + 2(\partial_y u^y)\tau u^\tau \\
&+ (u^y)^2 ((\partial_x u^x)\tau u^\tau - (\partial_y u^y)\tau u^\tau + 2) \\
&- (\partial_\tau u^y)\tau (u^y)^3\Big)\Big) \\
&+ \tau\Big(-(\partial_\tau u^x)\nu^x (u^\tau)^2 u^y \Big((u^y)^2 + 1\Big) \\
&+ (\partial_\tau u^y)(u^\tau)^2 \Big(n(u^\tau)^2 + \nu^x u^x \Big((u^y)^2 + 1\Big)\Big) \\
&+ u^\tau \Big((\partial_x u^y) n (u^\tau)^2 u^x + (\partial_y u^y) n (u^\tau)^2 u^y \\
&+ (\partial_y u^y)\nu^x u^x (u^y)^3 \\
&+ (\partial_x u^y)\nu^x \Big((u^x)^2 + 1\Big)\Big((u^y)^2 + 1\Big) \\
&- \nu^x u^y \Big((u^y)^2 + 1\Big)((\partial_x u^x)u^x + (\partial_y u^x)u^y)\Big)\Big)\Big) \\
S_4 = \frac{1}{6T\tau (u^\tau)^4} & \Big(-2D_s(\partial_x u^x) I_{31}\tau + 4D_s(\partial_y u^y) I_{31}\tau - 2\mathcal{B}'(T) D_s(\partial_x T)\nu^x \tau \\
&+ 4\mathcal{B}'(T) D_s(\partial_y T)\nu^y \tau - 2\mathcal{C} D_s(\partial_x u^x)\Pi_Q \tau + 4\mathcal{C} D_s(\partial_y u^y)\Pi_Q \tau \\
&+ 4\mathcal{C} D_s(\partial_x u^y)\pi_Q^{xy}\tau - 2\mathcal{C} D_s(\partial_y u^x)\pi_Q^{xy}\tau + 5\mathcal{C} D_s(\partial_x u^x)\pi_Q^{yy}\tau \\
&+ 7\mathcal{C} D_s(\partial_y u^y)\pi_Q^{yy}\tau + 2\mathcal{C} D_s(\partial_x u^x)\pi_Q{}^\eta{}_\eta \tau + 6\pi_Q^{yy} T\tau - 2D_s I_{31} u^\tau - 2\mathcal{C} D_s \Pi_Q u^\tau \\
&+ 3\mathcal{C} D_s \pi_Q^{yy} u^\tau - 2\mathcal{C} D_s \pi_Q{}^\eta{}_\eta u^\tau - 2\mathcal{A} D_s(\partial_\tau u^x)\nu^x \tau u^\tau - 2\mathcal{B} D_s(\partial_\tau u^x)\nu^x \tau u^\tau \\
&+ 4\mathcal{A} D_s(\partial_\tau u^y)\nu^y \tau u^\tau + 4\mathcal{B} D_s(\partial_\tau u^y)\nu^y \tau u^\tau + 3\mathcal{C}'(T) D_s(\partial_\tau T)\pi_Q^{yy}\tau u^\tau \\
&- 2\mathcal{A} D_s(\partial_x u^x)\nu^x \tau u^x + 4\mathcal{A} D_s(\partial_x u^y)\nu^y \tau u^x + 6\mathcal{B} D_s(\partial_x u^y)\nu^y \tau u^x \\
&+ 3\mathcal{C}'(T) D_s(\partial_x T)\pi_Q^{yy}\tau u^x - 2\mathcal{B} D_s \nu^x u^\tau u^x - 2D_s(\partial_\tau u^x) I_{31}\tau u^\tau u^x \\
&- 2\mathcal{B}'(T) D_s(\partial_\tau T)\nu^x \tau u^\tau u^x - 2\mathcal{C} D_s(\partial_\tau u^x)\Pi_Q \tau u^\tau u^x + 4\mathcal{C} D_s(\partial_\tau u^y)\pi_Q^{xy}\tau u^\tau u^x \\
&+ 5\mathcal{C} D_s(\partial_\tau u^x)\pi_Q^{yy}\tau u^\tau u^x + 2\mathcal{C} D_s(\partial_\tau u^x)\pi_Q{}^\eta{}_\eta \tau u^\tau u^x - 4D_s(\partial_x u^x) I_{31}\tau (u^x)^2 \\
&+ 8D_s(\partial_y u^y) I_{31}\tau (u^x)^2 - 4\mathcal{B}'(T) D_s(\partial_x T)\nu^x \tau (u^x)^2 \\
&+ 8\mathcal{B}'(T) D_s(\partial_y T)\nu^y \tau (u^x)^2 - 4\mathcal{C} D_s(\partial_x u^x)\Pi_Q \tau (u^x)^2 + 8\mathcal{C} D_s(\partial_y u^y)\Pi_Q \tau (u^x)^2 \\
&+ 8\mathcal{C} D_s(\partial_x u^y)\pi_Q^{xy}\tau (u^x)^2 - 2\mathcal{C} D_s(\partial_y u^x)\pi_Q^{xy}\tau (u^x)^2 + 10\mathcal{C} D_s(\partial_x u^x)\pi_Q^{yy}\tau (u^x)^2 \\
&+ 14\mathcal{C} D_s(\partial_y u^y)\pi_Q^{yy}\tau (u^x)^2 + 4\mathcal{C} D_s(\partial_x u^x)\pi_Q{}^\eta{}_\eta \tau (u^x)^2 + 12\pi_Q^{yy} T\tau (u^x)^2 \\
&- 4D_s I_{31} u^\tau (u^x)^2 - 4\mathcal{C} D_s \Pi_Q u^\tau (u^x)^2 + 6\mathcal{C} D_s \pi_Q^{yy} u^\tau (u^x)^2 - 4\mathcal{C} D_s \pi_Q{}^\eta{}_\eta u^\tau (u^x)^2 \\
&- 2\mathcal{A} D_s(\partial_\tau u^x)\nu^x \tau u^\tau (u^x)^2 + 8\mathcal{A} D_s(\partial_\tau u^y)\nu^y \tau u^\tau (u^x)^2 \\
&+ 10\mathcal{B} D_s(\partial_\tau u^y)\nu^y \tau u^\tau (u^x)^2 + 6\mathcal{C}'(T) D_s(\partial_\tau T)\pi_Q^{yy}\tau u^\tau (u^x)^2 \\
&- 2\mathcal{A} D_s(\partial_x u^x)\nu^x \tau (u^x)^3 + 8\mathcal{A} D_s(\partial_x u^y)\nu^y \tau (u^x)^3 + 12\mathcal{B} D_s(\partial_x u^y)\nu^y \tau (u^x)^3
\end{aligned}
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\begin{aligned}
&+ 6\mathcal{C}'(T)D_s(\partial_x T)\pi_Q^{yy}\tau\left(u^x\right)^3 - 2\mathcal{B}D_s\nu^x u^\tau\left(u^x\right)^3 - 2D_s(\partial_\tau u^x)I_{31}\tau u^\tau\left(u^x\right)^3 \\
&- 2\mathcal{B}'(T)D_s(\partial_\tau T)\nu^x\tau u^\tau\left(u^x\right)^3 - 2\mathcal{C}D_s(\partial_\tau u^x)\Pi_Q\tau u^\tau\left(u^x\right)^3 \\
&+ 4\mathcal{C}D_s(\partial_\tau u^y)\pi_Q^{xy}\tau u^\tau\left(u^x\right)^3 + 5\mathcal{C}D_s(\partial_\tau u^x)\pi_Q^{yy}\tau u^\tau\left(u^x\right)^3 \\
&+ 2\mathcal{C}D_s(\partial_\tau u^x)\pi_Q{}^\eta{}_\eta\tau u^\tau\left(u^x\right)^3 - 2D_s(\partial_x u^x)I_{31}\tau\left(u^x\right)^4 + 4D_s(\partial_y u^y)I_{31}\tau\left(u^x\right)^4 \\
&- 2\mathcal{B}'(T)D_s(\partial_x T)\nu^x\tau\left(u^x\right)^4 + 4\mathcal{B}'(T)D_s(\partial_y T)\nu^y\tau\left(u^x\right)^4 \\
&- 2\mathcal{C}D_s(\partial_x u^x)\Pi_Q\tau\left(u^x\right)^4 + 4\mathcal{C}D_s(\partial_y u^y)\Pi_Q\tau\left(u^x\right)^4 + 4\mathcal{C}D_s(\partial_x u^y)\pi_Q^{xy}\tau\left(u^x\right)^4 \\
&+ 5\mathcal{C}D_s(\partial_x u^x)\pi_Q^{yy}\tau\left(u^x\right)^4 + 7\mathcal{C}D_s(\partial_y u^y)\pi_Q^{yy}\tau\left(u^x\right)^4 + 2\mathcal{C}D_s(\partial_x u^x)\pi_Q{}^\eta{}_\eta\tau\left(u^x\right)^4 \\
&+ 6\pi_Q^{yy}T\tau\left(u^x\right)^4 - 2D_s I_{31}u^\tau\left(u^x\right)^4 - 2\mathcal{C}D_s\Pi_Q u^\tau\left(u^x\right)^4 + 3\mathcal{C}D_s\pi_Q^{yy}u^\tau\left(u^x\right)^4 \\
&- 2\mathcal{C}D_s\pi_Q{}^\eta{}_\eta u^\tau\left(u^x\right)^4 + 4\mathcal{A}D_s(\partial_\tau u^y)\nu^y\tau u^\tau\left(u^x\right)^4 + 6\mathcal{B}D_s(\partial_\tau u^y)\nu^y\tau u^\tau\left(u^x\right)^4 \\
&+ 3\mathcal{C}'(T)D_s(\partial_\tau T)\pi_Q^{yy}\tau u^\tau\left(u^x\right)^4 + 4\mathcal{A}D_s(\partial_x u^y)\nu^y\tau\left(u^x\right)^5 \\
&+ 6\mathcal{B}D_s(\partial_x u^y)\nu^y\tau\left(u^x\right)^5 + 3\mathcal{C}'(T)D_s(\partial_x T)\pi_Q^{yy}\tau\left(u^x\right)^5 - 2\mathcal{A}D_s(\partial_y u^x)\nu^x\tau u^y \\
&- 6\mathcal{B}D_s(\partial_y u^x)\nu^x\tau u^y + 4\mathcal{A}D_s(\partial_y u^y)\nu^y\tau u^y + 3\mathcal{C}'(T)D_s(\partial_y T)\pi_Q^{yy}\tau u^y \\
&- 2\mathcal{B}D_s\nu^y u^\tau u^y + 4D_s(\partial_\tau u^y)I_{31}\tau u^\tau u^y + 4\mathcal{B}'(T)D_s(\partial_\tau T)\nu^y\tau u^\tau u^y \\
&+ 4\mathcal{C}D_s(\partial_\tau u^y)\Pi_Q\tau u^\tau u^y - 8\mathcal{C}D_s(\partial_\tau u^x)\pi_Q^{xy}\tau u^\tau u^y + \mathcal{C}D_s(\partial_\tau u^y)\pi_Q^{yy}\tau u^\tau u^y \\
&+ 6D_s(\partial_x u^y)I_{31}\tau u^x u^y + 6\mathcal{B}'(T)D_s(\partial_x T)\nu^y\tau u^x u^y + 6\mathcal{C}D_s(\partial_x u^y)\Pi_Q\tau u^x u^y \\
&- 8\mathcal{C}D_s(\partial_x u^x)\pi_Q^{xy}\tau u^x u^y + 2\mathcal{C}D_s(\partial_y u^y)\pi_Q^{xy}\tau u^x u^y - 8\mathcal{C}D_s(\partial_x u^y)\pi_Q^{yy}\tau u^x u^y \\
&+ 2\mathcal{C}D_s(\partial_y u^x)\pi_Q^{yy}\tau u^x u^y - 2\mathcal{C}D_s(\partial_x u^y)\pi_Q{}^\eta{}_\eta\tau u^x u^y + 2\mathcal{A}D_s(\partial_\tau u^y)\nu^x\tau u^\tau u^x u^y \\
&+ 2\mathcal{B}D_s(\partial_\tau u^y)\nu^x\tau u^\tau u^x u^y + 2\mathcal{A}D_s(\partial_\tau u^x)\nu^y\tau u^\tau u^x u^y + 2\mathcal{B}D_s(\partial_\tau u^x)\nu^y\tau u^\tau u^x u^y \\
&+ 2\mathcal{A}D_s(\partial_x u^y)\nu^x\tau\left(u^x\right)^2 u^y - 2\mathcal{A}D_s(\partial_y u^x)\nu^x\tau\left(u^x\right)^2 u^y \\
&- 6\mathcal{B}D_s(\partial_y u^x)\nu^x\tau\left(u^x\right)^2 u^y + 2\mathcal{A}D_s(\partial_x u^x)\nu^y\tau\left(u^x\right)^2 u^y \\
&+ 8\mathcal{A}D_s(\partial_y u^y)\nu^y\tau\left(u^x\right)^2 u^y + 6\mathcal{C}'(T)D_s(\partial_y T)\pi_Q^{yy}\tau\left(u^x\right)^2 u^y \\
&- 2\mathcal{B}D_s\nu^y u^\tau\left(u^x\right)^2 u^y + 10D_s(\partial_\tau u^y)I_{31}\tau u^\tau\left(u^x\right)^2 u^y \\
&+ 10\mathcal{B}'(T)D_s(\partial_\tau T)\nu^y\tau u^\tau\left(u^x\right)^2 u^y + 10\mathcal{C}D_s(\partial_\tau u^y)\Pi_Q\tau u^\tau\left(u^x\right)^2 u^y \\
&- 8\mathcal{C}D_s(\partial_\tau u^x)\pi_Q^{xy}\tau u^\tau\left(u^x\right)^2 u^y - 7\mathcal{C}D_s(\partial_\tau u^y)\pi_Q^{yy}\tau u^\tau\left(u^x\right)^2 u^y \\
&- 2\mathcal{C}D_s(\partial_\tau u^y)\pi_Q{}^\eta{}_\eta\tau u^\tau\left(u^x\right)^2 u^y + 12D_s(\partial_x u^y)I_{31}\tau\left(u^x\right)^3 u^y \\
&+ 12\mathcal{B}'(T)D_s(\partial_x T)\nu^y\tau\left(u^x\right)^3 u^y + 12\mathcal{C}D_s(\partial_x u^y)\Pi_Q\tau\left(u^x\right)^3 u^y \\
&- 8\mathcal{C}D_s(\partial_x u^x)\pi_Q^{xy}\tau\left(u^x\right)^3 u^y + 2\mathcal{C}D_s(\partial_y u^y)\pi_Q^{xy}\tau\left(u^x\right)^3 u^y \\
&- 16\mathcal{C}D_s(\partial_x u^y)\pi_Q^{yy}\tau\left(u^x\right)^3 u^y + 2\mathcal{C}D_s(\partial_y u^x)\pi_Q^{yy}\tau\left(u^x\right)^3 u^y \\
&- 4\mathcal{C}D_s(\partial_x u^y)\pi_Q{}^\eta{}_\eta\tau\left(u^x\right)^3 u^y + 2\mathcal{A}D_s(\partial_\tau u^y)\nu^x\tau u^\tau\left(u^x\right)^3 u^y \\
&+ 2\mathcal{A}D_s(\partial_\tau u^x)\nu^y\tau u^\tau\left(u^x\right)^3 u^y + 2\mathcal{A}D_s(\partial_x u^y)\nu^x\tau\left(u^x\right)^4 u^y \\
&+ 2\mathcal{A}D_s(\partial_x u^x)\nu^y\tau\left(u^x\right)^4 u^y + 4\mathcal{A}D_s(\partial_y u^y)\nu^y\tau\left(u^x\right)^4 u^y \\
&+ 3\mathcal{C}'(T)D_s(\partial_y T)\pi_Q^{yy}\tau\left(u^x\right)^4 u^y + 6D_s(\partial_\tau u^y)I_{31}\tau u^\tau\left(u^x\right)^4 u^y \\
&+ 6\mathcal{B}'(T)D_s(\partial_\tau T)\nu^y\tau u^\tau\left(u^x\right)^4 u^y + 6\mathcal{C}D_s(\partial_\tau u^y)\Pi_Q\tau u^\tau\left(u^x\right)^4 u^y \\
&- 8\mathcal{C}D_s(\partial_\tau u^y)\pi_Q^{yy}\tau u^\tau\left(u^x\right)^4 u^y - 2\mathcal{C}D_s(\partial_\tau u^y)\pi_Q{}^\eta{}_\eta\tau u^\tau\left(u^x\right)^4 u^y \\
&+ 6D_s(\partial_x u^y)I_{31}\tau\left(u^x\right)^5 u^y + 6\mathcal{B}'(T)D_s(\partial_x T)\nu^y\tau\left(u^x\right)^5 u^y \\
&+ 6\mathcal{C}D_s(\partial_x u^y)\Pi_Q\tau\left(u^x\right)^5 u^y - 8\mathcal{C}D_s(\partial_x u^y)\pi_Q^{yy}\tau\left(u^x\right)^5 u^y \\
&- 2\mathcal{C}D_s(\partial_x u^y)\pi_Q{}^\eta{}_\eta\tau\left(u^x\right)^5 u^y - 6D_s(\partial_x u^x)I_{31}\tau\left(u^y\right)^2 + 12D_s(\partial_y u^y)I_{31}\tau\left(u^y\right)^2 \\
&- 6\mathcal{B}'(T)D_s(\partial_x T)\nu^x\tau\left(u^y\right)^2 + 12\mathcal{B}'(T)D_s(\partial_y T)\nu^y\tau\left(u^y\right)^2 \\
&- 6\mathcal{C}D_s(\partial_x u^x)\Pi_Q\tau\left(u^y\right)^2 + 12\mathcal{C}D_s(\partial_y u^y)\Pi_Q\tau\left(u^y\right)^2 + 8\mathcal{C}D_s(\partial_x u^y)\pi_Q^{xy}\tau\left(u^y\right)^2 \\
&- 12\mathcal{C}D_s(\partial_y u^x)\pi_Q^{xy}\tau\left(u^y\right)^2 + 10\mathcal{C}D_s(\partial_x u^x)\pi_Q^{yy}\tau\left(u^y\right)^2 \\
&+ 8\mathcal{C}D_s(\partial_y u^y)\pi_Q^{yy}\tau\left(u^y\right)^2 + 6\mathcal{C}D_s(\partial_x u^x)\pi_Q{}^\eta{}_\eta\tau\left(u^y\right)^2 + 12\pi_Q^{yy}T\tau\left(u^y\right)^2 \\
&- 6D_s I_{31}u^\tau\left(u^y\right)^2 - 6\mathcal{C}D_s\Pi_Q u^\tau\left(u^y\right)^2 + 6\mathcal{C}D_s\pi_Q^{yy}u^\tau\left(u^y\right)^2 - 6\mathcal{C}D_s\pi_Q{}^\eta{}_\eta u^\tau\left(u^y\right)^2
\end{aligned}
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\begin{aligned}
&- 6\mathcal{A}D_s(\partial_\tau u^x)\nu^x\tau u^\tau\left(u^y\right)^2 - 10\mathcal{B}D_s(\partial_\tau u^x)\nu^x\tau u^\tau\left(u^y\right)^2\\
&+ 8\mathcal{A}D_s(\partial_\tau u^y)\nu^y\tau u^\tau\left(u^y\right)^2 + 4\mathcal{B}D_s(\partial_\tau u^y)\nu^y\tau u^\tau\left(u^y\right)^2\\
&+ 6\mathcal{C}'(T)D_s(\partial_\tau T)\pi_Q^{yy}\tau u^\tau\left(u^y\right)^2 - 6\mathcal{A}D_s(\partial_x u^x)\nu^x\tau u^x\left(u^y\right)^2\\
&- 6\mathcal{B}D_s(\partial_x u^x)\nu^x\tau u^x\left(u^y\right)^2 + 2\mathcal{A}D_s(\partial_y u^y)\nu^x\tau u^x\left(u^y\right)^2\\
&+ 6\mathcal{B}D_s(\partial_y u^y)\nu^x\tau u^x\left(u^y\right)^2 + 8\mathcal{A}D_s(\partial_x u^y)\nu^y\tau u^x\left(u^y\right)^2\\
&+ 6\mathcal{B}D_s(\partial_x u^y)\nu^y\tau u^x\left(u^y\right)^2 + 2\mathcal{A}D_s(\partial_y u^x)\nu^y\tau u^x\left(u^y\right)^2\\
&+ 6\mathcal{B}D_s(\partial_y u^x)\nu^y\tau u^x\left(u^y\right)^2 + 6\mathcal{C}'(T)D_s(\partial_x T)\pi_Q^{yy}\tau u^x\left(u^y\right)^2\\
&- 4\mathcal{B}D_s\nu^x u^\tau u^x\left(u^y\right)^2 - 4D_s(\partial_\tau u^x)I_{31}\tau u^\tau u^x\left(u^y\right)^2\\
&- 4\mathcal{B}'(T)D_s(\partial_\tau T)\nu^x\tau u^\tau u^x\left(u^y\right)^2 - 4\mathcal{C}D_s(\partial_\tau u^x)\Pi_Q\tau u^\tau u^x\left(u^y\right)^2\\
&+ 12\mathcal{C}D_s(\partial_\tau u^y)\pi_Q^{xy}\tau u^\tau u^x\left(u^y\right)^2 + 13\mathcal{C}D_s(\partial_\tau u^x)\pi_Q^{yy}\tau u^\tau u^x\left(u^y\right)^2\\
&+ 4\mathcal{C}D_s(\partial_\tau u^x)\pi_Q{}^\eta{}_\eta\tau u^\tau u^x\left(u^y\right)^2 - 8D_s(\partial_x u^x)I_{31}\tau\left(u^x\right)^2\left(u^y\right)^2\\
&+ 16D_s(\partial_y u^y)I_{31}\tau\left(u^x\right)^2\left(u^y\right)^2 - 8\mathcal{B}'(T)D_s(\partial_x T)\nu^x\tau\left(u^x\right)^2\left(u^y\right)^2\\
&+ 16\mathcal{B}'(T)D_s(\partial_y T)\nu^y\tau\left(u^x\right)^2\left(u^y\right)^2 - 8\mathcal{C}D_s(\partial_x u^x)\Pi_Q\tau\left(u^x\right)^2\left(u^y\right)^2\\
&+ 16\mathcal{C}D_s(\partial_y u^y)\Pi_Q\tau\left(u^x\right)^2\left(u^y\right)^2 + 16\mathcal{C}D_s(\partial_x u^y)\pi_Q^{xy}\tau\left(u^x\right)^2\left(u^y\right)^2\\
&- 10\mathcal{C}D_s(\partial_y u^x)\pi_Q^{xy}\tau\left(u^x\right)^2\left(u^y\right)^2 + 18\mathcal{C}D_s(\partial_x u^x)\pi_Q^{yy}\tau\left(u^x\right)^2\left(u^y\right)^2\\
&+ 8\mathcal{C}D_s(\partial_x u^x)\pi_Q{}^\eta{}_\eta\tau\left(u^x\right)^2\left(u^y\right)^2 + 12\pi_Q^{yy}T\tau\left(u^x\right)^2\left(u^y\right)^2\\
&- 8D_sI_{31}u^\tau\left(u^x\right)^2\left(u^y\right)^2 - 8\mathcal{C}D_s\Pi_Q u^\tau\left(u^x\right)^2\left(u^y\right)^2 + 6\mathcal{C}D_s\pi_Q^{yy}u^\tau\left(u^x\right)^2\left(u^y\right)^2\\
&- 8\mathcal{C}D_s\pi_Q{}^\eta{}_\eta u^\tau\left(u^x\right)^2\left(u^y\right)^2 - 4\mathcal{A}D_s(\partial_\tau u^x)\nu^x\tau u^\tau\left(u^x\right)^2\left(u^y\right)^2\\
&- 6\mathcal{B}D_s(\partial_\tau u^x)\nu^x\tau u^\tau\left(u^x\right)^2\left(u^y\right)^2 + 6\mathcal{A}D_s(\partial_\tau u^y)\nu^y\tau u^\tau\left(u^x\right)^2\left(u^y\right)^2\\
&- 2\mathcal{B}D_s(\partial_\tau u^y)\nu^y\tau u^\tau\left(u^x\right)^2\left(u^y\right)^2 + 6\mathcal{C}'(T)D_s(\partial_\tau T)\pi_Q^{yy}\tau u^\tau\left(u^x\right)^2\left(u^y\right)^2\\
&- 4\mathcal{A}D_s(\partial_x u^x)\nu^x\tau\left(u^x\right)^3\left(u^y\right)^2 - 6\mathcal{B}D_s(\partial_x u^x)\nu^x\tau\left(u^x\right)^3\left(u^y\right)^2\\
&+ 2\mathcal{A}D_s(\partial_y u^y)\nu^x\tau\left(u^x\right)^3\left(u^y\right)^2 + 6\mathcal{B}D_s(\partial_y u^y)\nu^x\tau\left(u^x\right)^3\left(u^y\right)^2\\
&+ 6\mathcal{A}D_s(\partial_x u^y)\nu^y\tau\left(u^x\right)^3\left(u^y\right)^2 + 2\mathcal{A}D_s(\partial_y u^x)\nu^y\tau\left(u^x\right)^3\left(u^y\right)^2\\
&+ 6\mathcal{B}D_s(\partial_y u^x)\nu^y\tau\left(u^x\right)^3\left(u^y\right)^2 + 6\mathcal{C}'(T)D_s(\partial_x T)\pi_Q^{yy}\tau\left(u^x\right)^3\left(u^y\right)^2\\
&- 2\mathcal{B}D_s\nu^x u^\tau\left(u^x\right)^3\left(u^y\right)^2 - 2D_s(\partial_\tau u^x)I_{31}\tau u^\tau\left(u^x\right)^3\left(u^y\right)^2\\
&- 2\mathcal{B}'(T)D_s(\partial_\tau T)\nu^x\tau u^\tau\left(u^x\right)^3\left(u^y\right)^2 - 2\mathcal{C}D_s(\partial_\tau u^x)\Pi_Q\tau u^\tau\left(u^x\right)^3\left(u^y\right)^2\\
&+ 8\mathcal{C}D_s(\partial_\tau u^y)\pi_Q^{xy}\tau u^\tau\left(u^x\right)^3\left(u^y\right)^2 + 8\mathcal{C}D_s(\partial_\tau u^x)\pi_Q^{yy}\tau u^\tau\left(u^x\right)^3\left(u^y\right)^2\\
&+ 2\mathcal{C}D_s(\partial_\tau u^x)\pi_Q{}^\eta{}_\eta\tau u^\tau\left(u^x\right)^3\left(u^y\right)^2 - 2D_s(\partial_x u^x)I_{31}\tau\left(u^x\right)^4\left(u^y\right)^2\\
&+ 4D_s(\partial_y u^y)I_{31}\tau\left(u^x\right)^4\left(u^y\right)^2 - 2\mathcal{B}'(T)D_s(\partial_x T)\nu^x\tau\left(u^x\right)^4\left(u^y\right)^2\\
&+ 4\mathcal{B}'(T)D_s(\partial_y T)\nu^y\tau\left(u^x\right)^4\left(u^y\right)^2 - 2\mathcal{C}D_s(\partial_x u^x)\Pi_Q\tau\left(u^x\right)^4\left(u^y\right)^2\\
&+ 4\mathcal{C}D_s(\partial_y u^y)\Pi_Q\tau\left(u^x\right)^4\left(u^y\right)^2 + 8\mathcal{C}D_s(\partial_x u^y)\pi_Q^{xy}\tau\left(u^x\right)^4\left(u^y\right)^2\\
&+ 8\mathcal{C}D_s(\partial_x u^x)\pi_Q^{yy}\tau\left(u^x\right)^4\left(u^y\right)^2 - 8\mathcal{C}D_s(\partial_y u^y)\pi_Q^{yy}\tau\left(u^x\right)^4\left(u^y\right)^2\\
&+ 2\mathcal{C}D_s(\partial_x u^x)\pi_Q{}^\eta{}_\eta\tau\left(u^x\right)^4\left(u^y\right)^2 - 2D_sI_{31}u^\tau\left(u^x\right)^4\left(u^y\right)^2\\
&- 2\mathcal{C}D_s\Pi_Q u^\tau\left(u^x\right)^4\left(u^y\right)^2 - 2\mathcal{C}D_s\pi_Q{}^\eta{}_\eta u^\tau\left(u^x\right)^4\left(u^y\right)^2\\
&- 2\mathcal{A}D_s(\partial_\tau u^y)\nu^y\tau u^\tau\left(u^x\right)^4\left(u^y\right)^2 - 6\mathcal{B}D_s(\partial_\tau u^y)\nu^y\tau u^\tau\left(u^x\right)^4\left(u^y\right)^2\\
&- 2\mathcal{A}D_s(\partial_x u^y)\nu^y\tau\left(u^x\right)^5\left(u^y\right)^2 - 6\mathcal{B}D_s(\partial_x u^y)\nu^y\tau\left(u^x\right)^5\left(u^y\right)^2\\
&- 6\mathcal{A}D_s(\partial_y u^x)\nu^x\tau\left(u^y\right)^3 - 18\mathcal{B}D_s(\partial_y u^x)\nu^x\tau\left(u^y\right)^3 + 8\mathcal{A}D_s(\partial_y u^y)\nu^y\tau\left(u^y\right)^3\\
&+ 6\mathcal{C}'(T)D_s(\partial_y T)\pi_Q^{yy}\tau\left(u^y\right)^3 - 4\mathcal{B}D_s\nu^y u^\tau\left(u^y\right)^3 + 8D_s(\partial_\tau u^y)I_{31}\tau u^\tau\left(u^y\right)^3\\
&+ 8\mathcal{B}'(T)D_s(\partial_\tau T)\nu^y\tau u^\tau\left(u^y\right)^3 + 8\mathcal{C}D_s(\partial_\tau u^y)\Pi_Q\tau u^\tau\left(u^y\right)^3\\
&- 16\mathcal{C}D_s(\partial_\tau u^x)\pi_Q^{xy}\tau u^\tau\left(u^y\right)^3 + \mathcal{C}D_s(\partial_\tau u^y)\pi_Q^{yy}\tau u^\tau\left(u^y\right)^3\\
&+ 12D_s(\partial_x u^y)I_{31}\tau u^x\left(u^y\right)^3 + 12\mathcal{B}'(T)D_s(\partial_x T)\nu^y\tau u^x\left(u^y\right)^3\\
&+ 12\mathcal{C}D_s(\partial_x u^y)\Pi_Q\tau u^x\left(u^y\right)^3 - 16\mathcal{C}D_s(\partial_x u^x)\pi_Q^{xy}\tau u^x\left(u^y\right)^3
\end{aligned}
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\begin{aligned}
&+ 10\mathcal{C}D_s(\partial_y u^y)\pi_Q^{xy}\tau u^x\left(u^y\right)^3 - 8\mathcal{C}D_s(\partial_x u^y)\pi_Q^{yy}\tau u^x\left(u^y\right)^3 \\
&+ 10\mathcal{C}D_s(\partial_y u^x)\pi_Q^{yy}\tau u^x\left(u^y\right)^3 - 4\mathcal{C}D_s(\partial_x u^y)\pi_Q{}^{\eta}{}_{\eta}\tau u^x\left(u^y\right)^3 \\
&+ 4\mathcal{A}D_s(\partial_\tau u^y)\nu^x\tau u^\tau u^x\left(u^y\right)^3 + 8\mathcal{B}D_s(\partial_\tau u^y)\nu^x\tau u^\tau u^x\left(u^y\right)^3 \\
&+ 4\mathcal{A}D_s(\partial_\tau u^x)\nu^y\tau u^\tau u^x\left(u^y\right)^3 + 8\mathcal{B}D_s(\partial_\tau u^x)\nu^y\tau u^\tau u^x\left(u^y\right)^3 \\
&+ 4\mathcal{A}D_s(\partial_x u^y)\nu^x\tau\left(u^x\right)^2\left(u^y\right)^3 + 6\mathcal{B}D_s(\partial_x u^y)\nu^x\tau\left(u^x\right)^2\left(u^y\right)^3 \\
&- 4\mathcal{A}D_s(\partial_y u^x)\nu^x\tau\left(u^x\right)^2\left(u^y\right)^3 - 12\mathcal{B}D_s(\partial_y u^x)\nu^x\tau\left(u^x\right)^2\left(u^y\right)^3 \\
&+ 4\mathcal{A}D_s(\partial_x u^x)\nu^y\tau\left(u^x\right)^2\left(u^y\right)^3 + 6\mathcal{B}D_s(\partial_x u^x)\nu^y\tau\left(u^x\right)^2\left(u^y\right)^3 \\
&+ 6\mathcal{A}D_s(\partial_y u^y)\nu^y\tau\left(u^x\right)^2\left(u^y\right)^3 - 6\mathcal{B}D_s(\partial_y u^y)\nu^y\tau\left(u^x\right)^2\left(u^y\right)^3 \\
&+ 6\mathcal{C}'(T)D_s(\partial_y T)\pi_Q^{yy}\tau\left(u^x\right)^2\left(u^y\right)^3 - 2\mathcal{B}D_s\nu^y u^\tau\left(u^x\right)^2\left(u^y\right)^3 \\
&+ 10D_s(\partial_\tau u^y)I_{31}\tau u^\tau\left(u^x\right)^2\left(u^y\right)^3 + 10\mathcal{B}'(T)D_s(\partial_\tau T)\nu^y\tau u^\tau\left(u^x\right)^2\left(u^y\right)^3 \\
&+ 10\mathcal{C}D_s(\partial_\tau u^y)\Pi_Q\tau u^\tau\left(u^x\right)^2\left(u^y\right)^3 - 8\mathcal{C}D_s(\partial_\tau u^x)\pi_Q^{xy}\tau u^\tau\left(u^x\right)^2\left(u^y\right)^3 \\
&- 8\mathcal{C}D_s(\partial_\tau u^y)\pi_Q^{yy}\tau u^\tau\left(u^x\right)^2\left(u^y\right)^3 - 2\mathcal{C}D_s(\partial_\tau u^y)\pi_Q{}^{\eta}{}_{\eta}\tau u^\tau\left(u^x\right)^2\left(u^y\right)^3 \\
&+ 12D_s(\partial_x u^y)I_{31}\tau\left(u^x\right)^3\left(u^y\right)^3 + 12\mathcal{B}'(T)D_s(\partial_x T)\nu^y\tau\left(u^x\right)^3\left(u^y\right)^3 \\
&+ 12\mathcal{C}D_s(\partial_x u^y)\Pi_Q\tau\left(u^x\right)^3\left(u^y\right)^3 - 8\mathcal{C}D_s(\partial_x u^x)\pi_Q^{xy}\tau\left(u^x\right)^3\left(u^y\right)^3 \\
&+ 8\mathcal{C}D_s(\partial_y u^y)\pi_Q^{xy}\tau\left(u^x\right)^3\left(u^y\right)^3 - 8\mathcal{C}D_s(\partial_x u^y)\pi_Q^{yy}\tau\left(u^x\right)^3\left(u^y\right)^3 \\
&+ 8\mathcal{C}D_s(\partial_y u^x)\pi_Q^{yy}\tau\left(u^x\right)^3\left(u^y\right)^3 - 4\mathcal{C}D_s(\partial_x u^y)\pi_Q{}^{\eta}{}_{\eta}\tau\left(u^x\right)^3\left(u^y\right)^3 \\
&+ 2\mathcal{A}D_s(\partial_\tau u^y)\nu^x\tau u^\tau\left(u^x\right)^3\left(u^y\right)^3 + 6\mathcal{B}D_s(\partial_\tau u^y)\nu^x\tau u^\tau\left(u^x\right)^3\left(u^y\right)^3 \\
&+ 2\mathcal{A}D_s(\partial_\tau u^x)\nu^y\tau u^\tau\left(u^x\right)^3\left(u^y\right)^3 + 6\mathcal{B}D_s(\partial_\tau u^x)\nu^y\tau u^\tau\left(u^x\right)^3\left(u^y\right)^3 \\
&+ 2\mathcal{A}D_s(\partial_x u^y)\nu^x\tau\left(u^x\right)^4\left(u^y\right)^3 + 6\mathcal{B}D_s(\partial_x u^y)\nu^x\tau\left(u^x\right)^4\left(u^y\right)^3 \\
&+ 2\mathcal{A}D_s(\partial_x u^x)\nu^y\tau\left(u^x\right)^4\left(u^y\right)^3 + 6\mathcal{B}D_s(\partial_x u^x)\nu^y\tau\left(u^x\right)^4\left(u^y\right)^3 \\
&- 2\mathcal{A}D_s(\partial_y u^y)\nu^y\tau\left(u^x\right)^4\left(u^y\right)^3 - 6\mathcal{B}D_s(\partial_y u^y)\nu^y\tau\left(u^x\right)^4\left(u^y\right)^3 \\
&- 6D_s(\partial_x u^x)I_{31}\tau\left(u^y\right)^4 + 12D_s(\partial_y u^y)I_{31}\tau\left(u^y\right)^4 - 6\mathcal{B}'(T)D_s(\partial_x T)\nu^x\tau\left(u^y\right)^4 \\
&+ 12\mathcal{B}'(T)D_s(\partial_y T)\nu^y\tau\left(u^y\right)^4 - 6\mathcal{C}D_s(\partial_x u^x)\Pi_Q\tau\left(u^y\right)^4 \\
&+ 12\mathcal{C}D_s(\partial_y u^y)\Pi_Q\tau\left(u^y\right)^4 + 4\mathcal{C}D_s(\partial_x u^y)\pi_Q^{xy}\tau\left(u^y\right)^4 - 18\mathcal{C}D_s(\partial_y u^x)\pi_Q^{xy}\tau\left(u^y\right)^4 \\
&+ 5\mathcal{C}D_s(\partial_x u^x)\pi_Q^{yy}\tau\left(u^y\right)^4 + \mathcal{C}D_s(\partial_y u^y)\pi_Q^{yy}\tau\left(u^y\right)^4 + 6\mathcal{C}D_s(\partial_x u^x)\pi_Q{}^{\eta}{}_{\eta}\tau\left(u^y\right)^4 \\
&+ 6\pi_Q^{yy}T\tau\left(u^y\right)^4 - 6D_s I_{31}u^\tau\left(u^y\right)^4 - 6\mathcal{C}D_s\Pi_Q u^\tau\left(u^y\right)^4 + 3\mathcal{C}D_s\pi_Q^{yy}u^\tau\left(u^y\right)^4 \\
&- 6\mathcal{C}D_s\pi_Q{}^{\eta}{}_{\eta}u^\tau\left(u^y\right)^4 - 6\mathcal{A}D_s(\partial_\tau u^x)\nu^x\tau u^\tau\left(u^y\right)^4 - 14\mathcal{B}D_s(\partial_\tau u^x)\nu^x\tau u^\tau\left(u^y\right)^4 \\
&+ 4\mathcal{A}D_s(\partial_\tau u^y)\nu^y\tau u^\tau\left(u^y\right)^4 + 3\mathcal{C}'(T)D_s(\partial_\tau T)\pi_Q^{yy}\tau u^\tau\left(u^y\right)^4 \\
&- 6\mathcal{A}D_s(\partial_x u^x)\nu^x\tau u^x\left(u^y\right)^4 - 12\mathcal{B}D_s(\partial_x u^x)\nu^x\tau u^x\left(u^y\right)^4 \\
&+ 4\mathcal{A}D_s(\partial_y u^y)\nu^x\tau u^x\left(u^y\right)^4 + 12\mathcal{B}D_s(\partial_y u^y)\nu^x\tau u^x\left(u^y\right)^4 \\
&+ 4\mathcal{A}D_s(\partial_x u^y)\nu^y\tau u^x\left(u^y\right)^4 + 4\mathcal{A}D_s(\partial_y u^x)\nu^y\tau u^x\left(u^y\right)^4 \\
&+ 12\mathcal{B}D_s(\partial_y u^x)\nu^y\tau u^x\left(u^y\right)^4 + 3\mathcal{C}'(T)D_s(\partial_x T)\pi_Q^{yy}\tau u^x\left(u^y\right)^4 \\
&- 2\mathcal{B}D_s\nu^x u^\tau u^x\left(u^y\right)^4 - 2D_s(\partial_\tau u^x)I_{31}\tau u^\tau u^x\left(u^y\right)^4 \\
&- 2\mathcal{B}'(T)D_s(\partial_\tau T)\nu^x\tau u^\tau u^x\left(u^y\right)^4 - 2\mathcal{C}D_s(\partial_\tau u^x)\Pi_Q\tau u^\tau u^x\left(u^y\right)^4 \\
&+ 8\mathcal{C}D_s(\partial_\tau u^y)\pi_Q^{xy}\tau u^\tau u^x\left(u^y\right)^4 + 8\mathcal{C}D_s(\partial_\tau u^x)\pi_Q^{yy}\tau u^\tau u^x\left(u^y\right)^4 \\
&+ 2\mathcal{C}D_s(\partial_\tau u^x)\pi_Q{}^{\eta}{}_{\eta}\tau u^\tau u^x\left(u^y\right)^4 - 4D_s(\partial_x u^x)I_{31}\tau\left(u^x\right)^2\left(u^y\right)^4 \\
&+ 8D_s(\partial_y u^y)I_{31}\tau\left(u^x\right)^2\left(u^y\right)^4 - 4\mathcal{B}'(T)D_s(\partial_x T)\nu^x\tau\left(u^x\right)^2\left(u^y\right)^4 \\
&+ 8\mathcal{B}'(T)D_s(\partial_y T)\nu^y\tau\left(u^x\right)^2\left(u^y\right)^4 - 4\mathcal{C}D_s(\partial_x u^x)\Pi_Q\tau\left(u^x\right)^2\left(u^y\right)^4 \\
&+ 8\mathcal{C}D_s(\partial_y u^y)\Pi_Q\tau\left(u^x\right)^2\left(u^y\right)^4 + 8\mathcal{C}D_s(\partial_x u^y)\pi_Q^{xy}\tau\left(u^x\right)^2\left(u^y\right)^4 \\
&- 8\mathcal{C}D_s(\partial_y u^x)\pi_Q^{xy}\tau\left(u^x\right)^2\left(u^y\right)^4 + 8\mathcal{C}D_s(\partial_x u^x)\pi_Q^{yy}\tau\left(u^x\right)^2\left(u^y\right)^4 \\
&- 8\mathcal{C}D_s(\partial_y u^y)\pi_Q^{yy}\tau\left(u^x\right)^2\left(u^y\right)^4 + 4\mathcal{C}D_s(\partial_x u^x)\pi_Q{}^{\eta}{}_{\eta}\tau\left(u^x\right)^2\left(u^y\right)^4 \\
&- 4D_s I_{31}u^\tau\left(u^x\right)^2\left(u^y\right)^4 - 4\mathcal{C}D_s\Pi_Q u^\tau\left(u^x\right)^2\left(u^y\right)^4 - 4\mathcal{C}D_s\pi_Q{}^{\eta}{}_{\eta}u^\tau\left(u^x\right)^2\left(u^y\right)^4
\end{aligned}
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\begin{aligned}
&- 2\mathcal{A}D_s(\partial_\tau u^x)\nu^x\tau u^\tau\left(u^x\right)^2\left(u^y\right)^4 - 6\mathcal{B}D_s(\partial_\tau u^x)\nu^x\tau u^\tau\left(u^x\right)^2\left(u^y\right)^4\\
&- 2\mathcal{A}D_s(\partial_\tau u^y)\nu^y\tau u^\tau\left(u^x\right)^2\left(u^y\right)^4 - 6\mathcal{B}D_s(\partial_\tau u^y)\nu^y\tau u^\tau\left(u^x\right)^2\left(u^y\right)^4\\
&- 2\mathcal{A}D_s(\partial_x u^x)\nu^x\tau\left(u^x\right)^3\left(u^y\right)^4 - 6\mathcal{B}D_s(\partial_x u^x)\nu^x\tau\left(u^x\right)^3\left(u^y\right)^4\\
&+ 2\mathcal{A}D_s(\partial_y u^y)\nu^x\tau\left(u^x\right)^3\left(u^y\right)^4 + 6\mathcal{B}D_s(\partial_y u^y)\nu^x\tau\left(u^x\right)^3\left(u^y\right)^4\\
&- 2\mathcal{A}D_s(\partial_x u^y)\nu^y\tau\left(u^x\right)^3\left(u^y\right)^4 - 6\mathcal{B}D_s(\partial_x u^y)\nu^y\tau\left(u^x\right)^3\left(u^y\right)^4\\
&+ 2\mathcal{A}D_s(\partial_y u^x)\nu^y\tau\left(u^x\right)^3\left(u^y\right)^4 + 6\mathcal{B}D_s(\partial_y u^x)\nu^y\tau\left(u^x\right)^3\left(u^y\right)^4\\
&- 6\mathcal{A}D_s(\partial_y u^x)\nu^x\tau\left(u^y\right)^5 - 18\mathcal{B}D_s(\partial_y u^x)\nu^x\tau\left(u^y\right)^5 + 4\mathcal{A}D_s(\partial_y u^y)\nu^y\tau\left(u^y\right)^5\\
&+ 3\mathcal{C}'(T)D_s(\partial_y T)\pi_Q^{yy}\tau\left(u^y\right)^5 - 2\mathcal{B}D_s\nu^y u^\tau\left(u^y\right)^5 + 4D_s(\partial_\tau u^y)I_{31}\tau u^\tau\left(u^y\right)^5\\
&+ 4\mathcal{B}'(T)D_s(\partial_\tau T)\nu^y\tau u^\tau\left(u^y\right)^5 + 4\mathcal{C}D_s(\partial_\tau u^y)\Pi_Q\tau u^\tau\left(u^y\right)^5\\
&- 8\mathcal{C}D_s(\partial_\tau u^x)\pi_Q^{xy}\tau u^\tau\left(u^y\right)^5 + 6D_s(\partial_x u^y)I_{31}\tau u^x\left(u^y\right)^5\\
&+ 6\mathcal{B}'(T)D_s(\partial_x T)\nu^y\tau u^x\left(u^y\right)^5 + 6\mathcal{C}D_s(\partial_x u^y)\Pi_Q\tau u^x\left(u^y\right)^5\\
&- 8\mathcal{C}D_s(\partial_x u^x)\pi_Q^{xy}\tau u^x\left(u^y\right)^5 + 8\mathcal{C}D_s(\partial_y u^y)\pi_Q^{xy}\tau u^x\left(u^y\right)^5\\
&+ 8\mathcal{C}D_s(\partial_y u^x)\pi_Q^{yy}\tau u^x\left(u^y\right)^5 - 2\mathcal{C}D_s(\partial_x u^y)\pi_Q{}^\eta{}_\eta\tau u^x\left(u^y\right)^5\\
&+ 2\mathcal{A}D_s(\partial_\tau u^y)\nu^x\tau u^\tau u^x\left(u^y\right)^5 + 6\mathcal{B}D_s(\partial_\tau u^y)\nu^x\tau u^\tau u^x\left(u^y\right)^5\\
&+ 2\mathcal{A}D_s(\partial_\tau u^x)\nu^y\tau u^\tau u^x\left(u^y\right)^5 + 6\mathcal{B}D_s(\partial_\tau u^x)\nu^y\tau u^\tau u^x\left(u^y\right)^5\\
&+ 2\mathcal{A}D_s(\partial_x u^y)\nu^x\tau\left(u^x\right)^2\left(u^y\right)^5 + 6\mathcal{B}D_s(\partial_x u^y)\nu^x\tau\left(u^x\right)^2\left(u^y\right)^5\\
&- 2\mathcal{A}D_s(\partial_y u^x)\nu^x\tau\left(u^x\right)^2\left(u^y\right)^5 - 6\mathcal{B}D_s(\partial_y u^x)\nu^x\tau\left(u^x\right)^2\left(u^y\right)^5\\
&+ 2\mathcal{A}D_s(\partial_x u^x)\nu^y\tau\left(u^x\right)^2\left(u^y\right)^5 + 6\mathcal{B}D_s(\partial_x u^x)\nu^y\tau\left(u^x\right)^2\left(u^y\right)^5\\
&- 2\mathcal{A}D_s(\partial_y u^y)\nu^y\tau\left(u^x\right)^2\left(u^y\right)^5 - 6\mathcal{B}D_s(\partial_y u^y)\nu^y\tau\left(u^x\right)^2\left(u^y\right)^5\\
&- 2D_s(\partial_x u^x)I_{31}\tau\left(u^y\right)^6 + 4D_s(\partial_y u^y)I_{31}\tau\left(u^y\right)^6 - 2\mathcal{B}'(T)D_s(\partial_x T)\nu^x\tau\left(u^y\right)^6\\
&+ 4\mathcal{B}'(T)D_s(\partial_y T)\nu^y\tau\left(u^y\right)^6 - 2\mathcal{C}D_s(\partial_x u^x)\Pi_Q\tau\left(u^y\right)^6 + 4\mathcal{C}D_s(\partial_y u^y)\Pi_Q\tau\left(u^y\right)^6\\
&- 8\mathcal{C}D_s(\partial_y u^x)\pi_Q^{xy}\tau\left(u^y\right)^6 + 2\mathcal{C}D_s(\partial_x u^x)\pi_Q{}^\eta{}_\eta\tau\left(u^y\right)^6 - 2D_s I_{31}u^\tau\left(u^y\right)^6\\
&- 2\mathcal{C}D_s\Pi_Q u^\tau\left(u^y\right)^6 - 2\mathcal{C}D_s\pi_Q{}^\eta{}_\eta u^\tau\left(u^y\right)^6 - 2\mathcal{A}D_s(\partial_\tau u^x)\nu^x\tau u^\tau\left(u^y\right)^6\\
&- 6\mathcal{B}D_s(\partial_\tau u^x)\nu^x\tau u^\tau\left(u^y\right)^6 - 2\mathcal{A}D_s(\partial_x u^x)\nu^x\tau u^x\left(u^y\right)^6\\
&- 6\mathcal{B}D_s(\partial_x u^x)\nu^x\tau u^x\left(u^y\right)^6 + 2\mathcal{A}D_s(\partial_y u^y)\nu^x\tau u^x\left(u^y\right)^6\\
&+ 6\mathcal{B}D_s(\partial_y u^y)\nu^x\tau u^x\left(u^y\right)^6 + 2\mathcal{A}D_s(\partial_y u^x)\nu^y\tau u^x\left(u^y\right)^6\\
&+ 6\mathcal{B}D_s(\partial_y u^x)\nu^y\tau u^x\left(u^y\right)^6 - 2\mathcal{A}D_s(\partial_y u^x)\nu^x\tau\left(u^y\right)^7 - 6\mathcal{B}D_s(\partial_y u^x)\nu^x\tau\left(u^y\right)^7\Big)
\end{aligned}
$$

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\begin{aligned}
S_5 = \frac{1}{6T\tau\left(u^\tau\right)^4\left(\left(u^y\right)^2+1\right)}\Big(&-2D_s(\partial_x u^x)I_{31}\tau - 2D_s(\partial_y u^y)I_{31}\tau - 2\mathcal{B}'(T)D_s(\partial_x T)\nu^x\tau\\
&- 2\mathcal{B}'(T)D_s(\partial_y T)\nu^y\tau - 2\mathcal{C}D_s(\partial_x u^x)\Pi_Q\tau - 2\mathcal{C}D_s(\partial_y u^y)\Pi_Q\tau\\
&- 2\mathcal{C}D_s(\partial_x u^y)\pi_Q^{xy}\tau - 2\mathcal{C}D_s(\partial_y u^x)\pi_Q^{xy}\tau + 2\mathcal{C}D_s(\partial_x u^x)\pi_Q^{yy}\tau\\
&- 2\mathcal{C}D_s(\partial_y u^y)\pi_Q^{yy}\tau + 5\mathcal{C}D_s(\partial_x u^x)\pi_Q{}^\eta{}_\eta\tau + 3\mathcal{C}D_s(\partial_y u^y)\pi_Q{}^\eta{}_\eta\tau\\
&+ 6\pi_Q{}^\eta{}_\eta T\tau + 4D_s I_{31}u^\tau + 4\mathcal{C}D_s\Pi_Q u^\tau + 7\mathcal{C}D_s\pi_Q{}^\eta{}_\eta u^\tau\\
&- 2\mathcal{A}D_s(\partial_\tau u^x)\nu^x\tau u^\tau - 2\mathcal{B}D_s(\partial_\tau u^x)\nu^x\tau u^\tau - 2\mathcal{A}D_s(\partial_\tau u^y)\nu^y\tau u^\tau\\
&- 2\mathcal{B}D_s(\partial_\tau u^y)\nu^y\tau u^\tau + 3\mathcal{C}'(T)D_s(\partial_\tau T)\pi_Q{}^\eta{}_\eta\tau u^\tau\\
&- 2\mathcal{A}D_s(\partial_x u^x)\nu^x\tau u^x - 2\mathcal{A}D_s(\partial_x u^y)\nu^y\tau u^x\\
&+ 3\mathcal{C}'(T)D_s(\partial_x T)\pi_Q{}^\eta{}_\eta\tau u^x + 4\mathcal{B}D_s\nu^x u^\tau u^x - 2D_s(\partial_\tau u^x)I_{31}\tau u^\tau u^x\\
&- 2\mathcal{B}'(T)D_s(\partial_\tau T)\nu^x\tau u^\tau u^x - 2\mathcal{C}D_s(\partial_\tau u^x)\Pi_Q\tau u^\tau u^x\\
&- 2\mathcal{C}D_s(\partial_\tau u^y)\pi_Q^{xy}\tau u^\tau u^x + 2\mathcal{C}D_s(\partial_\tau u^x)\pi_Q^{yy}\tau u^\tau u^x\\
&+ 5\mathcal{C}D_s(\partial_\tau u^x)\pi_Q{}^\eta{}_\eta\tau u^\tau u^x - 4D_s(\partial_x u^x)I_{31}\tau\left(u^x\right)^2\\
&- 4D_s(\partial_y u^y)I_{31}\tau\left(u^x\right)^2 - 4\mathcal{B}'(T)D_s(\partial_x T)\nu^x\tau\left(u^x\right)^2\\
&- 4\mathcal{B}'(T)D_s(\partial_y T)\nu^y\tau\left(u^x\right)^2 - 4\mathcal{C}D_s(\partial_x u^x)\Pi_Q\tau\left(u^x\right)^2\\
&- 4\mathcal{C}D_s(\partial_y u^y)\Pi_Q\tau\left(u^x\right)^2 - 4\mathcal{C}D_s(\partial_x u^y)\pi_Q^{xy}\tau\left(u^x\right)^2
\end{aligned}
$$

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\begin{aligned}
&- 2\mathcal{C}D_s(\partial_y u^x)\pi_Q^{xy}\tau\left(u^x\right)^2 + 4\mathcal{C}D_s(\partial_x u^x)\pi_Q^{yy}\tau\left(u^x\right)^2 \\
&- 4\mathcal{C}D_s(\partial_y u^y)\pi_Q^{yy}\tau\left(u^x\right)^2 + 10\mathcal{C}D_s(\partial_x u^x)\pi_Q{}^\eta{}_\eta\tau\left(u^x\right)^2 \\
&+ 6\mathcal{C}D_s(\partial_y u^y)\pi_Q{}^\eta{}_\eta\tau\left(u^x\right)^2 + 12\pi_Q{}^\eta{}_\eta T\tau\left(u^x\right)^2 + 8D_s I_{31}u^\tau\left(u^x\right)^2 \\
&+ 8\mathcal{C}D_s\Pi_Q u^\tau\left(u^x\right)^2 + 14\mathcal{C}D_s\pi_Q{}^\eta{}_\eta u^\tau\left(u^x\right)^2 \\
&- 2\mathcal{A}D_s(\partial_\tau u^x)\nu^x\tau u^\tau\left(u^x\right)^2 - 4\mathcal{A}D_s(\partial_\tau u^y)\nu^y\tau u^\tau\left(u^x\right)^2 \\
&- 2\mathcal{B}D_s(\partial_\tau u^y)\nu^y\tau u^\tau\left(u^x\right)^2 + 6\mathcal{C}'(T)D_s(\partial_\tau T)\pi_Q{}^\eta{}_\eta\tau u^\tau\left(u^x\right)^2 \\
&- 2\mathcal{A}D_s(\partial_x u^x)\nu^x\tau\left(u^x\right)^3 - 4\mathcal{A}D_s(\partial_x u^y)\nu^y\tau\left(u^x\right)^3 \\
&+ 6\mathcal{C}'(T)D_s(\partial_x T)\pi_Q{}^\eta{}_\eta\tau\left(u^x\right)^3 + 4\mathcal{B}D_s\nu^x u^\tau\left(u^x\right)^3 \\
&- 2D_s(\partial_\tau u^x)I_{31}\tau u^\tau\left(u^x\right)^3 - 2\mathcal{B}'(T)D_s(\partial_\tau T)\nu^x\tau u^\tau\left(u^x\right)^3 \\
&- 2\mathcal{C}D_s(\partial_\tau u^x)\Pi_Q\tau u^\tau\left(u^x\right)^3 - 2\mathcal{C}D_s(\partial_\tau u^y)\pi_Q^{xy}\tau u^\tau\left(u^x\right)^3 \\
&+ 2\mathcal{C}D_s(\partial_\tau u^x)\pi_Q^{yy}\tau u^\tau\left(u^x\right)^3 + 5\mathcal{C}D_s(\partial_\tau u^x)\pi_Q{}^\eta{}_\eta\tau u^\tau\left(u^x\right)^3 \\
&- 2D_s(\partial_x u^x)I_{31}\tau\left(u^x\right)^4 - 2D_s(\partial_y u^y)I_{31}\tau\left(u^x\right)^4 \\
&- 2\mathcal{B}'(T)D_s(\partial_x T)\nu^x\tau\left(u^x\right)^4 - 2\mathcal{B}'(T)D_s(\partial_y T)\nu^y\tau\left(u^x\right)^4 \\
&- 2\mathcal{C}D_s(\partial_x u^x)\Pi_Q\tau\left(u^x\right)^4 - 2\mathcal{C}D_s(\partial_y u^y)\Pi_Q\tau\left(u^x\right)^4 \\
&- 2\mathcal{C}D_s(\partial_x u^y)\pi_Q^{xy}\tau\left(u^x\right)^4 + 2\mathcal{C}D_s(\partial_x u^x)\pi_Q^{yy}\tau\left(u^x\right)^4 \\
&- 2\mathcal{C}D_s(\partial_y u^y)\pi_Q^{yy}\tau\left(u^x\right)^4 + 5\mathcal{C}D_s(\partial_x u^x)\pi_Q{}^\eta{}_\eta\tau\left(u^x\right)^4 \\
&+ 3\mathcal{C}D_s(\partial_y u^y)\pi_Q{}^\eta{}_\eta\tau\left(u^x\right)^4 + 6\pi_Q{}^\eta{}_\eta T\tau\left(u^x\right)^4 + 4D_s I_{31}u^\tau\left(u^x\right)^4 \\
&+ 4\mathcal{C}D_s\Pi_Q u^\tau\left(u^x\right)^4 + 7\mathcal{C}D_s\pi_Q{}^\eta{}_\eta u^\tau\left(u^x\right)^4 - 2\mathcal{A}D_s(\partial_\tau u^y)\nu^y\tau u^\tau\left(u^x\right)^4 \\
&+ 3\mathcal{C}'(T)D_s(\partial_\tau T)\pi_Q{}^\eta{}_\eta\tau u^\tau\left(u^x\right)^4 - 2\mathcal{A}D_s(\partial_x u^y)\nu^y\tau\left(u^x\right)^5 \\
&+ 3\mathcal{C}'(T)D_s(\partial_x T)\pi_Q{}^\eta{}_\eta\tau\left(u^x\right)^5 - 2\mathcal{A}D_s(\partial_y u^x)\nu^x\tau u^y \\
&- 2\mathcal{A}D_s(\partial_y u^y)\nu^y\tau u^y + 3\mathcal{C}'(T)D_s(\partial_y T)\pi_Q{}^\eta{}_\eta\tau u^y + 4\mathcal{B}D_s\nu^y u^\tau u^y \\
&- 2D_s(\partial_\tau u^y)I_{31}\tau u^\tau u^y - 2\mathcal{B}'(T)D_s(\partial_\tau T)\nu^y\tau u^\tau u^y \\
&- 2\mathcal{C}D_s(\partial_\tau u^y)\Pi_Q\tau u^\tau u^y - 2\mathcal{C}D_s(\partial_\tau u^x)\pi_Q^{xy}\tau u^\tau u^y \\
&- 2\mathcal{C}D_s(\partial_\tau u^y)\pi_Q^{yy}\tau u^\tau u^y + 3\mathcal{C}D_s(\partial_\tau u^y)\pi_Q{}^\eta{}_\eta\tau u^\tau u^y \\
&- 2\mathcal{C}D_s(\partial_x u^x)\pi_Q^{xy}\tau u^x u^y + 2\mathcal{C}D_s(\partial_y u^y)\pi_Q^{xy}\tau u^x u^y \\
&- 2\mathcal{C}D_s(\partial_x u^y)\pi_Q^{yy}\tau u^x u^y + 2\mathcal{C}D_s(\partial_y u^x)\pi_Q^{yy}\tau u^x u^y \\
&- 2\mathcal{C}D_s(\partial_x u^y)\pi_Q{}^\eta{}_\eta\tau u^x u^y + 2\mathcal{A}D_s(\partial_\tau u^y)\nu^x\tau u^\tau u^x u^y \\
&+ 2\mathcal{B}D_s(\partial_\tau u^y)\nu^x\tau u^\tau u^x u^y + 2\mathcal{A}D_s(\partial_\tau u^x)\nu^y\tau u^\tau u^x u^y \\
&+ 2\mathcal{B}D_s(\partial_\tau u^x)\nu^y\tau u^\tau u^x u^y + 2\mathcal{A}D_s(\partial_x u^y)\nu^x\tau\left(u^x\right)^2 u^y \\
&- 2\mathcal{A}D_s(\partial_y u^x)\nu^x\tau\left(u^x\right)^2 u^y + 2\mathcal{A}D_s(\partial_x u^x)\nu^y\tau\left(u^x\right)^2 u^y \\
&- 4\mathcal{A}D_s(\partial_y u^y)\nu^y\tau\left(u^x\right)^2 u^y + 6\mathcal{C}'(T)D_s(\partial_y T)\pi_Q{}^\eta{}_\eta\tau\left(u^x\right)^2 u^y \\
&+ 4\mathcal{B}D_s\nu^y u^\tau\left(u^x\right)^2 u^y - 2D_s(\partial_\tau u^y)I_{31}\tau u^\tau\left(u^x\right)^2 u^y \\
&- 2\mathcal{B}'(T)D_s(\partial_\tau T)\nu^y\tau u^\tau\left(u^x\right)^2 u^y - 2\mathcal{C}D_s(\partial_\tau u^y)\Pi_Q\tau u^\tau\left(u^x\right)^2 u^y \\
&- 2\mathcal{C}D_s(\partial_\tau u^x)\pi_Q^{xy}\tau u^\tau\left(u^x\right)^2 u^y - 4\mathcal{C}D_s(\partial_\tau u^y)\pi_Q^{yy}\tau u^\tau\left(u^x\right)^2 u^y \\
&+ \mathcal{C}D_s(\partial_\tau u^y)\pi_Q{}^\eta{}_\eta\tau u^\tau\left(u^x\right)^2 u^y - 2\mathcal{C}D_s(\partial_x u^x)\pi_Q^{xy}\tau\left(u^x\right)^3 u^y \\
&+ 2\mathcal{C}D_s(\partial_y u^y)\pi_Q^{xy}\tau\left(u^x\right)^3 u^y - 4\mathcal{C}D_s(\partial_x u^y)\pi_Q^{yy}\tau\left(u^x\right)^3 u^y \\
&+ 2\mathcal{C}D_s(\partial_y u^x)\pi_Q^{yy}\tau\left(u^x\right)^3 u^y - 4\mathcal{C}D_s(\partial_x u^y)\pi_Q{}^\eta{}_\eta\tau\left(u^x\right)^3 u^y \\
&+ 2\mathcal{A}D_s(\partial_\tau u^y)\nu^x\tau u^\tau\left(u^x\right)^3 u^y + 2\mathcal{A}D_s(\partial_\tau u^x)\nu^y\tau u^\tau\left(u^x\right)^3 u^y \\
&+ 2\mathcal{A}D_s(\partial_x u^y)\nu^x\tau\left(u^x\right)^4 u^y + 2\mathcal{A}D_s(\partial_x u^x)\nu^y\tau\left(u^x\right)^4 u^y \\
&- 2\mathcal{A}D_s(\partial_y u^y)\nu^y\tau\left(u^x\right)^4 u^y + 3\mathcal{C}'(T)D_s(\partial_y T)\pi_Q{}^\eta{}_\eta\tau\left(u^x\right)^4 u^y \\
&- 2\mathcal{C}D_s(\partial_\tau u^y)\pi_Q^{yy}\tau u^\tau\left(u^x\right)^4 u^y - 2\mathcal{C}D_s(\partial_\tau u^y)\pi_Q{}^\eta{}_\eta\tau u^\tau\left(u^x\right)^4 u^y \\
&- 2\mathcal{C}D_s(\partial_x u^y)\pi_Q^{yy}\tau\left(u^x\right)^5 u^y - 2\mathcal{C}D_s(\partial_x u^y)\pi_Q{}^\eta{}_\eta\tau\left(u^x\right)^5 u^y \\
&- 6D_s(\partial_x u^x)I_{31}\tau\left(u^y\right)^2 - 6D_s(\partial_y u^y)I_{31}\tau\left(u^y\right)^2
\end{aligned}
$$

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\begin{aligned}
&- 6\mathcal{B}'(T)D_s(\partial_x T)\nu^x\tau\left(u^y\right)^2 - 6\mathcal{B}'(T)D_s(\partial_y T)\nu^y\tau\left(u^y\right)^2 \\
&- 6\mathcal{C}D_s(\partial_x u^x)\Pi_Q\tau\left(u^y\right)^2 - 6\mathcal{C}D_s(\partial_y u^y)\Pi_Q\tau\left(u^y\right)^2 \\
&- 4\mathcal{C}D_s(\partial_x u^y)\pi_Q^{xy}\tau\left(u^y\right)^2 - 6\mathcal{C}D_s(\partial_y u^x)\pi_Q^{xy}\tau\left(u^y\right)^2 \\
&+ 4\mathcal{C}D_s(\partial_x u^x)\pi_Q^{yy}\tau\left(u^y\right)^2 - 4\mathcal{C}D_s(\partial_y u^y)\pi_Q^{yy}\tau\left(u^y\right)^2 \\
&+ 15\mathcal{C}D_s(\partial_x u^x)\pi_Q{}^\eta{}_\eta\tau\left(u^y\right)^2 + 9\mathcal{C}D_s(\partial_y u^y)\pi_Q{}^\eta{}_\eta\tau\left(u^y\right)^2 \\
&+ 18\pi_Q{}^\eta{}_\eta T\tau\left(u^y\right)^2 + 12D_s I_{31}u^\tau\left(u^y\right)^2 + 12\mathcal{C}D_s\Pi_Q u^\tau\left(u^y\right)^2 \\
&+ 21\mathcal{C}D_s\pi_Q{}^\eta{}_\eta u^\tau\left(u^y\right)^2 - 6\mathcal{A}D_s(\partial_\tau u^x)\nu^x\tau u^\tau\left(u^y\right)^2 \\
&- 4\mathcal{B}D_s(\partial_\tau u^x)\nu^x\tau u^\tau\left(u^y\right)^2 - 4\mathcal{A}D_s(\partial_\tau u^y)\nu^y\tau u^\tau\left(u^y\right)^2 \\
&- 2\mathcal{B}D_s(\partial_\tau u^y)\nu^y\tau u^\tau\left(u^y\right)^2 + 9\mathcal{C}'(T)D_s(\partial_\tau T)\pi_Q{}^\eta{}_\eta\tau u^\tau\left(u^y\right)^2 \\
&- 6\mathcal{A}D_s(\partial_x u^x)\nu^x\tau u^x\left(u^y\right)^2 + 2\mathcal{A}D_s(\partial_y u^y)\nu^x\tau u^x\left(u^y\right)^2 \\
&- 4\mathcal{A}D_s(\partial_x u^y)\nu^y\tau u^x\left(u^y\right)^2 + 2\mathcal{A}D_s(\partial_y u^x)\nu^y\tau u^x\left(u^y\right)^2 \\
&+ 9\mathcal{C}'(T)D_s(\partial_x T)\pi_Q{}^\eta{}_\eta\tau u^x\left(u^y\right)^2 + 8\mathcal{B}D_s\nu^x u^\tau u^x\left(u^y\right)^2 \\
&- 4D_s(\partial_\tau u^x)I_{31}\tau u^\tau u^x\left(u^y\right)^2 - 4\mathcal{B}'(T)D_s(\partial_\tau T)\nu^x\tau u^\tau u^x\left(u^y\right)^2 \\
&- 4\mathcal{C}D_s(\partial_\tau u^x)\Pi_Q\tau u^\tau u^x\left(u^y\right)^2 + 4\mathcal{C}D_s(\partial_\tau u^x)\pi_Q^{yy}\tau u^\tau u^x\left(u^y\right)^2 \\
&+ 10\mathcal{C}D_s(\partial_\tau u^x)\pi_Q{}^\eta{}_\eta\tau u^\tau u^x\left(u^y\right)^2 - 8D_s(\partial_x u^x)I_{31}\tau\left(u^x\right)^2\left(u^y\right)^2 \\
&- 8D_s(\partial_y u^y)I_{31}\tau\left(u^x\right)^2\left(u^y\right)^2 - 8\mathcal{B}'(T)D_s(\partial_x T)\nu^x\tau\left(u^x\right)^2\left(u^y\right)^2 \\
&- 8\mathcal{B}'(T)D_s(\partial_y T)\nu^y\tau\left(u^x\right)^2\left(u^y\right)^2 - 8\mathcal{C}D_s(\partial_x u^x)\Pi_Q\tau\left(u^x\right)^2\left(u^y\right)^2 \\
&- 8\mathcal{C}D_s(\partial_y u^y)\Pi_Q\tau\left(u^x\right)^2\left(u^y\right)^2 - 2\mathcal{C}D_s(\partial_x u^y)\pi_Q^{xy}\tau\left(u^x\right)^2\left(u^y\right)^2 \\
&- 4\mathcal{C}D_s(\partial_y u^x)\pi_Q^{xy}\tau\left(u^x\right)^2\left(u^y\right)^2 + 6\mathcal{C}D_s(\partial_x u^x)\pi_Q^{yy}\tau\left(u^x\right)^2\left(u^y\right)^2 \\
&- 6\mathcal{C}D_s(\partial_y u^y)\pi_Q^{yy}\tau\left(u^x\right)^2\left(u^y\right)^2 + 20\mathcal{C}D_s(\partial_x u^x)\pi_Q{}^\eta{}_\eta\tau\left(u^x\right)^2\left(u^y\right)^2 \\
&+ 12\mathcal{C}D_s(\partial_y u^y)\pi_Q{}^\eta{}_\eta\tau\left(u^x\right)^2\left(u^y\right)^2 + 24\pi_Q{}^\eta{}_\eta T\tau\left(u^x\right)^2\left(u^y\right)^2 \\
&+ 16D_s I_{31}u^\tau\left(u^x\right)^2\left(u^y\right)^2 + 16\mathcal{C}D_s\Pi_Q u^\tau\left(u^x\right)^2\left(u^y\right)^2 \\
&+ 28\mathcal{C}D_s\pi_Q{}^\eta{}_\eta u^\tau\left(u^x\right)^2\left(u^y\right)^2 - 4\mathcal{A}D_s(\partial_\tau u^x)\nu^x\tau u^\tau\left(u^x\right)^2\left(u^y\right)^2 \\
&- 6\mathcal{A}D_s(\partial_\tau u^y)\nu^y\tau u^\tau\left(u^x\right)^2\left(u^y\right)^2 - 2\mathcal{B}D_s(\partial_\tau u^y)\nu^y\tau u^\tau\left(u^x\right)^2\left(u^y\right)^2 \\
&+ 12\mathcal{C}'(T)D_s(\partial_\tau T)\pi_Q{}^\eta{}_\eta\tau u^\tau\left(u^x\right)^2\left(u^y\right)^2 \\
&- 4\mathcal{A}D_s(\partial_x u^x)\nu^x\tau\left(u^x\right)^3\left(u^y\right)^2 + 2\mathcal{A}D_s(\partial_y u^y)\nu^x\tau\left(u^x\right)^3\left(u^y\right)^2 \\
&- 6\mathcal{A}D_s(\partial_x u^y)\nu^y\tau\left(u^x\right)^3\left(u^y\right)^2 + 2\mathcal{A}D_s(\partial_y u^x)\nu^y\tau\left(u^x\right)^3\left(u^y\right)^2 \\
&+ 12\mathcal{C}'(T)D_s(\partial_x T)\pi_Q{}^\eta{}_\eta\tau\left(u^x\right)^3\left(u^y\right)^2 + 4\mathcal{B}D_s\nu^x u^\tau\left(u^x\right)^3\left(u^y\right)^2 \\
&- 2D_s(\partial_\tau u^x)I_{31}\tau u^\tau\left(u^x\right)^3\left(u^y\right)^2 \\
&- 2\mathcal{B}'(T)D_s(\partial_\tau T)\nu^x\tau u^\tau\left(u^x\right)^3\left(u^y\right)^2 \\
&- 2\mathcal{C}D_s(\partial_\tau u^x)\Pi_Q\tau u^\tau\left(u^x\right)^3\left(u^y\right)^2 \\
&+ 2\mathcal{C}D_s(\partial_\tau u^y)\pi_Q^{xy}\tau u^\tau\left(u^x\right)^3\left(u^y\right)^2 \\
&+ 2\mathcal{C}D_s(\partial_\tau u^x)\pi_Q^{yy}\tau u^\tau\left(u^x\right)^3\left(u^y\right)^2 \\
&+ 5\mathcal{C}D_s(\partial_\tau u^x)\pi_Q{}^\eta{}_\eta\tau u^\tau\left(u^x\right)^3\left(u^y\right)^2 - 2D_s(\partial_x u^x)I_{31}\tau\left(u^x\right)^4\left(u^y\right)^2 \\
&- 2D_s(\partial_y u^y)I_{31}\tau\left(u^x\right)^4\left(u^y\right)^2 - 2\mathcal{B}'(T)D_s(\partial_x T)\nu^x\tau\left(u^x\right)^4\left(u^y\right)^2 \\
&- 2\mathcal{B}'(T)D_s(\partial_y T)\nu^y\tau\left(u^x\right)^4\left(u^y\right)^2 - 2\mathcal{C}D_s(\partial_x u^x)\Pi_Q\tau\left(u^x\right)^4\left(u^y\right)^2 \\
&- 2\mathcal{C}D_s(\partial_y u^y)\Pi_Q\tau\left(u^x\right)^4\left(u^y\right)^2 + 2\mathcal{C}D_s(\partial_x u^y)\pi_Q^{xy}\tau\left(u^x\right)^4\left(u^y\right)^2 \\
&+ 2\mathcal{C}D_s(\partial_x u^x)\pi_Q^{yy}\tau\left(u^x\right)^4\left(u^y\right)^2 - 2\mathcal{C}D_s(\partial_y u^y)\pi_Q^{yy}\tau\left(u^x\right)^4\left(u^y\right)^2 \\
&+ 5\mathcal{C}D_s(\partial_x u^x)\pi_Q{}^\eta{}_\eta\tau\left(u^x\right)^4\left(u^y\right)^2 + 3\mathcal{C}D_s(\partial_y u^y)\pi_Q{}^\eta{}_\eta\tau\left(u^x\right)^4\left(u^y\right)^2 \\
&+ 6\pi_Q{}^\eta{}_\eta T\tau\left(u^x\right)^4\left(u^y\right)^2 + 4D_s I_{31}u^\tau\left(u^x\right)^4\left(u^y\right)^2 \\
&+ 4\mathcal{C}D_s\Pi_Q u^\tau\left(u^x\right)^4\left(u^y\right)^2 + 7\mathcal{C}D_s\pi_Q{}^\eta{}_\eta u^\tau\left(u^x\right)^4\left(u^y\right)^2 \\
&- 2\mathcal{A}D_s(\partial_\tau u^y)\nu^y\tau u^\tau\left(u^x\right)^4\left(u^y\right)^2
\end{aligned}
$$

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\begin{aligned}
&+ 3\mathcal{C}'(T) D_s(\partial_\tau T) {\pi_Q}^\eta{}_\eta \tau u^\tau \left(u^x\right)^4 \left(u^y\right)^2 \\
&- 2\mathcal{A} D_s(\partial_x u^y) \nu^y \tau \left(u^x\right)^5 \left(u^y\right)^2 \\
&+ 3\mathcal{C}'(T) D_s(\partial_x T) {\pi_Q}^\eta{}_\eta \tau \left(u^x\right)^5 \left(u^y\right)^2 - 6\mathcal{A} D_s(\partial_y u^x) \nu^x \tau \left(u^y\right)^3 \\
&- 4\mathcal{A} D_s(\partial_y u^y) \nu^y \tau \left(u^y\right)^3 + 9\mathcal{C}'(T) D_s(\partial_y T) {\pi_Q}^\eta{}_\eta \tau \left(u^y\right)^3 \\
&+ 8\mathcal{B} D_s \nu^y u^\tau \left(u^y\right)^3 - 4 D_s(\partial_\tau u^y) I_{31} \tau u^\tau \left(u^y\right)^3 \\
&- 4\mathcal{B}'(T) D_s(\partial_\tau T) \nu^y \tau u^\tau \left(u^y\right)^3 - 4\mathcal{C} D_s(\partial_\tau u^y) \Pi_Q \tau u^\tau \left(u^y\right)^3 \\
&- 4\mathcal{C} D_s(\partial_\tau u^x) \pi_Q^{xy} \tau u^\tau \left(u^y\right)^3 - 2\mathcal{C} D_s(\partial_\tau u^y) \pi_Q^{yy} \tau u^\tau \left(u^y\right)^3 \\
&+ 6\mathcal{C} D_s(\partial_\tau u^y) {\pi_Q}^\eta{}_\eta \tau u^\tau \left(u^y\right)^3 - 4\mathcal{C} D_s(\partial_x u^x) \pi_Q^{xy} \tau u^x \left(u^y\right)^3 \\
&+ 4\mathcal{C} D_s(\partial_y u^y) \pi_Q^{xy} \tau u^x \left(u^y\right)^3 - 2\mathcal{C} D_s(\partial_x u^y) \pi_Q^{yy} \tau u^x \left(u^y\right)^3 \\
&+ 4\mathcal{C} D_s(\partial_y u^x) \pi_Q^{yy} \tau u^x \left(u^y\right)^3 - 4\mathcal{C} D_s(\partial_x u^y) {\pi_Q}^\eta{}_\eta \tau u^x \left(u^y\right)^3 \\
&+ 4\mathcal{A} D_s(\partial_\tau u^y) \nu^x \tau u^\tau u^x \left(u^y\right)^3 + 2\mathcal{B} D_s(\partial_\tau u^y) \nu^x \tau u^\tau u^x \left(u^y\right)^3 \\
&+ 4\mathcal{A} D_s(\partial_\tau u^x) \nu^y \tau u^\tau u^x \left(u^y\right)^3 + 2\mathcal{B} D_s(\partial_\tau u^x) \nu^y \tau u^\tau u^x \left(u^y\right)^3 \\
&+ 4\mathcal{A} D_s(\partial_x u^y) \nu^x \tau \left(u^x\right)^2 \left(u^y\right)^3 - 4\mathcal{A} D_s(\partial_y u^x) \nu^x \tau \left(u^x\right)^2 \left(u^y\right)^3 \\
&+ 4\mathcal{A} D_s(\partial_x u^x) \nu^y \tau \left(u^x\right)^2 \left(u^y\right)^3 - 6\mathcal{A} D_s(\partial_y u^y) \nu^y \tau \left(u^x\right)^2 \left(u^y\right)^3 \\
&+ 12\mathcal{C}'(T) D_s(\partial_y T) {\pi_Q}^\eta{}_\eta \tau \left(u^x\right)^2 \left(u^y\right)^3 + 4\mathcal{B} D_s \nu^y u^\tau \left(u^x\right)^2 \left(u^y\right)^3 \\
&- 2 D_s(\partial_\tau u^y) I_{31} \tau u^\tau \left(u^x\right)^2 \left(u^y\right)^3 \\
&- 2\mathcal{B}'(T) D_s(\partial_\tau T) \nu^y \tau u^\tau \left(u^x\right)^2 \left(u^y\right)^3 \\
&- 2\mathcal{C} D_s(\partial_\tau u^y) \Pi_Q \tau u^\tau \left(u^x\right)^2 \left(u^y\right)^3 \\
&- 2\mathcal{C} D_s(\partial_\tau u^x) \pi_Q^{xy} \tau u^\tau \left(u^x\right)^2 \left(u^y\right)^3 \\
&- 2\mathcal{C} D_s(\partial_\tau u^y) \pi_Q^{yy} \tau u^\tau \left(u^x\right)^2 \left(u^y\right)^3 \\
&+ \mathcal{C} D_s(\partial_\tau u^y) {\pi_Q}^\eta{}_\eta \tau u^\tau \left(u^x\right)^2 \left(u^y\right)^3 - 2\mathcal{C} D_s(\partial_x u^x) \pi_Q^{xy} \tau \left(u^x\right)^3 \left(u^y\right)^3 \\
&+ 2\mathcal{C} D_s(\partial_y u^y) \pi_Q^{xy} \tau \left(u^x\right)^3 \left(u^y\right)^3 - 2\mathcal{C} D_s(\partial_x u^y) \pi_Q^{yy} \tau \left(u^x\right)^3 \left(u^y\right)^3 \\
&+ 2\mathcal{C} D_s(\partial_y u^x) \pi_Q^{yy} \tau \left(u^x\right)^3 \left(u^y\right)^3 - 4\mathcal{C} D_s(\partial_x u^y) {\pi_Q}^\eta{}_\eta \tau \left(u^x\right)^3 \left(u^y\right)^3 \\
&+ 2\mathcal{A} D_s(\partial_\tau u^y) \nu^x \tau u^\tau \left(u^x\right)^3 \left(u^y\right)^3 + 2\mathcal{A} D_s(\partial_\tau u^x) \nu^y \tau u^\tau \left(u^x\right)^3 \left(u^y\right)^3 \\
&+ 2\mathcal{A} D_s(\partial_x u^y) \nu^x \tau \left(u^x\right)^4 \left(u^y\right)^3 + 2\mathcal{A} D_s(\partial_x u^x) \nu^y \tau \left(u^x\right)^4 \left(u^y\right)^3 \\
&- 2\mathcal{A} D_s(\partial_y u^y) \nu^y \tau \left(u^x\right)^4 \left(u^y\right)^3 \\
&+ 3\mathcal{C}'(T) D_s(\partial_y T) {\pi_Q}^\eta{}_\eta \tau \left(u^x\right)^4 \left(u^y\right)^3 - 6 D_s(\partial_x u^x) I_{31} \tau \left(u^y\right)^4 \\
&- 6 D_s(\partial_y u^y) I_{31} \tau \left(u^y\right)^4 - 6\mathcal{B}'(T) D_s(\partial_x T) \nu^x \tau \left(u^y\right)^4 \\
&- 6\mathcal{B}'(T) D_s(\partial_y T) \nu^y \tau \left(u^y\right)^4 - 6\mathcal{C} D_s(\partial_x u^x) \Pi_Q \tau \left(u^y\right)^4 \\
&- 6\mathcal{C} D_s(\partial_y u^y) \Pi_Q \tau \left(u^y\right)^4 - 2\mathcal{C} D_s(\partial_x u^y) \pi_Q^{xy} \tau \left(u^y\right)^4 \\
&- 6\mathcal{C} D_s(\partial_y u^x) \pi_Q^{xy} \tau \left(u^y\right)^4 + 2\mathcal{C} D_s(\partial_x u^x) \pi_Q^{yy} \tau \left(u^y\right)^4 \\
&- 2\mathcal{C} D_s(\partial_y u^y) \pi_Q^{yy} \tau \left(u^y\right)^4 + 15\mathcal{C} D_s(\partial_x u^x) {\pi_Q}^\eta{}_\eta \tau \left(u^y\right)^4 \\
&+ 9\mathcal{C} D_s(\partial_y u^y) {\pi_Q}^\eta{}_\eta \tau \left(u^y\right)^4 + 18 {\pi_Q}^\eta{}_\eta T \tau \left(u^y\right)^4 + 12 D_s I_{31} u^\tau \left(u^y\right)^4 \\
&+ 12\mathcal{C} D_s \Pi_Q u^\tau \left(u^y\right)^4 + 21\mathcal{C} D_s {\pi_Q}^\eta{}_\eta u^\tau \left(u^y\right)^4 \\
&- 6\mathcal{A} D_s(\partial_\tau u^x) \nu^x \tau u^\tau \left(u^y\right)^4 - 2\mathcal{B} D_s(\partial_\tau u^x) \nu^x \tau u^\tau \left(u^y\right)^4 \\
&- 2\mathcal{A} D_s(\partial_\tau u^y) \nu^y \tau u^\tau \left(u^y\right)^4 + 9\mathcal{C}'(T) D_s(\partial_\tau T) {\pi_Q}^\eta{}_\eta \tau u^\tau \left(u^y\right)^4 \\
&- 6\mathcal{A} D_s(\partial_x u^x) \nu^x \tau u^x \left(u^y\right)^4 + 4\mathcal{A} D_s(\partial_y u^y) \nu^x \tau u^x \left(u^y\right)^4 \\
&- 2\mathcal{A} D_s(\partial_x u^y) \nu^y \tau u^x \left(u^y\right)^4 + 4\mathcal{A} D_s(\partial_y u^x) \nu^y \tau u^x \left(u^y\right)^4 \\
&+ 9\mathcal{C}'(T) D_s(\partial_x T) {\pi_Q}^\eta{}_\eta \tau u^x \left(u^y\right)^4 + 4\mathcal{B} D_s \nu^x u^\tau u^x \left(u^y\right)^4 \\
&- 2 D_s(\partial_\tau u^x) I_{31} \tau u^\tau u^x \left(u^y\right)^4 - 2\mathcal{B}'(T) D_s(\partial_\tau T) \nu^x \tau u^\tau u^x \left(u^y\right)^4 \\
&- 2\mathcal{C} D_s(\partial_\tau u^x) \Pi_Q \tau u^\tau u^x \left(u^y\right)^4 + 2\mathcal{C} D_s(\partial_\tau u^y) \pi_Q^{xy} \tau u^\tau u^x \left(u^y\right)^4 \\
&+ 2\mathcal{C} D_s(\partial_\tau u^x) \pi_Q^{yy} \tau u^\tau u^x \left(u^y\right)^4 + 5\mathcal{C} D_s(\partial_\tau u^x) {\pi_Q}^\eta{}_\eta \tau u^\tau u^x \left(u^y\right)^4
\end{aligned}
$$

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\begin{aligned}
&- 4D_s(\partial_x u^x) I_{31}\tau \left(u^x\right)^2 \left(u^y\right)^4 - 4D_s(\partial_y u^y) I_{31}\tau \left(u^x\right)^2 \left(u^y\right)^4 \\
&- 4\mathcal{B}'(T) D_s(\partial_x T)\nu^x \tau \left(u^x\right)^2 \left(u^y\right)^4 \\
&- 4\mathcal{B}'(T) D_s(\partial_y T)\nu^y \tau \left(u^x\right)^2 \left(u^y\right)^4 - 4\mathcal{C} D_s(\partial_x u^x)\Pi_Q \tau \left(u^x\right)^2 \left(u^y\right)^4 \\
&- 4\mathcal{C} D_s(\partial_y u^y)\Pi_Q \tau \left(u^x\right)^2 \left(u^y\right)^4 + 2\mathcal{C} D_s(\partial_x u^y)\pi_Q^{xy} \tau \left(u^x\right)^2 \left(u^y\right)^4 \\
&- 2\mathcal{C} D_s(\partial_y u^x)\pi_Q^{xy} \tau \left(u^x\right)^2 \left(u^y\right)^4 + 2\mathcal{C} D_s(\partial_x u^x)\pi_Q^{yy} \tau \left(u^x\right)^2 \left(u^y\right)^4 \\
&- 2\mathcal{C} D_s(\partial_y u^y)\pi_Q^{yy} \tau \left(u^x\right)^2 \left(u^y\right)^4 + 10\mathcal{C} D_s(\partial_x u^x)\pi_Q{}^{\eta}{}_{\eta} \tau \left(u^x\right)^2 \left(u^y\right)^4 \\
&+ 6\mathcal{C} D_s(\partial_y u^y)\pi_Q{}^{\eta}{}_{\eta} \tau \left(u^x\right)^2 \left(u^y\right)^4 + 12\pi_Q{}^{\eta}{}_{\eta} T\tau \left(u^x\right)^2 \left(u^y\right)^4 \\
&+ 8D_s I_{31} u^\tau \left(u^x\right)^2 \left(u^y\right)^4 + 8\mathcal{C} D_s \Pi_Q u^\tau \left(u^x\right)^2 \left(u^y\right)^4 \\
&+ 14\mathcal{C} D_s \pi_Q{}^{\eta}{}_{\eta} u^\tau \left(u^x\right)^2 \left(u^y\right)^4 - 2\mathcal{A} D_s(\partial_\tau u^x)\nu^x \tau u^\tau \left(u^x\right)^2 \left(u^y\right)^4 \\
&- 2\mathcal{A} D_s(\partial_\tau u^y)\nu^y \tau u^\tau \left(u^x\right)^2 \left(u^y\right)^4 \\
&+ 6\mathcal{C}'(T) D_s(\partial_\tau T)\pi_Q{}^{\eta}{}_{\eta} \tau u^\tau \left(u^x\right)^2 \left(u^y\right)^4 \\
&- 2\mathcal{A} D_s(\partial_x u^x)\nu^x \tau \left(u^x\right)^3 \left(u^y\right)^4 + 2\mathcal{A} D_s(\partial_y u^y)\nu^x \tau \left(u^x\right)^3 \left(u^y\right)^4 \\
&- 2\mathcal{A} D_s(\partial_x u^y)\nu^y \tau \left(u^x\right)^3 \left(u^y\right)^4 + 2\mathcal{A} D_s(\partial_y u^x)\nu^y \tau \left(u^x\right)^3 \left(u^y\right)^4 \\
&+ 6\mathcal{C}'(T) D_s(\partial_x T)\pi_Q{}^{\eta}{}_{\eta} \tau \left(u^x\right)^3 \left(u^y\right)^4 - 6\mathcal{A} D_s(\partial_y u^x)\nu^x \tau \left(u^y\right)^5 \\
&- 2\mathcal{A} D_s(\partial_y u^y)\nu^y \tau \left(u^y\right)^5 + 9\mathcal{C}'(T) D_s(\partial_y T)\pi_Q{}^{\eta}{}_{\eta} \tau \left(u^y\right)^5 \\
&+ 4\mathcal{B} D_s \nu^y u^\tau \left(u^y\right)^5 - 2D_s(\partial_\tau u^y) I_{31}\tau u^\tau \left(u^y\right)^5 \\
&- 2\mathcal{B}'(T) D_s(\partial_\tau T)\nu^y \tau u^\tau \left(u^y\right)^5 - 2\mathcal{C} D_s(\partial_\tau u^y)\Pi_Q \tau u^\tau \left(u^y\right)^5 \\
&- 2\mathcal{C} D_s(\partial_\tau u^x)\pi_Q^{xy} \tau u^\tau \left(u^y\right)^5 + 3\mathcal{C} D_s(\partial_\tau u^y)\pi_Q{}^{\eta}{}_{\eta} \tau u^\tau \left(u^y\right)^5 \\
&- 2\mathcal{C} D_s(\partial_x u^x)\pi_Q^{xy} \tau u^x \left(u^y\right)^5 + 2\mathcal{C} D_s(\partial_y u^y)\pi_Q^{xy} \tau u^x \left(u^y\right)^5 \\
&+ 2\mathcal{C} D_s(\partial_y u^x)\pi_Q^{yy} \tau u^x \left(u^y\right)^5 - 2\mathcal{C} D_s(\partial_x u^y)\pi_Q{}^{\eta}{}_{\eta} \tau u^x \left(u^y\right)^5 \\
&+ 2\mathcal{A} D_s(\partial_\tau u^y)\nu^x \tau u^\tau u^x \left(u^y\right)^5 + 2\mathcal{A} D_s(\partial_\tau u^x)\nu^y \tau u^\tau u^x \left(u^y\right)^5 \\
&+ 2\mathcal{A} D_s(\partial_x u^y)\nu^x \tau \left(u^x\right)^2 \left(u^y\right)^5 - 2\mathcal{A} D_s(\partial_y u^x)\nu^x \tau \left(u^x\right)^2 \left(u^y\right)^5 \\
&+ 2\mathcal{A} D_s(\partial_x u^x)\nu^y \tau \left(u^x\right)^2 \left(u^y\right)^5 - 2\mathcal{A} D_s(\partial_y u^y)\nu^y \tau \left(u^x\right)^2 \left(u^y\right)^5 \\
&+ 6\mathcal{C}'(T) D_s(\partial_y T)\pi_Q{}^{\eta}{}_{\eta} \tau \left(u^x\right)^2 \left(u^y\right)^5 - 2D_s(\partial_x u^x) I_{31}\tau \left(u^y\right)^6 \\
&- 2D_s(\partial_y u^y) I_{31}\tau \left(u^y\right)^6 - 2\mathcal{B}'(T) D_s(\partial_x T)\nu^x \tau \left(u^y\right)^6 \\
&- 2\mathcal{B}'(T) D_s(\partial_y T)\nu^y \tau \left(u^y\right)^6 - 2\mathcal{C} D_s(\partial_x u^x)\Pi_Q \tau \left(u^y\right)^6 \\
&- 2\mathcal{C} D_s(\partial_y u^y)\Pi_Q \tau \left(u^y\right)^6 - 2\mathcal{C} D_s(\partial_y u^x)\pi_Q^{xy} \tau \left(u^y\right)^6 \\
&+ 5\mathcal{C} D_s(\partial_x u^x)\pi_Q{}^{\eta}{}_{\eta} \tau \left(u^y\right)^6 + 3\mathcal{C} D_s(\partial_y u^y)\pi_Q{}^{\eta}{}_{\eta} \tau \left(u^y\right)^6 \\
&+ 6\pi_Q{}^{\eta}{}_{\eta} T\tau \left(u^y\right)^6 + 4D_s I_{31} u^\tau \left(u^y\right)^6 + 4\mathcal{C} D_s \Pi_Q u^\tau \left(u^y\right)^6 \\
&+ 7\mathcal{C} D_s \pi_Q{}^{\eta}{}_{\eta} u^\tau \left(u^y\right)^6 - 2\mathcal{A} D_s(\partial_\tau u^x)\nu^x \tau u^\tau \left(u^y\right)^6 \\
&+ 3\mathcal{C}'(T) D_s(\partial_\tau T)\pi_Q{}^{\eta}{}_{\eta} \tau u^\tau \left(u^y\right)^6 - 2\mathcal{A} D_s(\partial_x u^x)\nu^x \tau u^x \left(u^y\right)^6 \\
&+ 2\mathcal{A} D_s(\partial_y u^y)\nu^x \tau u^x \left(u^y\right)^6 + 2\mathcal{A} D_s(\partial_y u^x)\nu^y \tau u^x \left(u^y\right)^6 \\
&+ 3\mathcal{C}'(T) D_s(\partial_x T)\pi_Q{}^{\eta}{}_{\eta} \tau u^x \left(u^y\right)^6 - 2\mathcal{A} D_s(\partial_y u^x)\nu^x \tau \left(u^y\right)^7 \\
&+ 3\mathcal{C}'(T) D_s(\partial_y T)\pi_Q{}^{\eta}{}_{\eta} \tau \left(u^y\right)^7 \Big)
\end{aligned}
$$

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\begin{aligned}
S_6 = \frac{1}{6T\tau \left(u^\tau\right)^4 \left(\left(u^y\right)^2 + 1\right)} \Big(& 3D_s(\partial_x u^y) I_{31}\tau + 3D_s(\partial_y u^x) I_{31}\tau + 3\mathcal{B}'(T) D_s(\partial_y T)\nu^x \tau \\
&+ 3\mathcal{B}'(T) D_s(\partial_x T)\nu^y \tau + 3\mathcal{C} D_s(\partial_x u^y)\Pi_Q \tau + 3\mathcal{C} D_s(\partial_y u^x)\Pi_Q \tau \\
&+ 6\mathcal{C} D_s(\partial_x u^x)\pi_Q^{xy} \tau + 6\mathcal{C} D_s(\partial_y u^y)\pi_Q^{xy} \tau - 3\mathcal{C} D_s(\partial_x u^y)\pi_Q^{yy} \tau \\
&+ 3\mathcal{C} D_s(\partial_y u^x)\pi_Q^{yy} \tau - 3\mathcal{C} D_s(\partial_x u^y)\pi_Q{}^{\eta}{}_{\eta} \tau + 6\pi_Q^{xy} T\tau + 3\mathcal{C} D_s \pi_Q^{xy} u^\tau \\
&+ 3\mathcal{A} D_s(\partial_\tau u^y)\nu^x \tau u^\tau + 3\mathcal{B} D_s(\partial_\tau u^y)\nu^x \tau u^\tau + 3\mathcal{A} D_s(\partial_\tau u^x)\nu^y \tau u^\tau \\
&+ 3\mathcal{B} D_s(\partial_\tau u^x)\nu^y \tau u^\tau + 3\mathcal{C}'(T) D_s(\partial_\tau T)\pi_Q^{xy} \tau u^\tau \\
&+ 3\mathcal{A} D_s(\partial_x u^y)\nu^x \tau u^x + 3\mathcal{B} D_s(\partial_x u^y)\nu^x \tau u^x - 3\mathcal{B} D_s(\partial_y u^x)\nu^x \tau u^x \\
&+ 3\mathcal{A} D_s(\partial_x u^x)\nu^y \tau u^x + 3\mathcal{B} D_s(\partial_x u^x)\nu^y \tau u^x - 3\mathcal{B} D_s(\partial_y u^y)\nu^y \tau u^x
\end{aligned}
$$

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\begin{aligned}
&+ 3\mathcal{C}'(T)D_s(\partial_x T)\pi_Q^{xy}\tau u^x + 3D_s(\partial_\tau u^y)I_{31}\tau u^\tau u^x \\
&+ 3\mathcal{B}'(T)D_s(\partial_\tau T)\nu^y\tau u^\tau u^x + 3\mathcal{C}D_s(\partial_\tau u^y)\Pi_Q\tau u^\tau u^x \\
&+ 3\mathcal{C}D_s(\partial_\tau u^x)\pi_Q^{xy}\tau u^\tau u^x - 6\mathcal{C}D_s(\partial_\tau u^y)\pi_Q^{yy}\tau u^\tau u^x \\
&- 3\mathcal{C}D_s(\partial_\tau u^y)\pi_Q{}^\eta{}_\eta\tau u^\tau u^x + 9D_s(\partial_x u^y)I_{31}\tau\left(u^x\right)^2 \\
&+ 6D_s(\partial_y u^x)I_{31}\tau\left(u^x\right)^2 + 6\mathcal{B}'(T)D_s(\partial_y T)\nu^x\tau\left(u^x\right)^2 \\
&+ 9\mathcal{B}'(T)D_s(\partial_x T)\nu^y\tau\left(u^x\right)^2 + 9\mathcal{C}D_s(\partial_x u^y)\Pi_Q\tau\left(u^x\right)^2 \\
&+ 6\mathcal{C}D_s(\partial_y u^x)\Pi_Q\tau\left(u^x\right)^2 + 9\mathcal{C}D_s(\partial_x u^x)\pi_Q^{xy}\tau\left(u^x\right)^2 \\
&+ 12\mathcal{C}D_s(\partial_y u^y)\pi_Q^{xy}\tau\left(u^x\right)^2 - 12\mathcal{C}D_s(\partial_x u^y)\pi_Q^{yy}\tau\left(u^x\right)^2 \\
&+ 6\mathcal{C}D_s(\partial_y u^x)\pi_Q^{yy}\tau\left(u^x\right)^2 - 9\mathcal{C}D_s(\partial_x u^y)\pi_Q{}^\eta{}_\eta\tau\left(u^x\right)^2 \\
&+ 12\pi_Q^{xy}T\tau\left(u^x\right)^2 + 6\mathcal{C}D_s\pi_Q^{xy}u^\tau\left(u^x\right)^2 + 6\mathcal{A}D_s(\partial_\tau u^y)\nu^x\tau u^\tau\left(u^x\right)^2 \\
&+ 6\mathcal{B}D_s(\partial_\tau u^y)\nu^x\tau u^\tau\left(u^x\right)^2 + 6\mathcal{A}D_s(\partial_\tau u^x)\nu^y\tau u^\tau\left(u^x\right)^2 \\
&+ 6\mathcal{B}D_s(\partial_\tau u^x)\nu^y\tau u^\tau\left(u^x\right)^2 + 6\mathcal{C}'(T)D_s(\partial_\tau T)\pi_Q^{xy}\tau u^\tau\left(u^x\right)^2 \\
&+ 6\mathcal{A}D_s(\partial_x u^y)\nu^x\tau\left(u^x\right)^3 + 6\mathcal{B}D_s(\partial_x u^y)\nu^x\tau\left(u^x\right)^3 \\
&- 3\mathcal{B}D_s(\partial_y u^x)\nu^x\tau\left(u^x\right)^3 + 6\mathcal{A}D_s(\partial_x u^x)\nu^y\tau\left(u^x\right)^3 \\
&+ 6\mathcal{B}D_s(\partial_x u^x)\nu^y\tau\left(u^x\right)^3 - 6\mathcal{B}D_s(\partial_y u^y)\nu^y\tau\left(u^x\right)^3 \\
&+ 6\mathcal{C}'(T)D_s(\partial_x T)\pi_Q^{xy}\tau\left(u^x\right)^3 + 6D_s(\partial_\tau u^y)I_{31}\tau u^\tau\left(u^x\right)^3 \\
&+ 6\mathcal{B}'(T)D_s(\partial_\tau T)\nu^y\tau u^\tau\left(u^x\right)^3 + 6\mathcal{C}D_s(\partial_\tau u^y)\Pi_Q\tau u^\tau\left(u^x\right)^3 \\
&+ 3\mathcal{C}D_s(\partial_\tau u^x)\pi_Q^{xy}\tau u^\tau\left(u^x\right)^3 - 12\mathcal{C}D_s(\partial_\tau u^y)\pi_Q^{yy}\tau u^\tau\left(u^x\right)^3 \\
&- 6\mathcal{C}D_s(\partial_\tau u^y)\pi_Q{}^\eta{}_\eta\tau u^\tau\left(u^x\right)^3 + 9D_s(\partial_x u^y)I_{31}\tau\left(u^x\right)^4 \\
&+ 3D_s(\partial_y u^x)I_{31}\tau\left(u^x\right)^4 + 3\mathcal{B}'(T)D_s(\partial_y T)\nu^x\tau\left(u^x\right)^4 \\
&+ 9\mathcal{B}'(T)D_s(\partial_x T)\nu^y\tau\left(u^x\right)^4 + 9\mathcal{C}D_s(\partial_x u^y)\Pi_Q\tau\left(u^x\right)^4 \\
&+ 3\mathcal{C}D_s(\partial_y u^x)\Pi_Q\tau\left(u^x\right)^4 + 3\mathcal{C}D_s(\partial_x u^x)\pi_Q^{xy}\tau\left(u^x\right)^4 \\
&+ 6\mathcal{C}D_s(\partial_y u^y)\pi_Q^{xy}\tau\left(u^x\right)^4 - 15\mathcal{C}D_s(\partial_x u^y)\pi_Q^{yy}\tau\left(u^x\right)^4 \\
&+ 3\mathcal{C}D_s(\partial_y u^x)\pi_Q^{yy}\tau\left(u^x\right)^4 - 9\mathcal{C}D_s(\partial_x u^y)\pi_Q{}^\eta{}_\eta\tau\left(u^x\right)^4 \\
&+ 6\pi_Q^{xy}T\tau\left(u^x\right)^4 + 3\mathcal{C}D_s\pi_Q^{xy}u^\tau\left(u^x\right)^4 + 3\mathcal{A}D_s(\partial_\tau u^y)\nu^x\tau u^\tau\left(u^x\right)^4 \\
&+ 3\mathcal{B}D_s(\partial_\tau u^y)\nu^x\tau u^\tau\left(u^x\right)^4 + 3\mathcal{A}D_s(\partial_\tau u^x)\nu^y\tau u^\tau\left(u^x\right)^4 \\
&+ 3\mathcal{B}D_s(\partial_\tau u^x)\nu^y\tau u^\tau\left(u^x\right)^4 + 3\mathcal{C}'(T)D_s(\partial_\tau T)\pi_Q^{xy}\tau u^\tau\left(u^x\right)^4 \\
&+ 3\mathcal{A}D_s(\partial_x u^y)\nu^x\tau\left(u^x\right)^5 + 3\mathcal{B}D_s(\partial_x u^y)\nu^x\tau\left(u^x\right)^5 \\
&+ 3\mathcal{A}D_s(\partial_x u^x)\nu^y\tau\left(u^x\right)^5 + 3\mathcal{B}D_s(\partial_x u^x)\nu^y\tau\left(u^x\right)^5 \\
&- 3\mathcal{B}D_s(\partial_y u^y)\nu^y\tau\left(u^x\right)^5 + 3\mathcal{C}'(T)D_s(\partial_x T)\pi_Q^{xy}\tau\left(u^x\right)^5 \\
&+ 3D_s(\partial_\tau u^y)I_{31}\tau u^\tau\left(u^x\right)^5 + 3\mathcal{B}'(T)D_s(\partial_\tau T)\nu^y\tau u^\tau\left(u^x\right)^5 \\
&+ 3\mathcal{C}D_s(\partial_\tau u^y)\Pi_Q\tau u^\tau\left(u^x\right)^5 - 6\mathcal{C}D_s(\partial_\tau u^y)\pi_Q^{yy}\tau u^\tau\left(u^x\right)^5 \\
&- 3\mathcal{C}D_s(\partial_\tau u^y)\pi_Q{}^\eta{}_\eta\tau u^\tau\left(u^x\right)^5 + 3D_s(\partial_x u^y)I_{31}\tau\left(u^x\right)^6 \\
&+ 3\mathcal{B}'(T)D_s(\partial_x T)\nu^y\tau\left(u^x\right)^6 + 3\mathcal{C}D_s(\partial_x u^y)\Pi_Q\tau\left(u^x\right)^6 \\
&- 6\mathcal{C}D_s(\partial_x u^y)\pi_Q^{yy}\tau\left(u^x\right)^6 - 3\mathcal{C}D_s(\partial_x u^y)\pi_Q{}^\eta{}_\eta\tau\left(u^x\right)^6 \\
&- 3\mathcal{B}D_s(\partial_x u^x)\nu^x\tau u^y + 3\mathcal{A}D_s(\partial_y u^y)\nu^x\tau u^y + 3\mathcal{B}D_s(\partial_y u^y)\nu^x\tau u^y \\
&- 3\mathcal{B}D_s(\partial_x u^y)\nu^y\tau u^y + 3\mathcal{A}D_s(\partial_y u^x)\nu^y\tau u^y + 3\mathcal{B}D_s(\partial_y u^x)\nu^y\tau u^y \\
&+ 3\mathcal{C}'(T)D_s(\partial_y T)\pi_Q^{xy}\tau u^y + 3D_s(\partial_\tau u^x)I_{31}\tau u^\tau u^y \\
&+ 3\mathcal{B}'(T)D_s(\partial_\tau T)\nu^x\tau u^\tau u^y + 3\mathcal{C}D_s(\partial_\tau u^x)\Pi_Q\tau u^\tau u^y \\
&+ 3\mathcal{C}D_s(\partial_\tau u^y)\pi_Q^{xy}\tau u^\tau u^y + 6\mathcal{C}D_s(\partial_\tau u^x)\pi_Q^{yy}\tau u^\tau u^y \\
&+ 3\mathcal{C}D_s(\partial_\tau u^x)\pi_Q{}^\eta{}_\eta\tau u^\tau u^y + D_s(\partial_x u^x)I_{31}\tau u^x u^y \\
&+ D_s(\partial_y u^y)I_{31}\tau u^x u^y + \mathcal{B}'(T)D_s(\partial_x T)\nu^x\tau u^x u^y \\
&+ \mathcal{B}'(T)D_s(\partial_y T)\nu^y\tau u^x u^y + \mathcal{C}D_s(\partial_x u^x)\Pi_Q\tau u^x u^y
\end{aligned}
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\begin{aligned}
&+ \mathcal{C}D_s(\partial_y u^y)\Pi_Q \tau u^x u^y + \mathcal{C}D_s(\partial_x u^y)\pi_Q^{xy}\tau u^x u^y \\
&- 5\mathcal{C}D_s(\partial_y u^x)\pi_Q^{xy}\tau u^x u^y + 5\mathcal{C}D_s(\partial_x u^x)\pi_Q^{yy}\tau u^x u^y \\
&- 5\mathcal{C}D_s(\partial_y u^y)\pi_Q^{yy}\tau u^x u^y + 5\mathcal{C}D_s(\partial_x u^x)\pi_Q{}^{\eta}{}_{\eta}\tau u^x u^y \\
&- 2D_s I_{31} u^\tau u^x u^y - 2\mathcal{C}D_s\Pi_Q u^\tau u^x u^y - 2\mathcal{C}D_s\pi_Q{}^{\eta}{}_{\eta} u^\tau u^x u^y \\
&- 2\mathcal{A}D_s(\partial_\tau u^x)\nu^x \tau u^\tau u^x u^y - 8\mathcal{B}D_s(\partial_\tau u^x)\nu^x\tau u^\tau u^x u^y \\
&- 2\mathcal{A}D_s(\partial_\tau u^y)\nu^y \tau u^\tau u^x u^y - 8\mathcal{B}D_s(\partial_\tau u^y)\nu^y\tau u^\tau u^x u^y \\
&- 2\mathcal{A}D_s(\partial_x u^x)\nu^x \tau \left(u^x\right)^2 u^y - 9\mathcal{B}D_s(\partial_x u^x)\nu^x\tau \left(u^x\right)^2 u^y \\
&+ 6\mathcal{A}D_s(\partial_y u^y)\nu^x \tau \left(u^x\right)^2 u^y + 9\mathcal{B}D_s(\partial_y u^y)\nu^x\tau \left(u^x\right)^2 u^y \\
&- 2\mathcal{A}D_s(\partial_x u^y)\nu^y \tau \left(u^x\right)^2 u^y - 12\mathcal{B}D_s(\partial_x u^y)\nu^y\tau \left(u^x\right)^2 u^y \\
&+ 6\mathcal{A}D_s(\partial_y u^x)\nu^y \tau \left(u^x\right)^2 u^y + 9\mathcal{B}D_s(\partial_y u^x)\nu^y\tau \left(u^x\right)^2 u^y \\
&+ 6\mathcal{C}'(T)D_s(\partial_y T)\pi_Q^{xy}\tau \left(u^x\right)^2 u^y - 2\mathcal{B}D_s\nu^x u^\tau \left(u^x\right)^2 u^y \\
&+ 4D_s(\partial_\tau u^x) I_{31}\tau u^\tau \left(u^x\right)^2 u^y + 4\mathcal{B}'(T)D_s(\partial_\tau T)\nu^x\tau u^\tau \left(u^x\right)^2 u^y \\
&+ 4\mathcal{C}D_s(\partial_\tau u^x)\Pi_Q\tau u^\tau \left(u^x\right)^2 u^y + 7\mathcal{C}D_s(\partial_\tau u^y)\pi_Q^{xy}\tau u^\tau \left(u^x\right)^2 u^y \\
&+ 14\mathcal{C}D_s(\partial_\tau u^x)\pi_Q^{yy}\tau u^\tau \left(u^x\right)^2 u^y + 8\mathcal{C}D_s(\partial_\tau u^x)\pi_Q{}^{\eta}{}_{\eta}\tau u^\tau \left(u^x\right)^2 u^y \\
&+ 2D_s(\partial_x u^x) I_{31}\tau \left(u^x\right)^3 u^y + 2D_s(\partial_y u^y) I_{31}\tau \left(u^x\right)^3 u^y \\
&+ 2\mathcal{B}'(T)D_s(\partial_x T)\nu^x\tau \left(u^x\right)^3 u^y + 2\mathcal{B}'(T)D_s(\partial_y T)\nu^y\tau \left(u^x\right)^3 u^y \\
&+ 2\mathcal{C}D_s(\partial_x u^x)\Pi_Q\tau \left(u^x\right)^3 u^y + 2\mathcal{C}D_s(\partial_y u^y)\Pi_Q\tau \left(u^x\right)^3 u^y \\
&+ 5\mathcal{C}D_s(\partial_x u^y)\pi_Q^{xy}\tau \left(u^x\right)^3 u^y - 5\mathcal{C}D_s(\partial_y u^x)\pi_Q^{xy}\tau \left(u^x\right)^3 u^y \\
&+ 13\mathcal{C}D_s(\partial_x u^x)\pi_Q^{yy}\tau \left(u^x\right)^3 u^y - 10\mathcal{C}D_s(\partial_y u^y)\pi_Q^{yy}\tau \left(u^x\right)^3 u^y \\
&+ 10\mathcal{C}D_s(\partial_x u^x)\pi_Q{}^{\eta}{}_{\eta}\tau \left(u^x\right)^3 u^y - 4D_s I_{31} u^\tau \left(u^x\right)^3 u^y \\
&- 4\mathcal{C}D_s\Pi_Q u^\tau \left(u^x\right)^3 u^y - 4\mathcal{C}D_s\pi_Q{}^{\eta}{}_{\eta} u^\tau \left(u^x\right)^3 u^y \\
&- 2\mathcal{A}D_s(\partial_\tau u^x)\nu^x\tau u^\tau \left(u^x\right)^3 u^y - 6\mathcal{B}D_s(\partial_\tau u^x)\nu^x\tau u^\tau \left(u^x\right)^3 u^y \\
&- 4\mathcal{A}D_s(\partial_\tau u^y)\nu^y\tau u^\tau \left(u^x\right)^3 u^y - 14\mathcal{B}D_s(\partial_\tau u^y)\nu^y\tau u^\tau \left(u^x\right)^3 u^y \\
&- 2\mathcal{A}D_s(\partial_x u^x)\nu^x \tau \left(u^x\right)^4 u^y - 6\mathcal{B}D_s(\partial_x u^x)\nu^x\tau \left(u^x\right)^4 u^y \\
&+ 3\mathcal{A}D_s(\partial_y u^y)\nu^x \tau \left(u^x\right)^4 u^y + 6\mathcal{B}D_s(\partial_y u^y)\nu^x\tau \left(u^x\right)^4 u^y \\
&- 4\mathcal{A}D_s(\partial_x u^y)\nu^y \tau \left(u^x\right)^4 u^y - 15\mathcal{B}D_s(\partial_x u^y)\nu^y\tau \left(u^x\right)^4 u^y \\
&+ 3\mathcal{A}D_s(\partial_y u^x)\nu^y \tau \left(u^x\right)^4 u^y + 6\mathcal{B}D_s(\partial_y u^x)\nu^y\tau \left(u^x\right)^4 u^y \\
&+ 3\mathcal{C}'(T)D_s(\partial_y T)\pi_Q^{xy}\tau \left(u^x\right)^4 u^y - 2\mathcal{B}D_s\nu^x u^\tau \left(u^x\right)^4 u^y \\
&+ D_s(\partial_\tau u^x) I_{31}\tau u^\tau \left(u^x\right)^4 u^y + \mathcal{B}'(T)D_s(\partial_\tau T)\nu^x\tau u^\tau \left(u^x\right)^4 u^y \\
&+ \mathcal{C}D_s(\partial_\tau u^x)\Pi_Q\tau u^\tau \left(u^x\right)^4 u^y + 4\mathcal{C}D_s(\partial_\tau u^y)\pi_Q^{xy}\tau u^\tau \left(u^x\right)^4 u^y \\
&+ 8\mathcal{C}D_s(\partial_\tau u^x)\pi_Q^{yy}\tau u^\tau \left(u^x\right)^4 u^y + 5\mathcal{C}D_s(\partial_\tau u^x)\pi_Q{}^{\eta}{}_{\eta}\tau u^\tau \left(u^x\right)^4 u^y \\
&+ D_s(\partial_x u^x) I_{31}\tau \left(u^x\right)^5 u^y + D_s(\partial_y u^y) I_{31}\tau \left(u^x\right)^5 u^y \\
&+ \mathcal{B}'(T)D_s(\partial_x T)\nu^x\tau \left(u^x\right)^5 u^y + \mathcal{B}'(T)D_s(\partial_y T)\nu^y\tau \left(u^x\right)^5 u^y \\
&+ \mathcal{C}D_s(\partial_x u^x)\Pi_Q\tau \left(u^x\right)^5 u^y + \mathcal{C}D_s(\partial_y u^y)\Pi_Q\tau \left(u^x\right)^5 u^y \\
&+ 4\mathcal{C}D_s(\partial_x u^y)\pi_Q^{xy}\tau \left(u^x\right)^5 u^y + 8\mathcal{C}D_s(\partial_x u^x)\pi_Q^{yy}\tau \left(u^x\right)^5 u^y \\
&- 5\mathcal{C}D_s(\partial_y u^y)\pi_Q^{yy}\tau \left(u^x\right)^5 u^y + 5\mathcal{C}D_s(\partial_x u^x)\pi_Q{}^{\eta}{}_{\eta}\tau \left(u^x\right)^5 u^y \\
&- 2D_s I_{31} u^\tau \left(u^x\right)^5 u^y - 2\mathcal{C}D_s\Pi_Q u^\tau \left(u^x\right)^5 u^y - 2\mathcal{C}D_s\pi_Q{}^{\eta}{}_{\eta} u^\tau \left(u^x\right)^5 u^y \\
&- 2\mathcal{A}D_s(\partial_\tau u^y)\nu^y\tau u^\tau \left(u^x\right)^5 u^y - 6\mathcal{B}D_s(\partial_\tau u^y)\nu^y\tau u^\tau \left(u^x\right)^5 u^y \\
&- 2\mathcal{A}D_s(\partial_x u^y)\nu^y \tau \left(u^x\right)^6 u^y - 6\mathcal{B}D_s(\partial_x u^y)\nu^y\tau \left(u^x\right)^6 u^y \\
&+ 9D_s(\partial_x u^y) I_{31}\tau \left(u^y\right)^2 + 12D_s(\partial_y u^x) I_{31}\tau \left(u^y\right)^2 \\
&+ 12\mathcal{B}'(T)D_s(\partial_y T)\nu^x\tau \left(u^y\right)^2 + 9\mathcal{B}'(T)D_s(\partial_x T)\nu^y\tau \left(u^y\right)^2 \\
&+ 9\mathcal{C}D_s(\partial_x u^y)\Pi_Q\tau \left(u^y\right)^2 + 12\mathcal{C}D_s(\partial_y u^x)\Pi_Q\tau \left(u^y\right)^2 \\
&+ 18\mathcal{C}D_s(\partial_x u^x)\pi_Q^{xy}\tau \left(u^y\right)^2 + 15\mathcal{C}D_s(\partial_y u^y)\pi_Q^{xy}\tau \left(u^y\right)^2
\end{aligned}
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\begin{aligned}
&- 6\mathcal{C}D_s(\partial_x u^y)\pi_Q^{yy}\tau\left(u^y\right)^2 + 12\mathcal{C}D_s(\partial_y u^x)\pi_Q^{yy}\tau\left(u^y\right)^2 \\
&- 9\mathcal{C}D_s(\partial_x u^y)\pi_Q{}^{\eta}{}_{\eta}\tau\left(u^y\right)^2 + 3\mathcal{C}D_s(\partial_y u^x)\pi_Q{}^{\eta}{}_{\eta}\tau\left(u^y\right)^2 \\
&+ 18\pi_Q^{xy}T\tau\left(u^y\right)^2 + 9\mathcal{C}D_s\pi_Q^{xy}u^\tau\left(u^y\right)^2 + 9\mathcal{A}D_s(\partial_\tau u^y)\nu^x\tau u^\tau\left(u^y\right)^2 \\
&+ 9\mathcal{B}D_s(\partial_\tau u^y)\nu^x\tau u^\tau\left(u^y\right)^2 + 9\mathcal{A}D_s(\partial_\tau u^x)\nu^y\tau u^\tau\left(u^y\right)^2 \\
&+ 9\mathcal{B}D_s(\partial_\tau u^x)\nu^y\tau u^\tau\left(u^y\right)^2 + 9\mathcal{C}'(T)D_s(\partial_\tau T)\pi_Q^{xy}\tau u^\tau\left(u^y\right)^2 \\
&+ 9\mathcal{A}D_s(\partial_x u^y)\nu^x\tau u^x\left(u^y\right)^2 + 12\mathcal{B}D_s(\partial_x u^y)\nu^x\tau u^x\left(u^y\right)^2 \\
&- 2\mathcal{A}D_s(\partial_y u^x)\nu^x\tau u^x\left(u^y\right)^2 - 15\mathcal{B}D_s(\partial_y u^x)\nu^x\tau u^x\left(u^y\right)^2 \\
&+ 9\mathcal{A}D_s(\partial_x u^x)\nu^y\tau u^x\left(u^y\right)^2 + 12\mathcal{B}D_s(\partial_x u^x)\nu^y\tau u^x\left(u^y\right)^2 \\
&- 2\mathcal{A}D_s(\partial_y u^y)\nu^y\tau u^x\left(u^y\right)^2 - 12\mathcal{B}D_s(\partial_y u^y)\nu^y\tau u^x\left(u^y\right)^2 \\
&+ 9\mathcal{C}'(T)D_s(\partial_x T)\pi_Q^{xy}\tau u^x\left(u^y\right)^2 - 2\mathcal{B}D_s\nu^y u^\tau u^x\left(u^y\right)^2 \\
&+ 7D_s(\partial_\tau u^y)I_{31}\tau u^\tau u^x\left(u^y\right)^2 + 7\mathcal{B}'(T)D_s(\partial_\tau T)\nu^y\tau u^\tau u^x\left(u^y\right)^2 \\
&+ 7\mathcal{C}D_s(\partial_\tau u^y)\Pi_Q\tau u^\tau u^x\left(u^y\right)^2 - 2\mathcal{C}D_s(\partial_\tau u^x)\pi_Q^{xy}\tau u^\tau u^x\left(u^y\right)^2 \\
&- 14\mathcal{C}D_s(\partial_\tau u^y)\pi_Q^{yy}\tau u^\tau u^x\left(u^y\right)^2 - 9\mathcal{C}D_s(\partial_\tau u^y)\pi_Q{}^{\eta}{}_{\eta}\tau u^\tau u^x\left(u^y\right)^2 \\
&+ 21D_s(\partial_x u^y)I_{31}\tau\left(u^x\right)^2\left(u^y\right)^2 + 18D_s(\partial_y u^x)I_{31}\tau\left(u^x\right)^2\left(u^y\right)^2 \\
&+ 18\mathcal{B}'(T)D_s(\partial_y T)\nu^x\tau\left(u^x\right)^2\left(u^y\right)^2 \\
&+ 21\mathcal{B}'(T)D_s(\partial_x T)\nu^y\tau\left(u^x\right)^2\left(u^y\right)^2 \\
&+ 21\mathcal{C}D_s(\partial_x u^y)\Pi_Q\tau\left(u^x\right)^2\left(u^y\right)^2 + 18\mathcal{C}D_s(\partial_y u^x)\Pi_Q\tau\left(u^x\right)^2\left(u^y\right)^2 \\
&+ 10\mathcal{C}D_s(\partial_x u^x)\pi_Q^{xy}\tau\left(u^x\right)^2\left(u^y\right)^2 + 23\mathcal{C}D_s(\partial_y u^y)\pi_Q^{xy}\tau\left(u^x\right)^2\left(u^y\right)^2 \\
&- 23\mathcal{C}D_s(\partial_x u^y)\pi_Q^{yy}\tau\left(u^x\right)^2\left(u^y\right)^2 + 23\mathcal{C}D_s(\partial_y u^x)\pi_Q^{yy}\tau\left(u^x\right)^2\left(u^y\right)^2 \\
&- 23\mathcal{C}D_s(\partial_x u^y)\pi_Q{}^{\eta}{}_{\eta}\tau\left(u^x\right)^2\left(u^y\right)^2 + 6\mathcal{C}D_s(\partial_y u^x)\pi_Q{}^{\eta}{}_{\eta}\tau\left(u^x\right)^2\left(u^y\right)^2 \\
&+ 24\pi_Q^{xy}T\tau\left(u^x\right)^2\left(u^y\right)^2 + 12\mathcal{C}D_s\pi_Q^{xy}u^\tau\left(u^x\right)^2\left(u^y\right)^2 \\
&+ 14\mathcal{A}D_s(\partial_\tau u^y)\nu^x\tau u^\tau\left(u^x\right)^2\left(u^y\right)^2 \\
&+ 20\mathcal{B}D_s(\partial_\tau u^y)\nu^x\tau u^\tau\left(u^x\right)^2\left(u^y\right)^2 \\
&+ 14\mathcal{A}D_s(\partial_\tau u^x)\nu^y\tau u^\tau\left(u^x\right)^2\left(u^y\right)^2 \\
&+ 20\mathcal{B}D_s(\partial_\tau u^x)\nu^y\tau u^\tau\left(u^x\right)^2\left(u^y\right)^2 \\
&+ 12\mathcal{C}'(T)D_s(\partial_\tau T)\pi_Q^{xy}\tau u^\tau\left(u^x\right)^2\left(u^y\right)^2 \\
&+ 14\mathcal{A}D_s(\partial_x u^y)\nu^x\tau\left(u^x\right)^3\left(u^y\right)^2 + 21\mathcal{B}D_s(\partial_x u^y)\nu^x\tau\left(u^x\right)^3\left(u^y\right)^2 \\
&- 2\mathcal{A}D_s(\partial_y u^x)\nu^x\tau\left(u^x\right)^3\left(u^y\right)^2 - 12\mathcal{B}D_s(\partial_y u^x)\nu^x\tau\left(u^x\right)^3\left(u^y\right)^2 \\
&+ 14\mathcal{A}D_s(\partial_x u^x)\nu^y\tau\left(u^x\right)^3\left(u^y\right)^2 + 21\mathcal{B}D_s(\partial_x u^x)\nu^y\tau\left(u^x\right)^3\left(u^y\right)^2 \\
&- 4\mathcal{A}D_s(\partial_y u^y)\nu^y\tau\left(u^x\right)^3\left(u^y\right)^2 - 21\mathcal{B}D_s(\partial_y u^y)\nu^y\tau\left(u^x\right)^3\left(u^y\right)^2 \\
&+ 12\mathcal{C}'(T)D_s(\partial_x T)\pi_Q^{xy}\tau\left(u^x\right)^3\left(u^y\right)^2 - 2\mathcal{B}D_s\nu^y u^\tau\left(u^x\right)^3\left(u^y\right)^2 \\
&+ 10D_s(\partial_\tau u^y)I_{31}\tau u^\tau\left(u^x\right)^3\left(u^y\right)^2 \\
&+ 10\mathcal{B}'(T)D_s(\partial_\tau T)\nu^y\tau u^\tau\left(u^x\right)^3\left(u^y\right)^2 \\
&+ 10\mathcal{C}D_s(\partial_\tau u^y)\Pi_Q\tau u^\tau\left(u^x\right)^3\left(u^y\right)^2 \\
&- 5\mathcal{C}D_s(\partial_\tau u^x)\pi_Q^{xy}\tau u^\tau\left(u^x\right)^3\left(u^y\right)^2 \\
&- 22\mathcal{C}D_s(\partial_\tau u^y)\pi_Q^{yy}\tau u^\tau\left(u^x\right)^3\left(u^y\right)^2 \\
&- 14\mathcal{C}D_s(\partial_\tau u^y)\pi_Q{}^{\eta}{}_{\eta}\tau u^\tau\left(u^x\right)^3\left(u^y\right)^2 \\
&+ 15D_s(\partial_x u^y)I_{31}\tau\left(u^x\right)^4\left(u^y\right)^2 + 6D_s(\partial_y u^x)I_{31}\tau\left(u^x\right)^4\left(u^y\right)^2 \\
&+ 6\mathcal{B}'(T)D_s(\partial_y T)\nu^x\tau\left(u^x\right)^4\left(u^y\right)^2 \\
&+ 15\mathcal{B}'(T)D_s(\partial_x T)\nu^y\tau\left(u^x\right)^4\left(u^y\right)^2 \\
&+ 15\mathcal{C}D_s(\partial_x u^y)\Pi_Q\tau\left(u^x\right)^4\left(u^y\right)^2 + 6\mathcal{C}D_s(\partial_y u^x)\Pi_Q\tau\left(u^x\right)^4\left(u^y\right)^2 \\
&- 5\mathcal{C}D_s(\partial_x u^x)\pi_Q^{xy}\tau\left(u^x\right)^4\left(u^y\right)^2 + 8\mathcal{C}D_s(\partial_y u^y)\pi_Q^{xy}\tau\left(u^x\right)^4\left(u^y\right)^2
\end{aligned}
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\begin{aligned}
&- 25\mathcal{C}D_s(\partial_x u^y)\pi_Q^{yy}\tau\left(u^x\right)^4\left(u^y\right)^2 + 11\mathcal{C}D_s(\partial_y u^x)\pi_Q^{yy}\tau\left(u^x\right)^4\left(u^y\right)^2\\
&- 19\mathcal{C}D_s(\partial_x u^y)\pi_Q{}^{\eta}{}_{\eta}\tau\left(u^x\right)^4\left(u^y\right)^2 + 3\mathcal{C}D_s(\partial_y u^x)\pi_Q{}^{\eta}{}_{\eta}\tau\left(u^x\right)^4\left(u^y\right)^2\\
&+ 6\pi_Q^{xy}T\tau\left(u^x\right)^4\left(u^y\right)^2 + 3\mathcal{C}D_s\pi_Q^{xy}u^\tau\left(u^x\right)^4\left(u^y\right)^2\\
&+ 5\mathcal{A}D_s(\partial_\tau u^y)\nu^x\tau u^\tau\left(u^x\right)^4\left(u^y\right)^2 + 9\mathcal{B}D_s(\partial_\tau u^y)\nu^x\tau u^\tau\left(u^x\right)^4\left(u^y\right)^2\\
&+ 5\mathcal{A}D_s(\partial_\tau u^x)\nu^y\tau u^\tau\left(u^x\right)^4\left(u^y\right)^2 + 9\mathcal{B}D_s(\partial_\tau u^x)\nu^y\tau u^\tau\left(u^x\right)^4\left(u^y\right)^2\\
&+ 3\mathcal{C}'(T)D_s(\partial_\tau T)\pi_Q^{xy}\tau u^\tau\left(u^x\right)^4\left(u^y\right)^2\\
&+ 5\mathcal{A}D_s(\partial_x u^y)\nu^x\tau\left(u^x\right)^5\left(u^y\right)^2 + 9\mathcal{B}D_s(\partial_x u^y)\nu^x\tau\left(u^x\right)^5\left(u^y\right)^2\\
&+ 5\mathcal{A}D_s(\partial_x u^x)\nu^y\tau\left(u^x\right)^5\left(u^y\right)^2 + 9\mathcal{B}D_s(\partial_x u^x)\nu^y\tau\left(u^x\right)^5\left(u^y\right)^2\\
&- 2\mathcal{A}D_s(\partial_y u^y)\nu^y\tau\left(u^x\right)^5\left(u^y\right)^2 - 9\mathcal{B}D_s(\partial_y u^y)\nu^y\tau\left(u^x\right)^5\left(u^y\right)^2\\
&+ 3\mathcal{C}'(T)D_s(\partial_x T)\pi_Q^{xy}\tau\left(u^x\right)^5\left(u^y\right)^2\\
&+ 3D_s(\partial_\tau u^y)I_{31}\tau u^\tau\left(u^x\right)^5\left(u^y\right)^2\\
&+ 3\mathcal{B}'(T)D_s(\partial_\tau T)\nu^y\tau u^\tau\left(u^x\right)^5\left(u^y\right)^2\\
&+ 3\mathcal{C}D_s(\partial_\tau u^y)\Pi_Q\tau u^\tau\left(u^x\right)^5\left(u^y\right)^2\\
&- 8\mathcal{C}D_s(\partial_\tau u^y)\pi_Q^{yy}\tau u^\tau\left(u^x\right)^5\left(u^y\right)^2\\
&- 5\mathcal{C}D_s(\partial_\tau u^y)\pi_Q{}^{\eta}{}_{\eta}\tau u^\tau\left(u^x\right)^5\left(u^y\right)^2 + 3D_s(\partial_x u^y)I_{31}\tau\left(u^x\right)^6\left(u^y\right)^2\\
&+ 3\mathcal{B}'(T)D_s(\partial_x T)\nu^y\tau\left(u^x\right)^6\left(u^y\right)^2 + 3\mathcal{C}D_s(\partial_x u^y)\Pi_Q\tau\left(u^x\right)^6\left(u^y\right)^2\\
&- 8\mathcal{C}D_s(\partial_x u^y)\pi_Q^{yy}\tau\left(u^x\right)^6\left(u^y\right)^2 - 5\mathcal{C}D_s(\partial_x u^y)\pi_Q{}^{\eta}{}_{\eta}\tau\left(u^x\right)^6\left(u^y\right)^2\\
&- 9\mathcal{B}D_s(\partial_x u^x)\nu^x\tau\left(u^y\right)^3 + 9\mathcal{A}D_s(\partial_y u^y)\nu^x\tau\left(u^y\right)^3\\
&+ 9\mathcal{B}D_s(\partial_y u^y)\nu^x\tau\left(u^y\right)^3 - 6\mathcal{B}D_s(\partial_x u^y)\nu^y\tau\left(u^y\right)^3\\
&+ 9\mathcal{A}D_s(\partial_y u^x)\nu^y\tau\left(u^y\right)^3 + 9\mathcal{B}D_s(\partial_y u^x)\nu^y\tau\left(u^y\right)^3\\
&+ 9\mathcal{C}'(T)D_s(\partial_y T)\pi_Q^{xy}\tau\left(u^y\right)^3 + 9D_s(\partial_\tau u^x)I_{31}\tau u^\tau\left(u^y\right)^3\\
&+ 9\mathcal{B}'(T)D_s(\partial_\tau T)\nu^x\tau u^\tau\left(u^y\right)^3 + 9\mathcal{C}D_s(\partial_\tau u^x)\Pi_Q\tau u^\tau\left(u^y\right)^3\\
&+ 6\mathcal{C}D_s(\partial_\tau u^y)\pi_Q^{xy}\tau u^\tau\left(u^y\right)^3 + 12\mathcal{C}D_s(\partial_\tau u^x)\pi_Q^{yy}\tau u^\tau\left(u^y\right)^3\\
&+ 9\mathcal{C}D_s(\partial_\tau u^x)\pi_Q{}^{\eta}{}_{\eta}\tau u^\tau\left(u^y\right)^3 + 3D_s(\partial_x u^x)I_{31}\tau u^x\left(u^y\right)^3\\
&+ 3D_s(\partial_y u^y)I_{31}\tau u^x\left(u^y\right)^3 + 3\mathcal{B}'(T)D_s(\partial_x T)\nu^x\tau u^x\left(u^y\right)^3\\
&+ 3\mathcal{B}'(T)D_s(\partial_y T)\nu^y\tau u^x\left(u^y\right)^3 + 3\mathcal{C}D_s(\partial_x u^x)\Pi_Q\tau u^x\left(u^y\right)^3\\
&+ 3\mathcal{C}D_s(\partial_y u^y)\Pi_Q\tau u^x\left(u^y\right)^3 + 2\mathcal{C}D_s(\partial_x u^y)\pi_Q^{xy}\tau u^x\left(u^y\right)^3\\
&- 18\mathcal{C}D_s(\partial_y u^x)\pi_Q^{xy}\tau u^x\left(u^y\right)^3 + 10\mathcal{C}D_s(\partial_x u^x)\pi_Q^{yy}\tau u^x\left(u^y\right)^3\\
&- 13\mathcal{C}D_s(\partial_y u^y)\pi_Q^{yy}\tau u^x\left(u^y\right)^3 + 15\mathcal{C}D_s(\partial_x u^x)\pi_Q{}^{\eta}{}_{\eta}\tau u^x\left(u^y\right)^3\\
&- 3\mathcal{C}D_s(\partial_y u^y)\pi_Q{}^{\eta}{}_{\eta}\tau u^x\left(u^y\right)^3 - 6D_s I_{31}u^\tau u^x\left(u^y\right)^3\\
&- 6\mathcal{C}D_s\Pi_Q u^\tau u^x\left(u^y\right)^3 - 6\mathcal{C}D_s\pi_Q{}^{\eta}{}_{\eta}u^\tau u^x\left(u^y\right)^3\\
&- 6\mathcal{A}D_s(\partial_\tau u^x)\nu^x\tau u^\tau u^x\left(u^y\right)^3 - 22\mathcal{B}D_s(\partial_\tau u^x)\nu^x\tau u^\tau u^x\left(u^y\right)^3\\
&- 4\mathcal{A}D_s(\partial_\tau u^y)\nu^y\tau u^\tau u^x\left(u^y\right)^3 - 14\mathcal{B}D_s(\partial_\tau u^y)\nu^y\tau u^\tau u^x\left(u^y\right)^3\\
&- 6\mathcal{A}D_s(\partial_x u^x)\nu^x\tau\left(u^x\right)^2\left(u^y\right)^3 - 24\mathcal{B}D_s(\partial_x u^x)\nu^x\tau\left(u^x\right)^2\left(u^y\right)^3\\
&+ 14\mathcal{A}D_s(\partial_y u^y)\nu^x\tau\left(u^x\right)^2\left(u^y\right)^3 + 24\mathcal{B}D_s(\partial_y u^y)\nu^x\tau\left(u^x\right)^2\left(u^y\right)^3\\
&- 4\mathcal{A}D_s(\partial_x u^y)\nu^y\tau\left(u^x\right)^2\left(u^y\right)^3 - 21\mathcal{B}D_s(\partial_x u^y)\nu^y\tau\left(u^x\right)^2\left(u^y\right)^3\\
&+ 14\mathcal{A}D_s(\partial_y u^x)\nu^y\tau\left(u^x\right)^2\left(u^y\right)^3 + 24\mathcal{B}D_s(\partial_y u^x)\nu^y\tau\left(u^x\right)^2\left(u^y\right)^3\\
&+ 12\mathcal{C}'(T)D_s(\partial_y T)\pi_Q^{xy}\tau\left(u^x\right)^2\left(u^y\right)^3 - 4\mathcal{B}D_s\nu^x u^\tau\left(u^x\right)^2\left(u^y\right)^3\\
&+ 8D_s(\partial_\tau u^x)I_{31}\tau u^\tau\left(u^x\right)^2\left(u^y\right)^3\\
&+ 8\mathcal{B}'(T)D_s(\partial_\tau T)\nu^x\tau u^\tau\left(u^x\right)^2\left(u^y\right)^3\\
&+ 8\mathcal{C}D_s(\partial_\tau u^x)\Pi_Q\tau u^\tau\left(u^x\right)^2\left(u^y\right)^3\\
&+ 15\mathcal{C}D_s(\partial_\tau u^y)\pi_Q^{xy}\tau u^\tau\left(u^x\right)^2\left(u^y\right)^3
\end{aligned}
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\begin{aligned}
&+ 22\mathcal{C}D_s(\partial_\tau u^x)\pi_Q^{yy}\tau u^\tau\left(u^x\right)^2\left(u^y\right)^3\\
&+ 16\mathcal{C}D_s(\partial_\tau u^x)\pi_Q{}^\eta{}_\eta\tau u^\tau\left(u^x\right)^2\left(u^y\right)^3 + 4D_s(\partial_x u^x)I_{31}\tau\left(u^x\right)^3\left(u^y\right)^3\\
&+ 4D_s(\partial_y u^y)I_{31}\tau\left(u^x\right)^3\left(u^y\right)^3 + 4\mathcal{B}'(T)D_s(\partial_x T)\nu^x\tau\left(u^x\right)^3\left(u^y\right)^3\\
&+ 4\mathcal{B}'(T)D_s(\partial_y T)\nu^y\tau\left(u^x\right)^3\left(u^y\right)^3 + 4\mathcal{C}D_s(\partial_x u^x)\Pi_Q\tau\left(u^x\right)^3\left(u^y\right)^3\\
&+ 4\mathcal{C}D_s(\partial_y u^y)\Pi_Q\tau\left(u^x\right)^3\left(u^y\right)^3 + 13\mathcal{C}D_s(\partial_x u^y)\pi_Q^{xy}\tau\left(u^x\right)^3\left(u^y\right)^3\\
&- 13\mathcal{C}D_s(\partial_y u^x)\pi_Q^{xy}\tau\left(u^x\right)^3\left(u^y\right)^3 + 21\mathcal{C}D_s(\partial_x u^x)\pi_Q^{yy}\tau\left(u^x\right)^3\left(u^y\right)^3\\
&- 21\mathcal{C}D_s(\partial_y u^y)\pi_Q^{yy}\tau\left(u^x\right)^3\left(u^y\right)^3\\
&+ 20\mathcal{C}D_s(\partial_x u^x)\pi_Q{}^\eta{}_\eta\tau\left(u^x\right)^3\left(u^y\right)^3 - 6\mathcal{C}D_s(\partial_y u^y)\pi_Q{}^\eta{}_\eta\tau\left(u^x\right)^3\left(u^y\right)^3\\
&- 8D_s I_{31}u^\tau\left(u^x\right)^3\left(u^y\right)^3 - 8\mathcal{C}D_s\Pi_Q u^\tau\left(u^x\right)^3\left(u^y\right)^3\\
&- 8\mathcal{C}D_s\pi_Q{}^\eta{}_\eta u^\tau\left(u^x\right)^3\left(u^y\right)^3 - 4\mathcal{A}D_s(\partial_\tau u^x)\nu^x\tau u^\tau\left(u^x\right)^3\left(u^y\right)^3\\
&- 12\mathcal{B}D_s(\partial_\tau u^x)\nu^x\tau u^\tau\left(u^x\right)^3\left(u^y\right)^3\\
&- 6\mathcal{A}D_s(\partial_\tau u^y)\nu^y\tau u^\tau\left(u^x\right)^3\left(u^y\right)^3\\
&- 20\mathcal{B}D_s(\partial_\tau u^y)\nu^y\tau u^\tau\left(u^x\right)^3\left(u^y\right)^3 - 4\mathcal{A}D_s(\partial_x u^x)\nu^x\tau\left(u^x\right)^4\left(u^y\right)^3\\
&- 12\mathcal{B}D_s(\partial_x u^x)\nu^x\tau\left(u^x\right)^4\left(u^y\right)^3 + 5\mathcal{A}D_s(\partial_y u^y)\nu^x\tau\left(u^x\right)^4\left(u^y\right)^3\\
&+ 12\mathcal{B}D_s(\partial_y u^y)\nu^x\tau\left(u^x\right)^4\left(u^y\right)^3 - 6\mathcal{A}D_s(\partial_x u^y)\nu^y\tau\left(u^x\right)^4\left(u^y\right)^3\\
&- 21\mathcal{B}D_s(\partial_x u^y)\nu^y\tau\left(u^x\right)^4\left(u^y\right)^3 + 5\mathcal{A}D_s(\partial_y u^x)\nu^y\tau\left(u^x\right)^4\left(u^y\right)^3\\
&+ 12\mathcal{B}D_s(\partial_y u^x)\nu^y\tau\left(u^x\right)^4\left(u^y\right)^3\\
&+ 3\mathcal{C}'(T)D_s(\partial_y T)\pi_Q^{xy}\tau\left(u^x\right)^4\left(u^y\right)^3 - 2\mathcal{B}D_s\nu^x u^\tau\left(u^x\right)^4\left(u^y\right)^3\\
&+ D_s(\partial_\tau u^x)I_{31}\tau u^\tau\left(u^x\right)^4\left(u^y\right)^3\\
&+ \mathcal{B}'(T)D_s(\partial_\tau T)\nu^x\tau u^\tau\left(u^x\right)^4\left(u^y\right)^3\\
&+ \mathcal{C}D_s(\partial_\tau u^x)\Pi_Q\tau u^\tau\left(u^x\right)^4\left(u^y\right)^3 + 8\mathcal{C}D_s(\partial_\tau u^y)\pi_Q^{xy}\tau u^\tau\left(u^x\right)^4\left(u^y\right)^3\\
&+ 8\mathcal{C}D_s(\partial_\tau u^x)\pi_Q^{yy}\tau u^\tau\left(u^x\right)^4\left(u^y\right)^3\\
&+ 5\mathcal{C}D_s(\partial_\tau u^x)\pi_Q{}^\eta{}_\eta\tau u^\tau\left(u^x\right)^4\left(u^y\right)^3 + D_s(\partial_x u^x)I_{31}\tau\left(u^x\right)^5\left(u^y\right)^3\\
&+ D_s(\partial_y u^y)I_{31}\tau\left(u^x\right)^5\left(u^y\right)^3 + \mathcal{B}'(T)D_s(\partial_x T)\nu^x\tau\left(u^x\right)^5\left(u^y\right)^3\\
&+ \mathcal{B}'(T)D_s(\partial_y T)\nu^y\tau\left(u^x\right)^5\left(u^y\right)^3 + \mathcal{C}D_s(\partial_x u^x)\Pi_Q\tau\left(u^x\right)^5\left(u^y\right)^3\\
&+ \mathcal{C}D_s(\partial_y u^y)\Pi_Q\tau\left(u^x\right)^5\left(u^y\right)^3 + 8\mathcal{C}D_s(\partial_x u^y)\pi_Q^{xy}\tau\left(u^x\right)^5\left(u^y\right)^3\\
&+ 8\mathcal{C}D_s(\partial_x u^x)\pi_Q^{yy}\tau\left(u^x\right)^5\left(u^y\right)^3 - 8\mathcal{C}D_s(\partial_y u^y)\pi_Q^{yy}\tau\left(u^x\right)^5\left(u^y\right)^3\\
&+ 5\mathcal{C}D_s(\partial_x u^x)\pi_Q{}^\eta{}_\eta\tau\left(u^x\right)^5\left(u^y\right)^3 - 3\mathcal{C}D_s(\partial_y u^y)\pi_Q{}^\eta{}_\eta\tau\left(u^x\right)^5\left(u^y\right)^3\\
&- 2D_s I_{31}u^\tau\left(u^x\right)^5\left(u^y\right)^3 - 2\mathcal{C}D_s\Pi_Q u^\tau\left(u^x\right)^5\left(u^y\right)^3\\
&- 2\mathcal{C}D_s\pi_Q{}^\eta{}_\eta u^\tau\left(u^x\right)^5\left(u^y\right)^3 - 2\mathcal{A}D_s(\partial_\tau u^y)\nu^y\tau u^\tau\left(u^x\right)^5\left(u^y\right)^3\\
&- 6\mathcal{B}D_s(\partial_\tau u^y)\nu^y\tau u^\tau\left(u^x\right)^5\left(u^y\right)^3 - 2\mathcal{A}D_s(\partial_x u^y)\nu^y\tau\left(u^x\right)^6\left(u^y\right)^3\\
&- 6\mathcal{B}D_s(\partial_x u^y)\nu^y\tau\left(u^x\right)^6\left(u^y\right)^3 + 9D_s(\partial_x u^y)I_{31}\tau\left(u^y\right)^4\\
&+ 18D_s(\partial_y u^x)I_{31}\tau\left(u^y\right)^4 + 18\mathcal{B}'(T)D_s(\partial_y T)\nu^x\tau\left(u^y\right)^4\\
&+ 9\mathcal{B}'(T)D_s(\partial_x T)\nu^y\tau\left(u^y\right)^4 + 9\mathcal{C}D_s(\partial_x u^y)\Pi_Q\tau\left(u^y\right)^4\\
&+ 18\mathcal{C}D_s(\partial_y u^x)\Pi_Q\tau\left(u^y\right)^4 + 18\mathcal{C}D_s(\partial_x u^x)\pi_Q^{xy}\tau\left(u^y\right)^4\\
&+ 12\mathcal{C}D_s(\partial_y u^y)\pi_Q^{xy}\tau\left(u^y\right)^4 - 3\mathcal{C}D_s(\partial_x u^y)\pi_Q^{yy}\tau\left(u^y\right)^4\\
&+ 15\mathcal{C}D_s(\partial_y u^x)\pi_Q^{yy}\tau\left(u^y\right)^4 - 9\mathcal{C}D_s(\partial_x u^y)\pi_Q{}^\eta{}_\eta\tau\left(u^y\right)^4\\
&+ 9\mathcal{C}D_s(\partial_y u^x)\pi_Q{}^\eta{}_\eta\tau\left(u^y\right)^4 + 18\pi_Q^{xy}T\tau\left(u^y\right)^4 + 9\mathcal{C}D_s\pi_Q^{xy}u^\tau\left(u^y\right)^4\\
&+ 9\mathcal{A}D_s(\partial_\tau u^y)\nu^x\tau u^\tau\left(u^y\right)^4 + 9\mathcal{B}D_s(\partial_\tau u^y)\nu^x\tau u^\tau\left(u^y\right)^4\\
&+ 9\mathcal{A}D_s(\partial_\tau u^x)\nu^y\tau u^\tau\left(u^y\right)^4 + 9\mathcal{B}D_s(\partial_\tau u^x)\nu^y\tau u^\tau\left(u^y\right)^4\\
&+ 9\mathcal{C}'(T)D_s(\partial_\tau T)\pi_Q^{xy}\tau u^\tau\left(u^y\right)^4 + 9\mathcal{A}D_s(\partial_x u^y)\nu^x\tau u^x\left(u^y\right)^4\\
&+ 15\mathcal{B}D_s(\partial_x u^y)\nu^x\tau u^x\left(u^y\right)^4 - 6\mathcal{A}D_s(\partial_y u^x)\nu^x\tau u^x\left(u^y\right)^4
\end{aligned}
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\begin{aligned}
&- 27\mathcal{B}D_s(\partial_y u^x)\nu^x\tau u^x\left(u^y\right)^4 + 9\mathcal{A}D_s(\partial_x u^x)\nu^y\tau u^x\left(u^y\right)^4 \\
&+ 15\mathcal{B}D_s(\partial_x u^x)\nu^y\tau u^x\left(u^y\right)^4 - 4\mathcal{A}D_s(\partial_y u^y)\nu^y\tau u^x\left(u^y\right)^4 \\
&- 15\mathcal{B}D_s(\partial_y u^y)\nu^y\tau u^x\left(u^y\right)^4 + 9\mathcal{C}'(T)D_s(\partial_x T)\pi_Q^{xy}\tau u^x\left(u^y\right)^4 \\
&- 4\mathcal{B}D_s\nu^y u^\tau u^x\left(u^y\right)^4 + 5D_s(\partial_\tau u^y)I_{31}\tau u^\tau u^x\left(u^y\right)^4 \\
&+ 5\mathcal{B}'(T)D_s(\partial_\tau T)\nu^y\tau u^\tau u^x\left(u^y\right)^4 + 5\mathcal{C}D_s(\partial_\tau u^y)\Pi_Q\tau u^\tau u^x\left(u^y\right)^4 \\
&- 13\mathcal{C}D_s(\partial_\tau u^x)\pi_Q^{xy}\tau u^\tau u^x\left(u^y\right)^4 - 8\mathcal{C}D_s(\partial_\tau u^y)\pi_Q^{yy}\tau u^\tau u^x\left(u^y\right)^4 \\
&- 9\mathcal{C}D_s(\partial_\tau u^y)\pi_Q{}^\eta{}_\eta\tau u^\tau u^x\left(u^y\right)^4 + 15D_s(\partial_x u^y)I_{31}\tau\left(u^x\right)^2\left(u^y\right)^4 \\
&+ 18D_s(\partial_y u^x)I_{31}\tau\left(u^x\right)^2\left(u^y\right)^4 \\
&+ 18\mathcal{B}'(T)D_s(\partial_y T)\nu^x\tau\left(u^x\right)^2\left(u^y\right)^4 \\
&+ 15\mathcal{B}'(T)D_s(\partial_x T)\nu^y\tau\left(u^x\right)^2\left(u^y\right)^4 \\
&+ 15\mathcal{C}D_s(\partial_x u^y)\Pi_Q\tau\left(u^x\right)^2\left(u^y\right)^4 + 18\mathcal{C}D_s(\partial_y u^x)\Pi_Q\tau\left(u^x\right)^2\left(u^y\right)^4 \\
&- 7\mathcal{C}D_s(\partial_x u^x)\pi_Q^{xy}\tau\left(u^x\right)^2\left(u^y\right)^4 + 19\mathcal{C}D_s(\partial_y u^y)\pi_Q^{xy}\tau\left(u^x\right)^2\left(u^y\right)^4 \\
&- 11\mathcal{C}D_s(\partial_x u^y)\pi_Q^{yy}\tau\left(u^x\right)^2\left(u^y\right)^4 + 25\mathcal{C}D_s(\partial_y u^x)\pi_Q^{yy}\tau\left(u^x\right)^2\left(u^y\right)^4 \\
&- 19\mathcal{C}D_s(\partial_x u^y)\pi_Q{}^\eta{}_\eta\tau\left(u^x\right)^2\left(u^y\right)^4 \\
&+ 12\mathcal{C}D_s(\partial_y u^x)\pi_Q{}^\eta{}_\eta\tau\left(u^x\right)^2\left(u^y\right)^4 + 12\pi_Q^{xy}T\tau\left(u^x\right)^2\left(u^y\right)^4 \\
&+ 6\mathcal{C}D_s\pi_Q^{xy}u^\tau\left(u^x\right)^2\left(u^y\right)^4 + 10\mathcal{A}D_s(\partial_\tau u^y)\nu^x\tau u^\tau\left(u^x\right)^2\left(u^y\right)^4 \\
&+ 20\mathcal{B}D_s(\partial_\tau u^y)\nu^x\tau u^\tau\left(u^x\right)^2\left(u^y\right)^4 \\
&+ 10\mathcal{A}D_s(\partial_\tau u^x)\nu^y\tau u^\tau\left(u^x\right)^2\left(u^y\right)^4 \\
&+ 20\mathcal{B}D_s(\partial_\tau u^x)\nu^y\tau u^\tau\left(u^x\right)^2\left(u^y\right)^4 \\
&+ 6\mathcal{C}'(T)D_s(\partial_\tau T)\pi_Q^{xy}\tau u^\tau\left(u^x\right)^2\left(u^y\right)^4 \\
&+ 10\mathcal{A}D_s(\partial_x u^y)\nu^x\tau\left(u^x\right)^3\left(u^y\right)^4 + 21\mathcal{B}D_s(\partial_x u^y)\nu^x\tau\left(u^x\right)^3\left(u^y\right)^4 \\
&- 4\mathcal{A}D_s(\partial_y u^x)\nu^x\tau\left(u^x\right)^3\left(u^y\right)^4 - 15\mathcal{B}D_s(\partial_y u^x)\nu^x\tau\left(u^x\right)^3\left(u^y\right)^4 \\
&+ 10\mathcal{A}D_s(\partial_x u^x)\nu^y\tau\left(u^x\right)^3\left(u^y\right)^4 + 21\mathcal{B}D_s(\partial_x u^x)\nu^y\tau\left(u^x\right)^3\left(u^y\right)^4 \\
&- 6\mathcal{A}D_s(\partial_y u^y)\nu^y\tau\left(u^x\right)^3\left(u^y\right)^4 - 21\mathcal{B}D_s(\partial_y u^y)\nu^y\tau\left(u^x\right)^3\left(u^y\right)^4 \\
&+ 6\mathcal{C}'(T)D_s(\partial_x T)\pi_Q^{xy}\tau\left(u^x\right)^3\left(u^y\right)^4 - 2\mathcal{B}D_s\nu^y u^\tau\left(u^x\right)^3\left(u^y\right)^4 \\
&+ 4D_s(\partial_\tau u^y)I_{31}\tau u^\tau\left(u^x\right)^3\left(u^y\right)^4 \\
&+ 4\mathcal{B}'(T)D_s(\partial_\tau T)\nu^y\tau u^\tau\left(u^x\right)^3\left(u^y\right)^4 \\
&+ 4\mathcal{C}D_s(\partial_\tau u^y)\Pi_Q\tau u^\tau\left(u^x\right)^3\left(u^y\right)^4 \\
&- 8\mathcal{C}D_s(\partial_\tau u^x)\pi_Q^{xy}\tau u^\tau\left(u^x\right)^3\left(u^y\right)^4 \\
&- 8\mathcal{C}D_s(\partial_\tau u^y)\pi_Q^{yy}\tau u^\tau\left(u^x\right)^3\left(u^y\right)^4 \\
&- 8\mathcal{C}D_s(\partial_\tau u^y)\pi_Q{}^\eta{}_\eta\tau u^\tau\left(u^x\right)^3\left(u^y\right)^4 + 6D_s(\partial_x u^y)I_{31}\tau\left(u^x\right)^4\left(u^y\right)^4 \\
&+ 3D_s(\partial_y u^x)I_{31}\tau\left(u^x\right)^4\left(u^y\right)^4 + 3\mathcal{B}'(T)D_s(\partial_y T)\nu^x\tau\left(u^x\right)^4\left(u^y\right)^4 \\
&+ 6\mathcal{B}'(T)D_s(\partial_x T)\nu^y\tau\left(u^x\right)^4\left(u^y\right)^4 + 6\mathcal{C}D_s(\partial_x u^y)\Pi_Q\tau\left(u^x\right)^4\left(u^y\right)^4 \\
&+ 3\mathcal{C}D_s(\partial_y u^x)\Pi_Q\tau\left(u^x\right)^4\left(u^y\right)^4 - 8\mathcal{C}D_s(\partial_x u^x)\pi_Q^{xy}\tau\left(u^x\right)^4\left(u^y\right)^4 \\
&+ 8\mathcal{C}D_s(\partial_y u^y)\pi_Q^{xy}\tau\left(u^x\right)^4\left(u^y\right)^4 - 8\mathcal{C}D_s(\partial_x u^y)\pi_Q^{yy}\tau\left(u^x\right)^4\left(u^y\right)^4 \\
&+ 8\mathcal{C}D_s(\partial_y u^x)\pi_Q^{yy}\tau\left(u^x\right)^4\left(u^y\right)^4 - 10\mathcal{C}D_s(\partial_x u^y)\pi_Q{}^\eta{}_\eta\tau\left(u^x\right)^4\left(u^y\right)^4 \\
&+ 3\mathcal{C}D_s(\partial_y u^x)\pi_Q{}^\eta{}_\eta\tau\left(u^x\right)^4\left(u^y\right)^4 + 2\mathcal{A}D_s(\partial_\tau u^y)\nu^x\tau u^\tau\left(u^x\right)^4\left(u^y\right)^4 \\
&+ 6\mathcal{B}D_s(\partial_\tau u^y)\nu^x\tau u^\tau\left(u^x\right)^4\left(u^y\right)^4 + 2\mathcal{A}D_s(\partial_\tau u^x)\nu^y\tau u^\tau\left(u^x\right)^4\left(u^y\right)^4 \\
&+ 6\mathcal{B}D_s(\partial_\tau u^x)\nu^y\tau u^\tau\left(u^x\right)^4\left(u^y\right)^4 + 2\mathcal{A}D_s(\partial_x u^y)\nu^x\tau\left(u^x\right)^5\left(u^y\right)^4 \\
&+ 6\mathcal{B}D_s(\partial_x u^y)\nu^x\tau\left(u^x\right)^5\left(u^y\right)^4 + 2\mathcal{A}D_s(\partial_x u^x)\nu^y\tau\left(u^x\right)^5\left(u^y\right)^4 \\
&+ 6\mathcal{B}D_s(\partial_x u^x)\nu^y\tau\left(u^x\right)^5\left(u^y\right)^4 - 2\mathcal{A}D_s(\partial_y u^y)\nu^y\tau\left(u^x\right)^5\left(u^y\right)^4 \\
&- 6\mathcal{B}D_s(\partial_y u^y)\nu^y\tau\left(u^x\right)^5\left(u^y\right)^4 - 9\mathcal{B}D_s(\partial_x u^x)\nu^x\tau\left(u^y\right)^5
\end{aligned}
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\begin{aligned}
&+ 9\mathcal{A}D_s(\partial_y u^y)\nu^x\tau\left(u^y\right)^5 + 9\mathcal{B}D_s(\partial_y u^y)\nu^x\tau\left(u^y\right)^5 \\
&- 3\mathcal{B}D_s(\partial_x u^y)\nu^y\tau\left(u^y\right)^5 + 9\mathcal{A}D_s(\partial_y u^x)\nu^y\tau\left(u^y\right)^5 \\
&+ 9\mathcal{B}D_s(\partial_y u^x)\nu^y\tau\left(u^y\right)^5 + 9\mathcal{C}'(T)D_s(\partial_y T)\pi_Q^{xy}\tau\left(u^y\right)^5 \\
&+ 9D_s(\partial_\tau u^x)I_{31}\tau u^\tau\left(u^y\right)^5 + 9\mathcal{B}'(T)D_s(\partial_\tau T)\nu^x\tau u^\tau\left(u^y\right)^5 \\
&+ 9\mathcal{C}D_s(\partial_\tau u^x)\Pi_Q\tau u^\tau\left(u^y\right)^5 + 3\mathcal{C}D_s(\partial_\tau u^y)\pi_Q^{xy}\tau u^\tau\left(u^y\right)^5 \\
&+ 6\mathcal{C}D_s(\partial_\tau u^x)\pi_Q^{yy}\tau u^\tau\left(u^y\right)^5 + 9\mathcal{C}D_s(\partial_\tau u^x)\pi_Q{}^\eta{}_\eta\tau u^\tau\left(u^y\right)^5 \\
&+ 3D_s(\partial_x u^x)I_{31}\tau u^x\left(u^y\right)^5 + 3D_s(\partial_y u^y)I_{31}\tau u^x\left(u^y\right)^5 \\
&+ 3\mathcal{B}'(T)D_s(\partial_x T)\nu^x\tau u^x\left(u^y\right)^5 + 3\mathcal{B}'(T)D_s(\partial_y T)\nu^y\tau u^x\left(u^y\right)^5 \\
&+ 3\mathcal{C}D_s(\partial_x u^x)\Pi_Q\tau u^x\left(u^y\right)^5 + 3\mathcal{C}D_s(\partial_y u^y)\Pi_Q\tau u^x\left(u^y\right)^5 \\
&+ \mathcal{C}D_s(\partial_x u^y)\pi_Q^{xy}\tau u^x\left(u^y\right)^5 - 21\mathcal{C}D_s(\partial_y u^x)\pi_Q^{xy}\tau u^x\left(u^y\right)^5 \\
&+ 5\mathcal{C}D_s(\partial_x u^x)\pi_Q^{yy}\tau u^x\left(u^y\right)^5 - 8\mathcal{C}D_s(\partial_y u^y)\pi_Q^{yy}\tau u^x\left(u^y\right)^5 \\
&+ 15\mathcal{C}D_s(\partial_x u^x)\pi_Q{}^\eta{}_\eta\tau u^x\left(u^y\right)^5 - 6\mathcal{C}D_s(\partial_y u^y)\pi_Q{}^\eta{}_\eta\tau u^x\left(u^y\right)^5 \\
&- 6D_s I_{31}u^\tau u^x\left(u^y\right)^5 - 6\mathcal{C}D_s\Pi_Q u^\tau u^x\left(u^y\right)^5 - 6\mathcal{C}D_s\pi_Q{}^\eta{}_\eta u^\tau u^x\left(u^y\right)^5 \\
&- 6\mathcal{A}D_s(\partial_\tau u^x)\nu^x\tau u^\tau u^x\left(u^y\right)^5 - 20\mathcal{B}D_s(\partial_\tau u^x)\nu^x\tau u^\tau u^x\left(u^y\right)^5 \\
&- 2\mathcal{A}D_s(\partial_\tau u^y)\nu^y\tau u^\tau u^x\left(u^y\right)^5 - 6\mathcal{B}D_s(\partial_\tau u^y)\nu^y\tau u^\tau u^x\left(u^y\right)^5 \\
&- 6\mathcal{A}D_s(\partial_x u^x)\nu^x\tau\left(u^x\right)^2\left(u^y\right)^5 - 21\mathcal{B}D_s(\partial_x u^x)\nu^x\tau\left(u^x\right)^2\left(u^y\right)^5 \\
&+ 10\mathcal{A}D_s(\partial_y u^y)\nu^x\tau\left(u^x\right)^2\left(u^y\right)^5 + 21\mathcal{B}D_s(\partial_y u^y)\nu^x\tau\left(u^x\right)^2\left(u^y\right)^5 \\
&- 2\mathcal{A}D_s(\partial_x u^y)\nu^y\tau\left(u^x\right)^2\left(u^y\right)^5 - 9\mathcal{B}D_s(\partial_x u^y)\nu^y\tau\left(u^x\right)^2\left(u^y\right)^5 \\
&+ 10\mathcal{A}D_s(\partial_y u^x)\nu^y\tau\left(u^x\right)^2\left(u^y\right)^5 + 21\mathcal{B}D_s(\partial_y u^x)\nu^y\tau\left(u^x\right)^2\left(u^y\right)^5 \\
&+ 6\mathcal{C}'(T)D_s(\partial_y T)\pi_Q^{xy}\tau\left(u^x\right)^2\left(u^y\right)^5 - 2\mathcal{B}D_s\nu^x u^\tau\left(u^x\right)^2\left(u^y\right)^5 \\
&+ 4D_s(\partial_\tau u^x)I_{31}\tau u^\tau\left(u^x\right)^2\left(u^y\right)^5 \\
&+ 4\mathcal{B}'(T)D_s(\partial_\tau T)\nu^x\tau u^\tau\left(u^x\right)^2\left(u^y\right)^5 \\
&+ 4\mathcal{C}D_s(\partial_\tau u^x)\Pi_Q\tau u^\tau\left(u^x\right)^2\left(u^y\right)^5 \\
&+ 8\mathcal{C}D_s(\partial_\tau u^y)\pi_Q^{xy}\tau u^\tau\left(u^x\right)^2\left(u^y\right)^5 \\
&+ 8\mathcal{C}D_s(\partial_\tau u^x)\pi_Q^{yy}\tau u^\tau\left(u^x\right)^2\left(u^y\right)^5 \\
&+ 8\mathcal{C}D_s(\partial_\tau u^x)\pi_Q{}^\eta{}_\eta\tau u^\tau\left(u^x\right)^2\left(u^y\right)^5 + 2D_s(\partial_x u^x)I_{31}\tau\left(u^x\right)^3\left(u^y\right)^5 \\
&+ 2D_s(\partial_y u^y)I_{31}\tau\left(u^x\right)^3\left(u^y\right)^5 + 2\mathcal{B}'(T)D_s(\partial_x T)\nu^x\tau\left(u^x\right)^3\left(u^y\right)^5 \\
&+ 2\mathcal{B}'(T)D_s(\partial_y T)\nu^y\tau\left(u^x\right)^3\left(u^y\right)^5 + 2\mathcal{C}D_s(\partial_x u^x)\Pi_Q\tau\left(u^x\right)^3\left(u^y\right)^5 \\
&+ 2\mathcal{C}D_s(\partial_y u^y)\Pi_Q\tau\left(u^x\right)^3\left(u^y\right)^5 + 8\mathcal{C}D_s(\partial_x u^y)\pi_Q^{xy}\tau\left(u^x\right)^3\left(u^y\right)^5 \\
&- 8\mathcal{C}D_s(\partial_y u^x)\pi_Q^{xy}\tau\left(u^x\right)^3\left(u^y\right)^5 + 8\mathcal{C}D_s(\partial_x u^x)\pi_Q^{yy}\tau\left(u^x\right)^3\left(u^y\right)^5 \\
&- 8\mathcal{C}D_s(\partial_y u^y)\pi_Q^{yy}\tau\left(u^x\right)^3\left(u^y\right)^5 + 10\mathcal{C}D_s(\partial_x u^x)\pi_Q{}^\eta{}_\eta\tau\left(u^x\right)^3\left(u^y\right)^5 \\
&- 6\mathcal{C}D_s(\partial_y u^y)\pi_Q{}^\eta{}_\eta\tau\left(u^x\right)^3\left(u^y\right)^5 - 4D_s I_{31}u^\tau\left(u^x\right)^3\left(u^y\right)^5 \\
&- 4\mathcal{C}D_s\Pi_Q u^\tau\left(u^x\right)^3\left(u^y\right)^5 - 4\mathcal{C}D_s\pi_Q{}^\eta{}_\eta u^\tau\left(u^x\right)^3\left(u^y\right)^5 \\
&- 2\mathcal{A}D_s(\partial_\tau u^x)\nu^x\tau u^\tau\left(u^x\right)^3\left(u^y\right)^5 - 6\mathcal{B}D_s(\partial_\tau u^x)\nu^x\tau u^\tau\left(u^x\right)^3\left(u^y\right)^5 \\
&- 2\mathcal{A}D_s(\partial_\tau u^y)\nu^y\tau u^\tau\left(u^x\right)^3\left(u^y\right)^5 - 6\mathcal{B}D_s(\partial_\tau u^y)\nu^y\tau u^\tau\left(u^x\right)^3\left(u^y\right)^5 \\
&- 2\mathcal{A}D_s(\partial_x u^x)\nu^x\tau\left(u^x\right)^4\left(u^y\right)^5 - 6\mathcal{B}D_s(\partial_x u^x)\nu^x\tau\left(u^x\right)^4\left(u^y\right)^5 \\
&+ 2\mathcal{A}D_s(\partial_y u^y)\nu^x\tau\left(u^x\right)^4\left(u^y\right)^5 + 6\mathcal{B}D_s(\partial_y u^y)\nu^x\tau\left(u^x\right)^4\left(u^y\right)^5 \\
&- 2\mathcal{A}D_s(\partial_x u^y)\nu^y\tau\left(u^x\right)^4\left(u^y\right)^5 - 6\mathcal{B}D_s(\partial_x u^y)\nu^y\tau\left(u^x\right)^4\left(u^y\right)^5 \\
&+ 2\mathcal{A}D_s(\partial_y u^x)\nu^y\tau\left(u^x\right)^4\left(u^y\right)^5 + 6\mathcal{B}D_s(\partial_y u^x)\nu^y\tau\left(u^x\right)^4\left(u^y\right)^5 \\
&+ 3D_s(\partial_x u^y)I_{31}\tau\left(u^y\right)^6 + 12D_s(\partial_y u^x)I_{31}\tau\left(u^y\right)^6 \\
&+ 12\mathcal{B}'(T)D_s(\partial_y T)\nu^x\tau\left(u^y\right)^6 + 3\mathcal{B}'(T)D_s(\partial_x T)\nu^y\tau\left(u^y\right)^6 \\
&+ 3\mathcal{C}D_s(\partial_x u^y)\Pi_Q\tau\left(u^y\right)^6 + 12\mathcal{C}D_s(\partial_y u^x)\Pi_Q\tau\left(u^y\right)^6
\end{aligned}
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\begin{aligned}
&+ 6\mathcal{C}D_s(\partial_x u^x)\pi_Q^{xy}\tau\left(u^y\right)^6 + 3\mathcal{C}D_s(\partial_y u^y)\pi_Q^{xy}\tau\left(u^y\right)^6 \\
&+ 6\mathcal{C}D_s(\partial_y u^x)\pi_Q^{yy}\tau\left(u^y\right)^6 - 3\mathcal{C}D_s(\partial_x u^y)\pi_Q{}^{\eta}{}_{\eta}\tau\left(u^y\right)^6 \\
&+ 9\mathcal{C}D_s(\partial_y u^x)\pi_Q{}^{\eta}{}_{\eta}\tau\left(u^y\right)^6 + 6\pi_Q^{xy}T\tau\left(u^y\right)^6 + 3\mathcal{C}D_s\pi_Q^{xy}u^{\tau}\left(u^y\right)^6 \\
&+ 3\mathcal{A}D_s(\partial_\tau u^y)\nu^x\tau u^{\tau}\left(u^y\right)^6 + 3\mathcal{B}D_s(\partial_\tau u^y)\nu^x\tau u^{\tau}\left(u^y\right)^6 \\
&+ 3\mathcal{A}D_s(\partial_\tau u^x)\nu^y\tau u^{\tau}\left(u^y\right)^6 + 3\mathcal{B}D_s(\partial_\tau u^x)\nu^y\tau u^{\tau}\left(u^y\right)^6 \\
&+ 3\mathcal{C}'(T)D_s(\partial_\tau T)\pi_Q^{xy}\tau u^{\tau}\left(u^y\right)^6 + 3\mathcal{A}D_s(\partial_x u^y)\nu^x\tau u^x\left(u^y\right)^6 \\
&+ 6\mathcal{B}D_s(\partial_x u^y)\nu^x\tau u^x\left(u^y\right)^6 - 6\mathcal{A}D_s(\partial_y u^x)\nu^x\tau u^x\left(u^y\right)^6 \\
&- 21\mathcal{B}D_s(\partial_y u^x)\nu^x\tau u^x\left(u^y\right)^6 + 3\mathcal{A}D_s(\partial_x u^x)\nu^y\tau u^x\left(u^y\right)^6 \\
&+ 6\mathcal{B}D_s(\partial_x u^x)\nu^y\tau u^x\left(u^y\right)^6 - 2\mathcal{A}D_s(\partial_y u^y)\nu^y\tau u^x\left(u^y\right)^6 \\
&- 6\mathcal{B}D_s(\partial_y u^y)\nu^y\tau u^x\left(u^y\right)^6 + 3\mathcal{C}'(T)D_s(\partial_x T)\pi_Q^{xy}\tau u^x\left(u^y\right)^6 \\
&- 2\mathcal{B}D_s\nu^y u^{\tau}u^x\left(u^y\right)^6 + D_s(\partial_\tau u^y)I_{31}\tau u^{\tau}u^x\left(u^y\right)^6 \\
&+ \mathcal{B}'(T)D_s(\partial_\tau T)\nu^y\tau u^{\tau}u^x\left(u^y\right)^6 + \mathcal{C}D_s(\partial_\tau u^y)\Pi_Q\tau u^{\tau}u^x\left(u^y\right)^6 \\
&- 8\mathcal{C}D_s(\partial_\tau u^x)\pi_Q^{xy}\tau u^{\tau}u^x\left(u^y\right)^6 - 3\mathcal{C}D_s(\partial_\tau u^y)\pi_Q{}^{\eta}{}_{\eta}\tau u^{\tau}u^x\left(u^y\right)^6 \\
&+ 3D_s(\partial_x u^y)I_{31}\tau\left(u^x\right)^2\left(u^y\right)^6 + 6D_s(\partial_y u^x)I_{31}\tau\left(u^x\right)^2\left(u^y\right)^6 \\
&+ 6\mathcal{B}'(T)D_s(\partial_y T)\nu^x\tau\left(u^x\right)^2\left(u^y\right)^6 \\
&+ 3\mathcal{B}'(T)D_s(\partial_x T)\nu^y\tau\left(u^x\right)^2\left(u^y\right)^6 + 3\mathcal{C}D_s(\partial_x u^y)\Pi_Q\tau\left(u^x\right)^2\left(u^y\right)^6 \\
&+ 6\mathcal{C}D_s(\partial_y u^x)\Pi_Q\tau\left(u^x\right)^2\left(u^y\right)^6 - 8\mathcal{C}D_s(\partial_x u^x)\pi_Q^{xy}\tau\left(u^x\right)^2\left(u^y\right)^6 \\
&+ 8\mathcal{C}D_s(\partial_y u^y)\pi_Q^{xy}\tau\left(u^x\right)^2\left(u^y\right)^6 + 8\mathcal{C}D_s(\partial_y u^x)\pi_Q^{yy}\tau\left(u^x\right)^2\left(u^y\right)^6 \\
&- 5\mathcal{C}D_s(\partial_x u^y)\pi_Q{}^{\eta}{}_{\eta}\tau\left(u^x\right)^2\left(u^y\right)^6 + 6\mathcal{C}D_s(\partial_y u^x)\pi_Q{}^{\eta}{}_{\eta}\tau\left(u^x\right)^2\left(u^y\right)^6 \\
&+ 2\mathcal{A}D_s(\partial_\tau u^y)\nu^x\tau u^{\tau}\left(u^x\right)^2\left(u^y\right)^6 + 6\mathcal{B}D_s(\partial_\tau u^y)\nu^x\tau u^{\tau}\left(u^x\right)^2\left(u^y\right)^6 \\
&+ 2\mathcal{A}D_s(\partial_\tau u^x)\nu^y\tau u^{\tau}\left(u^x\right)^2\left(u^y\right)^6 + 6\mathcal{B}D_s(\partial_\tau u^x)\nu^y\tau u^{\tau}\left(u^x\right)^2\left(u^y\right)^6 \\
&+ 2\mathcal{A}D_s(\partial_x u^y)\nu^x\tau\left(u^x\right)^3\left(u^y\right)^6 + 6\mathcal{B}D_s(\partial_x u^y)\nu^x\tau\left(u^x\right)^3\left(u^y\right)^6 \\
&- 2\mathcal{A}D_s(\partial_y u^x)\nu^x\tau\left(u^x\right)^3\left(u^y\right)^6 - 6\mathcal{B}D_s(\partial_y u^x)\nu^x\tau\left(u^x\right)^3\left(u^y\right)^6 \\
&+ 2\mathcal{A}D_s(\partial_x u^x)\nu^y\tau\left(u^x\right)^3\left(u^y\right)^6 + 6\mathcal{B}D_s(\partial_x u^x)\nu^y\tau\left(u^x\right)^3\left(u^y\right)^6 \\
&- 2\mathcal{A}D_s(\partial_y u^y)\nu^y\tau\left(u^x\right)^3\left(u^y\right)^6 - 6\mathcal{B}D_s(\partial_y u^y)\nu^y\tau\left(u^x\right)^3\left(u^y\right)^6 \\
&- 3\mathcal{B}D_s(\partial_x u^x)\nu^x\tau\left(u^y\right)^7 + 3\mathcal{A}D_s(\partial_y u^y)\nu^x\tau\left(u^y\right)^7 \\
&+ 3\mathcal{B}D_s(\partial_y u^y)\nu^x\tau\left(u^y\right)^7 + 3\mathcal{A}D_s(\partial_y u^x)\nu^y\tau\left(u^y\right)^7 \\
&+ 3\mathcal{B}D_s(\partial_y u^x)\nu^y\tau\left(u^y\right)^7 + 3\mathcal{C}'(T)D_s(\partial_y T)\pi_Q^{xy}\tau\left(u^y\right)^7 \\
&+ 3D_s(\partial_\tau u^x)I_{31}\tau u^{\tau}\left(u^y\right)^7 + 3\mathcal{B}'(T)D_s(\partial_\tau T)\nu^x\tau u^{\tau}\left(u^y\right)^7 \\
&+ 3\mathcal{C}D_s(\partial_\tau u^x)\Pi_Q\tau u^{\tau}\left(u^y\right)^7 + 3\mathcal{C}D_s(\partial_\tau u^x)\pi_Q{}^{\eta}{}_{\eta}\tau u^{\tau}\left(u^y\right)^7 \\
&+ D_s(\partial_x u^x)I_{31}\tau u^x\left(u^y\right)^7 + D_s(\partial_y u^y)I_{31}\tau u^x\left(u^y\right)^7 \\
&+ \mathcal{B}'(T)D_s(\partial_x T)\nu^x\tau u^x\left(u^y\right)^7 + \mathcal{B}'(T)D_s(\partial_y T)\nu^y\tau u^x\left(u^y\right)^7 \\
&+ \mathcal{C}D_s(\partial_x u^x)\Pi_Q\tau u^x\left(u^y\right)^7 + \mathcal{C}D_s(\partial_y u^y)\Pi_Q\tau u^x\left(u^y\right)^7 \\
&- 8\mathcal{C}D_s(\partial_y u^x)\pi_Q^{xy}\tau u^x\left(u^y\right)^7 + 5\mathcal{C}D_s(\partial_x u^x)\pi_Q{}^{\eta}{}_{\eta}\tau u^x\left(u^y\right)^7 \\
&- 3\mathcal{C}D_s(\partial_y u^y)\pi_Q{}^{\eta}{}_{\eta}\tau u^x\left(u^y\right)^7 - 2D_s I_{31}u^{\tau}u^x\left(u^y\right)^7 \\
&- 2\mathcal{C}D_s\Pi_Q u^{\tau}u^x\left(u^y\right)^7 - 2\mathcal{C}D_s\pi_Q{}^{\eta}{}_{\eta}u^{\tau}u^x\left(u^y\right)^7 \\
&- 2\mathcal{A}D_s(\partial_\tau u^x)\nu^x\tau u^{\tau}u^x\left(u^y\right)^7 - 6\mathcal{B}D_s(\partial_\tau u^x)\nu^x\tau u^{\tau}u^x\left(u^y\right)^7 \\
&- 2\mathcal{A}D_s(\partial_x u^x)\nu^x\tau\left(u^x\right)^2\left(u^y\right)^7 - 6\mathcal{B}D_s(\partial_x u^x)\nu^x\tau\left(u^x\right)^2\left(u^y\right)^7 \\
&+ 2\mathcal{A}D_s(\partial_y u^y)\nu^x\tau\left(u^x\right)^2\left(u^y\right)^7 + 6\mathcal{B}D_s(\partial_y u^y)\nu^x\tau\left(u^x\right)^2\left(u^y\right)^7 \\
&+ 2\mathcal{A}D_s(\partial_y u^x)\nu^y\tau\left(u^x\right)^2\left(u^y\right)^7 + 6\mathcal{B}D_s(\partial_y u^x)\nu^y\tau\left(u^x\right)^2\left(u^y\right)^7 \\
&+ 3D_s(\partial_y u^x)I_{31}\tau\left(u^y\right)^8 + 3\mathcal{B}'(T)D_s(\partial_y T)\nu^x\tau\left(u^y\right)^8 \\
&+ 3\mathcal{C}D_s(\partial_y u^x)\Pi_Q\tau\left(u^y\right)^8 + 3\mathcal{C}D_s(\partial_y u^x)\pi_Q{}^{\eta}{}_{\eta}\tau\left(u^y\right)^8
\end{aligned}
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\begin{aligned}
&- 2\mathcal{A}D_s(\partial_y u^x)\nu^x\tau u^x\left(u^y\right)^8 - 6\mathcal{B}D_s(\partial_y u^x)\nu^x\tau u^x\left(u^y\right)^8\Big)\\
S_7 = &\frac{1}{6T\tau\left(u^\tau\right)^4\left(\left(u^y\right)^2+1\right)}\Big(5D_s(\partial_x u^x)I_{31}\tau + 5D_s(\partial_y u^y)I_{31}\tau + 5\mathcal{B}'(T)D_s(\partial_x T)\nu^x\tau\\
&+ 5\mathcal{B}'(T)D_s(\partial_y T)\nu^y\tau + 5\mathcal{C}D_s(\partial_x u^x)\Pi_Q\tau + 5\mathcal{C}D_s(\partial_y u^y)\Pi_Q\tau\\
&+ 2\mathcal{C}D_s(\partial_x u^y)\pi_Q^{xy}\tau + 2\mathcal{C}D_s(\partial_y u^x)\pi_Q^{xy}\tau - 2\mathcal{C}D_s(\partial_x u^x)\pi_Q^{yy}\tau\\
&+ 2\mathcal{C}D_s(\partial_y u^y)\pi_Q^{yy}\tau - 2\mathcal{C}D_s(\partial_x u^x)\pi_Q{}^\eta{}_\eta\tau + 6\Pi_Q T\tau + 5D_s I_{31}u^\tau\\
&+ 5\mathcal{C}D_s\Pi_Q u^\tau + 2\mathcal{C}D_s\pi_Q{}^\eta{}_\eta u^\tau + 3(\partial_T I_{31})D_s(\partial_\tau T)\tau u^\tau\\
&+ 2\mathcal{A}D_s(\partial_\tau u^x)\nu^x\tau u^\tau + 5\mathcal{B}D_s(\partial_\tau u^x)\nu^x\tau u^\tau + 2\mathcal{A}D_s(\partial_\tau u^y)\nu^y\tau u^\tau\\
&+ 5\mathcal{B}D_s(\partial_\tau u^y)\nu^y\tau u^\tau + 3\mathcal{C}'(T)D_s(\partial_\tau T)\Pi_Q\tau u^\tau\\
&+ 3(\partial_T I_{31})D_s(\partial_x T)\tau u^x + 2\mathcal{A}D_s(\partial_x u^x)\nu^x\tau u^x\\
&+ 2\mathcal{A}D_s(\partial_x u^y)\nu^y\tau u^x + 3\mathcal{C}'(T)D_s(\partial_x T)\Pi_Q\tau u^x + 5\mathcal{B}D_s\nu^x u^\tau u^x\\
&+ 5D_s(\partial_\tau u^x)I_{31}\tau u^\tau u^x + 5\mathcal{B}'(T)D_s(\partial_\tau T)\nu^x\tau u^\tau u^x\\
&+ 5\mathcal{C}D_s(\partial_\tau u^x)\Pi_Q\tau u^\tau u^x + 2\mathcal{C}D_s(\partial_\tau u^y)\pi_Q^{xy}\tau u^\tau u^x\\
&- 2\mathcal{C}D_s(\partial_\tau u^x)\pi_Q^{yy}\tau u^\tau u^x - 2\mathcal{C}D_s(\partial_\tau u^x)\pi_Q{}^\eta{}_\eta\tau u^\tau u^x\\
&+ 10D_s(\partial_x u^x)I_{31}\tau\left(u^x\right)^2 + 10D_s(\partial_y u^y)I_{31}\tau\left(u^x\right)^2\\
&+ 10\mathcal{B}'(T)D_s(\partial_x T)\nu^x\tau\left(u^x\right)^2 + 10\mathcal{B}'(T)D_s(\partial_y T)\nu^y\tau\left(u^x\right)^2\\
&+ 10\mathcal{C}D_s(\partial_x u^x)\Pi_Q\tau\left(u^x\right)^2 + 10\mathcal{C}D_s(\partial_y u^y)\Pi_Q\tau\left(u^x\right)^2\\
&+ 4\mathcal{C}D_s(\partial_x u^y)\pi_Q^{xy}\tau\left(u^x\right)^2 + 2\mathcal{C}D_s(\partial_y u^x)\pi_Q^{xy}\tau\left(u^x\right)^2\\
&- 4\mathcal{C}D_s(\partial_x u^x)\pi_Q^{yy}\tau\left(u^x\right)^2 + 4\mathcal{C}D_s(\partial_y u^y)\pi_Q^{yy}\tau\left(u^x\right)^2\\
&- 4\mathcal{C}D_s(\partial_x u^x)\pi_Q{}^\eta{}_\eta\tau\left(u^x\right)^2 + 12\Pi_Q T\tau\left(u^x\right)^2 + 10D_s I_{31}u^\tau\left(u^x\right)^2\\
&+ 10\mathcal{C}D_s\Pi_Q u^\tau\left(u^x\right)^2 + 4\mathcal{C}D_s\pi_Q{}^\eta{}_\eta u^\tau\left(u^x\right)^2\\
&+ 6(\partial_T I_{31})D_s(\partial_\tau T)\tau u^\tau\left(u^x\right)^2 + 2\mathcal{A}D_s(\partial_\tau u^x)\nu^x\tau u^\tau\left(u^x\right)^2\\
&+ 4\mathcal{A}D_s(\partial_\tau u^y)\nu^y\tau u^\tau\left(u^x\right)^2 + 5\mathcal{B}D_s(\partial_\tau u^y)\nu^y\tau u^\tau\left(u^x\right)^2\\
&+ 6\mathcal{C}'(T)D_s(\partial_\tau T)\Pi_Q\tau u^\tau\left(u^x\right)^2 + 6(\partial_T I_{31})D_s(\partial_x T)\tau\left(u^x\right)^3\\
&+ 2\mathcal{A}D_s(\partial_x u^x)\nu^x\tau\left(u^x\right)^3 + 4\mathcal{A}D_s(\partial_x u^y)\nu^y\tau\left(u^x\right)^3\\
&+ 6\mathcal{C}'(T)D_s(\partial_x T)\Pi_Q\tau\left(u^x\right)^3 + 5\mathcal{B}D_s\nu^x u^\tau\left(u^x\right)^3\\
&+ 5D_s(\partial_\tau u^x)I_{31}\tau u^\tau\left(u^x\right)^3 + 5\mathcal{B}'(T)D_s(\partial_\tau T)\nu^x\tau u^\tau\left(u^x\right)^3\\
&+ 5\mathcal{C}D_s(\partial_\tau u^x)\Pi_Q\tau u^\tau\left(u^x\right)^3 + 2\mathcal{C}D_s(\partial_\tau u^y)\pi_Q^{xy}\tau u^\tau\left(u^x\right)^3\\
&- 2\mathcal{C}D_s(\partial_\tau u^x)\pi_Q^{yy}\tau u^\tau\left(u^x\right)^3 - 2\mathcal{C}D_s(\partial_\tau u^x)\pi_Q{}^\eta{}_\eta\tau u^\tau\left(u^x\right)^3\\
&+ 5D_s(\partial_x u^x)I_{31}\tau\left(u^x\right)^4 + 5D_s(\partial_y u^y)I_{31}\tau\left(u^x\right)^4\\
&+ 5\mathcal{B}'(T)D_s(\partial_x T)\nu^x\tau\left(u^x\right)^4 + 5\mathcal{B}'(T)D_s(\partial_y T)\nu^y\tau\left(u^x\right)^4\\
&+ 5\mathcal{C}D_s(\partial_x u^x)\Pi_Q\tau\left(u^x\right)^4 + 5\mathcal{C}D_s(\partial_y u^y)\Pi_Q\tau\left(u^x\right)^4\\
&+ 2\mathcal{C}D_s(\partial_x u^y)\pi_Q^{xy}\tau\left(u^x\right)^4 - 2\mathcal{C}D_s(\partial_x u^x)\pi_Q^{yy}\tau\left(u^x\right)^4\\
&+ 2\mathcal{C}D_s(\partial_y u^y)\pi_Q^{yy}\tau\left(u^x\right)^4 - 2\mathcal{C}D_s(\partial_x u^x)\pi_Q{}^\eta{}_\eta\tau\left(u^x\right)^4\\
&+ 6\Pi_Q T\tau\left(u^x\right)^4 + 5D_s I_{31}u^\tau\left(u^x\right)^4 + 5\mathcal{C}D_s\Pi_Q u^\tau\left(u^x\right)^4\\
&+ 2\mathcal{C}D_s\pi_Q{}^\eta{}_\eta u^\tau\left(u^x\right)^4 + 3(\partial_T I_{31})D_s(\partial_\tau T)\tau u^\tau\left(u^x\right)^4\\
&+ 2\mathcal{A}D_s(\partial_\tau u^y)\nu^y\tau u^\tau\left(u^x\right)^4 + 3\mathcal{C}'(T)D_s(\partial_\tau T)\Pi_Q\tau u^\tau\left(u^x\right)^4\\
&+ 3(\partial_T I_{31})D_s(\partial_x T)\tau\left(u^x\right)^5 + 2\mathcal{A}D_s(\partial_x u^y)\nu^y\tau\left(u^x\right)^5\\
&+ 3\mathcal{C}'(T)D_s(\partial_x T)\Pi_Q\tau\left(u^x\right)^5 + 3(\partial_T I_{31})D_s(\partial_y T)\tau u^y\\
&+ 2\mathcal{A}D_s(\partial_y u^x)\nu^x\tau u^y + 2\mathcal{A}D_s(\partial_y u^y)\nu^y\tau u^y\\
&+ 3\mathcal{C}'(T)D_s(\partial_y T)\Pi_Q\tau u^y + 5\mathcal{B}D_s\nu^y u^\tau u^y + 5D_s(\partial_\tau u^y)I_{31}\tau u^\tau u^y\\
&+ 5\mathcal{B}'(T)D_s(\partial_\tau T)\nu^y\tau u^\tau u^y + 5\mathcal{C}D_s(\partial_\tau u^y)\Pi_Q\tau u^\tau u^y\\
&+ 2\mathcal{C}D_s(\partial_\tau u^x)\pi_Q^{xy}\tau u^\tau u^y + 2\mathcal{C}D_s(\partial_\tau u^y)\pi_Q^{yy}\tau u^\tau u^y\\
&+ 2\mathcal{C}D_s(\partial_x u^x)\pi_Q^{xy}\tau u^x u^y - 2\mathcal{C}D_s(\partial_y u^y)\pi_Q^{xy}\tau u^x u^y
\end{aligned}
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\begin{aligned}
&+ 2\mathcal{C}D_s(\partial_x u^y)\pi_Q^{yy}\tau u^x u^y - 2\mathcal{C}D_s(\partial_y u^x)\pi_Q^{yy}\tau u^x u^y \\
&+ 2\mathcal{C}D_s(\partial_x u^y)\pi_Q{}^{\eta}{}_{\eta}\tau u^x u^y - 2\mathcal{A}D_s(\partial_\tau u^y)\nu^x\tau u^\tau u^x u^y \\
&- 5\mathcal{B}D_s(\partial_\tau u^y)\nu^x\tau u^\tau u^x u^y - 2\mathcal{A}D_s(\partial_\tau u^x)\nu^y\tau u^\tau u^x u^y \\
&- 5\mathcal{B}D_s(\partial_\tau u^x)\nu^y\tau u^\tau u^x u^y + 6(\partial_T I_{31})D_s(\partial_y T)\tau\left(u^x\right)^2 u^y \\
&- 2\mathcal{A}D_s(\partial_x u^y)\nu^x\tau\left(u^x\right)^2 u^y + 2\mathcal{A}D_s(\partial_y u^x)\nu^x\tau\left(u^x\right)^2 u^y \\
&- 2\mathcal{A}D_s(\partial_x u^x)\nu^y\tau\left(u^x\right)^2 u^y + 4\mathcal{A}D_s(\partial_y u^y)\nu^y\tau\left(u^x\right)^2 u^y \\
&+ 6\mathcal{C}'(T)D_s(\partial_y T)\Pi_Q\tau\left(u^x\right)^2 u^y + 5\mathcal{B}D_s\nu^y u^\tau\left(u^x\right)^2 u^y \\
&+ 5D_s(\partial_\tau u^y)I_{31}\tau u^\tau\left(u^x\right)^2 u^y + 5\mathcal{B}'(T)D_s(\partial_\tau T)\nu^y\tau u^\tau\left(u^x\right)^2 u^y \\
&+ 5\mathcal{C}D_s(\partial_\tau u^y)\Pi_Q\tau u^\tau\left(u^x\right)^2 u^y + 2\mathcal{C}D_s(\partial_\tau u^x)\pi_Q^{xy}\tau u^\tau\left(u^x\right)^2 u^y \\
&+ 4\mathcal{C}D_s(\partial_\tau u^y)\pi_Q^{yy}\tau u^\tau\left(u^x\right)^2 u^y + 2\mathcal{C}D_s(\partial_\tau u^y)\pi_Q{}^{\eta}{}_{\eta}\tau u^\tau\left(u^x\right)^2 u^y \\
&+ 2\mathcal{C}D_s(\partial_x u^x)\pi_Q^{xy}\tau\left(u^x\right)^3 u^y - 2\mathcal{C}D_s(\partial_y u^y)\pi_Q^{xy}\tau\left(u^x\right)^3 u^y \\
&+ 4\mathcal{C}D_s(\partial_x u^y)\pi_Q^{yy}\tau\left(u^x\right)^3 u^y - 2\mathcal{C}D_s(\partial_y u^x)\pi_Q^{yy}\tau\left(u^x\right)^3 u^y \\
&+ 4\mathcal{C}D_s(\partial_x u^y)\pi_Q{}^{\eta}{}_{\eta}\tau\left(u^x\right)^3 u^y - 2\mathcal{A}D_s(\partial_\tau u^y)\nu^x\tau u^\tau\left(u^x\right)^3 u^y \\
&- 2\mathcal{A}D_s(\partial_\tau u^x)\nu^y\tau u^\tau\left(u^x\right)^3 u^y + 3(\partial_T I_{31})D_s(\partial_y T)\tau\left(u^x\right)^4 u^y \\
&- 2\mathcal{A}D_s(\partial_x u^y)\nu^x\tau\left(u^x\right)^4 u^y - 2\mathcal{A}D_s(\partial_x u^x)\nu^y\tau\left(u^x\right)^4 u^y \\
&+ 2\mathcal{A}D_s(\partial_y u^y)\nu^y\tau\left(u^x\right)^4 u^y + 3\mathcal{C}'(T)D_s(\partial_y T)\Pi_Q\tau\left(u^x\right)^4 u^y \\
&+ 2\mathcal{C}D_s(\partial_\tau u^y)\pi_Q^{yy}\tau u^\tau\left(u^x\right)^4 u^y + 2\mathcal{C}D_s(\partial_\tau u^y)\pi_Q{}^{\eta}{}_{\eta}\tau u^\tau\left(u^x\right)^4 u^y \\
&+ 2\mathcal{C}D_s(\partial_x u^y)\pi_Q^{yy}\tau\left(u^x\right)^5 u^y + 2\mathcal{C}D_s(\partial_x u^y)\pi_Q{}^{\eta}{}_{\eta}\tau\left(u^x\right)^5 u^y \\
&+ 15D_s(\partial_x u^x)I_{31}\tau\left(u^y\right)^2 + 15D_s(\partial_y u^y)I_{31}\tau\left(u^y\right)^2 \\
&+ 15\mathcal{B}'(T)D_s(\partial_x T)\nu^x\tau\left(u^y\right)^2 + 15\mathcal{B}'(T)D_s(\partial_y T)\nu^y\tau\left(u^y\right)^2 \\
&+ 15\mathcal{C}D_s(\partial_x u^x)\Pi_Q\tau\left(u^y\right)^2 + 15\mathcal{C}D_s(\partial_y u^y)\Pi_Q\tau\left(u^y\right)^2 \\
&+ 4\mathcal{C}D_s(\partial_x u^y)\pi_Q^{xy}\tau\left(u^y\right)^2 + 6\mathcal{C}D_s(\partial_y u^x)\pi_Q^{xy}\tau\left(u^y\right)^2 \\
&- 4\mathcal{C}D_s(\partial_x u^x)\pi_Q^{yy}\tau\left(u^y\right)^2 + 4\mathcal{C}D_s(\partial_y u^y)\pi_Q^{yy}\tau\left(u^y\right)^2 \\
&- 6\mathcal{C}D_s(\partial_x u^x)\pi_Q{}^{\eta}{}_{\eta}\tau\left(u^y\right)^2 + 18\Pi_Q T\tau\left(u^y\right)^2 + 15D_s I_{31}u^\tau\left(u^y\right)^2 \\
&+ 15\mathcal{C}D_s\Pi_Q u^\tau\left(u^y\right)^2 + 6\mathcal{C}D_s\pi_Q{}^{\eta}{}_{\eta}u^\tau\left(u^y\right)^2 \\
&+ 9(\partial_T I_{31})D_s(\partial_\tau T)\tau u^\tau\left(u^y\right)^2 + 6\mathcal{A}D_s(\partial_\tau u^x)\nu^x\tau u^\tau\left(u^y\right)^2 \\
&+ 10\mathcal{B}D_s(\partial_\tau u^x)\nu^x\tau u^\tau\left(u^y\right)^2 + 4\mathcal{A}D_s(\partial_\tau u^y)\nu^y\tau u^\tau\left(u^y\right)^2 \\
&+ 5\mathcal{B}D_s(\partial_\tau u^y)\nu^y\tau u^\tau\left(u^y\right)^2 + 9\mathcal{C}'(T)D_s(\partial_\tau T)\Pi_Q\tau u^\tau\left(u^y\right)^2 \\
&+ 9(\partial_T I_{31})D_s(\partial_x T)\tau u^x\left(u^y\right)^2 + 6\mathcal{A}D_s(\partial_x u^x)\nu^x\tau u^x\left(u^y\right)^2 \\
&- 2\mathcal{A}D_s(\partial_y u^y)\nu^x\tau u^x\left(u^y\right)^2 + 4\mathcal{A}D_s(\partial_x u^y)\nu^y\tau u^x\left(u^y\right)^2 \\
&- 2\mathcal{A}D_s(\partial_y u^x)\nu^y\tau u^x\left(u^y\right)^2 + 9\mathcal{C}'(T)D_s(\partial_x T)\Pi_Q\tau u^x\left(u^y\right)^2 \\
&+ 10\mathcal{B}D_s\nu^x u^\tau u^x\left(u^y\right)^2 + 10D_s(\partial_\tau u^x)I_{31}\tau u^\tau u^x\left(u^y\right)^2 \\
&+ 10\mathcal{B}'(T)D_s(\partial_\tau T)\nu^x\tau u^\tau u^x\left(u^y\right)^2 + 10\mathcal{C}D_s(\partial_\tau u^x)\Pi_Q\tau u^\tau u^x\left(u^y\right)^2 \\
&- 4\mathcal{C}D_s(\partial_\tau u^x)\pi_Q^{yy}\tau u^\tau u^x\left(u^y\right)^2 - 4\mathcal{C}D_s(\partial_\tau u^x)\pi_Q{}^{\eta}{}_{\eta}\tau u^\tau u^x\left(u^y\right)^2 \\
&+ 20D_s(\partial_x u^x)I_{31}\tau\left(u^x\right)^2\left(u^y\right)^2 + 20D_s(\partial_y u^y)I_{31}\tau\left(u^x\right)^2\left(u^y\right)^2 \\
&+ 20\mathcal{B}'(T)D_s(\partial_x T)\nu^x\tau\left(u^x\right)^2\left(u^y\right)^2 \\
&+ 20\mathcal{B}'(T)D_s(\partial_y T)\nu^y\tau\left(u^x\right)^2\left(u^y\right)^2 \\
&+ 20\mathcal{C}D_s(\partial_x u^x)\Pi_Q\tau\left(u^x\right)^2\left(u^y\right)^2 + 20\mathcal{C}D_s(\partial_y u^y)\Pi_Q\tau\left(u^x\right)^2\left(u^y\right)^2 \\
&+ 2\mathcal{C}D_s(\partial_x u^y)\pi_Q^{xy}\tau\left(u^x\right)^2\left(u^y\right)^2 + 4\mathcal{C}D_s(\partial_y u^x)\pi_Q^{xy}\tau\left(u^x\right)^2\left(u^y\right)^2 \\
&- 6\mathcal{C}D_s(\partial_x u^x)\pi_Q^{yy}\tau\left(u^x\right)^2\left(u^y\right)^2 + 6\mathcal{C}D_s(\partial_y u^y)\pi_Q^{yy}\tau\left(u^x\right)^2\left(u^y\right)^2 \\
&- 8\mathcal{C}D_s(\partial_x u^x)\pi_Q{}^{\eta}{}_{\eta}\tau\left(u^x\right)^2\left(u^y\right)^2 + 24\Pi_Q T\tau\left(u^x\right)^2\left(u^y\right)^2 \\
&+ 20D_s I_{31}u^\tau\left(u^x\right)^2\left(u^y\right)^2 + 20\mathcal{C}D_s\Pi_Q u^\tau\left(u^x\right)^2\left(u^y\right)^2
\end{aligned}
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\begin{aligned}
&+ 8\mathcal{C}D_s{\pi_Q}^{\eta}{}_{\eta}u^{\tau}\left(u^x\right)^2\left(u^y\right)^2 + 12(\partial_T I_{31})D_s(\partial_\tau T)\tau u^{\tau}\left(u^x\right)^2\left(u^y\right)^2\\
&+ 4\mathcal{A}D_s(\partial_\tau u^x)\nu^x\tau u^{\tau}\left(u^x\right)^2\left(u^y\right)^2 + 6\mathcal{A}D_s(\partial_\tau u^y)\nu^y\tau u^{\tau}\left(u^x\right)^2\left(u^y\right)^2\\
&+ 5\mathcal{B}D_s(\partial_\tau u^y)\nu^y\tau u^{\tau}\left(u^x\right)^2\left(u^y\right)^2\\
&+ 12\mathcal{C}'(T)D_s(\partial_\tau T)\Pi_Q\tau u^{\tau}\left(u^x\right)^2\left(u^y\right)^2\\
&+ 12(\partial_T I_{31})D_s(\partial_x T)\tau\left(u^x\right)^3\left(u^y\right)^2 + 4\mathcal{A}D_s(\partial_x u^x)\nu^x\tau\left(u^x\right)^3\left(u^y\right)^2\\
&- 2\mathcal{A}D_s(\partial_y u^y)\nu^x\tau\left(u^x\right)^3\left(u^y\right)^2 + 6\mathcal{A}D_s(\partial_x u^y)\nu^y\tau\left(u^x\right)^3\left(u^y\right)^2\\
&- 2\mathcal{A}D_s(\partial_y u^x)\nu^y\tau\left(u^x\right)^3\left(u^y\right)^2\\
&+ 12\mathcal{C}'(T)D_s(\partial_x T)\Pi_Q\tau\left(u^x\right)^3\left(u^y\right)^2 + 5\mathcal{B}D_s\nu^x u^{\tau}\left(u^x\right)^3\left(u^y\right)^2\\
&+ 5D_s(\partial_\tau u^x)I_{31}\tau u^{\tau}\left(u^x\right)^3\left(u^y\right)^2\\
&+ 5\mathcal{B}'(T)D_s(\partial_\tau T)\nu^x\tau u^{\tau}\left(u^x\right)^3\left(u^y\right)^2\\
&+ 5\mathcal{C}D_s(\partial_\tau u^x)\Pi_Q\tau u^{\tau}\left(u^x\right)^3\left(u^y\right)^2\\
&- 2\mathcal{C}D_s(\partial_\tau u^y)\pi_Q^{xy}\tau u^{\tau}\left(u^x\right)^3\left(u^y\right)^2\\
&- 2\mathcal{C}D_s(\partial_\tau u^x)\pi_Q^{yy}\tau u^{\tau}\left(u^x\right)^3\left(u^y\right)^2\\
&- 2\mathcal{C}D_s(\partial_\tau u^x){\pi_Q}^{\eta}{}_{\eta}\tau u^{\tau}\left(u^x\right)^3\left(u^y\right)^2 + 5D_s(\partial_x u^x)I_{31}\tau\left(u^x\right)^4\left(u^y\right)^2\\
&+ 5D_s(\partial_y u^y)I_{31}\tau\left(u^x\right)^4\left(u^y\right)^2 + 5\mathcal{B}'(T)D_s(\partial_x T)\nu^x\tau\left(u^x\right)^4\left(u^y\right)^2\\
&+ 5\mathcal{B}'(T)D_s(\partial_y T)\nu^y\tau\left(u^x\right)^4\left(u^y\right)^2 + 5\mathcal{C}D_s(\partial_x u^x)\Pi_Q\tau\left(u^x\right)^4\left(u^y\right)^2\\
&+ 5\mathcal{C}D_s(\partial_y u^y)\Pi_Q\tau\left(u^x\right)^4\left(u^y\right)^2 - 2\mathcal{C}D_s(\partial_x u^y)\pi_Q^{xy}\tau\left(u^x\right)^4\left(u^y\right)^2\\
&- 2\mathcal{C}D_s(\partial_x u^x)\pi_Q^{yy}\tau\left(u^x\right)^4\left(u^y\right)^2 + 2\mathcal{C}D_s(\partial_y u^y)\pi_Q^{yy}\tau\left(u^x\right)^4\left(u^y\right)^2\\
&- 2\mathcal{C}D_s(\partial_x u^x){\pi_Q}^{\eta}{}_{\eta}\tau\left(u^x\right)^4\left(u^y\right)^2 + 6\Pi_Q T\tau\left(u^x\right)^4\left(u^y\right)^2\\
&+ 5D_s I_{31}u^{\tau}\left(u^x\right)^4\left(u^y\right)^2 + 5\mathcal{C}D_s\Pi_Q u^{\tau}\left(u^x\right)^4\left(u^y\right)^2\\
&+ 2\mathcal{C}D_s{\pi_Q}^{\eta}{}_{\eta}u^{\tau}\left(u^x\right)^4\left(u^y\right)^2 + 3(\partial_T I_{31})D_s(\partial_\tau T)\tau u^{\tau}\left(u^x\right)^4\left(u^y\right)^2\\
&+ 2\mathcal{A}D_s(\partial_\tau u^y)\nu^y\tau u^{\tau}\left(u^x\right)^4\left(u^y\right)^2\\
&+ 3\mathcal{C}'(T)D_s(\partial_\tau T)\Pi_Q\tau u^{\tau}\left(u^x\right)^4\left(u^y\right)^2\\
&+ 3(\partial_T I_{31})D_s(\partial_x T)\tau\left(u^x\right)^5\left(u^y\right)^2 + 2\mathcal{A}D_s(\partial_x u^y)\nu^y\tau\left(u^x\right)^5\left(u^y\right)^2\\
&+ 3\mathcal{C}'(T)D_s(\partial_x T)\Pi_Q\tau\left(u^x\right)^5\left(u^y\right)^2 + 9(\partial_T I_{31})D_s(\partial_y T)\tau\left(u^y\right)^3\\
&+ 6\mathcal{A}D_s(\partial_y u^x)\nu^x\tau\left(u^y\right)^3 + 4\mathcal{A}D_s(\partial_y u^y)\nu^y\tau\left(u^y\right)^3\\
&+ 9\mathcal{C}'(T)D_s(\partial_y T)\Pi_Q\tau\left(u^y\right)^3 + 10\mathcal{B}D_s\nu^y u^{\tau}\left(u^y\right)^3\\
&+ 10D_s(\partial_\tau u^y)I_{31}\tau u^{\tau}\left(u^y\right)^3 + 10\mathcal{B}'(T)D_s(\partial_\tau T)\nu^y\tau u^{\tau}\left(u^y\right)^3\\
&+ 10\mathcal{C}D_s(\partial_\tau u^y)\Pi_Q\tau u^{\tau}\left(u^y\right)^3 + 4\mathcal{C}D_s(\partial_\tau u^x)\pi_Q^{xy}\tau u^{\tau}\left(u^y\right)^3\\
&+ 2\mathcal{C}D_s(\partial_\tau u^y)\pi_Q^{yy}\tau u^{\tau}\left(u^y\right)^3 + 4\mathcal{C}D_s(\partial_x u^x)\pi_Q^{xy}\tau u^{x}\left(u^y\right)^3\\
&- 4\mathcal{C}D_s(\partial_y u^y)\pi_Q^{xy}\tau u^{x}\left(u^y\right)^3 + 2\mathcal{C}D_s(\partial_x u^y)\pi_Q^{yy}\tau u^{x}\left(u^y\right)^3\\
&- 4\mathcal{C}D_s(\partial_y u^x)\pi_Q^{yy}\tau u^{x}\left(u^y\right)^3 + 4\mathcal{C}D_s(\partial_x u^y){\pi_Q}^{\eta}{}_{\eta}\tau u^{x}\left(u^y\right)^3\\
&- 4\mathcal{A}D_s(\partial_\tau u^y)\nu^x\tau u^{\tau}u^{x}\left(u^y\right)^3 - 5\mathcal{B}D_s(\partial_\tau u^y)\nu^x\tau u^{\tau}u^{x}\left(u^y\right)^3\\
&- 4\mathcal{A}D_s(\partial_\tau u^x)\nu^y\tau u^{\tau}u^{x}\left(u^y\right)^3 - 5\mathcal{B}D_s(\partial_\tau u^x)\nu^y\tau u^{\tau}u^{x}\left(u^y\right)^3\\
&+ 12(\partial_T I_{31})D_s(\partial_y T)\tau\left(u^x\right)^2\left(u^y\right)^3 - 4\mathcal{A}D_s(\partial_x u^y)\nu^x\tau\left(u^x\right)^2\left(u^y\right)^3\\
&+ 4\mathcal{A}D_s(\partial_y u^x)\nu^x\tau\left(u^x\right)^2\left(u^y\right)^3 - 4\mathcal{A}D_s(\partial_x u^x)\nu^y\tau\left(u^x\right)^2\left(u^y\right)^3\\
&+ 6\mathcal{A}D_s(\partial_y u^y)\nu^y\tau\left(u^x\right)^2\left(u^y\right)^3\\
&+ 12\mathcal{C}'(T)D_s(\partial_y T)\Pi_Q\tau\left(u^x\right)^2\left(u^y\right)^3 + 5\mathcal{B}D_s\nu^y u^{\tau}\left(u^x\right)^2\left(u^y\right)^3\\
&+ 5D_s(\partial_\tau u^y)I_{31}\tau u^{\tau}\left(u^x\right)^2\left(u^y\right)^3\\
&+ 5\mathcal{B}'(T)D_s(\partial_\tau T)\nu^y\tau u^{\tau}\left(u^x\right)^2\left(u^y\right)^3\\
&+ 5\mathcal{C}D_s(\partial_\tau u^y)\Pi_Q\tau u^{\tau}\left(u^x\right)^2\left(u^y\right)^3\\
&+ 2\mathcal{C}D_s(\partial_\tau u^x)\pi_Q^{xy}\tau u^{\tau}\left(u^x\right)^2\left(u^y\right)^3
\end{aligned}
$$

$$
\begin{aligned}
&+ 2\mathcal{C}D_s(\partial_\tau u^y)\pi_Q^{yy}\tau u^\tau \left(u^x\right)^2\left(u^y\right)^3 \\
&+ 2\mathcal{C}D_s(\partial_\tau u^y)\pi_Q{}^\eta{}_\eta\tau u^\tau \left(u^x\right)^2\left(u^y\right)^3 + 2\mathcal{C}D_s(\partial_x u^x)\pi_Q^{xy}\tau\left(u^x\right)^3\left(u^y\right)^3 \\
&- 2\mathcal{C}D_s(\partial_y u^y)\pi_Q^{xy}\tau\left(u^x\right)^3\left(u^y\right)^3 + 2\mathcal{C}D_s(\partial_x u^y)\pi_Q^{yy}\tau\left(u^x\right)^3\left(u^y\right)^3 \\
&- 2\mathcal{C}D_s(\partial_y u^x)\pi_Q^{yy}\tau\left(u^x\right)^3\left(u^y\right)^3 + 4\mathcal{C}D_s(\partial_x u^y)\pi_Q{}^\eta{}_\eta\tau\left(u^x\right)^3\left(u^y\right)^3 \\
&- 2\mathcal{A}D_s(\partial_\tau u^y)\nu^x\tau u^\tau\left(u^x\right)^3\left(u^y\right)^3 - 2\mathcal{A}D_s(\partial_\tau u^x)\nu^y\tau u^\tau\left(u^x\right)^3\left(u^y\right)^3 \\
&+ 3(\partial_T I_{31})D_s(\partial_y T)\tau\left(u^x\right)^4\left(u^y\right)^3 - 2\mathcal{A}D_s(\partial_x u^y)\nu^x\tau\left(u^x\right)^4\left(u^y\right)^3 \\
&- 2\mathcal{A}D_s(\partial_x u^x)\nu^y\tau\left(u^x\right)^4\left(u^y\right)^3 + 2\mathcal{A}D_s(\partial_y u^y)\nu^y\tau\left(u^x\right)^4\left(u^y\right)^3 \\
&+ 3\mathcal{C}'(T)D_s(\partial_y T)\Pi_Q\tau\left(u^x\right)^4\left(u^y\right)^3 + 15D_s(\partial_x u^x)I_{31}\tau\left(u^y\right)^4 \\
&+ 15D_s(\partial_y u^y)I_{31}\tau\left(u^y\right)^4 + 15\mathcal{B}'(T)D_s(\partial_x T)\nu^x\tau\left(u^y\right)^4 \\
&+ 15\mathcal{B}'(T)D_s(\partial_y T)\nu^y\tau\left(u^y\right)^4 + 15\mathcal{C}D_s(\partial_x u^x)\Pi_Q\tau\left(u^y\right)^4 \\
&+ 15\mathcal{C}D_s(\partial_y u^y)\Pi_Q\tau\left(u^y\right)^4 + 2\mathcal{C}D_s(\partial_x u^y)\pi_Q^{xy}\tau\left(u^y\right)^4 \\
&+ 6\mathcal{C}D_s(\partial_y u^x)\pi_Q^{xy}\tau\left(u^y\right)^4 - 2\mathcal{C}D_s(\partial_x u^x)\pi_Q^{yy}\tau\left(u^y\right)^4 \\
&+ 2\mathcal{C}D_s(\partial_y u^y)\pi_Q^{yy}\tau\left(u^y\right)^4 - 6\mathcal{C}D_s(\partial_x u^x)\pi_Q{}^\eta{}_\eta\tau\left(u^y\right)^4 \\
&+ 18\Pi_Q T\tau\left(u^y\right)^4 + 15D_s I_{31}u^\tau\left(u^y\right)^4 + 15\mathcal{C}D_s\Pi_Q u^\tau\left(u^y\right)^4 \\
&+ 6\mathcal{C}D_s\pi_Q{}^\eta{}_\eta u^\tau\left(u^y\right)^4 + 9(\partial_T I_{31})D_s(\partial_\tau T)\tau u^\tau\left(u^y\right)^4 \\
&+ 6\mathcal{A}D_s(\partial_\tau u^x)\nu^x\tau u^\tau\left(u^y\right)^4 + 5\mathcal{B}D_s(\partial_\tau u^x)\nu^x\tau u^\tau\left(u^y\right)^4 \\
&+ 2\mathcal{A}D_s(\partial_\tau u^y)\nu^y\tau u^\tau\left(u^y\right)^4 + 9\mathcal{C}'(T)D_s(\partial_\tau T)\Pi_Q\tau u^\tau\left(u^y\right)^4 \\
&+ 9(\partial_T I_{31})D_s(\partial_x T)\tau u^x\left(u^y\right)^4 + 6\mathcal{A}D_s(\partial_x u^x)\nu^x\tau u^x\left(u^y\right)^4 \\
&- 4\mathcal{A}D_s(\partial_y u^y)\nu^x\tau u^x\left(u^y\right)^4 + 2\mathcal{A}D_s(\partial_x u^y)\nu^y\tau u^x\left(u^y\right)^4 \\
&- 4\mathcal{A}D_s(\partial_y u^x)\nu^y\tau u^x\left(u^y\right)^4 + 9\mathcal{C}'(T)D_s(\partial_x T)\Pi_Q\tau u^x\left(u^y\right)^4 \\
&+ 5\mathcal{B}D_s\nu^x u^\tau u^x\left(u^y\right)^4 + 5D_s(\partial_\tau u^x)I_{31}\tau u^\tau u^x\left(u^y\right)^4 \\
&+ 5\mathcal{B}'(T)D_s(\partial_\tau T)\nu^x\tau u^\tau u^x\left(u^y\right)^4 + 5\mathcal{C}D_s(\partial_\tau u^x)\Pi_Q\tau u^\tau u^x\left(u^y\right)^4 \\
&- 2\mathcal{C}D_s(\partial_\tau u^y)\pi_Q^{xy}\tau u^\tau u^x\left(u^y\right)^4 - 2\mathcal{C}D_s(\partial_\tau u^x)\pi_Q^{yy}\tau u^\tau u^x\left(u^y\right)^4 \\
&- 2\mathcal{C}D_s(\partial_\tau u^x)\pi_Q{}^\eta{}_\eta\tau u^\tau u^x\left(u^y\right)^4 + 10D_s(\partial_x u^x)I_{31}\tau\left(u^x\right)^2\left(u^y\right)^4 \\
&+ 10D_s(\partial_y u^y)I_{31}\tau\left(u^x\right)^2\left(u^y\right)^4 \\
&+ 10\mathcal{B}'(T)D_s(\partial_x T)\nu^x\tau\left(u^x\right)^2\left(u^y\right)^4 \\
&+ 10\mathcal{B}'(T)D_s(\partial_y T)\nu^y\tau\left(u^x\right)^2\left(u^y\right)^4 \\
&+ 10\mathcal{C}D_s(\partial_x u^x)\Pi_Q\tau\left(u^x\right)^2\left(u^y\right)^4 + 10\mathcal{C}D_s(\partial_y u^y)\Pi_Q\tau\left(u^x\right)^2\left(u^y\right)^4 \\
&- 2\mathcal{C}D_s(\partial_x u^y)\pi_Q^{xy}\tau\left(u^x\right)^2\left(u^y\right)^4 + 2\mathcal{C}D_s(\partial_y u^x)\pi_Q^{xy}\tau\left(u^x\right)^2\left(u^y\right)^4 \\
&- 2\mathcal{C}D_s(\partial_x u^x)\pi_Q^{yy}\tau\left(u^x\right)^2\left(u^y\right)^4 + 2\mathcal{C}D_s(\partial_y u^y)\pi_Q^{yy}\tau\left(u^x\right)^2\left(u^y\right)^4 \\
&- 4\mathcal{C}D_s(\partial_x u^x)\pi_Q{}^\eta{}_\eta\tau\left(u^x\right)^2\left(u^y\right)^4 + 12\Pi_Q T\tau\left(u^x\right)^2\left(u^y\right)^4 \\
&+ 10D_s I_{31}u^\tau\left(u^x\right)^2\left(u^y\right)^4 + 10\mathcal{C}D_s\Pi_Q u^\tau\left(u^x\right)^2\left(u^y\right)^4 \\
&+ 4\mathcal{C}D_s\pi_Q{}^\eta{}_\eta u^\tau\left(u^x\right)^2\left(u^y\right)^4 + 6(\partial_T I_{31})D_s(\partial_\tau T)\tau u^\tau\left(u^x\right)^2\left(u^y\right)^4 \\
&+ 2\mathcal{A}D_s(\partial_\tau u^x)\nu^x\tau u^\tau\left(u^x\right)^2\left(u^y\right)^4 + 2\mathcal{A}D_s(\partial_\tau u^y)\nu^y\tau u^\tau\left(u^x\right)^2\left(u^y\right)^4 \\
&+ 6\mathcal{C}'(T)D_s(\partial_\tau T)\Pi_Q\tau u^\tau\left(u^x\right)^2\left(u^y\right)^4 \\
&+ 6(\partial_T I_{31})D_s(\partial_x T)\tau\left(u^x\right)^3\left(u^y\right)^4 + 2\mathcal{A}D_s(\partial_x u^x)\nu^x\tau\left(u^x\right)^3\left(u^y\right)^4 \\
&- 2\mathcal{A}D_s(\partial_y u^y)\nu^x\tau\left(u^x\right)^3\left(u^y\right)^4 + 2\mathcal{A}D_s(\partial_x u^y)\nu^y\tau\left(u^x\right)^3\left(u^y\right)^4 \\
&- 2\mathcal{A}D_s(\partial_y u^x)\nu^y\tau\left(u^x\right)^3\left(u^y\right)^4 + 6\mathcal{C}'(T)D_s(\partial_x T)\Pi_Q\tau\left(u^x\right)^3\left(u^y\right)^4 \\
&+ 9(\partial_T I_{31})D_s(\partial_y T)\tau\left(u^y\right)^5 + 6\mathcal{A}D_s(\partial_y u^x)\nu^x\tau\left(u^y\right)^5 \\
&+ 2\mathcal{A}D_s(\partial_y u^y)\nu^y\tau\left(u^y\right)^5 + 9\mathcal{C}'(T)D_s(\partial_y T)\Pi_Q\tau\left(u^y\right)^5 \\
&+ 5\mathcal{B}D_s\nu^y u^\tau\left(u^y\right)^5 + 5D_s(\partial_\tau u^y)I_{31}\tau u^\tau\left(u^y\right)^5 \\
&+ 5\mathcal{B}'(T)D_s(\partial_\tau T)\nu^y\tau u^\tau\left(u^y\right)^5 + 5\mathcal{C}D_s(\partial_\tau u^y)\Pi_Q\tau u^\tau\left(u^y\right)^5
\end{aligned}
$$

$$
\begin{aligned}
&+ 2\mathcal{C}D_s(\partial_\tau u^x)\pi_Q^{xy}\tau u^\tau \left(u^y\right)^5 + 2\mathcal{C}D_s(\partial_x u^x)\pi_Q^{xy}\tau u^x \left(u^y\right)^5 \\
&- 2\mathcal{C}D_s(\partial_y u^y)\pi_Q^{xy}\tau u^x \left(u^y\right)^5 - 2\mathcal{C}D_s(\partial_y u^x)\pi_Q^{yy}\tau u^x \left(u^y\right)^5 \\
&+ 2\mathcal{C}D_s(\partial_x u^y)\pi_Q{}^\eta{}_\eta\tau u^x \left(u^y\right)^5 - 2\mathcal{A}D_s(\partial_\tau u^y)\nu^x\tau u^\tau u^x \left(u^y\right)^5 \\
&- 2\mathcal{A}D_s(\partial_\tau u^x)\nu^y\tau u^\tau u^x \left(u^y\right)^5 + 6(\partial_T I_{31})D_s(\partial_y T)\tau \left(u^x\right)^2 \left(u^y\right)^5 \\
&- 2\mathcal{A}D_s(\partial_x u^y)\nu^x\tau \left(u^x\right)^2 \left(u^y\right)^5 + 2\mathcal{A}D_s(\partial_y u^x)\nu^x\tau \left(u^x\right)^2 \left(u^y\right)^5 \\
&- 2\mathcal{A}D_s(\partial_x u^x)\nu^y\tau \left(u^x\right)^2 \left(u^y\right)^5 + 2\mathcal{A}D_s(\partial_y u^y)\nu^y\tau \left(u^x\right)^2 \left(u^y\right)^5 \\
&+ 6\mathcal{C}'(T)D_s(\partial_y T)\Pi_Q\tau \left(u^x\right)^2 \left(u^y\right)^5 + 5D_s(\partial_x u^x)I_{31}\tau \left(u^y\right)^6 \\
&+ 5D_s(\partial_y u^y)I_{31}\tau \left(u^y\right)^6 + 5\mathcal{B}'(T)D_s(\partial_x T)\nu^x\tau \left(u^y\right)^6 \\
&+ 5\mathcal{B}'(T)D_s(\partial_y T)\nu^y\tau \left(u^y\right)^6 + 5\mathcal{C}D_s(\partial_x u^x)\Pi_Q\tau \left(u^y\right)^6 \\
&+ 5\mathcal{C}D_s(\partial_y u^y)\Pi_Q\tau \left(u^y\right)^6 + 2\mathcal{C}D_s(\partial_y u^x)\pi_Q^{xy}\tau \left(u^y\right)^6 \\
&- 2\mathcal{C}D_s(\partial_x u^x)\pi_Q{}^\eta{}_\eta\tau \left(u^y\right)^6 + 6\Pi_Q T\tau \left(u^y\right)^6 + 5D_s I_{31}u^\tau \left(u^y\right)^6 \\
&+ 5\mathcal{C}D_s\Pi_Q u^\tau \left(u^y\right)^6 + 2\mathcal{C}D_s\pi_Q{}^\eta{}_\eta u^\tau \left(u^y\right)^6 \\
&+ 3(\partial_T I_{31})D_s(\partial_\tau T)\tau u^\tau \left(u^y\right)^6 + 2\mathcal{A}D_s(\partial_\tau u^x)\nu^x\tau u^\tau \left(u^y\right)^6 \\
&+ 3\mathcal{C}'(T)D_s(\partial_\tau T)\Pi_Q\tau u^\tau \left(u^y\right)^6 + 3(\partial_T I_{31})D_s(\partial_x T)\tau u^x \left(u^y\right)^6 \\
&+ 2\mathcal{A}D_s(\partial_x u^x)\nu^x\tau u^x \left(u^y\right)^6 - 2\mathcal{A}D_s(\partial_y u^y)\nu^x\tau u^x \left(u^y\right)^6 \\
&- 2\mathcal{A}D_s(\partial_y u^x)\nu^y\tau u^x \left(u^y\right)^6 + 3\mathcal{C}'(T)D_s(\partial_x T)\Pi_Q\tau u^x \left(u^y\right)^6 \\
&+ 3(\partial_T I_{31})D_s(\partial_y T)\tau \left(u^y\right)^7 + 2\mathcal{A}D_s(\partial_y u^x)\nu^x\tau \left(u^y\right)^7 \\
&+ 3\mathcal{C}'(T)D_s(\partial_y T)\Pi_Q\tau \left(u^y\right)^7\Big)
\end{aligned}
$$

A.2.3 S5, two-way

`two_d_two_way_matrix` in `src/Matrix/2d_two_way.jl`, fields:

$$
\phi = (T, u^x, u^y, \pi^{yy}, \pi^\eta{}_\eta, \pi^{xy}, \Pi, \alpha, \nu^x, \nu^y, \pi_Q^{yy}, \pi_Q{}^\eta{}_\eta, \pi_Q^{xy}, \Pi_Q)
$$

$$
\begin{aligned}
(A_t)_{1,1} &= -n - (\partial_T n)T + P''(T)T + (\partial_T n)h\left(u^\tau\right)^2 + h'(T)n\left(u^\tau\right)^2 + 2h'(T)\nu^x u^x \\
&\quad + P'(T)\left(u^x\right)^2 + P''(T)T\left(u^x\right)^2 + 2h'(T)\nu^y u^y + P'(T)\left(u^y\right)^2 + P''(T)T\left(u^y\right)^2 \\
(A_t)_{1,2} &= \frac{2}{\left(u^y\right)^2+1}\Big(-\pi^\eta{}_\eta u^x - \pi^{yy}u^x + P'(T)Tu^x + \pi^{xy}u^y + P'(T)Tu^x\left(u^y\right)^2 \\
&\quad + \Pi u^x\left(\left(u^y\right)^2+1\right) + h\left(\left(u^y\right)^2+1\right)\left(nu^x+\nu^x\right)\Big) \\
(A_t)_{1,3} &= \frac{2}{\left(\left(u^y\right)^2+1\right)^2}\Big(h\left(\left(u^y\right)^2+1\right)^2\left(nu^y+\nu^y\right) + \pi^{xy}\left(u^x - u^x\left(u^y\right)^2\right) \\
&\quad + u^y\Big(\pi^{yy} + P'(T)T + \pi^\eta{}_\eta\left(u^x\right)^2 + \pi^{yy}\left(u^x\right)^2 + 2P'(T)T\left(u^y\right)^2 \\
&\quad + P'(T)T\left(u^y\right)^4 + \Pi\left(\left(u^y\right)^2+1\right)^2\Big)\Big) \\
(A_t)_{1,4} &= \frac{\left(u^y\right)^2 - \left(u^x\right)^2}{\left(u^y\right)^2+1} \\
(A_t)_{1,5} &= -\frac{\left(u^x\right)^2}{\left(u^y\right)^2+1} \\
(A_t)_{1,6} &= \frac{2u^x u^y}{\left(u^y\right)^2+1} \\
(A_t)_{1,7} &= \left(u^x\right)^2 + \left(u^y\right)^2 \\
(A_t)_{1,8} &= (\partial_\alpha n)\left(h\left(u^\tau\right)^2 - T\right) \\
(A_t)_{1,9} &= 2hu^x \\
(A_t)_{1,10} &= 2hu^y
\end{aligned}
$$

$$
\begin{aligned}
(A_t)_{2,1} &= \frac{1}{u^\tau}\Big((u^\tau)^2 u^x((\partial_T n)h + P''(T)T + P'(T)) \\
&\quad + h'(T)\left(n\,(u^\tau)^2 u^x + \nu^x\left(2\,(u^x)^2 + (u^y)^2 + 1\right) + \nu^y u^x u^y \right)\Big) \\
(A_t)_{2,2} &= \frac{1}{(u^\tau)^3\left((u^y)^2+1\right)}\Big(-\pi^\eta{}_\eta - \pi^{yy} + P'(T)T - 3\pi^\eta{}_\eta\,(u^x)^2 - 3\pi^{yy}\,(u^x)^2 \\
&\quad + 3P'(T)T\,(u^x)^2 - 2\pi^\eta{}_\eta\,(u^x)^4 - 2\pi^{yy}\,(u^x)^4 + 2P'(T)T\,(u^x)^4 \\
&\quad + 3\pi^{xy}u^x u^y + 2\pi^{xy}\,(u^x)^3\,u^y - 2\pi^\eta{}_\eta\,(u^y)^2 - \pi^{yy}\,(u^y)^2 \\
&\quad + 3P'(T)T\,(u^y)^2 - 3\pi^\eta{}_\eta\,(u^x)^2\,(u^y)^2 - 3\pi^{yy}\,(u^x)^2\,(u^y)^2 \\
&\quad + 6P'(T)T\,(u^x)^2\,(u^y)^2 + 2P'(T)T\,(u^x)^4\,(u^y)^2 + 3\pi^{xy}u^x\,(u^y)^3 \\
&\quad - \pi^\eta{}_\eta\,(u^y)^4 + 3P'(T)T\,(u^y)^4 + 3P'(T)T\,(u^x)^2\,(u^y)^4 \\
&\quad + P'(T)T\,(u^y)^6 \\
&\quad + \Pi\left((u^y)^2+1\right)\left(2\,(u^x)^4 + 3\,(u^x)^2\left((u^y)^2+1\right) + \left((u^y)^2+1\right)^2\right) \\
&\quad + h\left((u^y)^2+1\right)\Big(\nu^y u^y\left((u^y)^2+1\right) \\
&\quad + \nu^x u^x\left(2\,(u^x)^2 + 3\,(u^y)^2 + 3\right) \\
&\quad + n\Big(2\,(u^x)^4 + 3\,(u^x)^2\left((u^y)^2+1\right) + \left((u^y)^2+1\right)^2\Big)\Big)\Big) \\
(A_t)_{2,3} &= \frac{1}{(u^\tau)^3\left((u^y)^2+1\right)^2}\Big(h\left((u^y)^2+1\right)^2\left(n\,(u^\tau)^2 u^x u^y + \nu^x u^y\left((u^y)^2+1\right) + \nu^y\left((u^x)^3 + u^x\right)\right) \\
&\quad + \pi^{xy}\Big(-2\,(u^x)^4\left((u^y)^2-1\right) + \left((u^y)^2+1\right)^2 \\
&\quad - 3\,(u^x)^2\left((u^y)^4-1\right)\Big) \\
&\quad + u^x u^y\Big(3\pi^{yy} + P'(T)T + 5\pi^{yy}\,(u^x)^2 + P'(T)T\,(u^x)^2 \\
&\quad + 2\pi^{yy}\,(u^x)^4 + 3\pi^{yy}\,(u^y)^2 + 3P'(T)T\,(u^y)^2 \\
&\quad + 3\pi^{yy}\,(u^x)^2\,(u^y)^2 + 2P'(T)T\,(u^x)^2\,(u^y)^2 \\
&\quad + 3P'(T)T\,(u^y)^4 + P'(T)T\,(u^x)^2\,(u^y)^4 \\
&\quad + P'(T)T\,(u^y)^6 + \Pi\,(u^\tau)^2\left((u^y)^2+1\right)^2 \\
&\quad + \pi^\eta{}_\eta\left(2\,(u^x)^4 + 3\,(u^x)^2\left((u^y)^2+1\right) + \left((u^y)^2+1\right)^2\right)\Big)\Big) \\
(A_t)_{2,4} &= -\frac{u^x\left((u^x)^2+1\right)}{u^\tau\left((u^y)^2+1\right)} \\
(A_t)_{2,5} &= -\frac{u^\tau u^x}{(u^y)^2+1} \\
(A_t)_{2,6} &= \frac{u^y\left(2\,(u^x)^2 + (u^y)^2 + 1\right)}{u^\tau\left((u^y)^2+1\right)} \\
(A_t)_{2,7} &= u^\tau u^x \\
(A_t)_{2,8} &= (\partial_\alpha n)h u^\tau u^x \\
(A_t)_{2,9} &= \frac{h\left(2\,(u^x)^2 + (u^y)^2 + 1\right)}{u^\tau} \\
(A_t)_{2,10} &= \frac{h u^x u^y}{u^\tau} \\
(A_t)_{3,1} &= \frac{1}{u^\tau}\Big((u^\tau)^2 u^y((\partial_T n)h + P''(T)T + P'(T)) \\
&\quad + h'(T)\left(n\,(u^\tau)^2 u^y + \nu^x u^x u^y + \nu^y\left((u^x)^2 + 2\,(u^y)^2 + 1\right)\right)\Big)
\end{aligned}
$$

$$
\begin{aligned}
(A_t)_{3,2} &= \frac{1}{(u^\tau)^3}\Big(u^x u^y\left(P'(T)T\,(u^\tau)^2+\Pi\,(u^\tau)^2-\pi^{yy}\right)+\pi^{xy}\left((u^y)^2+1\right)\\
&\quad + h\left(n\,(u^\tau)^2\,u^x u^y+\nu^x u^y\left((u^y)^2+1\right)+\nu^y\left((u^x)^3+u^x\right)\right)\Big)\\
(A_t)_{3,3} &= \frac{1}{(u^\tau)^3}\Big(\pi^{yy}+P'(T)T+\pi^{yy}\,(u^x)^2+2P'(T)T\,(u^x)^2+P'(T)T\,(u^x)^4-\pi^{xy}u^x u^y\\
&\quad + 3P'(T)T\,(u^y)^2+3P'(T)T\,(u^x)^2\,(u^y)^2+2P'(T)T\,(u^y)^4\\
&\quad + \Pi\left((u^x)^4+(u^x)^2\left(3\,(u^y)^2+2\right)+2\,(u^y)^4+3\,(u^y)^2+1\right)\\
&\quad + h\Big(\nu^x\left((u^x)^3+u^x\right)+\nu^y u^y\left(3\,(u^x)^2+2\,(u^y)^2+3\right)\\
&\quad + n\left((u^x)^4+(u^x)^2\left(3\,(u^y)^2+2\right)+2\,(u^y)^4+3\,(u^y)^2+1\right)\Big)\Big)\\
(A_t)_{3,4} &= \frac{u^y}{u^\tau}\\
(A_t)_{3,6} &= \frac{u^x}{u^\tau}\\
(A_t)_{3,7} &= u^\tau u^y\\
(A_t)_{3,8} &= (\partial_\alpha n)hu^\tau u^y\\
(A_t)_{3,9} &= \frac{hu^x u^y}{u^\tau}\\
(A_t)_{3,10} &= \frac{h\left((u^x)^2+2\,(u^y)^2+1\right)}{u^\tau}\\
(A_t)_{4,2} &= \frac{3\delta_{\pi\pi}\pi^{yy}u^x-6\pi^{xy}\tau_\pi u^y\left((u^y)^2+1\right)-2u^x\left(-3\pi^{yy}\tau_\pi\,(u^y)^2+(u^y)^2\,\eta+\eta\right)}{3u^\tau}\\
(A_t)_{4,3} &= \frac{u^y\left(3\delta_{\pi\pi}\pi^{yy}+6\pi^{xy}\tau_\pi u^x u^y-6\pi^{yy}\tau_\pi\left((u^x)^2+1\right)+6\,(u^x)^2\,\eta+4\,(u^y)^2\,\eta+4\eta\right)}{3u^\tau}\\
(A_t)_{4,4} &= \tau_\pi u^\tau\\
(A_t)_{5,2} &= \frac{u^x(3\delta_{\pi\pi}\pi^\eta{}_\eta-2\eta)}{3\tau^2u^\tau}\\
(A_t)_{5,3} &= \frac{u^y(3\delta_{\pi\pi}\pi^\eta{}_\eta-2\eta)}{3\tau^2u^\tau}\\
(A_t)_{5,5} &= \frac{\tau_\pi u^\tau}{\tau^2}\\
(A_t)_{6,2} &= \frac{1}{3u^\tau}\Big(3\delta_{\pi\pi}\pi^{xy}u^x-3\pi^{xy}\tau_\pi u^x\left(2\,(u^y)^2+1\right)\\
&\quad + u^y\left(3\pi^\eta{}_\eta\tau_\pi\,(u^\tau)^2+3\pi^{yy}\left(2\tau_\pi\,(u^x)^2+\tau_\pi\right)+\eta\left((u^x)^2+3\,(u^y)^2+3\right)\right)\Big)\\
(A_t)_{6,3} &= \frac{1}{3u^\tau\left((u^y)^2+1\right)}\Big(-3\pi^{xy}\tau_\pi u^y-3\pi^\eta{}_\eta\tau_\pi u^x\,(u^y)^2-3\pi^\eta{}_\eta\tau_\pi\,(u^x)^3\,(u^y)^2-3\pi^{xy}\tau_\pi\,(u^y)^3\\
&\quad + 6\pi^{xy}\tau_\pi\,(u^x)^2\,(u^y)^3-3\pi^\eta{}_\eta\tau_\pi u^x\,(u^y)^4+3\delta_{\pi\pi}\pi^{xy}u^y\left((u^y)^2+1\right)\\
&\quad - 3\pi^{yy}\tau_\pi u^x\left((u^x)^2+1\right)\left(2\,(u^y)^2+1\right)+3u^x\eta+3\,(u^x)^3\,\eta\\
&\quad + 4u^x\,(u^y)^2\,\eta+3\,(u^x)^3\,(u^y)^2\,\eta+u^x\,(u^y)^4\,\eta\Big)\\
(A_t)_{6,6} &= \tau_\pi u^\tau\\
(A_t)_{7,2} &= \frac{u^x\zeta}{u^\tau}\\
(A_t)_{7,3} &= \frac{u^y\zeta}{u^\tau}\\
(A_t)_{7,7} &= \tau_\Pi u^\tau\\
(A_t)_{8,1} &= (\partial_T n)u^\tau\\
(A_t)_{8,2} &= \frac{n\,(u^\tau)^2\,u^x+\nu^x\left((u^y)^2+1\right)-\nu^y u^x u^y}{(u^\tau)^3}
\end{aligned}
$$

$$(A_t)_{8,3} = \frac{n\left(u^\tau\right)^2 u^y - \nu^x u^x u^y + \nu^y\left(\left(u^x\right)^2+1\right)}{\left(u^\tau\right)^3}$$
$$(A_t)_{8,8} = (\partial_\alpha n)u^\tau$$
$$(A_t)_{8,9} = \frac{u^x}{u^\tau}$$
$$(A_t)_{8,10} = \frac{u^y}{u^\tau}$$
$$(A_t)_{9,1} = \frac{D_s u^\tau(h'(T)\nu^x + u^x((\partial_T n)T+n))}{T}$$
$$(A_t)_{9,2} = \frac{D_s h\left(n\left(u^\tau\right)^2 + \nu^x\left(u^x - u^x\left(u^y\right)^2\right) + \nu^y\left(\left(u^x\right)^2+1\right)u^y\right)}{Tu^\tau}$$
$$(A_t)_{9,3} = -\frac{D_s h\left(\left(u^x\right)^2+1\right)\left(\nu^y u^x - \nu^x u^y\right)}{Tu^\tau}$$
$$(A_t)_{9,8} = (\partial_\alpha n)D_s u^\tau u^x$$
$$(A_t)_{9,9} = \frac{D_s h u^\tau}{T}$$
$$(A_t)_{10,1} = \frac{D_s u^\tau(h'(T)\nu^y + u^y((\partial_T n)T+n))}{T}$$
$$(A_t)_{10,2} = \frac{D_s h\left(\left(u^y\right)^2+1\right)\left(\nu^y u^x - \nu^x u^y\right)}{Tu^\tau}$$
$$(A_t)_{10,3} = \frac{D_s h\left(n\left(u^\tau\right)^2 + \nu^x u^x\left(\left(u^y\right)^2+1\right) - \nu^y\left(\left(u^x\right)^2-1\right)u^y\right)}{Tu^\tau}$$
$$(A_t)_{10,8} = (\partial_\alpha n)D_s u^\tau u^y$$
$$(A_t)_{10,10} = \frac{D_s h u^\tau}{T}$$
$$\begin{aligned}(A_t)_{11,1} = {} & \frac{1}{6Tu^\tau}D_s\Big(3\mathcal{C}'(T)\pi_Q^{yy}\left(u^\tau\right)^2 - 2\mathcal{B}'(T)\nu^x u^x\left(\left(u^y\right)^2+1\right) \\ & + 2\mathcal{B}'(T)\nu^y u^y\left(3\left(u^x\right)^2 + 2\left(u^y\right)^2 + 2\right)\Big)\end{aligned}$$
$$\begin{aligned}(A_t)_{11,2} = {} & -\frac{1}{6T\left(u^\tau\right)^3}D_s\Big(2\mathcal{A}\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right)\left(\nu^x\left(u^y\right)^2 + \nu^x - \nu^y u^x u^y\right) \\ & + 2\mathcal{B}\left(\left(3\left(u^x\right)^2+4\right)\left(u^y\right)^2 + 3\left(u^y\right)^4 + 1\right)\left(\nu^x\left(u^y\right)^2 + \nu^x - \nu^y u^x u^y\right) \\ & + \left(u^\tau\right)^2\Big(2I_{31}u^x\left(\left(u^y\right)^2+1\right) \\ & + \mathcal{C}\Big(2\Pi_Q u^x\left(\left(u^y\right)^2+1\right) \\ & - 2\left(\left(u^y\right)^2+1\right)\left(\pi_Q{}^\eta{}_\eta u^x - 4\pi_Q^{xy}u^y\right) \\ & - \pi_Q^{yy}u^x\left(8\left(u^y\right)^2+5\right)\Big)\Big)\Big)\end{aligned}$$
$$\begin{aligned}(A_t)_{11,3} = {} & \frac{1}{6T\left(u^\tau\right)^3}D_s\Big(2\mathcal{B}\nu^x u^x u^y\left(\left(3\left(u^x\right)^2+4\right)\left(u^y\right)^2 + 3\left(u^y\right)^4 + 1\right) \\ & - 2\mathcal{A}\left(u^\tau\right)^2\Big(\nu^y\left(u^x\right)^2\left(\left(u^y\right)^2-2\right) - 2\nu^y\left(\left(u^y\right)^2+1\right) \\ & - \nu^x u^x u^y\left(\left(u^y\right)^2+1\right)\Big) \\ & - 2\mathcal{B}\nu^y\Big(3\left(u^x\right)^4\left(\left(u^y\right)^2-1\right) - 2\left(\left(u^y\right)^2+1\right) \\ & + \left(u^x\right)^2\left(3\left(u^y\right)^4 + \left(u^y\right)^2 - 5\right)\Big) \\ & + \left(u^\tau\right)^2\Big(2I_{31}u^y\left(3\left(u^x\right)^2 + 2\left(u^y\right)^2 + 2\right) + 4\mathcal{C}\pi_Q^{xy}\left(2u^x\left(u^y\right)^2 + u^x\right) \\ & + \mathcal{C}u^y\Big(\pi_Q^{yy} - 8\pi_Q^{yy}\left(u^x\right)^2 - 2\pi_Q{}^\eta{}_\eta\left(u^x\right)^2 \\ & + \Pi_Q\left(6\left(u^x\right)^2 + 4\left(u^y\right)^2 + 4\right)\Big)\Big)\Big)\end{aligned}$$

$$
\begin{aligned}
(A_t)_{11,9} &= -\frac{\mathcal{B}D_s u^x \left(\left(u^y\right)^2+1\right)}{3Tu^\tau} \\
(A_t)_{11,10} &= \frac{\mathcal{B}D_s u^y \left(3\left(u^x\right)^2+2\left(u^y\right)^2+2\right)}{3Tu^\tau} \\
(A_t)_{11,11} &= \frac{\mathcal{C}D_s u^\tau}{2T} \\
(A_t)_{12,1} &= \frac{D_s\left(3\mathcal{C}'(T){\pi_Q}^\eta{}_\eta\left(u^\tau\right)^2-2\mathcal{B}'(T)(\nu^x u^x+\nu^y u^y)\right)}{6Tu^\tau} \\
(A_t)_{12,2} &= -\frac{1}{6T\left(u^\tau\right)^3}D_s\Big(2\mathcal{B}\left(\nu^x\left(u^y\right)^2+\nu^x-\nu^y u^x u^y\right)+2\mathcal{A}\left(u^\tau\right)^2\left(\nu^x\left(u^y\right)^2+\nu^x-\nu^y u^x u^y\right) \\
&\quad +\left(u^\tau\right)^2\left(\mathcal{C}(2\Pi_Q u^x+2\pi_Q^{xy}u^y-2\pi_Q^{yy}u^x-5{\pi_Q}^\eta{}_\eta u^x)+2I_{31}u^x\right)\Big) \\
(A_t)_{12,3} &= -\frac{1}{6T\left(u^\tau\right)^3\left(\left(u^y\right)^2+1\right)}D_s\Big(2\mathcal{B}\left(\left(u^y\right)^2+1\right)\left(-\nu^x u^x u^y+\nu^y\left(u^x\right)^2+\nu^y\right) \\
&\quad +2\mathcal{A}\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right)\left(-\nu^x u^x u^y+\nu^y\left(u^x\right)^2+\nu^y\right) \\
&\quad +\left(u^\tau\right)^2\Big(-2\mathcal{C}\pi_Q^{xy}u^x\left(\left(u^y\right)^2-1\right)+2I_{31}u^y\left(\left(u^y\right)^2+1\right) \\
&\quad +\mathcal{C}u^y\Big(2\pi_Q^{yy}\left(\left(u^x\right)^2+1\right) \\
&\quad +{\pi_Q}^\eta{}_\eta\left(2\left(u^x\right)^2-3\left(u^y\right)^2-3\right)+2\Pi_Q\left(\left(u^y\right)^2+1\right)\Big)\Big)\Big) \\
(A_t)_{12,9} &= -\frac{\mathcal{B}D_s u^x}{3Tu^\tau} \\
(A_t)_{12,10} &= -\frac{\mathcal{B}D_s u^y}{3Tu^\tau} \\
(A_t)_{12,12} &= \frac{\mathcal{C}D_s u^\tau}{2T} \\
(A_t)_{13,1} &= \frac{1}{6Tu^\tau}D_s\Big(3\mathcal{C}'(T)\pi_Q^{xy}\left(u^\tau\right)^2+\mathcal{B}'(T)\nu^y u^x\left(3\left(u^x\right)^2+\left(u^y\right)^2+3\right) \\
&\quad +\mathcal{B}'(T)\nu^x u^y\left(\left(u^x\right)^2+3\left(u^y\right)^2+3\right)\Big) \\
(A_t)_{13,2} &= \frac{1}{6T\left(u^\tau\right)^3}D_s\Big(\mathcal{A}\left(u^\tau\right)^2\left(3\nu^y\left(\left(u^y\right)^2+1\right)-2\nu^x u^x u^y\left(\left(u^y\right)^2+1\right)\right. \\
&\quad \left.+\nu^y\left(u^x\right)^2\left(2\left(u^y\right)^2+3\right)\right) \\
&\quad +\left(u^\tau\right)^2\Big(\mathcal{C}\pi_Q^{xy}u^x\left(3-8\left(u^y\right)^2\right)+I_{31}u^y\left(\left(u^x\right)^2+3\left(u^y\right)^2+3\right) \\
&\quad +\mathcal{C}u^y\Big(\pi_Q^{yy}\left(8\left(u^x\right)^2+6\right)+\Pi_Q\left(\left(u^x\right)^2+3\left(u^y\right)^2+3\right) \\
&\quad +{\pi_Q}^\eta{}_\eta\left(5\left(u^x\right)^2+3\left(u^y\right)^2+3\right)\Big)\Big) \\
&\quad +\mathcal{B}\Big(-2\nu^x u^x u^y\left(\left(u^y\right)^2+1\right)\left(3\left(u^x\right)^2+3\left(u^y\right)^2+4\right) \\
&\quad +\nu^y\Big(3\left(\left(u^y\right)^2+1\right)^2+\left(u^x\right)^4\left(6\left(u^y\right)^2+3\right) \\
&\quad +2\left(u^x\right)^2\left(3\left(u^y\right)^4+7\left(u^y\right)^2+3\right)\Big)\Big)\Big) \\
(A_t)_{13,3} &= \frac{1}{6T\left(u^\tau\right)^3\left(\left(u^y\right)^2+1\right)}D_s\Big(\mathcal{A}\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right)\left(-2\nu^y u^x\left(\left(u^x\right)^2+1\right)u^y+3\nu^x\left(\left(u^y\right)^2+1\right)\right. \\
&\quad \left.+\nu^x\left(u^x\right)^2\left(2\left(u^y\right)^2+3\right)\right) \\
&\quad +\left(u^\tau\right)^2\Big(I_{31}u^x\left(\left(u^y\right)^2+1\right)\left(3\left(u^x\right)^2+\left(u^y\right)^2+3\right) \\
&\quad +\mathcal{C}\Big(-3{\pi_Q}^\eta{}_\eta u^x-3{\pi_Q}^\eta{}_\eta\left(u^x\right)^3+3\pi_Q^{xy}u^y+4\pi_Q^{xy}\left(u^x\right)^2u^y \\
&\quad -6{\pi_Q}^\eta{}_\eta u^x\left(u^y\right)^2-5{\pi_Q}^\eta{}_\eta\left(u^x\right)^3\left(u^y\right)^2+3\pi_Q^{xy}\left(u^y\right)^3 \\
&\quad +8\pi_Q^{xy}\left(u^x\right)^2\left(u^y\right)^3-3{\pi_Q}^\eta{}_\eta u^x\left(u^y\right)^4
\end{aligned}
$$

$$
\begin{aligned}
&+ \Pi_Q u^x \left(\left(u^y\right)^2+1\right)\left(3\left(u^x\right)^2+\left(u^y\right)^2+3\right)\\
&- 2\pi_Q^{yy} u^x \left(\left(u^x\right)^2+1\right)\left(4\left(u^y\right)^2+3\right)\Big)\Big)\\
&+ \mathcal{B}\left(\left(u^y\right)^2+1\right)\Big(-2\nu^y u^x\left(\left(u^x\right)^2+1\right)u^y\left(3\left(u^x\right)^2+3\left(u^y\right)^2+4\right)\\
&+ \nu^x\Big(3\left(\left(u^y\right)^2+1\right)^2+\left(u^x\right)^4\left(6\left(u^y\right)^2+3\right)\\
&+ 2\left(u^x\right)^2\left(3\left(u^y\right)^4+7\left(u^y\right)^2+3\right)\Big)\Big)\Big)\\
(A_t)_{13,9} &= \frac{\mathcal{B}D_s u^y\left(\left(u^x\right)^2+3\left(u^y\right)^2+3\right)}{6Tu^\tau}\\
(A_t)_{13,10} &= \frac{\mathcal{B}D_s u^x\left(3\left(u^x\right)^2+\left(u^y\right)^2+3\right)}{6Tu^\tau}\\
(A_t)_{13,13} &= \frac{\mathcal{C}D_s u^\tau}{2T}\\
(A_t)_{14,1} &= \frac{D_s\left(5\mathcal{B}'(T)(\nu^x u^x+\nu^y u^y)+3\mathcal{C}'(T)\Pi_Q\left(u^\tau\right)^2+3(\partial_T I_{31})\left(u^\tau\right)^2\right)}{6Tu^\tau}\\
(A_t)_{14,2} &= \frac{1}{6T\left(u^\tau\right)^3}D_s\Big(5\mathcal{B}\left(\nu^x\left(u^y\right)^2+\nu^x-\nu^y u^x u^y\right)+2\mathcal{A}\left(u^\tau\right)^2\left(\nu^x\left(u^y\right)^2+\nu^x-\nu^y u^x u^y\right)\\
&+ \left(u^\tau\right)^2\left(\mathcal{C}(5\Pi_Q u^x+2\pi_Q^{xy}u^y-2\pi_Q^{yy}u^x-2{\pi_Q}^{\eta}{}_{\eta}u^x)+5I_{31}u^x\right)\Big)\\
(A_t)_{14,3} &= \frac{1}{6T\left(u^\tau\right)^3\left(\left(u^y\right)^2+1\right)}D_s\Big(5\mathcal{B}\left(\left(u^y\right)^2+1\right)\left(-\nu^x u^x u^y+\nu^y\left(u^x\right)^2+\nu^y\right)\\
&+ 2\mathcal{A}\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right)\left(-\nu^x u^x u^y+\nu^y\left(u^x\right)^2+\nu^y\right)\\
&+ \left(u^\tau\right)^2\left(5I_{31}u^y\left(\left(u^y\right)^2+1\right)\right.\\
&+ \mathcal{C}\Big(2u^y\left(\pi_Q^{yy}\left(u^x\right)^2+\pi_Q^{yy}+{\pi_Q}^{\eta}{}_{\eta}\left(u^x\right)^2\right)\\
&- 2\pi_Q^{xy}u^x\left(\left(u^y\right)^2-1\right)+5\Pi_Q u^y\left(\left(u^y\right)^2+1\right)\Big)\Big)\Big)\\
(A_t)_{14,8} &= \frac{(\partial_\alpha I_{31})D_s u^\tau}{2T}\\
(A_t)_{14,9} &= \frac{5\mathcal{B}D_s u^x}{6Tu^\tau}\\
(A_t)_{14,10} &= \frac{5\mathcal{B}D_s u^y}{6Tu^\tau}\\
(A_t)_{14,14} &= \frac{\mathcal{C}D_s u^\tau}{2T}\\
(A_x)_{1,1} &= \frac{1}{u^\tau}\Big(\left(u^\tau\right)^2 u^x((\partial_T n)h+P''(T)T+P'(T))\\
&+ h'(T)\left(n\left(u^\tau\right)^2u^x+\nu^x\left(2\left(u^x\right)^2+\left(u^y\right)^2+1\right)+\nu^y u^x u^y\right)\Big)\\
(A_x)_{1,2} &= \frac{1}{\left(u^\tau\right)^3\left(\left(u^y\right)^2+1\right)}\Big(-\pi^\eta{}_\eta-\pi^{yy}+P'(T)T-3\pi^\eta{}_\eta\left(u^x\right)^2-3\pi^{yy}\left(u^x\right)^2\\
&+ 3P'(T)T\left(u^x\right)^2-2\pi^\eta{}_\eta\left(u^x\right)^4-2\pi^{yy}\left(u^x\right)^4+2P'(T)T\left(u^x\right)^4\\
&+ 3\pi^{xy}u^x u^y+2\pi^{xy}\left(u^x\right)^3u^y-2\pi^\eta{}_\eta\left(u^y\right)^2-\pi^{yy}\left(u^y\right)^2\\
&+ 3P'(T)T\left(u^y\right)^2-3\pi^\eta{}_\eta\left(u^x\right)^2\left(u^y\right)^2-3\pi^{yy}\left(u^x\right)^2\left(u^y\right)^2\\
&+ 6P'(T)T\left(u^x\right)^2\left(u^y\right)^2+2P'(T)T\left(u^x\right)^4\left(u^y\right)^2+3\pi^{xy}u^x\left(u^y\right)^3\\
&- \pi^\eta{}_\eta\left(u^y\right)^4+3P'(T)T\left(u^y\right)^4+3P'(T)T\left(u^x\right)^2\left(u^y\right)^4\\
&+ P'(T)T\left(u^y\right)^6\\
&+ \Pi\left(\left(u^y\right)^2+1\right)\left(2\left(u^x\right)^4+3\left(u^x\right)^2\left(\left(u^y\right)^2+1\right)+\left(\left(u^y\right)^2+1\right)^2\right)\\
&+ h\left(\left(u^y\right)^2+1\right)\left(\nu^y u^y\left(\left(u^y\right)^2+1\right)\right.
\end{aligned}
$$

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\begin{aligned}
&+ \nu^x u^x \left(2\left(u^x\right)^2 + 3\left(u^y\right)^2 + 3\right) \\
&+ n\Big(2\left(u^x\right)^4 + 3\left(u^x\right)^2\left(\left(u^y\right)^2 + 1\right) + \left(\left(u^y\right)^2 + 1\right)^2\Big)\Big)\Big) \\
(A_x)_{1,3} = &\frac{1}{\left(u^\tau\right)^3\left(\left(u^y\right)^2+1\right)^2}\Big(h\left(\left(u^y\right)^2+1\right)^2\left(n\left(u^\tau\right)^2 u^x u^y + \nu^x u^y\left(\left(u^y\right)^2+1\right) + \nu^y\left(\left(u^x\right)^3 + u^x\right)\right) \\
&+ \pi^{xy}\Big(-2\left(u^x\right)^4\left(\left(u^y\right)^2-1\right) + \left(\left(u^y\right)^2+1\right)^2 \\
&- 3\left(u^x\right)^2\left(\left(u^y\right)^4-1\right)\Big) \\
&+ u^x u^y\Big(3\pi^{yy} + P'(T)T + 5\pi^{yy}\left(u^x\right)^2 + P'(T)T\left(u^x\right)^2 \\
&+ 2\pi^{yy}\left(u^x\right)^4 + 3\pi^{yy}\left(u^y\right)^2 + 3P'(T)T\left(u^y\right)^2 \\
&+ 3\pi^{yy}\left(u^x\right)^2\left(u^y\right)^2 + 2P'(T)T\left(u^x\right)^2\left(u^y\right)^2 \\
&+ 3P'(T)T\left(u^y\right)^4 + P'(T)T\left(u^x\right)^2\left(u^y\right)^4 \\
&+ P'(T)T\left(u^y\right)^6 + \Pi\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right)^2 \\
&+ \pi^\eta{}_\eta\left(2\left(u^x\right)^4 + 3\left(u^x\right)^2\left(\left(u^y\right)^2+1\right) + \left(\left(u^y\right)^2+1\right)^2\right)\Big)\Big) \\
(A_x)_{1,4} = &-\frac{u^x\left(\left(u^x\right)^2+1\right)}{u^\tau\left(\left(u^y\right)^2+1\right)} \\
(A_x)_{1,5} = &-\frac{u^\tau u^x}{\left(u^y\right)^2+1} \\
(A_x)_{1,6} = &\frac{u^y\left(2\left(u^x\right)^2 + \left(u^y\right)^2 + 1\right)}{u^\tau\left(\left(u^y\right)^2+1\right)} \\
(A_x)_{1,7} = &\, u^\tau u^x \\
(A_x)_{1,8} = &\, (\partial_\alpha n) h u^\tau u^x \\
(A_x)_{1,9} = &\frac{h\left(2\left(u^x\right)^2 + \left(u^y\right)^2 + 1\right)}{u^\tau} \\
(A_x)_{1,10} = &\frac{h u^x u^y}{u^\tau} \\
(A_x)_{2,1} = &\, P'(T) + n + (\partial_T n)T + 2h'(T)\nu^x u^x + P'(T)\left(u^x\right)^2 + (\partial_T n)h\left(u^x\right)^2 + h'(T)n\left(u^x\right)^2 \\
&+ P''(T)T\left(u^x\right)^2 \\
(A_x)_{2,2} = &\, 2\left(P'(T)Tu^x + hnu^x + h\nu^x + \Pi u^x - \frac{\pi^\eta{}_\eta u^x - \pi^{xy}u^y + \pi^{yy}u^x}{\left(u^y\right)^2+1}\right) \\
(A_x)_{2,3} = &\frac{2\left(u^y\left(\pi^\eta{}_\eta\left(u^x\right)^2 + \pi^{yy}\left(u^x\right)^2 + \pi^{yy}\right) + \pi^{xy}\left(u^x - u^x\left(u^y\right)^2\right)\right)}{\left(\left(u^y\right)^2+1\right)^2} \\
(A_x)_{2,4} = &-\frac{\left(u^x\right)^2+1}{\left(u^y\right)^2+1} \\
(A_x)_{2,5} = &-\frac{\left(u^\tau\right)^2}{\left(u^y\right)^2+1} \\
(A_x)_{2,6} = &\frac{2u^x u^y}{\left(u^y\right)^2+1} \\
(A_x)_{2,7} = &\left(u^x\right)^2 + 1 \\
(A_x)_{2,8} = &\, (\partial_\alpha n)\left(h\left(u^x\right)^2 + T\right) \\
(A_x)_{2,9} = &\, 2hu^x \\
(A_x)_{3,1} = &\, h'(T)u^y(nu^x + \nu^x) + h'(T)\nu^y u^x + u^x u^y((\partial_T n)h + P''(T)T + P'(T))
\end{aligned}
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\begin{aligned}
(A_x)_{3,2} &= u^y(P'(T)T + \Pi) + h(nu^y + \nu^y) \\
(A_x)_{3,3} &= u^x(P'(T)T + \Pi) + h(nu^x + \nu^x) \\
(A_x)_{3,6} &= 1 \\
(A_x)_{3,7} &= u^x u^y \\
(A_x)_{3,8} &= (\partial_\alpha n) h u^x u^y \\
(A_x)_{3,9} &= h u^y \\
(A_x)_{3,10} &= h u^x \\
(A_x)_{4,2} &= \delta_{\pi\pi}\pi^{yy} - \frac{2\pi^{xy}\tau_\pi u^x \left(\left(u^y\right)^2 + 1\right) u^y}{\left(u^\tau\right)^2} + \frac{2\pi^{yy}\tau_\pi \left(u^x\right)^2 \left(u^y\right)^2}{\left(u^\tau\right)^2} - \frac{2}{3}\left(\left(u^y\right)^2 + 1\right)\eta \\
(A_x)_{4,3} &= \frac{2u^x u^y \left(\pi^{xy}\tau_\pi u^x u^y - \pi^{yy}\tau_\pi \left(\left(u^x\right)^2 + 1\right) + \left(u^\tau\right)^2 \eta\right)}{\left(u^\tau\right)^2} \\
(A_x)_{4,4} &= \tau_\pi u^x \\
(A_x)_{5,2} &= \frac{3\delta_{\pi\pi}\pi^\eta{}_\eta - 2\eta}{3\tau^2} \\
(A_x)_{5,5} &= \frac{\tau_\pi u^x}{\tau^2} \\
(A_x)_{6,2} &= \delta_{\pi\pi}\pi^{xy} \\
&\quad + \frac{u^x \left(u^y \left(3\pi^\eta{}_\eta \tau_\pi \left(u^\tau\right)^2 + 3\pi^{yy}\left(2\tau_\pi \left(u^x\right)^2 + \tau_\pi\right) + \left(u^\tau\right)^2 \eta\right) - 3\pi^{xy}\tau_\pi u^x \left(2\left(u^y\right)^2 + 1\right)\right)}{3\left(u^\tau\right)^2} \\
(A_x)_{6,3} &= \frac{1}{\left(u^\tau\right)^2\left(\left(u^y\right)^2 + 1\right)}\Big(-\pi^{yy}\tau_\pi \left(u^x\right)^2 \left(\left(u^x\right)^2 + 1\right)\left(2\left(u^y\right)^2 + 1\right) \\
&\quad + \pi^{xy}\tau_\pi u^x u^y \left(\left(2\left(u^x\right)^2 - 1\right)\left(u^y\right)^2 - 1\right) \\
&\quad - \left(u^\tau\right)^2 \left(\pi^\eta{}_\eta \tau_\pi \left(u^x\right)^2 \left(u^y\right)^2 - \left(\left(u^x\right)^2 + 1\right)\left(\left(u^y\right)^2 + 1\right)\eta\right)\Big) \\
(A_x)_{6,6} &= \tau_\pi u^x \\
(A_x)_{7,2} &= \zeta \\
(A_x)_{7,7} &= \tau_\Pi u^x \\
(A_x)_{8,1} &= (\partial_T n) u^x \\
(A_x)_{8,2} &= n \\
(A_x)_{8,8} &= (\partial_\alpha n) u^x \\
(A_x)_{8,9} &= 1 \\
(A_x)_{9,1} &= \frac{D_s\left(h'(T)\nu^x u^x + (\partial_T n)T\left(\left(u^x\right)^2 + 1\right) + n\left(\left(u^x\right)^2 + 1\right)\right)}{T} \\
(A_x)_{9,2} &= \frac{D_s h\left(n\left(u^\tau\right)^2 u^x + \nu^x\left(\left(u^x\right)^2\left(-\left(u^y\right)^2\right) + \left(u^x\right)^2 + 2\left(u^y\right)^2 + 2\right) + \nu^y\left(u^x\right)^3 u^y\right)}{T\left(u^\tau\right)^2} \\
(A_x)_{9,3} &= -\frac{D_s h\left(u^x\right)^2\left(-\nu^x u^x u^y + \nu^y\left(u^x\right)^2 + \nu^y\right)}{T\left(u^\tau\right)^2} \\
(A_x)_{9,8} &= (\partial_\alpha n) D_s\left(\left(u^x\right)^2 + 1\right) \\
(A_x)_{9,9} &= \frac{D_s h u^x}{T} \\
(A_x)_{10,1} &= \frac{D_s u^x(h'(T)\nu^y + u^y((\partial_T n)T + n))}{T} \\
(A_x)_{10,2} &= \frac{D_s h\left(\left(u^y\right)^2 + 1\right)\left(-\nu^x u^x u^y + \nu^y\left(u^x\right)^2 + \nu^y\right)}{T\left(u^\tau\right)^2} \\
(A_x)_{10,3} &= \frac{D_s h\left(n\left(u^\tau\right)^2 u^x + \nu^x\left(\left(u^x\right)^2 + 1\right)\left(\left(u^y\right)^2 + 1\right) - \nu^y u^x\left(\left(u^x\right)^2 + 1\right)u^y\right)}{T\left(u^\tau\right)^2}
\end{aligned}
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\begin{aligned}
(A_x)_{10,8} &= (\partial_\alpha n) D_s u^x u^y \\
(A_x)_{10,10} &= \frac{D_s h u^x}{T} \\
(A_x)_{11,1} &= \frac{D_s \left(3\mathcal{C}'(T)\pi_Q^{yy} u^x - 2\mathcal{B}'(T)\left(\nu^x \left(u^y\right)^2 + \nu^x - 3\nu^y u^x u^y\right)\right)}{6T} \\
(A_x)_{11,2} &= -\frac{1}{6T\left(u^\tau\right)^2} D_s \Big(2I_{31}\left(u^\tau\right)^2 \left(\left(u^y\right)^2 + 1\right) \\
&\quad + 2u^x \left(\mathcal{A}\left(u^y\right)^2 + \mathcal{A} + 3\mathcal{B}\left(u^y\right)^2\right)\left(\nu^x \left(u^y\right)^2 + \nu^x - \nu^y u^x u^y\right) \\
&\quad + \mathcal{C}\Big(2\Pi_Q \left(u^\tau\right)^2 \left(\left(u^y\right)^2 + 1\right) \\
&\quad - 2\left(\left(u^y\right)^2 + 1\right)\left(\pi_Q{}^\eta{}_\eta \left(u^\tau\right)^2 - 4\pi_Q^{xy} u^x u^y\right) \\
&\quad - \pi_Q^{yy}\left(\left(u^x\right)^2 \left(8\left(u^y\right)^2 + 5\right) + 5\left(\left(u^y\right)^2 + 1\right)\right)\Big)\Big) \\
(A_x)_{11,3} &= \frac{1}{3T\left(u^\tau\right)^2} D_s \Big(u^x \Big(3I_{31}\left(u^\tau\right)^2 u^y - \mathcal{A}\nu^y \left(u^x\right)^2 \left(\left(u^y\right)^2 - 2\right) + 2\mathcal{A}\nu^y \left(\left(u^y\right)^2 + 1\right) \\
&\quad + \mathcal{A}\nu^x u^x u^y \left(\left(u^y\right)^2 + 1\right) \\
&\quad + 3\mathcal{B}\left(\nu^x u^x \left(u^y\right)^3 - \nu^y \left(u^x\right)^2 \left(\left(u^y\right)^2 - 1\right) + \nu^y\right)\Big) \\
&\quad + \mathcal{C}\Big(-u^x u^y \left(-3\Pi_Q \left(u^\tau\right)^2 + 4\pi_Q^{yy}\left(\left(u^x\right)^2 + 1\right) + \pi_Q{}^\eta{}_\eta \left(u^\tau\right)^2\right) \\
&\quad + 2\pi_Q^{xy}\left(\left(u^x\right)^2 \left(2\left(u^y\right)^2 + 1\right) + \left(u^y\right)^2 + 1\right)\Big)\Big) \\
(A_x)_{11,9} &= -\frac{\mathcal{B} D_s \left(\left(u^y\right)^2 + 1\right)}{3T} \\
(A_x)_{11,10} &= \frac{\mathcal{B} D_s u^x u^y}{T} \\
(A_x)_{11,11} &= \frac{\mathcal{C} D_s u^x}{2T} \\
(A_x)_{12,1} &= -\frac{D_s(2\mathcal{B}'(T)\nu^x - 3\mathcal{C}'(T)\pi_Q{}^\eta{}_\eta u^x)}{6T} \\
(A_x)_{12,2} &= -\frac{1}{6T\left(u^\tau\right)^2} D_s \Big(2I_{31}\left(u^\tau\right)^2 + 2\mathcal{A}u^x \left(\nu^x \left(u^y\right)^2 + \nu^x - \nu^y u^x u^y\right) \\
&\quad + \mathcal{C}\Big(-5\pi_Q{}^\eta{}_\eta + 2\Pi_Q \left(u^\tau\right)^2 - 5\pi_Q{}^\eta{}_\eta \left(u^x\right)^2 - 2\pi_Q^{yy}\left(\left(u^x\right)^2 + 1\right) \\
&\quad + 2\pi_Q^{xy} u^x u^y - 5\pi_Q{}^\eta{}_\eta \left(u^y\right)^2\Big)\Big) \\
(A_x)_{12,3} &= \frac{1}{3T\left(u^\tau\right)^2\left(\left(u^y\right)^2 + 1\right)} D_s \Big(-\mathcal{C}u^x u^y \left(\pi_Q^{yy}\left(\left(u^x\right)^2 + 1\right) + \pi_Q{}^\eta{}_\eta \left(u^\tau\right)^2\right) \\
&\quad - \mathcal{A}u^x \left(\left(u^y\right)^2 + 1\right)\left(-\nu^x u^x u^y + \nu^y \left(u^x\right)^2 + \nu^y\right) \\
&\quad + \mathcal{C}\pi_Q^{xy}\left(\left(u^x\right)^2 \left(\left(u^y\right)^2 - 1\right) - \left(u^y\right)^2 - 1\right)\Big) \\
(A_x)_{12,9} &= -\frac{\mathcal{B} D_s}{3T} \\
(A_x)_{12,12} &= \frac{\mathcal{C} D_s u^x}{2T} \\
(A_x)_{13,1} &= \frac{D_s \left(\mathcal{B}'(T)\nu^x u^x u^y + 3\mathcal{B}'(T)\nu^y \left(\left(u^x\right)^2 + 1\right) + 3\mathcal{C}'(T)\pi_Q^{xy} u^x\right)}{6T} \\
(A_x)_{13,2} &= \frac{1}{6T\left(u^\tau\right)^2} D_s \Big(3\mathcal{B}\nu^y u^x + 3\mathcal{B}\nu^y \left(u^x\right)^3 - 3\mathcal{B}\nu^x u^y + I_{31} u^x u^y - 6\mathcal{B}\nu^x \left(u^x\right)^2 u^y + I_{31}\left(u^x\right)^3 u^y \\
&\quad + 6\mathcal{B}\nu^y u^x \left(u^y\right)^2 + 6\mathcal{B}\nu^y \left(u^x\right)^3 \left(u^y\right)^2 - 3\mathcal{B}\nu^x \left(u^y\right)^3 + I_{31} u^x \left(u^y\right)^3 \\
&\quad - 6\mathcal{B}\nu^x \left(u^x\right)^2 \left(u^y\right)^3 \\
&\quad + \mathcal{A}u^x \Big(3\nu^y \left(\left(u^y\right)^2 + 1\right) - 2\nu^x u^x u^y \left(\left(u^y\right)^2 + 1\right)
\end{aligned}
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\begin{aligned}
&+ \nu^y \left(u^x\right)^2 \left(2\left(u^y\right)^2+3\right)\Big)\\
&+ \mathcal{C}\Big(u^x u^y \left(\Pi_Q \left(u^\tau\right)^2 + \pi_Q^{yy}\left(8\left(u^x\right)^2+5\right) + 5{\pi_Q}^\eta{}_\eta \left(u^\tau\right)^2\right)\\
&+ \pi_Q^{xy}\left(\left(u^x\right)^2\left(3-8\left(u^y\right)^2\right)+6\left(\left(u^y\right)^2+1\right)\right)\Big)\Big)\\
(A_x)_{13,3} = &\frac{1}{6T\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right)} D_s\Big(3I_{31}\left(u^\tau\right)^2\left(\left(u^x\right)^2+1\right)\left(\left(u^y\right)^2+1\right)\\
&+ \mathcal{C}\Big(-3{\pi_Q}^\eta{}_\eta - 6{\pi_Q}^\eta{}_\eta \left(u^x\right)^2 - 3{\pi_Q}^\eta{}_\eta \left(u^x\right)^4 + \pi_Q^{xy} u^x u^y\\
&+ 4\pi_Q^{xy}\left(u^x\right)^3 u^y - 6{\pi_Q}^\eta{}_\eta \left(u^y\right)^2 - 11{\pi_Q}^\eta{}_\eta \left(u^x\right)^2\left(u^y\right)^2\\
&- 5{\pi_Q}^\eta{}_\eta \left(u^x\right)^4\left(u^y\right)^2 + \pi_Q^{xy} u^x \left(u^y\right)^3 + 8\pi_Q^{xy}\left(u^x\right)^3\left(u^y\right)^3\\
&- 3{\pi_Q}^\eta{}_\eta \left(u^y\right)^4 - 5{\pi_Q}^\eta{}_\eta \left(u^x\right)^2\left(u^y\right)^4\\
&+ 3\Pi_Q\left(u^\tau\right)^2\left(\left(u^x\right)^2+1\right)\left(\left(u^y\right)^2+1\right)\\
&- \pi_Q^{yy}\left(\left(u^x\right)^2+1\right)\left(\left(u^x\right)^2\left(8\left(u^y\right)^2+6\right)+3\left(\left(u^y\right)^2+1\right)\right)\Big)\\
&+ \left(u^y\right)^2+1\Big(\mathcal{A}u^x\Big(-2\nu^y u^x\left(\left(u^x\right)^2+1\right)u^y + 3\nu^x\left(\left(u^y\right)^2+1\right)\\
&+ \nu^x\left(u^x\right)^2\left(2\left(u^y\right)^2+3\right)\Big)\\
&+ 3\mathcal{B}\left(\left(u^x\right)^2+1\right)\left(\nu^x\left(2u^x\left(u^y\right)^2+u^x\right) - \nu^y\left(2\left(u^x\right)^2+1\right)u^y\right)\Big)\Big)\\
(A_x)_{13,9} = &\frac{\mathcal{B}D_s u^x u^y}{6T}\\
(A_x)_{13,10} = &\frac{\mathcal{B}D_s\left(\left(u^x\right)^2+1\right)}{2T}\\
(A_x)_{13,13} = &\frac{\mathcal{C}D_s u^x}{2T}\\
(A_x)_{14,1} = &\frac{D_s(5\mathcal{B}'(T)\nu^x + 3u^x(\mathcal{C}'(T)\Pi_Q + (\partial_T I_{31})))}{6T}\\
(A_x)_{14,2} = &\frac{1}{6T\left(u^\tau\right)^2} D_s\Big(5I_{31}\left(u^\tau\right)^2 + 5\mathcal{C}\Pi_Q\left(u^\tau\right)^2 + 2\mathcal{A}u^x\left(\nu^x\left(u^y\right)^2 + \nu^x - \nu^y u^x u^y\right)\\
&- 2\mathcal{C}\left(-\pi_Q^{xy}u^x u^y + \pi_Q^{yy}\left(u^x\right)^2 + \pi_Q^{yy} + {\pi_Q}^\eta{}_\eta\left(u^x\right)^2 + {\pi_Q}^\eta{}_\eta\left(u^y\right)^2 + {\pi_Q}^\eta{}_\eta\right)\Big)\\
(A_x)_{14,3} = &\frac{1}{3T\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right)} D_s\Big(\mathcal{C}u^x u^y\left(\pi_Q^{yy}\left(\left(u^x\right)^2+1\right) + {\pi_Q}^\eta{}_\eta\left(u^\tau\right)^2\right)\\
&+ \mathcal{A}u^x\left(\left(u^y\right)^2+1\right)\left(-\nu^x u^x u^y + \nu^y\left(u^x\right)^2 + \nu^y\right)\\
&+ \mathcal{C}\pi_Q^{xy}\left(-\left(\left(u^x\right)^2\left(\left(u^y\right)^2-1\right)\right) + \left(u^y\right)^2+1\right)\Big)\\
(A_x)_{14,8} = &\frac{(\partial_\alpha I_{31})D_s u^x}{2T}\\
(A_x)_{14,9} = &\frac{5\mathcal{B}D_s}{6T}\\
(A_x)_{14,14} = &\frac{\mathcal{C}D_s u^x}{2T}\\
(A_y)_{1,1} = &\frac{1}{u^\tau}\Big(\left(u^\tau\right)^2 u^y((\partial_T n)h + P''(T)T + P'(T))\\
&+ h'(T)\left(n\left(u^\tau\right)^2 u^y + \nu^x u^x u^y + \nu^y\left(\left(u^x\right)^2 + 2\left(u^y\right)^2 + 1\right)\right)\Big)\\
(A_y)_{1,2} = &\frac{1}{\left(u^\tau\right)^3}\Big(u^x u^y\left(P'(T)T\left(u^\tau\right)^2 + \Pi\left(u^\tau\right)^2 - \pi^{yy}\right) + \pi^{xy}\left(\left(u^y\right)^2+1\right)\\
&+ h\left(n\left(u^\tau\right)^2 u^x u^y + \nu^x u^y\left(\left(u^y\right)^2+1\right) + \nu^y\left(\left(u^x\right)^3 + u^x\right)\right)\Big)\\
(A_y)_{1,3} = &\frac{1}{\left(u^\tau\right)^3}\Big(\pi^{yy} + P'(T)T + \pi^{yy}\left(u^x\right)^2 + 2P'(T)T\left(u^x\right)^2 + P'(T)T\left(u^x\right)^4 - \pi^{xy}u^x u^y\\
&+ 3P'(T)T\left(u^y\right)^2 + 3P'(T)T\left(u^x\right)^2\left(u^y\right)^2 + 2P'(T)T\left(u^y\right)^4
\end{aligned}
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\begin{aligned}
&+ \Pi \left((u^x)^4 + (u^x)^2 \left(3 (u^y)^2 + 2 \right) + 2 (u^y)^4 + 3 (u^y)^2 + 1 \right) \\
&+ h \Big(\nu^x \left((u^x)^3 + u^x \right) + \nu^y u^y \left(3 (u^x)^2 + 2 (u^y)^2 + 3 \right) \\
&+ n \left((u^x)^4 + (u^x)^2 \left(3 (u^y)^2 + 2 \right) + 2 (u^y)^4 + 3 (u^y)^2 + 1 \right) \Big) \Big) \\
(A_y)_{1,4} &= \frac{u^y}{u^\tau} \\
(A_y)_{1,6} &= \frac{u^x}{u^\tau} \\
(A_y)_{1,7} &= u^\tau u^y \\
(A_y)_{1,8} &= (\partial_\alpha n) h u^\tau u^y \\
(A_y)_{1,9} &= \frac{h u^x u^y}{u^\tau} \\
(A_y)_{1,10} &= \frac{h \left((u^x)^2 + 2 (u^y)^2 + 1 \right)}{u^\tau} \\
(A_y)_{2,1} &= h'(T) u^y (n u^x + \nu^x) + h'(T) \nu^y u^x + u^x u^y ((\partial_T n) h + P''(T) T + P'(T)) \\
(A_y)_{2,2} &= u^y (P'(T) T + \Pi) + h (n u^y + \nu^y) \\
(A_y)_{2,3} &= u^x (P'(T) T + \Pi) + h (n u^x + \nu^x) \\
(A_y)_{2,6} &= 1 \\
(A_y)_{2,7} &= u^x u^y \\
(A_y)_{2,8} &= (\partial_\alpha n) h u^x u^y \\
(A_y)_{2,9} &= h u^y \\
(A_y)_{2,10} &= h u^x \\
(A_y)_{3,1} &= P'(T) + n + (\partial_T n) T + 2 h'(T) \nu^y u^y + P'(T) (u^y)^2 + (\partial_T n) h (u^y)^2 + h'(T) n (u^y)^2 \\
&+ P''(T) T (u^y)^2 \\
(A_y)_{3,3} &= 2 (u^y (P'(T) T + \Pi) + h (n u^y + \nu^y)) \\
(A_y)_{3,4} &= 1 \\
(A_y)_{3,7} &= (u^y)^2 + 1 \\
(A_y)_{3,8} &= (\partial_\alpha n) \left(h (u^y)^2 + T \right) \\
(A_y)_{3,10} &= 2 h u^y \\
(A_y)_{4,2} &= - \frac{2 \tau_\pi (u^y)^2 \left(\pi^{xy} (u^y)^2 + \pi^{xy} - \pi^{yy} u^x u^y \right)}{(u^\tau)^2} \\
(A_y)_{4,3} &= \frac{1}{3 (u^\tau)^2} \Big(3 \delta_{\pi\pi} \pi^{yy} (u^\tau)^2 - 6 \pi^{yy} \tau_\pi \left((u^x)^2 + 1 \right) (u^y)^2 + 6 \pi^{xy} \tau_\pi u^x (u^y)^3 + 4 \eta \\
&+ 4 (u^x)^2 \eta + 8 (u^y)^2 \eta + 4 (u^x)^2 (u^y)^2 \eta + 4 (u^y)^4 \eta \Big) \\
(A_y)_{4,4} &= \tau_\pi u^y \\
(A_y)_{5,3} &= \frac{3 \delta_{\pi\pi} \pi^\eta{}_\eta - 2 \eta}{3 \tau^2} \\
(A_y)_{5,5} &= \frac{\tau_\pi u^y}{\tau^2} \\
(A_y)_{6,2} &= \frac{1}{(u^\tau)^2} \Big(\pi^{yy} \tau_\pi (u^y)^2 + \pi^\eta{}_\eta \tau_\pi (u^\tau)^2 (u^y)^2 + 2 \pi^{yy} \tau_\pi (u^x)^2 (u^y)^2 \\
&- \pi^{xy} \tau_\pi u^x u^y \left(2 (u^y)^2 + 1 \right) + \eta + (u^x)^2 \eta + 2 (u^y)^2 \eta + (u^x)^2 (u^y)^2 \eta + (u^y)^4 \eta \Big) \\
(A_y)_{6,3} &= \delta_{\pi\pi} \pi^{xy} \\
&- \frac{1}{3 (u^\tau)^2 \left((u^y)^2 + 1 \right)} u^y \Big(3 \pi^{yy} \tau_\pi u^x \left((u^x)^2 + 1 \right) \left(2 (u^y)^2 + 1 \right) \\
&+ 3 \pi^{xy} \tau_\pi u^y \left(\left(1 - 2 (u^x)^2 \right) (u^y)^2 + 1 \right) \\
&+ (u^\tau)^2 u^x \left(3 \pi^\eta{}_\eta \tau_\pi (u^y)^2 - \left((u^y)^2 + 1 \right) \eta \right) \Big)
\end{aligned}
$$

$$
\begin{aligned}
(A_y)_{6,6} &= \tau_\pi u^y \\
(A_y)_{7,3} &= \zeta \\
(A_y)_{7,7} &= \tau_\Pi u^y \\
(A_y)_{8,1} &= (\partial_T n) u^y \\
(A_y)_{8,3} &= n \\
(A_y)_{8,8} &= (\partial_\alpha n) u^y \\
(A_y)_{8,10} &= 1 \\
(A_y)_{9,1} &= \frac{D_s u^y (h'(T)\nu^x + u^x((\partial_T n)T + n))}{T} \\
(A_y)_{9,2} &= \frac{D_s h \left(n \left(u^\tau\right)^2 u^y - \nu^x u^x u^y \left(\left(u^y\right)^2 + 1 \right) + \nu^y \left(\left(u^x\right)^2 + 1 \right) \left(\left(u^y\right)^2 + 1 \right) \right)}{T \left(u^\tau\right)^2} \\
(A_y)_{9,3} &= \frac{D_s h \left(\left(u^x\right)^2 + 1 \right) \left(\nu^x \left(u^y\right)^2 + \nu^x - \nu^y u^x u^y \right)}{T \left(u^\tau\right)^2} \\
(A_y)_{9,8} &= (\partial_\alpha n) D_s u^x u^y \\
(A_y)_{9,9} &= \frac{D_s h u^y}{T} \\
(A_y)_{10,1} &= \frac{D_s \left(h'(T)\nu^y u^y + (\partial_T n) T \left(\left(u^y\right)^2 + 1 \right) + n \left(\left(u^y\right)^2 + 1 \right) \right)}{T} \\
(A_y)_{10,2} &= -\frac{D_s h \left(u^y\right)^2 \left(\nu^x \left(u^y\right)^2 + \nu^x - \nu^y u^x u^y \right)}{T \left(u^\tau\right)^2} \\
(A_y)_{10,3} &= \frac{D_s h \left(n \left(u^\tau\right)^2 u^y + \nu^x u^x \left(u^y\right)^3 + \nu^y \left(- \left(\left(u^x\right)^2 \left(\left(u^y\right)^2 - 2 \right) \right) + \left(u^y\right)^2 + 2 \right) \right)}{T \left(u^\tau\right)^2} \\
(A_y)_{10,8} &= (\partial_\alpha n) D_s \left(\left(u^y\right)^2 + 1 \right) \\
(A_y)_{10,10} &= \frac{D_s h u^y}{T} \\
(A_y)_{11,1} &= \frac{D_s \left(4\mathcal{B}'(T)\nu^y \left(\left(u^y\right)^2 + 1 \right) + 3\mathcal{C}'(T)\pi_Q^{yy} u^y \right)}{6T} \\
(A_y)_{11,2} &= -\frac{1}{3T \left(u^\tau\right)^2} D_s \Big(u^y \left(\left(u^y\right)^2 + 1 \right) (\mathcal{A} + 3\mathcal{B}) \left(\nu^x \left(u^y\right)^2 + \nu^x - \nu^y u^x u^y \right) \\
&\quad + \mathcal{C} \left(4 \left(u^y\right)^2 + 1 \right) \left(\pi_Q^{xy} \left(u^y\right)^2 + \pi_Q^{xy} - \pi_Q^{yy} u^x u^y \right) \Big) \\
(A_y)_{11,3} &= \frac{1}{6T \left(u^\tau\right)^2} D_s \Big(4 I_{31} \left(u^\tau\right)^2 \left(\left(u^y\right)^2 + 1 \right) \\
&\quad + \mathcal{C} \Big(4\Pi_Q \left(u^\tau\right)^2 \left(\left(u^y\right)^2 + 1 \right) + 2\pi_Q^{xy} u^x u^y \left(4 \left(u^y\right)^2 + 1 \right) \\
&\quad + \pi_Q^{yy} \left(\left(u^x\right)^2 \left(7 - 8 \left(u^y\right)^2 \right) + \left(u^y\right)^2 + 7 \right) \Big) \\
&\quad + 2u^y \Big(3\mathcal{B} u^x u^y \left(\nu^x \left(u^y\right)^2 + \nu^x - \nu^y u^x u^y \right) \\
&\quad + \mathcal{A} \Big(-\nu^y \left(u^x\right)^2 \left(\left(u^y\right)^2 - 2 \right) + 2\nu^y \left(\left(u^y\right)^2 + 1 \right) \\
&\quad + \nu^x u^x u^y \left(\left(u^y\right)^2 + 1 \right) \Big) \Big) \Big) \\
(A_y)_{11,10} &= \frac{2\mathcal{B} D_s \left(\left(u^y\right)^2 + 1 \right)}{3T} \\
(A_y)_{11,11} &= \frac{\mathcal{C} D_s u^y}{2T} \\
(A_y)_{12,1} &= -\frac{D_s (2\mathcal{B}'(T)\nu^y - 3\mathcal{C}'(T)\pi_Q{}^\eta{}_\eta u^y)}{6T} \\
(A_y)_{12,2} &= -\frac{D_s \left(\mathcal{A} u^y \left(\nu^x \left(u^y\right)^2 + \nu^x - \nu^y u^x u^y \right) + \mathcal{C} \left(\pi_Q^{xy} \left(u^y\right)^2 + \pi_Q^{xy} - \pi_Q^{yy} u^x u^y \right) \right)}{3T \left(u^\tau\right)^2}
\end{aligned}
$$

$$
\begin{aligned}
(A_y)_{12,3} &= -\frac{1}{6T\left(u^\tau\right)^2} D_s \Big(2I_{31}\left(u^\tau\right)^2 + 2\mathcal{A}u^y \left(-\nu^x u^x u^y + \nu^y \left(u^x\right)^2 + \nu^y \right) \\
&\quad + \mathcal{C}\Big(-3{\pi_Q}^\eta{}_\eta + 2\Pi_Q \left(u^\tau\right)^2 - 3{\pi_Q}^\eta{}_\eta \left(u^x\right)^2 + 2\pi_Q^{yy}\left(\left(u^x\right)^2 + 1 \right) \\
&\quad - 2\pi_Q^{xy} u^x u^y - 3{\pi_Q}^\eta{}_\eta \left(u^y\right)^2 \Big) \Big) \\
(A_y)_{12,10} &= -\frac{\mathcal{B}D_s}{3T} \\
(A_y)_{12,12} &= \frac{\mathcal{C}D_s u^y}{2T} \\
(A_y)_{13,1} &= \frac{D_s\left(3\mathcal{B}'(T)\nu^x \left(\left(u^y\right)^2 + 1 \right) + \mathcal{B}'(T)\nu^y u^x u^y + 3\mathcal{C}'(T)\pi_Q^{xy} u^y \right)}{6T} \\
(A_y)_{13,2} &= \frac{1}{6T\left(u^\tau\right)^2} D_s \Big(-3\mathcal{B}\nu^x u^x + 3\mathcal{A}\nu^y u^y + 3\mathcal{B}\nu^y u^y + 3\mathcal{A}\nu^y \left(u^x\right)^2 u^y + 6\mathcal{B}\nu^y \left(u^x\right)^2 u^y \\
&\quad - 2\mathcal{A}\nu^x u^x \left(u^y\right)^2 - 9\mathcal{B}\nu^x u^x \left(u^y\right)^2 + 3\mathcal{A}\nu^y \left(u^y\right)^3 + 3\mathcal{B}\nu^y \left(u^y\right)^3 \\
&\quad + 2\mathcal{A}\nu^y \left(u^x\right)^2 \left(u^y\right)^3 + 6\mathcal{B}\nu^y \left(u^x\right)^2 \left(u^y\right)^3 - 2\mathcal{A}\nu^x u^x \left(u^y\right)^4 - 6\mathcal{B}\nu^x u^x \left(u^y\right)^4 \\
&\quad + 3I_{31}\left(u^\tau\right)^2 \left(\left(u^y\right)^2 + 1 \right) \\
&\quad + \mathcal{C}\Big(3\Pi_Q \left(u^\tau\right)^2 \left(\left(u^y\right)^2 + 1 \right) \\
&\quad + \pi_Q^{yy} \left(\left(u^x\right)^2 \left(8\left(u^y\right)^2 + 3 \right) + 6\left(u^y\right)^2 + 3 \right) \\
&\quad + u^y \left(3{\pi_Q}^\eta{}_\eta \left(u^\tau\right)^2 u^y - \pi_Q^{xy} u^x \left(8\left(u^y\right)^2 + 5 \right) \right) \Big) \Big) \\
(A_y)_{13,3} &= \frac{1}{6T\left(u^\tau\right)^2 \left(\left(u^y\right)^2 + 1 \right)} D_s \Big(\left(\left(u^y\right)^2 + 1 \Big(I_{31}\left(u^\tau\right)^2 u^x u^y \right. \right. \\
&\quad + 3\mathcal{B}\nu^x \left(2\left(u^x\right)^2 + 1 \right) u^y \left(\left(u^y\right)^2 + 1 \right) \\
&\quad - 3\mathcal{B}\nu^y u^x \left(\left(u^x\right)^2 + 1 \right) \left(2\left(u^y\right)^2 + 1 \right) \\
&\quad + \mathcal{A}u^y \Big(-2\nu^y u^x \left(\left(u^x\right)^2 + 1 \right) u^y + 3\nu^x \left(\left(u^y\right)^2 + 1 \right) \\
&\quad \left. + \nu^x \left(u^x\right)^2 \left(2\left(u^y\right)^2 + 3 \right) \right) \Big) \\
&\quad + \mathcal{C}\Big(u^x u^y \Big(-3{\pi_Q}^\eta{}_\eta \left(u^\tau\right)^2 \left(u^y\right)^2 + \Pi_Q \left(u^\tau\right)^2 \left(\left(u^y\right)^2 + 1 \right) \\
&\quad - \pi_Q^{yy} \left(\left(u^x\right)^2 + 1 \right) \left(8\left(u^y\right)^2 + 5 \right) \Big) \\
&\quad + \pi_Q^{xy} \Big(3\left(\left(u^y\right)^4 + 3\left(u^y\right)^2 + 2 \right) \\
&\quad \left. \left. + \left(u^x\right)^2 \left(8\left(u^y\right)^4 + 8\left(u^y\right)^2 + 6 \right) \right) \right) \Big) \\
(A_y)_{13,9} &= \frac{\mathcal{B}D_s \left(\left(u^y\right)^2 + 1 \right)}{2T} \\
(A_y)_{13,10} &= \frac{\mathcal{B}D_s u^x u^y}{6T} \\
(A_y)_{13,13} &= \frac{\mathcal{C}D_s u^y}{2T} \\
(A_y)_{14,1} &= \frac{D_s(5\mathcal{B}'(T)\nu^y + 3u^y(\mathcal{C}'(T)\Pi_Q + (\partial_T I_{31})))}{6T} \\
(A_y)_{14,2} &= \frac{D_s\left(\mathcal{A}u^y \left(\nu^x \left(u^y\right)^2 + \nu^x - \nu^y u^x u^y \right) + \mathcal{C}\left(\pi_Q^{xy}\left(u^y\right)^2 + \pi_Q^{xy} - \pi_Q^{yy} u^x u^y \right) \right)}{3T\left(u^\tau\right)^2} \\
(A_y)_{14,3} &= \frac{1}{6T\left(u^\tau\right)^2} D_s \Big(5I_{31}\left(u^\tau\right)^2 + 5\mathcal{C}\Pi_Q \left(u^\tau\right)^2 + 2\mathcal{A}u^y \left(-\nu^x u^x u^y + \nu^y \left(u^x\right)^2 + \nu^y \right) \\
&\quad + 2\mathcal{C}\left(-\pi_Q^{xy} u^x u^y + \pi_Q^{yy}\left(u^x\right)^2 + \pi_Q^{yy} \right) \Big) \\
(A_y)_{14,8} &= \frac{(\partial_\alpha I_{31}) D_s u^y}{2T}
\end{aligned}
$$

$$
\begin{aligned}
(A_y)_{14,10} &= \frac{5\mathcal{B}D_s}{6T} \\
(A_y)_{14,14} &= \frac{\mathcal{C}D_s u^y}{2T} \\
S_1 &= \frac{1}{\tau\left(\left(u^y\right)^2+1\right)}\Big(\pi^\eta{}_\eta + P'(T)T - \pi^\eta{}_\eta\left(u^x\right)^2 - \pi^{yy}\left(u^x\right)^2 + P'(T)T\left(u^x\right)^2 + 2\pi^{xy}u^x u^y \\
&\quad + \pi^\eta{}_\eta\left(u^y\right)^2 + \pi^{yy}\left(u^y\right)^2 + 2P'(T)T\left(u^y\right)^2 + P'(T)T\left(u^x\right)^2\left(u^y\right)^2 \\
&\quad + P'(T)T\left(u^y\right)^4 + \Pi\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right) \\
&\quad + h\left(\left(u^y\right)^2+1\right)\left(n\left(u^\tau\right)^2 + 2(\nu^x u^x + \nu^y u^y)\right)\Big) \\
S_2 &= \frac{1}{\tau u^\tau\left(\left(u^y\right)^2+1\right)}\Big(-\pi^\eta{}_\eta u^x - \pi^{yy}u^x + P'(T)Tu^x - \pi^\eta{}_\eta\left(u^x\right)^3 - \pi^{yy}\left(u^x\right)^3 \\
&\quad + P'(T)T\left(u^x\right)^3 + \pi^{xy}u^y + 2\pi^{xy}\left(u^x\right)^2 u^y - \pi^\eta{}_\eta u^x\left(u^y\right)^2 \\
&\quad + 2P'(T)Tu^x\left(u^y\right)^2 + P'(T)T\left(u^x\right)^3\left(u^y\right)^2 + \pi^{xy}\left(u^y\right)^3 \\
&\quad + P'(T)Tu^x\left(u^y\right)^4 + \Pi\left(u^\tau\right)^2 u^x\left(\left(u^y\right)^2+1\right) \\
&\quad + h\left(\left(u^y\right)^2+1\right)\left(n\left(u^\tau\right)^2 u^x + \nu^x\left(2\left(u^x\right)^2 + \left(u^y\right)^2 + 1\right) + \nu^y u^x u^y\right)\Big) \\
S_3 &= \frac{1}{\tau u^\tau}\Big(\pi^{xy}u^x + u^y\left(P'(T)T\left(u^\tau\right)^2 + \Pi\left(u^\tau\right)^2 + \pi^{yy}\right) \\
&\quad + h\left(n\left(u^\tau\right)^2 u^y + \nu^x u^x u^y + \nu^y\left(\left(u^x\right)^2 + 2\left(u^y\right)^2 + 1\right)\right)\Big) \\
S_4 &= \frac{\delta_{\pi\pi}\pi^{yy}u^\tau}{\tau} + \pi^{yy} - \frac{2u^\tau\left(\left(u^y\right)^2+1\right)\eta}{3\tau} \\
S_5 &= \frac{3\pi^\eta{}_\eta(\delta_{\pi\pi}u^\tau + \tau) + 4u^\tau\eta}{3\tau^3} \\
S_6 &= \frac{\delta_{\pi\pi}\pi^{xy}u^\tau}{\tau} + \pi^{xy} - \frac{2u^\tau u^x u^y\eta}{3\tau} \\
S_7 &= \Pi + \frac{u^\tau\zeta}{\tau} \\
S_8 &= \frac{n\left(u^\tau\right)^2 + \nu^x u^x + \nu^y u^y}{\tau u^\tau} \\
S_9 &= \frac{D_s h\nu^x u^\tau}{T\tau} + \nu^x \\
S_{10} &= \frac{D_s h\nu^y u^\tau}{T\tau} + \nu^y \\
S_{11} &= \pi_Q^{yy} \\
&\quad - \frac{1}{6T\tau u^\tau}D_s\Big(2I_{31}\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right) + 2\mathcal{B}\left(\left(u^y\right)^2+1\right)(\nu^x u^x + \nu^y u^y) \\
&\quad + \mathcal{C}\left(u^\tau\right)^2\left(2\Pi_Q\left(\left(u^y\right)^2+1\right) - 3\pi_Q^{yy} + 2\pi_Q{}^\eta{}_\eta\left(\left(u^y\right)^2+1\right)\right)\Big) \\
S_{12} &= \frac{D_s\left(4\mathcal{B}(\nu^x u^x + \nu^y u^y) + \mathcal{C}\left(u^\tau\right)^2(4\Pi_Q + 7\pi_Q{}^\eta{}_\eta) + 4I_{31}\left(u^\tau\right)^2\right)}{6T\tau u^\tau} + \pi_Q{}^\eta{}_\eta \\
S_{13} &= -\frac{D_s u^x u^y\left(\mathcal{B}(\nu^x u^x + \nu^y u^y) + I_{31}\left(u^\tau\right)^2\right)}{3T\tau u^\tau} + \frac{\mathcal{C}D_s u^\tau(3\pi_Q^{xy} - 2u^x u^y(\Pi_Q + \pi_Q{}^\eta{}_\eta))}{6T\tau} + \pi_Q^{xy} \\
S_{14} &= \frac{D_s\left(5\mathcal{B}(\nu^x u^x + \nu^y u^y) + \mathcal{C}\left(u^\tau\right)^2(5\Pi_Q + 2\pi_Q{}^\eta{}_\eta) + 5I_{31}\left(u^\tau\right)^2\right)}{6T\tau u^\tau} + \Pi_Q
\end{aligned}
$$

A.2.4 S5, two-way, conformal

`two_d_two_way_matrix_conformal` in `src/Matrix/2d_two_way.jl`, fields:

$$\phi = (T, u^x, u^y, \pi^{yy}, \pi^\eta{}_\eta, \pi^{xy}, \alpha, \nu^x, \nu^y, \pi_Q^{yy}, \pi_Q{}^\eta{}_\eta, \pi_Q^{xy}, \Pi_Q)$$

$$
\begin{aligned}
(A_t)_{1,1} &= -n - (\partial_T n)T + (\partial_T n)h\left(u^\tau\right)^2 + h'(T)n\left(u^\tau\right)^2 + 2h'(T)(\nu^x u^x + \nu^y u^y) \\
&\quad + 4a_{\mathrm{SB}}T^3\left(4\left(u^x\right)^2 + 4\left(u^y\right)^2 + 3\right) \\
(A_t)_{1,2} &= \frac{2\left(4a_{\mathrm{SB}}T^4u^x\left(u^y\right)^2 + 4a_{\mathrm{SB}}T^4u^x + h\left(\left(u^y\right)^2+1\right)(nu^x+\nu^x) - \pi^\eta{}_\eta u^x + \pi^{xy}u^y - \pi^{yy}u^x\right)}{\left(u^y\right)^2+1} \\
(A_t)_{1,3} &= \frac{2}{\left(\left(u^y\right)^2+1\right)^2}\Big(h\left(\left(u^y\right)^2+1\right)^2(nu^y+\nu^y) + \pi^{xy}\left(u^x - u^x\left(u^y\right)^2\right) \\
&\quad + u^y\left(4a_{\mathrm{SB}}T^4\left(\left(u^y\right)^2+1\right)^2 + \pi^\eta{}_\eta\left(u^x\right)^2 + \pi^{yy}\left(\left(u^x\right)^2+1\right)\right)\Big) \\
(A_t)_{1,4} &= \frac{\left(u^y\right)^2 - \left(u^x\right)^2}{\left(u^y\right)^2+1} \\
(A_t)_{1,5} &= -\frac{\left(u^x\right)^2}{\left(u^y\right)^2+1} \\
(A_t)_{1,6} &= \frac{2u^xu^y}{\left(u^y\right)^2+1} \\
(A_t)_{1,7} &= (\partial_\alpha n)\left(h\left(u^\tau\right)^2 - T\right) \\
(A_t)_{1,8} &= 2hu^x \\
(A_t)_{1,9} &= 2hu^y \\
(A_t)_{2,1} &= \frac{\left(u^\tau\right)^2u^x\left(16a_{\mathrm{SB}}T^3 + (\partial_T n)h\right) + h'(T)\left(n\left(u^\tau\right)^2u^x + \nu^x\left(2\left(u^x\right)^2 + \left(u^y\right)^2 + 1\right) + \nu^y u^x u^y\right)}{u^\tau} \\
(A_t)_{2,2} &= \frac{1}{\left(u^\tau\right)^3\left(\left(u^y\right)^2+1\right)}\Big(-\pi^{yy} + 4a_{\mathrm{SB}}T^4 - 3\pi^{yy}\left(u^x\right)^2 + 12a_{\mathrm{SB}}T^4\left(u^x\right)^2 - 2\pi^{yy}\left(u^x\right)^4 \\
&\quad + 8a_{\mathrm{SB}}T^4\left(u^x\right)^4 + 3\pi^{xy}u^xu^y + 2\pi^{xy}\left(u^x\right)^3u^y - \pi^{yy}\left(u^y\right)^2 \\
&\quad + 12a_{\mathrm{SB}}T^4\left(u^y\right)^2 - 3\pi^{yy}\left(u^x\right)^2\left(u^y\right)^2 + 24a_{\mathrm{SB}}T^4\left(u^x\right)^2\left(u^y\right)^2 \\
&\quad + 8a_{\mathrm{SB}}T^4\left(u^x\right)^4\left(u^y\right)^2 + 3\pi^{xy}u^x\left(u^y\right)^3 + 12a_{\mathrm{SB}}T^4\left(u^y\right)^4 \\
&\quad + 12a_{\mathrm{SB}}T^4\left(u^x\right)^2\left(u^y\right)^4 + 4a_{\mathrm{SB}}T^4\left(u^y\right)^6 \\
&\quad - \pi^\eta{}_\eta\left(2\left(u^x\right)^4 + 3\left(u^x\right)^2\left(\left(u^y\right)^2+1\right) + \left(\left(u^y\right)^2+1\right)^2\right) \\
&\quad + h\left(\left(u^y\right)^2+1\right)\left(\nu^y u^y\left(\left(u^y\right)^2+1\right)\right. \\
&\quad + \nu^x u^x\left(2\left(u^x\right)^2 + 3\left(u^y\right)^2 + 3\right) \\
&\quad \left. + n\Big(2\left(u^x\right)^4 + 3\left(u^x\right)^2\left(\left(u^y\right)^2+1\right) + \left(\left(u^y\right)^2+1\right)^2\Big)\right)\Big) \\
(A_t)_{2,3} &= \frac{1}{\left(u^\tau\right)^3\left(\left(u^y\right)^2+1\right)^2}\Big(h\left(\left(u^y\right)^2+1\right)^2\left(n\left(u^\tau\right)^2u^xu^y + \nu^x u^y\left(\left(u^y\right)^2+1\right) + \nu^y\left(\left(u^x\right)^3 + u^x\right)\right) \\
&\quad + \pi^{xy}\Big(-2\left(u^x\right)^4\left(\left(u^y\right)^2-1\right) + \left(\left(u^y\right)^2+1\right)^2 \\
&\quad - 3\left(u^x\right)^2\left(\left(u^y\right)^4-1\right)\Big) \\
&\quad + u^xu^y\Big(4a_{\mathrm{SB}}T^4\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right)^2 \\
&\quad + \pi^{yy}\left(\left(u^x\right)^2+1\right)\left(2\left(u^x\right)^2 + 3\left(u^y\right)^2 + 3\right) \\
&\quad + \pi^\eta{}_\eta\left(2\left(u^x\right)^4 + 3\left(u^x\right)^2\left(\left(u^y\right)^2+1\right) + \left(\left(u^y\right)^2+1\right)^2\right)\Big)\Big) \\
(A_t)_{2,4} &= -\frac{u^x\left(\left(u^x\right)^2+1\right)}{u^\tau\left(\left(u^y\right)^2+1\right)}
\end{aligned}
$$

$$
\begin{aligned}
(A_t)_{2,5} &= -\frac{u^\tau u^x}{(u^y)^2+1} \\
(A_t)_{2,6} &= \frac{u^y\left(2\left(u^x\right)^2+\left(u^y\right)^2+1\right)}{u^\tau\left(\left(u^y\right)^2+1\right)} \\
(A_t)_{2,7} &= (\partial_\alpha n)hu^\tau u^x \\
(A_t)_{2,8} &= \frac{h\left(2\left(u^x\right)^2+\left(u^y\right)^2+1\right)}{u^\tau} \\
(A_t)_{2,9} &= \frac{hu^xu^y}{u^\tau} \\
(A_t)_{3,1} &= \frac{\left(u^\tau\right)^2u^y\left(16a_{\mathrm{SB}}T^3+(\partial_T n)h\right)+h'(T)\left(n\left(u^\tau\right)^2u^y+\nu^xu^xu^y+\nu^y\left(\left(u^x\right)^2+2\left(u^y\right)^2+1\right)\right)}{u^\tau} \\
(A_t)_{3,2} &= \frac{1}{\left(u^\tau\right)^3}\Big(u^xu^y\left(4a_{\mathrm{SB}}T^4\left(u^\tau\right)^2-\pi^{yy}\right)+\pi^{xy}\left(\left(u^y\right)^2+1\right) \\
&\quad + h\left(n\left(u^\tau\right)^2u^xu^y+\nu^xu^y\left(\left(u^y\right)^2+1\right)+\nu^y\left(\left(u^x\right)^3+u^x\right)\right)\Big) \\
(A_t)_{3,3} &= \frac{1}{\left(u^\tau\right)^3}\Big(\pi^{yy}+4a_{\mathrm{SB}}T^4+\pi^{yy}\left(u^x\right)^2+8a_{\mathrm{SB}}T^4\left(u^x\right)^2+4a_{\mathrm{SB}}T^4\left(u^x\right)^4-\pi^{xy}u^xu^y \\
&\quad + 12a_{\mathrm{SB}}T^4\left(u^y\right)^2+12a_{\mathrm{SB}}T^4\left(u^x\right)^2\left(u^y\right)^2+8a_{\mathrm{SB}}T^4\left(u^y\right)^4 \\
&\quad + h\Big(\nu^x\left(\left(u^x\right)^3+u^x\right)+\nu^yu^y\left(3\left(u^x\right)^2+2\left(u^y\right)^2+3\right) \\
&\quad + n\left(\left(u^x\right)^4+\left(u^x\right)^2\left(3\left(u^y\right)^2+2\right)+2\left(u^y\right)^4+3\left(u^y\right)^2+1\right)\Big)\Big) \\
(A_t)_{3,4} &= \frac{u^y}{u^\tau} \\
(A_t)_{3,6} &= \frac{u^x}{u^\tau} \\
(A_t)_{3,7} &= (\partial_\alpha n)hu^\tau u^y \\
(A_t)_{3,8} &= \frac{hu^xu^y}{u^\tau} \\
(A_t)_{3,9} &= \frac{h\left(\left(u^x\right)^2+2\left(u^y\right)^2+1\right)}{u^\tau} \\
(A_t)_{4,2} &= \frac{3\delta_{\pi\pi}\pi^{yy}u^x-6\pi^{xy}\tau_\pi u^y\left(\left(u^y\right)^2+1\right)-2u^x\left(-3\pi^{yy}\tau_\pi\left(u^y\right)^2+\left(u^y\right)^2\eta+\eta\right)}{3u^\tau} \\
(A_t)_{4,3} &= \frac{u^y\left(3\delta_{\pi\pi}\pi^{yy}+6\pi^{xy}\tau_\pi u^xu^y-6\pi^{yy}\tau_\pi\left(\left(u^x\right)^2+1\right)+6\left(u^x\right)^2\eta+4\left(u^y\right)^2\eta+4\eta\right)}{3u^\tau} \\
(A_t)_{4,4} &= \tau_\pi u^\tau \\
(A_t)_{5,2} &= \frac{u^x(3\delta_{\pi\pi}\pi^\eta{}_\eta-2\eta)}{3\tau^2u^\tau} \\
(A_t)_{5,3} &= \frac{u^y(3\delta_{\pi\pi}\pi^\eta{}_\eta-2\eta)}{3\tau^2u^\tau} \\
(A_t)_{5,5} &= \frac{\tau_\pi u^\tau}{\tau^2} \\
(A_t)_{6,2} &= \frac{1}{3u^\tau}\Big(3\delta_{\pi\pi}\pi^{xy}u^x-3\pi^{xy}\tau_\pi u^x\left(2\left(u^y\right)^2+1\right) \\
&\quad + u^y\left(3\pi^\eta{}_\eta\tau_\pi\left(u^\tau\right)^2+3\pi^{yy}\left(2\tau_\pi\left(u^x\right)^2+\tau_\pi\right)+\eta\left(\left(u^x\right)^2+3\left(u^y\right)^2+3\right)\right)\Big) \\
(A_t)_{6,3} &= \frac{1}{3u^\tau\left(\left(u^y\right)^2+1\right)}\Big(-3\pi^{xy}\tau_\pi u^y-3\pi^\eta{}_\eta\tau_\pi u^x\left(u^y\right)^2-3\pi^\eta{}_\eta\tau_\pi\left(u^x\right)^3\left(u^y\right)^2-3\pi^{xy}\tau_\pi\left(u^y\right)^3 \\
&\quad + 6\pi^{xy}\tau_\pi\left(u^x\right)^2\left(u^y\right)^3-3\pi^\eta{}_\eta\tau_\pi u^x\left(u^y\right)^4+3\delta_{\pi\pi}\pi^{xy}u^y\left(\left(u^y\right)^2+1\right) \\
&\quad - 3\pi^{yy}\tau_\pi u^x\left(\left(u^x\right)^2+1\right)\left(2\left(u^y\right)^2+1\right)+3u^x\eta+3\left(u^x\right)^3\eta \\
&\quad + 4u^x\left(u^y\right)^2\eta+3\left(u^x\right)^3\left(u^y\right)^2\eta+u^x\left(u^y\right)^4\eta\Big)
\end{aligned}
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\begin{aligned}
(A_t)_{6,6} &= \tau_\pi u^\tau \\
(A_t)_{7,1} &= (\partial_T n) u^\tau \\
(A_t)_{7,2} &= \frac{n\left(u^\tau\right)^2 u^x + \nu^x\left(\left(u^y\right)^2+1\right) - \nu^y u^x u^y}{\left(u^\tau\right)^3} \\
(A_t)_{7,3} &= \frac{n\left(u^\tau\right)^2 u^y - \nu^x u^x u^y + \nu^y\left(\left(u^x\right)^2+1\right)}{\left(u^\tau\right)^3} \\
(A_t)_{7,7} &= (\partial_\alpha n) u^\tau \\
(A_t)_{7,8} &= \frac{u^x}{u^\tau} \\
(A_t)_{7,9} &= \frac{u^y}{u^\tau} \\
(A_t)_{8,1} &= \frac{D_s u^\tau (h'(T)\nu^x + u^x((\partial_T n)T + n))}{T} \\
(A_t)_{8,2} &= \frac{D_s h\left(n\left(u^\tau\right)^2 + \nu^x\left(u^x - u^x\left(u^y\right)^2\right) + \nu^y\left(\left(u^x\right)^2+1\right)u^y\right)}{T u^\tau} \\
(A_t)_{8,3} &= -\frac{D_s h\left(\left(u^x\right)^2+1\right)(\nu^y u^x - \nu^x u^y)}{T u^\tau} \\
(A_t)_{8,7} &= (\partial_\alpha n) D_s u^\tau u^x \\
(A_t)_{8,8} &= \frac{D_s h u^\tau}{T} \\
(A_t)_{9,1} &= \frac{D_s u^\tau (h'(T)\nu^y + u^y((\partial_T n)T + n))}{T} \\
(A_t)_{9,2} &= \frac{D_s h\left(\left(u^y\right)^2+1\right)(\nu^y u^x - \nu^x u^y)}{T u^\tau} \\
(A_t)_{9,3} &= \frac{D_s h\left(n\left(u^\tau\right)^2 + \nu^x u^x\left(\left(u^y\right)^2+1\right) - \nu^y\left(\left(u^x\right)^2-1\right)u^y\right)}{T u^\tau} \\
(A_t)_{9,7} &= (\partial_\alpha n) D_s u^\tau u^y \\
(A_t)_{9,9} &= \frac{D_s h u^\tau}{T} \\
(A_t)_{10,1} &= \frac{1}{6Tu^\tau} D_s\Big(3\mathcal{C}'(T)\pi_Q^{yy}\left(u^\tau\right)^2 - 2\mathcal{B}'(T)\nu^x u^x\left(\left(u^y\right)^2+1\right) \\
&\quad + 2\mathcal{B}'(T)\nu^y u^y\left(3\left(u^x\right)^2 + 2\left(u^y\right)^2 + 2\right)\Big) \\
(A_t)_{10,2} &= -\frac{1}{6T\left(u^\tau\right)^3} D_s\Big(2\mathcal{A}\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right)\left(\nu^x\left(u^y\right)^2 + \nu^x - \nu^y u^x u^y\right) \\
&\quad + 2\mathcal{B}\left(\left(3\left(u^x\right)^2+4\right)\left(u^y\right)^2 + 3\left(u^y\right)^4 + 1\right)\left(\nu^x\left(u^y\right)^2 + \nu^x - \nu^y u^x u^y\right) \\
&\quad + \left(u^\tau\right)^2\Big(2I_{31}u^x\left(\left(u^y\right)^2+1\right) \\
&\quad + \mathcal{C}\Big(2\Pi_Q u^x\left(\left(u^y\right)^2+1\right) \\
&\quad - 2\left(\left(u^y\right)^2+1\right)(\pi_Q{}^\eta{}_\eta u^x - 4\pi_Q^{xy}u^y) \\
&\quad - \pi_Q^{yy}u^x\left(8\left(u^y\right)^2+5\right)\Big)\Big)\Big) \\
(A_t)_{10,3} &= \frac{1}{6T\left(u^\tau\right)^3} D_s\Big(2\mathcal{B}\nu^x u^x u^y\left(\left(3\left(u^x\right)^2+4\right)\left(u^y\right)^2 + 3\left(u^y\right)^4 + 1\right) \\
&\quad - 2\mathcal{A}\left(u^\tau\right)^2\left(\nu^y\left(u^x\right)^2\left(\left(u^y\right)^2-2\right) - 2\nu^y\left(\left(u^y\right)^2+1\right)\right. \\
&\quad \left. - \nu^x u^x u^y\left(\left(u^y\right)^2+1\right)\right) \\
&\quad - 2\mathcal{B}\nu^y\Big(3\left(u^x\right)^4\left(\left(u^y\right)^2-1\right) - 2\left(\left(u^y\right)^2+1\right) \\
&\quad + \left(u^x\right)^2\left(3\left(u^y\right)^4 + \left(u^y\right)^2 - 5\right)\Big)
\end{aligned}
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\begin{aligned}
&+ (u^\tau)^2 \left(2I_{31}u^y \left(3\left(u^x\right)^2 + 2\left(u^y\right)^2 + 2\right) + 4\mathcal{C}\pi_Q^{xy} \left(2u^x \left(u^y\right)^2 + u^x\right)\right.\\
&+ \mathcal{C}u^y \Big(\pi_Q^{yy} - 8\pi_Q^{yy} \left(u^x\right)^2 - 2{\pi_Q}^\eta{}_\eta \left(u^x\right)^2\\
&\left.+ \Pi_Q \left(6\left(u^x\right)^2 + 4\left(u^y\right)^2 + 4\right)\Big)\right)\Big)\\
(A_t)_{10,8} &= -\frac{\mathcal{B}D_s u^x \left(\left(u^y\right)^2 + 1\right)}{3Tu^\tau}\\
(A_t)_{10,9} &= \frac{\mathcal{B}D_s u^y \left(3\left(u^x\right)^2 + 2\left(u^y\right)^2 + 2\right)}{3Tu^\tau}\\
(A_t)_{10,10} &= \frac{\mathcal{C}D_s u^\tau}{2T}\\
(A_t)_{11,1} &= \frac{D_s \left(3\mathcal{C}'(T){\pi_Q}^\eta{}_\eta \left(u^\tau\right)^2 - 2\mathcal{B}'(T)(\nu^x u^x + \nu^y u^y)\right)}{6Tu^\tau}\\
(A_t)_{11,2} &= -\frac{1}{6T\left(u^\tau\right)^3} D_s \Big(2\mathcal{B}\left(\nu^x \left(u^y\right)^2 + \nu^x - \nu^y u^x u^y\right) + 2\mathcal{A}\left(u^\tau\right)^2 \left(\nu^x \left(u^y\right)^2 + \nu^x - \nu^y u^x u^y\right)\\
&+ \left(u^\tau\right)^2 \left(\mathcal{C}(2\Pi_Q u^x + 2\pi_Q^{xy} u^y - 2\pi_Q^{yy} u^x - 5{\pi_Q}^\eta{}_\eta u^x) + 2I_{31}u^x\right)\Big)\\
(A_t)_{11,3} &= -\frac{1}{6T\left(u^\tau\right)^3 \left(\left(u^y\right)^2 + 1\right)} D_s \Big(2\mathcal{B}\left(\left(u^y\right)^2 + 1\right)\left(-\nu^x u^x u^y + \nu^y \left(u^x\right)^2 + \nu^y\right)\\
&+ 2\mathcal{A}\left(u^\tau\right)^2 \left(\left(u^y\right)^2 + 1\right)\left(-\nu^x u^x u^y + \nu^y \left(u^x\right)^2 + \nu^y\right)\\
&+ \left(u^\tau\right)^2 \left(-2\mathcal{C}\pi_Q^{xy} u^x \left(\left(u^y\right)^2 - 1\right) + 2I_{31}u^y \left(\left(u^y\right)^2 + 1\right)\right.\\
&+ \mathcal{C}u^y \Big(2\pi_Q^{yy} \left(\left(u^x\right)^2 + 1\right)\\
&\left.+ {\pi_Q}^\eta{}_\eta \left(2\left(u^x\right)^2 - 3\left(u^y\right)^2 - 3\right) + 2\Pi_Q \left(\left(u^y\right)^2 + 1\right)\Big)\right)\Big)\\
(A_t)_{11,8} &= -\frac{\mathcal{B}D_s u^x}{3Tu^\tau}\\
(A_t)_{11,9} &= -\frac{\mathcal{B}D_s u^y}{3Tu^\tau}\\
(A_t)_{11,11} &= \frac{\mathcal{C}D_s u^\tau}{2T}\\
(A_t)_{12,1} &= \frac{1}{6Tu^\tau} D_s \Big(3\mathcal{C}'(T)\pi_Q^{xy} \left(u^\tau\right)^2 + \mathcal{B}'(T)\nu^y u^x \left(3\left(u^x\right)^2 + \left(u^y\right)^2 + 3\right)\\
&+ \mathcal{B}'(T)\nu^x u^y \left(\left(u^x\right)^2 + 3\left(u^y\right)^2 + 3\right)\Big)\\
(A_t)_{12,2} &= \frac{1}{6T\left(u^\tau\right)^3} D_s \Big(\mathcal{A}\left(u^\tau\right)^2 \left(3\nu^y \left(\left(u^y\right)^2 + 1\right) - 2\nu^x u^x u^y \left(\left(u^y\right)^2 + 1\right)\right.\\
&\left.+ \nu^y \left(u^x\right)^2 \left(2\left(u^y\right)^2 + 3\right)\right)\\
&+ \left(u^\tau\right)^2 \left(\mathcal{C}\pi_Q^{xy} u^x \left(3 - 8\left(u^y\right)^2\right) + I_{31}u^y \left(\left(u^x\right)^2 + 3\left(u^y\right)^2 + 3\right)\right.\\
&+ \mathcal{C}u^y \Big(\pi_Q^{yy} \left(8\left(u^x\right)^2 + 6\right) + \Pi_Q \left(\left(u^x\right)^2 + 3\left(u^y\right)^2 + 3\right)\\
&\left.+ {\pi_Q}^\eta{}_\eta \left(5\left(u^x\right)^2 + 3\left(u^y\right)^2 + 3\right)\Big)\right)\\
&+ \mathcal{B}\Big(-2\nu^x u^x u^y \left(\left(u^y\right)^2 + 1\right)\left(3\left(u^x\right)^2 + 3\left(u^y\right)^2 + 4\right)\\
&+ \nu^y \Big(3\left(\left(u^y\right)^2 + 1\right)^2 + \left(u^x\right)^4 \left(6\left(u^y\right)^2 + 3\right)\\
&+ 2\left(u^x\right)^2 \left(3\left(u^y\right)^4 + 7\left(u^y\right)^2 + 3\right)\Big)\Big)\Big)\\
(A_t)_{12,3} &= \frac{1}{6T\left(u^\tau\right)^3 \left(\left(u^y\right)^2 + 1\right)} D_s \Big(\mathcal{A}\left(u^\tau\right)^2 \left(\left(u^y\right)^2 + 1\right)\left(-2\nu^y u^x \left(\left(u^x\right)^2 + 1\right)u^y + 3\nu^x \left(\left(u^y\right)^2 + 1\right)\right.\\
&\left.+ \nu^x \left(u^x\right)^2 \left(2\left(u^y\right)^2 + 3\right)\right)
\end{aligned}
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\begin{aligned}
&+ (u^\tau)^2 \left(I_{31} u^x \left((u^y)^2+1\right)\left(3(u^x)^2+(u^y)^2+3\right)\right.\\
&+ \mathcal{C}\Big(-3{\pi_Q}^{\eta}{}_{\eta} u^x - 3{\pi_Q}^{\eta}{}_{\eta}(u^x)^3 + 3\pi_Q^{xy} u^y + 4\pi_Q^{xy}(u^x)^2 u^y\\
&- 6{\pi_Q}^{\eta}{}_{\eta} u^x (u^y)^2 - 5{\pi_Q}^{\eta}{}_{\eta}(u^x)^3 (u^y)^2 + 3\pi_Q^{xy}(u^y)^3\\
&+ 8\pi_Q^{xy}(u^x)^2(u^y)^3 - 3{\pi_Q}^{\eta}{}_{\eta} u^x (u^y)^4\\
&+ \Pi_Q u^x \left((u^y)^2+1\right)\left(3(u^x)^2+(u^y)^2+3\right)\\
&\left.- 2\pi_Q^{yy} u^x \left((u^x)^2+1\right)\left(4(u^y)^2+3\right)\right)\Big)\\
&+ \mathcal{B}\left((u^y)^2+1\right)\Big(-2\nu^y u^x \left((u^x)^2+1\right) u^y \left(3(u^x)^2+3(u^y)^2+4\right)\\
&+ \nu^x \Big(3\left((u^y)^2+1\right)^2 + (u^x)^4\left(6(u^y)^2+3\right)\\
&+ 2(u^x)^2\left(3(u^y)^4+7(u^y)^2+3\right)\Big)\Big)\Big)\\
(A_t)_{12,8} &= \frac{\mathcal{B} D_s u^y \left((u^x)^2+3(u^y)^2+3\right)}{6Tu^\tau}\\
(A_t)_{12,9} &= \frac{\mathcal{B} D_s u^x \left(3(u^x)^2+(u^y)^2+3\right)}{6Tu^\tau}\\
(A_t)_{12,12} &= \frac{\mathcal{C} D_s u^\tau}{2T}\\
(A_t)_{13,1} &= \frac{D_s\left(5\mathcal{B}'(T)(\nu^x u^x + \nu^y u^y) + 3\mathcal{C}'(T)\Pi_Q (u^\tau)^2 + 3(\partial_T I_{31})(u^\tau)^2\right)}{6Tu^\tau}\\
(A_t)_{13,2} &= \frac{1}{6T(u^\tau)^3} D_s\Big(5\mathcal{B}\left(\nu^x (u^y)^2 + \nu^x - \nu^y u^x u^y\right) + 2\mathcal{A}(u^\tau)^2\left(\nu^x (u^y)^2 + \nu^x - \nu^y u^x u^y\right)\\
&+ (u^\tau)^2\left(\mathcal{C}(5\Pi_Q u^x + 2\pi_Q^{xy} u^y - 2\pi_Q^{yy} u^x - 2{\pi_Q}^{\eta}{}_{\eta} u^x) + 5I_{31} u^x\right)\Big)\\
(A_t)_{13,3} &= \frac{1}{6T(u^\tau)^3\left((u^y)^2+1\right)} D_s\Big(5\mathcal{B}\left((u^y)^2+1\right)\left(-\nu^x u^x u^y + \nu^y (u^x)^2 + \nu^y\right)\\
&+ 2\mathcal{A}(u^\tau)^2\left((u^y)^2+1\right)\left(-\nu^x u^x u^y + \nu^y (u^x)^2 + \nu^y\right)\\
&+ (u^\tau)^2\Big(5I_{31} u^y\left((u^y)^2+1\right)\\
&+ \mathcal{C}\Big(2u^y\left(\pi_Q^{yy}(u^x)^2 + \pi_Q^{yy} + {\pi_Q}^{\eta}{}_{\eta}(u^x)^2\right)\\
&- 2\pi_Q^{xy} u^x\left((u^y)^2-1\right) + 5\Pi_Q u^y\left((u^y)^2+1\right)\Big)\Big)\Big)\\
(A_t)_{13,7} &= \frac{(\partial_\alpha I_{31}) D_s u^\tau}{2T}\\
(A_t)_{13,8} &= \frac{5\mathcal{B} D_s u^x}{6Tu^\tau}\\
(A_t)_{13,9} &= \frac{5\mathcal{B} D_s u^y}{6Tu^\tau}\\
(A_t)_{13,13} &= \frac{\mathcal{C} D_s u^\tau}{2T}\\
(A_x)_{1,1} &= \frac{(u^\tau)^2 u^x\left(16 a_{\mathrm{SB}} T^3 + (\partial_T n)h\right) + h'(T)\left(n(u^\tau)^2 u^x + \nu^x\left(2(u^x)^2 + (u^y)^2 + 1\right) + \nu^y u^x u^y\right)}{u^\tau}\\
(A_x)_{1,2} &= \frac{1}{(u^\tau)^3\left((u^y)^2+1\right)}\Big(-\pi^{yy} + 4a_{\mathrm{SB}}T^4 - 3\pi^{yy}(u^x)^2 + 12 a_{\mathrm{SB}} T^4 (u^x)^2 - 2\pi^{yy}(u^x)^4\\
&+ 8 a_{\mathrm{SB}} T^4 (u^x)^4 + 3\pi^{xy} u^x u^y + 2\pi^{xy}(u^x)^3 u^y - \pi^{yy}(u^y)^2\\
&+ 12 a_{\mathrm{SB}} T^4 (u^y)^2 - 3\pi^{yy}(u^x)^2 (u^y)^2 + 24 a_{\mathrm{SB}} T^4 (u^x)^2 (u^y)^2\\
&+ 8 a_{\mathrm{SB}} T^4 (u^x)^4 (u^y)^2 + 3\pi^{xy} u^x (u^y)^3 + 12 a_{\mathrm{SB}} T^4 (u^y)^4\\
&+ 12 a_{\mathrm{SB}} T^4 (u^x)^2 (u^y)^4 + 4 a_{\mathrm{SB}} T^4 (u^y)^6
\end{aligned}
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\begin{aligned}
&- \pi^{\eta}{}_{\eta}\left(2\left(u^{x}\right)^{4}+3\left(u^{x}\right)^{2}\left(\left(u^{y}\right)^{2}+1\right)+\left(\left(u^{y}\right)^{2}+1\right)^{2}\right)\\
&+ h\left(\left(u^{y}\right)^{2}+1\right)\left(\nu^{y}u^{y}\left(\left(u^{y}\right)^{2}+1\right)\right.\\
&+ \nu^{x}u^{x}\left(2\left(u^{x}\right)^{2}+3\left(u^{y}\right)^{2}+3\right)\\
&\left.+ n\left(2\left(u^{x}\right)^{4}+3\left(u^{x}\right)^{2}\left(\left(u^{y}\right)^{2}+1\right)+\left(\left(u^{y}\right)^{2}+1\right)^{2}\right)\right)\\
(A_x)_{1,3} = &\frac{1}{\left(u^{\tau}\right)^{3}\left(\left(u^{y}\right)^{2}+1\right)^{2}}\left(h\left(\left(u^{y}\right)^{2}+1\right)^{2}\left(n\left(u^{\tau}\right)^{2}u^{x}u^{y}+\nu^{x}u^{y}\left(\left(u^{y}\right)^{2}+1\right)+\nu^{y}\left(\left(u^{x}\right)^{3}+u^{x}\right)\right)\right.\\
&+ \pi^{xy}\left(-2\left(u^{x}\right)^{4}\left(\left(u^{y}\right)^{2}-1\right)+\left(\left(u^{y}\right)^{2}+1\right)^{2}\right.\\
&\left.- 3\left(u^{x}\right)^{2}\left(\left(u^{y}\right)^{4}-1\right)\right)\\
&+ u^{x}u^{y}\left(4a_{\mathrm{SB}}T^{4}\left(u^{\tau}\right)^{2}\left(\left(u^{y}\right)^{2}+1\right)^{2}\right.\\
&+ \pi^{yy}\left(\left(u^{x}\right)^{2}+1\right)\left(2\left(u^{x}\right)^{2}+3\left(u^{y}\right)^{2}+3\right)\\
&\left.+ \pi^{\eta}{}_{\eta}\left(2\left(u^{x}\right)^{4}+3\left(u^{x}\right)^{2}\left(\left(u^{y}\right)^{2}+1\right)+\left(\left(u^{y}\right)^{2}+1\right)^{2}\right)\right)\right)\\
(A_x)_{1,4} = &-\frac{u^{x}\left(\left(u^{x}\right)^{2}+1\right)}{u^{\tau}\left(\left(u^{y}\right)^{2}+1\right)}\\
(A_x)_{1,5} = &-\frac{u^{\tau}u^{x}}{\left(u^{y}\right)^{2}+1}\\
(A_x)_{1,6} = &\frac{u^{y}\left(2\left(u^{x}\right)^{2}+\left(u^{y}\right)^{2}+1\right)}{u^{\tau}\left(\left(u^{y}\right)^{2}+1\right)}\\
(A_x)_{1,7} = &(\partial_{\alpha}n)hu^{\tau}u^{x}\\
(A_x)_{1,8} = &\frac{h\left(2\left(u^{x}\right)^{2}+\left(u^{y}\right)^{2}+1\right)}{u^{\tau}}\\
(A_x)_{1,9} = &\frac{hu^{x}u^{y}}{u^{\tau}}\\
(A_x)_{2,1} = &16a_{\mathrm{SB}}T^{3}\left(u^{x}\right)^{2}+4a_{\mathrm{SB}}T^{3}+h'(T)n\left(u^{x}\right)^{2}+2h'(T)\nu^{x}u^{x}+(\partial_{T}n)h\left(u^{x}\right)^{2}+(\partial_{T}n)T+n\\
(A_x)_{2,2} = &2\left(4a_{\mathrm{SB}}T^{4}u^{x}+hnu^{x}+h\nu^{x}-\frac{\pi^{\eta}{}_{\eta}u^{x}-\pi^{xy}u^{y}+\pi^{yy}u^{x}}{\left(u^{y}\right)^{2}+1}\right)\\
(A_x)_{2,3} = &\frac{2\left(u^{y}\left(\pi^{\eta}{}_{\eta}\left(u^{x}\right)^{2}+\pi^{yy}\left(u^{x}\right)^{2}+\pi^{yy}\right)+\pi^{xy}\left(u^{x}-u^{x}\left(u^{y}\right)^{2}\right)\right)}{\left(\left(u^{y}\right)^{2}+1\right)^{2}}\\
(A_x)_{2,4} = &-\frac{\left(u^{x}\right)^{2}+1}{\left(u^{y}\right)^{2}+1}\\
(A_x)_{2,5} = &-\frac{\left(u^{\tau}\right)^{2}}{\left(u^{y}\right)^{2}+1}\\
(A_x)_{2,6} = &\frac{2u^{x}u^{y}}{\left(u^{y}\right)^{2}+1}\\
(A_x)_{2,7} = &(\partial_{\alpha}n)\left(h\left(u^{x}\right)^{2}+T\right)\\
(A_x)_{2,8} = &2hu^{x}\\
(A_x)_{3,1} = &u^{x}u^{y}\left(16a_{\mathrm{SB}}T^{3}+(\partial_{T}n)h\right)+h'(T)u^{y}(nu^{x}+\nu^{x})+h'(T)\nu^{y}u^{x}\\
(A_x)_{3,2} = &4a_{\mathrm{SB}}T^{4}u^{y}+h(nu^{y}+\nu^{y})\\
(A_x)_{3,3} = &4a_{\mathrm{SB}}T^{4}u^{x}+h(nu^{x}+\nu^{x})\\
(A_x)_{3,6} = &1
\end{aligned}
$$

$$(A_x)_{3,7} = (\partial_\alpha n) h u^x u^y$$
$$(A_x)_{3,8} = h u^y$$
$$(A_x)_{3,9} = h u^x$$
$$(A_x)_{4,2} = \delta_{\pi\pi}\pi^{yy} - \frac{2\pi^{xy}\tau_\pi u^x \left(\left(u^y\right)^2+1\right) u^y}{\left(u^\tau\right)^2} + \frac{2\pi^{yy}\tau_\pi \left(u^x\right)^2 \left(u^y\right)^2}{\left(u^\tau\right)^2} - \frac{2}{3}\left(\left(u^y\right)^2+1\right)\eta$$
$$(A_x)_{4,3} = \frac{2u^x u^y \left(\pi^{xy}\tau_\pi u^x u^y - \pi^{yy}\tau_\pi \left(\left(u^x\right)^2+1\right) + \left(u^\tau\right)^2 \eta\right)}{\left(u^\tau\right)^2}$$
$$(A_x)_{4,4} = \tau_\pi u^x$$
$$(A_x)_{5,2} = \frac{3\delta_{\pi\pi}\pi^\eta{}_\eta - 2\eta}{3\tau^2}$$
$$(A_x)_{5,5} = \frac{\tau_\pi u^x}{\tau^2}$$
$$(A_x)_{6,2} = \delta_{\pi\pi}\pi^{xy} + \frac{u^x \left(u^y \left(3\pi^\eta{}_\eta \tau_\pi \left(u^\tau\right)^2 + 3\pi^{yy}\left(2\tau_\pi \left(u^x\right)^2 + \tau_\pi\right) + \left(u^\tau\right)^2 \eta\right) - 3\pi^{xy}\tau_\pi u^x \left(2\left(u^y\right)^2+1\right)\right)}{3\left(u^\tau\right)^2}$$
$$(A_x)_{6,3} = \frac{1}{\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right)}\Big(-\pi^{yy}\tau_\pi \left(u^x\right)^2 \left(\left(u^x\right)^2+1\right)\left(2\left(u^y\right)^2+1\right) + \pi^{xy}\tau_\pi u^x u^y \left(\left(2\left(u^x\right)^2-1\right)\left(u^y\right)^2-1\right) - \left(u^\tau\right)^2\left(\pi^\eta{}_\eta \tau_\pi \left(u^x\right)^2 \left(u^y\right)^2 - \left(\left(u^x\right)^2+1\right)\left(\left(u^y\right)^2+1\right)\eta\right)\Big)$$
$$(A_x)_{6,6} = \tau_\pi u^x$$
$$(A_x)_{7,1} = (\partial_T n) u^x$$
$$(A_x)_{7,2} = n$$
$$(A_x)_{7,7} = (\partial_\alpha n) u^x$$
$$(A_x)_{7,8} = 1$$
$$(A_x)_{8,1} = \frac{D_s\left(h'(T)\nu^x u^x + (\partial_T n) T\left(\left(u^x\right)^2+1\right) + n\left(\left(u^x\right)^2+1\right)\right)}{T}$$
$$(A_x)_{8,2} = \frac{D_s h\left(n\left(u^\tau\right)^2 u^x + \nu^x\left(\left(u^x\right)^2\left(-\left(u^y\right)^2\right) + \left(u^x\right)^2 + 2\left(u^y\right)^2 + 2\right) + \nu^y\left(u^x\right)^3 u^y\right)}{T\left(u^\tau\right)^2}$$
$$(A_x)_{8,3} = -\frac{D_s h \left(u^x\right)^2\left(-\nu^x u^x u^y + \nu^y\left(u^x\right)^2 + \nu^y\right)}{T\left(u^\tau\right)^2}$$
$$(A_x)_{8,7} = (\partial_\alpha n) D_s\left(\left(u^x\right)^2+1\right)$$
$$(A_x)_{8,8} = \frac{D_s h u^x}{T}$$
$$(A_x)_{9,1} = \frac{D_s u^x (h'(T)\nu^y + u^y((\partial_T n)T + n))}{T}$$
$$(A_x)_{9,2} = \frac{D_s h\left(\left(u^y\right)^2+1\right)\left(-\nu^x u^x u^y + \nu^y\left(u^x\right)^2 + \nu^y\right)}{T\left(u^\tau\right)^2}$$
$$(A_x)_{9,3} = \frac{D_s h\left(n\left(u^\tau\right)^2 u^x + \nu^x\left(\left(u^x\right)^2+1\right)\left(\left(u^y\right)^2+1\right) - \nu^y u^x\left(\left(u^x\right)^2+1\right)u^y\right)}{T\left(u^\tau\right)^2}$$
$$(A_x)_{9,7} = (\partial_\alpha n) D_s u^x u^y$$
$$(A_x)_{9,9} = \frac{D_s h u^x}{T}$$
$$(A_x)_{10,1} = \frac{D_s\left(3\mathcal{C}'(T)\pi_Q^{yy} u^x - 2\mathcal{B}'(T)\left(\nu^x\left(u^y\right)^2 + \nu^x - 3\nu^y u^x u^y\right)\right)}{6T}$$

$$
\begin{aligned}
(A_x)_{10,2} = &-\frac{1}{6T\left(u^\tau\right)^2} D_s \Big(2I_{31} \left(u^\tau\right)^2 \left(\left(u^y\right)^2 + 1 \right) \\
&+ 2u^x \left(\mathcal{A} \left(u^y\right)^2 + \mathcal{A} + 3\mathcal{B} \left(u^y\right)^2 \right) \left(\nu^x \left(u^y\right)^2 + \nu^x - \nu^y u^x u^y \right) \\
&+ \mathcal{C} \Big(2\Pi_Q \left(u^\tau\right)^2 \left(\left(u^y\right)^2 + 1 \right) \\
&- 2 \left(\left(u^y\right)^2 + 1 \right) \left({\pi_Q}^\eta{}_\eta \left(u^\tau\right)^2 - 4\pi_Q^{xy} u^x u^y \right) \\
&- \pi_Q^{yy} \left(\left(u^x\right)^2 \left(8 \left(u^y\right)^2 + 5 \right) + 5 \left(\left(u^y\right)^2 + 1 \right) \right) \Big) \Big) \\
(A_x)_{10,3} = &\frac{1}{3T\left(u^\tau\right)^2} D_s \Big(u^x \Big(3I_{31} \left(u^\tau\right)^2 u^y - \mathcal{A}\nu^y \left(u^x\right)^2 \left(\left(u^y\right)^2 - 2 \right) + 2\mathcal{A}\nu^y \left(\left(u^y\right)^2 + 1 \right) \\
&+ \mathcal{A}\nu^x u^x u^y \left(\left(u^y\right)^2 + 1 \right) \\
&+ 3\mathcal{B} \left(\nu^x u^x \left(u^y\right)^3 - \nu^y \left(u^x\right)^2 \left(\left(u^y\right)^2 - 1 \right) + \nu^y \right) \Big) \\
&+ \mathcal{C} \Big(-u^x u^y \left(-3\Pi_Q \left(u^\tau\right)^2 + 4\pi_Q^{yy} \left(\left(u^x\right)^2 + 1 \right) + {\pi_Q}^\eta{}_\eta \left(u^\tau\right)^2 \right) \\
&+ 2\pi_Q^{xy} \left(\left(u^x\right)^2 \left(2 \left(u^y\right)^2 + 1 \right) + \left(u^y\right)^2 + 1 \right) \Big) \Big) \\
(A_x)_{10,8} = &-\frac{\mathcal{B} D_s \left(\left(u^y\right)^2 + 1 \right)}{3T} \\
(A_x)_{10,9} = &\frac{\mathcal{B} D_s u^x u^y}{T} \\
(A_x)_{10,10} = &\frac{\mathcal{C} D_s u^x}{2T} \\
(A_x)_{11,1} = &-\frac{D_s (2\mathcal{B}'(T)\nu^x - 3\mathcal{C}'(T){\pi_Q}^\eta{}_\eta u^x)}{6T} \\
(A_x)_{11,2} = &-\frac{1}{6T\left(u^\tau\right)^2} D_s \Big(2I_{31} \left(u^\tau\right)^2 + 2\mathcal{A}u^x \left(\nu^x \left(u^y\right)^2 + \nu^x - \nu^y u^x u^y \right) \\
&+ \mathcal{C} \Big(-5{\pi_Q}^\eta{}_\eta + 2\Pi_Q \left(u^\tau\right)^2 - 5{\pi_Q}^\eta{}_\eta \left(u^x\right)^2 - 2\pi_Q^{yy} \left(\left(u^x\right)^2 + 1 \right) \\
&+ 2\pi_Q^{xy} u^x u^y - 5{\pi_Q}^\eta{}_\eta \left(u^y\right)^2 \Big) \Big) \\
(A_x)_{11,3} = &\frac{1}{3T\left(u^\tau\right)^2 \left(\left(u^y\right)^2 + 1 \right)} D_s \Big(-\mathcal{C} u^x u^y \left(\pi_Q^{yy} \left(\left(u^x\right)^2 + 1 \right) + {\pi_Q}^\eta{}_\eta \left(u^\tau\right)^2 \right) \\
&- \mathcal{A}u^x \left(\left(u^y\right)^2 + 1 \right) \left(-\nu^x u^x u^y + \nu^y \left(u^x\right)^2 + \nu^y \right) \\
&+ \mathcal{C}\pi_Q^{xy} \left(\left(u^x\right)^2 \left(\left(u^y\right)^2 - 1 \right) - \left(u^y\right)^2 - 1 \right) \Big) \\
(A_x)_{11,8} = &-\frac{\mathcal{B} D_s}{3T} \\
(A_x)_{11,11} = &\frac{\mathcal{C} D_s u^x}{2T} \\
(A_x)_{12,1} = &\frac{D_s \left(\mathcal{B}'(T)\nu^x u^x u^y + 3\mathcal{B}'(T)\nu^y \left(\left(u^x\right)^2 + 1 \right) + 3\mathcal{C}'(T)\pi_Q^{xy} u^x \right)}{6T} \\
(A_x)_{12,2} = &\frac{1}{6T\left(u^\tau\right)^2} D_s \Big(3\mathcal{B}\nu^y u^x + 3\mathcal{B}\nu^y \left(u^x\right)^3 - 3\mathcal{B}\nu^x u^y + I_{31} u^x u^y - 6\mathcal{B}\nu^x \left(u^x\right)^2 u^y + I_{31} \left(u^x\right)^3 u^y \\
&+ 6\mathcal{B}\nu^y u^x \left(u^y\right)^2 + 6\mathcal{B}\nu^y \left(u^x\right)^3 \left(u^y\right)^2 - 3\mathcal{B}\nu^x \left(u^y\right)^3 + I_{31} u^x \left(u^y\right)^3 \\
&- 6\mathcal{B}\nu^x \left(u^x\right)^2 \left(u^y\right)^3 \\
&+ \mathcal{A}u^x \Big(3\nu^y \left(\left(u^y\right)^2 + 1 \right) - 2\nu^x u^x u^y \left(\left(u^y\right)^2 + 1 \right) \\
&+ \nu^y \left(u^x\right)^2 \left(2 \left(u^y\right)^2 + 3 \right) \Big) \\
&+ \mathcal{C} \Big(u^x u^y \left(\Pi_Q \left(u^\tau\right)^2 + \pi_Q^{yy} \left(8 \left(u^x\right)^2 + 5 \right) + 5{\pi_Q}^\eta{}_\eta \left(u^\tau\right)^2 \right) \\
&+ \pi_Q^{xy} \left(\left(u^x\right)^2 \left(3 - 8 \left(u^y\right)^2 \right) + 6 \left(\left(u^y\right)^2 + 1 \right) \right) \Big) \Big)
\end{aligned}
$$

$$
\begin{aligned}
(A_x)_{12,3} = &\frac{1}{6T\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right)}D_s\Big(3I_{31}\left(u^\tau\right)^2\left(\left(u^x\right)^2+1\right)\left(\left(u^y\right)^2+1\right)\\
&+\mathcal{C}\Big(-3{\pi_Q}^\eta{}_\eta-6{\pi_Q}^\eta{}_\eta\left(u^x\right)^2-3{\pi_Q}^\eta{}_\eta\left(u^x\right)^4+\pi_Q^{xy}u^xu^y\\
&+4\pi_Q^{xy}\left(u^x\right)^3u^y-6{\pi_Q}^\eta{}_\eta\left(u^y\right)^2-11{\pi_Q}^\eta{}_\eta\left(u^x\right)^2\left(u^y\right)^2\\
&-5{\pi_Q}^\eta{}_\eta\left(u^x\right)^4\left(u^y\right)^2+\pi_Q^{xy}u^x\left(u^y\right)^3+8\pi_Q^{xy}\left(u^x\right)^3\left(u^y\right)^3\\
&-3{\pi_Q}^\eta{}_\eta\left(u^y\right)^4-5{\pi_Q}^\eta{}_\eta\left(u^x\right)^2\left(u^y\right)^4\\
&+3\Pi_Q\left(u^\tau\right)^2\left(\left(u^x\right)^2+1\right)\left(\left(u^y\right)^2+1\right)\\
&-\pi_Q^{yy}\left(\left(u^x\right)^2+1\right)\left(\left(u^x\right)^2\left(8\left(u^y\right)^2+6\right)+3\left(\left(u^y\right)^2+1\right)\right)\Big)\\
&+\left(u^y\right)^2+1\Big(\mathcal{A}u^x\Big(-2\nu^yu^x\left(\left(u^x\right)^2+1\right)u^y+3\nu^x\left(\left(u^y\right)^2+1\right)\\
&+\nu^x\left(u^x\right)^2\left(2\left(u^y\right)^2+3\right)\Big)\\
&+3\mathcal{B}\left(\left(u^x\right)^2+1\right)\left(\nu^x\left(2u^x\left(u^y\right)^2+u^x\right)-\nu^y\left(2\left(u^x\right)^2+1\right)u^y\right)\Big)\Big)\\
(A_x)_{12,8} = &\frac{\mathcal{B}D_su^xu^y}{6T}\\
(A_x)_{12,9} = &\frac{\mathcal{B}D_s\left(\left(u^x\right)^2+1\right)}{2T}\\
(A_x)_{12,12} = &\frac{\mathcal{C}D_su^x}{2T}\\
(A_x)_{13,1} = &\frac{D_s(5\mathcal{B}'(T)\nu^x+3u^x(\mathcal{C}'(T)\Pi_Q+(\partial_TI_{31})))}{6T}\\
(A_x)_{13,2} = &\frac{1}{6T\left(u^\tau\right)^2}D_s\Big(5I_{31}\left(u^\tau\right)^2+5\mathcal{C}\Pi_Q\left(u^\tau\right)^2+2\mathcal{A}u^x\left(\nu^x\left(u^y\right)^2+\nu^x-\nu^yu^xu^y\right)\\
&-2\mathcal{C}\left(-\pi_Q^{xy}u^xu^y+\pi_Q^{yy}\left(u^x\right)^2+\pi_Q^{yy}+{\pi_Q}^\eta{}_\eta\left(u^x\right)^2+{\pi_Q}^\eta{}_\eta\left(u^y\right)^2+{\pi_Q}^\eta{}_\eta\right)\Big)\\
(A_x)_{13,3} = &\frac{1}{3T\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right)}D_s\Big(\mathcal{C}u^xu^y\left(\pi_Q^{yy}\left(\left(u^x\right)^2+1\right)+{\pi_Q}^\eta{}_\eta\left(u^\tau\right)^2\right)\\
&+\mathcal{A}u^x\left(\left(u^y\right)^2+1\right)\left(-\nu^xu^xu^y+\nu^y\left(u^x\right)^2+\nu^y\right)\\
&+\mathcal{C}\pi_Q^{xy}\left(-\left(\left(u^x\right)^2\left(\left(u^y\right)^2-1\right)\right)+\left(u^y\right)^2+1\right)\Big)\\
(A_x)_{13,7} = &\frac{(\partial_\alpha I_{31})D_su^x}{2T}\\
(A_x)_{13,8} = &\frac{5\mathcal{B}D_s}{6T}\\
(A_x)_{13,13} = &\frac{\mathcal{C}D_su^x}{2T}\\
(A_y)_{1,1} = &\frac{\left(u^\tau\right)^2u^y\left(16a_{\mathrm{SB}}T^3+(\partial_Tn)h\right)+h'(T)\left(n\left(u^\tau\right)^2u^y+\nu^xu^xu^y+\nu^y\left(\left(u^x\right)^2+2\left(u^y\right)^2+1\right)\right)}{u^\tau}\\
(A_y)_{1,2} = &\frac{1}{\left(u^\tau\right)^3}\Big(u^xu^y\left(4a_{\mathrm{SB}}T^4\left(u^\tau\right)^2-\pi^{yy}\right)+\pi^{xy}\left(\left(u^y\right)^2+1\right)\\
&+h\left(n\left(u^\tau\right)^2u^xu^y+\nu^xu^y\left(\left(u^y\right)^2+1\right)+\nu^y\left(\left(u^x\right)^3+u^x\right)\right)\Big)\\
(A_y)_{1,3} = &\frac{1}{\left(u^\tau\right)^3}\Big(\pi^{yy}+4a_{\mathrm{SB}}T^4+\pi^{yy}\left(u^x\right)^2+8a_{\mathrm{SB}}T^4\left(u^x\right)^2+4a_{\mathrm{SB}}T^4\left(u^x\right)^4-\pi^{xy}u^xu^y\\
&+12a_{\mathrm{SB}}T^4\left(u^y\right)^2+12a_{\mathrm{SB}}T^4\left(u^x\right)^2\left(u^y\right)^2+8a_{\mathrm{SB}}T^4\left(u^y\right)^4\\
&+h\Big(\nu^x\left(\left(u^x\right)^3+u^x\right)+\nu^yu^y\left(3\left(u^x\right)^2+2\left(u^y\right)^2+3\right)\\
&+n\left(\left(u^x\right)^4+\left(u^x\right)^2\left(3\left(u^y\right)^2+2\right)+2\left(u^y\right)^4+3\left(u^y\right)^2+1\right)\Big)\Big)\\
(A_y)_{1,4} = &\frac{u^y}{u^\tau}
\end{aligned}
$$

$$(A_y)_{1,6} = \frac{u^x}{u^\tau}$$
$$(A_y)_{1,7} = (\partial_\alpha n) h u^\tau u^y$$
$$(A_y)_{1,8} = \frac{h u^x u^y}{u^\tau}$$
$$(A_y)_{1,9} = \frac{h\left((u^x)^2 + 2\,(u^y)^2 + 1\right)}{u^\tau}$$
$$(A_y)_{2,1} = u^x u^y \left(16 a_{\mathrm{SB}} T^3 + (\partial_T n) h\right) + h'(T) u^y (n u^x + \nu^x) + h'(T) \nu^y u^x$$
$$(A_y)_{2,2} = 4 a_{\mathrm{SB}} T^4 u^y + h(n u^y + \nu^y)$$
$$(A_y)_{2,3} = 4 a_{\mathrm{SB}} T^4 u^x + h(n u^x + \nu^x)$$
$$(A_y)_{2,6} = 1$$
$$(A_y)_{2,7} = (\partial_\alpha n) h u^x u^y$$
$$(A_y)_{2,8} = h u^y$$
$$(A_y)_{2,9} = h u^x$$
$$(A_y)_{3,1} = 16 a_{\mathrm{SB}} T^3 (u^y)^2 + 4 a_{\mathrm{SB}} T^3 + h'(T) n (u^y)^2 + 2 h'(T) \nu^y u^y + (\partial_T n) h (u^y)^2 + (\partial_T n) T + n$$
$$(A_y)_{3,3} = 8 a_{\mathrm{SB}} T^4 u^y + 2h(n u^y + \nu^y)$$
$$(A_y)_{3,4} = 1$$
$$(A_y)_{3,7} = (\partial_\alpha n)\left(h (u^y)^2 + T\right)$$
$$(A_y)_{3,9} = 2 h u^y$$
$$(A_y)_{4,2} = -\frac{2\tau_\pi (u^y)^2 \left(\pi^{xy} (u^y)^2 + \pi^{xy} - \pi^{yy} u^x u^y\right)}{(u^\tau)^2}$$
$$(A_y)_{4,3} = \frac{1}{3\,(u^\tau)^2}\Big(3\delta_{\pi\pi}\pi^{yy} (u^\tau)^2 - 6\pi^{yy}\tau_\pi \left((u^x)^2 + 1\right)(u^y)^2 + 6\pi^{xy}\tau_\pi u^x (u^y)^3 + 4\eta$$
$$+ 4\,(u^x)^2\,\eta + 8\,(u^y)^2\,\eta + 4\,(u^x)^2\,(u^y)^2\,\eta + 4\,(u^y)^4\,\eta\Big)$$
$$(A_y)_{4,4} = \tau_\pi u^y$$
$$(A_y)_{5,3} = \frac{3\delta_{\pi\pi}\pi^\eta{}_\eta - 2\eta}{3\tau^2}$$
$$(A_y)_{5,5} = \frac{\tau_\pi u^y}{\tau^2}$$
$$(A_y)_{6,2} = \frac{1}{(u^\tau)^2}\Big(\pi^{yy}\tau_\pi (u^y)^2 + \pi^\eta{}_\eta \tau_\pi (u^\tau)^2 (u^y)^2 + 2\pi^{yy}\tau_\pi (u^x)^2 (u^y)^2$$
$$- \pi^{xy}\tau_\pi u^x u^y \left(2\,(u^y)^2 + 1\right) + \eta + (u^x)^2\,\eta + 2\,(u^y)^2\,\eta + (u^x)^2\,(u^y)^2\,\eta + (u^y)^4\,\eta\Big)$$
$$(A_y)_{6,3} = \delta_{\pi\pi}\pi^{xy}$$
$$- \frac{1}{3\,(u^\tau)^2\left((u^y)^2 + 1\right)} u^y \Big(3\pi^{yy}\tau_\pi u^x \left((u^x)^2 + 1\right)\left(2\,(u^y)^2 + 1\right)$$
$$+ 3\pi^{xy}\tau_\pi u^y \left(\left(1 - 2\,(u^x)^2\right)(u^y)^2 + 1\right)$$
$$+ (u^\tau)^2 u^x \left(3\pi^\eta{}_\eta \tau_\pi (u^y)^2 - \left((u^y)^2 + 1\right)\eta\right)\Big)$$
$$(A_y)_{6,6} = \tau_\pi u^y$$
$$(A_y)_{7,1} = (\partial_T n) u^y$$
$$(A_y)_{7,3} = n$$
$$(A_y)_{7,7} = (\partial_\alpha n) u^y$$
$$(A_y)_{7,9} = 1$$
$$(A_y)_{8,1} = \frac{D_s u^y (h'(T)\nu^x + u^x((\partial_T n) T + n))}{T}$$
$$(A_y)_{8,2} = \frac{D_s h\left(n\,(u^\tau)^2\,u^y - \nu^x u^x u^y \left((u^y)^2 + 1\right) + \nu^y \left((u^x)^2 + 1\right)\left((u^y)^2 + 1\right)\right)}{T\,(u^\tau)^2}$$

$$(A_y)_{8,3} = \frac{D_s h \left((u^x)^2 + 1 \right) \left(\nu^x (u^y)^2 + \nu^x - \nu^y u^x u^y \right)}{T (u^\tau)^2}$$
$$(A_y)_{8,7} = (\partial_\alpha n) D_s u^x u^y$$
$$(A_y)_{8,8} = \frac{D_s h u^y}{T}$$
$$(A_y)_{9,1} = \frac{D_s \left(h'(T) \nu^y u^y + (\partial_T n) T \left((u^y)^2 + 1 \right) + n \left((u^y)^2 + 1 \right) \right)}{T}$$
$$(A_y)_{9,2} = -\frac{D_s h (u^y)^2 \left(\nu^x (u^y)^2 + \nu^x - \nu^y u^x u^y \right)}{T (u^\tau)^2}$$
$$(A_y)_{9,3} = \frac{D_s h \left(n (u^\tau)^2 u^y + \nu^x u^x (u^y)^3 + \nu^y \left(- \left((u^x)^2 \left((u^y)^2 - 2 \right) \right) + (u^y)^2 + 2 \right) \right)}{T (u^\tau)^2}$$
$$(A_y)_{9,7} = (\partial_\alpha n) D_s \left((u^y)^2 + 1 \right)$$
$$(A_y)_{9,9} = \frac{D_s h u^y}{T}$$
$$(A_y)_{10,1} = \frac{D_s \left(4\mathcal{B}'(T) \nu^y \left((u^y)^2 + 1 \right) + 3\mathcal{C}'(T) \pi_Q^{yy} u^y \right)}{6T}$$
$$\begin{aligned}(A_y)_{10,2} = &-\frac{1}{3T (u^\tau)^2} D_s \Big(u^y \left((u^y)^2 + 1 \right) (\mathcal{A} + 3\mathcal{B}) \left(\nu^x (u^y)^2 + \nu^x - \nu^y u^x u^y \right) \\ &+ \mathcal{C} \left(4 (u^y)^2 + 1 \right) \left(\pi_Q^{xy} (u^y)^2 + \pi_Q^{xy} - \pi_Q^{yy} u^x u^y \right) \Big)\end{aligned}$$
$$\begin{aligned}(A_y)_{10,3} = &\frac{1}{6T (u^\tau)^2} D_s \Big(4 I_{31} (u^\tau)^2 \left((u^y)^2 + 1 \right) \\ &+ \mathcal{C} \Big(4\Pi_Q (u^\tau)^2 \left((u^y)^2 + 1 \right) + 2\pi_Q^{xy} u^x u^y \left(4 (u^y)^2 + 1 \right) \\ &+ \pi_Q^{yy} \left((u^x)^2 \left(7 - 8 (u^y)^2 \right) + (u^y)^2 + 7 \right) \Big) \\ &+ 2u^y \Big(3\mathcal{B} u^x u^y \left(\nu^x (u^y)^2 + \nu^x - \nu^y u^x u^y \right) \\ &+ \mathcal{A} \Big(-\nu^y (u^x)^2 \left((u^y)^2 - 2 \right) + 2\nu^y \left((u^y)^2 + 1 \right) \\ &+ \nu^x u^x u^y \left((u^y)^2 + 1 \right) \Big) \Big) \Big) \Big)\end{aligned}$$
$$(A_y)_{10,9} = \frac{2\mathcal{B} D_s \left((u^y)^2 + 1 \right)}{3T}$$
$$(A_y)_{10,10} = \frac{\mathcal{C} D_s u^y}{2T}$$
$$(A_y)_{11,1} = -\frac{D_s (2\mathcal{B}'(T) \nu^y - 3\mathcal{C}'(T) \pi_Q{}^\eta{}_\eta u^y)}{6T}$$
$$(A_y)_{11,2} = -\frac{D_s \left(\mathcal{A} u^y \left(\nu^x (u^y)^2 + \nu^x - \nu^y u^x u^y \right) + \mathcal{C} \left(\pi_Q^{xy} (u^y)^2 + \pi_Q^{xy} - \pi_Q^{yy} u^x u^y \right) \right)}{3T (u^\tau)^2}$$
$$\begin{aligned}(A_y)_{11,3} = &-\frac{1}{6T (u^\tau)^2} D_s \Big(2 I_{31} (u^\tau)^2 + 2\mathcal{A} u^y \left(-\nu^x u^x u^y + \nu^y (u^x)^2 + \nu^y \right) \\ &+ \mathcal{C} \Big(-3\pi_Q{}^\eta{}_\eta + 2\Pi_Q (u^\tau)^2 - 3\pi_Q{}^\eta{}_\eta (u^x)^2 + 2\pi_Q^{yy} \left((u^x)^2 + 1 \right) \\ &- 2\pi_Q^{xy} u^x u^y - 3\pi_Q{}^\eta{}_\eta (u^y)^2 \Big) \Big)\end{aligned}$$
$$(A_y)_{11,9} = -\frac{\mathcal{B} D_s}{3T}$$
$$(A_y)_{11,11} = \frac{\mathcal{C} D_s u^y}{2T}$$
$$(A_y)_{12,1} = \frac{D_s \left(3\mathcal{B}'(T) \nu^x \left((u^y)^2 + 1 \right) + \mathcal{B}'(T) \nu^y u^x u^y + 3\mathcal{C}'(T) \pi_Q^{xy} u^y \right)}{6T}$$

$$
\begin{aligned}
(A_y)_{12,2} = &\frac{1}{6T\left(u^\tau\right)^2} D_s\Big(-3\mathcal{B}\nu^x u^x + 3\mathcal{A}\nu^y u^y + 3\mathcal{B}\nu^y u^y + 3\mathcal{A}\nu^y \left(u^x\right)^2 u^y + 6\mathcal{B}\nu^y \left(u^x\right)^2 u^y \\
&- 2\mathcal{A}\nu^x u^x \left(u^y\right)^2 - 9\mathcal{B}\nu^x u^x \left(u^y\right)^2 + 3\mathcal{A}\nu^y \left(u^y\right)^3 + 3\mathcal{B}\nu^y \left(u^y\right)^3 \\
&+ 2\mathcal{A}\nu^y \left(u^x\right)^2 \left(u^y\right)^3 + 6\mathcal{B}\nu^y \left(u^x\right)^2 \left(u^y\right)^3 - 2\mathcal{A}\nu^x u^x \left(u^y\right)^4 - 6\mathcal{B}\nu^x u^x \left(u^y\right)^4 \\
&+ 3I_{31} \left(u^\tau\right)^2 \Big(\left(u^y\right)^2 + 1\Big) \\
&+ \mathcal{C}\Big(3\Pi_Q \left(u^\tau\right)^2 \Big(\left(u^y\right)^2 + 1\Big) \\
&+ \pi_Q^{yy} \Big(\left(u^x\right)^2 \Big(8\left(u^y\right)^2 + 3\Big) + 6\left(u^y\right)^2 + 3\Big) \\
&+ u^y \Big(3\pi_Q{}^\eta{}_\eta \left(u^\tau\right)^2 u^y - \pi_Q^{xy} u^x \Big(8\left(u^y\right)^2 + 5\Big)\Big)\Big)\Big) \\
(A_y)_{12,3} = &\frac{1}{6T\left(u^\tau\right)^2 \Big(\left(u^y\right)^2 + 1\Big)} D_s\Big(\left(u^y\right)^2 + 1\Big(I_{31} \left(u^\tau\right)^2 u^x u^y \\
&+ 3\mathcal{B}\nu^x \Big(2\left(u^x\right)^2 + 1\Big) u^y \Big(\left(u^y\right)^2 + 1\Big) \\
&- 3\mathcal{B}\nu^y u^x \Big(\left(u^x\right)^2 + 1\Big) \Big(2\left(u^y\right)^2 + 1\Big) \\
&+ \mathcal{A}u^y \Big(-2\nu^y u^x \Big(\left(u^x\right)^2 + 1\Big) u^y + 3\nu^x \Big(\left(u^y\right)^2 + 1\Big) \\
&+ \nu^x \left(u^x\right)^2 \Big(2\left(u^y\right)^2 + 3\Big)\Big)\Big) \\
&+ \mathcal{C}\Big(u^x u^y \Big(-3\pi_Q{}^\eta{}_\eta \left(u^\tau\right)^2 \left(u^y\right)^2 + \Pi_Q \left(u^\tau\right)^2 \Big(\left(u^y\right)^2 + 1\Big) \\
&- \pi_Q^{yy} \Big(\left(u^x\right)^2 + 1\Big) \Big(8\left(u^y\right)^2 + 5\Big)\Big) \\
&+ \pi_Q^{xy} \Big(3\Big(\left(u^y\right)^4 + 3\left(u^y\right)^2 + 2\Big) \\
&+ \left(u^x\right)^2 \Big(8\left(u^y\right)^4 + 8\left(u^y\right)^2 + 6\Big)\Big)\Big)\Big) \\
(A_y)_{12,8} = &\frac{\mathcal{B}D_s \Big(\left(u^y\right)^2 + 1\Big)}{2T} \\
(A_y)_{12,9} = &\frac{\mathcal{B}D_s u^x u^y}{6T} \\
(A_y)_{12,12} = &\frac{\mathcal{C}D_s u^y}{2T} \\
(A_y)_{13,1} = &\frac{D_s(5\mathcal{B}'(T)\nu^y + 3u^y(\mathcal{C}'(T)\Pi_Q + (\partial_T I_{31})))}{6T} \\
(A_y)_{13,2} = &\frac{D_s \left(\mathcal{A}u^y \left(\nu^x \left(u^y\right)^2 + \nu^x - \nu^y u^x u^y\right) + \mathcal{C}\left(\pi_Q^{xy} \left(u^y\right)^2 + \pi_Q^{xy} - \pi_Q^{yy} u^x u^y\right)\right)}{3T\left(u^\tau\right)^2} \\
(A_y)_{13,3} = &\frac{1}{6T\left(u^\tau\right)^2} D_s\Big(5I_{31} \left(u^\tau\right)^2 + 5\mathcal{C}\Pi_Q \left(u^\tau\right)^2 + 2\mathcal{A}u^y \Big(-\nu^x u^x u^y + \nu^y \left(u^x\right)^2 + \nu^y\Big) \\
&+ 2\mathcal{C}\Big(-\pi_Q^{xy} u^x u^y + \pi_Q^{yy} \left(u^x\right)^2 + \pi_Q^{yy}\Big)\Big) \\
(A_y)_{13,7} = &\frac{(\partial_\alpha I_{31}) D_s u^y}{2T} \\
(A_y)_{13,9} = &\frac{5\mathcal{B}D_s}{6T} \\
(A_y)_{13,13} = &\frac{\mathcal{C}D_s u^y}{2T} \\
S_1 = &\frac{1}{\tau\left(\left(u^y\right)^2 + 1\right)} \Big(4a_{\mathrm{SB}}T^4 - \pi^{yy} \left(u^x\right)^2 + 4a_{\mathrm{SB}}T^4 \left(u^x\right)^2 + 2\pi^{xy} u^x u^y + \pi^{yy} \left(u^y\right)^2 \\
&+ 8a_{\mathrm{SB}}T^4 \left(u^y\right)^2 + 4a_{\mathrm{SB}}T^4 \left(u^x\right)^2 \left(u^y\right)^2 + 4a_{\mathrm{SB}}T^4 \left(u^y\right)^4 \\
&+ \pi^\eta{}_\eta \Big(-\left(u^x\right)^2 + \left(u^y\right)^2 + 1\Big) \\
&+ h\Big(\left(u^y\right)^2 + 1\Big) \Big(n\left(u^\tau\right)^2 + 2(\nu^x u^x + \nu^y u^y)\Big)\Big)
\end{aligned}
$$

$$
\begin{aligned}
S_2 &= \frac{1}{\tau u^\tau \left((u^y)^2+1\right)}\Big(-\pi^{yy}u^x + 4a_{\mathrm{SB}}T^4u^x - \pi^\eta{}_\eta\,(u^\tau)^2\,u^x - \pi^{yy}\,(u^x)^3 + 4a_{\mathrm{SB}}T^4\,(u^x)^3 \\
&\quad + \pi^{xy}u^y + 2\pi^{xy}\,(u^x)^2\,u^y + 8a_{\mathrm{SB}}T^4u^x\,(u^y)^2 + 4a_{\mathrm{SB}}T^4\,(u^x)^3\,(u^y)^2 \\
&\quad + \pi^{xy}\,(u^y)^3 + 4a_{\mathrm{SB}}T^4u^x\,(u^y)^4 \\
&\quad + h\left((u^y)^2+1\right)\left(n\,(u^\tau)^2\,u^x + \nu^x\left(2\,(u^x)^2 + (u^y)^2 + 1\right) + \nu^y u^x u^y\right)\Big) \\
S_3 &= \frac{u^y\left(4a_{\mathrm{SB}}T^4\,(u^\tau)^2 + \pi^{yy}\right) + h\left(n\,(u^\tau)^2\,u^y + \nu^x u^x u^y + \nu^y\left((u^x)^2 + 2\,(u^y)^2 + 1\right)\right) + \pi^{xy}u^x}{\tau u^\tau} \\
S_4 &= \frac{\delta_{\pi\pi}\pi^{yy}u^\tau}{\tau} + \pi^{yy} - \frac{2u^\tau\left((u^y)^2+1\right)\eta}{3\tau} \\
S_5 &= \frac{3\pi^\eta{}_\eta(\delta_{\pi\pi}u^\tau + \tau) + 4u^\tau\eta}{3\tau^3} \\
S_6 &= \frac{\delta_{\pi\pi}\pi^{xy}u^\tau}{\tau} + \pi^{xy} - \frac{2u^\tau u^x u^y\eta}{3\tau} \\
S_7 &= \frac{n\,(u^\tau)^2 + \nu^x u^x + \nu^y u^y}{\tau u^\tau} \\
S_8 &= \frac{D_s h\nu^x u^\tau}{T\tau} + \nu^x \\
S_9 &= \frac{D_s h\nu^y u^\tau}{T\tau} + \nu^y \\
S_{10} &= \pi_Q^{yy} \\
&\quad - \frac{1}{6T\tau u^\tau}D_s\Big(2I_{31}\,(u^\tau)^2\left((u^y)^2+1\right) + 2\mathcal{B}\left((u^y)^2+1\right)(\nu^x u^x + \nu^y u^y) \\
&\quad + \mathcal{C}\,(u^\tau)^2\left(2\Pi_Q\left((u^y)^2+1\right) - 3\pi_Q^{yy} + 2\pi_Q{}^\eta{}_\eta\left((u^y)^2+1\right)\right)\Big) \\
S_{11} &= \frac{D_s\left(4\mathcal{B}(\nu^x u^x + \nu^y u^y) + \mathcal{C}\,(u^\tau)^2\,(4\Pi_Q + 7\pi_Q{}^\eta{}_\eta) + 4I_{31}\,(u^\tau)^2\right)}{6T\tau u^\tau} + \pi_Q{}^\eta{}_\eta \\
S_{12} &= -\frac{D_s u^x u^y\left(\mathcal{B}(\nu^x u^x + \nu^y u^y) + I_{31}\,(u^\tau)^2\right)}{3T\tau u^\tau} + \frac{\mathcal{C}D_s u^\tau(3\pi_Q^{xy} - 2u^x u^y(\Pi_Q + \pi_Q{}^\eta{}_\eta))}{6T\tau} + \pi_Q^{xy} \\
S_{13} &= \frac{D_s\left(5\mathcal{B}(\nu^x u^x + \nu^y u^y) + \mathcal{C}\,(u^\tau)^2\,(5\Pi_Q + 2\pi_Q{}^\eta{}_\eta) + 5I_{31}\,(u^\tau)^2\right)}{6T\tau u^\tau} + \Pi_Q
\end{aligned}
$$

A.2.5 S6, fully coupled

`two_d_two_way_cM_matrix` in `src/Matrix/2d_two_way_cM.jl`, fields:

$$
\phi = (T, u^x, u^y, \pi^{yy}, \pi^\eta{}_\eta, \pi^{xy}, \Pi, \alpha, \nu^x, \nu^y, \pi_Q^{yy}, \pi_Q{}^\eta{}_\eta, \pi_Q^{xy}, \Pi_Q)
$$

$$
\begin{aligned}
(A_t)_{1,1} &= -n - (\partial_T n)T + P''(T)T + (\partial_T n)h\,(u^\tau)^2 + h'(T)n\,(u^\tau)^2 + 2h'(T)\nu^x u^x \\
&\quad + P'(T)\,(u^x)^2 + P''(T)T\,(u^x)^2 + 2h'(T)\nu^y u^y + P'(T)\,(u^y)^2 + P''(T)T\,(u^y)^2 \\
(A_t)_{1,2} &= \frac{2}{(u^y)^2+1}\Big(\Pi_Q u^x - \pi^\eta{}_\eta u^x - \pi^{yy}u^x - \pi_Q^{yy}u^x - \pi_Q{}^\eta{}_\eta u^x + P'(T)Tu^x + \pi^{xy}u^y + \pi_Q^{xy}u^y \\
&\quad + \Pi_Q u^x\,(u^y)^2 + P'(T)Tu^x\,(u^y)^2 + \Pi u^x\left((u^y)^2+1\right) \\
&\quad + h\left((u^y)^2+1\right)(nu^x + \nu^x)\Big) \\
(A_t)_{1,3} &= \frac{2}{\left((u^y)^2+1\right)^2}\Big(\pi_Q^{xy}u^x + \Pi u^y + \Pi_Q u^y + \pi^{yy}u^y + \pi_Q^{yy}u^y + P'(T)Tu^y + \pi^\eta{}_\eta\,(u^x)^2\,u^y \\
&\quad + \pi^{yy}\,(u^x)^2\,u^y + \pi_Q^{yy}\,(u^x)^2\,u^y + \pi_Q{}^\eta{}_\eta\,(u^x)^2\,u^y - \pi_Q^{xy}u^x\,(u^y)^2 + 2\Pi\,(u^y)^3 \\
&\quad + 2\Pi_Q\,(u^y)^3 + 2P'(T)T\,(u^y)^3 + \Pi\,(u^y)^5 + \Pi_Q\,(u^y)^5 + P'(T)T\,(u^y)^5 \\
&\quad + h\left((u^y)^2+1\right)^2(nu^y + \nu^y) + \pi^{xy}\left(u^x - u^x\,(u^y)^2\right)\Big)
\end{aligned}
$$

$$
\begin{aligned}
(A_t)_{1,4} &= \frac{(u^y)^2 - (u^x)^2}{(u^y)^2 + 1} \\
(A_t)_{1,5} &= -\frac{(u^x)^2}{(u^y)^2 + 1} \\
(A_t)_{1,6} &= \frac{2u^x u^y}{(u^y)^2 + 1} \\
(A_t)_{1,7} &= (u^x)^2 + (u^y)^2 \\
(A_t)_{1,8} &= (\partial_\alpha n)\left(h\,(u^\tau)^2 - T\right) \\
(A_t)_{1,9} &= 2hu^x \\
(A_t)_{1,10} &= 2hu^y \\
(A_t)_{1,11} &= \frac{(u^y)^2 - (u^x)^2}{(u^y)^2 + 1} \\
(A_t)_{1,12} &= -\frac{(u^x)^2}{(u^y)^2 + 1} \\
(A_t)_{1,13} &= \frac{2u^x u^y}{(u^y)^2 + 1} \\
(A_t)_{1,14} &= (u^x)^2 + (u^y)^2 \\
(A_t)_{2,1} &= \frac{1}{u^\tau}\Big((u^\tau)^2 u^x((\partial_T n)h + P''(T)T + P'(T)) \\
&\quad + h'(T)\left(n\,(u^\tau)^2 u^x + \nu^x\left(2\,(u^x)^2 + (u^y)^2 + 1\right) + \nu^y u^x u^y\right)\Big) \\
(A_t)_{2,2} &= \frac{1}{(u^\tau)^3\left((u^y)^2+1\right)}\Big(\Pi_Q - \pi^\eta{}_\eta - \pi^{yy} - \pi_Q^{yy} - \pi_Q{}^\eta{}_\eta + P'(T)T + 3\Pi_Q\,(u^x)^2 \\
&\quad - 3\pi^\eta{}_\eta\,(u^x)^2 - 3\pi^{yy}\,(u^x)^2 - 3\pi_Q^{yy}\,(u^x)^2 - 3\pi_Q{}^\eta{}_\eta\,(u^x)^2 \\
&\quad + 3P'(T)T\,(u^x)^2 + 2\Pi_Q\,(u^x)^4 - 2\pi^\eta{}_\eta\,(u^x)^4 - 2\pi^{yy}\,(u^x)^4 \\
&\quad - 2\pi_Q^{yy}\,(u^x)^4 - 2\pi_Q{}^\eta{}_\eta\,(u^x)^4 + 2P'(T)T\,(u^x)^4 + 3\pi^{xy}u^x u^y \\
&\quad + 3\pi_Q^{xy}u^x u^y + 2\pi^{xy}\,(u^x)^3\,u^y + 2\pi_Q^{xy}\,(u^x)^3\,u^y + 3\Pi_Q\,(u^y)^2 \\
&\quad - 2\pi^\eta{}_\eta\,(u^y)^2 - \pi^{yy}\,(u^y)^2 - \pi_Q^{yy}\,(u^y)^2 - 2\pi_Q{}^\eta{}_\eta\,(u^y)^2 \\
&\quad + 3P'(T)T\,(u^y)^2 + 6\Pi_Q\,(u^x)^2\,(u^y)^2 - 3\pi^\eta{}_\eta\,(u^x)^2\,(u^y)^2 \\
&\quad - 3\pi^{yy}\,(u^x)^2\,(u^y)^2 - 3\pi_Q^{yy}\,(u^x)^2\,(u^y)^2 - 3\pi_Q{}^\eta{}_\eta\,(u^x)^2\,(u^y)^2 \\
&\quad + 6P'(T)T\,(u^x)^2\,(u^y)^2 + 2\Pi_Q\,(u^x)^4\,(u^y)^2 + 2P'(T)T\,(u^x)^4\,(u^y)^2 \\
&\quad + 3\pi^{xy}u^x\,(u^y)^3 + 3\pi_Q^{xy}u^x\,(u^y)^3 + 3\Pi_Q\,(u^y)^4 - \pi^\eta{}_\eta\,(u^y)^4 \\
&\quad - \pi_Q{}^\eta{}_\eta\,(u^y)^4 + 3P'(T)T\,(u^y)^4 + 3\Pi_Q\,(u^x)^2\,(u^y)^4 \\
&\quad + 3P'(T)T\,(u^x)^2\,(u^y)^4 + \Pi_Q\,(u^y)^6 + P'(T)T\,(u^y)^6 \\
&\quad + \Pi\left((u^y)^2+1\right)\left(2\,(u^x)^4 + 3\,(u^x)^2\left((u^y)^2+1\right) + \left((u^y)^2+1\right)^2\right) \\
&\quad + h\left((u^y)^2+1\right)\Big(\nu^y u^y\left((u^y)^2+1\right) \\
&\quad + \nu^x u^x\left(2\,(u^x)^2 + 3\,(u^y)^2 + 3\right) \\
&\quad + n\Big(2\,(u^x)^4 + 3\,(u^x)^2\left((u^y)^2+1\right) + \left((u^y)^2+1\right)^2\Big)\Big)\Big) \\
(A_t)_{2,3} &= \frac{1}{(u^\tau)^3\left((u^y)^2+1\right)^2}\Big(\Pi_Q\,(u^\tau)^2\,u^x u^y\left((u^y)^2+1\right)^2 + n\,(u^\tau)^2\,u^x u^y\left((u^y)^2+1\right)^2(h-T) \\
&\quad + nT\,(u^\tau)^2\,u^x u^y\left((u^y)^2+1\right)^2 \\
&\quad + (u^\tau)^2\,u^x u^y\left((u^y)^2+1\right)^2(P'(T)T + \Pi)
\end{aligned}
$$

$$
\begin{aligned}
&+ h\left(\left(u^y\right)^2+1\right)^2\left(\nu^x u^y\left(\left(u^y\right)^2+1\right)+\nu^y\left(\left(u^x\right)^3+u^x\right)\right)\\
&- u^y\left(\left(u^y\right)^2+1\right)\left(-\pi^\eta{}_\eta\left(u^\tau\right)^2 u^x-\pi^{yy}\left(\left(u^x\right)^3+u^x\right)\right.\\
&\left.+\pi^{xy}u^y\left(2\left(u^x\right)^2+\left(u^y\right)^2+1\right)\right)\\
&- u^y\left(\left(u^y\right)^2+1\right)\left(-\pi_Q{}^\eta{}_\eta\left(u^\tau\right)^2 u^x-\pi_Q^{yy}\left(\left(u^x\right)^3+u^x\right)\right.\\
&\left.+\pi_Q^{xy}u^y\left(2\left(u^x\right)^2+\left(u^y\right)^2+1\right)\right)\\
&+\left(u^\tau\right)^2\left(2u^x u^y\left(\pi^\eta{}_\eta\left(u^x\right)^2+\pi^{yy}\left(u^x\right)^2+\pi^{yy}\right)\right.\\
&\left.+\pi^{xy}\left(\left(\left(u^y\right)^2+1\right)^2-2\left(u^x\right)^2\left(\left(u^y\right)^2-1\right)\right)\right)\\
&+\left(u^\tau\right)^2\left(2u^x u^y\left(\pi_Q^{yy}\left(u^x\right)^2+\pi_Q^{yy}+\pi_Q{}^\eta{}_\eta\left(u^x\right)^2\right)\right.\\
&\left.+\pi_Q^{xy}\left(\left(\left(u^y\right)^2+1\right)^2-2\left(u^x\right)^2\left(\left(u^y\right)^2-1\right)\right)\right)\Big)\\
(A_t)_{2,4} &= -\frac{u^x\left(\left(u^x\right)^2+1\right)}{u^\tau\left(\left(u^y\right)^2+1\right)}\\
(A_t)_{2,5} &= -\frac{u^\tau u^x}{\left(u^y\right)^2+1}\\
(A_t)_{2,6} &= \frac{u^y\left(2\left(u^x\right)^2+\left(u^y\right)^2+1\right)}{u^\tau\left(\left(u^y\right)^2+1\right)}\\
(A_t)_{2,7} &= u^\tau u^x\\
(A_t)_{2,8} &= (\partial_\alpha n)hu^\tau u^x\\
(A_t)_{2,9} &= \frac{h\left(2\left(u^x\right)^2+\left(u^y\right)^2+1\right)}{u^\tau}\\
(A_t)_{2,10} &= \frac{hu^x u^y}{u^\tau}\\
(A_t)_{2,11} &= -\frac{u^x\left(\left(u^x\right)^2+1\right)}{u^\tau\left(\left(u^y\right)^2+1\right)}\\
(A_t)_{2,12} &= -\frac{u^\tau u^x}{\left(u^y\right)^2+1}\\
(A_t)_{2,13} &= \frac{u^y\left(2\left(u^x\right)^2+\left(u^y\right)^2+1\right)}{u^\tau\left(\left(u^y\right)^2+1\right)}\\
(A_t)_{2,14} &= u^\tau u^x\\
(A_t)_{3,1} &= \frac{1}{u^\tau}\Big(\left(u^\tau\right)^2 u^y((\partial_T n)h+P''(T)T+P'(T))\\
&\quad + h'(T)\left(n\left(u^\tau\right)^2 u^y+\nu^x u^x u^y+\nu^y\left(\left(u^x\right)^2+2\left(u^y\right)^2+1\right)\right)\Big)\\
(A_t)_{3,2} &= \frac{1}{\left(u^\tau\right)^3}\Big(\pi^{xy}+\pi_Q^{xy}+h\nu^y u^x+h\nu^y\left(u^x\right)^3+h\nu^x u^y+hnu^x u^y+\Pi u^x u^y+\Pi_Q u^x u^y\\
&\quad -\pi^{yy}u^x u^y-\pi_Q^{yy}u^x u^y+P'(T)Tu^x u^y+hn\left(u^x\right)^3 u^y+\Pi\left(u^x\right)^3 u^y+\Pi_Q\left(u^x\right)^3 u^y\\
&\quad +P'(T)T\left(u^x\right)^3 u^y+\pi^{xy}\left(u^y\right)^2+\pi_Q^{xy}\left(u^y\right)^2+h\nu^x\left(u^y\right)^3+hnu^x\left(u^y\right)^3\\
&\quad +\Pi u^x\left(u^y\right)^3+\Pi_Q u^x\left(u^y\right)^3+P'(T)Tu^x\left(u^y\right)^3\Big)\\
(A_t)_{3,3} &= \frac{1}{\left(u^\tau\right)^3}\Big(\Pi_Q+\pi^{yy}+\pi_Q^{yy}+P'(T)T+2\Pi_Q\left(u^x\right)^2+\pi^{yy}\left(u^x\right)^2+\pi_Q^{yy}\left(u^x\right)^2\\
&\quad +2P'(T)T\left(u^x\right)^2+\Pi_Q\left(u^x\right)^4+P'(T)T\left(u^x\right)^4-\pi^{xy}u^x u^y-\pi_Q^{xy}u^x u^y+3\Pi_Q\left(u^y\right)^2
\end{aligned}
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\begin{aligned}
&+ 3P'(T)T\left(u^y\right)^2 + 3\Pi_Q\left(u^x\right)^2\left(u^y\right)^2 + 3P'(T)T\left(u^x\right)^2\left(u^y\right)^2 + 2\Pi_Q\left(u^y\right)^4 \\
&+ 2P'(T)T\left(u^y\right)^4 + \Pi\left(\left(u^x\right)^4 + \left(u^x\right)^2\left(3\left(u^y\right)^2 + 2\right) + 2\left(u^y\right)^4 + 3\left(u^y\right)^2 + 1\right) \\
&+ h\Big(\nu^x\left(\left(u^x\right)^3 + u^x\right) + \nu^y u^y\left(3\left(u^x\right)^2 + 2\left(u^y\right)^2 + 3\right) \\
&+ n\left(\left(u^x\right)^4 + \left(u^x\right)^2\left(3\left(u^y\right)^2 + 2\right) + 2\left(u^y\right)^4 + 3\left(u^y\right)^2 + 1\right)\Big)\Big) \\
(A_t)_{3,4} &= \frac{u^y}{u^\tau} \\
(A_t)_{3,6} &= \frac{u^x}{u^\tau} \\
(A_t)_{3,7} &= u^\tau u^y \\
(A_t)_{3,8} &= (\partial_\alpha n) h u^\tau u^y \\
(A_t)_{3,9} &= \frac{h u^x u^y}{u^\tau} \\
(A_t)_{3,10} &= \frac{h\left(\left(u^x\right)^2 + 2\left(u^y\right)^2 + 1\right)}{u^\tau} \\
(A_t)_{3,11} &= \frac{u^y}{u^\tau} \\
(A_t)_{3,13} &= \frac{u^x}{u^\tau} \\
(A_t)_{3,14} &= u^\tau u^y \\
(A_t)_{4,2} &= \frac{3\delta_{\pi\pi}\pi^{yy}u^x - 6\pi^{xy}\tau_\pi u^y\left(\left(u^y\right)^2 + 1\right) - 2u^x\left(-3\pi^{yy}\tau_\pi\left(u^y\right)^2 + \left(u^y\right)^2\eta + \eta\right)}{3u^\tau} \\
(A_t)_{4,3} &= \frac{u^y\left(3\delta_{\pi\pi}\pi^{yy} + 6\pi^{xy}\tau_\pi u^x u^y - 6\pi^{yy}\tau_\pi\left(\left(u^x\right)^2 + 1\right) + 6\left(u^x\right)^2\eta + 4\left(u^y\right)^2\eta + 4\eta\right)}{3u^\tau} \\
(A_t)_{4,4} &= \tau_\pi u^\tau \\
(A_t)_{5,2} &= \frac{u^x(3\delta_{\pi\pi}\pi^\eta{}_\eta - 2\eta)}{3\tau^2 u^\tau} \\
(A_t)_{5,3} &= \frac{u^y(3\delta_{\pi\pi}\pi^\eta{}_\eta - 2\eta)}{3\tau^2 u^\tau} \\
(A_t)_{5,5} &= \frac{\tau_\pi u^\tau}{\tau^2} \\
(A_t)_{6,2} &= \frac{1}{3u^\tau}\Big(3\delta_{\pi\pi}\pi^{xy}u^x - 3\pi^{xy}\tau_\pi u^x\left(2\left(u^y\right)^2 + 1\right) \\
&+ u^y\left(3\pi^\eta{}_\eta\tau_\pi\left(u^\tau\right)^2 + 3\pi^{yy}\left(2\tau_\pi\left(u^x\right)^2 + \tau_\pi\right) + \eta\left(\left(u^x\right)^2 + 3\left(u^y\right)^2 + 3\right)\right)\Big) \\
(A_t)_{6,3} &= \frac{1}{3u^\tau\left(\left(u^y\right)^2 + 1\right)}\Big(-3\pi^{xy}\tau_\pi u^y - 3\pi^\eta{}_\eta\tau_\pi u^x\left(u^y\right)^2 - 3\pi^\eta{}_\eta\tau_\pi\left(u^x\right)^3\left(u^y\right)^2 - 3\pi^{xy}\tau_\pi\left(u^y\right)^3 \\
&+ 6\pi^{xy}\tau_\pi\left(u^x\right)^2\left(u^y\right)^3 - 3\pi^\eta{}_\eta\tau_\pi u^x\left(u^y\right)^4 + 3\delta_{\pi\pi}\pi^{xy}u^y\left(\left(u^y\right)^2 + 1\right) \\
&- 3\pi^{yy}\tau_\pi u^x\left(\left(u^x\right)^2 + 1\right)\left(2\left(u^y\right)^2 + 1\right) + 3u^x\eta + 3\left(u^x\right)^3\eta \\
&+ 4u^x\left(u^y\right)^2\eta + 3\left(u^x\right)^3\left(u^y\right)^2\eta + u^x\left(u^y\right)^4\eta\Big) \\
(A_t)_{6,6} &= \tau_\pi u^\tau \\
(A_t)_{7,2} &= \frac{u^x\zeta}{u^\tau} \\
(A_t)_{7,3} &= \frac{u^y\zeta}{u^\tau} \\
(A_t)_{7,7} &= \tau_\Pi u^\tau \\
(A_t)_{8,1} &= (\partial_T n) u^\tau \\
(A_t)_{8,2} &= \frac{n\left(u^\tau\right)^2 u^x + \nu^x\left(\left(u^y\right)^2 + 1\right) - \nu^y u^x u^y}{\left(u^\tau\right)^3}
\end{aligned}
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\begin{aligned}
(A_t)_{8,3} &= \frac{n\left(u^\tau\right)^2 u^y - \nu^x u^x u^y + \nu^y\left(\left(u^x\right)^2+1\right)}{\left(u^\tau\right)^3}\\
(A_t)_{8,8} &= (\partial_\alpha n)u^\tau\\
(A_t)_{8,9} &= \frac{u^x}{u^\tau}\\
(A_t)_{8,10} &= \frac{u^y}{u^\tau}\\
(A_t)_{9,1} &= \frac{D_s u^\tau(h'(T)\nu^x + u^x((\partial_T n)T+n))}{T}\\
(A_t)_{9,2} &= \frac{1}{T\left(u^\tau\right)^3\left(\left(u^y\right)^2+1\right)} D_s\Big(-\pi_Q^{yy} - \pi_Q{}^\eta{}_\eta - 2\pi_Q^{yy}\left(u^x\right)^2 - 2\pi_Q{}^\eta{}_\eta\left(u^x\right)^2 - \pi_Q^{yy}\left(u^x\right)^4\\
&\quad - \pi_Q{}^\eta{}_\eta\left(u^x\right)^4 + 2\pi_Q^{xy}u^x u^y + \pi_Q^{xy}\left(u^x\right)^3 u^y - \pi_Q^{yy}\left(u^y\right)^2\\
&\quad - 2\pi_Q{}^\eta{}_\eta\left(u^y\right)^2 - \pi_Q^{yy}\left(u^x\right)^2\left(u^y\right)^2 - \pi_Q{}^\eta{}_\eta\left(u^x\right)^2\left(u^y\right)^2\\
&\quad + \pi_Q^{yy}\left(u^x\right)^4\left(u^y\right)^2 + \pi_Q{}^\eta{}_\eta\left(u^x\right)^4\left(u^y\right)^2 + \pi_Q^{xy}u^x\left(u^y\right)^3\\
&\quad - \pi_Q^{xy}\left(u^x\right)^3\left(u^y\right)^3 - \pi_Q{}^\eta{}_\eta\left(u^y\right)^4 + \pi_Q^{yy}\left(u^x\right)^2\left(u^y\right)^4\\
&\quad + \pi_Q{}^\eta{}_\eta\left(u^x\right)^2\left(u^y\right)^4 - \pi_Q^{xy}u^x\left(u^y\right)^5 + \Pi_Q\left(u^\tau\right)^4\left(\left(u^y\right)^2+1\right)\\
&\quad + h\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right)\left(n\left(u^\tau\right)^2 + \nu^x\left(u^x - u^x\left(u^y\right)^2\right) + \nu^y\left(\left(u^x\right)^2+1\right)u^y\right)\Big)\\
(A_t)_{9,3} &= \frac{1}{T\left(u^\tau\right)^3\left(\left(u^y\right)^2+1\right)^2} D_s\Big(-h\left(u^\tau\right)^2\left(\left(u^x\right)^2+1\right)\left(\left(u^y\right)^2+1\right)^2\left(\nu^y u^x - \nu^x u^y\right)\\
&\quad + \pi_Q^{xy}\Big(\left(u^x\right)^4\left(\left(u^y\right)^2-1\right)^2 + \left(\left(u^y\right)^2+1\right)^2\\
&\quad + \left(u^x\right)^2\left(\left(u^y\right)^6 - 2\left(u^y\right)^4 - \left(u^y\right)^2 + 2\right)\Big)\\
&\quad + u^x u^y\Big(-\pi_Q^{yy}\left(\left(u^x\right)^2+1\right)\left(\left(u^x\right)^2\left(\left(u^y\right)^2-1\right) + \left(u^y\right)^4 - \left(u^y\right)^2 - 2\right)\\
&\quad + \pi_Q{}^\eta{}_\eta\Big(-\left(u^x\right)^4\left(\left(u^y\right)^2-1\right) + \left(\left(u^y\right)^2+1\right)^2\\
&\quad + \left(u^x\right)^2\left(-\left(u^y\right)^4 + \left(u^y\right)^2 + 2\right)\Big)\Big)\Big)\\
(A_t)_{9,8} &= (\partial_\alpha n)D_s u^\tau u^x\\
(A_t)_{9,9} &= \frac{D_s h u^\tau}{T}\\
(A_t)_{9,11} &= -\frac{D_s u^x\left(\left(u^x\right)^2+1\right)}{Tu^\tau\left(\left(u^y\right)^2+1\right)}\\
(A_t)_{9,12} &= -\frac{D_s u^\tau u^x}{T\left(u^y\right)^2 + T}\\
(A_t)_{9,13} &= \frac{D_s\left(2\left(u^x\right)^2 u^y + \left(u^y\right)^3 + u^y\right)}{Tu^\tau\left(\left(u^y\right)^2+1\right)}\\
(A_t)_{9,14} &= \frac{D_s u^\tau u^x}{T}\\
(A_t)_{10,1} &= \frac{D_s u^\tau(h'(T)\nu^y + u^y((\partial_T n)T+n))}{T}\\
(A_t)_{10,2} &= \frac{1}{T\left(u^\tau\right)^3} D_s\Big(h\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right)\left(\nu^y u^x - \nu^x u^y\right) - \pi_Q^{xy}\left(\left(u^x\right)^2\left(u^y\right)^2 + \left(u^y\right)^4 - 1\right)\\
&\quad + u^x u^y\left(\pi_Q^{yy}\left(\left(u^x\right)^2 + \left(u^y\right)^2\right) + \pi_Q{}^\eta{}_\eta\left(u^\tau\right)^2\right)\Big)\\
(A_t)_{10,3} &= \frac{1}{T\left(u^\tau\right)^3\left(\left(u^y\right)^2+1\right)} D_s\Big(\pi_Q^{yy} + \pi_Q^{yy}\left(u^x\right)^2 - 2\pi_Q^{xy}u^x u^y - \pi_Q^{xy}\left(u^x\right)^3 u^y - \pi_Q^{yy}\left(u^x\right)^2\left(u^y\right)^2\\
&\quad - \pi_Q{}^\eta{}_\eta\left(u^x\right)^2\left(u^y\right)^2 - \pi_Q^{yy}\left(u^x\right)^4\left(u^y\right)^2 - \pi_Q{}^\eta{}_\eta\left(u^x\right)^4\left(u^y\right)^2
\end{aligned}
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\begin{aligned}
&- \pi_Q^{xy} u^x \left(u^y\right)^3 + \pi_Q^{xy} \left(u^x\right)^3 \left(u^y\right)^3 - \pi_Q^{yy} \left(u^y\right)^4 - \pi_Q^{yy} \left(u^x\right)^2 \left(u^y\right)^4 \\
&- \pi_Q{}^{\eta}{}_{\eta} \left(u^x\right)^2 \left(u^y\right)^4 + \pi_Q^{xy} u^x \left(u^y\right)^5 + \Pi_Q \left(u^\tau\right)^4 \left(\left(u^y\right)^2 + 1\right) \\
&+ h \left(u^\tau\right)^2 \left(\left(u^y\right)^2 + 1\right) \left(n \left(u^\tau\right)^2 + \nu^x u^x \left(\left(u^y\right)^2 + 1\right) - \nu^y \left(\left(u^x\right)^2 - 1\right) u^y\right)\Big) \\
(A_t)_{10,8} &= (\partial_\alpha n) D_s u^\tau u^y \\
(A_t)_{10,10} &= \frac{D_s h u^\tau}{T} \\
(A_t)_{10,11} &= \frac{D_s u^y}{T u^\tau} \\
(A_t)_{10,13} &= \frac{D_s u^x}{T u^\tau} \\
(A_t)_{10,14} &= \frac{D_s u^\tau u^y}{T} \\
(A_t)_{11,1} &= \frac{1}{6 T u^\tau} D_s \Big(3 \mathcal{C}'(T) \pi_Q^{yy} \left(u^\tau\right)^2 - 2 \mathcal{B}'(T) \nu^x u^x \left(\left(u^y\right)^2 + 1\right) \\
&+ 2 \mathcal{B}'(T) \nu^y u^y \left(3 \left(u^x\right)^2 + 2 \left(u^y\right)^2 + 2\right)\Big) \\
(A_t)_{11,2} &= -\frac{1}{6 T \left(u^\tau\right)^3} D_s \Big(2 \mathcal{A} \left(u^\tau\right)^2 \left(\left(u^y\right)^2 + 1\right) \left(\nu^x \left(u^y\right)^2 + \nu^x - \nu^y u^x u^y\right) \\
&+ 2 \mathcal{B} \left(\left(3 \left(u^x\right)^2 + 4\right) \left(u^y\right)^2 + 3 \left(u^y\right)^4 + 1\right) \left(\nu^x \left(u^y\right)^2 + \nu^x - \nu^y u^x u^y\right) \\
&+ \left(u^\tau\right)^2 \Big(2 I_{31} u^x \left(\left(u^y\right)^2 + 1\right) \\
&+ \mathcal{C} \Big(2 \Pi_Q u^x \left(\left(u^y\right)^2 + 1\right) \\
&- 2 \left(\left(u^y\right)^2 + 1\right) \left(\pi_Q{}^{\eta}{}_{\eta} u^x - 4 \pi_Q^{xy} u^y\right) \\
&- \pi_Q^{yy} u^x \left(8 \left(u^y\right)^2 + 5\right)\Big)\Big)\Big) \\
(A_t)_{11,3} &= \frac{1}{6 T \left(u^\tau\right)^3} D_s \Big(2 \mathcal{B} \nu^x u^x u^y \left(\left(3 \left(u^x\right)^2 + 4\right) \left(u^y\right)^2 + 3 \left(u^y\right)^4 + 1\right) \\
&- 2 \mathcal{A} \left(u^\tau\right)^2 \left(\nu^y \left(u^x\right)^2 \left(\left(u^y\right)^2 - 2\right) - 2 \nu^y \left(\left(u^y\right)^2 + 1\right) \right. \\
&\left. - \nu^x u^x u^y \left(\left(u^y\right)^2 + 1\right)\right) \\
&- 2 \mathcal{B} \nu^y \Big(3 \left(u^x\right)^4 \left(\left(u^y\right)^2 - 1\right) - 2 \left(\left(u^y\right)^2 + 1\right) \\
&+ \left(u^x\right)^2 \left(3 \left(u^y\right)^4 + \left(u^y\right)^2 - 5\right)\Big) \\
&+ \left(u^\tau\right)^2 \left(2 I_{31} u^y \left(3 \left(u^x\right)^2 + 2 \left(u^y\right)^2 + 2\right) + 4 \mathcal{C} \pi_Q^{xy} \left(2 u^x \left(u^y\right)^2 + u^x\right)\right. \\
&+ \mathcal{C} u^y \Big(\pi_Q^{yy} - 8 \pi_Q^{yy} \left(u^x\right)^2 - 2 \pi_Q{}^{\eta}{}_{\eta} \left(u^x\right)^2 \\
&\left.+ \Pi_Q \left(6 \left(u^x\right)^2 + 4 \left(u^y\right)^2 + 4\right)\Big)\Big)\Big) \\
(A_t)_{11,9} &= -\frac{\mathcal{B} D_s u^x \left(\left(u^y\right)^2 + 1\right)}{3 T u^\tau} \\
(A_t)_{11,10} &= \frac{\mathcal{B} D_s u^y \left(3 \left(u^x\right)^2 + 2 \left(u^y\right)^2 + 2\right)}{3 T u^\tau} \\
(A_t)_{11,11} &= \frac{\mathcal{C} D_s u^\tau}{2 T} \\
(A_t)_{12,1} &= \frac{D_s \left(3 \mathcal{C}'(T) \pi_Q{}^{\eta}{}_{\eta} \left(u^\tau\right)^2 - 2 \mathcal{B}'(T) (\nu^x u^x + \nu^y u^y)\right)}{6 T u^\tau} \\
(A_t)_{12,2} &= -\frac{1}{6 T \left(u^\tau\right)^3} D_s \Big(2 \mathcal{B} \left(\nu^x \left(u^y\right)^2 + \nu^x - \nu^y u^x u^y\right) + 2 \mathcal{A} \left(u^\tau\right)^2 \left(\nu^x \left(u^y\right)^2 + \nu^x - \nu^y u^x u^y\right) \\
&+ \left(u^\tau\right)^2 \left(\mathcal{C} (2 \Pi_Q u^x + 2 \pi_Q^{xy} u^y - 2 \pi_Q^{yy} u^x - 5 \pi_Q{}^{\eta}{}_{\eta} u^x) + 2 I_{31} u^x\right)\Big)
\end{aligned}
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\begin{aligned}
(A_t)_{12,3} = &-\frac{1}{6T\left(u^\tau\right)^3\left(\left(u^y\right)^2+1\right)}D_s\Big(2\mathcal{B}\left(\left(u^y\right)^2+1\right)\left(-\nu^x u^x u^y+\nu^y\left(u^x\right)^2+\nu^y\right)\\
&+2\mathcal{A}\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right)\left(-\nu^x u^x u^y+\nu^y\left(u^x\right)^2+\nu^y\right)\\
&+\left(u^\tau\right)^2\Big(-2\mathcal{C}\pi_Q^{xy}u^x\left(\left(u^y\right)^2-1\right)+2I_{31}u^y\left(\left(u^y\right)^2+1\right)\\
&+\mathcal{C}u^y\Big(2\pi_Q^{yy}\left(\left(u^x\right)^2+1\right)\\
&+{\pi_Q}^\eta{}_\eta\left(2\left(u^x\right)^2-3\left(u^y\right)^2-3\right)+2\Pi_Q\left(\left(u^y\right)^2+1\right)\Big)\Big)\Big)\\
(A_t)_{12,9} = &-\frac{\mathcal{B}D_s u^x}{3Tu^\tau}\\
(A_t)_{12,10} = &-\frac{\mathcal{B}D_s u^y}{3Tu^\tau}\\
(A_t)_{12,12} = &\frac{\mathcal{C}D_s u^\tau}{2T}\\
(A_t)_{13,1} = &\frac{1}{6Tu^\tau}D_s\Big(3\mathcal{C}'(T)\pi_Q^{xy}\left(u^\tau\right)^2+\mathcal{B}'(T)\nu^y u^x\left(3\left(u^x\right)^2+\left(u^y\right)^2+3\right)\\
&+\mathcal{B}'(T)\nu^x u^y\left(\left(u^x\right)^2+3\left(u^y\right)^2+3\right)\Big)\\
(A_t)_{13,2} = &\frac{1}{6T\left(u^\tau\right)^3}D_s\Big(\mathcal{A}\left(u^\tau\right)^2\Big(3\nu^y\left(\left(u^y\right)^2+1\right)-2\nu^x u^x u^y\left(\left(u^y\right)^2+1\right)\\
&+\nu^y\left(u^x\right)^2\left(2\left(u^y\right)^2+3\right)\Big)\\
&+\left(u^\tau\right)^2\Big(\mathcal{C}\pi_Q^{xy}u^x\left(3-8\left(u^y\right)^2\right)+I_{31}u^y\left(\left(u^x\right)^2+3\left(u^y\right)^2+3\right)\\
&+\mathcal{C}u^y\Big(\pi_Q^{yy}\left(8\left(u^x\right)^2+6\right)+\Pi_Q\left(\left(u^x\right)^2+3\left(u^y\right)^2+3\right)\\
&+{\pi_Q}^\eta{}_\eta\left(5\left(u^x\right)^2+3\left(u^y\right)^2+3\right)\Big)\Big)\\
&+\mathcal{B}\Big(-2\nu^x u^x u^y\left(\left(u^y\right)^2+1\right)\left(3\left(u^x\right)^2+3\left(u^y\right)^2+4\right)\\
&+\nu^y\Big(3\left(\left(u^y\right)^2+1\right)^2+\left(u^x\right)^4\left(6\left(u^y\right)^2+3\right)\\
&+2\left(u^x\right)^2\left(3\left(u^y\right)^4+7\left(u^y\right)^2+3\right)\Big)\Big)\Big)\\
(A_t)_{13,3} = &\frac{1}{6T\left(u^\tau\right)^3\left(\left(u^y\right)^2+1\right)}D_s\Big(\mathcal{A}\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right)\Big(-2\nu^y u^x\left(\left(u^x\right)^2+1\right)u^y+3\nu^x\left(\left(u^y\right)^2+1\right)\\
&+\nu^x\left(u^x\right)^2\left(2\left(u^y\right)^2+3\right)\Big)\\
&+\left(u^\tau\right)^2\Big(I_{31}u^x\left(\left(u^y\right)^2+1\right)\left(3\left(u^x\right)^2+\left(u^y\right)^2+3\right)\\
&+\mathcal{C}\Big(-3{\pi_Q}^\eta{}_\eta u^x-3{\pi_Q}^\eta{}_\eta\left(u^x\right)^3+3\pi_Q^{xy}u^y+4\pi_Q^{xy}\left(u^x\right)^2u^y\\
&-6{\pi_Q}^\eta{}_\eta u^x\left(u^y\right)^2-5{\pi_Q}^\eta{}_\eta\left(u^x\right)^3\left(u^y\right)^2+3\pi_Q^{xy}\left(u^y\right)^3\\
&+8\pi_Q^{xy}\left(u^x\right)^2\left(u^y\right)^3-3{\pi_Q}^\eta{}_\eta u^x\left(u^y\right)^4\\
&+\Pi_Q u^x\left(\left(u^y\right)^2+1\right)\left(3\left(u^x\right)^2+\left(u^y\right)^2+3\right)\\
&-2\pi_Q^{yy}u^x\left(\left(u^x\right)^2+1\right)\left(4\left(u^y\right)^2+3\right)\Big)\Big)\\
&+\mathcal{B}\left(\left(u^y\right)^2+1\right)\Big(-2\nu^y u^x\left(\left(u^x\right)^2+1\right)u^y\left(3\left(u^x\right)^2+3\left(u^y\right)^2+4\right)\\
&+\nu^x\Big(3\left(\left(u^y\right)^2+1\right)^2+\left(u^x\right)^4\left(6\left(u^y\right)^2+3\right)\\
&+2\left(u^x\right)^2\left(3\left(u^y\right)^4+7\left(u^y\right)^2+3\right)\Big)\Big)\Big)\\
(A_t)_{13,9} = &\frac{\mathcal{B}D_s u^y\left(\left(u^x\right)^2+3\left(u^y\right)^2+3\right)}{6Tu^\tau}\\
(A_t)_{13,10} = &\frac{\mathcal{B}D_s u^x\left(3\left(u^x\right)^2+\left(u^y\right)^2+3\right)}{6Tu^\tau}
\end{aligned}
$$

$$
\begin{aligned}
(A_t)_{13,13} &= \frac{\mathcal{C}D_s u^\tau}{2T} \\
(A_t)_{14,1} &= \frac{D_s\left(5\mathcal{B}'(T)(\nu^x u^x + \nu^y u^y) + 3\mathcal{C}'(T)\Pi_Q\left(u^\tau\right)^2 + 3(\partial_T I_{31})\left(u^\tau\right)^2\right)}{6Tu^\tau} \\
(A_t)_{14,2} &= \frac{1}{6T\left(u^\tau\right)^3} D_s\Big(5\mathcal{B}\left(\nu^x\left(u^y\right)^2 + \nu^x - \nu^y u^x u^y\right) + 2\mathcal{A}\left(u^\tau\right)^2\left(\nu^x\left(u^y\right)^2 + \nu^x - \nu^y u^x u^y\right) \\
&\quad + \left(u^\tau\right)^2\left(\mathcal{C}(5\Pi_Q u^x + 2\pi_Q^{xy} u^y - 2\pi_Q^{yy} u^x - 2{\pi_Q}^\eta{}_\eta u^x) + 5I_{31}u^x\right)\Big) \\
(A_t)_{14,3} &= \frac{1}{6T\left(u^\tau\right)^3\left(\left(u^y\right)^2+1\right)} D_s\Big(5\mathcal{B}\left(\left(u^y\right)^2+1\right)\left(-\nu^x u^x u^y + \nu^y\left(u^x\right)^2 + \nu^y\right) \\
&\quad + 2\mathcal{A}\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right)\left(-\nu^x u^x u^y + \nu^y\left(u^x\right)^2 + \nu^y\right) \\
&\quad + \left(u^\tau\right)^2\Big(5I_{31}u^y\left(\left(u^y\right)^2+1\right) \\
&\quad + \mathcal{C}\Big(2u^y\left(\pi_Q^{yy}\left(u^x\right)^2 + \pi_Q^{yy} + {\pi_Q}^\eta{}_\eta\left(u^x\right)^2\right) \\
&\quad - 2\pi_Q^{xy}u^x\left(\left(u^y\right)^2-1\right) + 5\Pi_Q u^y\left(\left(u^y\right)^2+1\right)\Big)\Big)\Big) \\
(A_t)_{14,8} &= \frac{(\partial_\alpha I_{31})D_s u^\tau}{2T} \\
(A_t)_{14,9} &= \frac{5\mathcal{B}D_s u^x}{6Tu^\tau} \\
(A_t)_{14,10} &= \frac{5\mathcal{B}D_s u^y}{6Tu^\tau} \\
(A_t)_{14,14} &= \frac{\mathcal{C}D_s u^\tau}{2T} \\
(A_x)_{1,1} &= \frac{1}{u^\tau}\Big(\left(u^\tau\right)^2 u^x((\partial_T n)h + P''(T)T + P'(T)) \\
&\quad + h'(T)\left(n\left(u^\tau\right)^2 u^x + \nu^x\left(2\left(u^x\right)^2 + \left(u^y\right)^2 + 1\right) + \nu^y u^x u^y\right)\Big) \\
(A_x)_{1,2} &= \frac{1}{\left(u^\tau\right)^3\left(\left(u^y\right)^2+1\right)}\Big(\Pi_Q - {\pi^\eta}_\eta - \pi^{yy} - \pi_Q^{yy} - {\pi_Q}^\eta{}_\eta + P'(T)T + 3\Pi_Q\left(u^x\right)^2 \\
&\quad - 3{\pi^\eta}_\eta\left(u^x\right)^2 - 3\pi^{yy}\left(u^x\right)^2 - 3\pi_Q^{yy}\left(u^x\right)^2 - 3{\pi_Q}^\eta{}_\eta\left(u^x\right)^2 \\
&\quad + 3P'(T)T\left(u^x\right)^2 + 2\Pi_Q\left(u^x\right)^4 - 2{\pi^\eta}_\eta\left(u^x\right)^4 - 2\pi^{yy}\left(u^x\right)^4 \\
&\quad - 2\pi_Q^{yy}\left(u^x\right)^4 - 2{\pi_Q}^\eta{}_\eta\left(u^x\right)^4 + 2P'(T)T\left(u^x\right)^4 + 3\pi^{xy}u^x u^y \\
&\quad + 3\pi_Q^{xy}u^x u^y + 2\pi^{xy}\left(u^x\right)^3 u^y + 2\pi_Q^{xy}\left(u^x\right)^3 u^y + 3\Pi_Q\left(u^y\right)^2 \\
&\quad - 2{\pi^\eta}_\eta\left(u^y\right)^2 - \pi^{yy}\left(u^y\right)^2 - \pi_Q^{yy}\left(u^y\right)^2 - 2{\pi_Q}^\eta{}_\eta\left(u^y\right)^2 \\
&\quad + 3P'(T)T\left(u^y\right)^2 + 6\Pi_Q\left(u^x\right)^2\left(u^y\right)^2 - 3{\pi^\eta}_\eta\left(u^x\right)^2\left(u^y\right)^2 \\
&\quad - 3\pi^{yy}\left(u^x\right)^2\left(u^y\right)^2 - 3\pi_Q^{yy}\left(u^x\right)^2\left(u^y\right)^2 - 3{\pi_Q}^\eta{}_\eta\left(u^x\right)^2\left(u^y\right)^2 \\
&\quad + 6P'(T)T\left(u^x\right)^2\left(u^y\right)^2 + 2\Pi_Q\left(u^x\right)^4\left(u^y\right)^2 + 2P'(T)T\left(u^x\right)^4\left(u^y\right)^2 \\
&\quad + 3\pi^{xy}u^x\left(u^y\right)^3 + 3\pi_Q^{xy}u^x\left(u^y\right)^3 + 3\Pi_Q\left(u^y\right)^4 - {\pi^\eta}_\eta\left(u^y\right)^4 \\
&\quad - {\pi_Q}^\eta{}_\eta\left(u^y\right)^4 + 3P'(T)T\left(u^y\right)^4 + 3\Pi_Q\left(u^x\right)^2\left(u^y\right)^4 \\
&\quad + 3P'(T)T\left(u^x\right)^2\left(u^y\right)^4 + \Pi_Q\left(u^y\right)^6 + P'(T)T\left(u^y\right)^6 \\
&\quad + \Pi\left(\left(u^y\right)^2+1\right)\left(2\left(u^x\right)^4 + 3\left(u^x\right)^2\left(\left(u^y\right)^2+1\right) + \left(\left(u^y\right)^2+1\right)^2\right) \\
&\quad + h\left(\left(u^y\right)^2+1\right)\Big(\nu^y u^y\left(\left(u^y\right)^2+1\right) \\
&\quad + \nu^x u^x\left(2\left(u^x\right)^2 + 3\left(u^y\right)^2 + 3\right) \\
&\quad + n\Big(2\left(u^x\right)^4 + 3\left(u^x\right)^2\left(\left(u^y\right)^2+1\right) + \left(\left(u^y\right)^2+1\right)^2\Big)\Big)\Big) \\
(A_x)_{1,3} &= \frac{1}{\left(u^\tau\right)^3\left(\left(u^y\right)^2+1\right)^2}\Big(\Pi_Q\left(u^\tau\right)^2 u^x u^y\left(\left(u^y\right)^2+1\right)^2 + n\left(u^\tau\right)^2 u^x u^y\left(\left(u^y\right)^2+1\right)^2(h-T)
\end{aligned}
$$

$$
\begin{aligned}
&+ nT\left(u^{\tau}\right)^{2}u^{x}u^{y}\left(\left(u^{y}\right)^{2}+1\right)^{2}\\
&+ \left(u^{\tau}\right)^{2}u^{x}u^{y}\left(\left(u^{y}\right)^{2}+1\right)^{2}\left(P'(T)T+\Pi\right)\\
&+ h\left(\left(u^{y}\right)^{2}+1\right)^{2}\left(\nu^{x}u^{y}\left(\left(u^{y}\right)^{2}+1\right)+\nu^{y}\left(\left(u^{x}\right)^{3}+u^{x}\right)\right)\\
&- u^{y}\left(\left(u^{y}\right)^{2}+1\right)\left(-\pi^{\eta}{}_{\eta}\left(u^{\tau}\right)^{2}u^{x}-\pi^{yy}\left(\left(u^{x}\right)^{3}+u^{x}\right)\right.\\
&\left.+ \pi^{xy}u^{y}\left(2\left(u^{x}\right)^{2}+\left(u^{y}\right)^{2}+1\right)\right)\\
&- u^{y}\left(\left(u^{y}\right)^{2}+1\right)\left(-\pi_{Q}{}^{\eta}{}_{\eta}\left(u^{\tau}\right)^{2}u^{x}-\pi_{Q}^{yy}\left(\left(u^{x}\right)^{3}+u^{x}\right)\right.\\
&\left.+ \pi_{Q}^{xy}u^{y}\left(2\left(u^{x}\right)^{2}+\left(u^{y}\right)^{2}+1\right)\right)\\
&+ \left(u^{\tau}\right)^{2}\left(2u^{x}u^{y}\left(\pi^{\eta}{}_{\eta}\left(u^{x}\right)^{2}+\pi^{yy}\left(u^{x}\right)^{2}+\pi^{yy}\right)\right.\\
&\left.+ \pi^{xy}\left(\left(\left(u^{y}\right)^{2}+1\right)^{2}-2\left(u^{x}\right)^{2}\left(\left(u^{y}\right)^{2}-1\right)\right)\right)\\
&+ \left(u^{\tau}\right)^{2}\left(2u^{x}u^{y}\left(\pi_{Q}^{yy}\left(u^{x}\right)^{2}+\pi_{Q}^{yy}+\pi_{Q}{}^{\eta}{}_{\eta}\left(u^{x}\right)^{2}\right)\right.\\
&\left.+ \pi_{Q}^{xy}\left(\left(\left(u^{y}\right)^{2}+1\right)^{2}-2\left(u^{x}\right)^{2}\left(\left(u^{y}\right)^{2}-1\right)\right)\right)\Big)\\
(A_x)_{1,4} &= -\frac{u^{x}\left(\left(u^{x}\right)^{2}+1\right)}{u^{\tau}\left(\left(u^{y}\right)^{2}+1\right)}\\
(A_x)_{1,5} &= -\frac{u^{\tau}u^{x}}{\left(u^{y}\right)^{2}+1}\\
(A_x)_{1,6} &= \frac{u^{y}\left(2\left(u^{x}\right)^{2}+\left(u^{y}\right)^{2}+1\right)}{u^{\tau}\left(\left(u^{y}\right)^{2}+1\right)}\\
(A_x)_{1,7} &= u^{\tau}u^{x}\\
(A_x)_{1,8} &= (\partial_{\alpha}n)hu^{\tau}u^{x}\\
(A_x)_{1,9} &= \frac{h\left(2\left(u^{x}\right)^{2}+\left(u^{y}\right)^{2}+1\right)}{u^{\tau}}\\
(A_x)_{1,10} &= \frac{hu^{x}u^{y}}{u^{\tau}}\\
(A_x)_{1,11} &= -\frac{u^{x}\left(\left(u^{x}\right)^{2}+1\right)}{u^{\tau}\left(\left(u^{y}\right)^{2}+1\right)}\\
(A_x)_{1,12} &= -\frac{u^{\tau}u^{x}}{\left(u^{y}\right)^{2}+1}\\
(A_x)_{1,13} &= \frac{u^{y}\left(2\left(u^{x}\right)^{2}+\left(u^{y}\right)^{2}+1\right)}{u^{\tau}\left(\left(u^{y}\right)^{2}+1\right)}\\
(A_x)_{1,14} &= u^{\tau}u^{x}\\
(A_x)_{2,1} &= P'(T)+n+(\partial_{T}n)T+2h'(T)\nu^{x}u^{x}+P'(T)\left(u^{x}\right)^{2}+(\partial_{T}n)h\left(u^{x}\right)^{2}+h'(T)n\left(u^{x}\right)^{2}\\
&\quad+P''(T)T\left(u^{x}\right)^{2}\\
(A_x)_{2,2} &= \frac{2}{\left(u^{y}\right)^{2}+1}\Big(\Pi_{Q}u^{x}-\pi^{\eta}{}_{\eta}u^{x}-\pi^{yy}u^{x}-\pi_{Q}^{yy}u^{x}-\pi_{Q}{}^{\eta}{}_{\eta}u^{x}+P'(T)Tu^{x}+\pi^{xy}u^{y}+\pi_{Q}^{xy}u^{y}\\
&\quad+\Pi_{Q}u^{x}\left(u^{y}\right)^{2}+P'(T)Tu^{x}\left(u^{y}\right)^{2}+\Pi u^{x}\left(\left(u^{y}\right)^{2}+1\right)\\
&\quad+h\left(\left(u^{y}\right)^{2}+1\right)\left(nu^{x}+\nu^{x}\right)\Big)\\
(A_x)_{2,3} &= \frac{2}{\left(\left(u^{y}\right)^{2}+1\right)^{2}}\Big(u^{y}\left(\left(u^{x}\right)^{2}\left(\pi^{\eta}{}_{\eta}+\pi_{Q}{}^{\eta}{}_{\eta}\right)+\pi^{yy}\left(\left(u^{x}\right)^{2}+1\right)+\pi_{Q}^{yy}\left(\left(u^{x}\right)^{2}+1\right)\right)
\end{aligned}
$$

$$+\pi^{xy}\left(u^x-u^x\left(u^y\right)^2\right)+\pi_Q^{xy}\left(u^x-u^x\left(u^y\right)^2\right)\Big)$$

$$(A_x)_{2,4}=-\frac{\left(u^x\right)^2+1}{\left(u^y\right)^2+1}$$

$$(A_x)_{2,5}=-\frac{\left(u^\tau\right)^2}{\left(u^y\right)^2+1}$$

$$(A_x)_{2,6}=\frac{2u^xu^y}{\left(u^y\right)^2+1}$$

$$(A_x)_{2,7}=\left(u^x\right)^2+1$$

$$(A_x)_{2,8}=\left(\partial_\alpha n\right)\left(h\left(u^x\right)^2+T\right)$$

$$(A_x)_{2,9}=2hu^x$$

$$(A_x)_{2,11}=-\frac{\left(u^x\right)^2+1}{\left(u^y\right)^2+1}$$

$$(A_x)_{2,12}=-\frac{\left(u^\tau\right)^2}{\left(u^y\right)^2+1}$$

$$(A_x)_{2,13}=\frac{2u^xu^y}{\left(u^y\right)^2+1}$$

$$(A_x)_{2,14}=\left(u^x\right)^2+1$$

$$(A_x)_{3,1}=h'(T)u^y(nu^x+\nu^x)+h'(T)\nu^yu^x+u^xu^y((\partial_Tn)h+P''(T)T+P'(T))$$

$$(A_x)_{3,2}=u^y(P'(T)T+\Pi+\Pi_Q)+h(nu^y+\nu^y)$$

$$(A_x)_{3,3}=u^x(P'(T)T+\Pi+\Pi_Q)+h(nu^x+\nu^x)$$

$$(A_x)_{3,6}=1$$

$$(A_x)_{3,7}=u^xu^y$$

$$(A_x)_{3,8}=(\partial_\alpha n)hu^xu^y$$

$$(A_x)_{3,9}=hu^y$$

$$(A_x)_{3,10}=hu^x$$

$$(A_x)_{3,13}=1$$

$$(A_x)_{3,14}=u^xu^y$$

$$(A_x)_{4,2}=\delta_{\pi\pi}\pi^{yy}-\frac{2\pi^{xy}\tau_\pi u^x\left(\left(u^y\right)^2+1\right)u^y}{\left(u^\tau\right)^2}+\frac{2\pi^{yy}\tau_\pi\left(u^x\right)^2\left(u^y\right)^2}{\left(u^\tau\right)^2}-\frac{2}{3}\left(\left(u^y\right)^2+1\right)\eta$$

$$(A_x)_{4,3}=\frac{2u^xu^y\left(\pi^{xy}\tau_\pi u^xu^y-\pi^{yy}\tau_\pi\left(\left(u^x\right)^2+1\right)+\left(u^\tau\right)^2\eta\right)}{\left(u^\tau\right)^2}$$

$$(A_x)_{4,4}=\tau_\pi u^x$$

$$(A_x)_{5,2}=\frac{3\delta_{\pi\pi}\pi^\eta{}_\eta-2\eta}{3\tau^2}$$

$$(A_x)_{5,5}=\frac{\tau_\pi u^x}{\tau^2}$$

$$(A_x)_{6,2}=\delta_{\pi\pi}\pi^{xy}+\frac{u^x\left(u^y\left(3\pi^\eta{}_\eta\tau_\pi\left(u^\tau\right)^2+3\pi^{yy}\left(2\tau_\pi\left(u^x\right)^2+\tau_\pi\right)+\left(u^\tau\right)^2\eta\right)-3\pi^{xy}\tau_\pi u^x\left(2\left(u^y\right)^2+1\right)\right)}{3\left(u^\tau\right)^2}$$

$$(A_x)_{6,3}=\frac{1}{\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right)}\Big(-\pi^{yy}\tau_\pi\left(u^x\right)^2\left(\left(u^x\right)^2+1\right)\left(2\left(u^y\right)^2+1\right)+\pi^{xy}\tau_\pi u^xu^y\left(\left(2\left(u^x\right)^2-1\right)\left(u^y\right)^2-1\right)-\left(u^\tau\right)^2\left(\pi^\eta{}_\eta\tau_\pi\left(u^x\right)^2\left(u^y\right)^2-\left(\left(u^x\right)^2+1\right)\left(\left(u^y\right)^2+1\right)\eta\right)\Big)$$

$$(A_x)_{6,6}=\tau_\pi u^x$$

$$(A_x)_{7,2}=\zeta$$

$$
\begin{aligned}
(A_x)_{7,7} &= \tau_\Pi u^x \\
(A_x)_{8,1} &= (\partial_T n) u^x \\
(A_x)_{8,2} &= n \\
(A_x)_{8,8} &= (\partial_\alpha n) u^x \\
(A_x)_{8,9} &= 1 \\
(A_x)_{9,1} &= \frac{D_s \left(h'(T) \nu^x u^x + (\partial_T n) T \left((u^x)^2 + 1 \right) + n \left((u^x)^2 + 1 \right) \right)}{T} \\
(A_x)_{9,2} &= -\frac{1}{T (u^\tau)^2 \left((u^y)^2 + 1 \right)} D_s \Big(\pi_Q^{yy} u^x + {\pi_Q}^\eta{}_\eta u^x + \pi_Q^{yy} (u^x)^3 + {\pi_Q}^\eta{}_\eta (u^x)^3 - 2\pi_Q^{xy} u^y \\
&\quad - \pi_Q^{xy} (u^x)^2 u^y + \pi_Q^{yy} u^x (u^y)^2 - \pi_Q^{yy} (u^x)^3 (u^y)^2 \\
&\quad - {\pi_Q}^\eta{}_\eta (u^x)^3 (u^y)^2 - 2\pi_Q^{xy} (u^y)^3 + \pi_Q^{xy} (u^x)^2 (u^y)^3 \\
&\quad - {\pi_Q}^\eta{}_\eta u^x (u^y)^4 - \Pi_Q (u^\tau)^2 u^x \left((u^y)^2 + 1 \right) \\
&\quad + h \left((u^y)^2 + 1 \right) \left(-n (u^\tau)^2 u^x - \nu^y (u^x)^3 u^y + \nu^x (u^x)^2 \left((u^y)^2 - 1 \right) \right. \\
&\quad \left. - 2\nu^x \left((u^y)^2 + 1 \right) \right) \Big) \\
(A_x)_{9,3} &= \frac{1}{T (u^\tau)^2 \left((u^y)^2 + 1 \right)^2} D_s \Big(-{\pi_Q}^\eta{}_\eta (u^\tau)^2 (u^x)^2 u^y \left((u^y)^2 - 1 \right) \\
&\quad - h (u^x)^2 \left((u^y)^2 + 1 \right)^2 \left(-\nu^x u^x u^y + \nu^y (u^x)^2 + \nu^y \right) \\
&\quad + \pi_Q^{xy} u^x \left((u^x)^2 \left((u^y)^2 - 1 \right)^2 - 3 (u^y)^4 - 2 (u^y)^2 + 1 \right) \\
&\quad - \pi_Q^{yy} \left((u^x)^2 + 1 \right) u^y \left((u^x)^2 \left((u^y)^2 - 1 \right) - 2 \left((u^y)^2 + 1 \right) \right) \Big) \\
(A_x)_{9,8} &= (\partial_\alpha n) D_s \left((u^x)^2 + 1 \right) \\
(A_x)_{9,9} &= \frac{D_s h u^x}{T} \\
(A_x)_{9,11} &= -\frac{D_s (u^x)^2 + D_s}{T (u^y)^2 + T} \\
(A_x)_{9,12} &= -\frac{D_s (u^\tau)^2}{T \left((u^y)^2 + 1 \right)} \\
(A_x)_{9,13} &= \frac{2 D_s u^x u^y}{T (u^y)^2 + T} \\
(A_x)_{9,14} &= \frac{D_s \left((u^x)^2 + 1 \right)}{T} \\
(A_x)_{10,1} &= \frac{D_s u^x (h'(T) \nu^y + u^y ((\partial_T n) T + n))}{T} \\
(A_x)_{10,2} &= \frac{1}{T (u^\tau)^2} D_s \Big(h \left((u^y)^2 + 1 \right) \left(-\nu^x u^x u^y + \nu^y (u^x)^2 + \nu^y \right) \\
&\quad + u^y \left(-\pi_Q^{xy} u^x u^y + \pi_Q^{yy} (u^x)^2 + \pi_Q^{yy} + {\pi_Q}^\eta{}_\eta (u^x)^2 + {\pi_Q}^\eta{}_\eta (u^y)^2 + {\pi_Q}^\eta{}_\eta \right) \Big) \\
(A_x)_{10,3} &= \frac{1}{T (u^\tau)^2 \left((u^y)^2 + 1 \right)} D_s \Big(\Pi_Q (u^\tau)^2 u^x \left((u^y)^2 + 1 \right) \\
&\quad + h \left((u^y)^2 + 1 \right) \left(n (u^\tau)^2 u^x - \nu^y u^x \left((u^x)^2 + 1 \right) u^y \right. \\
&\quad \left. + \nu^x \left((u^x)^2 + 1 \right) \left((u^y)^2 + 1 \right) \right) \\
&\quad - u^y \Big(u^x u^y \left(\pi_Q^{yy} \left((u^x)^2 + 1 \right) + {\pi_Q}^\eta{}_\eta (u^\tau)^2 \right) \\
&\quad + \pi_Q^{xy} \left(- \left((u^x)^2 \left((u^y)^2 - 1 \right) \right) + (u^y)^2 + 1 \right) \Big) \Big)
\end{aligned}
$$

$$
\begin{aligned}
(A_x)_{10,8} &= (\partial_\alpha n) D_s u^x u^y \\
(A_x)_{10,10} &= \frac{D_s h u^x}{T} \\
(A_x)_{10,13} &= \frac{D_s}{T} \\
(A_x)_{10,14} &= \frac{D_s u^x u^y}{T} \\
(A_x)_{11,1} &= \frac{D_s \left(3\mathcal{C}'(T)\pi_Q^{yy} u^x - 2\mathcal{B}'(T)\left(\nu^x \left(u^y\right)^2 + \nu^x - 3\nu^y u^x u^y\right)\right)}{6T} \\
(A_x)_{11,2} &= -\frac{1}{6T\left(u^\tau\right)^2} D_s \Big(2I_{31}\left(u^\tau\right)^2 \left(\left(u^y\right)^2 + 1\right) \\
&\quad + 2u^x \left(\mathcal{A}\left(u^y\right)^2 + \mathcal{A} + 3\mathcal{B}\left(u^y\right)^2\right)\left(\nu^x \left(u^y\right)^2 + \nu^x - \nu^y u^x u^y\right) \\
&\quad + \mathcal{C}\Big(2\Pi_Q \left(u^\tau\right)^2 \left(\left(u^y\right)^2 + 1\right) \\
&\quad - 2\left(\left(u^y\right)^2 + 1\right)\left(\pi_Q{}^\eta{}_\eta \left(u^\tau\right)^2 - 4\pi_Q^{xy} u^x u^y\right) \\
&\quad - \pi_Q^{yy}\left(\left(u^x\right)^2 \left(8\left(u^y\right)^2 + 5\right) + 5\left(\left(u^y\right)^2 + 1\right)\right)\Big)\Big) \\
(A_x)_{11,3} &= \frac{1}{3T\left(u^\tau\right)^2} D_s \Big(u^x \Big(3I_{31}\left(u^\tau\right)^2 u^y - \mathcal{A}\nu^y \left(u^x\right)^2 \left(\left(u^y\right)^2 - 2\right) + 2\mathcal{A}\nu^y \left(\left(u^y\right)^2 + 1\right) \\
&\quad + \mathcal{A}\nu^x u^x u^y \left(\left(u^y\right)^2 + 1\right) \\
&\quad + 3\mathcal{B}\left(\nu^x u^x \left(u^y\right)^3 - \nu^y \left(u^x\right)^2 \left(\left(u^y\right)^2 - 1\right) + \nu^y\right)\Big) \\
&\quad + \mathcal{C}\Big(-u^x u^y \left(-3\Pi_Q \left(u^\tau\right)^2 + 4\pi_Q^{yy}\left(\left(u^x\right)^2 + 1\right) + \pi_Q{}^\eta{}_\eta \left(u^\tau\right)^2\right) \\
&\quad + 2\pi_Q^{xy}\left(\left(u^x\right)^2 \left(2\left(u^y\right)^2 + 1\right) + \left(u^y\right)^2 + 1\right)\Big)\Big) \\
(A_x)_{11,9} &= -\frac{\mathcal{B} D_s \left(\left(u^y\right)^2 + 1\right)}{3T} \\
(A_x)_{11,10} &= \frac{\mathcal{B} D_s u^x u^y}{T} \\
(A_x)_{11,11} &= \frac{\mathcal{C} D_s u^x}{2T} \\
(A_x)_{12,1} &= -\frac{D_s(2\mathcal{B}'(T)\nu^x - 3\mathcal{C}'(T)\pi_Q{}^\eta{}_\eta u^x)}{6T} \\
(A_x)_{12,2} &= -\frac{1}{6T\left(u^\tau\right)^2} D_s \Big(2I_{31}\left(u^\tau\right)^2 + 2\mathcal{A}u^x \left(\nu^x \left(u^y\right)^2 + \nu^x - \nu^y u^x u^y\right) \\
&\quad + \mathcal{C}\Big(-5\pi_Q{}^\eta{}_\eta + 2\Pi_Q \left(u^\tau\right)^2 - 5\pi_Q{}^\eta{}_\eta \left(u^x\right)^2 - 2\pi_Q^{yy}\left(\left(u^x\right)^2 + 1\right) \\
&\quad + 2\pi_Q^{xy} u^x u^y - 5\pi_Q{}^\eta{}_\eta \left(u^y\right)^2\Big)\Big) \\
(A_x)_{12,3} &= \frac{1}{3T\left(u^\tau\right)^2\left(\left(u^y\right)^2 + 1\right)} D_s \Big(-\mathcal{C}u^x u^y \left(\pi_Q^{yy}\left(\left(u^x\right)^2 + 1\right) + \pi_Q{}^\eta{}_\eta \left(u^\tau\right)^2\right) \\
&\quad - \mathcal{A}u^x \left(\left(u^y\right)^2 + 1\right)\left(-\nu^x u^x u^y + \nu^y \left(u^x\right)^2 + \nu^y\right) \\
&\quad + \mathcal{C}\pi_Q^{xy}\left(\left(u^x\right)^2 \left(\left(u^y\right)^2 - 1\right) - \left(u^y\right)^2 - 1\right)\Big) \\
(A_x)_{12,9} &= -\frac{\mathcal{B} D_s}{3T} \\
(A_x)_{12,12} &= \frac{\mathcal{C} D_s u^x}{2T} \\
(A_x)_{13,1} &= \frac{D_s \left(\mathcal{B}'(T)\nu^x u^x u^y + 3\mathcal{B}'(T)\nu^y \left(\left(u^x\right)^2 + 1\right) + 3\mathcal{C}'(T)\pi_Q^{xy} u^x\right)}{6T} \\
(A_x)_{13,2} &= \frac{1}{6T\left(u^\tau\right)^2} D_s \Big(3\mathcal{B}\nu^y u^x + 3\mathcal{B}\nu^y \left(u^x\right)^3 - 3\mathcal{B}\nu^x u^y + I_{31} u^x u^y - 6\mathcal{B}\nu^x \left(u^x\right)^2 u^y + I_{31}\left(u^x\right)^3 u^y
\end{aligned}
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\begin{aligned}
&+ 6\mathcal{B}\nu^y u^x \left(u^y\right)^2 + 6\mathcal{B}\nu^y \left(u^x\right)^3 \left(u^y\right)^2 - 3\mathcal{B}\nu^x \left(u^y\right)^3 + I_{31} u^x \left(u^y\right)^3 \\
&- 6\mathcal{B}\nu^x \left(u^x\right)^2 \left(u^y\right)^3 \\
&+ \mathcal{A}u^x \Big(3\nu^y \left(\left(u^y\right)^2 + 1\right) - 2\nu^x u^x u^y \left(\left(u^y\right)^2 + 1\right) \\
&+ \nu^y \left(u^x\right)^2 \left(2\left(u^y\right)^2 + 3\right)\Big) \\
&+ \mathcal{C}\Big(u^x u^y \left(\Pi_Q \left(u^\tau\right)^2 + \pi_Q^{yy} \left(8\left(u^x\right)^2 + 5\right) + 5{\pi_Q}^\eta{}_\eta \left(u^\tau\right)^2\right) \\
&+ \pi_Q^{xy} \left(\left(u^x\right)^2 \left(3 - 8\left(u^y\right)^2\right) + 6\left(\left(u^y\right)^2 + 1\right)\right)\Big)\Big) \\
(A_x)_{13,3} = &\frac{1}{6T\left(u^\tau\right)^2 \left(\left(u^y\right)^2 + 1\right)} D_s \Big(3I_{31} \left(u^\tau\right)^2 \left(\left(u^x\right)^2 + 1\right) \left(\left(u^y\right)^2 + 1\right) \\
&+ \mathcal{C}\Big(-3{\pi_Q}^\eta{}_\eta - 6{\pi_Q}^\eta{}_\eta \left(u^x\right)^2 - 3{\pi_Q}^\eta{}_\eta \left(u^x\right)^4 + \pi_Q^{xy} u^x u^y \\
&+ 4\pi_Q^{xy} \left(u^x\right)^3 u^y - 6{\pi_Q}^\eta{}_\eta \left(u^y\right)^2 - 11{\pi_Q}^\eta{}_\eta \left(u^x\right)^2 \left(u^y\right)^2 \\
&- 5{\pi_Q}^\eta{}_\eta \left(u^x\right)^4 \left(u^y\right)^2 + \pi_Q^{xy} u^x \left(u^y\right)^3 + 8\pi_Q^{xy} \left(u^x\right)^3 \left(u^y\right)^3 \\
&- 3{\pi_Q}^\eta{}_\eta \left(u^y\right)^4 - 5{\pi_Q}^\eta{}_\eta \left(u^x\right)^2 \left(u^y\right)^4 \\
&+ 3\Pi_Q \left(u^\tau\right)^2 \left(\left(u^x\right)^2 + 1\right) \left(\left(u^y\right)^2 + 1\right) \\
&- \pi_Q^{yy} \left(\left(u^x\right)^2 + 1\right) \left(\left(u^x\right)^2 \left(8\left(u^y\right)^2 + 6\right) + 3\left(\left(u^y\right)^2 + 1\right)\right)\Big) \\
&+ \left(u^y\right)^2 + 1\Big(\mathcal{A}u^x \Big(-2\nu^y u^x \left(\left(u^x\right)^2 + 1\right) u^y + 3\nu^x \left(\left(u^y\right)^2 + 1\right) \\
&+ \nu^x \left(u^x\right)^2 \left(2\left(u^y\right)^2 + 3\right)\Big) \\
&+ 3\mathcal{B}\left(\left(u^x\right)^2 + 1\right) \left(\nu^x \left(2u^x \left(u^y\right)^2 + u^x\right) - \nu^y \left(2\left(u^x\right)^2 + 1\right) u^y\right)\Big)\Big) \\
(A_x)_{13,9} = &\frac{\mathcal{B}D_s u^x u^y}{6T} \\
(A_x)_{13,10} = &\frac{\mathcal{B}D_s \left(\left(u^x\right)^2 + 1\right)}{2T} \\
(A_x)_{13,13} = &\frac{\mathcal{C}D_s u^x}{2T} \\
(A_x)_{14,1} = &\frac{D_s(5\mathcal{B}'(T)\nu^x + 3u^x(\mathcal{C}'(T)\Pi_Q + (\partial_T I_{31})))}{6T} \\
(A_x)_{14,2} = &\frac{1}{6T\left(u^\tau\right)^2} D_s \Big(5I_{31} \left(u^\tau\right)^2 + 5\mathcal{C}\Pi_Q \left(u^\tau\right)^2 + 2\mathcal{A}u^x \left(\nu^x \left(u^y\right)^2 + \nu^x - \nu^y u^x u^y\right) \\
&- 2\mathcal{C}\left(-\pi_Q^{xy} u^x u^y + \pi_Q^{yy} \left(u^x\right)^2 + \pi_Q^{yy} + \pi_Q{}^\eta{}_\eta \left(u^x\right)^2 + \pi_Q{}^\eta{}_\eta \left(u^y\right)^2 + \pi_Q{}^\eta{}_\eta\right)\Big) \\
(A_x)_{14,3} = &\frac{1}{3T\left(u^\tau\right)^2 \left(\left(u^y\right)^2 + 1\right)} D_s \Big(\mathcal{C}u^x u^y \left(\pi_Q^{yy} \left(\left(u^x\right)^2 + 1\right) + \pi_Q{}^\eta{}_\eta \left(u^\tau\right)^2\right) \\
&+ \mathcal{A}u^x \left(\left(u^y\right)^2 + 1\right) \left(-\nu^x u^x u^y + \nu^y \left(u^x\right)^2 + \nu^y\right) \\
&+ \mathcal{C}\pi_Q^{xy} \left(-\left(\left(u^x\right)^2 \left(\left(u^y\right)^2 - 1\right)\right) + \left(u^y\right)^2 + 1\right)\Big) \\
(A_x)_{14,8} = &\frac{(\partial_\alpha I_{31}) D_s u^x}{2T} \\
(A_x)_{14,9} = &\frac{5\mathcal{B}D_s}{6T} \\
(A_x)_{14,14} = &\frac{\mathcal{C}D_s u^x}{2T} \\
(A_y)_{1,1} = &\frac{1}{u^\tau}\Big(\left(u^\tau\right)^2 u^y((\partial_T n)h + P''(T)T + P'(T)) \\
&+ h'(T)\left(n\left(u^\tau\right)^2 u^y + \nu^x u^x u^y + \nu^y \left(\left(u^x\right)^2 + 2\left(u^y\right)^2 + 1\right)\right)\Big) \\
(A_y)_{1,2} = &\frac{1}{\left(u^\tau\right)^3}\Big(\pi^{xy} + \pi_Q^{xy} + h\nu^y u^x + h\nu^y \left(u^x\right)^3 + h\nu^x u^y + hnu^x u^y + \Pi u^x u^y + \Pi_Q u^x u^y
\end{aligned}
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\begin{aligned}
&- \pi^{yy}u^x u^y - \pi_Q^{yy}u^x u^y + P'(T)Tu^x u^y + hn\left(u^x\right)^3 u^y + \Pi\left(u^x\right)^3 u^y + \Pi_Q\left(u^x\right)^3 u^y \\
&+ P'(T)T\left(u^x\right)^3 u^y + \pi^{xy}\left(u^y\right)^2 + \pi_Q^{xy}\left(u^y\right)^2 + h\nu^x\left(u^y\right)^3 + hnu^x\left(u^y\right)^3 \\
&+ \Pi u^x\left(u^y\right)^3 + \Pi_Q u^x\left(u^y\right)^3 + P'(T)Tu^x\left(u^y\right)^3\Big) \\
(A_y)_{1,3} = &\frac{1}{\left(u^\tau\right)^3}\Big(\Pi_Q + \pi^{yy} + \pi_Q^{yy} + P'(T)T + 2\Pi_Q\left(u^x\right)^2 + \pi^{yy}\left(u^x\right)^2 + \pi_Q^{yy}\left(u^x\right)^2 \\
&+ 2P'(T)T\left(u^x\right)^2 + \Pi_Q\left(u^x\right)^4 + P'(T)T\left(u^x\right)^4 - \pi^{xy}u^x u^y - \pi_Q^{xy}u^x u^y + 3\Pi_Q\left(u^y\right)^2 \\
&+ 3P'(T)T\left(u^y\right)^2 + 3\Pi_Q\left(u^x\right)^2\left(u^y\right)^2 + 3P'(T)T\left(u^x\right)^2\left(u^y\right)^2 + 2\Pi_Q\left(u^y\right)^4 \\
&+ 2P'(T)T\left(u^y\right)^4 + \Pi\Big(\left(u^x\right)^4 + \left(u^x\right)^2\Big(3\left(u^y\right)^2 + 2\Big) + 2\left(u^y\right)^4 + 3\left(u^y\right)^2 + 1\Big) \\
&+ h\Big(\nu^x\Big(\left(u^x\right)^3 + u^x\Big) + \nu^y u^y\Big(3\left(u^x\right)^2 + 2\left(u^y\right)^2 + 3\Big) \\
&+ n\Big(\left(u^x\right)^4 + \left(u^x\right)^2\Big(3\left(u^y\right)^2 + 2\Big) + 2\left(u^y\right)^4 + 3\left(u^y\right)^2 + 1\Big)\Big)\Big) \\
(A_y)_{1,4} = &\frac{u^y}{u^\tau} \\
(A_y)_{1,6} = &\frac{u^x}{u^\tau} \\
(A_y)_{1,7} = &u^\tau u^y \\
(A_y)_{1,8} = &(\partial_\alpha n)hu^\tau u^y \\
(A_y)_{1,9} = &\frac{hu^x u^y}{u^\tau} \\
(A_y)_{1,10} = &\frac{h\left(\left(u^x\right)^2 + 2\left(u^y\right)^2 + 1\right)}{u^\tau} \\
(A_y)_{1,11} = &\frac{u^y}{u^\tau} \\
(A_y)_{1,13} = &\frac{u^x}{u^\tau} \\
(A_y)_{1,14} = &u^\tau u^y \\
(A_y)_{2,1} = &h'(T)u^y(nu^x + \nu^x) + h'(T)\nu^y u^x + u^x u^y((\partial_T n)h + P''(T)T + P'(T)) \\
(A_y)_{2,2} = &u^y(P'(T)T + \Pi + \Pi_Q) + h(nu^y + \nu^y) \\
(A_y)_{2,3} = &u^x(P'(T)T + \Pi + \Pi_Q) + h(nu^x + \nu^x) \\
(A_y)_{2,6} = &1 \\
(A_y)_{2,7} = &u^x u^y \\
(A_y)_{2,8} = &(\partial_\alpha n)hu^x u^y \\
(A_y)_{2,9} = &hu^y \\
(A_y)_{2,10} = &hu^x \\
(A_y)_{2,13} = &1 \\
(A_y)_{2,14} = &u^x u^y \\
(A_y)_{3,1} = &P'(T) + n + (\partial_T n)T + 2h'(T)\nu^y u^y + P'(T)\left(u^y\right)^2 + (\partial_T n)h\left(u^y\right)^2 + h'(T)n\left(u^y\right)^2 \\
&+ P''(T)T\left(u^y\right)^2 \\
(A_y)_{3,3} = &2(u^y(P'(T)T + \Pi + \Pi_Q) + h(nu^y + \nu^y)) \\
(A_y)_{3,4} = &1 \\
(A_y)_{3,7} = &\left(u^y\right)^2 + 1 \\
(A_y)_{3,8} = &(\partial_\alpha n)\left(h\left(u^y\right)^2 + T\right) \\
(A_y)_{3,10} = &2hu^y \\
(A_y)_{3,11} = &1 \\
(A_y)_{3,14} = &\left(u^y\right)^2 + 1 \\
(A_y)_{4,2} = &-\frac{2\tau_\pi\left(u^y\right)^2\left(\pi^{xy}\left(u^y\right)^2 + \pi^{xy} - \pi^{yy}u^x u^y\right)}{\left(u^\tau\right)^2}
\end{aligned}
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\begin{aligned}
(A_y)_{4,3} &= \frac{1}{3\left(u^\tau\right)^2}\Big(3\delta_{\pi\pi}\pi^{yy}\left(u^\tau\right)^2 - 6\pi^{yy}\tau_\pi\left(\left(u^x\right)^2+1\right)\left(u^y\right)^2 + 6\pi^{xy}\tau_\pi u^x\left(u^y\right)^3 + 4\eta \\
&\quad + 4\left(u^x\right)^2\eta + 8\left(u^y\right)^2\eta + 4\left(u^x\right)^2\left(u^y\right)^2\eta + 4\left(u^y\right)^4\eta\Big) \\
(A_y)_{4,4} &= \tau_\pi u^y \\
(A_y)_{5,3} &= \frac{3\delta_{\pi\pi}\pi^\eta{}_\eta - 2\eta}{3\tau^2} \\
(A_y)_{5,5} &= \frac{\tau_\pi u^y}{\tau^2} \\
(A_y)_{6,2} &= \frac{1}{\left(u^\tau\right)^2}\Big(\pi^{yy}\tau_\pi\left(u^y\right)^2 + \pi^\eta{}_\eta\tau_\pi\left(u^\tau\right)^2\left(u^y\right)^2 + 2\pi^{yy}\tau_\pi\left(u^x\right)^2\left(u^y\right)^2 \\
&\quad - \pi^{xy}\tau_\pi u^x u^y\left(2\left(u^y\right)^2+1\right) + \eta + \left(u^x\right)^2\eta + 2\left(u^y\right)^2\eta + \left(u^x\right)^2\left(u^y\right)^2\eta + \left(u^y\right)^4\eta\Big) \\
(A_y)_{6,3} &= \delta_{\pi\pi}\pi^{xy} \\
&\quad - \frac{1}{3\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right)}u^y\Big(3\pi^{yy}\tau_\pi u^x\left(\left(u^x\right)^2+1\right)\left(2\left(u^y\right)^2+1\right) \\
&\quad + 3\pi^{xy}\tau_\pi u^y\left(\left(1-2\left(u^x\right)^2\right)\left(u^y\right)^2+1\right) \\
&\quad + \left(u^\tau\right)^2 u^x\left(3\pi^\eta{}_\eta\tau_\pi\left(u^y\right)^2 - \left(\left(u^y\right)^2+1\right)\eta\right)\Big) \\
(A_y)_{6,6} &= \tau_\pi u^y \\
(A_y)_{7,3} &= \zeta \\
(A_y)_{7,7} &= \tau_\Pi u^y \\
(A_y)_{8,1} &= (\partial_T n)u^y \\
(A_y)_{8,3} &= n \\
(A_y)_{8,8} &= (\partial_\alpha n)u^y \\
(A_y)_{8,10} &= 1 \\
(A_y)_{9,1} &= \frac{D_s u^y(h'(T)\nu^x + u^x((\partial_T n)T + n))}{T} \\
(A_y)_{9,2} &= \frac{1}{T\left(u^\tau\right)^2}D_s\Big(\Pi_Q\left(u^\tau\right)^2 u^y + \pi_Q^{yy}\left(u^x\right)^2 u^y - \pi_Q^{xy}u^x\left(\left(u^y\right)^2+1\right) \\
&\quad + h\left(n\left(u^\tau\right)^2 u^y - \nu^x u^x u^y\left(\left(u^y\right)^2+1\right) + \nu^y\left(\left(u^x\right)^2+1\right)\left(\left(u^y\right)^2+1\right)\right)\Big) \\
(A_y)_{9,3} &= \frac{D_s\left(h\left(\left(u^x\right)^2+1\right)\left(\nu^x\left(u^y\right)^2 + \nu^x - \nu^y u^x u^y\right) - u^x\left(-\pi_Q^{xy}u^x u^y + \pi_Q^{yy}\left(u^x\right)^2 + \pi_Q^{yy}\right)\right)}{T\left(u^\tau\right)^2} \\
(A_y)_{9,8} &= (\partial_\alpha n)D_s u^x u^y \\
(A_y)_{9,9} &= \frac{D_s h u^y}{T} \\
(A_y)_{9,13} &= \frac{D_s}{T} \\
(A_y)_{9,14} &= \frac{D_s u^x u^y}{T} \\
(A_y)_{10,1} &= \frac{D_s\left(h'(T)\nu^y u^y + (\partial_T n)T\left(\left(u^y\right)^2+1\right) + n\left(\left(u^y\right)^2+1\right)\right)}{T} \\
(A_y)_{10,2} &= -\frac{D_s u^y\left(h u^y\left(\nu^x\left(u^y\right)^2 + \nu^x - \nu^y u^x u^y\right) + \pi_Q^{xy}\left(\left(u^y\right)^2+1\right) - \pi_Q^{yy}u^x u^y\right)}{T\left(u^\tau\right)^2} \\
(A_y)_{10,3} &= \frac{1}{T\left(u^\tau\right)^2}D_s\Big(u^y\left(\Pi_Q\left(u^\tau\right)^2 + \pi_Q^{xy}u^x u^y - \pi_Q^{yy}\left(\left(u^x\right)^2+1\right)\right) \\
&\quad + h\left(n\left(u^\tau\right)^2 u^y + \nu^x u^x\left(u^y\right)^3 + \nu^y\left(-\left(\left(u^x\right)^2\left(\left(u^y\right)^2-2\right)\right) + \left(u^y\right)^2+2\right)\right)\Big) \\
(A_y)_{10,8} &= (\partial_\alpha n)D_s\left(\left(u^y\right)^2+1\right) \\
(A_y)_{10,10} &= \frac{D_s h u^y}{T}
\end{aligned}
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\begin{aligned}
(A_y)_{10,11} &= \frac{D_s}{T} \\
(A_y)_{10,14} &= \frac{D_s\left(\left(u^y\right)^2+1\right)}{T} \\
(A_y)_{11,1} &= \frac{D_s\left(4\mathcal{B}'(T)\nu^y\left(\left(u^y\right)^2+1\right)+3\mathcal{C}'(T)\pi_Q^{yy}u^y\right)}{6T} \\
(A_y)_{11,2} &= -\frac{1}{3T\left(u^\tau\right)^2}D_s\Big(u^y\left(\left(u^y\right)^2+1\right)(\mathcal{A}+3\mathcal{B})\left(\nu^x\left(u^y\right)^2+\nu^x-\nu^y u^x u^y\right) \\
&\quad +\mathcal{C}\left(4\left(u^y\right)^2+1\right)\left(\pi_Q^{xy}\left(u^y\right)^2+\pi_Q^{xy}-\pi_Q^{yy}u^x u^y\right)\Big) \\
(A_y)_{11,3} &= \frac{1}{6T\left(u^\tau\right)^2}D_s\Big(4I_{31}\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right) \\
&\quad +\mathcal{C}\Big(4\Pi_Q\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right)+2\pi_Q^{xy}u^x u^y\left(4\left(u^y\right)^2+1\right) \\
&\quad +\pi_Q^{yy}\left(\left(u^x\right)^2\left(7-8\left(u^y\right)^2\right)+\left(u^y\right)^2+7\right)\Big) \\
&\quad +2u^y\Big(3\mathcal{B}u^x u^y\left(\nu^x\left(u^y\right)^2+\nu^x-\nu^y u^x u^y\right) \\
&\quad +\mathcal{A}\Big(-\nu^y\left(u^x\right)^2\left(\left(u^y\right)^2-2\right)+2\nu^y\left(\left(u^y\right)^2+1\right) \\
&\quad +\nu^x u^x u^y\left(\left(u^y\right)^2+1\right)\Big)\Big)\Big) \\
(A_y)_{11,10} &= \frac{2\mathcal{B}D_s\left(\left(u^y\right)^2+1\right)}{3T} \\
(A_y)_{11,11} &= \frac{\mathcal{C}D_s u^y}{2T} \\
(A_y)_{12,1} &= -\frac{D_s(2\mathcal{B}'(T)\nu^y-3\mathcal{C}'(T)\pi_Q{}^\eta{}_\eta u^y)}{6T} \\
(A_y)_{12,2} &= -\frac{D_s\left(\mathcal{A}u^y\left(\nu^x\left(u^y\right)^2+\nu^x-\nu^y u^x u^y\right)+\mathcal{C}\left(\pi_Q^{xy}\left(u^y\right)^2+\pi_Q^{xy}-\pi_Q^{yy}u^x u^y\right)\right)}{3T\left(u^\tau\right)^2} \\
(A_y)_{12,3} &= -\frac{1}{6T\left(u^\tau\right)^2}D_s\Big(2I_{31}\left(u^\tau\right)^2+2\mathcal{A}u^y\left(-\nu^x u^x u^y+\nu^y\left(u^x\right)^2+\nu^y\right) \\
&\quad +\mathcal{C}\Big(-3\pi_Q{}^\eta{}_\eta+2\Pi_Q\left(u^\tau\right)^2-3\pi_Q{}^\eta{}_\eta\left(u^x\right)^2+2\pi_Q^{yy}\left(\left(u^x\right)^2+1\right) \\
&\quad -2\pi_Q^{xy}u^x u^y-3\pi_Q{}^\eta{}_\eta\left(u^y\right)^2\Big)\Big) \\
(A_y)_{12,10} &= -\frac{\mathcal{B}D_s}{3T} \\
(A_y)_{12,12} &= \frac{\mathcal{C}D_s u^y}{2T} \\
(A_y)_{13,1} &= \frac{D_s\left(3\mathcal{B}'(T)\nu^x\left(\left(u^y\right)^2+1\right)+\mathcal{B}'(T)\nu^y u^x u^y+3\mathcal{C}'(T)\pi_Q^{xy}u^y\right)}{6T} \\
(A_y)_{13,2} &= \frac{1}{6T\left(u^\tau\right)^2}D_s\Big(-3\mathcal{B}\nu^x u^x+3\mathcal{A}\nu^y u^y+3\mathcal{B}\nu^y u^y+3\mathcal{A}\nu^y\left(u^x\right)^2u^y+6\mathcal{B}\nu^y\left(u^x\right)^2u^y \\
&\quad -2\mathcal{A}\nu^x u^x\left(u^y\right)^2-9\mathcal{B}\nu^x u^x\left(u^y\right)^2+3\mathcal{A}\nu^y\left(u^y\right)^3+3\mathcal{B}\nu^y\left(u^y\right)^3 \\
&\quad +2\mathcal{A}\nu^y\left(u^x\right)^2\left(u^y\right)^3+6\mathcal{B}\nu^y\left(u^x\right)^2\left(u^y\right)^3-2\mathcal{A}\nu^x u^x\left(u^y\right)^4-6\mathcal{B}\nu^x u^x\left(u^y\right)^4 \\
&\quad +3I_{31}\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right) \\
&\quad +\mathcal{C}\Big(3\Pi_Q\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right) \\
&\quad +\pi_Q^{yy}\left(\left(u^x\right)^2\left(8\left(u^y\right)^2+3\right)+6\left(u^y\right)^2+3\right) \\
&\quad +u^y\left(3\pi_Q{}^\eta{}_\eta\left(u^\tau\right)^2u^y-\pi_Q^{xy}u^x\left(8\left(u^y\right)^2+5\right)\right)\Big)\Big) \\
(A_y)_{13,3} &= \frac{1}{6T\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right)}D_s\Big(\left(u^y\right)^2+1\Big(I_{31}\left(u^\tau\right)^2u^x u^y
\end{aligned}
$$

$$
\begin{aligned}
&+ 3\mathcal{B}\nu^x\left(2\left(u^x\right)^2+1\right)u^y\left(\left(u^y\right)^2+1\right)\\
&- 3\mathcal{B}\nu^y u^x\left(\left(u^x\right)^2+1\right)\left(2\left(u^y\right)^2+1\right)\\
&+ \mathcal{A}u^y\Big(-2\nu^y u^x\left(\left(u^x\right)^2+1\right)u^y+3\nu^x\left(\left(u^y\right)^2+1\right)\\
&+ \nu^x\left(u^x\right)^2\left(2\left(u^y\right)^2+3\right)\Big)\Big)\\
&+ \mathcal{C}\Big(u^x u^y\Big(-3\pi_Q{}^\eta{}_\eta\left(u^\tau\right)^2\left(u^y\right)^2+\Pi_Q\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right)\\
&- \pi_Q^{yy}\left(\left(u^x\right)^2+1\right)\left(8\left(u^y\right)^2+5\right)\Big)\\
&+ \pi_Q^{xy}\Big(3\left(\left(u^y\right)^4+3\left(u^y\right)^2+2\right)\\
&+ \left(u^x\right)^2\left(8\left(u^y\right)^4+8\left(u^y\right)^2+6\right)\Big)\Big)\Big)\\
(A_y)_{13,9} &= \frac{\mathcal{B}D_s\left(\left(u^y\right)^2+1\right)}{2T}\\
(A_y)_{13,10} &= \frac{\mathcal{B}D_s u^x u^y}{6T}\\
(A_y)_{13,13} &= \frac{\mathcal{C}D_s u^y}{2T}\\
(A_y)_{14,1} &= \frac{D_s(5\mathcal{B}'(T)\nu^y+3u^y(\mathcal{C}'(T)\Pi_Q+(\partial_T I_{31})))}{6T}\\
(A_y)_{14,2} &= \frac{D_s\left(\mathcal{A}u^y\left(\nu^x\left(u^y\right)^2+\nu^x-\nu^y u^x u^y\right)+\mathcal{C}\left(\pi_Q^{xy}\left(u^y\right)^2+\pi_Q^{xy}-\pi_Q^{yy}u^x u^y\right)\right)}{3T\left(u^\tau\right)^2}\\
(A_y)_{14,3} &= \frac{1}{6T\left(u^\tau\right)^2}D_s\Big(5I_{31}\left(u^\tau\right)^2+5\mathcal{C}\Pi_Q\left(u^\tau\right)^2+2\mathcal{A}u^y\left(-\nu^x u^x u^y+\nu^y\left(u^x\right)^2+\nu^y\right)\\
&\quad + 2\mathcal{C}\left(-\pi_Q^{xy}u^x u^y+\pi_Q^{yy}\left(u^x\right)^2+\pi_Q^{yy}\right)\Big)\\
(A_y)_{14,8} &= \frac{(\partial_\alpha I_{31})D_s u^y}{2T}\\
(A_y)_{14,10} &= \frac{5\mathcal{B}D_s}{6T}\\
(A_y)_{14,14} &= \frac{\mathcal{C}D_s u^y}{2T}\\
S_1 &= \frac{1}{\tau\left(\left(u^y\right)^2+1\right)}\Big(\Pi_Q+\pi^\eta{}_\eta+\pi_Q{}^\eta{}_\eta+P'(T)T+\Pi_Q\left(u^x\right)^2-\pi^\eta{}_\eta\left(u^x\right)^2-\pi^{yy}\left(u^x\right)^2\\
&\quad - \pi_Q^{yy}\left(u^x\right)^2-\pi_Q{}^\eta{}_\eta\left(u^x\right)^2+P'(T)T\left(u^x\right)^2+2\pi^{xy}u^x u^y+2\pi_Q^{xy}u^x u^y\\
&\quad + 2\Pi_Q\left(u^y\right)^2+\pi^\eta{}_\eta\left(u^y\right)^2+\pi^{yy}\left(u^y\right)^2+\pi_Q^{yy}\left(u^y\right)^2+\pi_Q{}^\eta{}_\eta\left(u^y\right)^2\\
&\quad + 2P'(T)T\left(u^y\right)^2+\Pi_Q\left(u^x\right)^2\left(u^y\right)^2+P'(T)T\left(u^x\right)^2\left(u^y\right)^2+\Pi_Q\left(u^y\right)^4\\
&\quad + P'(T)T\left(u^y\right)^4+\Pi\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right)\\
&\quad + h\left(\left(u^y\right)^2+1\right)\left(n\left(u^\tau\right)^2+2(\nu^x u^x+\nu^y u^y)\right)\Big)\\
S_2 &= \frac{1}{\tau u^\tau\left(\left(u^y\right)^2+1\right)}\Big(\Pi_Q u^x-\pi^\eta{}_\eta u^x-\pi^{yy}u^x-\pi_Q^{yy}u^x-\pi_Q{}^\eta{}_\eta u^x+P'(T)Tu^x+\Pi_Q\left(u^x\right)^3\\
&\quad - \pi^\eta{}_\eta\left(u^x\right)^3-\pi^{yy}\left(u^x\right)^3-\pi_Q^{yy}\left(u^x\right)^3-\pi_Q{}^\eta{}_\eta\left(u^x\right)^3+P'(T)T\left(u^x\right)^3\\
&\quad + \pi^{xy}u^y+\pi_Q^{xy}u^y+2\pi^{xy}\left(u^x\right)^2u^y+2\pi_Q^{xy}\left(u^x\right)^2u^y+2\Pi_Q u^x\left(u^y\right)^2\\
&\quad - \pi^\eta{}_\eta u^x\left(u^y\right)^2-\pi_Q{}^\eta{}_\eta u^x\left(u^y\right)^2+2P'(T)Tu^x\left(u^y\right)^2+\Pi_Q\left(u^x\right)^3\left(u^y\right)^2\\
&\quad + P'(T)T\left(u^x\right)^3\left(u^y\right)^2+\pi^{xy}\left(u^y\right)^3+\pi_Q^{xy}\left(u^y\right)^3+\Pi_Q u^x\left(u^y\right)^4\\
&\quad + P'(T)Tu^x\left(u^y\right)^4+\Pi\left(u^\tau\right)^2u^x\left(\left(u^y\right)^2+1\right)\\
&\quad + h\left(\left(u^y\right)^2+1\right)\left(n\left(u^\tau\right)^2u^x+\nu^x\left(2\left(u^x\right)^2+\left(u^y\right)^2+1\right)+\nu^y u^x u^y\right)\Big)
\end{aligned}
$$

$$
\begin{aligned}
S_3 &= \frac{1}{\tau u^\tau}\Big(\pi^{xy}u^x + \pi_Q^{xy}u^x + \Pi u^y + \Pi_Q u^y + \pi^{yy}u^y + \pi_Q^{yy}u^y + P'(T)Tu^y + \Pi\left(u^x\right)^2 u^y \\
&\quad + \Pi_Q\left(u^x\right)^2 u^y + P'(T)T\left(u^x\right)^2 u^y + \Pi\left(u^y\right)^3 + \Pi_Q\left(u^y\right)^3 + P'(T)T\left(u^y\right)^3 \\
&\quad + h\left(n\left(u^\tau\right)^2 u^y + \nu^x u^x u^y + \nu^y\left(\left(u^x\right)^2 + 2\left(u^y\right)^2 + 1\right)\right)\Big) \\
S_4 &= \frac{\delta_{\pi\pi}\pi^{yy}u^\tau}{\tau} + \pi^{yy} - \frac{2u^\tau\left(\left(u^y\right)^2+1\right)\eta}{3\tau} \\
S_5 &= \frac{3\pi^\eta{}_\eta(\delta_{\pi\pi}u^\tau + \tau) + 4u^\tau\eta}{3\tau^3} \\
S_6 &= \frac{\delta_{\pi\pi}\pi^{xy}u^\tau}{\tau} + \pi^{xy} - \frac{2u^\tau u^x u^y\eta}{3\tau} \\
S_7 &= \Pi + \frac{u^\tau\zeta}{\tau} \\
S_8 &= \frac{n\left(u^\tau\right)^2 + \nu^x u^x + \nu^y u^y}{\tau u^\tau} \\
S_9 &= \nu^x \\
&\quad + \frac{1}{T\tau u^\tau\left(\left(u^y\right)^2+1\right)}D_s\Big(-2\pi_Q{}^\eta{}_\eta u^x - 2\pi_Q{}^\eta{}_\eta\left(u^x\right)^3 - \pi_Q^{yy}\left(\left(u^x\right)^3 + u^x\right) + \pi_Q^{xy}u^y \\
&\quad + 2\pi_Q^{xy}\left(u^x\right)^2 u^y - 3\pi_Q{}^\eta{}_\eta u^x\left(u^y\right)^2 - \pi_Q{}^\eta{}_\eta\left(u^x\right)^3\left(u^y\right)^2 + \pi_Q^{xy}\left(u^y\right)^3 \\
&\quad - \pi_Q{}^\eta{}_\eta u^x\left(u^y\right)^4 + h\nu^x\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right)\Big) \\
S_{10} &= \frac{D_s\left(h\nu^y\left(u^\tau\right)^2 + \pi_Q^{xy}u^x + u^y\left(\pi_Q^{yy} - \pi_Q{}^\eta{}_\eta\left(u^\tau\right)^2\right)\right)}{T\tau u^\tau} + \nu^y \\
S_{11} &= \pi_Q^{yy} \\
&\quad - \frac{1}{6T\tau u^\tau}D_s\Big(2I_{31}\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right) + 2\mathcal{B}\left(\left(u^y\right)^2+1\right)\left(\nu^x u^x + \nu^y u^y\right) \\
&\quad + \mathcal{C}\left(u^\tau\right)^2\left(2\Pi_Q\left(\left(u^y\right)^2+1\right) - 3\pi_Q^{yy} + 2\pi_Q{}^\eta{}_\eta\left(\left(u^y\right)^2+1\right)\right)\Big) \\
S_{12} &= \frac{D_s\left(4\mathcal{B}(\nu^x u^x + \nu^y u^y) + \mathcal{C}\left(u^\tau\right)^2(4\Pi_Q + 7\pi_Q{}^\eta{}_\eta) + 4I_{31}\left(u^\tau\right)^2\right)}{6T\tau u^\tau} + \pi_Q{}^\eta{}_\eta \\
S_{13} &= -\frac{D_s u^x u^y\left(\mathcal{B}(\nu^x u^x + \nu^y u^y) + I_{31}\left(u^\tau\right)^2\right)}{3T\tau u^\tau} + \frac{\mathcal{C}D_s u^\tau(3\pi_Q^{xy} - 2u^x u^y(\Pi_Q + \pi_Q{}^\eta{}_\eta))}{6T\tau} + \pi_Q^{xy} \\
S_{14} &= \frac{D_s\left(5\mathcal{B}(\nu^x u^x + \nu^y u^y) + \mathcal{C}\left(u^\tau\right)^2(5\Pi_Q + 2\pi_Q{}^\eta{}_\eta) + 5I_{31}\left(u^\tau\right)^2\right)}{6T\tau u^\tau} + \Pi_Q
\end{aligned}
$$

A.2.6 S6, fully coupled, conformal

`two_d_two_way_cM_matrix_conformal` in `src/Matrix/2d_two_way_cM.jl`, fields:

$$
\phi = (T, u^x, u^y, \pi^{yy}, \pi^\eta{}_\eta, \pi^{xy}, \alpha, \nu^x, \nu^y, \pi_Q^{yy}, \pi_Q{}^\eta{}_\eta, \pi_Q^{xy}, \Pi_Q)
$$

$$
\begin{aligned}
(A_t)_{1,1} &= -n - (\partial_T n)T + (\partial_T n)h\left(u^\tau\right)^2 + h'(T)n\left(u^\tau\right)^2 + 2h'(T)(\nu^x u^x + \nu^y u^y) \\
&\quad + 4a_{\mathrm{SB}}T^3\left(4\left(u^x\right)^2 + 4\left(u^y\right)^2 + 3\right) \\
(A_t)_{1,2} &= \frac{2}{\left(u^y\right)^2+1}\Big(-\pi^\eta{}_\eta u^x - \pi^{yy}u^x - \pi_Q^{yy}u^x - \pi_Q{}^\eta{}_\eta u^x + 4a_{\mathrm{SB}}T^4u^x + \pi^{xy}u^y + \pi_Q^{xy}u^y \\
&\quad + 4a_{\mathrm{SB}}T^4u^x\left(u^y\right)^2 + \Pi_Q u^x\left(\left(u^y\right)^2+1\right) + h\left(\left(u^y\right)^2+1\right)(nu^x + \nu^x)\Big) \\
(A_t)_{1,3} &= \frac{2}{\left(\left(u^y\right)^2+1\right)^2}\Big(\pi_Q^{xy}u^x + \Pi_Q u^y + \pi^{yy}u^y + \pi_Q^{yy}u^y + 4a_{\mathrm{SB}}T^4u^y + \pi^\eta{}_\eta\left(u^x\right)^2 u^y \\
&\quad + \pi^{yy}\left(u^x\right)^2 u^y + \pi_Q^{yy}\left(u^x\right)^2 u^y + \pi_Q{}^\eta{}_\eta\left(u^x\right)^2 u^y - \pi_Q^{xy}u^x\left(u^y\right)^2 + 2\Pi_Q\left(u^y\right)^3 \\
&\quad + 8a_{\mathrm{SB}}T^4\left(u^y\right)^3 + \Pi_Q\left(u^y\right)^5 + 4a_{\mathrm{SB}}T^4\left(u^y\right)^5 + h\left(\left(u^y\right)^2+1\right)^2(nu^y + \nu^y)
\end{aligned}
$$

$$
\begin{aligned}
&+ \pi^{xy}\left(u^x - u^x\left(u^y\right)^2\right)\Big)\\
(A_t)_{1,4} &= \frac{\left(u^y\right)^2 - \left(u^x\right)^2}{\left(u^y\right)^2 + 1}\\
(A_t)_{1,5} &= -\frac{\left(u^x\right)^2}{\left(u^y\right)^2 + 1}\\
(A_t)_{1,6} &= \frac{2u^x u^y}{\left(u^y\right)^2 + 1}\\
(A_t)_{1,7} &= \left(\partial_\alpha n\right)\left(h\left(u^\tau\right)^2 - T\right)\\
(A_t)_{1,8} &= 2hu^x\\
(A_t)_{1,9} &= 2hu^y\\
(A_t)_{1,10} &= \frac{\left(u^y\right)^2 - \left(u^x\right)^2}{\left(u^y\right)^2 + 1}\\
(A_t)_{1,11} &= -\frac{\left(u^x\right)^2}{\left(u^y\right)^2 + 1}\\
(A_t)_{1,12} &= \frac{2u^x u^y}{\left(u^y\right)^2 + 1}\\
(A_t)_{1,13} &= \left(u^x\right)^2 + \left(u^y\right)^2\\
(A_t)_{2,1} &= \frac{\left(u^\tau\right)^2 u^x\left(16a_{\mathrm{SB}}T^3 + \left(\partial_T n\right)h\right) + h'(T)\left(n\left(u^\tau\right)^2 u^x + \nu^x\left(2\left(u^x\right)^2 + \left(u^y\right)^2 + 1\right) + \nu^y u^x u^y\right)}{u^\tau}\\
(A_t)_{2,2} &= \frac{1}{\left(u^\tau\right)^3\left(\left(u^y\right)^2 + 1\right)}\Big(-{\pi^\eta}_\eta - \pi^{yy} - \pi_Q^{yy} - {\pi_Q}^\eta{}_\eta + 4a_{\mathrm{SB}}T^4 - 3{\pi^\eta}_\eta\left(u^x\right)^2 - 3\pi^{yy}\left(u^x\right)^2\\
&- 3\pi_Q^{yy}\left(u^x\right)^2 - 3{\pi_Q}^\eta{}_\eta\left(u^x\right)^2 + 12a_{\mathrm{SB}}T^4\left(u^x\right)^2 - 2{\pi^\eta}_\eta\left(u^x\right)^4\\
&- 2\pi^{yy}\left(u^x\right)^4 - 2\pi_Q^{yy}\left(u^x\right)^4 - 2{\pi_Q}^\eta{}_\eta\left(u^x\right)^4 + 8a_{\mathrm{SB}}T^4\left(u^x\right)^4\\
&+ 3\pi^{xy}u^x u^y + 3\pi_Q^{xy}u^x u^y + 2\pi^{xy}\left(u^x\right)^3 u^y + 2\pi_Q^{xy}\left(u^x\right)^3 u^y\\
&- 2{\pi^\eta}_\eta\left(u^y\right)^2 - \pi^{yy}\left(u^y\right)^2 - \pi_Q^{yy}\left(u^y\right)^2 - 2{\pi_Q}^\eta{}_\eta\left(u^y\right)^2\\
&+ 12a_{\mathrm{SB}}T^4\left(u^y\right)^2 - 3{\pi^\eta}_\eta\left(u^x\right)^2\left(u^y\right)^2 - 3\pi^{yy}\left(u^x\right)^2\left(u^y\right)^2\\
&- 3\pi_Q^{yy}\left(u^x\right)^2\left(u^y\right)^2 - 3{\pi_Q}^\eta{}_\eta\left(u^x\right)^2\left(u^y\right)^2 + 24a_{\mathrm{SB}}T^4\left(u^x\right)^2\left(u^y\right)^2\\
&+ 8a_{\mathrm{SB}}T^4\left(u^x\right)^4\left(u^y\right)^2 + 3\pi^{xy}u^x\left(u^y\right)^3 + 3\pi_Q^{xy}u^x\left(u^y\right)^3 - {\pi^\eta}_\eta\left(u^y\right)^4\\
&- {\pi_Q}^\eta{}_\eta\left(u^y\right)^4 + 12a_{\mathrm{SB}}T^4\left(u^y\right)^4 + 12a_{\mathrm{SB}}T^4\left(u^x\right)^2\left(u^y\right)^4 + 4a_{\mathrm{SB}}T^4\left(u^y\right)^6\\
&+ \Pi_Q\left(\left(u^y\right)^2 + 1\right)\left(2\left(u^x\right)^4 + 3\left(u^x\right)^2\left(\left(u^y\right)^2 + 1\right) + \left(\left(u^y\right)^2 + 1\right)^2\right)\\
&+ h\left(\left(u^y\right)^2 + 1\right)\Big(\nu^y u^y\left(\left(u^y\right)^2 + 1\right)\\
&+ \nu^x u^x\left(2\left(u^x\right)^2 + 3\left(u^y\right)^2 + 3\right)\\
&+ n\Big(2\left(u^x\right)^4 + 3\left(u^x\right)^2\left(\left(u^y\right)^2 + 1\right) + \left(\left(u^y\right)^2 + 1\right)^2\Big)\Big)\Big)\\
(A_t)_{2,3} &= \frac{1}{\left(u^\tau\right)^3\left(\left(u^y\right)^2 + 1\right)^2}\Big(\Pi_Q\left(u^\tau\right)^2 u^x u^y\left(\left(u^y\right)^2 + 1\right)^2 + n\left(u^\tau\right)^2 u^x u^y\left(\left(u^y\right)^2 + 1\right)^2(h - T)\\
&+ nT\left(u^\tau\right)^2 u^x u^y\left(\left(u^y\right)^2 + 1\right)^2 + 4a_{\mathrm{SB}}T^4\left(u^\tau\right)^2 u^x u^y\left(\left(u^y\right)^2 + 1\right)^2\\
&+ h\left(\left(u^y\right)^2 + 1\right)^2\left(\nu^x u^y\left(\left(u^y\right)^2 + 1\right) + \nu^y\left(\left(u^x\right)^3 + u^x\right)\right)\\
&- u^y\left(\left(u^y\right)^2 + 1\right)\Big(-{\pi^\eta}_\eta\left(u^\tau\right)^2 u^x - \pi^{yy}\left(\left(u^x\right)^3 + u^x\right)\\
&+ \pi^{xy}u^y\left(2\left(u^x\right)^2 + \left(u^y\right)^2 + 1\right)\Big)\\
&- u^y\left(\left(u^y\right)^2 + 1\right)\Big(-{\pi_Q}^\eta{}_\eta\left(u^\tau\right)^2 u^x - \pi_Q^{yy}\left(\left(u^x\right)^3 + u^x\right)
\end{aligned}
$$

$$
\begin{aligned}
&+ \pi_Q^{xy} u^y \left(2\left(u^x\right)^2 + \left(u^y\right)^2 + 1\right)\Big) \\
&+ \left(u^\tau\right)^2 \Big(2u^x u^y \left(\pi^\eta{}_\eta \left(u^x\right)^2 + \pi^{yy}\left(u^x\right)^2 + \pi^{yy}\right) \\
&+ \pi^{xy}\left(\left(\left(u^y\right)^2 + 1\right)^2 - 2\left(u^x\right)^2\left(\left(u^y\right)^2 - 1\right)\right)\Big) \\
&+ \left(u^\tau\right)^2 \Big(2u^x u^y \left(\pi_Q^{yy}\left(u^x\right)^2 + \pi_Q^{yy} + \pi_Q{}^\eta{}_\eta \left(u^x\right)^2\right) \\
&+ \pi_Q^{xy}\left(\left(\left(u^y\right)^2 + 1\right)^2 - 2\left(u^x\right)^2\left(\left(u^y\right)^2 - 1\right)\right)\Big)\Big) \\
(A_t)_{2,4} &= -\frac{u^x\left(\left(u^x\right)^2 + 1\right)}{u^\tau\left(\left(u^y\right)^2 + 1\right)} \\
(A_t)_{2,5} &= -\frac{u^\tau u^x}{\left(u^y\right)^2 + 1} \\
(A_t)_{2,6} &= \frac{u^y\left(2\left(u^x\right)^2 + \left(u^y\right)^2 + 1\right)}{u^\tau\left(\left(u^y\right)^2 + 1\right)} \\
(A_t)_{2,7} &= (\partial_\alpha n) h u^\tau u^x \\
(A_t)_{2,8} &= \frac{h\left(2\left(u^x\right)^2 + \left(u^y\right)^2 + 1\right)}{u^\tau} \\
(A_t)_{2,9} &= \frac{h u^x u^y}{u^\tau} \\
(A_t)_{2,10} &= -\frac{u^x\left(\left(u^x\right)^2 + 1\right)}{u^\tau\left(\left(u^y\right)^2 + 1\right)} \\
(A_t)_{2,11} &= -\frac{u^\tau u^x}{\left(u^y\right)^2 + 1} \\
(A_t)_{2,12} &= \frac{u^y\left(2\left(u^x\right)^2 + \left(u^y\right)^2 + 1\right)}{u^\tau\left(\left(u^y\right)^2 + 1\right)} \\
(A_t)_{2,13} &= u^\tau u^x \\
(A_t)_{3,1} &= \frac{\left(u^\tau\right)^2 u^y\left(16 a_{\mathrm{SB}} T^3 + (\partial_T n) h\right) + h'(T)\left(n\left(u^\tau\right)^2 u^y + \nu^x u^x u^y + \nu^y\left(\left(u^x\right)^2 + 2\left(u^y\right)^2 + 1\right)\right)}{u^\tau} \\
(A_t)_{3,2} &= \frac{1}{\left(u^\tau\right)^3}\Big(\pi^{xy} + \pi_Q^{xy} + h\nu^y u^x + h\nu^y\left(u^x\right)^3 + h\nu^x u^y + h n u^x u^y + \Pi_Q u^x u^y - \pi^{yy} u^x u^y \\
&- \pi_Q^{yy} u^x u^y + 4 a_{\mathrm{SB}} T^4 u^x u^y + h n\left(u^x\right)^3 u^y + \Pi_Q\left(u^x\right)^3 u^y + 4 a_{\mathrm{SB}} T^4\left(u^x\right)^3 u^y \\
&+ \pi^{xy}\left(u^y\right)^2 + \pi_Q^{xy}\left(u^y\right)^2 + h\nu^x\left(u^y\right)^3 + h n u^x\left(u^y\right)^3 + \Pi_Q u^x\left(u^y\right)^3 \\
&+ 4 a_{\mathrm{SB}} T^4 u^x\left(u^y\right)^3\Big) \\
(A_t)_{3,3} &= \frac{1}{\left(u^\tau\right)^3}\Big(\pi^{yy} + \pi_Q^{yy} + 4 a_{\mathrm{SB}} T^4 + \pi^{yy}\left(u^x\right)^2 + \pi_Q^{yy}\left(u^x\right)^2 + 8 a_{\mathrm{SB}} T^4\left(u^x\right)^2 + 4 a_{\mathrm{SB}} T^4\left(u^x\right)^4 \\
&- \pi^{xy} u^x u^y - \pi_Q^{xy} u^x u^y + 12 a_{\mathrm{SB}} T^4\left(u^y\right)^2 + 12 a_{\mathrm{SB}} T^4\left(u^x\right)^2\left(u^y\right)^2 + 8 a_{\mathrm{SB}} T^4\left(u^y\right)^4 \\
&+ \Pi_Q\left(\left(u^x\right)^4 + \left(u^x\right)^2\left(3\left(u^y\right)^2 + 2\right) + 2\left(u^y\right)^4 + 3\left(u^y\right)^2 + 1\right) \\
&+ h\Big(\nu^x\left(\left(u^x\right)^3 + u^x\right) + \nu^y u^y\left(3\left(u^x\right)^2 + 2\left(u^y\right)^2 + 3\right) \\
&+ n\left(\left(u^x\right)^4 + \left(u^x\right)^2\left(3\left(u^y\right)^2 + 2\right) + 2\left(u^y\right)^4 + 3\left(u^y\right)^2 + 1\right)\Big)\Big) \\
(A_t)_{3,4} &= \frac{u^y}{u^\tau} \\
(A_t)_{3,6} &= \frac{u^x}{u^\tau} \\
(A_t)_{3,7} &= (\partial_\alpha n) h u^\tau u^y
\end{aligned}
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\begin{aligned}
(A_t)_{3,8} &= \frac{hu^x u^y}{u^\tau} \\
(A_t)_{3,9} &= \frac{h\left((u^x)^2 + 2\,(u^y)^2 + 1\right)}{u^\tau} \\
(A_t)_{3,10} &= \frac{u^y}{u^\tau} \\
(A_t)_{3,12} &= \frac{u^x}{u^\tau} \\
(A_t)_{3,13} &= u^\tau u^y \\
(A_t)_{4,2} &= \frac{3\delta_{\pi\pi}\pi^{yy}u^x - 6\pi^{xy}\tau_\pi u^y\left((u^y)^2+1\right) - 2u^x\left(-3\pi^{yy}\tau_\pi\,(u^y)^2 + (u^y)^2\,\eta + \eta\right)}{3u^\tau} \\
(A_t)_{4,3} &= \frac{u^y\left(3\delta_{\pi\pi}\pi^{yy} + 6\pi^{xy}\tau_\pi u^x u^y - 6\pi^{yy}\tau_\pi\left((u^x)^2+1\right) + 6\,(u^x)^2\,\eta + 4\,(u^y)^2\,\eta + 4\eta\right)}{3u^\tau} \\
(A_t)_{4,4} &= \tau_\pi u^\tau \\
(A_t)_{5,2} &= \frac{u^x(3\delta_{\pi\pi}\pi^\eta{}_\eta - 2\eta)}{3\tau^2 u^\tau} \\
(A_t)_{5,3} &= \frac{u^y(3\delta_{\pi\pi}\pi^\eta{}_\eta - 2\eta)}{3\tau^2 u^\tau} \\
(A_t)_{5,5} &= \frac{\tau_\pi u^\tau}{\tau^2} \\
(A_t)_{6,2} &= \frac{1}{3u^\tau}\Big(3\delta_{\pi\pi}\pi^{xy}u^x - 3\pi^{xy}\tau_\pi u^x\left(2\,(u^y)^2+1\right) \\
&\quad + u^y\left(3\pi^\eta{}_\eta\tau_\pi\,(u^\tau)^2 + 3\pi^{yy}\left(2\tau_\pi\,(u^x)^2 + \tau_\pi\right) + \eta\left((u^x)^2 + 3\,(u^y)^2 + 3\right)\right)\Big) \\
(A_t)_{6,3} &= \frac{1}{3u^\tau\left((u^y)^2+1\right)}\Big(-3\pi^{xy}\tau_\pi u^y - 3\pi^\eta{}_\eta\tau_\pi u^x\,(u^y)^2 - 3\pi^\eta{}_\eta\tau_\pi\,(u^x)^3\,(u^y)^2 - 3\pi^{xy}\tau_\pi\,(u^y)^3 \\
&\quad + 6\pi^{xy}\tau_\pi\,(u^x)^2\,(u^y)^3 - 3\pi^\eta{}_\eta\tau_\pi u^x\,(u^y)^4 + 3\delta_{\pi\pi}\pi^{xy}u^y\left((u^y)^2+1\right) \\
&\quad - 3\pi^{yy}\tau_\pi u^x\left((u^x)^2+1\right)\left(2\,(u^y)^2+1\right) + 3u^x\eta + 3\,(u^x)^3\,\eta \\
&\quad + 4u^x\,(u^y)^2\,\eta + 3\,(u^x)^3\,(u^y)^2\,\eta + u^x\,(u^y)^4\,\eta\Big) \\
(A_t)_{6,6} &= \tau_\pi u^\tau \\
(A_t)_{7,1} &= (\partial_T n)u^\tau \\
(A_t)_{7,2} &= \frac{n\,(u^\tau)^2\,u^x + \nu^x\left((u^y)^2+1\right) - \nu^y u^x u^y}{(u^\tau)^3} \\
(A_t)_{7,3} &= \frac{n\,(u^\tau)^2\,u^y - \nu^x u^x u^y + \nu^y\left((u^x)^2+1\right)}{(u^\tau)^3} \\
(A_t)_{7,7} &= (\partial_\alpha n)u^\tau \\
(A_t)_{7,8} &= \frac{u^x}{u^\tau} \\
(A_t)_{7,9} &= \frac{u^y}{u^\tau} \\
(A_t)_{8,1} &= \frac{D_s u^\tau(h'(T)\nu^x + u^x((\partial_T n)T + n))}{T} \\
(A_t)_{8,2} &= \frac{1}{T\,(u^\tau)^3\left((u^y)^2+1\right)}D_s\Big(-\pi_Q^{yy} - \pi_Q{}^\eta{}_\eta - 2\pi_Q^{yy}\,(u^x)^2 - 2\pi_Q{}^\eta{}_\eta\,(u^x)^2 - \pi_Q^{yy}\,(u^x)^4 \\
&\quad - \pi_Q{}^\eta{}_\eta\,(u^x)^4 + 2\pi_Q^{xy}u^x u^y + \pi_Q^{xy}\,(u^x)^3\,u^y - \pi_Q^{yy}\,(u^y)^2 \\
&\quad - 2\pi_Q{}^\eta{}_\eta\,(u^y)^2 - \pi_Q^{yy}\,(u^x)^2\,(u^y)^2 - \pi_Q{}^\eta{}_\eta\,(u^x)^2\,(u^y)^2 \\
&\quad + \pi_Q^{yy}\,(u^x)^4\,(u^y)^2 + \pi_Q{}^\eta{}_\eta\,(u^x)^4\,(u^y)^2 + \pi_Q^{xy}u^x\,(u^y)^3 \\
&\quad - \pi_Q^{xy}\,(u^x)^3\,(u^y)^3 - \pi_Q{}^\eta{}_\eta\,(u^y)^4 + \pi_Q^{yy}\,(u^x)^2\,(u^y)^4
\end{aligned}
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\begin{aligned}
&+ \pi_Q{}^{\eta}{}_{\eta} \left(u^x\right)^2 \left(u^y\right)^4 - \pi_Q^{xy} u^x \left(u^y\right)^5 + \Pi_Q \left(u^\tau\right)^4 \left(\left(u^y\right)^2 + 1\right) \\
&+ h \left(u^\tau\right)^2 \left(\left(u^y\right)^2 + 1\right) \left(n \left(u^\tau\right)^2 + \nu^x \left(u^x - u^x \left(u^y\right)^2\right) + \nu^y \left(\left(u^x\right)^2 + 1\right) u^y\right)\Big) \\
(A_t)_{8,3} = &\frac{1}{T \left(u^\tau\right)^3 \left(\left(u^y\right)^2 + 1\right)^2} D_s \Big(-h \left(u^\tau\right)^2 \left(\left(u^x\right)^2 + 1\right) \left(\left(u^y\right)^2 + 1\right)^2 \left(\nu^y u^x - \nu^x u^y\right) \\
&+ \pi_Q^{xy} \Big(\left(u^x\right)^4 \left(\left(u^y\right)^2 - 1\right)^2 + \left(\left(u^y\right)^2 + 1\right)^2 \\
&+ \left(u^x\right)^2 \left(\left(u^y\right)^6 - 2 \left(u^y\right)^4 - \left(u^y\right)^2 + 2\right)\Big) \\
&+ u^x u^y \Big(-\pi_Q^{yy} \left(\left(u^x\right)^2 + 1\right) \left(\left(u^x\right)^2 \left(\left(u^y\right)^2 - 1\right) + \left(u^y\right)^4 - \left(u^y\right)^2 - 2\right) \\
&+ \pi_Q{}^{\eta}{}_{\eta} \Big(- \left(u^x\right)^4 \left(\left(u^y\right)^2 - 1\right) + \left(\left(u^y\right)^2 + 1\right)^2 \\
&+ \left(u^x\right)^2 \left(- \left(u^y\right)^4 + \left(u^y\right)^2 + 2\right)\Big)\Big)\Big) \\
(A_t)_{8,7} = &(\partial_\alpha n) D_s u^\tau u^x \\
(A_t)_{8,8} = &\frac{D_s h u^\tau}{T} \\
(A_t)_{8,10} = &-\frac{D_s u^x \left(\left(u^x\right)^2 + 1\right)}{T u^\tau \left(\left(u^y\right)^2 + 1\right)} \\
(A_t)_{8,11} = &-\frac{D_s u^\tau u^x}{T \left(u^y\right)^2 + T} \\
(A_t)_{8,12} = &\frac{D_s \left(2 \left(u^x\right)^2 u^y + \left(u^y\right)^3 + u^y\right)}{T u^\tau \left(\left(u^y\right)^2 + 1\right)} \\
(A_t)_{8,13} = &\frac{D_s u^\tau u^x}{T} \\
(A_t)_{9,1} = &\frac{D_s u^\tau (h'(T) \nu^y + u^y ((\partial_T n) T + n))}{T} \\
(A_t)_{9,2} = &\frac{1}{T \left(u^\tau\right)^3} D_s \Big(h \left(u^\tau\right)^2 \left(\left(u^y\right)^2 + 1\right) \left(\nu^y u^x - \nu^x u^y\right) - \pi_Q^{xy} \left(\left(u^x\right)^2 \left(u^y\right)^2 + \left(u^y\right)^4 - 1\right) \\
&+ u^x u^y \left(\pi_Q^{yy} \left(\left(u^x\right)^2 + \left(u^y\right)^2\right) + \pi_Q{}^{\eta}{}_{\eta} \left(u^\tau\right)^2\right)\Big) \\
(A_t)_{9,3} = &\frac{1}{T \left(u^\tau\right)^3 \left(\left(u^y\right)^2 + 1\right)} D_s \Big(\pi_Q^{yy} + \pi_Q^{yy} \left(u^x\right)^2 - 2 \pi_Q^{xy} u^x u^y - \pi_Q^{xy} \left(u^x\right)^3 u^y - \pi_Q^{yy} \left(u^x\right)^2 \left(u^y\right)^2 \\
&- \pi_Q{}^{\eta}{}_{\eta} \left(u^x\right)^2 \left(u^y\right)^2 - \pi_Q^{yy} \left(u^x\right)^4 \left(u^y\right)^2 - \pi_Q{}^{\eta}{}_{\eta} \left(u^x\right)^4 \left(u^y\right)^2 \\
&- \pi_Q^{xy} u^x \left(u^y\right)^3 + \pi_Q^{xy} \left(u^x\right)^3 \left(u^y\right)^3 - \pi_Q^{yy} \left(u^y\right)^4 - \pi_Q^{yy} \left(u^x\right)^2 \left(u^y\right)^4 \\
&- \pi_Q{}^{\eta}{}_{\eta} \left(u^x\right)^2 \left(u^y\right)^4 + \pi_Q^{xy} u^x \left(u^y\right)^5 + \Pi_Q \left(u^\tau\right)^4 \left(\left(u^y\right)^2 + 1\right) \\
&+ h \left(u^\tau\right)^2 \left(\left(u^y\right)^2 + 1\right) \left(n \left(u^\tau\right)^2 + \nu^x u^x \left(\left(u^y\right)^2 + 1\right) - \nu^y \left(\left(u^x\right)^2 - 1\right) u^y\right)\Big) \\
(A_t)_{9,7} = &(\partial_\alpha n) D_s u^\tau u^y \\
(A_t)_{9,9} = &\frac{D_s h u^\tau}{T} \\
(A_t)_{9,10} = &\frac{D_s u^y}{T u^\tau} \\
(A_t)_{9,12} = &\frac{D_s u^x}{T u^\tau} \\
(A_t)_{9,13} = &\frac{D_s u^\tau u^y}{T} \\
(A_t)_{10,1} = &\frac{1}{6 T u^\tau} D_s \Big(3 \mathcal{C}'(T) \pi_Q^{yy} \left(u^\tau\right)^2 - 2 \mathcal{B}'(T) \nu^x u^x \left(\left(u^y\right)^2 + 1\right) \\
&+ 2 \mathcal{B}'(T) \nu^y u^y \left(3 \left(u^x\right)^2 + 2 \left(u^y\right)^2 + 2\right)\Big)
\end{aligned}
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\begin{aligned}
(A_t)_{10,2} = &-\frac{1}{6T\left(u^\tau\right)^3} D_s\Big(2\mathcal{A}\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right)\left(\nu^x\left(u^y\right)^2+\nu^x-\nu^y u^x u^y\right)\\
&+2\mathcal{B}\left(\left(3\left(u^x\right)^2+4\right)\left(u^y\right)^2+3\left(u^y\right)^4+1\right)\left(\nu^x\left(u^y\right)^2+\nu^x-\nu^y u^x u^y\right)\\
&+\left(u^\tau\right)^2\Big(2I_{31}u^x\left(\left(u^y\right)^2+1\right)\\
&+\mathcal{C}\Big(2\Pi_Q u^x\left(\left(u^y\right)^2+1\right)\\
&-2\left(\left(u^y\right)^2+1\right)\left(\pi_Q{}^\eta{}_\eta u^x-4\pi_Q^{xy}u^y\right)\\
&-\pi_Q^{yy}u^x\left(8\left(u^y\right)^2+5\right)\Big)\Big)\Big)\\
(A_t)_{10,3} = &\frac{1}{6T\left(u^\tau\right)^3} D_s\Big(2\mathcal{B}\nu^x u^x u^y\left(\left(3\left(u^x\right)^2+4\right)\left(u^y\right)^2+3\left(u^y\right)^4+1\right)\\
&-2\mathcal{A}\left(u^\tau\right)^2\left(\nu^y\left(u^x\right)^2\left(\left(u^y\right)^2-2\right)-2\nu^y\left(\left(u^y\right)^2+1\right)\right.\\
&\left.-\nu^x u^x u^y\left(\left(u^y\right)^2+1\right)\right)\\
&-2\mathcal{B}\nu^y\Big(3\left(u^x\right)^4\left(\left(u^y\right)^2-1\right)-2\left(\left(u^y\right)^2+1\right)\\
&+\left(u^x\right)^2\left(3\left(u^y\right)^4+\left(u^y\right)^2-5\right)\Big)\\
&+\left(u^\tau\right)^2\Big(2I_{31}u^y\left(3\left(u^x\right)^2+2\left(u^y\right)^2+2\right)+4\mathcal{C}\pi_Q^{xy}\left(2u^x\left(u^y\right)^2+u^x\right)\\
&+\mathcal{C}u^y\Big(\pi_Q^{yy}-8\pi_Q^{yy}\left(u^x\right)^2-2\pi_Q{}^\eta{}_\eta\left(u^x\right)^2\\
&+\Pi_Q\left(6\left(u^x\right)^2+4\left(u^y\right)^2+4\right)\Big)\Big)\Big)\\
(A_t)_{10,8} = &-\frac{\mathcal{B}D_s u^x\left(\left(u^y\right)^2+1\right)}{3Tu^\tau}\\
(A_t)_{10,9} = &\frac{\mathcal{B}D_s u^y\left(3\left(u^x\right)^2+2\left(u^y\right)^2+2\right)}{3Tu^\tau}\\
(A_t)_{10,10} = &\frac{\mathcal{C}D_s u^\tau}{2T}\\
(A_t)_{11,1} = &\frac{D_s\left(3\mathcal{C}'(T)\pi_Q{}^\eta{}_\eta\left(u^\tau\right)^2-2\mathcal{B}'(T)(\nu^x u^x+\nu^y u^y)\right)}{6Tu^\tau}\\
(A_t)_{11,2} = &-\frac{1}{6T\left(u^\tau\right)^3} D_s\Big(2\mathcal{B}\left(\nu^x\left(u^y\right)^2+\nu^x-\nu^y u^x u^y\right)+2\mathcal{A}\left(u^\tau\right)^2\left(\nu^x\left(u^y\right)^2+\nu^x-\nu^y u^x u^y\right)\\
&+\left(u^\tau\right)^2\left(\mathcal{C}(2\Pi_Q u^x+2\pi_Q^{xy}u^y-2\pi_Q^{yy}u^x-5\pi_Q{}^\eta{}_\eta u^x)+2I_{31}u^x\right)\Big)\\
(A_t)_{11,3} = &-\frac{1}{6T\left(u^\tau\right)^3\left(\left(u^y\right)^2+1\right)} D_s\Big(2\mathcal{B}\left(\left(u^y\right)^2+1\right)\left(-\nu^x u^x u^y+\nu^y\left(u^x\right)^2+\nu^y\right)\\
&+2\mathcal{A}\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right)\left(-\nu^x u^x u^y+\nu^y\left(u^x\right)^2+\nu^y\right)\\
&+\left(u^\tau\right)^2\Big(-2\mathcal{C}\pi_Q^{xy}u^x\left(\left(u^y\right)^2-1\right)+2I_{31}u^y\left(\left(u^y\right)^2+1\right)\\
&+\mathcal{C}u^y\Big(2\pi_Q^{yy}\left(\left(u^x\right)^2+1\right)\\
&+\pi_Q{}^\eta{}_\eta\left(2\left(u^x\right)^2-3\left(u^y\right)^2-3\right)+2\Pi_Q\left(\left(u^y\right)^2+1\right)\Big)\Big)\Big)\\
(A_t)_{11,8} = &-\frac{\mathcal{B}D_s u^x}{3Tu^\tau}\\
(A_t)_{11,9} = &-\frac{\mathcal{B}D_s u^y}{3Tu^\tau}\\
(A_t)_{11,11} = &\frac{\mathcal{C}D_s u^\tau}{2T}\\
(A_t)_{12,1} = &\frac{1}{6Tu^\tau} D_s\Big(3\mathcal{C}'(T)\pi_Q^{xy}\left(u^\tau\right)^2+\mathcal{B}'(T)\nu^y u^x\left(3\left(u^x\right)^2+\left(u^y\right)^2+3\right)\\
&+\mathcal{B}'(T)\nu^x u^y\left(\left(u^x\right)^2+3\left(u^y\right)^2+3\right)\Big)
\end{aligned}
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\begin{aligned}
(A_t)_{12,2} = &\frac{1}{6T\left(u^\tau\right)^3} D_s\Big(\mathcal{A}\left(u^\tau\right)^2\Big(3\nu^y\left(\left(u^y\right)^2+1\right)-2\nu^x u^x u^y\left(\left(u^y\right)^2+1\right)\\
&+\nu^y\left(u^x\right)^2\left(2\left(u^y\right)^2+3\right)\Big)\\
&+\left(u^\tau\right)^2\Big(\mathcal{C}\pi_Q^{xy}u^x\left(3-8\left(u^y\right)^2\right)+I_{31}u^y\left(\left(u^x\right)^2+3\left(u^y\right)^2+3\right)\\
&+\mathcal{C}u^y\Big(\pi_Q^{yy}\left(8\left(u^x\right)^2+6\right)+\Pi_Q\left(\left(u^x\right)^2+3\left(u^y\right)^2+3\right)\\
&+{\pi_Q}^\eta{}_\eta\left(5\left(u^x\right)^2+3\left(u^y\right)^2+3\right)\Big)\Big)\\
&+\mathcal{B}\Big(-2\nu^x u^x u^y\left(\left(u^y\right)^2+1\right)\left(3\left(u^x\right)^2+3\left(u^y\right)^2+4\right)\\
&+\nu^y\Big(3\left(\left(u^y\right)^2+1\right)^2+\left(u^x\right)^4\left(6\left(u^y\right)^2+3\right)\\
&+2\left(u^x\right)^2\left(3\left(u^y\right)^4+7\left(u^y\right)^2+3\right)\Big)\Big)\Big)\\
(A_t)_{12,3} = &\frac{1}{6T\left(u^\tau\right)^3\left(\left(u^y\right)^2+1\right)} D_s\Big(\mathcal{A}\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right)\left(-2\nu^y u^x\left(\left(u^x\right)^2+1\right)u^y+3\nu^x\left(\left(u^y\right)^2+1\right)\right.\\
&\left.+\nu^x\left(u^x\right)^2\left(2\left(u^y\right)^2+3\right)\right)\\
&+\left(u^\tau\right)^2\Big(I_{31}u^x\left(\left(u^y\right)^2+1\right)\left(3\left(u^x\right)^2+\left(u^y\right)^2+3\right)\\
&+\mathcal{C}\Big(-3{\pi_Q}^\eta{}_\eta u^x-3{\pi_Q}^\eta{}_\eta\left(u^x\right)^3+3\pi_Q^{xy}u^y+4\pi_Q^{xy}\left(u^x\right)^2u^y\\
&-6{\pi_Q}^\eta{}_\eta u^x\left(u^y\right)^2-5{\pi_Q}^\eta{}_\eta\left(u^x\right)^3\left(u^y\right)^2+3\pi_Q^{xy}\left(u^y\right)^3\\
&+8\pi_Q^{xy}\left(u^x\right)^2\left(u^y\right)^3-3{\pi_Q}^\eta{}_\eta u^x\left(u^y\right)^4\\
&+\Pi_Q u^x\left(\left(u^y\right)^2+1\right)\left(3\left(u^x\right)^2+\left(u^y\right)^2+3\right)\\
&-2\pi_Q^{yy}u^x\left(\left(u^x\right)^2+1\right)\left(4\left(u^y\right)^2+3\right)\Big)\Big)\\
&+\mathcal{B}\left(\left(u^y\right)^2+1\right)\Big(-2\nu^y u^x\left(\left(u^x\right)^2+1\right)u^y\left(3\left(u^x\right)^2+3\left(u^y\right)^2+4\right)\\
&+\nu^x\Big(3\left(\left(u^y\right)^2+1\right)^2+\left(u^x\right)^4\left(6\left(u^y\right)^2+3\right)\\
&+2\left(u^x\right)^2\left(3\left(u^y\right)^4+7\left(u^y\right)^2+3\right)\Big)\Big)\Big)\\
(A_t)_{12,8} = &\frac{\mathcal{B}D_s u^y\left(\left(u^x\right)^2+3\left(u^y\right)^2+3\right)}{6Tu^\tau}\\
(A_t)_{12,9} = &\frac{\mathcal{B}D_s u^x\left(3\left(u^x\right)^2+\left(u^y\right)^2+3\right)}{6Tu^\tau}\\
(A_t)_{12,12} = &\frac{\mathcal{C}D_s u^\tau}{2T}\\
(A_t)_{13,1} = &\frac{D_s\left(5\mathcal{B}'(T)(\nu^x u^x+\nu^y u^y)+3\mathcal{C}'(T)\Pi_Q\left(u^\tau\right)^2+3(\partial_T I_{31})\left(u^\tau\right)^2\right)}{6Tu^\tau}\\
(A_t)_{13,2} = &\frac{1}{6T\left(u^\tau\right)^3} D_s\Big(5\mathcal{B}\left(\nu^x\left(u^y\right)^2+\nu^x-\nu^y u^x u^y\right)+2\mathcal{A}\left(u^\tau\right)^2\left(\nu^x\left(u^y\right)^2+\nu^x-\nu^y u^x u^y\right)\\
&+\left(u^\tau\right)^2\left(\mathcal{C}(5\Pi_Q u^x+2\pi_Q^{xy}u^y-2\pi_Q^{yy}u^x-2{\pi_Q}^\eta{}_\eta u^x)+5I_{31}u^x\right)\Big)\\
(A_t)_{13,3} = &\frac{1}{6T\left(u^\tau\right)^3\left(\left(u^y\right)^2+1\right)} D_s\Big(5\mathcal{B}\left(\left(u^y\right)^2+1\right)\left(-\nu^x u^x u^y+\nu^y\left(u^x\right)^2+\nu^y\right)\\
&+2\mathcal{A}\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right)\left(-\nu^x u^x u^y+\nu^y\left(u^x\right)^2+\nu^y\right)\\
&+\left(u^\tau\right)^2\Big(5I_{31}u^y\left(\left(u^y\right)^2+1\right)\\
&+\mathcal{C}\Big(2u^y\left(\pi_Q^{yy}\left(u^x\right)^2+\pi_Q^{yy}+{\pi_Q}^\eta{}_\eta\left(u^x\right)^2\right)\\
&-2\pi_Q^{xy}u^x\left(\left(u^y\right)^2-1\right)+5\Pi_Q u^y\left(\left(u^y\right)^2+1\right)\Big)\Big)\Big)
\end{aligned}
$$

$$(A_t)_{13,7} = \frac{(\partial_\alpha I_{31}) D_s u^\tau}{2T}$$
$$(A_t)_{13,8} = \frac{5\mathcal{B} D_s u^x}{6T u^\tau}$$
$$(A_t)_{13,9} = \frac{5\mathcal{B} D_s u^y}{6T u^\tau}$$
$$(A_t)_{13,13} = \frac{\mathcal{C} D_s u^\tau}{2T}$$
$$(A_x)_{1,1} = \frac{\left(u^\tau\right)^2 u^x \left(16 a_{\mathrm{SB}} T^3 + (\partial_T n) h\right) + h'(T) \left(n \left(u^\tau\right)^2 u^x + \nu^x \left(2 \left(u^x\right)^2 + \left(u^y\right)^2 + 1\right) + \nu^y u^x u^y\right)}{u^\tau}$$
$$\begin{aligned}(A_x)_{1,2} = \frac{1}{\left(u^\tau\right)^3 \left(\left(u^y\right)^2 + 1\right)} \Big(& -\pi^\eta{}_\eta - \pi^{yy} - \pi_Q^{yy} - \pi_Q{}^\eta{}_\eta + 4 a_{\mathrm{SB}} T^4 - 3 \pi^\eta{}_\eta \left(u^x\right)^2 - 3 \pi^{yy} \left(u^x\right)^2 \\ & - 3 \pi_Q^{yy} \left(u^x\right)^2 - 3 \pi_Q{}^\eta{}_\eta \left(u^x\right)^2 + 12 a_{\mathrm{SB}} T^4 \left(u^x\right)^2 - 2 \pi^\eta{}_\eta \left(u^x\right)^4 \\ & - 2 \pi^{yy} \left(u^x\right)^4 - 2 \pi_Q^{yy} \left(u^x\right)^4 - 2 \pi_Q{}^\eta{}_\eta \left(u^x\right)^4 + 8 a_{\mathrm{SB}} T^4 \left(u^x\right)^4 \\ & + 3 \pi^{xy} u^x u^y + 3 \pi_Q^{xy} u^x u^y + 2 \pi^{xy} \left(u^x\right)^3 u^y + 2 \pi_Q^{xy} \left(u^x\right)^3 u^y \\ & - 2 \pi^\eta{}_\eta \left(u^y\right)^2 - \pi^{yy} \left(u^y\right)^2 - \pi_Q^{yy} \left(u^y\right)^2 - 2 \pi_Q{}^\eta{}_\eta \left(u^y\right)^2 \\ & + 12 a_{\mathrm{SB}} T^4 \left(u^y\right)^2 - 3 \pi^\eta{}_\eta \left(u^x\right)^2 \left(u^y\right)^2 - 3 \pi^{yy} \left(u^x\right)^2 \left(u^y\right)^2 \\ & - 3 \pi_Q^{yy} \left(u^x\right)^2 \left(u^y\right)^2 - 3 \pi_Q{}^\eta{}_\eta \left(u^x\right)^2 \left(u^y\right)^2 + 24 a_{\mathrm{SB}} T^4 \left(u^x\right)^2 \left(u^y\right)^2 \\ & + 8 a_{\mathrm{SB}} T^4 \left(u^x\right)^4 \left(u^y\right)^2 + 3 \pi^{xy} u^x \left(u^y\right)^3 + 3 \pi_Q^{xy} u^x \left(u^y\right)^3 - \pi^\eta{}_\eta \left(u^y\right)^4 \\ & - \pi_Q{}^\eta{}_\eta \left(u^y\right)^4 + 12 a_{\mathrm{SB}} T^4 \left(u^y\right)^4 + 12 a_{\mathrm{SB}} T^4 \left(u^x\right)^2 \left(u^y\right)^4 + 4 a_{\mathrm{SB}} T^4 \left(u^y\right)^6 \\ & + \Pi_Q \left(\left(u^y\right)^2 + 1\right) \left(2 \left(u^x\right)^4 + 3 \left(u^x\right)^2 \left(\left(u^y\right)^2 + 1\right) + \left(\left(u^y\right)^2 + 1\right)^2\right) \\ & + h \left(\left(u^y\right)^2 + 1\right) \left(\nu^y u^y \left(\left(u^y\right)^2 + 1\right)\right. \\ & + \nu^x u^x \left(2 \left(u^x\right)^2 + 3 \left(u^y\right)^2 + 3\right) \\ & \left. + n \left(2 \left(u^x\right)^4 + 3 \left(u^x\right)^2 \left(\left(u^y\right)^2 + 1\right) + \left(\left(u^y\right)^2 + 1\right)^2\right)\right)\Big)\end{aligned}$$
$$\begin{aligned}(A_x)_{1,3} = \frac{1}{\left(u^\tau\right)^3 \left(\left(u^y\right)^2 + 1\right)^2} \Big(& \Pi_Q \left(u^\tau\right)^2 u^x u^y \left(\left(u^y\right)^2 + 1\right)^2 + n \left(u^\tau\right)^2 u^x u^y \left(\left(u^y\right)^2 + 1\right)^2 (h - T) \\ & + n T \left(u^\tau\right)^2 u^x u^y \left(\left(u^y\right)^2 + 1\right)^2 + 4 a_{\mathrm{SB}} T^4 \left(u^\tau\right)^2 u^x u^y \left(\left(u^y\right)^2 + 1\right)^2 \\ & + h \left(\left(u^y\right)^2 + 1\right)^2 \left(\nu^x u^y \left(\left(u^y\right)^2 + 1\right) + \nu^y \left(\left(u^x\right)^3 + u^x\right)\right) \\ & - u^y \left(\left(u^y\right)^2 + 1\right) \left(-\pi^\eta{}_\eta \left(u^\tau\right)^2 u^x - \pi^{yy} \left(\left(u^x\right)^3 + u^x\right)\right. \\ & \left. + \pi^{xy} u^y \left(2 \left(u^x\right)^2 + \left(u^y\right)^2 + 1\right)\right) \\ & - u^y \left(\left(u^y\right)^2 + 1\right) \left(-\pi_Q{}^\eta{}_\eta \left(u^\tau\right)^2 u^x - \pi_Q^{yy} \left(\left(u^x\right)^3 + u^x\right)\right. \\ & \left. + \pi_Q^{xy} u^y \left(2 \left(u^x\right)^2 + \left(u^y\right)^2 + 1\right)\right) \\ & + \left(u^\tau\right)^2 \left(2 u^x u^y \left(\pi^\eta{}_\eta \left(u^x\right)^2 + \pi^{yy} \left(u^x\right)^2 + \pi^{yy}\right)\right. \\ & \left. + \pi^{xy} \left(\left(\left(u^y\right)^2 + 1\right)^2 - 2 \left(u^x\right)^2 \left(\left(u^y\right)^2 - 1\right)\right)\right) \\ & + \left(u^\tau\right)^2 \left(2 u^x u^y \left(\pi_Q^{yy} \left(u^x\right)^2 + \pi_Q^{yy} + \pi_Q{}^\eta{}_\eta \left(u^x\right)^2\right)\right. \\ & \left. + \pi_Q^{xy} \left(\left(\left(u^y\right)^2 + 1\right)^2 - 2 \left(u^x\right)^2 \left(\left(u^y\right)^2 - 1\right)\right)\right)\Big)\end{aligned}$$
$$(A_x)_{1,4} = -\frac{u^x \left(\left(u^x\right)^2 + 1\right)}{u^\tau \left(\left(u^y\right)^2 + 1\right)}$$

$$
\begin{aligned}
(A_x)_{1,5} &= -\frac{u^\tau u^x}{(u^y)^2+1}\\
(A_x)_{1,6} &= \frac{u^y\left(2\left(u^x\right)^2+\left(u^y\right)^2+1\right)}{u^\tau\left(\left(u^y\right)^2+1\right)}\\
(A_x)_{1,7} &= (\partial_\alpha n)hu^\tau u^x\\
(A_x)_{1,8} &= \frac{h\left(2\left(u^x\right)^2+\left(u^y\right)^2+1\right)}{u^\tau}\\
(A_x)_{1,9} &= \frac{hu^x u^y}{u^\tau}\\
(A_x)_{1,10} &= -\frac{u^x\left(\left(u^x\right)^2+1\right)}{u^\tau\left(\left(u^y\right)^2+1\right)}\\
(A_x)_{1,11} &= -\frac{u^\tau u^x}{(u^y)^2+1}\\
(A_x)_{1,12} &= \frac{u^y\left(2\left(u^x\right)^2+\left(u^y\right)^2+1\right)}{u^\tau\left(\left(u^y\right)^2+1\right)}\\
(A_x)_{1,13} &= u^\tau u^x\\
(A_x)_{2,1} &= 16a_{\mathrm{SB}}T^3\left(u^x\right)^2+4a_{\mathrm{SB}}T^3+h'(T)n\left(u^x\right)^2+2h'(T)\nu^x u^x+(\partial_T n)h\left(u^x\right)^2+(\partial_T n)T+n\\
(A_x)_{2,2} &= \frac{2}{(u^y)^2+1}\Big(-\pi^\eta{}_\eta u^x-\pi^{yy}u^x-\pi_Q^{yy}u^x-\pi_Q{}^\eta{}_\eta u^x+4a_{\mathrm{SB}}T^4u^x+\pi^{xy}u^y+\pi_Q^{xy}u^y\\
&\quad+4a_{\mathrm{SB}}T^4u^x\left(u^y\right)^2+\Pi_Q u^x\left(\left(u^y\right)^2+1\right)+h\left(\left(u^y\right)^2+1\right)\left(nu^x+\nu^x\right)\Big)\\
(A_x)_{2,3} &= \frac{2}{\left(\left(u^y\right)^2+1\right)^2}\Big(u^y\left(\left(u^x\right)^2\left(\pi^\eta{}_\eta+\pi_Q{}^\eta{}_\eta\right)+\pi^{yy}\left(\left(u^x\right)^2+1\right)+\pi_Q^{yy}\left(\left(u^x\right)^2+1\right)\right)\\
&\quad+\pi^{xy}\left(u^x-u^x\left(u^y\right)^2\right)+\pi_Q^{xy}\left(u^x-u^x\left(u^y\right)^2\right)\Big)\\
(A_x)_{2,4} &= -\frac{\left(u^x\right)^2+1}{\left(u^y\right)^2+1}\\
(A_x)_{2,5} &= -\frac{\left(u^\tau\right)^2}{\left(u^y\right)^2+1}\\
(A_x)_{2,6} &= \frac{2u^x u^y}{\left(u^y\right)^2+1}\\
(A_x)_{2,7} &= (\partial_\alpha n)\left(h\left(u^x\right)^2+T\right)\\
(A_x)_{2,8} &= 2hu^x\\
(A_x)_{2,10} &= -\frac{\left(u^x\right)^2+1}{\left(u^y\right)^2+1}\\
(A_x)_{2,11} &= -\frac{\left(u^\tau\right)^2}{\left(u^y\right)^2+1}\\
(A_x)_{2,12} &= \frac{2u^x u^y}{\left(u^y\right)^2+1}\\
(A_x)_{2,13} &= \left(u^x\right)^2+1\\
(A_x)_{3,1} &= u^x u^y\left(16a_{\mathrm{SB}}T^3+(\partial_T n)h\right)+h'(T)u^y(nu^x+\nu^x)+h'(T)\nu^y u^x\\
(A_x)_{3,2} &= u^y\left(4a_{\mathrm{SB}}T^4+\Pi_Q\right)+h(nu^y+\nu^y)\\
(A_x)_{3,3} &= u^x\left(4a_{\mathrm{SB}}T^4+\Pi_Q\right)+h(nu^x+\nu^x)\\
(A_x)_{3,6} &= 1\\
(A_x)_{3,7} &= (\partial_\alpha n)hu^x u^y\\
(A_x)_{3,8} &= hu^y
\end{aligned}
$$

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\begin{aligned}
(A_x)_{3,9} &= hu^x \\
(A_x)_{3,12} &= 1 \\
(A_x)_{3,13} &= u^x u^y \\
(A_x)_{4,2} &= \delta_{\pi\pi}\pi^{yy} - \frac{2\pi^{xy}\tau_\pi u^x \left(\left(u^y\right)^2+1\right) u^y}{\left(u^\tau\right)^2} + \frac{2\pi^{yy}\tau_\pi \left(u^x\right)^2 \left(u^y\right)^2}{\left(u^\tau\right)^2} - \frac{2}{3}\left(\left(u^y\right)^2+1\right)\eta \\
(A_x)_{4,3} &= \frac{2u^x u^y \left(\pi^{xy}\tau_\pi u^x u^y - \pi^{yy}\tau_\pi \left(\left(u^x\right)^2+1\right) + \left(u^\tau\right)^2 \eta\right)}{\left(u^\tau\right)^2} \\
(A_x)_{4,4} &= \tau_\pi u^x \\
(A_x)_{5,2} &= \frac{3\delta_{\pi\pi}\pi^\eta{}_\eta - 2\eta}{3\tau^2} \\
(A_x)_{5,5} &= \frac{\tau_\pi u^x}{\tau^2} \\
(A_x)_{6,2} &= \delta_{\pi\pi}\pi^{xy} \\
&\quad + \frac{u^x \left(u^y \left(3\pi^\eta{}_\eta \tau_\pi \left(u^\tau\right)^2 + 3\pi^{yy}\left(2\tau_\pi \left(u^x\right)^2 + \tau_\pi\right) + \left(u^\tau\right)^2 \eta\right) - 3\pi^{xy}\tau_\pi u^x \left(2\left(u^y\right)^2+1\right)\right)}{3\left(u^\tau\right)^2} \\
(A_x)_{6,3} &= \frac{1}{\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right)}\Big(-\pi^{yy}\tau_\pi \left(u^x\right)^2 \left(\left(u^x\right)^2+1\right)\left(2\left(u^y\right)^2+1\right) \\
&\quad + \pi^{xy}\tau_\pi u^x u^y \left(\left(2\left(u^x\right)^2-1\right)\left(u^y\right)^2 - 1\right) \\
&\quad - \left(u^\tau\right)^2 \left(\pi^\eta{}_\eta \tau_\pi \left(u^x\right)^2 \left(u^y\right)^2 - \left(\left(u^x\right)^2+1\right)\left(\left(u^y\right)^2+1\right)\eta\right)\Big) \\
(A_x)_{6,6} &= \tau_\pi u^x \\
(A_x)_{7,1} &= (\partial_T n) u^x \\
(A_x)_{7,2} &= n \\
(A_x)_{7,7} &= (\partial_\alpha n) u^x \\
(A_x)_{7,8} &= 1 \\
(A_x)_{8,1} &= \frac{D_s\left(h'(T)\nu^x u^x + (\partial_T n)T\left(\left(u^x\right)^2+1\right) + n\left(\left(u^x\right)^2+1\right)\right)}{T} \\
(A_x)_{8,2} &= -\frac{1}{T\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right)} D_s\Big(\pi_Q^{yy} u^x + \pi_Q{}^\eta{}_\eta u^x + \pi_Q^{yy}\left(u^x\right)^3 + \pi_Q{}^\eta{}_\eta \left(u^x\right)^3 - 2\pi_Q^{xy} u^y \\
&\quad - \pi_Q^{xy}\left(u^x\right)^2 u^y + \pi_Q^{yy} u^x \left(u^y\right)^2 - \pi_Q^{yy}\left(u^x\right)^3 \left(u^y\right)^2 \\
&\quad - \pi_Q{}^\eta{}_\eta \left(u^x\right)^3 \left(u^y\right)^2 - 2\pi_Q^{xy}\left(u^y\right)^3 + \pi_Q^{xy}\left(u^x\right)^2 \left(u^y\right)^3 \\
&\quad - \pi_Q{}^\eta{}_\eta u^x \left(u^y\right)^4 - \Pi_Q \left(u^\tau\right)^2 u^x \left(\left(u^y\right)^2+1\right) \\
&\quad + h\left(\left(u^y\right)^2+1\right)\Big(-n\left(u^\tau\right)^2 u^x - \nu^y \left(u^x\right)^3 u^y + \nu^x \left(u^x\right)^2 \left(\left(u^y\right)^2-1\right) \\
&\quad - 2\nu^x\left(\left(u^y\right)^2+1\right)\Big)\Big) \\
(A_x)_{8,3} &= \frac{1}{T\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right)^2} D_s\Big(-\pi_Q{}^\eta{}_\eta \left(u^\tau\right)^2 \left(u^x\right)^2 u^y \left(\left(u^y\right)^2-1\right) \\
&\quad - h\left(u^x\right)^2\left(\left(u^y\right)^2+1\right)^2\left(-\nu^x u^x u^y + \nu^y\left(u^x\right)^2 + \nu^y\right) \\
&\quad + \pi_Q^{xy} u^x \left(\left(u^x\right)^2\left(\left(u^y\right)^2-1\right)^2 - 3\left(u^y\right)^4 - 2\left(u^y\right)^2 + 1\right) \\
&\quad - \pi_Q^{yy}\left(\left(u^x\right)^2+1\right) u^y \left(\left(u^x\right)^2\left(\left(u^y\right)^2-1\right) - 2\left(\left(u^y\right)^2+1\right)\right)\Big) \\
(A_x)_{8,7} &= (\partial_\alpha n) D_s\left(\left(u^x\right)^2+1\right) \\
(A_x)_{8,8} &= \frac{D_s h u^x}{T}
\end{aligned}
$$

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\begin{aligned}
(A_x)_{8,10} &= -\frac{D_s \left(u^x\right)^2 + D_s}{T \left(u^y\right)^2 + T} \\
(A_x)_{8,11} &= -\frac{D_s \left(u^\tau\right)^2}{T \left(\left(u^y\right)^2 + 1\right)} \\
(A_x)_{8,12} &= \frac{2 D_s u^x u^y}{T \left(u^y\right)^2 + T} \\
(A_x)_{8,13} &= \frac{D_s \left(\left(u^x\right)^2 + 1\right)}{T} \\
(A_x)_{9,1} &= \frac{D_s u^x (h'(T) \nu^y + u^y ((\partial_T n) T + n))}{T} \\
(A_x)_{9,2} &= \frac{1}{T \left(u^\tau\right)^2} D_s \Big(h \left(\left(u^y\right)^2 + 1\right) \left(-\nu^x u^x u^y + \nu^y \left(u^x\right)^2 + \nu^y \right) \\
&\quad + u^y \left(-\pi_Q^{xy} u^x u^y + \pi_Q^{yy} \left(u^x\right)^2 + \pi_Q^{yy} + {\pi_Q}^\eta{}_\eta \left(u^x\right)^2 + {\pi_Q}^\eta{}_\eta \left(u^y\right)^2 + {\pi_Q}^\eta{}_\eta \right) \Big) \\
(A_x)_{9,3} &= \frac{1}{T \left(u^\tau\right)^2 \left(\left(u^y\right)^2 + 1\right)} D_s \Big(\Pi_Q \left(u^\tau\right)^2 u^x \left(\left(u^y\right)^2 + 1\right) \\
&\quad + h \left(\left(u^y\right)^2 + 1\right) \left(n \left(u^\tau\right)^2 u^x - \nu^y u^x \left(\left(u^x\right)^2 + 1\right) u^y \right. \\
&\quad \left. + \nu^x \left(\left(u^x\right)^2 + 1\right) \left(\left(u^y\right)^2 + 1\right) \right) \\
&\quad - u^y \Big(u^x u^y \left(\pi_Q^{yy} \left(\left(u^x\right)^2 + 1\right) + {\pi_Q}^\eta{}_\eta \left(u^\tau\right)^2 \right) \\
&\quad + \pi_Q^{xy} \left(- \left(\left(u^x\right)^2 \left(\left(u^y\right)^2 - 1\right)\right) + \left(u^y\right)^2 + 1 \right) \Big) \Big) \\
(A_x)_{9,7} &= (\partial_\alpha n) D_s u^x u^y \\
(A_x)_{9,9} &= \frac{D_s h u^x}{T} \\
(A_x)_{9,12} &= \frac{D_s}{T} \\
(A_x)_{9,13} &= \frac{D_s u^x u^y}{T} \\
(A_x)_{10,1} &= \frac{D_s \left(3\mathcal{C}'(T) \pi_Q^{yy} u^x - 2\mathcal{B}'(T) \left(\nu^x \left(u^y\right)^2 + \nu^x - 3\nu^y u^x u^y \right) \right)}{6T} \\
(A_x)_{10,2} &= -\frac{1}{6T \left(u^\tau\right)^2} D_s \Big(2 I_{31} \left(u^\tau\right)^2 \left(\left(u^y\right)^2 + 1\right) \\
&\quad + 2u^x \left(\mathcal{A} \left(u^y\right)^2 + \mathcal{A} + 3\mathcal{B} \left(u^y\right)^2 \right) \left(\nu^x \left(u^y\right)^2 + \nu^x - \nu^y u^x u^y \right) \\
&\quad + \mathcal{C} \Big(2\Pi_Q \left(u^\tau\right)^2 \left(\left(u^y\right)^2 + 1\right) \\
&\quad - 2 \left(\left(u^y\right)^2 + 1\right) \left({\pi_Q}^\eta{}_\eta \left(u^\tau\right)^2 - 4\pi_Q^{xy} u^x u^y \right) \\
&\quad - \pi_Q^{yy} \left(\left(u^x\right)^2 \left(8 \left(u^y\right)^2 + 5 \right) + 5 \left(\left(u^y\right)^2 + 1\right) \right) \Big) \Big) \\
(A_x)_{10,3} &= \frac{1}{3T \left(u^\tau\right)^2} D_s \Big(u^x \Big(3 I_{31} \left(u^\tau\right)^2 u^y - \mathcal{A} \nu^y \left(u^x\right)^2 \left(\left(u^y\right)^2 - 2\right) + 2\mathcal{A} \nu^y \left(\left(u^y\right)^2 + 1\right) \\
&\quad + \mathcal{A} \nu^x u^x u^y \left(\left(u^y\right)^2 + 1\right) \\
&\quad + 3\mathcal{B} \left(\nu^x u^x \left(u^y\right)^3 - \nu^y \left(u^x\right)^2 \left(\left(u^y\right)^2 - 1\right) + \nu^y \right) \Big) \\
&\quad + \mathcal{C} \Big(-u^x u^y \left(-3\Pi_Q \left(u^\tau\right)^2 + 4\pi_Q^{yy} \left(\left(u^x\right)^2 + 1\right) + {\pi_Q}^\eta{}_\eta \left(u^\tau\right)^2 \right) \\
&\quad + 2\pi_Q^{xy} \left(\left(u^x\right)^2 \left(2 \left(u^y\right)^2 + 1 \right) + \left(u^y\right)^2 + 1 \right) \Big) \Big) \\
(A_x)_{10,8} &= -\frac{\mathcal{B} D_s \left(\left(u^y\right)^2 + 1\right)}{3T} \\
(A_x)_{10,9} &= \frac{\mathcal{B} D_s u^x u^y}{T}
\end{aligned}
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\begin{aligned}
(A_x)_{10,10} &= \frac{\mathcal{C}D_s u^x}{2T} \\
(A_x)_{11,1} &= -\frac{D_s(2\mathcal{B}'(T)\nu^x - 3\mathcal{C}'(T){\pi_Q}^{\eta}{}_{\eta}u^x)}{6T} \\
(A_x)_{11,2} &= -\frac{1}{6T\left(u^{\tau}\right)^2}D_s\Big(2I_{31}\left(u^{\tau}\right)^2 + 2\mathcal{A}u^x\left(\nu^x\left(u^y\right)^2 + \nu^x - \nu^y u^x u^y\right) \\
&\quad + \mathcal{C}\Big(-5{\pi_Q}^{\eta}{}_{\eta} + 2\Pi_Q\left(u^{\tau}\right)^2 - 5{\pi_Q}^{\eta}{}_{\eta}\left(u^x\right)^2 - 2\pi_Q^{yy}\left(\left(u^x\right)^2+1\right) \\
&\quad + 2\pi_Q^{xy}u^x u^y - 5{\pi_Q}^{\eta}{}_{\eta}\left(u^y\right)^2\Big)\Big) \\
(A_x)_{11,3} &= \frac{1}{3T\left(u^{\tau}\right)^2\left(\left(u^y\right)^2+1\right)}D_s\Big(-\mathcal{C}u^x u^y\left(\pi_Q^{yy}\left(\left(u^x\right)^2+1\right) + {\pi_Q}^{\eta}{}_{\eta}\left(u^{\tau}\right)^2\right) \\
&\quad - \mathcal{A}u^x\left(\left(u^y\right)^2+1\right)\left(-\nu^x u^x u^y + \nu^y\left(u^x\right)^2 + \nu^y\right) \\
&\quad + \mathcal{C}\pi_Q^{xy}\left(\left(u^x\right)^2\left(\left(u^y\right)^2-1\right) - \left(u^y\right)^2 - 1\right)\Big) \\
(A_x)_{11,8} &= -\frac{\mathcal{B}D_s}{3T} \\
(A_x)_{11,11} &= \frac{\mathcal{C}D_s u^x}{2T} \\
(A_x)_{12,1} &= \frac{D_s\left(\mathcal{B}'(T)\nu^x u^x u^y + 3\mathcal{B}'(T)\nu^y\left(\left(u^x\right)^2+1\right) + 3\mathcal{C}'(T)\pi_Q^{xy}u^x\right)}{6T} \\
(A_x)_{12,2} &= \frac{1}{6T\left(u^{\tau}\right)^2}D_s\Big(3\mathcal{B}\nu^y u^x + 3\mathcal{B}\nu^y\left(u^x\right)^3 - 3\mathcal{B}\nu^x u^y + I_{31}u^x u^y - 6\mathcal{B}\nu^x\left(u^x\right)^2 u^y + I_{31}\left(u^x\right)^3 u^y \\
&\quad + 6\mathcal{B}\nu^y u^x\left(u^y\right)^2 + 6\mathcal{B}\nu^y\left(u^x\right)^3\left(u^y\right)^2 - 3\mathcal{B}\nu^x\left(u^y\right)^3 + I_{31}u^x\left(u^y\right)^3 \\
&\quad - 6\mathcal{B}\nu^x\left(u^x\right)^2\left(u^y\right)^3 \\
&\quad + \mathcal{A}u^x\Big(3\nu^y\left(\left(u^y\right)^2+1\right) - 2\nu^x u^x u^y\left(\left(u^y\right)^2+1\right) \\
&\quad + \nu^y\left(u^x\right)^2\left(2\left(u^y\right)^2+3\right)\Big) \\
&\quad + \mathcal{C}\Big(u^x u^y\left(\Pi_Q\left(u^{\tau}\right)^2 + \pi_Q^{yy}\left(8\left(u^x\right)^2+5\right) + 5{\pi_Q}^{\eta}{}_{\eta}\left(u^{\tau}\right)^2\right) \\
&\quad + \pi_Q^{xy}\left(\left(u^x\right)^2\left(3 - 8\left(u^y\right)^2\right) + 6\left(\left(u^y\right)^2+1\right)\right)\Big)\Big) \\
(A_x)_{12,3} &= \frac{1}{6T\left(u^{\tau}\right)^2\left(\left(u^y\right)^2+1\right)}D_s\Big(3I_{31}\left(u^{\tau}\right)^2\left(\left(u^x\right)^2+1\right)\left(\left(u^y\right)^2+1\right) \\
&\quad + \mathcal{C}\Big(-3{\pi_Q}^{\eta}{}_{\eta} - 6{\pi_Q}^{\eta}{}_{\eta}\left(u^x\right)^2 - 3{\pi_Q}^{\eta}{}_{\eta}\left(u^x\right)^4 + \pi_Q^{xy}u^x u^y \\
&\quad + 4\pi_Q^{xy}\left(u^x\right)^3 u^y - 6{\pi_Q}^{\eta}{}_{\eta}\left(u^y\right)^2 - 11{\pi_Q}^{\eta}{}_{\eta}\left(u^x\right)^2\left(u^y\right)^2 \\
&\quad - 5{\pi_Q}^{\eta}{}_{\eta}\left(u^x\right)^4\left(u^y\right)^2 + \pi_Q^{xy}u^x\left(u^y\right)^3 + 8\pi_Q^{xy}\left(u^x\right)^3\left(u^y\right)^3 \\
&\quad - 3{\pi_Q}^{\eta}{}_{\eta}\left(u^y\right)^4 - 5{\pi_Q}^{\eta}{}_{\eta}\left(u^x\right)^2\left(u^y\right)^4 \\
&\quad + 3\Pi_Q\left(u^{\tau}\right)^2\left(\left(u^x\right)^2+1\right)\left(\left(u^y\right)^2+1\right) \\
&\quad - \pi_Q^{yy}\left(\left(u^x\right)^2+1\right)\left(\left(u^x\right)^2\left(8\left(u^y\right)^2+6\right) + 3\left(\left(u^y\right)^2+1\right)\right)\Big) \\
&\quad + \left(u^y\right)^2 + 1\Big(\mathcal{A}u^x\Big(-2\nu^y u^x\left(\left(u^x\right)^2+1\right)u^y + 3\nu^x\left(\left(u^y\right)^2+1\right) \\
&\quad + \nu^x\left(u^x\right)^2\left(2\left(u^y\right)^2+3\right)\Big) \\
&\quad + 3\mathcal{B}\left(\left(u^x\right)^2+1\right)\left(\nu^x\left(2u^x\left(u^y\right)^2 + u^x\right) - \nu^y\left(2\left(u^x\right)^2+1\right)u^y\right)\Big)\Big) \\
(A_x)_{12,8} &= \frac{\mathcal{B}D_s u^x u^y}{6T} \\
(A_x)_{12,9} &= \frac{\mathcal{B}D_s\left(\left(u^x\right)^2+1\right)}{2T} \\
(A_x)_{12,12} &= \frac{\mathcal{C}D_s u^x}{2T}
\end{aligned}
$$

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\begin{aligned}
(A_x)_{13,1} &= \frac{D_s(5\mathcal{B}'(T)\nu^x + 3u^x(\mathcal{C}'(T)\Pi_Q + (\partial_T I_{31})))}{6T} \\
(A_x)_{13,2} &= \frac{1}{6T\left(u^\tau\right)^2} D_s\Big(5I_{31}\left(u^\tau\right)^2 + 5\mathcal{C}\Pi_Q\left(u^\tau\right)^2 + 2\mathcal{A}u^x\left(\nu^x\left(u^y\right)^2 + \nu^x - \nu^y u^x u^y\right) \\
&\quad - 2\mathcal{C}\left(-\pi_Q^{xy}u^x u^y + \pi_Q^{yy}\left(u^x\right)^2 + \pi_Q^{yy} + \pi_Q{}^\eta{}_\eta\left(u^x\right)^2 + \pi_Q{}^\eta{}_\eta\left(u^y\right)^2 + \pi_Q{}^\eta{}_\eta\right)\Big) \\
(A_x)_{13,3} &= \frac{1}{3T\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right)} D_s\Big(\mathcal{C}u^x u^y\left(\pi_Q^{yy}\left(\left(u^x\right)^2+1\right) + \pi_Q{}^\eta{}_\eta\left(u^\tau\right)^2\right) \\
&\quad + \mathcal{A}u^x\left(\left(u^y\right)^2+1\right)\left(-\nu^x u^x u^y + \nu^y\left(u^x\right)^2 + \nu^y\right) \\
&\quad + \mathcal{C}\pi_Q^{xy}\left(-\left(\left(u^x\right)^2\left(\left(u^y\right)^2-1\right)\right) + \left(u^y\right)^2+1\right)\Big) \\
(A_x)_{13,7} &= \frac{(\partial_\alpha I_{31})D_s u^x}{2T} \\
(A_x)_{13,8} &= \frac{5\mathcal{B}D_s}{6T} \\
(A_x)_{13,13} &= \frac{\mathcal{C}D_s u^x}{2T} \\
(A_y)_{1,1} &= \frac{\left(u^\tau\right)^2 u^y\left(16a_{\mathrm{SB}}T^3 + (\partial_T n)h\right) + h'(T)\left(n\left(u^\tau\right)^2 u^y + \nu^x u^x u^y + \nu^y\left(\left(u^x\right)^2 + 2\left(u^y\right)^2 + 1\right)\right)}{u^\tau} \\
(A_y)_{1,2} &= \frac{1}{\left(u^\tau\right)^3}\Big(\pi^{xy} + \pi_Q^{xy} + h\nu^y u^x + h\nu^y\left(u^x\right)^3 + h\nu^x u^y + hnu^x u^y + \Pi_Q u^x u^y - \pi^{yy}u^x u^y \\
&\quad - \pi_Q^{yy}u^x u^y + 4a_{\mathrm{SB}}T^4 u^x u^y + hn\left(u^x\right)^3 u^y + \Pi_Q\left(u^x\right)^3 u^y + 4a_{\mathrm{SB}}T^4\left(u^x\right)^3 u^y \\
&\quad + \pi^{xy}\left(u^y\right)^2 + \pi_Q^{xy}\left(u^y\right)^2 + h\nu^x\left(u^y\right)^3 + hnu^x\left(u^y\right)^3 + \Pi_Q u^x\left(u^y\right)^3 \\
&\quad + 4a_{\mathrm{SB}}T^4 u^x\left(u^y\right)^3\Big) \\
(A_y)_{1,3} &= \frac{1}{\left(u^\tau\right)^3}\Big(\pi^{yy} + \pi_Q^{yy} + 4a_{\mathrm{SB}}T^4 + \pi^{yy}\left(u^x\right)^2 + \pi_Q^{yy}\left(u^x\right)^2 + 8a_{\mathrm{SB}}T^4\left(u^x\right)^2 + 4a_{\mathrm{SB}}T^4\left(u^x\right)^4 \\
&\quad - \pi^{xy}u^x u^y - \pi_Q^{xy}u^x u^y + 12a_{\mathrm{SB}}T^4\left(u^y\right)^2 + 12a_{\mathrm{SB}}T^4\left(u^x\right)^2\left(u^y\right)^2 + 8a_{\mathrm{SB}}T^4\left(u^y\right)^4 \\
&\quad + \Pi_Q\left(\left(u^x\right)^4 + \left(u^x\right)^2\left(3\left(u^y\right)^2+2\right) + 2\left(u^y\right)^4 + 3\left(u^y\right)^2 + 1\right) \\
&\quad + h\Big(\nu^x\left(\left(u^x\right)^3 + u^x\right) + \nu^y u^y\left(3\left(u^x\right)^2 + 2\left(u^y\right)^2 + 3\right) \\
&\quad + n\left(\left(u^x\right)^4 + \left(u^x\right)^2\left(3\left(u^y\right)^2+2\right) + 2\left(u^y\right)^4 + 3\left(u^y\right)^2 + 1\right)\Big)\Big) \\
(A_y)_{1,4} &= \frac{u^y}{u^\tau} \\
(A_y)_{1,6} &= \frac{u^x}{u^\tau} \\
(A_y)_{1,7} &= (\partial_\alpha n)hu^\tau u^y \\
(A_y)_{1,8} &= \frac{hu^x u^y}{u^\tau} \\
(A_y)_{1,9} &= \frac{h\left(\left(u^x\right)^2 + 2\left(u^y\right)^2 + 1\right)}{u^\tau} \\
(A_y)_{1,10} &= \frac{u^y}{u^\tau} \\
(A_y)_{1,12} &= \frac{u^x}{u^\tau} \\
(A_y)_{1,13} &= u^\tau u^y \\
(A_y)_{2,1} &= u^x u^y\left(16a_{\mathrm{SB}}T^3 + (\partial_T n)h\right) + h'(T)u^y(nu^x + \nu^x) + h'(T)\nu^y u^x \\
(A_y)_{2,2} &= u^y\left(4a_{\mathrm{SB}}T^4 + \Pi_Q\right) + h(nu^y + \nu^y) \\
(A_y)_{2,3} &= u^x\left(4a_{\mathrm{SB}}T^4 + \Pi_Q\right) + h(nu^x + \nu^x) \\
(A_y)_{2,6} &= 1 \\
(A_y)_{2,7} &= (\partial_\alpha n)hu^x u^y \\
(A_y)_{2,8} &= hu^y
\end{aligned}
$$

$$
\begin{aligned}
(A_y)_{2,9} &= hu^x \\
(A_y)_{2,12} &= 1 \\
(A_y)_{2,13} &= u^x u^y \\
(A_y)_{3,1} &= 16a_{\mathrm{SB}}T^3 \left(u^y\right)^2 + 4a_{\mathrm{SB}}T^3 + h'(T)n\left(u^y\right)^2 + 2h'(T)\nu^y u^y + (\partial_T n)h\left(u^y\right)^2 + (\partial_T n)T + n \\
(A_y)_{3,3} &= 2\left(u^y\left(4a_{\mathrm{SB}}T^4 + \Pi_Q\right) + h(nu^y + \nu^y)\right) \\
(A_y)_{3,4} &= 1 \\
(A_y)_{3,7} &= (\partial_\alpha n)\left(h\left(u^y\right)^2 + T\right) \\
(A_y)_{3,9} &= 2hu^y \\
(A_y)_{3,10} &= 1 \\
(A_y)_{3,13} &= \left(u^y\right)^2 + 1 \\
(A_y)_{4,2} &= -\frac{2\tau_\pi\left(u^y\right)^2\left(\pi^{xy}\left(u^y\right)^2 + \pi^{xy} - \pi^{yy}u^x u^y\right)}{\left(u^\tau\right)^2} \\
(A_y)_{4,3} &= \frac{1}{3\left(u^\tau\right)^2}\Big(3\delta_{\pi\pi}\pi^{yy}\left(u^\tau\right)^2 - 6\pi^{yy}\tau_\pi\left(\left(u^x\right)^2 + 1\right)\left(u^y\right)^2 + 6\pi^{xy}\tau_\pi u^x\left(u^y\right)^3 + 4\eta \\
&\quad + 4\left(u^x\right)^2\eta + 8\left(u^y\right)^2\eta + 4\left(u^x\right)^2\left(u^y\right)^2\eta + 4\left(u^y\right)^4\eta\Big) \\
(A_y)_{4,4} &= \tau_\pi u^y \\
(A_y)_{5,3} &= \frac{3\delta_{\pi\pi}\pi^\eta{}_\eta - 2\eta}{3\tau^2} \\
(A_y)_{5,5} &= \frac{\tau_\pi u^y}{\tau^2} \\
(A_y)_{6,2} &= \frac{1}{\left(u^\tau\right)^2}\Big(\pi^{yy}\tau_\pi\left(u^y\right)^2 + \pi^\eta{}_\eta\tau_\pi\left(u^\tau\right)^2\left(u^y\right)^2 + 2\pi^{yy}\tau_\pi\left(u^x\right)^2\left(u^y\right)^2 \\
&\quad - \pi^{xy}\tau_\pi u^x u^y\left(2\left(u^y\right)^2 + 1\right) + \eta + \left(u^x\right)^2\eta + 2\left(u^y\right)^2\eta + \left(u^x\right)^2\left(u^y\right)^2\eta + \left(u^y\right)^4\eta\Big) \\
(A_y)_{6,3} &= \delta_{\pi\pi}\pi^{xy} \\
&\quad - \frac{1}{3\left(u^\tau\right)^2\left(\left(u^y\right)^2 + 1\right)}u^y\Big(3\pi^{yy}\tau_\pi u^x\left(\left(u^x\right)^2 + 1\right)\left(2\left(u^y\right)^2 + 1\right) \\
&\quad + 3\pi^{xy}\tau_\pi u^y\left(\left(1 - 2\left(u^x\right)^2\right)\left(u^y\right)^2 + 1\right) \\
&\quad + \left(u^\tau\right)^2 u^x\left(3\pi^\eta{}_\eta\tau_\pi\left(u^y\right)^2 - \left(\left(u^y\right)^2 + 1\right)\eta\right)\Big) \\
(A_y)_{6,6} &= \tau_\pi u^y \\
(A_y)_{7,1} &= (\partial_T n)u^y \\
(A_y)_{7,3} &= n \\
(A_y)_{7,7} &= (\partial_\alpha n)u^y \\
(A_y)_{7,9} &= 1 \\
(A_y)_{8,1} &= \frac{D_s u^y(h'(T)\nu^x + u^x((\partial_T n)T + n))}{T} \\
(A_y)_{8,2} &= \frac{1}{T\left(u^\tau\right)^2}D_s\Big(\Pi_Q\left(u^\tau\right)^2 u^y + \pi_Q^{yy}\left(u^x\right)^2 u^y - \pi_Q^{xy}u^x\left(\left(u^y\right)^2 + 1\right) \\
&\quad + h\left(n\left(u^\tau\right)^2 u^y - \nu^x u^x u^y\left(\left(u^y\right)^2 + 1\right) + \nu^y\left(\left(u^x\right)^2 + 1\right)\left(\left(u^y\right)^2 + 1\right)\right)\Big) \\
(A_y)_{8,3} &= \frac{D_s\left(h\left(\left(u^x\right)^2 + 1\right)\left(\nu^x\left(u^y\right)^2 + \nu^x - \nu^y u^x u^y\right) - u^x\left(-\pi_Q^{xy}u^x u^y + \pi_Q^{yy}\left(u^x\right)^2 + \pi_Q^{yy}\right)\right)}{T\left(u^\tau\right)^2} \\
(A_y)_{8,7} &= (\partial_\alpha n)D_s u^x u^y \\
(A_y)_{8,8} &= \frac{D_s h u^y}{T} \\
(A_y)_{8,12} &= \frac{D_s}{T} \\
(A_y)_{8,13} &= \frac{D_s u^x u^y}{T}
\end{aligned}
$$

$$(A_y)_{9,1} = \frac{D_s\left(h'(T)\nu^y u^y + (\partial_T n)T\left(\left(u^y\right)^2+1\right) + n\left(\left(u^y\right)^2+1\right)\right)}{T}$$

$$(A_y)_{9,2} = -\frac{D_s u^y\left(hu^y\left(\nu^x\left(u^y\right)^2 + \nu^x - \nu^y u^x u^y\right) + \pi_Q^{xy}\left(\left(u^y\right)^2+1\right) - \pi_Q^{yy}u^x u^y\right)}{T\left(u^\tau\right)^2}$$

$$\begin{aligned}(A_y)_{9,3} = {} & \frac{1}{T\left(u^\tau\right)^2}D_s\Big(u^y\left(\Pi_Q\left(u^\tau\right)^2 + \pi_Q^{xy}u^x u^y - \pi_Q^{yy}\left(\left(u^x\right)^2+1\right)\right)\\ & + h\left(n\left(u^\tau\right)^2 u^y + \nu^x u^x\left(u^y\right)^3 + \nu^y\left(-\left(\left(u^x\right)^2\left(\left(u^y\right)^2-2\right)\right) + \left(u^y\right)^2+2\right)\right)\Big)\end{aligned}$$

$$(A_y)_{9,7} = (\partial_\alpha n)D_s\left(\left(u^y\right)^2+1\right)$$

$$(A_y)_{9,9} = \frac{D_s h u^y}{T}$$

$$(A_y)_{9,10} = \frac{D_s}{T}$$

$$(A_y)_{9,13} = \frac{D_s\left(\left(u^y\right)^2+1\right)}{T}$$

$$(A_y)_{10,1} = \frac{D_s\left(4\mathcal{B}'(T)\nu^y\left(\left(u^y\right)^2+1\right) + 3\mathcal{C}'(T)\pi_Q^{yy}u^y\right)}{6T}$$

$$\begin{aligned}(A_y)_{10,2} = {} & -\frac{1}{3T\left(u^\tau\right)^2}D_s\Big(u^y\left(\left(u^y\right)^2+1\right)(\mathcal{A}+3\mathcal{B})\left(\nu^x\left(u^y\right)^2 + \nu^x - \nu^y u^x u^y\right)\\ & + \mathcal{C}\left(4\left(u^y\right)^2+1\right)\left(\pi_Q^{xy}\left(u^y\right)^2 + \pi_Q^{xy} - \pi_Q^{yy}u^x u^y\right)\Big)\end{aligned}$$

$$\begin{aligned}(A_y)_{10,3} = {} & \frac{1}{6T\left(u^\tau\right)^2}D_s\Big(4I_{31}\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right)\\ & + \mathcal{C}\Big(4\Pi_Q\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right) + 2\pi_Q^{xy}u^x u^y\left(4\left(u^y\right)^2+1\right)\\ & + \pi_Q^{yy}\left(\left(u^x\right)^2\left(7-8\left(u^y\right)^2\right) + \left(u^y\right)^2+7\right)\Big)\\ & + 2u^y\Big(3\mathcal{B}u^x u^y\left(\nu^x\left(u^y\right)^2 + \nu^x - \nu^y u^x u^y\right)\\ & + \mathcal{A}\Big(-\nu^y\left(u^x\right)^2\left(\left(u^y\right)^2-2\right) + 2\nu^y\left(\left(u^y\right)^2+1\right)\\ & + \nu^x u^x u^y\left(\left(u^y\right)^2+1\right)\Big)\Big)\Big)\Big)\end{aligned}$$

$$(A_y)_{10,9} = \frac{2\mathcal{B}D_s\left(\left(u^y\right)^2+1\right)}{3T}$$

$$(A_y)_{10,10} = \frac{\mathcal{C}D_s u^y}{2T}$$

$$(A_y)_{11,1} = -\frac{D_s(2\mathcal{B}'(T)\nu^y - 3\mathcal{C}'(T)\pi_Q{}^\eta{}_\eta u^y)}{6T}$$

$$(A_y)_{11,2} = -\frac{D_s\left(\mathcal{A}u^y\left(\nu^x\left(u^y\right)^2 + \nu^x - \nu^y u^x u^y\right) + \mathcal{C}\left(\pi_Q^{xy}\left(u^y\right)^2 + \pi_Q^{xy} - \pi_Q^{yy}u^x u^y\right)\right)}{3T\left(u^\tau\right)^2}$$

$$\begin{aligned}(A_y)_{11,3} = {} & -\frac{1}{6T\left(u^\tau\right)^2}D_s\Big(2I_{31}\left(u^\tau\right)^2 + 2\mathcal{A}u^y\left(-\nu^x u^x u^y + \nu^y\left(u^x\right)^2 + \nu^y\right)\\ & + \mathcal{C}\Big(-3\pi_Q{}^\eta{}_\eta + 2\Pi_Q\left(u^\tau\right)^2 - 3\pi_Q{}^\eta{}_\eta\left(u^x\right)^2 + 2\pi_Q^{yy}\left(\left(u^x\right)^2+1\right)\\ & - 2\pi_Q^{xy}u^x u^y - 3\pi_Q{}^\eta{}_\eta\left(u^y\right)^2\Big)\Big)\end{aligned}$$

$$(A_y)_{11,9} = -\frac{\mathcal{B}D_s}{3T}$$

$$(A_y)_{11,11} = \frac{\mathcal{C}D_s u^y}{2T}$$

$$(A_y)_{12,1} = \frac{D_s\left(3\mathcal{B}'(T)\nu^x\left(\left(u^y\right)^2+1\right) + \mathcal{B}'(T)\nu^y u^x u^y + 3\mathcal{C}'(T)\pi_Q^{xy}u^y\right)}{6T}$$

$$
\begin{aligned}
(A_y)_{12,2} = &\frac{1}{6T\left(u^\tau\right)^2} D_s \Big(-3\mathcal{B}\nu^x u^x + 3\mathcal{A}\nu^y u^y + 3\mathcal{B}\nu^y u^y + 3\mathcal{A}\nu^y \left(u^x\right)^2 u^y + 6\mathcal{B}\nu^y \left(u^x\right)^2 u^y \\
&- 2\mathcal{A}\nu^x u^x \left(u^y\right)^2 - 9\mathcal{B}\nu^x u^x \left(u^y\right)^2 + 3\mathcal{A}\nu^y \left(u^y\right)^3 + 3\mathcal{B}\nu^y \left(u^y\right)^3 \\
&+ 2\mathcal{A}\nu^y \left(u^x\right)^2 \left(u^y\right)^3 + 6\mathcal{B}\nu^y \left(u^x\right)^2 \left(u^y\right)^3 - 2\mathcal{A}\nu^x u^x \left(u^y\right)^4 - 6\mathcal{B}\nu^x u^x \left(u^y\right)^4 \\
&+ 3I_{31} \left(u^\tau\right)^2 \Big(\left(u^y\right)^2 + 1 \Big) \\
&+ \mathcal{C} \Big(3\Pi_Q \left(u^\tau\right)^2 \Big(\left(u^y\right)^2 + 1 \Big) \\
&+ \pi_Q^{yy} \Big(\left(u^x\right)^2 \Big(8\left(u^y\right)^2 + 3 \Big) + 6\left(u^y\right)^2 + 3 \Big) \\
&+ u^y \Big(3{\pi_Q}^\eta{}_\eta \left(u^\tau\right)^2 u^y - \pi_Q^{xy} u^x \Big(8\left(u^y\right)^2 + 5 \Big) \Big) \Big) \Big) \\
(A_y)_{12,3} = &\frac{1}{6T\left(u^\tau\right)^2 \Big(\left(u^y\right)^2 + 1 \Big)} D_s \Big(\left(u^y\right)^2 + 1 \Big(I_{31} \left(u^\tau\right)^2 u^x u^y \\
&+ 3\mathcal{B}\nu^x \Big(2\left(u^x\right)^2 + 1 \Big) u^y \Big(\left(u^y\right)^2 + 1 \Big) \\
&- 3\mathcal{B}\nu^y u^x \Big(\left(u^x\right)^2 + 1 \Big) \Big(2\left(u^y\right)^2 + 1 \Big) \\
&+ \mathcal{A}u^y \Big(-2\nu^y u^x \Big(\left(u^x\right)^2 + 1 \Big) u^y + 3\nu^x \Big(\left(u^y\right)^2 + 1 \Big) \\
&+ \nu^x \left(u^x\right)^2 \Big(2\left(u^y\right)^2 + 3 \Big) \Big) \Big) \\
&+ \mathcal{C} \Big(u^x u^y \Big(-3{\pi_Q}^\eta{}_\eta \left(u^\tau\right)^2 \left(u^y\right)^2 + \Pi_Q \left(u^\tau\right)^2 \Big(\left(u^y\right)^2 + 1 \Big) \\
&- \pi_Q^{yy} \Big(\left(u^x\right)^2 + 1 \Big) \Big(8\left(u^y\right)^2 + 5 \Big) \Big) \\
&+ \pi_Q^{xy} \Big(3\Big(\left(u^y\right)^4 + 3\left(u^y\right)^2 + 2 \Big) \\
&+ \left(u^x\right)^2 \Big(8\left(u^y\right)^4 + 8\left(u^y\right)^2 + 6 \Big) \Big) \Big) \Big) \\
(A_y)_{12,8} = &\frac{\mathcal{B}D_s \Big(\left(u^y\right)^2 + 1 \Big)}{2T} \\
(A_y)_{12,9} = &\frac{\mathcal{B}D_s u^x u^y}{6T} \\
(A_y)_{12,12} = &\frac{\mathcal{C}D_s u^y}{2T} \\
(A_y)_{13,1} = &\frac{D_s(5\mathcal{B}'(T)\nu^y + 3u^y(\mathcal{C}'(T)\Pi_Q + (\partial_T I_{31})))}{6T} \\
(A_y)_{13,2} = &\frac{D_s \Big(\mathcal{A}u^y \Big(\nu^x \left(u^y\right)^2 + \nu^x - \nu^y u^x u^y \Big) + \mathcal{C} \Big(\pi_Q^{xy} \left(u^y\right)^2 + \pi_Q^{xy} - \pi_Q^{yy} u^x u^y \Big) \Big)}{3T\left(u^\tau\right)^2} \\
(A_y)_{13,3} = &\frac{1}{6T\left(u^\tau\right)^2} D_s \Big(5I_{31} \left(u^\tau\right)^2 + 5\mathcal{C}\Pi_Q \left(u^\tau\right)^2 + 2\mathcal{A}u^y \Big(-\nu^x u^x u^y + \nu^y \left(u^x\right)^2 + \nu^y \Big) \\
&+ 2\mathcal{C} \Big(-\pi_Q^{xy} u^x u^y + \pi_Q^{yy} \left(u^x\right)^2 + \pi_Q^{yy} \Big) \Big) \\
(A_y)_{13,7} = &\frac{(\partial_\alpha I_{31}) D_s u^y}{2T} \\
(A_y)_{13,9} = &\frac{5\mathcal{B}D_s}{6T} \\
(A_y)_{13,13} = &\frac{\mathcal{C}D_s u^y}{2T} \\
S_1 = &\frac{1}{\tau \Big(\left(u^y\right)^2 + 1 \Big)} \Big(\pi^\eta{}_\eta + {\pi_Q}^\eta{}_\eta + 4a_{\mathrm{SB}} T^4 - \pi^\eta{}_\eta \left(u^x\right)^2 - \pi^{yy} \left(u^x\right)^2 - \pi_Q^{yy} \left(u^x\right)^2 \\
&- {\pi_Q}^\eta{}_\eta \left(u^x\right)^2 + 4a_{\mathrm{SB}} T^4 \left(u^x\right)^2 + 2\pi^{xy} u^x u^y + 2\pi_Q^{xy} u^x u^y + \pi^\eta{}_\eta \left(u^y\right)^2 \\
&+ \pi^{yy} \left(u^y\right)^2 + \pi_Q^{yy} \left(u^y\right)^2 + {\pi_Q}^\eta{}_\eta \left(u^y\right)^2 + 8a_{\mathrm{SB}} T^4 \left(u^y\right)^2 \\
&+ 4a_{\mathrm{SB}} T^4 \left(u^x\right)^2 \left(u^y\right)^2 + 4a_{\mathrm{SB}} T^4 \left(u^y\right)^4 + \Pi_Q \left(u^\tau\right)^2 \Big(\left(u^y\right)^2 + 1 \Big)
\end{aligned}
$$

$$
\begin{aligned}
&+ h\left(\left(u^y\right)^2+1\right)\left(n\left(u^\tau\right)^2+2(\nu^x u^x+\nu^y u^y)\right)\Big)\\
S_2 = &\frac{1}{\tau u^\tau\left(\left(u^y\right)^2+1\right)}\Big(-\pi^\eta{}_\eta u^x-\pi^{yy}u^x-\pi_Q^{yy}u^x-\pi_Q{}^\eta{}_\eta u^x+4a_{\mathrm{SB}}T^4u^x-\pi^\eta{}_\eta\left(u^x\right)^3\\
&-\pi^{yy}\left(u^x\right)^3-\pi_Q^{yy}\left(u^x\right)^3-\pi_Q{}^\eta{}_\eta\left(u^x\right)^3+4a_{\mathrm{SB}}T^4\left(u^x\right)^3+\pi^{xy}u^y+\pi_Q^{xy}u^y\\
&+2\pi^{xy}\left(u^x\right)^2u^y+2\pi_Q^{xy}\left(u^x\right)^2u^y-\pi^\eta{}_\eta u^x\left(u^y\right)^2-\pi_Q{}^\eta{}_\eta u^x\left(u^y\right)^2\\
&+8a_{\mathrm{SB}}T^4u^x\left(u^y\right)^2+4a_{\mathrm{SB}}T^4\left(u^x\right)^3\left(u^y\right)^2+\pi^{xy}\left(u^y\right)^3+\pi_Q^{xy}\left(u^y\right)^3\\
&+4a_{\mathrm{SB}}T^4u^x\left(u^y\right)^4+\Pi_Q\left(u^\tau\right)^2u^x\left(\left(u^y\right)^2+1\right)\\
&+h\left(\left(u^y\right)^2+1\right)\left(n\left(u^\tau\right)^2u^x+\nu^x\left(2\left(u^x\right)^2+\left(u^y\right)^2+1\right)+\nu^y u^x u^y\right)\Big)\\
S_3 = &\frac{1}{\tau u^\tau}\Big(\pi^{xy}u^x+\pi_Q^{xy}u^x+\Pi_Q u^y+\pi^{yy}u^y+\pi_Q^{yy}u^y+4a_{\mathrm{SB}}T^4u^y+\Pi_Q\left(u^x\right)^2u^y\\
&+4a_{\mathrm{SB}}T^4\left(u^x\right)^2u^y+\Pi_Q\left(u^y\right)^3+4a_{\mathrm{SB}}T^4\left(u^y\right)^3\\
&+h\left(n\left(u^\tau\right)^2u^y+\nu^x u^x u^y+\nu^y\left(\left(u^x\right)^2+2\left(u^y\right)^2+1\right)\right)\Big)\\
S_4 = &\frac{\delta_{\pi\pi}\pi^{yy}u^\tau}{\tau}+\pi^{yy}-\frac{2u^\tau\left(\left(u^y\right)^2+1\right)\eta}{3\tau}\\
S_5 = &\frac{3\pi^\eta{}_\eta(\delta_{\pi\pi}u^\tau+\tau)+4u^\tau\eta}{3\tau^3}\\
S_6 = &\frac{\delta_{\pi\pi}\pi^{xy}u^\tau}{\tau}+\pi^{xy}-\frac{2u^\tau u^x u^y\eta}{3\tau}\\
S_7 = &\frac{n\left(u^\tau\right)^2+\nu^x u^x+\nu^y u^y}{\tau u^\tau}\\
S_8 = &\nu^x\\
&+\frac{1}{T\tau u^\tau\left(\left(u^y\right)^2+1\right)}D_s\Big(-2\pi_Q{}^\eta{}_\eta u^x-2\pi_Q{}^\eta{}_\eta\left(u^x\right)^3-\pi_Q^{yy}\left(\left(u^x\right)^3+u^x\right)+\pi_Q^{xy}u^y\\
&+2\pi_Q^{xy}\left(u^x\right)^2u^y-3\pi_Q{}^\eta{}_\eta u^x\left(u^y\right)^2-\pi_Q{}^\eta{}_\eta\left(u^x\right)^3\left(u^y\right)^2+\pi_Q^{xy}\left(u^y\right)^3\\
&-\pi_Q{}^\eta{}_\eta u^x\left(u^y\right)^4+h\nu^x\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right)\Big)\\
S_9 = &\frac{D_s\left(h\nu^y\left(u^\tau\right)^2+\pi_Q^{xy}u^x+u^y\left(\pi_Q^{yy}-\pi_Q{}^\eta{}_\eta\left(u^\tau\right)^2\right)\right)}{T\tau u^\tau}+\nu^y\\
S_{10} = &\pi_Q^{yy}\\
&-\frac{1}{6T\tau u^\tau}D_s\Big(2I_{31}\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right)+2\mathcal{B}\left(\left(u^y\right)^2+1\right)(\nu^x u^x+\nu^y u^y)\\
&+\mathcal{C}\left(u^\tau\right)^2\left(2\Pi_Q\left(\left(u^y\right)^2+1\right)-3\pi_Q^{yy}+2\pi_Q{}^\eta{}_\eta\left(\left(u^y\right)^2+1\right)\right)\Big)\\
S_{11} = &\frac{D_s\left(4\mathcal{B}(\nu^x u^x+\nu^y u^y)+\mathcal{C}\left(u^\tau\right)^2(4\Pi_Q+7\pi_Q{}^\eta{}_\eta)+4I_{31}\left(u^\tau\right)^2\right)}{6T\tau u^\tau}+\pi_Q{}^\eta{}_\eta\\
S_{12} = &-\frac{D_s u^x u^y\left(\mathcal{B}(\nu^x u^x+\nu^y u^y)+I_{31}\left(u^\tau\right)^2\right)}{3T\tau u^\tau}+\frac{\mathcal{C}D_s u^\tau(3\pi_Q^{xy}-2u^x u^y(\Pi_Q+\pi_Q{}^\eta{}_\eta))}{6T\tau}+\pi_Q^{xy}\\
S_{13} = &\frac{D_s\left(5\mathcal{B}(\nu^x u^x+\nu^y u^y)+\mathcal{C}\left(u^\tau\right)^2(5\Pi_Q+2\pi_Q{}^\eta{}_\eta)+5I_{31}\left(u^\tau\right)^2\right)}{6T\tau u^\tau}+\Pi_Q
\end{aligned}
$$